

AS & A-Level Physics
Lecture Notes
(Year 2 / A2 Level)

BY

Yuhao Yang

Shanghai Easyday Education

June 25, 2023

THIS PAGE IS INTENTIONALLY LEFT BLANK

此页特意留空

Practical Issues

Last Update: June 25, 2023

Contacts

Email: colin-young@live.com

The latest update can be found via: <https://github.com/yuhao-yang-cy/a2physics>

About the Lecture Notes

This is a set of very concise lecture notes written for CIE A-level Physics (syllabus code 9702). Presumably the target audience of the notes are students studying the relevant course.

I have been teaching A-Level courses for many years, and I have been planning to typeset my collection of handwritten notes with $\text{\LaTeX}$ not long after my teaching career started. The project never appeared to get any close to completion for many years, so I found joy when I eventually finished this project.

These notes are supposed to be self-contained. I believe I have done my best to make the lecture notes reflect the spirit of the syllabus set by the Cambridge International Examination Board. Apart from the essential derivations and explanations, I also included a handful of worked examples and problem sets, so that you might get some rough idea about the styles of questions that you might encounter in the exams. If you are a student studying this course, I believe these notes could help you get well-prepared for the exams.

Throughout the notes, key concepts are marked red, key definitions and important formulas are boxed. But to be honest, the main reason that I wrote up these notes was not to serve any of my students, but just to give myself a goal. Since physics is such a rich and interesting subject, I cannot help sharing a small part of topics beyond the syllabus that I personally find interesting. If you see anything beyond the syllabus, they usually show up in the footnotes and are labelled with an asterisk sign (*).

实际问题最后更新：2023 年 6 月 25 日联系方式电子邮件：colin-young@live.com 最新更新可通过以下方式找到：<https://github.com/yuhao-yang-cy/a2physics> 关于讲义这是一套为 CIE A-level 物理（教学大纲代码 9702）编写的非常简洁的讲义。想必笔记的目标受众是学习相关课程的学生。我教A-Level课程很多年了，在我的教学生涯开始后不久，我就打算用LATEX排版我的手写笔记集。多年来，这个项目似乎从未接近完成，所以当我最终完成这个项目时，我感到很高兴。这些注释应该是独立的。我相信我已尽力使讲义反映剑桥国际考试委员会制定的教学大纲的精神。除了基本的推导和解释之外，我还提供了一些示例和习题集，以便您可以大致了解考试中可能遇到的问题类型。如果您是学习这门课程的学生，我相信这些笔记可以帮助您为考试做好充分准备。在整个注释中，关键概念被标记为红色，关键定义和重要公式被框起来。但说实话，我写这些笔记的主要原因并不是为了服务任何一个学生，只是为了给自己一个目标。由于物理是一门如此丰富而有趣的学科，我忍不住分享教学大纲之外的一小部分我个人认为有趣的主题。如果您看到教学大纲之外的任何内容，它们通常会出现脚注中，并标有星号 (*)。我

Note that there will be an update for the syllabus since 2022. I am still working on the updates so that the notes are tailored for the new syllabus. I am glad to see some *introductory astronomy and cosmology* being added into the new syllabus, so you will see one new chapter coming in.

The contents of *electronics* and *telecommunication* have been removed from the new syllabus. As those future engineers might want to look at these topics, so I kept the two chapters in the new edition nonetheless.

Also very importantly, I am certain that there are tons of typos in the notes. If you spot any errors, please let me know.

Literature

I borrow heavily from the following resources:

- Cambridge International AS and A Level Physics Coursebook, by *David Sang, Graham Jones, Richard Woodside* and *Gurinder Chadha*, Cambridge University Press
- International A Level Physics Revision Guide, by *Richard Woodside*, Hodder Education
- Longman Advanced Level Physics, by *Kwok Wai Loo*, Pearson Education South Asia
- Past Papers of Cambridge International A-Level Physics Examinations
- HyperPhysics Website: <http://hyperphysics.phy-astr.gsu.edu/hbase/index.html>
- Wikipedia Website: <https://en.wikipedia.org>

Copyright

This work is offered under a **CC BY-NC** (Creative Commons Attribution-Non-Commercial) license. You may remix, adapt, and build upon this work, as long as the attribution is given and the new work is non-commercial.

请注意，自 2022 年起，教学大纲将会更新。我仍在进行更新，以便笔记适合新教学大纲。我很高兴看到新课程大纲中添加了一些介绍性的天文学和宇宙学，所以你会看到一个新的章节即将到来。电子和电信的内容已从新课程大纲中删除。由于那些未来的工程师可能想要研究这些主题，所以我仍然在新版本中保留了这两章。同样非常重要的是，我确信笔记中存在大量错别字。如果您发现任何错误，请告诉我。文献我大量借用以下资源：- 剑桥国际 AS 和 A Level 物理课程手册，作者：David Sang、Graha m Jones、Richard Woodside 和 Gurinder Chadha，剑桥大学出版社 - 国际 A Lev el 物理修订指南，作者：Richard Woodside，Hodder Education - 朗文高级物理，作者：Kw ok Wai Loo，Pearson Education South Asia - 剑桥国际 A-Level 物理考试过往试卷 - HyperPhysics 网站：
<http://hyperphysics.phy-astr.gsu.edu/hbase/index.html> - 维基百科网站：
<https://en.wikipedia.org> 版权本作品根据 C C BY-NC（知识共享署名-非商业）许可提供。您可以对本作品进行重新混合、改编和构建，只要注明出处并且新作品是非商业性的。二、

CONTENTS

CHAPTER 1

Circular Motion

PAGE 1

- 1.1 angular quantities 1
angular displacement - 1 / angular velocity - 1 / uniform circular
motion - 2 / centripetal acceleration - 3
- 1.2 centripetal force 4

CHAPTER 2

Gravitational Fields

PAGE 8

- 2.1 gravitational forces 8
Newton's law of gravitation - 8 / planetary motion - 9 / apparent
weight - 10
- 2.2 gravitational fields 12
- 2.3 gravitational field strength 12
gravitational field strength - 12 / acceleration of free fall - 13 /
gravitational field lines - 13
- 2.4 gravitational potential & potential energy 14
potential energy - 14 / gravitational potential energy - 15 / gravitational
potential - 18

CHAPTER 3

Oscillation

PAGE 20

- 3.1 oscillatory motion 20
amplitude, period, frequency - 20 / phase angle - 20 / acceleration
& restoring force - 21
- 3.2 simple harmonic oscillation 22
equation of motion - 22 / examples of simple harmonics - 23 /
velocity & acceleration - 25 / vibrational energy - 28

目录内容第一章圆周运动第 1 页 1.1 角量 1 角位移 - 1 / 角速度 - 1 / 匀速圆周运动 - 2 / 向心加速度 - 3 1.2 向心力 4 第二章引力场第 8 页 2.1 引力 8 牛顿万有引力定律 - 8 / 行星运动 - 9 / 表观重量 - 10 2.2 引力场 12 2.3 引力场强度 12 引力场强度 - 12 / 自由落体加速度 - 13 / 引力场线 - 13 2.4 引力势和势能 14 势能 - 14 / 引力势能 - 15 / 引力势 - 18 第三章振荡第 20 页 3.1 振荡运动 20 振幅、周期、频率 - 20 / 相位角 - 20 / 加速度和恢复力 - 21 3.2 简谐振动 22 运动方程 - 22 / 简谐振动示例 - 23 / 速度和加速度 - 25 / 振动能量 - 28 iii

3.3	damped oscillations	29
-----	---------------------	----

light damping - 29 / heavy damping - 30 / critical damping - 30

3.4	forced oscillations	31
-----	---------------------	----

free & forced oscillation - 31 / resonance - 31 / damping & resonance - 32

CHAPTER 4

Ideal Gases

PAGE 33

4.1	gas molecules	33
-----	---------------	----

motion of gas particles - 33 / amount of molecules - 33 / pressure (qualitative view) - 34

4.2	ideal gas	35
-----	-----------	----

ideal gas equation - 35 / empirical laws - 37

4.3	kinetic theory of ideal gases	39
-----	-------------------------------	----

assumptions of ideal gas model - 39 / pressure (quantitative view) - 40 / kinetic energy - 41

CHAPTER 5

Thermodynamics

PAGE 44

5.1	thermal physics basics	44
-----	------------------------	----

temperature scales - 44 / kinetic theory of matter - 44 / specific latent heat - 45 / specific heat capacity - 46

5.2	internal energy	47
-----	-----------------	----

kinetic energy - 48 / potential energy - 48 / internal energy of ideal gas - 48 / change of states - 49 / first law of thermodynamics - 50

5.3	temperature	54
-----	-------------	----

temperature & thermal energy - 54 / absolute zero - 54 / thermometer - 55

3.3 阻尼振动 29 轻阻尼 - 29 / 重阻尼 - 30 / 临界阻尼 - 30 3.4 受迫振动 31 自由振动和受迫振动 - 31 / 共振 - 31 / 阻尼和共振 - 32 第 4 章理想气体第 33 页 4.1 气体分子 33 气体粒子的运动 - 33 / 分子数量 - 33 / 压力 (定性视图) - 34 4.2 理想气体 35 理想气体方程 - 35 / 经验定律 - 37 4.3 理想气体动力学理论 39 理想气体模型假设 - 39 / 压力 (定量观点) - 40 / 动能 - 41 第 5 章热力学第 44 页 5.1 热物理基础知识 44 温标 - 44 / 物质动力学理论 - 44 / 比潜热 - 45 / 比热容 - 46 5.2 内能 47 动能 - 48 / 势能 - 48 / 理想气体内能 - 48 / 状态变化 - 49 / 热力学第一定律 - 50 5.3 温度 54 温度与热能 - 54 / 绝对零 - 54 / 温度计 - 55 iv

- 6.1 electric forces 57
Coulomb's law - 57 / electric fields - 58
- 6.2 electric field strength 58
electric field strength - 58 / electric field lines - 60
- 6.3 potential & potential energy 61
electric potential energy - 61 / electric potential - 63 / equipotential lines - 65
- 6.4 further discussions on electric fields 67
comparison with gravitational fields - 67 / electric field inside conductors - 68 / field strength & potential - 69

- 7.1 capacitors: an introduction 74
capacitors - 74 / mutual capacitance - 75 / self-capacitance - 75
- 7.2 energy stored in a capacitor 77
- 7.3 capacitor networks 78
capacitors in parallel - 78 / capacitors in series - 79 / capacitor networks - 80 / capacitors & resistors - 81
- 7.4 charging & discharging capacitors 82
charging phase - 82 / discharging phase - 82 / time constant - 83

- 8.1 magnetism 84
magnets - 84 / magnetic field due to currents - 85 / solenoids & electromagnets - 86

第 6 章静电第 57 页 6.1 电力 57 库仑定律 - 57 / 电场 - 58 6.2 电场强度 58 电场强度 - 58 / 电场线 - 60 6.3 电势和势能 61 电势能 - 61 / 电势 - 63 / 等势线 - 65 6.4 进一步讨论电场 67 与引力场的比较 - 67 / 导体内电场 - 68 / 场强和电势 - 69 第 7 章电容器第 74 页 7.1 电容器：简介 74 电容器 - 74 / 互电容 - 75 / 自电容 - 75 7.2 电容器中存储的能量 77 7.3 电容器网络 78 个并联电容器 - 78 / 串联电容器 - 79 / 电容器网络 - 80 / 电容器和电阻器 - 81 7.4 充电和放电电容器 82 充电阶段 - 82 / 放电阶段 - 82 / 时间常数 - 83 第 8 章磁场第 84 页 8.1 磁性 84 个磁铁 - 84 / 电流产生的磁场 - 85 / 螺线管和电磁铁 - 86 V

8.2	magnetic force on current-carrying conductor	87
-----	--	----

magnetic force on current-carrying conductor - 87 / magnetic flux density - 88

8.3	magnetic force on charged particles	91
-----	-------------------------------------	----

magnetic force on charged particles - 91 / charged particle in uniform magnetic fields - 93 / mass spectrometer - 94 / cyclotron - 95 / velocity selector - 97

8.4	Hall effect	98
-----	-------------	----

CHAPTER 9

Electromagnetic Induction PAGE 100

9.1	magnetic flux	100
-----	---------------	-----

9.2	laws of electromagnetic induction	101
-----	-----------------------------------	-----

Faraday's law - 101 / Lenz's law - 101

9.3	applications of electromagnetic induction	104
-----	---	-----

magnetic braking - 104 / the generator - 106 / electromagnetic gun - 107

CHAPTER 10

Alternating Currents PAGE 108

10.1	sinusoidal a.c.	108
------	-----------------	-----

sinusoidal waveform - 108 / power - 109 / r.m.s. current & r.m.s. voltage - 110 / measurement of a.c. with an oscilloscope - 111

10.2	power supply systems	112
------	----------------------	-----

high-voltage transmission - 112 / transformers - 113 / rectifiers - 114 / smoothing - 116

CHAPTER 11

Quantum Physics Basics PAGE 118

11.1	classical theories	118
------	--------------------	-----

particle models - 118 / wave models - 119 / history of light - 119

8.2 载流导体上的磁力 87 载流导体上的磁力 - 87 / 磁通密度 - 88 8.3 带电粒子上的磁力 91 带电粒子上的磁力 - 91 / 均匀磁场中的带电粒子 - 93 / 质谱仪 - 94 / 回旋加速器 - 95 / 速度选择器 - 97 8.4 霍尔效应 98 第 9 章电磁感应第 100 页 9.1 磁通量 100 9.2 电磁感应定律 101 法拉第定律 - 101 / 楞次定律 - 101 9.3 电磁感应的应用 104 磁制动 - 104 / 发电机 - 106 / 电磁枪 - 107 第 10 章交流电第 108 页 10.1 正弦交流电 108 sinusoidal waveform - 108 / power - 109 / r.m.s. 电流和均方根值电压 - 110 / 交流电测量使用示波器 - 111 10.2 电源系统 112 高压传输 - 112 / 变压器 - 113 / 整流器 - 114 / 平滑 - 116 第 11 章量子物理基础第 118 页 11.1 经典理论 118 粒子模型 - 118 / 波模型 - 119 / 光的历史 - 119 vi

11.2	photon theory	120
------	---------------	-----

photoelectric effect - 120 / photon theory - 122 / photoelectric effect explained - 123 / photon momentum - 125

11.3	wave-particle duality	127
------	-----------------------	-----

matter waves - 127 / electron diffraction - 128 / wave-particle duality - 129

11.4	quantisation of electron energy levels	129
------	--	-----

hydrogen atom (*) - 130 / electron energy levels - 130 / atomic spectrum - 131

11.5	band theory (*)	136
------	-----------------	-----

energy bands - 136 / band theory & electrical conductivity - 137

CHAPTER 12

Nuclear Physics

PAGE 140

12.1	nuclear energy	140
------	----------------	-----

mass-energy equivalence - 140 / binding energy - 141 / nuclear stability - 142 / fusion - 143 / fission - 144 / energy release from nuclear reactions - 145

12.2	radioactive decay	147
------	-------------------	-----

radioactive decay - 147 / decay equations - 148 / half-life - 149 / measurement of radioactive decay - 151

CHAPTER 13

Medical Imaging

PAGE 154

13.1	X-ray imaging	154
------	---------------	-----

nature of X-rays - 154 / production of X-rays - 155 / X-ray spectrum - 156 / X-ray attenuation - 157 / X-ray imaging techniques - 160 / CT scans - 161

11.2 光子理论 120 光电效应 - 120 / 光子理论 - 122 / 光电效应解释 - 123 / 光子动量 - 125 11.3 波粒二象性
127 物质波 - 127 / 电子衍射 - 128 / 波粒二象性 - 129 11.4 电子能级的量子化 129 氢原子 (H) - 130 / 电子
能级 - 130 / 原子能谱 - 131 11.5 能带理论 (C) 136 能带 - 136 / 能带理论与电导率 - 137 第十二章核物理页
140 12.1 核能 140 质能当量 - 140 / 结合能 - 141 / 核稳定性 - 142 / 聚变 - 143 / 裂变 - 144 / 核反应释放的
能量 - 145 12.2 放射性衰变 147 放射性衰变 - 147 / 衰变方程 - 148 / 半衰期 - 149 / 放射性衰变的测量 - 151
第 13 章医学成像第 154 页 13.1 X 射线成像 154 X 射线的性质 - 154 / X 射线的产生 - 155 / X 射线光谱 -
156 / X 射线衰减 - 157 / X 射线成像技术 - 160 / CT 扫描 - 161 vii

13.2	ultrasonic scans	164
------	------------------	-----

nature of ultrasound - 164 / production of ultrasounds - 165 /
attenuation of ultrasounds - 166 / reflected & transmitted intensity
at boundaries of structures - 166 / ultrasound imaging - 168

13.3	MRI (*)	169
------	---------	-----

nuclear magnetic resonance - 169 / magnetic resonance imaging -
171

13.4	PET	173
------	-----	-----

positron - 173 / electron-positron annihilation - 173 / positron emis-
sion tomography (PET) - 174

CHAPTER 14

Astrophysics

PAGE 176

14.1	astronomical distances	176
------	------------------------	-----

unit of length in astronomy - 176 / luminosity - 176 / radiant flux
intensity - 177 / standard candles - 177

14.2	spectrum of stars	179
------	-------------------	-----

black body radiation - 179 / surface temperature - 180 / stellar radii
- 181

14.3	cosmology	181
------	-----------	-----

quick review: absorption spectrum - 181 / quick review: Doppler
effect - 182 / redshift - 182 / recession of galaxies - 183 / Big Bang
theory - 184 / fate of the universe (*) - 186

CHAPTER 15

Electronics (*)

PAGE 188

15.1	sensors	188
------	---------	-----

thermistor - 188 / LDR - 189 / strain gauge - 189 / piezo-electric
transducer - 190

13.2 超声波扫描 164 超声波的性质 - 164 / 超声波的产生 - 165 / 超声波的衰减 - 166 / 结构边界处的反射和透射强度 - 166 / 超声波成像 - 168 13.3 MRI (㉑) 169 核磁共振 - 169 / 磁共振成像 - 171 13.4 PET 173 正电子 - 173 / 电子 - 正电子湮灭 - 173 / 正电子发射断层扫描 (PET) - 174 第 14 章天体物理学第 176 页 14.1 天文距离 176 天文学中的长度单位 - 176 / 光度 - 176 / 辐射通量强度 - 177 / 标准烛光 - 177 14.2 恒星光谱 179 黑体辐射 - 179 / 表面温度 - 180 / 恒星半径 - 181 14.3 宇宙学 181 快速回顾：吸收光谱 - 181 / 快速回顾：多普勒效应 - 182 / 红移 - 182 / 星系衰退 - 183 / 大爆炸理论 - 184 / 宇宙的命运 (㉑) - 186 第 15 章电子器件 (㉑) 第 188 页 15.1 传感器 188 热敏电阻 - 188 / LDR - 189 / 应变计 - 189 / 压电传感器 - 190 viii

15.2	processors	191
------	------------	-----

potential divider - 191 / op-amp: basics - 192 / voltage comparator
- 194 / negative feedback - 195 / inverting amplifier - 197 / non-
inverting amplifier - 198

15.3	output devices	202
------	----------------	-----

LED - 202 / relay - 202 / calibrated meters - 203

15.4	practical electronic circuits	204
------	-------------------------------	-----

automatic illumination system - 204 / multi-range voltmeter - 205
/ digital signal regenerator - 206 / summing amplifier - 207

CHAPTER 16 **Telecommunication (*)** PAGE 209

16.1	modulation	209
------	------------	-----

amplitude modulation - 211 / frequency modulation - 214 / comparison
between AM and FM - 215

16.2	digital transmission	215
------	----------------------	-----

analogue & digital signals - 215 / regeneration - 216 / binary num-
bers - 217 / analogue-to-digital conversion - 218

16.3	communication channels	219
------	------------------------	-----

wire pairs - 220 / coaxial cables - 220 / optic fibres - 221 / radio
waves - 221 / microwaves - 223

16.4	the decibel scale	224
------	-------------------	-----

Appendix A. Glossary PAGE 227

Appendix B. List of Formula PAGE 240

Appendix C. Physical Constants PAGE 244

15.2 处理器 191 分压器 - 191 / 运算放大器：基础知识 - 192 / 电压比较器 - 194 / 负反馈 - 195 / 反相放大器 - 197 / 同相放大器 - 198 15.3 输出设备 202 LED - 202 / 继电器 - 202 / 校准仪表 - 203 15.4 实用电子器件电路 204 自动照明系统 - 204 / 多量程电压表 - 205 / 数字信号再生器 - 206 / 求和放大器 - 207 第 16 章电信 (C) 第 209 页 1 6.1 调制 209 幅度调制 - 211 / 频率调制 - 214 / AM 和 FM 之间的比较 - 215 16.2 数字传输 215 模拟和数字信号 - 215 / 再生 - 216 / 二进制数 - 217 / 模数转换 - 218 16.3 通信通道 219 线对 - 220 / 同轴电缆 - 220 / 光纤 - 221 / 无线电波 - 221 / 微波 - 223 16.4 分贝标度 224 附录 A. 术语表页 227 附录 B. 分子式列表第 240 页附录 C. 物理常数第 244 页 ix

CHAPTER 1

Circular Motion

1.1 angular quantities

movement or rotation of an object along a circular path is called **circular motion**

to describe a circular motion, we can use *angular quantities*, which turn out to be more useful than linear displacement, linear velocity, etc.

1.1.1 angular displacement

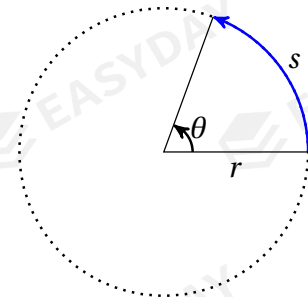
angular displacement is angle swiped out by object moving along circular

➤ unit: $[\theta] = \text{rad}$ (natural unit of measurement for angles)

conversion rule: $2\pi \text{ rad} = 360^\circ$

➤ if two radii form an angle of θ , then length of arc: $s = r\theta$

two radii subtending an arc of same length as radius form an angle of one **radian**



1.1.2 angular velocity

angular velocity describes how fast an object moves along a circular path

angular velocity is defined as angular displacement swiped out per unit time:

$$\omega = \frac{\Delta\theta}{\Delta t}$$

➤ unit of: $[\omega] = \text{rad s}^{-1}$, also in radian measures

➤ angular velocity is a *vector* quantity

this vector points in a direction normal to the plane of circular motion

杨宇浩 1 圆周运动第一章圆周运动 1.1 角量物体沿圆周路径的运动或旋转称为圆周运动来描述圆周运动，我们可以用角量来描述圆周运动，它比线性位移、线速度等更有用。

1.1.1 角位移角位移是物体沿圆周运动所扫出的角度 $r\theta$ s $\hat{a}$ 单位： $[\theta] = \text{rad}$ （自然单位角度的测量）转换规则： $2\pi \text{ rad} = 360^\circ$ $\hat{a}$ 如果两个半径形成 θ 角，则弧长： $s = r\theta$ 与半径相同长度的圆弧所对的两个半径形成一弧度的角

1.1.2 角速度角速度描述物体沿圆形路径移动的速度角速度定义为单位时间扫出的角位移： $\omega = \frac{\Delta\theta}{\Delta t}$ $\hat{a}$ 单位： $[\omega] = \text{rad s}^{-1}$ ，也以弧度为单位 $\hat{a}$ 角速度是一个矢量，该矢量指向垂直于圆周运动平面的方向 1

but in A-level course, we treat angular velocity as if it is a scalar

angular velocity and angular speed may be considered to be the same idea

➤ relation with linear velocity

in interval Δt , distance moved along arc $\Delta s = v\Delta t = r\Delta\theta \Rightarrow \omega = \frac{\Delta\theta}{\Delta t} = \frac{v}{r} \Rightarrow \boxed{v = \omega r}$

this relation between linear speed and angular speed holds at any instant

1.1.3 uniform circular motion

when studying linear motion, we started from motion with constant velocity v

consider the simplest possible circular motion $\rightarrow$ circular motion with constant ω

analogy with linear motion with constant v

uniform linear motion: $s = vt$

displacement $s \leftrightarrow \theta$, velocity $v \leftrightarrow \omega$

for uniform circular motion, one has: $\boxed{\theta = \omega t}$

➤ time taken for one complete revolution is called **period** T

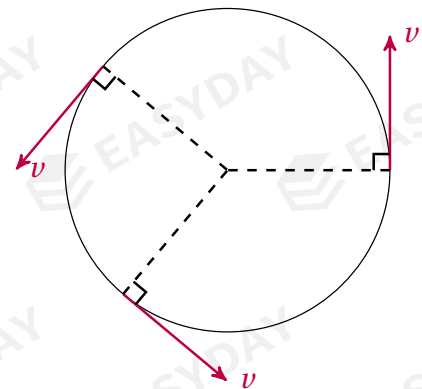
in one T , angle swiped is 2π , so $\boxed{\omega = \frac{2\pi}{T}}$

➤ uniform circular motion is still *accelerated* motion

speed is unchanged, but *velocity* is changing

direction of velocity always *tangential* to its path, so direction of velocity keeps changing

in general, any object moving along circular path is accelerating



Example 1.1 An object undergoes a uniform motion around a circular track of radius 2.5 m in 40 s, what is its angular speed and linear speed?

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{40} \approx 0.157 \text{ rad s}^{-1} \quad v = \omega r = 0.157 \times 2.5 \approx 0.39 \text{ m s}^{-1} \quad \square$$

Question 1.1 What is the angular velocity of the minute hand of a clock?

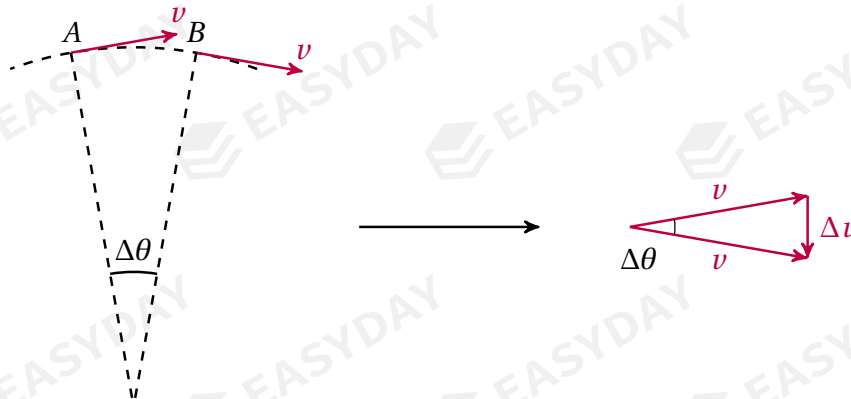
Question 1.2 A spacecraft moves around the earth in a circular orbit. The spacecraft has a speed of 7200 m s^{-1} at a height of 1300 km above the surface of the earth. Given that the radius of the earth is 6400 km. (a) What is the angular speed of this spacecraft? (b) What is its period?

杨宇浩 1 圆周运动但在 A-level 课程中，我们将角速度视为标量角速度，角速度可以认为是相同的想法 $\hat{\omega}$ 与区间 Δt 内的线速度的关系，沿弧线移动的距离 $\Delta s = v \Delta t = r \Delta \theta$ $\omega = \Delta \theta / \Delta t = v / r$ $v = \omega r$ 线速度和角速度之间的这种关系在任何情况下都成立瞬间 1.1.3 匀速圆周运动在研究直线运动时，我们从匀速 v 运动开始考虑最简单的圆周运动！匀速圆周运动 v v 与匀速直线运动类比： $s = vt$ 位移 s θ ，速度 v ω 对于匀速圆周运动，有： $\theta = \omega t$ $\hat{\omega}$ 完成一转所需的时间称为周期 T ，转动的角度为 2π ，所以 $\omega = 2\pi / T$ $\hat{\omega}$ 匀速圆周运动仍然是加速运动，速度不变，但速度在变化，速度方向总是与其路径相切，所以一般情况下，速度方向不断变化，任何沿圆周路径运动的物体都在加速例 1.1 一个物体在 40 s 内绕半径 2.5 m 的圆周轨迹做匀速运动，其角速度和线速度是多少？
 $\omega = 2\pi / T = 2\pi / 40 \approx 0.157 \text{ rad s}^{-1}$ $v = \omega r = 0.157 \times 2.5 \approx 0.39 \text{ m s}^{-1}$
 ä 问题 1.1 时钟分针的角速度是多少？问题 1.2 航天器沿圆形轨道绕地球运行。该航天器在距地球表面 1300 公里的高度处的速度为 7200 米·秒⁻¹。假设地球半径为 6400 公里。(a) 该航天器的角速度是多少？(b) 它的周期是多少？ 2

1.1.4 centripetal acceleration

centripetal acceleration is the acceleration due to the change in direction of velocity vector, it points toward the centre of circular path

consider motion along a circular path from A to B with constant speed v
under small (infinitesimal) duration of time Δt ^[1]



change in velocity: $\Delta v = 2v \sin \frac{\Delta\theta}{2} \approx v\Delta\theta$ (as $\Delta\theta \rightarrow 0$, $\sin \Delta\theta \approx \Delta\theta$)

acceleration: $a = \frac{\Delta v}{\Delta t} \approx v \frac{\Delta\theta}{\Delta t} = v\omega$ (as $\omega = \frac{\Delta\theta}{\Delta t}$)

recall relation $v = \omega r$, we find centripetal acceleration: $a_c = \frac{v^2}{r} = \omega^2 r$

- direction of centripetal acceleration: always towards centre of circular path
- centripetal acceleration is only responsible for the change in *direction* of velocity
change in *magnitude* of velocity will give rise to *tangential acceleration*

this is related to *angular acceleration*^[2], which is beyond the syllabus

Question 1.3 A racing car makes a 180° turn in 2.0 s. Assume the path is a semi-circle with a radius of 30 m and the car maintains a constant speed during the turn. (a) What is the angular velocity of the car? (b) What is the centripetal acceleration?

^[1] A more rigorous derivation can be given by using differentiation techniques

^[2] Angular acceleration is analogous to linear acceleration a , defined as rate of change of angular velocity: $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ (*). Similar to $v = \omega r = \frac{ds}{dt}$, the relation $a = \alpha r = \frac{dv}{dt}$ also holds.

杨宇浩 1 圆周运动 1.1.4 向心加速度向心加速度是由于速度矢量方向变化而产生的加速度，它指向圆周路径的中心。考虑在小（无穷小）持续时间 Δt 下，以恒定速度 v 从 A 到 B 的圆周路径运动 [1] $v \rightarrow v_B$ $\Delta \theta$ $v \rightarrow v_B$ Δv $\Delta \theta$ 速度变化： $\Delta v = 2v \sin \frac{\Delta \theta}{2} \approx v \Delta \theta$ (as $\Delta \theta \rightarrow 0$, $\sin \frac{\Delta \theta}{2} \approx \frac{\Delta \theta}{2}$) 加速度： $a = \frac{\Delta v}{\Delta t} \approx \frac{v \Delta \theta}{\Delta t}$ $\Delta t = \frac{v \Delta \theta}{a}$ (作为 $\omega = \frac{\Delta \theta}{\Delta t}$) 回忆关系 $v = \omega r$ ，我们发现向心加速度： $a_c = \frac{v^2}{r} = \omega^2 r$ $\hat{a}$ 向心加速度的方向：总是朝向圆形路径的中心 $\hat{a}$ 向心加速度只负责速度方向的变化，速度大小的变化会产生切向加速度，这与角加速度有关 [2]，这超出了教学大纲问题 1.3 一辆赛车在 2.0 秒内完成 180° 转弯。假设路径是半径为 30 m 的半圆，并且汽车在转弯过程中保持恒定速度。(a) 汽车的角速度是多少？(b) 向心加速度是多少？

[1] 通过使用微分技术可以给出更严格的推导 [2] 角加速度类似于线性加速度 a ，定义为角速度的变化率： $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ (C)。与 $v = \omega r = \frac{ds}{dt}$ 类似，关系 $a = \alpha r = \frac{dv}{dt}$ 也成立。

1.2 centripetal force

circular motion must involve change in velocity, so object is not in equilibrium

there must be a *net force* on an object performing circular motion

centripetal force (F_c) is the resultant force acting on an object moving along a circular path, and it is always directed towards centre of the circle

- centripetal force causes centripetal acceleration

using Newton's 2nd law: $F_c = m \frac{v^2}{r} = m\omega^2 r$

- F_c is not a new force by nature, it can have a variety of origins

F_c is a resultant of forces you learned before (weight, tension, contact force, friction, etc.)

- F_c acts at right angle to direction of velocity

or equivalently, if $F_{\text{net}} \perp v$ and F_{net} is of constant magnitude

then this net force provides centripetal force for circular motion

- effect of F_c : change *direction* of motion, or maintain circular orbits

to change *magnitude* of velocity, there requires a *tangential* component for the net force

again the idea of tangential force is beyond the syllabus

Example 1.2 A rock is able to orbit around the earth near the earth's surface. Let's ignore air resistance for this question, so the rock is acted by weight only. Given that radius of the earth $R = 6400$ km. (a) What is the orbital speed of the rock? (b) What is the orbital period?

- ✍ weight of object provides centripetal force: $mg = \frac{mv^2}{R}$

orbital speed: $v = \sqrt{gR} = \sqrt{9.81 \times 6.4 \times 10^6} \approx 7.9 \times 10^3 \text{ m s}^{-1}$

period: $T = \frac{2\pi R}{v} = \frac{2\pi \times 6.4 \times 10^6}{7.9 \times 10^3} \approx 5.1 \times 10^3 \text{ s} \approx 85 \text{ min}$ □

Example 1.3 A turntable can rotate freely about a vertical axis through its centre. A small object is placed on the turntable at distance $d = 40$ cm from the centre. The turntable is then set to rotate, and the angular speed of rotation is slowly increased. The coefficient of friction between the object and the turntable is $\mu = 0.30$. If the object does not slide off the turntable, find the maximum number of revolutions per minute.

杨宇豪 1 圆周运动 1.2 向心力圆周运动必然涉及速度的变化，因此物体不处于平衡状态，做圆周运动的物体上必然存在净力。向心力 (F_c) 是作用在沿圆周路径运动的物体上的合力，并且它总是指向圆心。向心力引起向心加速度，使用牛顿第二定律： $F_c = m v^2 / r = m \omega^2 r$ $\hat{F}_c$ 本质上不是一种新力，它可以有多种来源 F_c 是您之前学到的力（重量、张力、接触力、摩擦力等）的合成物 $\hat{F}_c$ 与速度方向成直角或等效地作用，如果 $F_{net} \perp v$ 和 F_{net} 大小恒定，则该合力为圆周运动提供向心力 $\hat{F}_c$ 的影响：改变运动方向，或保持圆形轨道以改变速度大小，则合力需要切向分量，再次，切向力的概念超出了教学大纲示例 1.2 一块岩石能够在地球表面附近绕地球运行。这个问题我们忽略空气阻力，因此岩石仅受重量作用。已知地球半径 $R = 6400$ 公里。(a) 岩石的轨道速度是多少？(b) 轨道周期是多少？- 物体重量提供向心力： $mg = m v^2 / R$ 轨道速度： $v = \sqrt{g R} = \sqrt{9.81 \times 6.4 \times 10^6} \approx 7.9 \times 10^3 \text{ m s}^{-1}$ 周期： $T = 2\pi R / v = 2\pi \times 6.4 \times 10^6 / 7.9 \times 10^3 \approx 5.1 \times 10^3 \text{ s} \approx 85 \text{ 分钟}$ 示例 1.3 转盘可以绕通过其中心的垂直轴自由旋转。将一个小物体放置在转盘上，距离中心 $d = 40$ 厘米。然后将转盘设置为旋转，并且旋转的角速度缓慢增加。物体与转盘之间的摩擦系数为 0.30。如果物体没有从转盘上滑落，则找出每分钟的最大转数。 4

if object stays on turntable, friction provides the centripetal force required: $f = m\omega^2 d$

increasing ω requires greater friction to provide centripetal force

but maximum limiting friction possible is: $f_{\text{lim}} = \mu N = \mu mg$, therefore

$$f \leq f_{\text{lim}} \Rightarrow m\omega^2 d \leq \mu mg \Rightarrow \omega^2 \leq \frac{\mu g}{d} \Rightarrow \omega_{\text{max}} = \sqrt{\frac{0.30 \times 9.81}{0.40}} \approx 2.71 \text{ rad s}^{-1}$$

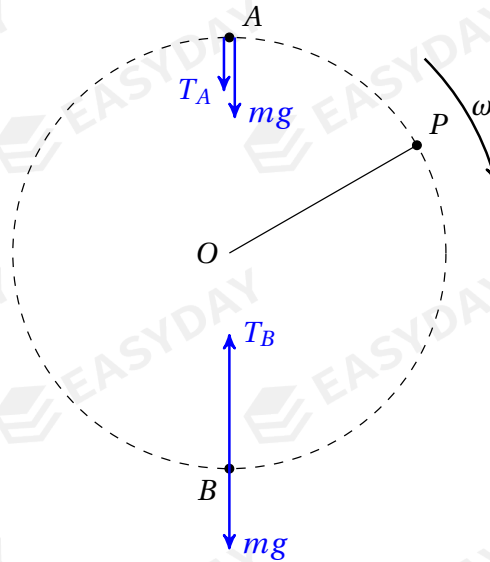
$$\text{period of revolution: } T_{\text{min}} = \frac{2\pi}{\omega_{\text{max}}} = \frac{2\pi}{2.71} \approx 2.32 \text{ s}$$

$$\text{number of revolutions in one minute: } n_{\text{max}} = \frac{t}{T_{\text{min}}} = \frac{60}{2.32} \approx 25.9$$

□

Example 1.4 Particle P of mass $m = 0.40 \text{ kg}$ is attached to one end of a light inextensible string of length $r = 0.80 \text{ m}$. The particle is whirled at a constant angular speed ω in a vertical plane.

(a) Given that the string never becomes slack, find the minimum value of ω . (b) Given instead that the string will break if the tension is greater than 20 N , find the maximum value of ω .



at top of circle (point A): $F_c = T_A + mg = m\omega^2 r \Rightarrow T_A = m\omega^2 r - mg$

at bottom of circle (point B): $F_c = T_B - mg = m\omega^2 r \Rightarrow T_B = m\omega^2 r + mg$

tension is minimum at A, but string being taut requires $T \geq 0$ at any point, so $T_A \geq 0$

$$m\omega^2 r - mg \geq 0 \Rightarrow \omega^2 \geq \frac{g}{r}$$

$$\omega_{\text{min}} = \sqrt{\frac{g}{r}} = \sqrt{\frac{9.81}{0.80}} \approx 3.5 \text{ rad s}^{-1}$$

tension is maximum at B, but string does not break requires $T \leq T_{\text{max}}$, so $T_B \leq T_{\text{max}}$

$$m\omega^2 r + mg \leq T_{\text{max}} \Rightarrow \omega^2 \leq \frac{T_{\text{max}}}{m} - \frac{g}{r}$$

$$\omega_{\text{max}} = \sqrt{\frac{T_{\text{max}}}{m} - \frac{g}{r}} = \sqrt{\frac{20}{0.40} - \frac{9.81}{0.80}} \approx 6.1 \text{ rad s}^{-1}$$

□

Yuhao Yang | 圆周运动 - 如果物体停留在转盘上，摩擦力提供所需的向心力： $f = m\omega^2 r$ 增加 ω 需要更大的摩擦力来提供向心力，但可能的最大限制摩擦力为： $f_{\text{lim}} = N = (mg)$ ，因此 $f \leq f_{\text{lim}} \Rightarrow m\omega^2 r \leq (mg) \Rightarrow \omega \leq \sqrt{g/r}$ $\omega_{\text{max}} = \sqrt{9.81/0.40} \approx 2.71 \text{ rad s}^{-1}$ 旋转周期： $T_{\text{min}} = 2\pi/\omega_{\text{max}} = 2\pi/2.71 \approx 2.32 \text{ s}$ 一分钟内的转数： $n_{\text{max}} = 60/T_{\text{min}} = 60/2.32 \approx 25.9$ 示例 1.4 质量为 $m = 0.40 \text{ kg}$ 的粒子 P 附着在灯的一端长度为 $r = 0.80 \text{ m}$ 的不可伸展弦。粒子在垂直平面内以恒定角速度 ω 旋转。(a) 假设绳子永远不会松弛，求 ω 的最小值。(b) 假定如果张力大于 20 N ，绳子就会断裂，求 ω 的最大值。

○ B A TB mg mgT A P

ω - 圆顶部 (A 点) : $F_c = T_A + mg = m\omega^2 r \Rightarrow T_A = m\omega^2 r - mg$ 圆底部 (B 点) : $F_c = T_B - mg = m\omega^2 r \Rightarrow T_B = m\omega^2 r + mg$ 张力在 A 处最小，但拉紧弦需要在任何点 $T \geq 0$ ，因此 $T_A \geq 0 \Rightarrow m\omega^2 r - mg \geq 0 \Rightarrow \omega^2 \geq gr/r$ $\omega_{\text{min}} = \sqrt{gr/r} = \sqrt{9.81/0.80} \approx 3.5 \text{ rad s}^{-1}$ 在 B 处张力最大，但弦不断裂需要 $T \leq T_{\text{max}}$ ，所以 $T_B \leq T_{\text{max}} \Rightarrow m\omega^2 r + mg \leq T_{\text{max}} \Rightarrow \omega^2 \leq (T_{\text{max}} - mg)/m - gr/r$ $\omega_{\text{max}} = \sqrt{(20 - 9.81 \cdot 0.40)/0.40 - 9.81/0.80} \approx 6.1 \text{ rad s}^{-1}$

5

Example 1.5 A pendulum bob of mass 120 g moves at constant speed and traces out a circle of radius $r = 10$ cm in a horizontal plane. The string makes an angle $\theta = 25^\circ$ to the vertical. (a) What is the tension in the string? (b) At what speed is the bob moving?

vertical component of tension T_y equals weight

$$T_y = mg \Rightarrow T \cos \theta = mg$$

$$T = \frac{mg}{\cos \theta} = \frac{0.12 \times 9.81}{\cos 25^\circ} \approx 1.3 \text{ N}$$

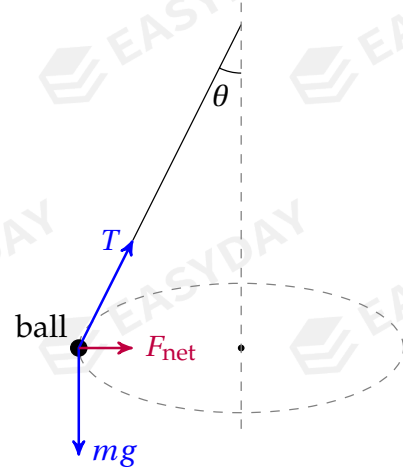
net force equals horizontal component of tension T_x

so component T_x provides centripetal force

$$F_c = T_x \Rightarrow T \sin \theta = \frac{mv^2}{r}$$

by eliminating T and m , one can find

$$v^2 = \frac{r \tan \theta}{g} = \frac{0.10 \times \tan 25^\circ}{9.81} \Rightarrow v \approx 0.069 \text{ m s}^{-1} \quad \square$$



Example 1.6 A small ball of mass m is attached to an inextensible string of length l . The ball is held with the string taut and horizontal and is then released from rest.

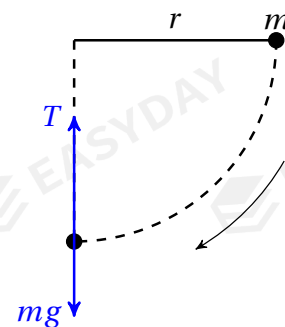
When the ball reaches lowest point, find its speed and the tension in the string in terms of m and l .

energy conservation: G.P.E. loss = K.E. gain

$$mgr = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gr}$$

at lowest point: $F_c = T - mg = m\frac{v^2}{r}$

$$T = mg + m\frac{v^2}{r} = mg + m\frac{2gr}{r} = 3mg \quad \square$$



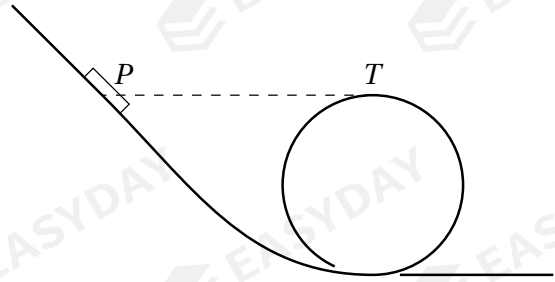
Question 1.4 Suggest what provides centripetal force in the following cases. (a) An athlete running on a curved track. (b) An aeroplane banking at a constant altitude. (c) A satellite moving around the earth.

Question 1.5 A turntable that can rotate freely in a horizontal plane is covered by dry mud. When the angular speed of rotation is gradually increased, state and explain whether the mud near edge of the plate or near the mud will first leave the plate?

Question 1.6 A bucket of water is swung at a constant speed and the motion describes a circle of radius $r = 1.0$ m in the vertical plane. If the water does not pour down from the bucket even when it is at the highest position, how fast do you need to swing the bucket?

Yuhao Yang | 圆周运动例 1.5 质量为 120 g 的摆锤以恒定速度运动，并在水平面上画出一个半径为 $r = 10$ cm 的圆。绳子与垂直方向成角度 $\theta = 25^\circ$ 。(a) 绳子的张力是多少？(b) 摆锤移动的速度是多少？球 mg T F_{net} θ - 张力的垂直分量 T_y 等于重量 $T_y = mg$ $T \cos \theta = mg$ $T = mg \cos \theta = 0.12 \times 9.81 \cos 25^\circ \approx 1.3$ N 净力等于张力 T_x 的水平分量，因此分量 T_x 提供向心力 $F_c = T_x$ $T \sin \theta = mv^2/r$ 通过消除 T 和 m ，可以得到 $v^2 = r \tan \theta g = 0.10 \times \tan 25^\circ \times 9.81$ $v \approx 0.069 \text{ m s}^{-1}$ 例 1.6 一个质量为 m 的小球附在一根长度为 l 的不可拉伸的绳子上。用拉紧且水平的绳子固定球，然后将球从静止状态释放。 mr T mg 当球到达最低点时，求其速度以及以 m 和 l 表示的绳子张力。 - 节能：G.P.E. 损失 = K.E. 增益 $mgr = \frac{1}{2}mv^2$ $v = \sqrt{2gr}$ 在最低点： $F_c = T - mg = mv^2/r$ $T = mg + mv^2/r = mg + m2gr/r = 3mg$ 问题 1.4 指出在下列情况下什么提供了向心力。(a) 运动员在弯曲的跑道上跑步。(b) 飞机在恒定高度倾斜。(c) 绕地球运行的卫星。问题 1.5 一个可以在水平面内自由旋转的转盘上覆盖着干泥。当旋转角速度逐渐增加时，请说明并解释靠近板边缘的泥浆或靠近泥浆的泥浆是否会首先离开板子？问题 1.6 一桶水以恒定速度摆动，其运动在垂直平面上画出半径为 $r = 1.0\text{m}$ 的圆。如果水桶在最高位置时水也没有从水桶中流下来，你需要以多快的速度摆动水桶？ 6

Question 1.7 This question is about the design of a roller-coaster. We consider a slider that starts from rest from a point P and slides along a frictionless circular track as sketched below. P is at the same height as the top of the track T . (a) Show that the slider cannot get to T . (b) As a designer for a roller-coaster, you have to make sure the slider can reach point T and continue to slide along the track, what is the minimum height for the point of release?



Yuhao Yang I CIRCULAR MOTION TP 问题1.7 本题是关于过山车的设计。我们考虑一个滑块，它从 P 点开始静止，沿着无摩擦的圆形轨道滑动，如下图所示。P 与轨道 T 的顶部处于相同高度。(a) 表明滑块无法到达 T。(b) 作为过山车的设计者，你必须确保滑块能够到达 T 点并继续沿着轨道滑动，释放点的最小高度是多少？ 7

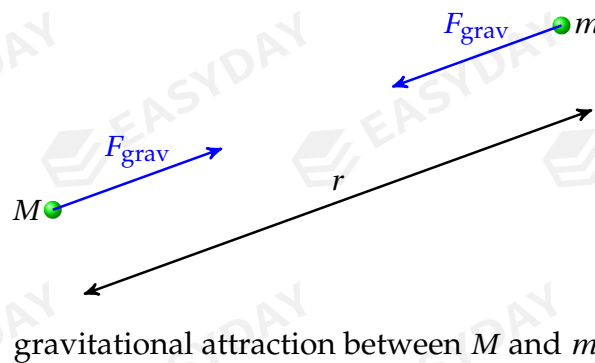
CHAPTER 2

Gravitational Fields

2.1 gravitational forces

2.1.1 Newton's law of gravitation

any object attracts any other object through the gravitational force



Newton's law of gravitation states that gravitational force between two *point* masses is proportional to the product of their masses and inversely proportional to the square of their distance $\left(F_{\text{grav}} \propto \frac{Mm}{r^2}\right)$

this law was formulated in *Issac Newton's* work 'The Principia', or 'Mathematical Principles of Natural Philosophy', first published in 1687

mathematically, gravitational force takes the form:

$$F_{\text{grav}} = \frac{GMm}{r^2}$$

$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ is the *gravitational constant*

- gravitational force is always *attractive*
- gravity is *universal*, i.e., gravitational attraction acts between *any* two masses
- Newton's law of gravitation refers to *point masses*

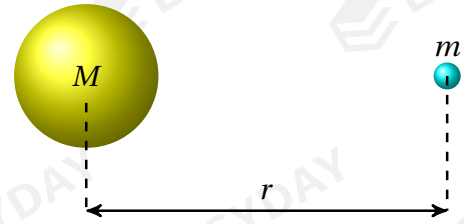
i.e., particles with no size, therefore distance r can be easily defined

杨宇浩 2 GRA 生命场第二章引力场 2.1 万有引力 2.1.1 牛顿万有引力定律任何物体通过万有引力吸引任何其他物体 M m r F_{grav} F_{grav} M 和 m 之间的万有引力牛顿万有引力定律指出，两点质量之间的万有引力与其质量的乘积成正比，与距离的平方成反比 ($F_{\text{grav}} \propto \frac{M m}{r^2}$) 该定律在伊萨克·牛顿 1687 年首次出版的著作《自然哲学的数学原理》中得到阐述，从数学上讲，引力的形式为： $F_{\text{grav}} = G \frac{M m}{r^2}$ $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ 是引力常数 $\hat{a}$ 引力总是有吸引力的重力是普遍存在的，即万有引力作用在任何两个质量之间牛顿万有引力定律指的是点质量，即没有大小的粒子，因此距离 r 可以很容易地定义 8

➤ a sphere with uniform mass distribution (e.g., stars, planets) can be treated as a *point model*

distance r is taken between centres of the spheres [3]

(see Example 2.8, field lines around a planet *seem* to point towards centre of planet)

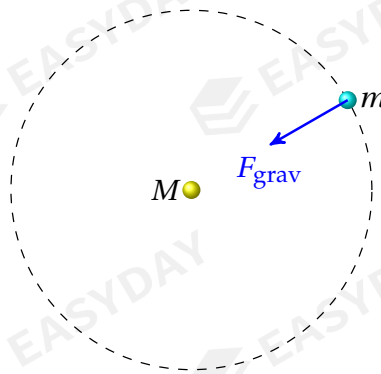


Example 2.1 The Earth can be thought as a uniform sphere of radius $R = 6.4 \times 10^6$ m and mass $M = 6.0 \times 10^{24}$ kg. Estimate the gravitational force on a man of 60 kg at sea level.

$$F = \frac{GMm}{R^2} = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 60}{(6.4 \times 10^6)^2} \approx 586 \text{ N} \quad \square$$

Question 2.1 Estimate the gravitational force between you and your deskmate.

2.1.2 planetary motion



a planet/satellite orbiting around a star/earth

a planet/satellite can move around a star/earth in circular orbit

circular motion requires centripetal force

for these objects, gravitational force provides centripetal force

$$F_{\text{grav}} = F_c \Rightarrow \boxed{\frac{GMm}{r^2} = \frac{mv^2}{r}} \quad \text{or} \quad \boxed{\frac{GMm}{r^2} = m\omega^2 r}$$

Example 2.2 GPS (Global Positioning System) satellites move in a circular orbits at about 20000 km above the earth's surface. The Earth has a radius $R = 6.4 \times 10^6$ m and mass $M = 6.0 \times 10^{24}$ kg.

(a) Find the speed of GPS satellites. (b) Find its orbital period.

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{6.4 \times 10^6 + 2.0 \times 10^7}} \approx 3.9 \times 10^3 \text{ m s}^{-1}$$

[3] This is known as *shell theorem*: a spherically symmetric shell (i.e., a hollow ball) affects external objects gravitationally as though all of its mass were concentrated at its centre, and it exerts no net gravitational force on any object inside, regardless of the object's location within the shell. (*)

Yuhao Yang 2 GRAVITATIONAL Fields $\hat{r}$ 一个具有均匀质量分布的球体（例如，恒星、行星）可以被视为一个点模型，在球体中心之间取距离 r [3]（参见例 2.8，行星周围的场线似乎指向行星的中心）例 2.1 地球可以被认为是一个半径为 $R = 6.4 \times 10^6 \text{ m}$ 且质量为 M 的均匀球体 $= 6.0 \times 10^{24} \text{ kg}$ 。估计 60 公斤重的人在水平面上受到的重力。 - $F = G \frac{Mm}{R^2} = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 60 / (6.4 \times 10^6)^2 = 586 \text{ N}$ 问题 2.1 估计你和你同桌之间的重力。 2.1. 2 行星运动 Mm F_{grav} 围绕恒星/地球运行的行星/卫星行星/卫星可以在圆形轨道上围绕恒星/地球运动这些物体的圆周运动需要向心力，重力提供向心力 $F_{\text{grav}} = F_c$) $G \frac{Mm}{r^2} = mv^2/r$ 或 $G \frac{Mm}{r^2} = m\omega^2 r$ 示例 2.2 GPS（全球定位）System）卫星在距离地球表面约 20000 公里的圆形轨道上运行。地球的半径 $R = 6.4 \times 10^6 \text{ m}$ ，质量 $M = 6.0 \times 10^{24} \text{ kg}$ 。(a) 求 GPS 卫星的速度。(b) 求其轨道周期。 - $G \frac{Mm}{r^2} = mv^2/r$) $v = \sqrt{GM/r} = \sqrt{6.67 \times 10^{-11} \times 6.0 \times 10^{24} / 6.4 \times 10^6 + 2.0 \times 10^7} = 3.9 \times 10^3 \text{ m s}^{-1}$ [3] 这称为壳定理：球对称壳（即空心球）会通过重力影响外部物体，就好像其所有质量都集中在其中心一样，并且无论物体在壳内的位置如何，它都不会对内部任何物体施加净重力。(Q)9

$$v = \frac{2\pi r}{T} \Rightarrow T = \frac{2\pi r}{v} = \frac{2\pi \times (6.4 \times 10^6 + 2.0 \times 10^7)}{3.9 \times 10^3} \approx 4.3 \times 10^4 \text{ s} \approx 11.8 \text{ hours} \quad \square$$

Example 2.3 A **geostationary satellite** moves in a circular orbit that appears motionless to ground observers. The satellite follows the Earth's rotation, so the satellite rotates from west to east above equator with an orbital period of 24 hours. Find the radius of this orbit.

$$\begin{aligned} \frac{GMm}{r^2} &= m\omega^2 r \Rightarrow \frac{GMm}{r^2} = m \left(\frac{2\pi}{T} \right)^2 r \Rightarrow r^3 = \frac{GMT^2}{4\pi^2} \\ r &= \left(\frac{GMT^2}{4\pi^2} \right)^{1/3} = \left(\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (24 \times 3600)^2}{4\pi^2} \right)^{1/3} \approx 4.23 \times 10^7 \text{ m} \quad \square \end{aligned}$$

Example 2.4 Assuming the planets in the solar system all move around the sun in circular orbits, show that the square of orbital period is proportional to the cube of orbital radius. [4]

$$\frac{GMm}{r^2} = m\omega^2 r \Rightarrow \frac{GMm}{r^2} = m \left(\frac{2\pi}{T} \right)^2 r \Rightarrow T^2 = \frac{4\pi^2}{GM} \cdot r^3$$

G is gravitational constant, M is mass of the sun, so $\frac{4\pi^2}{GM}$ is a constant, so $T^2 \propto r^3$ $\square$

Question 2.2 Given that it takes about 8.0 minutes for light to travel from the sun to the earth.

(a) What is the mass of the sun? (b) At what speed does the earth move around the sun?

2.1.3 apparent weight

an object's *actual weight* is the gravitational attraction exerted by the earth's gravity

an object's *apparent weight* is the upward force (e.g., normal contact force exerted by ground, tension in a spring balance, etc.) that opposes gravity and prevents the object from falling

apparent weight can be different from actual weight due to vertical acceleration or buoyancy

but if we consider rotation of the earth, this also causes apparent weight to be lessened

object resting on ground is actually rotating together with earth

resultant of gravitational force and contact force should provide centripetal force

$$\text{for object on equator: } F_{c,eq} = m\omega^2 R \Rightarrow F_{grav} - N_{eq} = m\omega^2 R \Rightarrow N_{eq} = \frac{GMm}{R^2} - m\omega^2 R$$

$$\text{for object at poles: } F_{c,pole} = 0 \Rightarrow F_{grav} - N_{pole} = 0 \Rightarrow N_{pole} = \frac{GMm}{R^2}$$

at lower latitudes, object describe larger circles, hence requires greater centripetal force

[4] This is known as *Kepler's 3rd law* for planetary motions. In the early 17th century, German astronomer Johannes Kepler discovered three scientific laws which describes how planets move around the sun. This $T^2 \propto r^3$ relation not only holds for circular orbits but are also correct for elliptical orbits.

Isaac Newton proved that Kepler's laws are consequences of his own law of universal gravitation, and therefore explained why the planets move in this way.

Yuhao Yang 2 GRAVITATIONAL Fields $v = 2\pi r / T$ $T = 2\pi r / v = 2\pi \times (6.4 \times 10^6 + 2.0 \times 10^7) / 3.9 \times 10^3 \approx 4.3 \times 10^4 \text{ s} \approx 11.8 \text{ 小时}$ $\ddot{a}$ 示例 2.3 一颗对地静止卫星在圆形轨道上运行，对地面观察者来说，该轨道看起来是静止的。卫星跟随地球自转，因此卫星在赤道上空自西向东自转，轨道周期为 24 小时。求该轨道的半径。

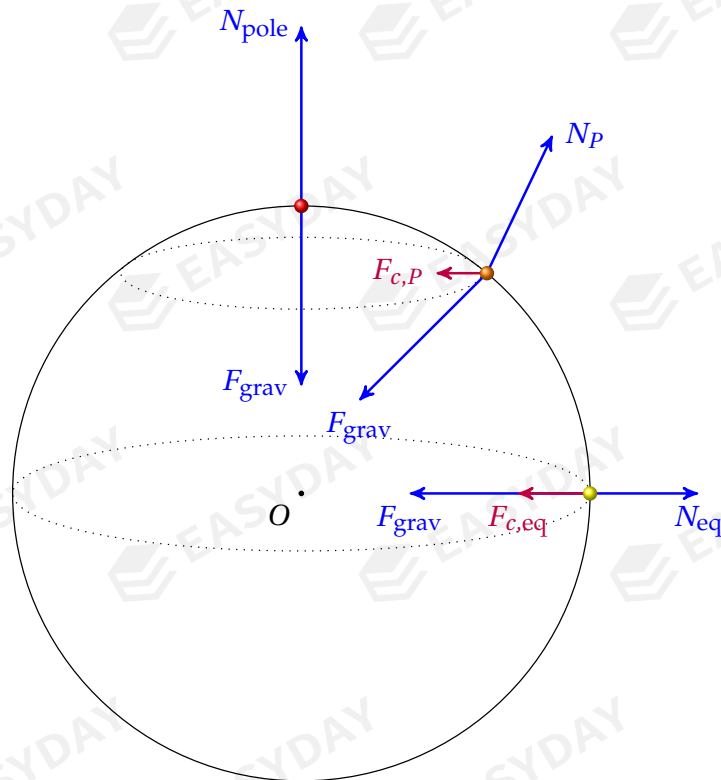
$-GMm/r^2 = m\omega^2 r$ $GMm/r^2 = m(2\pi/T)^2 r$ $r^3 = GM T^2 / 4\pi^2$ $r = (GM T^2 / 4\pi^2)^{1/3} = (6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (24 \times 3600)^2 / 4\pi^2)^{1/3} \approx 4.23 \times 10^7 \text{ m}$ $\ddot{a}$ 例 2.4 假设太阳系中的行星都以圆形轨道绕太阳运行，证明轨道周期的平方与轨道半径的立方成正比。

[4] $-GMm/r^2 = m\omega^2 r$ $GMm/r^2 = m(2\pi/T)^2 r$ $T^2 = 4\pi^2 GM / \omega^2 r^3$ G 是引力常数， M 是太阳的质量，所以 $4\pi^2 GM$ 是一个常数，所以 T^2 / r^3 $\ddot{a}$ 问题 2.2 假设光从太阳传播到地球大约需要 8.0 分钟。地球。

(a) 太阳的质量是多少？(b) 地球绕太阳转动的速度是多少？

2.1.3 表观重量物体的实际重量是地球引力所施加的引力。物体的表观重量是抵抗重力并防止物体下落的向上力（例如，地面施加的法向接触力、弹簧秤的拉力等）。由于垂直加速度或浮力，表观重量可能与实际重量不同，但如果我们考虑地球的自转，这也会导致表观重量减少。放在地面上的物体实际上一同旋转与地球引力和接触力的合力应为赤道上的物体提供向心力： $F_{c,eq} = m\omega^2 R$ $F_{grav} - N_{eq} = m\omega^2 R$ $N_{eq} = GMm/R^2 - m\omega^2 R$ 对于极点处的物体： $F_{c,pole} = 0$ $F_{grav} - N_{pole} = 0$ $N_{pole} = GMm/R^2$ 较低纬度越高，物体描述的圆圈越大，因此需要更大的向心力[4]。这被称为开普勒行星运动第三定律。

17 世纪初，德国天文学家约翰内斯·开普勒发现了描述行星如何绕太阳运行的三个科学定律。这种 T^2 / r^3 关系不仅适用于圆形轨道，而且适用于椭圆轨道。艾萨克·牛顿证明了开普勒定律是他自己的万有引力定律的结果，因此解释了为什么行星以这种方式运动。



apparent weight at various positions near earth's surface (not to scale)

this offsets the balancing normal force, so apparent weight decreases near the equator

Example 2.5 A stone of mass 5.0 kg is hung from a newton-meter near the equator. The Earth can be considered to be a uniform sphere of radius $R = 6370$ km and mass $M = 5.97 \times 10^{24}$ kg. (a) What is the gravitational force on the stone? (b) What is the reading on the meter?

✍ gravitational force: $F_{\text{grav}} = \frac{GMm}{R^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 5.0}{(6.37 \times 10^6)^2} \approx 49.07 \text{ N}$

centripetal force required: $F_c = m\omega^2 R = m \left(\frac{2\pi}{T} \right)^2 R = 5.0 \times \frac{4\pi^2}{(24 \times 3600)^2} \times 6.37 \times 10^6 \approx 0.17 \text{ N}$

apparent weight, or reading on meter: $N = F_{\text{grav}} - F_c = 49.07 - 0.17 \approx 48.90 \text{ N}$ $\square$

Question 2.3 Why astronauts in space stations are said to be *weightless*?

Question 2.4 How do you find the apparent weight of an object at an arbitrary latitude P ?

Does the apparent weight act vertically downwards? Give your reasons.

Yuhao Yang 2 GRAVITATIONAL Fields O Fgrav NeqFc,eq Fgrav Fc, P NP Fgrav Npole 在地球表面附近不同位置的表观重量（未按比例）这抵消了平衡法向力，因此在赤道附近表观重量减少示例 2.5 一块质量为 5.0 kg 的石头悬挂在赤道附近的牛顿米上。地球可以被认为是一个半径为 $R = 6370$ 公里、质量 $M = 5.97 \times 10^{24}$ 千克的均匀球体。(a) 石头上的重力是多少？(b) 仪表上的读数是多少？- 重力： $F_{\text{grav}} = G \frac{Mm}{R^2} = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 5.0 / (6.37 \times 10^6)^2 \approx 49.07 \text{ N}$ 所需向心力： $F_c = m\omega^2 R = m (2\pi / T)^2 R = 5.0 \times 4\pi^2 (24 \times 3600)^2 \times 6.37 \times 10^6 \approx 0.17 \text{ N}$ 表观重量，或仪表读数： $N = F_{\text{grav}} - F_c = 49.07 - 0.17 \approx 48.90 \text{ N}$ 问题 2.3 为什么说空间站中的宇航员是失重的？问题 2.4 如何求任意纬度 P 处物体的表观重量？表观重量是否垂直向下作用？给出你的理由。 11

2.2 gravitational fields

to explain how objects exert gravitational attraction upon one another at a distance, we introduce the concept of *force fields*

gravitational field is a region of space where a mass is acted by a force

any mass M (or several masses) can produce a gravitational field around it

a test mass m within this field will experience a gravitational force

to describe the effect on a small mass m in the field, we will further introduce

- *gravitational field strength*, to help us compute gravitational force on objects
- *gravitational potential*, to help us compute gravitational potential energy between objects

2.3 gravitational field strength

2.3.1 gravitational field strength

gravitational field strength is defined as gravitational force per unit mass:

$$g = \frac{F_{\text{grav}}}{m}$$

➤ unit of g : $[g] = \text{N kg}^{-1} = \text{m s}^{-2}$, same unit as acceleration

➤ field strength due to an isolated source of mass M

at distance r from the source, a test mass m is acted by a force: $F_{\text{grav}} = \frac{GMm}{r^2}$

field strength at this position: $g = \frac{F_{\text{grav}}}{m} = \Rightarrow g = \frac{GM}{r^2}$

note that the field is produced by the source M , so field strength g depends on M , not m

➤ field strength g is a *vector* quantity, it has a direction

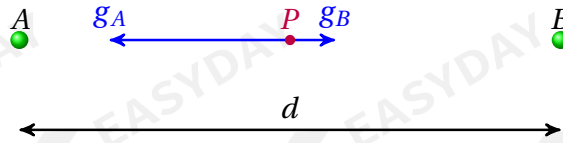
gravitation is *attractive*, so g points towards source mass

to compute combined field strength due to several sources, should perform *vector sum* of contributions from each individual

Example 2.6 Star A of mass 6.0×10^{30} kg and star B of mass 1.5×10^{30} kg are separated by a

杨宇豪 2 引力场 2.2 引力场为了解释物体如何在一定距离处相互施加引力，我们引入力场的概念。引力场是一个空间区域，其中一个质量受到力的作用，任何质量 M （或几个质量）可以在其周围产生一个引力场，测试质量 m 在这个场内将经历一个引力场。为了描述场中小质量 m 的作用，我们将进一步介绍 - 引力场强度，帮助我们计算物体上的引力 - 引力势，帮助我们计算物体之间的引力势能 2.3 引力场强度 2.3.1 引力场强度引力场强度定义为每单位质量的引力： $g = F_{\text{grav}} / m$ 单位： $[g] = \text{N kg}^{-1} = \text{m s}^{-2}$ ，与加速度相同单位 = 由于距离源为 r 处的孤立质量 M 源而产生的场强，测试质量 m 由力作用： $F_{\text{grav}} = G M m / r^2$ 此位置处的场强： $g = F_{\text{grav}} / m = G M / r^2$ 请注意，场由源 M 产生，因此场强 g 取决于 M ，而不是 m - 场强 g 是一个向量，它有一个引力方向，因此 g 指向源质量，以计算由于多个源而产生的组合场强，应执行每个单独贡献的向量和示例 2.6 质量为 $6.0 \times 10^{30} \text{ kg}$ 的恒星 A 和质量为 $1.5 \times 10^{30} \text{ kg}$ 的恒星 B 相隔 12

distance of 2.0×10^{12} m. (a) What is the field strength at the mid-point P of the two stars? (b) If a comet of mass 4.0×10^6 kg is at the mid-point, what force does it experience?



g_A acts towards A, g_B acts towards B, they are in opposite directions

$$g_P = g_A - g_B = \frac{GM_A}{r_A^2} - \frac{GM_B}{r_B^2} = 6.67 \times 10^{-11} \times \left[\frac{6.0 \times 10^{30}}{(1.0 \times 10^{12})^2} - \frac{1.5 \times 10^{30}}{(1.0 \times 10^{12})^2} \right] \approx 3.0 \times 10^{-4} \text{ N kg}^{-1}$$

$$\text{force on comet: } F = mg = 4.0 \times 10^6 \times 3.0 \times 10^{-4} \approx 1.2 \times 10^3 \text{ N}$$

□

2.3.2 acceleration of free fall

if field strength g is known, gravitational force on an object of mass m is: $F_{\text{grav}} = mg$

if the object is acted by gravity only, then $F_{\text{net}} = F_{\text{grav}} \Rightarrow ma = mg \Rightarrow a = g$ [5]

this shows gravitational field strength gives the acceleration of free fall!

Example 2.7 The earth has a radius of 6370 km. (a) Find the mass of the earth. [6] (b) Find the acceleration of free fall at the top of Mount Everest. (height of Mount Everest $H \approx 8.8$ km)

consider acceleration of free fall near surface of earth:

$$g_s = \frac{GM}{R^2} \Rightarrow 9.81 = \frac{6.67 \times 10^{-11} \times M}{(6.37 \times 10^6)^2} \Rightarrow M \approx 5.97 \times 10^{24} \text{ kg}$$

at top of Mount Everest:

$$g_{\text{ME}} = \frac{GM}{(R+H)^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.37 \times 10^6 + 8.8 \times 10^3)^2} \approx 9.78 \text{ N kg}^{-1} \Rightarrow a_{\text{ME}} \approx 9.78 \text{ m s}^{-2}$$

□

2.3.3 gravitational field lines

gravitational field lines are drawn to graphically represent the pattern of field strength

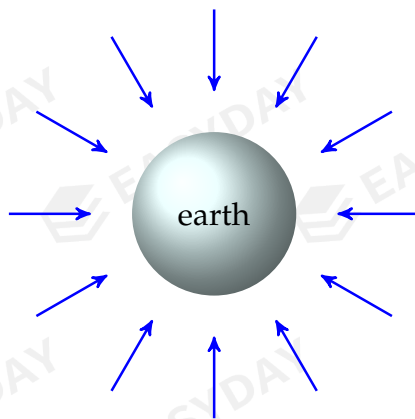
[5] Rigorously speaking, the two m 's are different concepts. There is the *inertia* mass, describing how much an object resists the change of state of motion. There is also the *gravitational* mass, describing the effect produced and experienced by the object in gravitational fields. Yet no experiment has ever demonstrated any significant difference between the two. The reason why the two masses are identical is very profound. We have shown here acceleration of free fall equals gravitational field strength, but Albert Einstein's *equivalence principle* suggests that it is actually impossible to distinguish between a uniform acceleration and a uniform gravitational field. This idea lies at the heart of the *general theory of relativity*, where I should probably stop going further.

[6] British scientist Henry Cavendish devised an experiment in 1798 to measure the gravitational force between masses in his laboratory. He was the first man to yield accurate values for the gravitational constant G . Then he was able to carry out this calculation, referred by himself as 'weighing the world'.

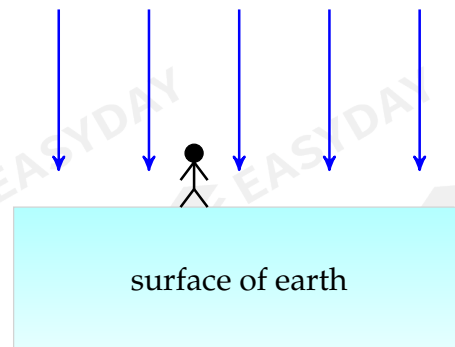
Yuhao Yang 2 GRAVITATIONAL Fields 距离 $2.0 \times 10^{12} \text{ m}$ 。(a) 两颗恒星中点 P 处的场强是多少？(b) 如果一颗质量为 4.0 英镑 106 千克的彗星位于中点，它受到的力是多少？

AB d g A gBP - g A 作用于 A，gB 作用于 B，方向相反 $g_P = g_A - g_B = G M_A r_A^2 - G M_B r_B^2 = 6.67 \times 10^{-11} \times \frac{6.0 \times 10^{30}}{(1.0 \times 10^{12})^2} - 1.5 \times 10^{30} (1.0 \times 10^{12})^2 = 3.0 \times 10^{-4} \text{ N kg}^{-1}$ 彗星上的力： $F = mg = 4.0 \times 106 \times 3.0 \times 10^{-4} = 1.2 \times 10^3 \text{ N}$ 2.3.2 自由落体加速度如果场强 g 已知，质量为 m 的物体上的引力为： $F_{\text{grav}} = mg$ 如果物体仅受重力作用，则 $F_{\text{net}} = F_{\text{grav}}$ $ma = mg$ $a = g$ [5] 这表明引力场强度给出了自由落体的加速度！例 2.7 地球半径为 6370 公里。(a) 求地球的质量。[6] (b) 求珠穆朗玛峰顶自由落体的加速度。(珠穆朗玛峰高度 $H = 8.8 \text{ km}$) - 考虑地球表面附近自由落体的加速度： $g_s = G M R^2$ $9.81 = 6.67 \times 10^{-11} \times M (6.37 \times 10^6)^2$ $M = 5.97 \times 10^{24} \text{ kg}$ 在珠穆朗玛峰顶部： $g_{ME} = G M (R + H)^2 = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} (6.37 \times 10^6 + 8.8 \times 10^3)^2 = 9.78 \text{ N kg}^{-1}$ $a_{ME} = 9.78 \text{ m s}^{-2}$ 2.3.3 引力场线绘制引力场线以图形方式表示场强模式[5]严格来说，这两个 m 是不同的概念。有惯性质量，描述物体抵抗运动状态变化的程度。还有引力质量，描述物体在引力场中产生和经历的效应。然而，没有任何实验证明两者之间有任何显著差异。两个质量相同的原因是非常深刻的。我们在这里证明了自由落体的加速度等于引力场强度，但阿尔伯特·爱因斯坦的等效原理表明，实际上不可能区分均匀加速度和均匀引力场。这个想法是广义相对论的核心，我可能应该停止进一步讨论。[6]英国科学家亨利·卡文迪什于 1798 年在他的实验室设计了一项测量质量之间引力的实验。他是第一个给出万有引力常数 G 精确值的人。然后他能够进行这项计算，他自己称之为“称量世界”。13

- *direction* of field lines show the *direction* of field strength in the field
- *spacing* between field lines indicates the *strength* of the gravitational field
- gravitational field lines always end up at a mass
this arises from the attractive nature of gravitation

Example 2.8 field around the earth

radial field (field lines normal to surface)

Example 2.9 field near earth's surface

almost a *uniform* field

(field lines are parallel and equally spaced)

2.4 gravitational potential & potential energy

2.4.1 potential energy

potential energy is the energy possessed by an object due to its position in a force field

work done *by* force field decreases P.E., and work done *against* a force field increases P.E.

let W be work by the force field, then we have: $W = -\Delta E_p$

Yuhao Yang 2 GRAVITATIONAL Fields - 场线的方向显示了场中场强的方向 - 场线之间的间距表示重力场的强度 - 重力场线总是以质量结束，这是由重力的吸引力性质产生的示例 2.8 围绕地球的场地球径向场（场线垂直于表面） 示例2.9 靠近地球表面的场地球表面几乎是一个均匀的场（力场线平行且等距） 2.4 引力势和势能 2.4. 1 势能势能是物体由于其在力场中的位置而拥有的能量，由力场所做的功会减小 P.E.，而对力场所做的功会增加 P.E.。令 W 为力场所做的功，则我们有： $W = \Delta E_p$ 14

to define potential energy of an object at a specific point X , we can

- (1) choose a position where potential energy is defined to be zero
- (2) find work done by force field to bring the object from zero P.E. point to X
- (3) consider change in P.E.: $\Delta E_p = E_{p,X} - E_{p,\text{initial}} = E_{p,X} - 0 = E_{p,X}$

but $\Delta E_p = -W$, so P.E. at point X is found: $E_{p,X} = -W$

so potential energy is equal to (negative) work done to move the object to a specific position

gravitational potential energy near earth's surface

we may choose a zero G.P.E. point, for example, $E_p(0) = 0$ at sea level

if mass m is moved up for a height h , work done by gravity is $W = -mgh$ ^[7]

this causes a change in gravitational potential energy $\Delta E_p = -W = mgh$

then at altitude h , G.P.E. can be given by $E_p(h) = mgh$

2.4.2 gravitational potential energy

we are now ready to derive an expression for G.P.E. between two masses M and m

we define $E_p = 0$ at $r = \infty$ (choice of zero potential energy, no force so no G.P.E.), then

gravitation potential energy is equal to the work done by gravitational force to bring a mass to a specific position from *infinity*

consider a mass m at infinity with zero energy and a source mass M at origin

let's find out how much work is done by gravitational force to pull m towards the origin



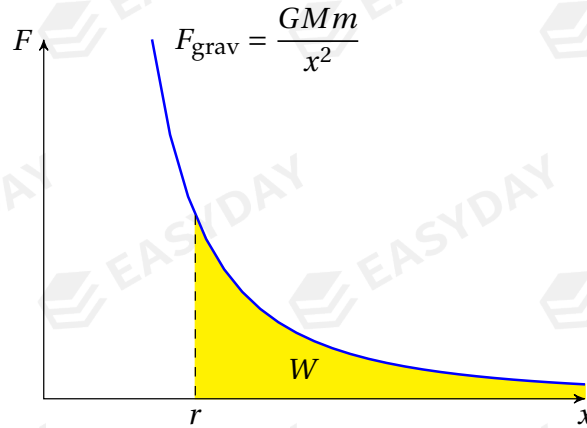
but F_{grav} varies as inverse square of separation x

so here we need to evaluate work done by a non-constant force

we can plot a $F-x$ graph, then magnitude of work done equals area under the graph

^[7] Negative sign because this is work against gravity.

Yuhao Yang 2 GRAVITATIONAL Fields 要定义物体在特定点 X 的势能，我们可以 (1) 选择势能定义为零的位置 (2) 求力场所做的功，使物体从零 P.E 开始。 (3) 考虑 P.E. 的变化： $\Delta E_p = E_{p,X} - E_{p,initial} = E_{p,X} - 0 = E_{p,X}$ 但 $\Delta E_p = -W$ ，因此 P.E. 在 X 点发现： $E_{p,X} = -W$ 因此势能等于将物体移动到地球表面附近特定位置的（负）功，我们可以选择零 G.P.E. 点，例如，在海平面 $E_p(0) = 0$ 如果质量 m 向上移动高度 h ，则重力所做的功为 $W = -mgh$ [7] 这会导致重力势能发生变化 $\Delta E_p = -W = mgh$ ，然后在海拔 h 处，G.P.E. 可以由 $E_p(h) = mgh$ 给出 2.4.2 重力势能我们现在准备导出 G.P.E 的表达式。在两个质量 M 和 m 之间，我们将 $E_p = 0$ 定义为 $r = 1$ （选择零势能，没有力，因此没有 G.P.E.），那么引力势能等于引力将质量从无穷远带到特定位置所做的功。将 m 拉向原点 M m/r W 1 但 F_{grav} 随分离 x 的平方反比变化，因此这里我们需要评估非恒定力所做的功，我们可以绘制 $F-x$ 图，则所做功的大小等于图下方的面积 [7] 负号，因为这是对抗重力的功。 15



integrate^[8] to evaluate the area: $W = \int_r^\infty \frac{GMm}{x^2} dx = -\frac{GMm}{x} \Big|_r^\infty = \frac{GMm}{r}$

$$\Delta E_p = -W \Rightarrow E_p(r) - E_p(\infty) = -\frac{GMm}{r}$$

but we have defined $E_p(\infty) = 0$, therefore: $E_p(r) = -\frac{GMm}{r}$

$E_p(r)$ gives G.P.E between masses M and m when they are at distance r from each other

- as $r \rightarrow \infty$, $E_p \rightarrow 0$, this agrees with our choice of zero G.P.E. point
- potential energy is a *scalar* quantity (negative sign cannot be dropped)
- G.P.E. is always *negative*, this is due to *attractive* nature of gravity

to separate masses, work must be done to overcome attraction

so G.P.E. increases with separation r , i.e., G.P.E. is maximum at infinity, which is zero

G.P.E. between masses at finite separation must be less than zero

- $E_p = mgh$ is only applicable near earth's surface where field is almost *uniform*

$E_p = -\frac{GMm}{r}$ is a more *general* formula for gravitational potential energy ^[9]

Example 2.10 A meteor is travelling towards a planet of mass M . When it is at a distance of r_1 from centre of M , it moves at speed v_1 . When it is r_2 from M , it moves at speed v_2 . Assume only gravitational force applies, establish a relationship between these quantities.

^[8] In general, work done by a non-constant force over large distance is $W = \int_{\text{initial}}^{\text{final}} F dx$.

For our case, x is the displacement away from the source, but gravitational force tends to pull the mass towards the source. F and x are in opposite directions, a negative sign is needed for F . Therefore the work done by gravity to bring mass m from infinity is: $W = \int_\infty^r F dx = \int_\infty^r \left(-\frac{GMm}{x^2} \right) dx = +\frac{GMm}{x} \Big|_\infty^r = \frac{GMm}{r}$.

^[9] One can recover $\Delta E_p = mg\Delta h$ from $E_p = -\frac{GMm}{r}$. Near the earth's surface, if $r_1 \approx r_2 \approx R$, and $r_1 > r_2$, then we have: $\Delta E_p = E_p(r_1) - E_p(r_2) = -GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = GMm \frac{r_1 - r_2}{r_1 r_2} \approx m \frac{GM}{R^2} \Delta r \stackrel{g=GM/R^2}{=} mg\Delta h$.

Yuhao Yang 2 GRAVITATIONAL Fields $F = -\frac{GMm}{x^2}$ 积分 [8] 来评估面积： $W = \int_{r_1}^{r_2} F dx = -GMm \int_{r_1}^{r_2} \frac{1}{x^2} dx = -GMm \left[-\frac{1}{x} \right]_{r_1}^{r_2} = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ 但我们已经定义了 $E_p(r_1) = 0$ ，因此： $E_p(r) = -GMm/r$ 给出当质量 M 和 m 彼此距离为 r 时，它们之间的 G.P.E. $E_p(r) = -GMm/r$ 。点 $\hat{a}$ 势能是一个标量（负号不能去掉） $\hat{a}$ G.P.E. 总是负数，这是由于重力对分离质量的吸引力，必须做功来克服吸引力，因此 G.P.E. 随着分离 r 的增加，即 G.P.E. 在无穷大处最大，此时 G.P.E 为零。有限间隔质量之间的距离必须小于零 $\hat{a}$ $E_p = mgh$ 仅适用于场几乎均匀的地球表面附近 $E_p = -GMm/r$ 是重力势能的更一般公式 [9] 例 2.10 一颗流星正向质量为 M 的行星行进。当它距 M 中心距离为 r_1 时，它以速度 v_1 移动。当距 M 为 r_2 时，以速度 v_2 移动。假设仅存在重力，建立这些量之间的关系。[8] 一般来说，长距离非恒定力所做的功为 $W = \int_{r_1}^{r_2} F dx$ 。对于我们的例子， x 是远离源的位移，但重力倾向于将质量拉向源。 F 和 x 方向相反， F 需要加负号。因此，重力使质量 m 从无穷大产生的功为： $W = \int_{r_1}^{r_2} F dx = \int_{r_1}^{r_2} \left(-\frac{GMm}{x^2} \right) dx = +GMm \left[\frac{1}{x} \right]_{r_1}^{r_2} = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ 。[9] 可以从 $E_p = -GMm/r$ 恢复 $\Delta E_p = mg \Delta h$ 。在地球表面附近，如果 $r_1 = r_2 = r$ 和 $r_1 \approx r_2$ ，则有： $\Delta E_p = E_p(r_1) - E_p(r_2) = -GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = GMm \left(\frac{1}{r_2} - \frac{1}{r_1} \right) \approx \frac{1}{2} m G M / R^2 \Delta r = mg \Delta h$ 。 16

✎ energy conservation: $\text{K.E.} + \text{G.P.E.} = \text{const} \Rightarrow \frac{1}{2}mv_1^2 + \left(-\frac{GMm}{r_1}\right) = \frac{1}{2}mv_2^2 + \left(-\frac{GMm}{r_2}\right)$ $\square$

Example 2.11 If an object is thrown from the surface of a planet at sufficiently high speed, it might escape from the influence of the planet's gravitational field. The minimum speed required is called the *escape velocity*. Using the data from previous examples, find the escape velocity from the surface of earth.

✎ assuming no energy loss to air resistance, then total energy is conserved

K.E. + G.P.E. at surface of planet = K.E. + G.P.E. at infinity

$$\frac{1}{2}mu^2 + \left(-\frac{GMm}{R}\right) = \frac{1}{2}mv^2 + 0 \xrightarrow{v \geq 0} u^2 \geq \frac{2GM}{R} \Rightarrow u_{\min} = \sqrt{\frac{2GM}{R}}$$

$$\text{for earth, escape velocity } u_{\min} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{6.4 \times 10^6}} \approx 1.12 \times 10^4 \text{ m s}^{-1} \quad \square$$

Question 2.5 A planet of uniform density distribution is of radius R and mass M . A rock falls from a height of $3R$ above the surface of the planet. Assume the planet has no atmosphere, show that the speed of the rock when it hits the ground is $v = \sqrt{\frac{3GM}{4R}}$.

Question 2.6 A space probe is travelling around a planet of mass M in a circular orbit of radius r . (a) Show that the total mechanical energy (sum of kinetic energy and gravitational energy) of the space probe is $E_{\text{total}} = -\frac{2GMm}{r}$. (b) If the space probe is subject to small resistive forces, state the change to its orbital radius and its orbiting speed.

Question 2.7 A *black hole* is a region of spacetime where gravitation is so strong that even light cannot escape from it. For a star of mass M to collapse and form a black hole, it has to be compressed below a certain radius. (a) Show that this radius is given by $R_S = \frac{2GM}{c^2}$, known as the *Schwarzschild radius*^[10]. (b) Show that the Schwarzschild radius for the sun is about 3 km.

^[10] When you deal with very strong gravitational fields, Newton's law of gravitation breaks down and effects of Einstein's *general theory of relativity* come into play. The radius of a *Newtonian* black hole being equal to the radius of a Schwarzschild black hole is a mere coincidence.

Yuhao Yang 2 GRAVITATIONAL Fields - 节能 : $K.E. + G.P.E. = \text{const}$) $\frac{1}{2}mv^2 + (-GMm/r) = \frac{1}{2}mv^2 + (-GMm/r)$ 例 2.11 如果一个物体以足够高的速度从行星表面抛出，它可能会逃脱行星引力场的影响。所需的最小速度称为逃逸速度。使用前面示例中的数据，求出地球表面的逃逸速度。 - 假设空气阻力没有能量损失，则总能量守恒 $K.E. + G.P.E. \text{在行星} = K.E. \text{的表面} + G.P.E. \text{无穷远}$ $\frac{1}{2}mv^2 + (-GMm/R) = \frac{1}{2}mv^2 + 0$ $v_0 = \sqrt{2GM/R}$ $u_{\min} = \sqrt{2GM/R}$ 对于地球，逃逸速度 $u_{\min} = \sqrt{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24} / 6.4 \times 10^6} = 1.12 \times 10^4 \text{ m s}^{-1}$ 问题 2.5 密度分布均匀的行星，半径为 R ，质量为 M 。一块岩石从距行星表面 $3R$ 的高度落下。假设行星没有大气层，证明岩石撞击地面时的速度为 $v = \sqrt{3GM/4R}$ 。问题 2.6 空间探测器在半径为 r 的圆形轨道上围绕质量为 M 的行星运行。(a) 证明空间探测器的总机械能（动能与引力能之和）为 $E_{\text{total}} = -2GMm/r$ 。(b) 如果空间探测器受到较小的阻力，请说明其轨道半径和轨道速度的变化。问题 2.7 黑洞是一个时空区域，其引力非常强，甚至连光也无法逃脱。质量为 M 的恒星要塌缩并形成黑洞，必须将其压缩到一定半径以下。(a) 表明该半径由 $R_S = 2GM/c^2$ 给出，称为史瓦西半径[10]。(b) 证明太阳的史瓦西半径约为 3 公里。[10]当你处理非常强的引力场时，牛顿万有引力定律就会失效，而爱因斯坦广义相对论的效应就会发挥作用。牛顿黑洞的半径与史瓦西黑洞的半径相等只是巧合。 17 号

2.4.3 gravitational potential

it is useful to introduce a quantity called *potential* at a specific point in a gravitational field

gravitational potential can be considered as the potential energy per unit mass: $\varphi = \frac{E_p}{m}$

gravitational potential at a point is defined as the work done to bring *unit* mass from *infinity* to that point

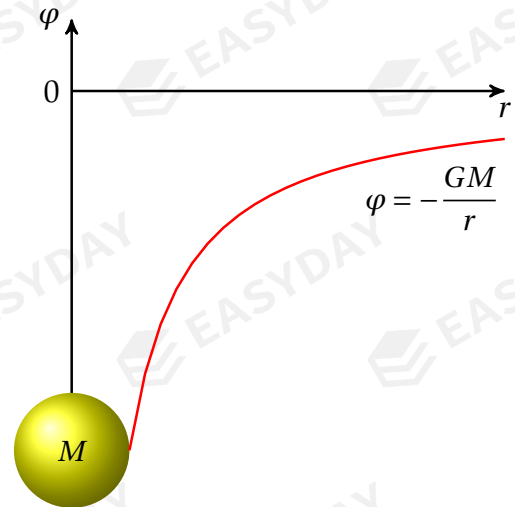
- unit: $[\varphi] = \text{J kg}^{-1}$
- gravitational potential due to an isolated source M

$$\varphi = \frac{E_p}{m} = \frac{-\frac{GMm}{r}}{m} \Rightarrow \boxed{\varphi = -\frac{GM}{r}}$$

- potential at infinity is zero: $\varphi_\infty = 0$
this is our choice of zero potential point
- gravitational potential is a *scalar*
combined potential due to several masses equals

scalar sum of potential of each individual

- gravitational potential is always *negative*
again this arises from attractive nature of gravity
work is done to pull unit mass away from source
farther from source means higher potential



Example 2.12 A star A of mass $M_A = 1.5 \times 10^{30} \text{ kg}$ and a planet B of mass $M_B = 6.0 \times 10^{26} \text{ kg}$ form an isolated astronomical system. Point P is between A and B , and is at distance $r_A = 2.0 \times 10^{12} \text{ m}$ from A , and distance $r_B = 8.0 \times 10^{10} \text{ m}$ from B . (a) Find the gravitational potential at P . (b) A meteor is initially at very large distance from the system with negligible speed. It then travels towards point P due to the gravitational attraction. Find its speed when it reaches P .

➤ gravitational potential at P : $\varphi_P = \varphi_A + \varphi_B = \left(-\frac{GM_A}{r_A}\right) + \left(-\frac{GM_B}{r_B}\right)$

$$\varphi_P = -6.67 \times 10^{-11} \times \left(\frac{1.5 \times 10^{30}}{2.0 \times 10^{12}} + \frac{6.0 \times 10^{26}}{8.0 \times 10^{10}}\right) \approx -5.05 \times 10^7 \text{ J kg}^{-1}$$

gain in K.E. = loss in G.P.E.: $\frac{1}{2}mv^2 = m\Delta\varphi \Rightarrow v^2 = 2(\varphi_\infty - \varphi_P) = -2\varphi_P$

$$v = \sqrt{-2 \times (-5.05 \times 10^7)} \approx 1.01 \times 10^4 \text{ m s}^{-1}$$

□

Yuhao Yang 2 GRA 生命场 2.4.3 引力势引入一个称为引力场中特定点势的量是有用的引力势可以被认为
 是每单位质量的势能： $\phi = E_p/m$ 一点处的引力势被定义为将单位质量从无穷大带到该点所做的功 $0 \rightarrow \phi$ M
 $\phi = -G M/r$ 单位： $[\phi] = \text{J kg}^{-1}$ 孤立源产生的重力势 $M \phi = E_p/m = -G M m/r$ $\phi = -G M/r$ 无
 穷远势为零： $\phi_{\infty} = 0$ 这是我们选择的零势点 ϕ 重力势是标量组合势，由于几个质量等于每个个体势的标量
 和 ϕ 重力势能始终为负值，这是由于重力的吸引力性质而产生的，将单位质量拉离源越远意味着势能越高
 。示例 2.12 质量为 $M_A = 1.5 \times 10^{30} \text{ kg}$ 的恒星 A 和质量为 $M_B = 6.0 \times 10^{26} \text{ kg}$ 的行星 B 形成一个孤立的天
 文系统。点 P 位于 A 和 B 之间，距 A 的距离为 $r_A = 2.0 \times 10^{12} \text{ m}$ ，距 B 的距离为 $r_B = 8.0 \times 10^{10} \text{ m}$ 。
 (a) 求 P 处的引力势。(b) 流星最初与系统的距离非常大，速度可忽略不计。然后由于万有引力的作用，它
 向 P 点移动。求其到达 P 时的速度。 $\phi_P = \phi_A + \phi_B = (-G M_A/r_A) + (-G M_B/r_B)$ $\phi_P =$
 $-6.67 \times 10^{-11} \times (1.5 \times 10^{30}/2.0 \times 10^{12} + 6.0 \times 10^{26}/8.0 \times 10^{10}) \text{ J kg}^{-1}$ $\phi_P = -5.05 \times 10^7 \text{ J kg}^{-1}$ K.E 增益 = G.P
 .E. 损失： $\frac{1}{2} m v^2 = m \Delta \phi$ $v^2 = 2(\phi_{\infty} - \phi_P) = -2\phi_P$ $v = \sqrt{-2 \times (-5.05 \times 10^7)} \text{ m s}^{-1} \approx 1.01 \times 10^4 \text{ m s}^{-1}$

Example 2.13 The Moon may be considered to be an isolated sphere of radius $R = 1.74 \times 10^3$ km. The gravitational potential at the surface of the moon is about $-2.82 \times 10^6 \text{ J kg}^{-1}$. (a) Find the mass of the moon. (b) A stone travels towards the moon such that its distance from the centre of the moon changes from $3R$ to $2R$. Determine the change in gravitational potential. (c) If the stone starts from rest, find its final speed.

$$\text{at surface: } \varphi(R) = -\frac{GM}{R} \Rightarrow -2.82 \times 10^6 = -\frac{6.67 \times 10^{-11} \times M}{1.74 \times 10^6} \Rightarrow M = 7.36 \times 10^{22} \text{ kg}$$

$$\text{from } 3R \text{ to } 2R: \Delta\varphi = \varphi_{(3R)} - \varphi_{(2R)} = \left(-\frac{GM}{3R}\right) - \left(-\frac{GM}{2R}\right) = \frac{GM}{6R} = \frac{2.82 \times 10^6}{6} \approx 4.70 \times 10^5 \text{ J kg}^{-1}$$

note this change is a *decrease* in gravitational potential

$$\text{gain in K.E.} = \text{loss in G.P.E.: } \frac{1}{2}mv^2 = m\Delta\varphi \Rightarrow v = \sqrt{2\Delta\varphi} = \sqrt{2 \times 4.70 \times 10^5} \approx 970 \text{ m s}^{-1} \quad \square$$

Question 2.8 Given that the moon is of radius 1700 km and mass $7.4 \times 10^{22} \text{ kg}$. (a) Find the change in gravitational potential when an object is moved from moon's surface to 800 km above the surface. (b) If a rock is projected vertically upwards with an initial speed of 1800 m s^{-1} from surface, find the rock's speed when it reaches a height of 800 km. (c) Suggest whether the rock can escape from the moon's gravitational field completely.

Yuhao Yang 2 GRAVITATIONAL Fields 示例 2.13 月球可以被认为是一个半径为 $R = 1.74 \times 10^3 \text{ km}$ 的孤立球体。月球表面的引力势约为 $-2.82 \times 10^6 \text{ J kg}^{-1}$ 。(a) 求月球的质量。(b) 一块石头向月球移动，其与月球中心的距离从 $3R$ 变为 $2R$ 。确定重力势的变化。(c) 如果石头从静止开始，求它的最终速度。- 表面： $\phi(R) = -GM/R = -2.82 \times 10^6 = -6.67 \times 10^{-11} \times M / 1.74 \times 10^6$) $M = 7.36 \times 10^{22} \text{ kg}$ 从 $3R$ 到 $2R$: $\Delta\phi = \phi(3R) - \phi(2R) = (-GM/3R) - (-GM/2R) = GM/6R = 2.82 \times 10^6 \times 6/4 = 4.70 \times 10^5 \text{ J kg}^{-1}$ 请注意，此变化是 K.E. 中重力势增益的减少。= G.P.E. 损失： $\frac{1}{2}mv^2 = m\Delta\phi$) $v = \sqrt{2\Delta\phi} = \sqrt{2 \times 4.70 \times 10^5} = 970 \text{ m s}^{-1}$ 问题 2.8 假设月球半径为 1700 km ，质量为 $7.4 \times 10^{22} \text{ kg}$ 。(a) 求当物体从月球表面移动到距表面 800 km 处时重力势的变化。(b) 如果一块岩石以 1800 m s^{-1} 的初始速度从表面垂直向上投射，求该岩石到达 800 km 高度时的速度。(c) 提出岩石是否能够完全逃离月球引力场。19 号

CHAPTER 3

Oscillation

3.1 oscillatory motion

oscillation refers to a repetitive back and forth motion about its *equilibrium position*

the equilibrium position is a point where all forces on oscillator are balanced

release an object from its equilibrium position from rest, it will stay at rest

examples of oscillation includes pendulum of a clock, vibrating string, swing, etc.

3.1.1 amplitude, period, frequency

to describe motion of an oscillator, we define the following quantities:

- **displacement** (x): distance from the equilibrium position
- **amplitude** (x_0): maximum displacement from the equilibrium position
- **period** (T): time for one complete oscillation
- **frequency** (f): number of oscillations per unit time

frequency is related to period as: $f = \frac{1}{T}$

displacement x varies with time t repetitively, for which we can plot an $x-t$ graph

amplitude x_0 and period T are labelled on the graph

3.1.2 phase angle

the point that an oscillator has reached within a complete cycle is called **phase angle** (ϕ)

- unit of phase angle: $[\phi] = \text{rad}$

it looks like an angle, but better think of it as a number telling fraction of a complete cycle

杨宇豪 3 振荡第三章振荡 3.1 振荡运动振荡是指围绕其平衡位置重复地来回运动。平衡位置是振荡器上所有力平衡的点，将物体从平衡位置释放到静止状态，它将保持静止状态。振荡的例子包括钟摆、振动弦、摆动等。

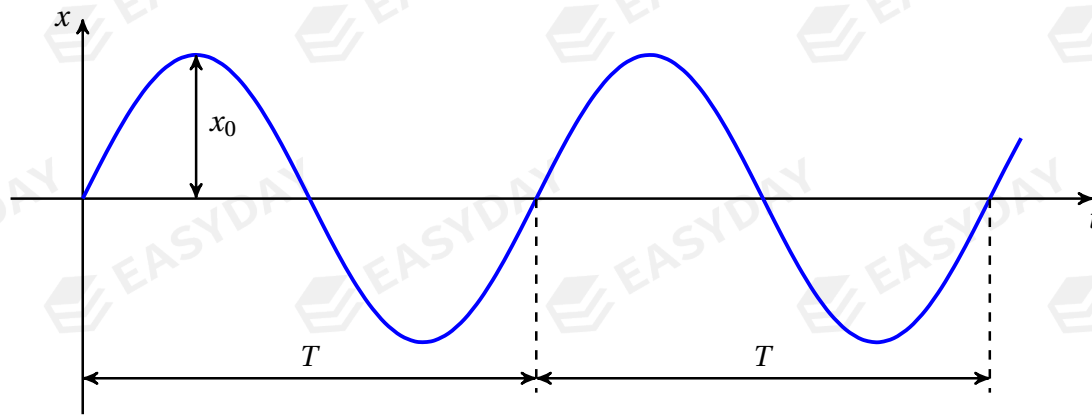
3.1.1 振幅、周期、频率来描述振荡器的运动，我们定义以下量：

(x): 距平衡位置的距离 振幅 (x_0): 距平衡位置的最大位移 周期 (T): 一次完整振荡的时间 频率 (f): 每单位时间的振荡次数 频率与周期的关系为： $f = 1/T$

位移 x 随时间 t 不断变化，我们可以绘制 $x-t$ 曲线图 振幅 x_0 和周期 T 在图上标记

3.1.2 相位角 振荡器在一个完整周期内达到的点称为相角 (ϕ)

相角单位： $[\phi] = \text{rad}$ 它看起来像一个角度，但最好将其视为完整周期的数字分数

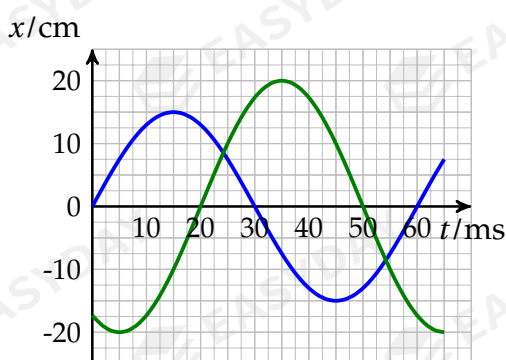


displacement-time graph for a typical oscillator

➤ we use **phase difference** $\Delta\phi$ to compare how much one oscillator is ahead of another

$\Delta\phi$ is found in terms of fraction of an oscillation: $\Delta\phi = \frac{\Delta t}{T} \times 2\pi$ (also measured in radians)

Example 3.1 Compare the two oscillations from the $x-t$ graph below.



both have period $T = 60$ ms

$$\text{frequency } f = \frac{1}{60 \times 10^{-3}} \approx 16.7 \text{ Hz}$$

they are of different amplitudes

one has $x_0 = 15$ cm, the other has $x_0 = 20$ cm

time difference: $\Delta t = 20$ ms

$$\text{phase difference: } \Delta\phi = \frac{\Delta t}{T} \times 2\pi = \frac{20}{60} \times 2\pi = \frac{2\pi}{3} \text{ rad}$$

3.1.3 acceleration & restoring force

for any oscillatory motion, consider its velocity and acceleration at various positions

its acceleration must be always pointing towards the equilibrium position

resultant force always acts in the direction to restore the system back to its equilibrium point,

this net force is known as the **restoring force**

if at equilibrium position, then no acceleration or restoring force

Yuhao Yang 3 典型振荡器的振荡 $x(t)$ 位移时间图 $\hat{a}$ 我们使用相位差 $\Delta\psi$ 来比较一个振荡器领先于另一个振荡器的程度 $\Delta\psi$ 以振荡分数的形式找到： $\Delta\psi = \Delta t T \times 2\pi$ （也以弧度测量）示例 3.1 比较下面 $x-t$ 图中的两个振荡。

t/ms	x/cm
10	20
20	30
30	40
40	50
50	60
60	-20
-10	0
10	20

两者的周期 $T = 60\text{ ms}$ 频率 $f = 1/60 \times 10^{-3} \approx 16.7\text{ Hz}$ 它们的幅度不同，一个具有 $x_0 = 15\text{ cm}$ ，另一个具有 $x_0 = 20\text{ cm}$ 时间差： $\Delta t = 20\text{ ms}$ 相位差： $\Delta\psi = \Delta t T \times 2\pi = 20/60 \times 2\pi = 2\pi/3\text{ rad}$

3.1.3 任何振荡运动的加速度和恢复力，考虑其在不同位置的速度和加速度，其加速度必须始终指向平衡位置合力始终作用于使系统恢复到平衡点的方向，该合力称为恢复力，如果处于平衡位置，则没有加速度或恢复力 21

3.2 simple harmonic oscillation

if an oscillator has an acceleration always proportional to its displacement from the equilibrium position, and acceleration is in opposite direction to displacement, then the oscillator is performing **simple harmonic motion**

many phenomena can be approximated by simple harmonics

examples are motion of a pendulum, molecular vibrations, etc.

complicated motions can be decomposed into a set of simple harmonics

simple harmonic motion provides a basis for the study of many complicated motions ^[11]

3.2.1 equation of motion

defining equation for simple harmonics can be written as $a = -\omega^2 x$

ω is some constant, so a is proportional to x

the minus sign shows a and x are in opposite directions

general solution to this this equation of motion ^[12] takes the form: $x = x_0 \sin(\omega t + \phi)$

x_0 represents the amplitude, ω is called the angular frequency, ϕ is the phase angle

angular frequency

➤ **angular frequency** satisfies the relation: $\omega = \frac{2\pi}{T} = 2\pi f$

➤ unit of angular frequency: $[\omega] = \text{rad} \cdot \text{s}^{-1}$

➤ angular frequency ω is determined by the system's *physical constants* only

if an object is set to oscillate *freely* with no external force, its period will always be the same

^[11] This can be done through a mathematical technique known as *Fourier analysis*. For example, a uniform circular motion can be considered as the combination of two simple harmonic motion in x - and y -directions.

^[12] You probably know that acceleration can be written as the second derivative of displacement: $a = \frac{d^2x}{dt^2}$, so $a = -\omega^2 x$ is equivalent to $\frac{d^2x}{dt^2} + \omega^2 x = 0$, which a *second-order differential equation*. If you do not know how to solve it, you may have the chance to study this in an advanced calculus course.

杨宇浩 3 振荡 3.2 简谐振动如果一个振子的加速度总是与其相对于平衡位置的位移成正比，且加速度与位移方向相反，则该振子正在进行简谐振动。很多现象都可以用简谐振动来近似，例如摆的运动、分子振动等。复杂的运动可以分解为一组简谐振动。简谐振动为许多复杂运动的研究提供了基础[11]

3.2.1 运动方程

简单谐振的定义方程可以写成 $a = -\omega^2 x$ ω 是某个常数，因此 a 与 x 成正比，负号表明 a 和 x 方向相反。该运动方程的通解 [12] 的形式为： $x = x_0 \sin(\omega t + \phi)$ x_0 表示幅度， ω 称为角频率， ϕ 是相位角角频率 $\hat{a}$ 角频率满足以下关系： $\omega = 2\pi T = 2\pi f$ $\hat{a}$ 角频率单位： $[\omega] = \text{rad} \cdot \text{s}^{-1}$ $\hat{a}$ 角频率 ω 由系统的物理常数决定，只有当物体被设置为在没有外力的情况下自由振荡时，其周期将始终相同 [11] 这可以通过称为傅里叶分析的数学技术来完成。例如，匀速圆周运动可以被视为 x 和 y 方向上两个简谐振动的组合。[12]你可能知道，加速度可以写成位移的二阶导数： $a = d^2x/dt^2$ ，因此 $a = -\omega^2 x$ 等价于 $d^2x/dt^2 + \omega^2 x = 0$ ，这是一个二阶微分方程。如果您不知道如何解决它，您可能有机会在高级微积分课程中学习这个问题。22 号

frequency of an free oscillatory system is called the **natural frequency**

phase angle

- phase angle ϕ is dependent on *initial conditions* (e.g. initial position and initial speed at $t = 0$?)
- in many cases, phase angle term can be avoided if a suitable trigonometric function is chosen

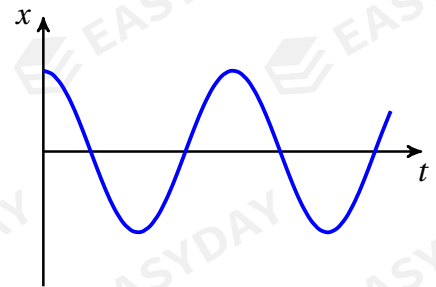
Example 3.2 A simple harmonic oscillator is displaced by 6.0 cm from its rest position and let go at $t = 0$. Given that the period of this system is 0.80 s, state an equation for its displacement-time relation.

🔗 angular frequency: $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.80} = \frac{5\pi}{2} \text{ rad s}^{-1}$

initial displacement $x(0) = +x_0 = 6.0 \text{ cm}$

for displacement-time relation, we use cosine function

$$x(t) = x_0 \cos \omega t \Rightarrow x = 6.0 \cos \left(\frac{5\pi}{2} t \right) \quad \square$$



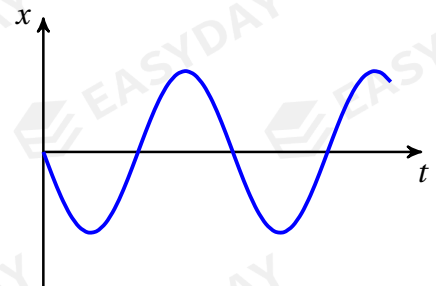
Example 3.3 A simple harmonic oscillator is initially at rest. At $t = 0$, it is given an initial speed in the negative direction. Given that the frequency is 1.5 Hz and the amplitude is 5.0 cm, state an equation for its displacement-time relation.

🔗 angular frequency: $\omega = 2\pi f = 2\pi \times 1.5 = 3\pi \text{ rad s}^{-1}$

initial displacement $x(0) = 0$

for displacement-time relation, we use sine function

$$x(t) = -x_0 \cos \omega t \Rightarrow x = -5.0 \sin(3\pi t) \quad \square$$



3.2.2 examples of simple harmonics

mass-spring oscillator

a mass-spring oscillator system consists of a block of mass m and an ideal spring

when a spring is stretched or compressed by a mass, the spring develops a restoring force

magnitude of this force obeys *Hooke's law*: $F = kx$

direction of this force is in opposite direction to displacement x

Yuhao Yang 3 自由振荡系统的振荡频率称为固有频率相位角 $\hat{\alpha}$ 相位角 ϕ 取决于初始条件（例如 $t = 0$ 时的初始位置和初始速度？） $\hat{\alpha}$ 在许多情况下，如果选择合适的三角函数，则可以避免相位角项

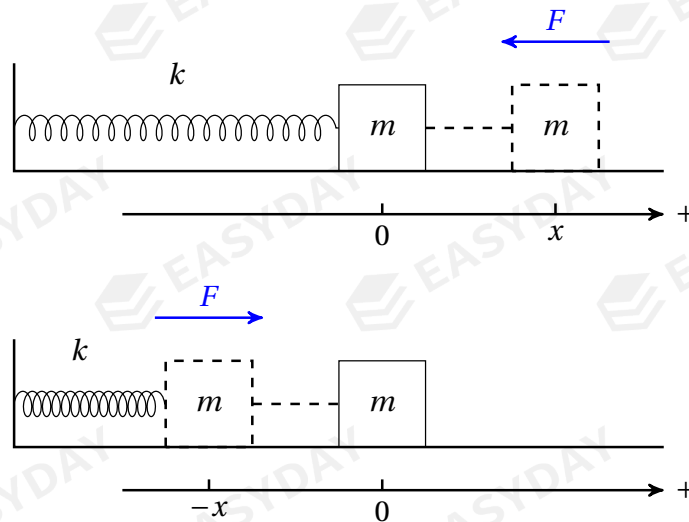
示例 3.2 简谐振子从其静止位置移动 6.0 cm ，并在 $t = 0$ 处释放。假设该系统的周期为 0.80 s ，说出其位移-时间关系的方程。

t x - 角频率： $\omega = 2\pi/T = 2\pi/0.80 = 5\pi/2 \text{ rad s}^{-1}$ 初始位移 $x(0) = +x_0 = 6.0 \text{ cm}$ 对于位移-时间关系，我们使用余弦函数 $x(t) = x_0 \cos \omega t$ $x = 6.0 \cos(5\pi/2 t)$ $\ddot{a}$

示例 3.3 简谐振子最初处于静止状态。在 $t = 0$ 处，给定负方向的初始速度。假设频率为 1.5 Hz ，振幅为 5.0 cm ，请写出其位移-时间关系的方程。

t x - 角频率： $\omega = 2\pi f = 2\pi \times 1.5 = 3\pi \text{ rad s}^{-1}$ 初始位移 $x(0) = 0$ 对于位移-时间关系，我们使用正弦函数 $x(t) = -x_0 \cos \omega t$ $x = -5.0 \sin(3\pi t)$ $\ddot{a}$

3.2.2 简谐波质量弹簧振荡器 α 的示例质量弹簧振荡器系统由质量为 m 的块和理想弹簧组成，当弹簧被质量拉伸或压缩时，弹簧产生恢复力，该力的大小遵循胡克定律： $F = kx$ 该力的方向与位移 x 的方向相反



restoring force acting on the ideal mass-spring oscillator

take vector nature of force into account, we find

$$F_{\text{net}} = ma \Rightarrow -kx = ma \Rightarrow a = -\frac{k}{m}x$$

spring constant k and mass m are constants, so $a \propto x$

negative sign shows a and x are in opposite directions

so mass-spring oscillator executes simple harmonic motion

compare with $a = -\omega^2 x \Rightarrow \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$

period of mass-spring oscillator: $T = \frac{2\pi}{\omega} \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$

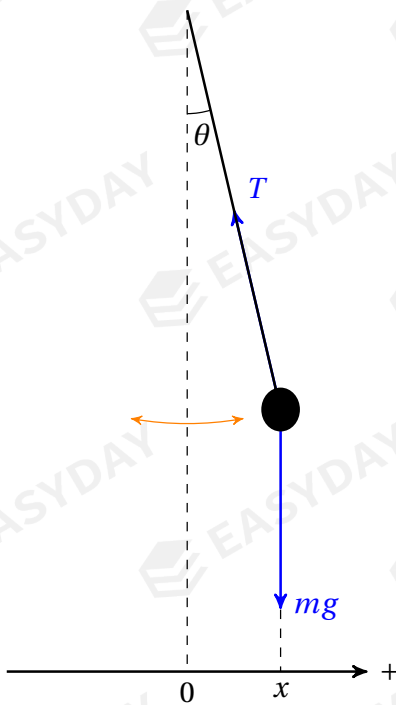
- period and frequency are solely determined by mass of oscillator m and spring constant k
- identical mass-spring systems will oscillate at same frequency no matter what amplitude
- $m \uparrow \Rightarrow T \uparrow$, greater mass means greater inertia, oscillation becomes slower
- $k \uparrow \Rightarrow T \downarrow$, greater k means stiffer spring, greater restoring force makes oscillation go faster

simple pendulum

a simple pendulum is set up by hanging a bob on a light cord from a fixed point

displace the bob by some angle and release from rest, it can swing freely

Yuhao Yang 3 振荡 $m \ddot{x} + kx = 0$ 作用在理想质量弹簧振子上的恢复力考虑到力的矢量性质，我们发现 $F_{\text{net}} = m\ddot{x} = -kx$ $\ddot{x} = -\frac{k}{m}x$ 弹簧常数 k 和质量 m 是常数，因此 $\ddot{x} / x$ 负号表示 $\ddot{x}$ 和 x 方向相反，因此质量弹簧振子与简谐振动 $\ddot{x} + \omega^2 x = 0$ $\omega^2 = \frac{k}{m}$ $\omega = \sqrt{\frac{k}{m}}$ 质量弹簧振荡器的周期： $T = \frac{2\pi}{\omega}$ $T = 2\pi \sqrt{\frac{m}{k}}$ 周期和频率仅由振荡器的质量 m 和弹簧常数 k 决定，相同的质量弹簧系统无论振幅如何，都会以相同的频率振荡 $\propto \frac{1}{\sqrt{k}}$ ，质量越大意味着惯性越大，振荡变得更慢 $\propto \sqrt{m}$ ， k 越大，弹簧越硬，恢复力越大，摆动得越快。简单摆简单摆是通过将一个摆锤悬挂在一根轻绳上，从一个固定点将摆锤移动一定的角度，然后从静止状态释放，它可以自由摆动 24



one can show this performs simple harmonic motion for *small-angle* oscillation

if angular displacement θ is small, then the pendulum has almost no vertical displacement, the motion can be considered to be purely horizontal

$$\text{vertically: } T \cos \theta \approx mg \quad \xrightarrow{\cos \theta \approx 1 \text{ as } \theta \rightarrow 0} \quad T \approx mg$$

$$\text{horizontally: } -T \sin \theta = ma \quad \xrightarrow{\sin \theta = x/L} \quad a \approx -\frac{g}{L}x$$

this shows simple pendulum undergoes simple harmonics
compare with defining equation for simple harmonics:

$$a = -\omega^2 x \quad \Rightarrow \quad \omega = \sqrt{\frac{g}{L}}$$

period for a simple pendulum:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

- period and frequency of a pendulum are determined by length of the string L only as long as displacement remains small, frequency does not depend on amplitude
- $L \uparrow \Rightarrow T \uparrow$, longer pendulums oscillate more slowly
- $g \downarrow \Rightarrow T \uparrow$, if there is no gravity ($g = 0$), then the bob will not move at all ($T \rightarrow \infty$)

Question 3.1 A cylindrical tube of total mass m and cross sectional area A floats upright in a liquid of density ρ . When the tube is given a small vertical displacement and released, the magnitude of the resultant force acting on the tube is related to its vertical displacement y by the expression: $F_{\text{net}} = \rho g A y$. (a) Show that the tube executes simple harmonic motion. (b) Find an expression for the frequency of the oscillation.

Question 3.2 A small glider moves along a horizontal air track and bounces off the buffers at the ends of the track. Assume the track is frictionless and the buffers are perfectly elastic, state and explain whether the glider describes simple harmonic motion.

3.2.3 velocity & acceleration

displacement of simple harmonic oscillator varies with time as: $x = x_0 \sin(\omega t + \phi)$

from this displacement-time relation, we can find velocity and acceleration relations

杨宇浩 3 振荡 0 θ mg $T+x$ 可以看出，如果角位移 θ 很小，那么摆体就进行小角度振荡的简谐振动，那么摆几乎没有垂直位移，可以认为运动是纯水平垂直运动： $T \cos \theta \approx mg \cos \theta / l \approx mg$ 水平： $-T \sin \theta = ma \sin \theta = x/L$ 这表明简单摆经历简单谐波，与简单谐波的方程进行比较： $a = -\omega^2 x$ $\omega = \sqrt{g/L}$ 单摆的周期： $T = 2\pi \sqrt{L/g}$ 摆的周期和频率仅由弦 L 的长度决定，只要位移保持较小，频率不取决于振幅 $\propto \sqrt{L}$ ，较长的摆振荡更慢 $\propto \sqrt{g}$ ，如果没有重力 ($g = 0$)，则摆锤根本不会移动 ($T \rightarrow \infty$) 问题 3.1 总质量为 m 且横截面积为 A 的圆柱管在密度为 ρ 的液体中直立漂浮。当管子受到较小的垂直位移并释放时，作用在管子上的合力的大小与其垂直位移 y 相关，表达式为： $F_{\text{net}} = \rho g A y$ 。(a) 表明管子执行简谐振动。(b) 找到振荡频率的表达式。问题 3.2 一个小型滑翔机沿着水平空气轨道移动，并在轨道末端的缓冲器上弹起。假设轨道是无摩擦的并且缓冲器是完全弹性的，说明并解释滑翔机是否描述简谐振动。3.2.3 简谐振子的速度和加速度简谐振子的位移随时间变化为： $x = x_0 \sin(\omega t + \phi)$ 从这个位移-时间关系，我们可以找到速度和加速度的关系 25

velocity

to find velocity-time relation, let's recall that velocity v is rate of change of displacement x

$$v = \frac{dx}{dt} = \frac{d}{dt} x_0 \sin(\omega t + \phi) \Rightarrow v(t) = \omega x_0 \cos(\omega t + \phi)$$

by taking $v^2 + \omega^2 x^2$, the sine and cosine terms can be eliminated, we find:

$$v^2 + \omega^2 x^2 = \omega^2 x_0^2 \cos^2(\dots) + \omega^2 x_0^2 \sin^2(\dots) = \omega^2 x_0^2$$

this gives velocity-displacement relation: $v(x) = \pm \omega \sqrt{x_0^2 - x^2}$

- at equilibrium position $x = 0$, speed is maximum: $v_{\max} = \omega x_0$
- when $x = \pm x_0$, oscillator is momentarily at rest: $v = 0$

acceleration

acceleration-time relation is found by further taking rate of change of velocity v

$$a = \frac{dv}{dt} = \frac{d}{dt} \omega x_0 \cos(\omega t + \phi) \Rightarrow a(t) = -\omega^2 x_0 \sin(\omega t + \phi)$$

this is actually unnecessary, if we compare this with $x(t) = x_0 \sin(\omega t + \phi)$, we have: $a = -\omega^2 x$

we have recovered the definition for simple harmonics

(if $a \propto x$ and in opposite directions to x , then simple harmonic motion)

so acceleration-displacement relation is given by the defining equation explicitly $a(x) = -\omega^2 x$

- at equilibrium position $x = 0$, zero acceleration
- when $x = \pm x_0$, acceleration is greatest: $a_{\max} = \omega^2 x_0$

let's take $x = x_0 \sin \omega t$ as example, changes of x , v , a over time are listed below

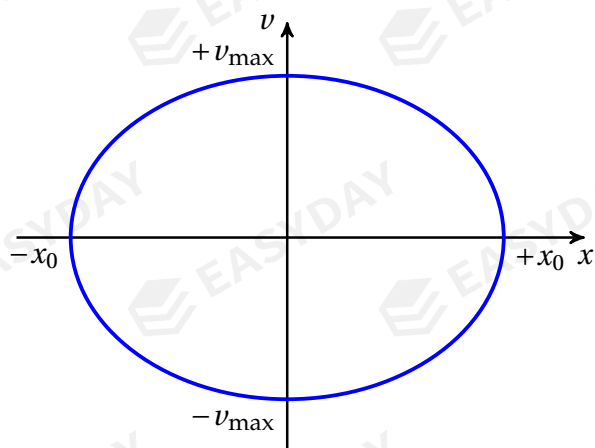
time t	0	$\frac{1}{4}T$	$\frac{1}{2}T$	$\frac{3}{4}T$	T
displacement: $x = x_0 \sin \omega t$	0	+max	0	-max	0
velocity: $v = \omega x_0 \cos \omega t$	+max	0	-max	0	+max
acceleration: $a = -\omega^2 x = -\omega^2 x_0 \sin \omega t$	0	-max	0	+max	0

Example 3.4 The motion of a simple pendulum is approximately simple harmonic. As the pendulum swings from one side to the other end, it moves through a distance of 6.0 cm and

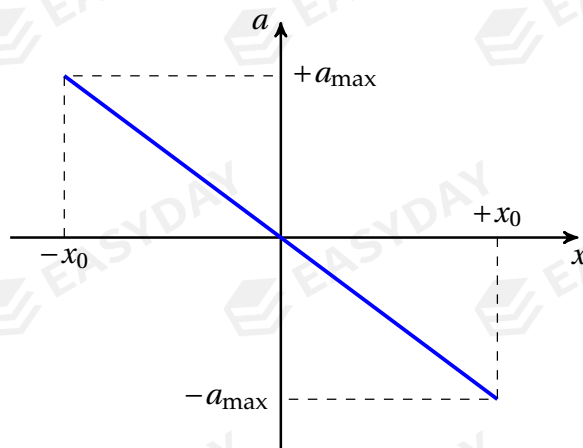
Yuhao Yang 3 振荡速度求速度-时间关系，我们回想一下，速度 v 是位移的变化率 $x \cdot v = dx/dt = d/dt x_0 \sin(\omega t + \phi)$ $v(t) = \omega x_0 \cos(\omega t + \phi)$ 通过取 $v^2 + \omega^2 x^2$ ，可以消除正弦和余弦项，我们发现： $v^2 + \omega^2 x^2 = \omega^2 x_0^2 \cos^2(\omega t + \phi) + \omega^2 x_0^2 \sin^2(\omega t + \phi) = \omega^2 x_0^2$ 给出速度-位移关系： $v(x) = \pm \omega \sqrt{x_0^2 - x^2}$ 在平衡位置 $x = 0$ 处，速度最大： $v_{\max} = \omega x_0$ 当 $x = \pm x_0$ 时，振子暂时静止： $v = 0$ 加速度通过进一步求速度变化率 $a = dv/dt = d/dt \omega x_0 \cos(\omega t + \phi)$ $a(t) = -\omega^2 x_0 \sin(\omega t + \phi)$ 这实际上是不必要的，如果我们将其与 $x(t) = x_0 \sin(\omega t + \phi)$ 进行比较，我们有： $a = -\omega^2 x$ 我们已经恢复了简谐运动的定义（如果 a/x 且与 x 方向相反，则为简谐运动），因此加速度-位移关系由定义方程明确给出 $a(x) = -\omega^2 x$ 在平衡位置 $x = 0$ ，零加速度 当 $x = \pm x_0$ 时，加速度最大： $a_{\max} = \omega^2 x_0$ 让我们取 $x = x_0 \sin \omega t$ 为例， x 、 v 、 a 随时间的变化如下：

时间 t	0	$1/4 T$	$1/2 T$	$3/4 T$	T
位移： $x = x_0 \sin \omega t$	0	$+x_0$	0	$-x_0$	0
速度： $v = \omega x_0 \cos \omega t$	$+x_0 \omega$	0	$-x_0 \omega$	0	$+x_0 \omega$
最大加速度： $a = -\omega^2 x = -\omega^2 x_0 \sin \omega t$	0	$-x_0 \omega^2$	0	$+x_0 \omega^2$	0

例3.4 单摆的运动近似为简谐振动。当单摆从一侧摆动到另一端时，它移动的距离为 6.0 厘米和 26



velocity-displacement graph



acceleration-displacement graph

the time taken is 1.0 s. (a) State the period and amplitude. (b) Find the greatest speed during the oscillation. (c) Find its speed when displacement $x = 1.2$ cm.

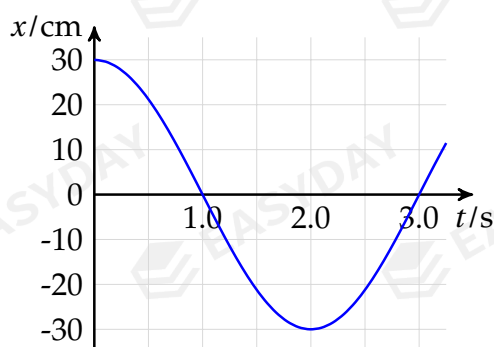
🔧 period: $T = 2 \times 1.0 = 2.0$ s, and amplitude: $x_0 = \frac{1}{2} \times 6.0 = 3.0$ cm

angular frequency: $\omega = \frac{2\pi}{T} = \frac{2\pi}{2.0} = \pi$ rad s⁻¹

greatest speed: $v_{\max} = \omega x_0 = \pi \times 3.0 \approx 9.4$ cm s⁻¹

speed at 1.2 cm: $v = \omega \sqrt{x_0^2 - x^2} = \pi \times \sqrt{3.0^2 - 1.2^2} \approx 8.6$ cm s⁻¹ □

Example 3.5 Given the x - t graph of a simple harmonic oscillator. (a) Find its speed at $t = 0$. (b) Find its greatest speed. (b) Find its acceleration at $t = 1.0$ s.



🔧 at $t = 0$, $x = +x_0 \Rightarrow v = 0$ (zero gradient)

from graph: amplitude $x_0 = 30$ cm, period $T = 4.0$ s

angular frequency: $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$ rad s⁻¹

greatest speed: $v_{\max} = \omega x_0 = \frac{\pi}{2} \times 30 \approx 47$ cm s⁻¹

at $t = 1.0$ s, $x = 0 \Rightarrow a = 0$

(equilibrium position so no acceleration) □

Question 3.3 Assume the motion of a car engine piston is simple harmonic. The piston completes 3000 oscillations per minute. The amplitude of the oscillation is 4.0 cm. (a) Find the greatest speed. (b) Find the greatest acceleration.

杨宇豪 3 振荡 x v $+x_0$ $-x_0$ $+v_{\max}$ $-v_{\max}$ 速度-位移图 x a $+x_0$ $-x_0$ $+a_{\max}$ $-a_{\max}$ 加速度-位移图所用时间为 1.0 s 。(a) 说出周期和幅度。(b) 求振荡过程中的最大速度。(c) 求位移 $x = 1.2\text{ cm}$ 时的速度。- 周期: $T = 2 \times 1.0 = 2.0\text{ s}$, 振幅: $x_0 = \frac{1}{2} \times 6.0 = 3.0\text{ cm}$ 角频率: $\omega = \frac{2\pi}{T} = \frac{2\pi}{2.0} = \pi\text{ rad s}^{-1}$ 最大速度: $v_{\max} = \omega x_0 = \pi \times 3.0 \approx 9.4\text{ cm s}^{-1}$ 速度 at 1.2 cm : $v = \omega \sqrt{x_0^2 - x^2} = \pi \times \sqrt{3.0^2 - 1.2^2} \approx 8.6\text{ cm s}^{-1}$ $\ddot{a}$ 示例 3.5 给出简谐振子的 x - t 图。(a) 求其在 $t = 0$ 时的速度。(b) 求其最大速度。(b) 求 $t = 1.0\text{ s}$ 时的加速度。 t/s x/cm 1.0 2.0 3.0 0 -30 -20 -10 0 10 20 30 - $t = 0, x = +x_0$ $v = 0$ (零梯度), 来自图表: 振幅 $x_0 = 30\text{ cm}$, 周期 $T = 4.0\text{ s}$ 角频率: $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}\text{ rad s}^{-1}$ 最大速度: $v_{\max} = \omega x_0 = \frac{\pi}{2} \times 30 \approx 47\text{ cm s}^{-1}$ at $t = 1.0\text{ s}, x = 0$ $a = 0$ (平衡位置, 因此没有加速度) $\ddot{a}$ 问题 3.3 假设汽车发动机活塞的运动是简谐运动。活塞每分钟完成 3000 次振荡。振动幅度为 4.0 cm 。(a) 求最大速度。(b) 求最大加速度。

3.2.4 vibrational energy

consider the *ideal* mass-spring oscillator, its vibrational energy consists of two parts:

- kinetic energy of the mass: $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 x_0^2 \cos^2 \omega t \stackrel{v=\pm\omega\sqrt{x_0^2-x^2}}{=} \frac{1}{2}m\omega^2(x_0^2 - x^2)$
- (elastic) potential energy in the spring: $E_p = \frac{1}{2}kx^2 \stackrel{\omega=\sqrt{\frac{k}{m}}}{=} \frac{1}{2}m\omega^2 x^2$

total energy of the oscillator: $E = E_k + E_p \Rightarrow E = \frac{1}{2}m\omega^2 x_0^2$

➤ although this formula is derived from the mass-spring model

$E = \frac{1}{2}m\omega^2 x_0^2$ can be used to compute vibrational energy of all simple harmonic oscillators

➤ for an ideal system, total energy remains constant

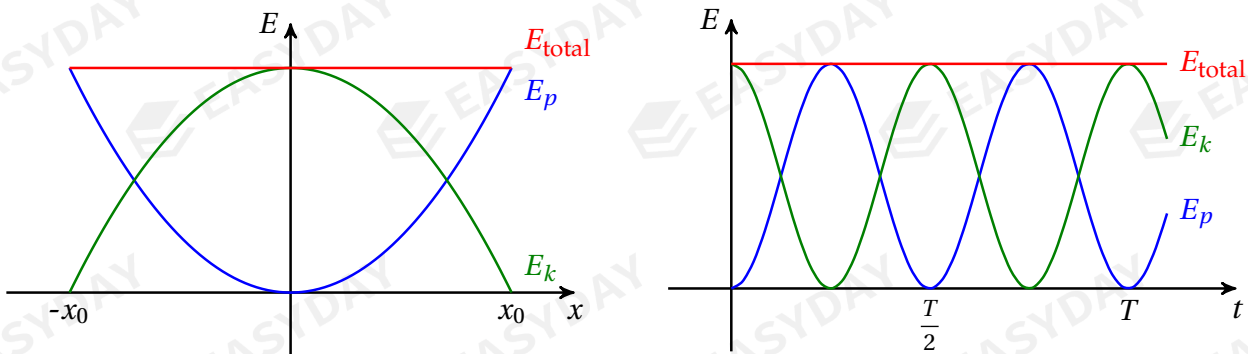
E_k and E_p keep changing, one transfers into another, but total energy is *conserved*

➤ when $x = 0$, $E_k = \max$, $E_p = 0$, vibrational energy is purely kinetic

$$E = E_{k,\max} = \frac{1}{2}mv_{\max}^2 \stackrel{v_{\max}=\omega x_0}{=} \frac{1}{2}m\omega^2 x_0^2$$

➤ when $x = \pm x_0$, $E_k = 0$, $E_p = \max$, vibrational energy is purely potential

$$E = E_{p,\max} = \frac{1}{2}kx_0^2 \stackrel{\omega=\sqrt{\frac{k}{m}}}{=} \frac{1}{2}m\omega^2 x_0^2$$



vibrational energy of a mass-spring oscillator

Example 3.6 A block of mass 150 g at the end of a spring oscillates with a period of 0.80 s. The maximum displacement from its rest position is 12 cm. Find the energy of the vibration.

$$E = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}m\left(\frac{2\pi}{T}\right)^2 x_0^2 = \frac{1}{2} \times 0.15 \times \frac{4\pi^2}{0.80^2} \times 0.12^2 \approx 6.7 \times 10^{-2} \text{ J} \quad \square$$

Question 3.4 An oscillator is given an energy of 20 mJ and starts to oscillate, it reaches an amplitude of 8.0 cm. If we want to double the amplitude, find the vibrational energy required.

Yuhao Yang 3 振荡 3.2.4 振动能考虑理想的质量弹簧振荡器，其振动能量由两部分组成：- 质量的动能： $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2x_0^2 \cos^2 \omega t$ $v = \omega x_0 \sin \omega t$ - 弹簧中的（弹性）势能： $E_p = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x_0^2 \sin^2 \omega t$ 振荡器的总能量： $E = E_k + E_p = \frac{1}{2}m\omega^2x_0^2$ 虽然该公式源自质量弹簧模型 $E = \frac{1}{2}m\omega^2x_0^2$ 可用于计算所有简谐振子的振动能量 对于理想系统，总能量保持恒定 E_k 和 E_p 不断变化，一个转移到另一个，但总能量守恒 当 $x = 0$ 、 $E_k = \frac{1}{2}mv_{\max}^2$ 、 $E_p = 0$ 时，振动能量是纯动能 $E = E_{k,\max} = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}m\omega^2x_0^2$ 当 $x = x_0$ 、 $E_k = 0$ 、 $E_p = \frac{1}{2}kx_0^2$ 时，振动能量为纯势能 $E = E_{p,\max} = \frac{1}{2}kx_0^2 = \frac{1}{2}m\omega^2x_0^2$ 质量弹簧振荡器的振动能量示例 3.6 弹簧末端质量为 150 g 的块以 0.80 s 的周期振荡。距静止位置的最大位移为 12 厘米。求振动的能量。 - $E = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2}m \left(\frac{2\pi}{T} \right)^2 x_0^2 = \frac{1}{2} \times 0.15 \times 4\pi^2 \times 0.80^2 \times 0.12^2 = 6.7 \times 10^{-2} \text{ J}$ 问题 3.4 一个振荡器被给予 20 mJ 的能量并开始振荡，它的振幅达到 8.0 厘米。如果我们想让振幅加倍，请找到所需的振动能量。 28

3.3 damped oscillations

total vibrational energy stays constant for an ideal system

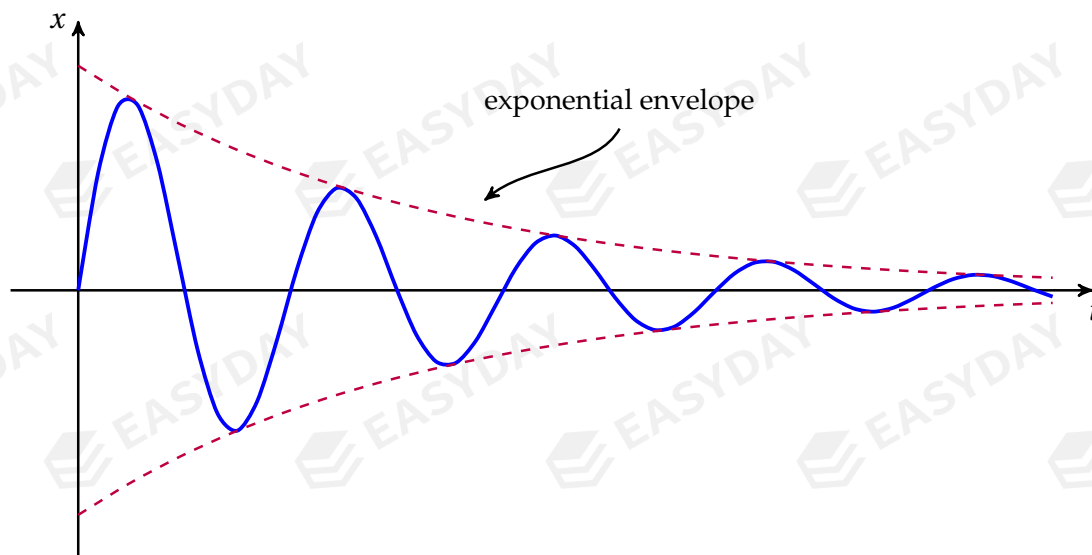
but in reality, there are friction, resistance and viscous forces that oppose motion

amplitude of an oscillator decreases due to energy loss to friction, this is called **damping**

3.3.1 light damping

for a **lightly-damped oscillator**, amplitude decreases *gradually*

oscillator will not stop moving back and forth after quite a few oscillations



- decrease in amplitude is *non-linear* in time (exponential decay in many cases)
- frequency and period are (almost) unchanged

Example 3.7 An oscillator is composed of a block of mass $m = 250$ g and a spring of $k = 1.6$ N/cm. It is displaced by 5.0 cm from its rest position and set free. (a) What is its angular frequency? (b) what is the initial vibrational energy? (c) After a few oscillations, 40% of its energy is lost due to damping. What is its new amplitude?

🔗 angular frequency: $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{160}{0.25}} \approx 25.3 \text{ rad s}^{-1}$

energy of oscillator: $E = \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} \times 0.25 \times 25.3^2 \times 0.050^2 = 0.20 \text{ J}$ ^[13]

^[13] An easier approach: $E = \frac{1}{2} k x_0^2 = \frac{1}{2} \times 160 \times 0.050^2 = 0.20 \text{ J}$.

杨宇浩 3 振荡 3.3 阻尼振荡理想系统的总振动能量保持恒定，但实际上，存在摩擦力、阻力和粘性力阻碍振荡器的运动，由于摩擦的能量损失，振荡器的振幅减小，这称为阻尼 3.3.1 轻阻尼振荡器的轻阻尼，振幅逐渐减小，在相当多次振荡后，振荡器不会停止来回移动 $t \times$ 指数包络 $\hat{a}$ 振幅的减小在时间上是非线性的（许多情况下呈指数衰减） 频率和周期（几乎）不变 示例 3.7 振荡器由质量 $m \leq 250 \text{ g}$ 的块和 $k \leq 1.6 \text{ N/cm}$ 的弹簧组成。它从静止位置移动 5.0 厘米并释放。(a) 其角频率是多少？(b) 初始振动能量是多少？(c) 经过几次振荡后，40% 的能量因阻尼而损失。它的新振幅是多少？ - 角频率： $\omega = \sqrt{k/m} = \sqrt{160/0.25} = 25.3 \text{ rad s}^{-1}$ 振荡器能量： $E = \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} \times 0.25 \times 25.3^2 \times 0.0502 = 0.20 \text{ J}$ [13] [13] 更简单的方法： $E = \frac{1}{2} k x^2 = \frac{1}{2} \times 160 \times 0.0502 = 0.20 \text{ J}$.29

since $E \propto x_0^2$, so: $\frac{E'}{E} = \frac{x_0'^2}{x_0^2} \Rightarrow 60\% = \frac{x_0'^2}{x_0^2} \Rightarrow x_0' = \sqrt{0.6}x_0 = \sqrt{0.6} \times 5.0 \approx 3.9 \text{ cm}$ $\square$

Question 3.5 A small toy boat of mass 360 g floats on surface of water. It is gently pushed down and then released. During the first four complete cycles of its oscillation, its amplitude decreased from 5.0 cm to 2.0 cm in a time of 6.0 s. Find the energy loss.

3.3.2 heavy damping

if resistive forces are too strong, there will be no oscillatory motion

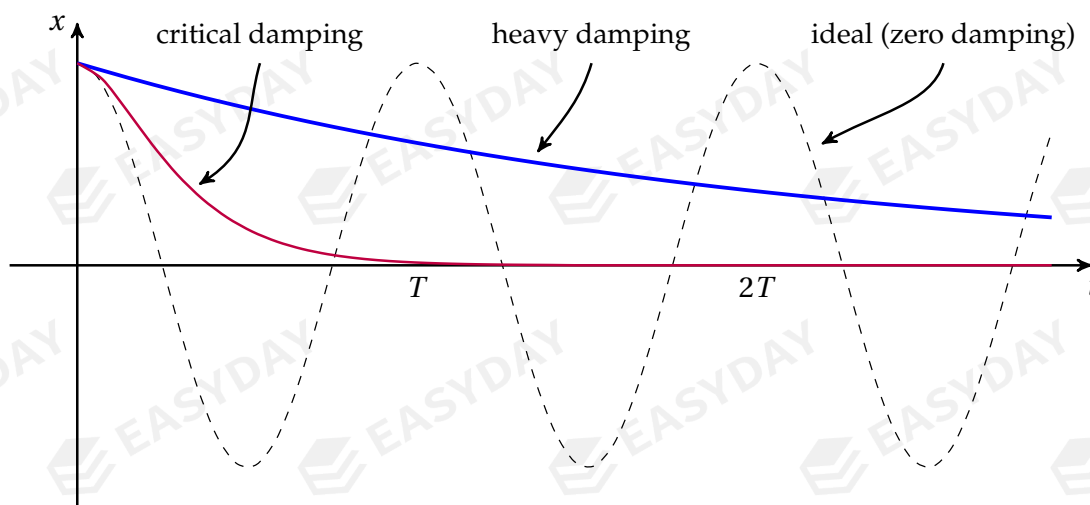
the system will return to the equilibrium position very slowly

this system is said to be **heavily damped**

3.3.3 critical damping

critical damping is the border between light damping and heavy damping

it occurs when system returns to equilibrium in *shortest* time without any oscillation



➤ critical damping is desirable in many engineering designs ^[14]

examples include door-closing mechanism, shock absorbers in vehicles and artillery, etc.

^[14] When a damped oscillator is required, critically-damped system provides the quickest approach to equilibrium without overshooting, while lightly-damped system reaches the zero position quickly but continues to oscillate, and heavily-damped system reaches zero position in very long time.

杨宇豪 3 自 E / x^2 以来的振荡，所以： $E_0 E = x_0^2 \times x_0^2 \times 60\% = x_0^2 \times x_0^2 \times 0.6 = p \times 0.6 \times 0 = p \times 0.6 \times 5.0 \times 3.9 \text{ cm}^2$ 问题 3.5 一艘质量为 360 g 的小玩具船漂浮在水面上。将其轻轻地向下推，然后释放。在前四个完整的振荡周期中，其振幅在 6.0 s 的时间内从 5.0 cm 减小到 2.0 cm 。求能量损失。

3.3.2 重阻尼 如果阻力太强，将不会产生振荡运动系统将非常缓慢地返回到平衡位置该系统被称为重阻尼

3.3.3 临界阻尼 临界阻尼是轻阻尼和重阻尼之间的边界它发生在系统在最短的时间内恢复平衡而没有任何振荡 $t \times$ 重阻尼 临界阻尼理想（零阻尼） $T/2\pi$ 临界阻尼在许多工程设计中是可取的 [14] 示例包括 [14] 当需要阻尼振荡器时，临界阻尼系统可以最快地接近平衡而不会超调，而轻阻尼系统可以快速到达零位，但会继续振荡，而重阻尼系统则需要很长时间才能到达零位。 30

3.4 forced oscillations

3.4.1 free & forced oscillation

an oscillator moving on its own with no gain or loss of energy is called **free oscillation**

amplitude of the oscillation is constant, its frequency called **natural frequency**

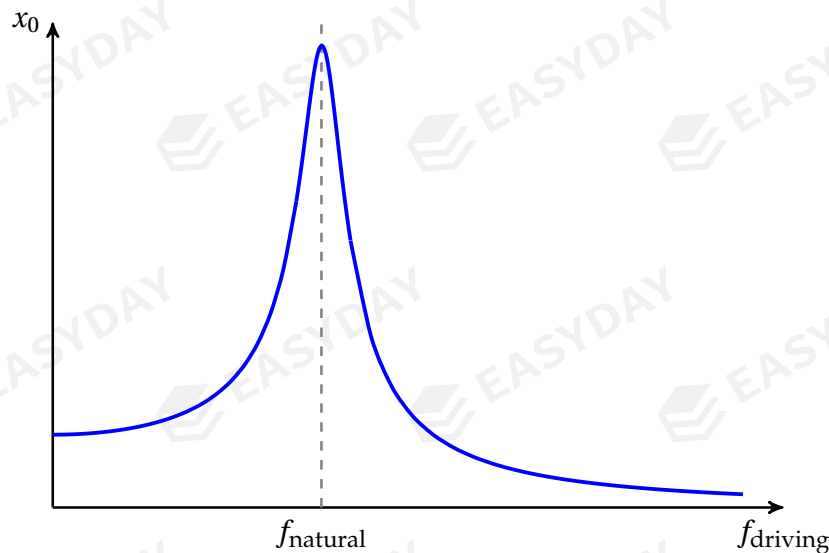
an oscillator may also move under an external driving force, it is **forced oscillation**

frequency of forced oscillator tends to driving frequency after sufficiently long time

3.4.2 resonance

for a forced oscillation system, when frequency of driving force f_{driving} is close to natural frequency f_{natural} , amplitude of oscillator increases rapidly

when driving frequency of external force equals natural frequency of the system, amplitude of the system becomes maximum, this phenomenon is called **resonance**



resonance is achieved when $f_{\text{driving}} = f_{\text{natural}}$
(amplitude tends to infinity if no damping)

➤ practical application of resonance

- microwave oven – water molecules resonate at microwave frequency and vibrate greatly
- MRI (magnetic resonance imaging) — precession of nuclei resonate at radio frequency,

杨宇浩 3 振荡 3.4 受迫振荡 3.4.1 自由与受迫振荡 一个不增加或损失能量而自行运动的振荡器称为自由振荡 振荡的振幅是恒定的，其频率称为固有频率 振荡器也可以在外部驱动力的作用下运动，它是受迫振荡 受迫 振荡器的频率在足够长的时间后趋向于驱动频率 3.4.2 受迫振荡系统的共振，当驱动力 f_{drive} 接近固有频率 f_{natural} 时，振荡器的振幅当外力的驱动频率等于系统的固有频率时，系统的振幅迅速增大，这种现象称为共振 f_{natural} f_{driven} 当 $f_{\text{driven}} = f_{\text{natural}}$ 时实现共振（如果无阻尼，振幅趋于无穷大） $\hat{a}$ 共振的实际应用 - 微波炉 - 水分子在微波频率下共振并剧烈振动 - MRI（磁共振成像） - 核进动在射频下共振，31

signals are processed to image nuclei of atoms inside a human body in detail

- radio/TV — RLC tuning circuits resonate at frequency of signals being received

➤ possible problems caused by resonance

- buildings during earthquake – resonate at frequency of shockwaves and collapse

- car suspension system – going over bumps may give large amplitude vibrations

- bridges and skyscrapers – resonance due to wind conditions

3.4.3 damping & resonance

an oscillation system can be subject to both driving force and resistive force

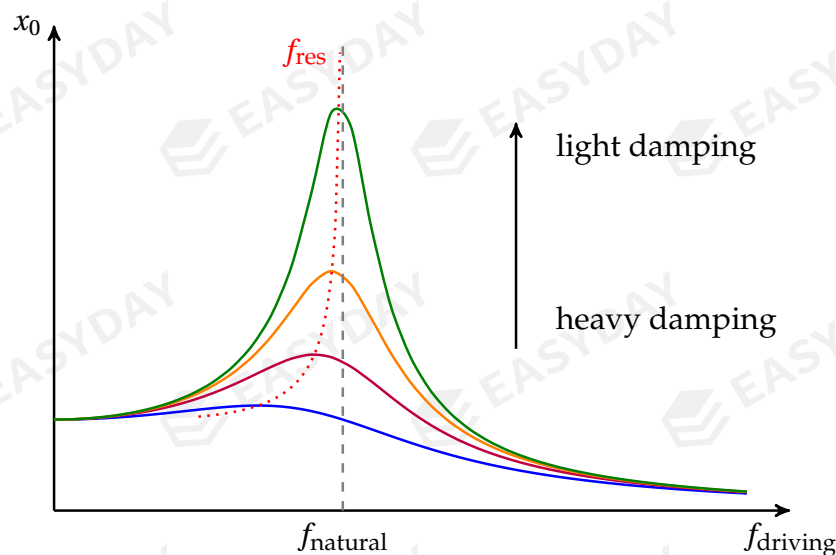
resonance behaviour will be changed by damping effects

➤ damping decreases amplitude of oscillation at all frequencies

greater damping causes resonance peak to become *flatter*

engineering systems are often deliberately damped to minimise resonance effect

➤ damping also shifts resonance frequency (slightly reduced for light damping)



resonance effect for various damping conditions

Yuhao Yang 3 振荡信号经过处理，可对人体内部原子核进行详细成像 - 广播/电视 - RLC 调谐电路在接收到的信号频率处产生共振 - 共振可能引起的问题 - 地震期间的建筑物 - 在冲击波和倒塌的频率下共振 - 汽车悬架系统 - 越过颠簸处可能会产生大幅振动 - 桥梁和摩天大楼 - 由于风力条件而产生共振 3.4.3 阻尼和共振 振荡系统可能会受到驱动力和共振的影响 阻尼效应将改变电阻力谐振行为 α 阻尼会降低所有频率下的振荡幅度 更大的阻尼会导致谐振峰值变得更平坦 工程系统通常会故意阻尼以尽量减少谐振效应 α 阻尼还会改变谐振频率（轻阻尼时会稍微降低） $f_{\text{natural}} f_{\text{drive}} \times 0 f_{\text{res}}$ 重阻尼 轻阻尼 各种阻尼条件下的谐振效应 32

CHAPTER 4

Ideal Gases

4.1 gas molecules

4.1.1 motion of gas particles

gas consists of a large number of molecules

gas molecules move *randomly* at high speeds

- randomness results from *collisions* of fast-moving molecules in the gas

for an individual molecule, its velocity changes constantly as it collides with other molecules

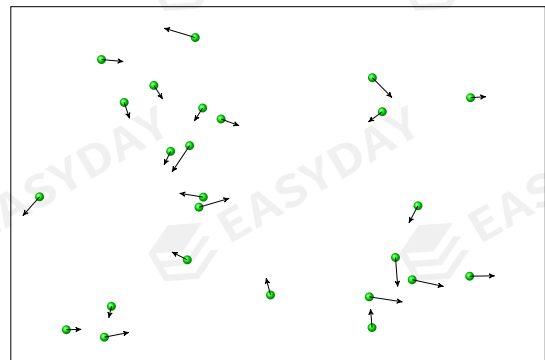
for the gas at any instant, there is a range of velocities for molecules

- experimental evidence of random motion: **Brownian motion**

dust or smoke particles in air undergo jerky random motion (viewed through microscope)

this is due to collisions with gas molecules that move randomly

- speed of gas molecules depend on temperature
molecules move faster at higher temperature^[15]



motion of gas molecules in a container

4.1.2 amount of molecules

there are a huge number of molecules in a gas

we introduce **amount of substance** to measure the size of a collection of particles

^[15] We will prove this statement later in this chapter.

杨宇豪 4 理想气体第四章理想气体 4.1 气体分子 4.1.1 气体粒子的运动容器中气体分子的运动气体由大量分子组成气体分子高速随机运动——对于单个分子来说，随机性是由于气体中快速运动的分子碰撞而产生的，对于气体来说，在任何时刻，它的速度随着与其他气体分子的碰撞而不断变化，分子存在一个速度范围——随机性的实验证据运动：布朗运动空气中的灰尘或烟雾颗粒经历不稳定的随机运动（通过显微镜观察），这是由于与随机移动的气体分子的碰撞造成的 $\hat{\alpha}$ 气体分子的速度取决于温度分子在较高温度下移动得更快 [15] 4.1.2 分子数量气体中有大量分子，我们引入物质数量来测量颗粒集合的大小 [15]我们将在本章后面证明这一说法。 33

- unit of amount of substance: $[n] = \text{mol}$

one **mole** is defined as the amount carbon-12 atoms in a sample of 12 grams

- 1 mole of substance contains 6.02×10^{23} particles

this number is called **Avogadro constant**: $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ [16]

conversion between number of molecules and amount of substance: $N = nN_A$

- it is useful to introduce the notion of molar mass M

molar mass of a substance is defined as the mass of a given sample divided by the

amount of substance: $M = \frac{m}{n}$

– amount of substance = $\frac{\text{mass of sample}}{\text{molar mass}}$, or $n = \frac{m}{M}$

– mass of single molecule = $\frac{\text{molar mass}}{\text{Avogadro constant}}$, or $m_0 = \frac{M}{N_A}$

Example 4.1 Find the number of molecules in 160 grams of argon-40 gas.

✎ amount of gas: $n = \frac{m}{M} = \frac{160 \text{ g}}{40 \text{ g mol}^{-1}} = 4.0 \text{ mol}$

number of gas molecules: $N = nN_A = 4.0 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} \approx 2.41 \times 10^{24}$ □

Question 4.1 Find the mass of a sample of uranium-235 that contains 6.0×10^{20} atoms.

4.1.3 pressure (qualitative view)

when gas molecules collide with walls of container and rebound, they are acted by a force by Newton's third law, gas molecules must exert a reaction force on container in return contributions from many molecules give rise to a pressure

Example 4.2 If a gas is heated with its volume fixed, how does the pressure change?

✎ at higher temperature, gas molecules move faster

they will collide *harder* and produce a greater force upon each collision

they will also collide more *frequently* with the container

[16] In 2018, IUPAC suggested a new definition of the mole, which is defined to contain exactly 6.02×10^{23} particles. This new definition fixed numerical value of the Avogadro constant, and emphasized that the quantity 'amount of substance' is concerned with counting number of particles rather than measuring the mass of a sample.

Yuhao Yang 4 理想气体 $\hat{a}$ 物质质量单位：[n] = mol 一摩尔被定义为 12 克样品中碳 12 原子的量 $\hat{a}$ 1 摩尔物质含有 6.02×10^{23} 颗粒，这个数字称为阿伏加德罗常数： $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ [16] 分子数与物质质量之间的换算： $N = n N_A$ $\hat{a}$ 引入摩尔质量的概念很有用 M 物质的摩尔质量定义为给定样品的质量除以物质的量： $M = m/n$ - 物质的量 = 样品质量 / 摩尔质量，或 $n = m/M$ - 单分子的质量 = 摩尔质量 / 阿伏加德罗常数，或 $m_0 = M / N_A$ 示例 4.1 求 160 克物质中的分子数氩 40 气体。 - 气体量： $n = m/M = 160 \text{ g} / 40 \text{ g mol}^{-1} = 4.0 \text{ mol}$ 气体分子数： $N = n N_A = 4.0 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 2.41 \times 10^{24}$ $\hat{a}$ 问题 4.1 求含有 6.0×10^{23} 样品的质量 1020 个原子。 4.1.3 压力（定性观点）当气体分子与容器壁碰撞并反弹时，根据牛顿第三定律，它们受到力的作用，气体分子必须对容器施加反作用力，作为回报，许多分子的贡献会产生压力 例 4.2 如果气体在体积固定的情况下被加热，压力如何变化？ - 在更高的温度下，气体分子移动得更快，它们会更强烈地碰撞，每次碰撞时产生更大的力，它们也会更频繁地与容器碰撞 [16] 2018 年，IUPAC 提出了摩尔的新定义，它被定义为恰好包含 6.02×10^{23} 个粒子。这个新定义固定了阿伏加德罗常数的数值，并强调“物质的量”涉及粒子数量的计数，而不是测量样品的质量。 34

so pressure of the gas will increase □

Question 4.2 If you pump gas into a bicycle tyre, state and explain how the pressure changes.

Question 4.3 A fixed amount of gas is allowed to expand at constant temperature, state and explain how the pressure changes.

4.2 ideal gas

4.2.1 ideal gas equation

a gas that satisfies the equation $pV = nRT$ or $pV = NkT$ at any pressure p , any volume V , and thermodynamic temperature T is called an **ideal gas**

molar gas constant: $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

Boltzmann constant: $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$

values of R and k apply for any ideal gas, i.e., they are *universal* constants

➤ recall conversion between number of molecules and amount of substance: $N = nN_A$

we have relation between the constants: $R = kN_A$, or $k = \frac{R}{N_A}$

➤ one must use *thermodynamic temperature* in the equation

thermodynamic temperature is measured in kelvins (K), so it is also called the *Kelvin scale*^[17]

conversion between Kelvin temperature and Celsius temperature: $T_K(\text{K}) \begin{matrix} \xrightarrow{-273} \\ \xleftarrow{+273} \end{matrix} T_C(^{\circ}\text{C})$

real gases

real gas behaves ideally at sufficiently high temperature and low pressure

- at very low temperatures, real gas will condense into liquid or solid
- at very high pressures, intermolecular forces become important

however, under normal conditions (room temperature $T \approx 300 \text{ K}$ and standard atmospheric pressure $p \approx 1.0 \times 10^5 \text{ Pa}$), there is no significant difference between a real gas and an ideal gas

so ideal gas approximation can be used with good accuracy for most of our applications

^[17] We will discuss in details about Kelvin scale in §5.1.1 and §5.3.2.

Yuhao Yang 4 理想气体，因此气体压力会增加。问题 4.2 如果将气体泵入自行车轮胎，请说明并解释压力如何变化。问题 4.3 允许一定量的气体在恒温下膨胀，说明并解释压力如何变化。

4.2 理想气体

4.2.1 理想气体方程

在任意压力 p 、任意体积 V 和热力学温度 T 下，满足方程 $pV = nRT$ 或 $pV = NkT$ 的气体称为理想气体。摩尔气体常数： $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ 玻尔兹曼常数： $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ R 和 k 的值适用于任何理想气体，即它们是通用常数。回想一下分子数和物质量之间的换算： $N = nN_A$ 我们有常数之间的关系： $R = kN_A$ 或 $k = R/N_A$ 。必须在方程中使用热力学温度。热力学温度以开尔文 (K) 为单位，因此也称为开尔文温标[17]。开尔文温度和摄氏温度之间的转换： $T(K) = T(^{\circ}\text{C}) + 273.15$ 。真实气体。真实气体在足够高的温度和低压下表现理想。在非常低的温度下，真实气体会凝结成液体或固体。在非常高的压力下，分子间力变得很重要，但是，在正常条件下（室温 $T \approx 300 \text{ K}$ 和标准大气压 $p \approx 1.0 \times 10^5 \text{ Pa}$ ），没有显著差异。真实气体和理想气体之间存在差异，因此理想气体近似可以在我们的大多数应用中以良好的精度使用[17]。我们将在第 5.1.1 节和第 5.3.2 节中详细讨论开尔文标度。

35

Example 4.3 A sealed cylinder of volume of 0.050 m^3 contains 75 g of air. The molar mass of air is 29 g mol^{-1} . (a) Find the air pressure when its temperature is 30°C . (b) The gas is allowed to expand with its pressure fixed. Find the temperature of the gas when the volume doubles.

amount of gas: $n = \frac{m}{M} = \frac{75}{29} \approx 2.59 \text{ mol}$

pressure at 30°C : $p = \frac{nRT_1}{V_1} = \frac{2.59 \times 8.31 \times (30 + 273)}{0.050} \approx 1.30 \times 10^5 \text{ Pa}$

pressure fixed, so $V \propto T \Rightarrow \frac{T_2}{T_1} = \frac{V_2}{V_1} = 2 \Rightarrow T_2 = 2 \times (30 + 273) = 606 \text{ K} = 333^\circ\text{C}$ $\square$

Example 4.4 A gas cylinder holding 5000 cm^3 of air at a temperature of 27°C and a pressure of $6.0 \times 10^5 \text{ Pa}$ is used to fill balloons. Each balloon contains 1000 cm^3 of air at 27°C and $1.0 \times 10^5 \text{ Pa}$ when filled. (a) Find the initial amount of gas in the cylinder. (b) Find the number of balloons that can be filled.

initial amount of gas in cylinder: $n_0 = \frac{p_0 V}{RT} = \frac{6.0 \times 10^5 \times 5000 \times 10^{-6}}{8.31 \times (27 + 273)} \approx 1.203 \text{ mol}$

final amount of gas in cylinder: $n_{\text{remain}} = \frac{pV}{RT} = \frac{1.0 \times 10^5 \times 5000 \times 10^{-6}}{8.31 \times (27 + 273)} \approx 0.201 \text{ mol}$ ^[18]

amount of gas in each balloon: $n_b = \frac{pV_b}{RT} = \frac{1.0 \times 10^5 \times 1000 \times 10^{-6}}{8.31 \times (27 + 273)} \approx 0.040 \text{ mol}$

number of balloons: $N = \frac{n_0 - n_{\text{remain}}}{n_b} = \frac{1.203 - 0.201}{0.040} \approx 25$ $\square$

Example 4.5 A storage cylinder has a volume of $5.0 \times 10^{-4} \text{ m}^3$. The gas is at a temperature of 300 K and a pressure of $4.0 \times 10^6 \text{ Pa}$. (a) Find the number of molecules in the cylinder. (b) The gas molecules slowly leak from the cylinder at a rate of $1.6 \times 10^{16} \text{ s}^{-1}$. Find the time, in days, after which the pressure will reduce by 5.0% .

initial number of molecules: $N_0 = \frac{p_0 V}{kT} = \frac{4.0 \times 10^6 \times 5.0 \times 10^{-4}}{1.38 \times 10^{-23} \times 300} \approx 4.83 \times 10^{23}$

volume fixed, so $N \propto p \Rightarrow \frac{\Delta N}{N_0} = \frac{\Delta p}{p_0} = 5.0\%$

number of molecules escaped: $\Delta N = 0.05 \times 4.83 \times 10^{23} \approx 2.42 \times 10^{22}$

time needed: $t = \frac{2.42 \times 10^{22}}{1.6 \times 10^{16}} \approx 1.51 \times 10^6 \text{ s} \approx 17.4 \text{ days}$ $\square$

[18] Air will leave the cylinder to fill balloons only if pressure inside the cylinder is higher than pressure of the balloon. When the two pressures become equal, no more balloons can be filled, there will be some air remain in cylinder.

Yuhao Yang 4 理想气体实施例4.3 体积为 0.050m^3 的密封圆筒含有 75g 空气。空气的摩尔质量为 29 g mol^{-1} 。(a) 求空气温度为 $30\pm\text{C}$ 时的压力。(b) 在压力固定的情况下允许气体膨胀。求体积加倍时气体的温度。

- 气体量： $n = m/M = 75/29 = 2.59$

$30\pm\text{C}$ 时的摩尔压力： $p = nRT/V = 2.59 \times 8.31 \times (30 + 273)/0.050 = 1.30 \times 10^5\text{ Pa}$

压力固定，因此 V/T $T_2/T_1 = V_2/V_1 = 2$ $T_2 = 2 \times (30 + 273) = 606\text{ K} = 333\pm\text{C}$

ä 示例 4.4 使用装有 5000 cm^3 空气、温度为 $27\pm\text{C}$ 、压力为 $6.0 \times 10^5\text{ Pa}$ 的气瓶来填充气球。每个气球充满后在 $27\pm\text{C}$ 和 $1.0 \times 10^5\text{ Pa}$ 下含有 1000 cm^3 的空气。(a) 求气瓶中的初始气体量。(b) 求可填充的气球数量。

- 气瓶中的初始气体量： $n_0 = p_0 V_0 / RT = 9.0 \times 10^5 \times 5000 \times 10^{-6} / (8.31 \times (27 + 273)) = 1.203\text{ mol}$

气瓶中的最终气体量： $n_{\text{remain}} = p V_{\text{remain}} / RT = 1.0 \times 10^5 \times 5000 \times 10^{-6} / (8.31 \times (27 + 273)) = 0.201\text{ mol}$

[18] 每个气球中的气体量： $n_b = p V_b / RT = 1.0 \times 10^5 \times 1000 \times 10^{-6} / (8.31 \times (27 + 273)) = 0.040\text{ mol}$

气球数量： $N = n_0 - n_{\text{remain}} = 1.203 - 0.201 = 1.002$

ä 示例 4.5 储罐的体积为 $5.0 \times 10^{-4}\text{ m}^3$ 。该气体的温度为 300 K ，压力为 $4.0 \times 10^6\text{ Pa}$ 。(a) 求圆柱体中的分子数。(b) 气体分子以 $1.6 \times 10^{16}\text{ s}^{-1}$ 的速率缓慢地从气缸中泄漏。求出压力将降低 5.0% 后的时间（以天为单位）。

- 初始分子数： $N_0 = p_0 V / kT = 4.0 \times 10^6 \times 5.0 \times 10^{-4} / (1.38 \times 10^{-23} \times 300) = 4.83 \times 10^{23}$

体积固定，因此 N/p $\Delta N / N_0 = \Delta p / p_0 = 5.0\%$

逃逸分子数： $\Delta N = 0.05 \times 4.83 \times 10^{23} = 2.42 \times 10^{22}$

所需时间： $t = \Delta N / (1.6 \times 10^{16}) = 1.51 \times 10^6\text{ s} = 17.4\text{ 天}$

ä [18] 只有当气缸内的压力高于气球的压力时，空气才会离开气缸来填充气球。当两个压力相等时，不再可以填充气球，气缸中将残留一些空气。

Question 4.4 Containers *A* has a volume of $2.5 \times 10^{-2} \text{ m}^3$ contains a gas at a temperature of 17°C and pressure of $1.3 \times 10^5 \text{ Pa}$ and . Another container *B* of same size holds a gas at same temperature and a pressure of $1.9 \times 10^5 \text{ Pa}$. The two containers are initially isolated from each another. (a) Find the total amount of molecules. (b) The two containers are now connected through a tube of negligible volume. Assume the temperature stays unchanged, find the final pressure of the gas.

Question 4.5 The air in a car tyre can be assumed to have a constant volume of $3.0 \times 10^{-2} \text{ m}^3$. The pressure of this air is $2.8 \times 10^5 \text{ Pa}$ at a temperature of 25°C . The pressure is to be increased using a pump. On each stroke 0.015 mol of air is forced into the tyre. If gas has a final pressure of $3.6 \times 10^5 \text{ Pa}$ and final temperature of 28°C . Find the number of strokes of the pump required.

4.2.2 empirical laws

historically, the ideal gas law was first stated by *Émile Clapeyron* in 1834:

for a fixed amount of gas, $\frac{PV}{T} = \text{const}$

his work was based on the empirical Boyle's law, Charles's law, and Gay-Lussac's law

we will next recover these laws from the ideal gas equation

Boyle's law

Boyle's law was discovered by *Robert Boyle* in 1662, based on experimental observations

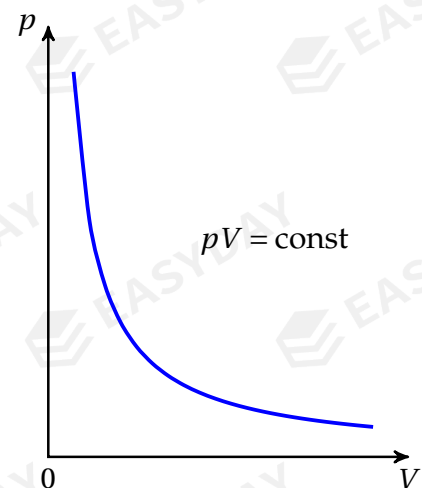
if temperature T remains constant, then

$$pV = \text{const}, \text{ or } p \propto \frac{1}{V}$$

pressure p of gas is inversely proportional to volume V

- for a gas with fixed temperature: $p_1 V_1 = p_2 V_2$
- a thermodynamic process for which temperature is kept constant is called an *isothermal* process

p - V relation for an isothermal process is shown

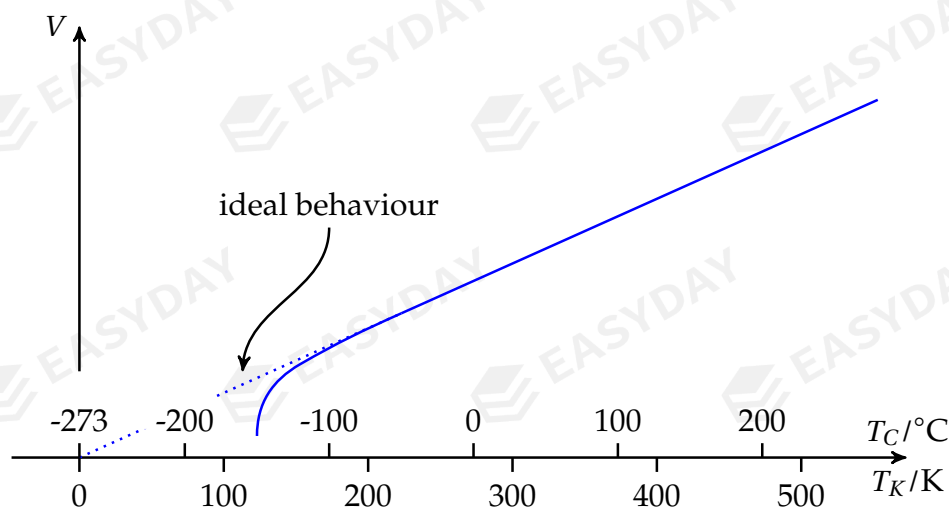


Yuhao Yang 4 理想气体问题 4.4 容器 A 的体积为 $2.5 \times 10^{-2} \text{ m}^3$ ，含有温度为 17°C 、压力为 $1.3 \times 10^5 \text{ Pa}$ 的气体。另一个相同尺寸的容器 B 容纳相同温度和 $1.9 \times 10^5 \text{ Pa}$ 压力的气体。两个容器最初彼此隔离。(a) 求分子总数。(b) 两个容器现在通过一根体积可以忽略不计的管子连接起来。假设温度保持不变，求气体的最终压力。问题 4.5 可以假设汽车轮胎中的空气具有 $3.0 \times 10^{-2} \text{ m}^3$ 的恒定体积。温度为 25°C 时，该空气的压力为 $2.8 \times 10^5 \text{ Pa}$ 。使用泵增加压力。每次冲程，0.015 摩尔的空气被压入轮胎。如果气体的最终压力为 $3.6 \times 10^5 \text{ Pa}$ ，最终温度为 28°C 。找出所需泵的冲程数。

4.2.2 经验定律历史上，理想气体定律由 Émile Clapeyron 于 1834 年首次提出：对于固定量的气体， $PVT = \text{const}$ 他的工作基于经验波义耳定律、查尔斯定律和盖-吕萨克定律，我们接下来将从理想气体方程波义耳定律恢复这些定律波义耳定律是由罗伯特波义耳于 1662 年发现的，基于实验观察 $p \propto 1/V$ $pV = \text{const}$ 如果温度 T 保持恒定，则 $pV = \text{const}$ 或 $p \propto 1/V$ 气体压力 p 与体积 V 成反比 $p_1 V_1 = p_2 V_2$ $\hat{=}$ 温度保持恒定的热力学过程称为等温过程等温过程的 p - V 关系如图 37 所示

Charles's law

Charles's law was discovered by *Jacques Charles* in 1787, based on experimental observations



if pressure p remains constant, then: $\frac{V}{T} = \text{const}$, or $V \propto T$

i.e., volume V of gas is directly proportional to its temperature T

- proportionality relation only applies if Kelvin scale is used
- a thermodynamic process for which pressure is kept constant is called an *isobaric* process

V - T relation for an isobaric process is shown

- Charles's law implies that volume of gas tends to zero at a certain temperature
historically this is how the idea of *absolute zero* first arose
- as $T \rightarrow 0$, a real gas condenses into solid
there will be deviation from ideal behaviour (dotted line)

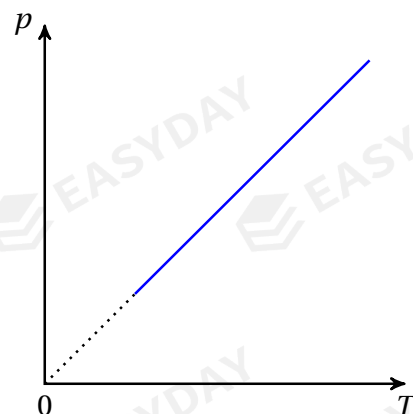
Gay-Lussac's law

Gay-Lussac's law was discovered by *Joseph Louis Gay-Lussac* between 1800 and 1802

if volume V remains constant, then

$$\frac{p}{T} = \text{const}, \text{ or } p \propto T$$

i.e., pressure p is directly proportional to temperature T



杨宇浩 4 理想气体查尔斯定律查尔斯定律是 Jacques Charles 于 1787 年根据实验观察发现的 $V \propto T$ /K TC / $\pm$ C-200 -100 0 100 200 0 100 200 300 400 500 -273 理想行为如果压力 p 保持恒定，则： $V/T = \text{const}$ ，或 $V \propto T$ 即，气体的体积 V 与其温度 T 成正比 - 比例关系仅在使用开尔文标度时适用 - 压力保持恒定的热力学过程称为等压过程，显示了等压过程的 V - T 关系 - 查尔斯定律意味着气体体积在历史上在某个温度趋于零，这就是绝对零的概念首次出现的原因 - 当 $T \rightarrow 0$ ，实际气体凝结成固体时，会偏离理想行为（虚线）盖吕萨克定律 $p \propto T$ 盖吕萨克定律由 Joseph Louis Gay-Lussac 在 1800 年至 1802 年间发现，如果体积 V 保持恒定，则 $p/T = \text{const}$ 或 $p \propto T$ 即压力 p 与温度 T 成正比 38

➤ a thermodynamic process for which volume is kept constant

is called an *isochoric* process, or *isometric* process

p - T relation for an isochoric process is shown

➤ behaviour of real gas again deviates from ideal behaviour (dotted line) as $T \rightarrow 0$

4.3 kinetic theory of ideal gases

kinetic model of gases: a theory based on microscopic motion of molecules of a gas that explains its macroscopic properties

4.3.1 assumptions of ideal gas model

kinetic theory of the ideal gas model is based on the following assumptions:

- gas molecules are in constant *random* motion
- *intermolecular separation* is much greater than size of molecules
- volume of molecules is negligible compared to volume occupied by gas
- *intermolecular forces* are negligible
- collisions between molecules are perfectly *elastic*, i.e., no kinetic energy lost
- molecules travel in straight line between collisions

Example 4.6 A mass of 20 g helium-4 at a temperature of 37°C has a pressure of 1.2×10^5 Pa. Each helium-4 atom has a diameter of 280 pm. (a) Find the volume occupied by the gas and the volume of atoms in this gas. (b) Compare the two volumes, suggest whether this gas can be considered as an ideal gas.

✎ number of helium molecules: $N = nN_A = \frac{m}{M} \times N_A = \frac{20}{4.0} \times 6.02 \times 10^{23} \approx 3.01 \times 10^{24}$

volume of gas: $V_{\text{gas}} = \frac{NkT}{p} = \frac{3.01 \times 10^{24} \times 1.38 \times 10^{-23} \times (37 + 273)}{1.2 \times 10^5} \approx 0.107 \text{ m}^3$

volume of one atom: $V_{\text{atom}} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times (140 \times 10^{-12})^3 \approx 1.15 \times 10^{-29} \text{ m}^3$

volume of all atoms: $V_{\text{atoms}} = NV_{\text{atom}} = 3.01 \times 10^{24} \times 1.15 \times 10^{-29} \text{ m}^3 \approx 3.46 \times 10^{-5} \text{ m}^3$

$V_{\text{gas}} \gg V_{\text{atoms}}$, so this gas can approximate to an ideal gas

□

Yuhao Yang 4 理想气体 $\hat{\alpha}$ 体积保持恒定的热力学过程称为等容过程，或显示等容过程的等距过程 p - T 关系 $\hat{\alpha}$ 实际气体的行为再次偏离理想行为（虚线），如 T ! 0 4.3 理想气体动力学理论 气体动力学模型：基于气体分子微观运动的理论，解释其宏观性质 4.3.1 理想气体模型的假设 理想气体模型动力学理论基于以下假设：- 气体分子处于恒定的随机运动 - 分子间分离远大于分子大小与气体所占的体积相比，分子体积可以忽略不计 - 分子间力可以忽略不计 - 分子之间的碰撞是完全弹性的，即，没有动能损失 - 分子在碰撞之间沿直线行进 示例 4.6 质量为 20 g 氦 4，温度为 $37 \pm C$ ，压力为 $1.2 \times 10^5 \text{ Pa}$ 。每个氦 4 原子的直径为 280 pm。(a) 求该气体所占的体积以及该气体中原子的体积。(b) 比较两个体积，表明该气体是否可以被视为理想气体。

- 氦分子数： $N = n N_A = m M \times N_A = 20 \text{ g} \times 6.02 \times 10^{23} \text{ mol}^{-1} / 4 \text{ g mol}^{-1} = 3.01 \times 10^{24}$

气体体积： $V_{\text{gas}} = N k T / p = 3.01 \times 10^{24} \times 1.38 \times 10^{-23} \text{ J K}^{-1} \times (37 + 273) \text{ K} / 1.2 \times 10^5 \text{ Pa} = 0.107 \text{ m}^3$

一个原子的体积： $V_{\text{atom}} = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \times (140 \times 10^{-12} \text{ m})^3 = 1.15 \times 10^{-29} \text{ m}^3$

所有原子的体积： $V_{\text{atoms}} = N V_{\text{atom}} = 3.01 \times 10^{24} \times 1.15 \times 10^{-29} \text{ m}^3 = 3.46 \times 10^{-5} \text{ m}^3$

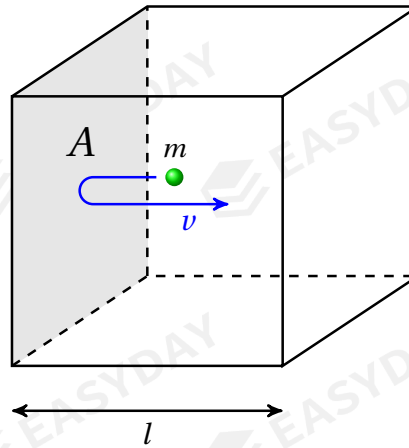
$V_{\text{gas}} \gg V_{\text{atoms}}$ ，因此该气体可以近似于理想气体 $\hat{\alpha}$ 39

4.3.2 pressure (quantitative view)

we are ready to derive a formula for pressure due to ideal gas

pressure of gas is due to collision of gas molecules with container

let's first consider the effect of one single molecule moving in one dimension only, and then generalise the result to a gas containing N molecules moving in all three dimensions



one gas molecule moving in 1-D

let's assume this single molecule only moves in x -direction (see figure)

change in momentum when colliding with wall: $\Delta P_x = mv_x - (-mv_x) = 2mv_x$ ^[19]

time interval between collisions: $\Delta t = \frac{2l}{v_x}$

average force acting: $F_x = \frac{\Delta P_x}{\Delta t} = \frac{2mv_x}{\frac{2l}{v_x}} = \frac{mv_x^2}{l}$

average pressure: $p_x = \frac{F}{A} = \frac{mv_x^2}{lA} \Rightarrow p_x = \frac{mv_x^2}{V}$

generalisation to N molecules moving in 3-D

- N molecules so N times the contributions to pressure

but there is a *distribution* of speeds for N molecules, so should take average of v^2

- in three-dimensional space, we have: $v^2 = v_x^2 + v_y^2 + v_z^2$

but molecules have no preference in any specific direction, so: $\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle = \frac{\langle v^2 \rangle}{3}$

pressure should be shared equally among three dimensions: $p = p_x = p_y = p_z$

therefore we find the pressure of an ideal gas is given by: $p = \frac{Nm \langle v^2 \rangle}{3V}$

^[19] In this section we use P for momentum of a particle and p for pressure of a gas to avoid confusion.

Yuhao Yang 4 理想气体 4.3.2 压力（定量视图） 我们已经准备好导出压力公式了，因为气体的理想气体压力是由于气体分子与容器的碰撞而产生的。我们首先考虑一个单一分子仅在一维运动的影响，然后将结果推广到包含 N 个分子在所有三个维度上运动的气体。考虑一个一维运动的气体分子，让我们假设该单一分子仅在 x 方向上运动（参见图）与墙壁碰撞时动量的变化： $\Delta P_x = mv_x - (-mv_x) = 2mv_x$ [19] 碰撞之间的时间间隔： $\Delta t = 2l/v_x$ 平均作用力： $F_x = \Delta P_x / \Delta t = 2mv_x / (2l/v_x) = mv_x^2 / l$ 平均压力： $p_x = F_x / A = mv_x^2 / (lA) = mv_x^2 / V$ 推广到在 3-D 中移动的 N 个分子—— N 个分子，因此 N 乘以对压力的贡献，但 N 个分子存在速度分布，因此应取 v^2 的平均值——在三维空间中，我们有： $v^2 = v_x^2 + v_y^2 + v_z^2$ 但分子在任何特定方向上没有偏好，因此： $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} = \overline{v^2} / 3$ 压力应在三个维度之间平均分配： $p = p_x = p_y = p_z$ 因此我们发现理想气体的压力由下式给出： $p = N m \overline{v^2} / 3V$ [19] 在本节中，我们使用 P 表示粒子的动量，使用 p 表示气体的压力以避免混淆。

➤ $\langle v^2 \rangle$ is the *mean square velocity* of gas molecules

we can further define r.m.s. (root mean square) velocity: $v_{\text{rms}} = \sqrt{\langle v^2 \rangle}$

gas molecules in random motion so there exists a range of velocities

we cannot tell exact velocity of a specific molecule, but can only tell mean values

➤ N is number of molecules, m is mass of one molecule

then Nm gives total mass of the gas, and $\frac{Nm}{V}$ gives gas density ρ

we can rewrite the pressure formula as: $p = \frac{1}{3} \rho \langle v^2 \rangle$

(pressure depends only on density and mean square speed of molecules)

➤ physical interpretation of the formula

- $N \uparrow \Rightarrow$ more molecules, more collisions $\Rightarrow p \uparrow$
- $m \uparrow \Rightarrow$ greater mass, greater force upon collision $\Rightarrow p \uparrow$
- $v \uparrow \Rightarrow$ strike container harder, also more often $\Rightarrow p \uparrow$
- $V \uparrow \Rightarrow$ spend more time in gas, less frequent collision with container $\Rightarrow p \downarrow$

4.3.3 kinetic energy

we now have two equations for ideal gases:

$$\begin{cases} pV = nRT, \text{ or } pV = NkT & \text{ideal gas law} \\ p = \frac{Nm \langle v^2 \rangle}{3V} & \text{pressure law} \end{cases}$$

compare the two equations: $pV = \frac{1}{3} Nm \langle v^2 \rangle = NkT \Rightarrow m \langle v^2 \rangle = 3kT$

mean kinetic energy of a single molecule in a gas is: $\langle E_k \rangle = \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} kT$

mean K.E. of ideal gas molecules is *proportional* to its thermodynamic temperature

➤ useful relation for molecular speeds: $v_{\text{rms}}^2 \propto T$

recall our statement in §4.1.1, higher temperature means higher speed for molecules

➤ we only talk about *translational* K.E. here

molecules have this energy because they are moving through space

total kinetic energy may also include *rotational* K.E. and *vibrational* K.E. [20]

[20] There is an important result in classical thermal physics, known as the *equipartition of energy theorem*. It states that the average energy per molecule is $\frac{1}{2} kT$ for each independent *degree of freedom*. A molecule

Yuhao Yang 4 理想气体 $\overline{v^2}$ 是气体分子的均方速度，我们可以进一步定义 r.m.s.（均方根）速度：
 $v_{rms} = \sqrt{\overline{v^2}}$ 随机运动的气体分子，因此存在一系列速度，我们无法得知特定分子的精确速度，但只能得知平均值 $\overline{v^2}$ 。N 是分子数，m 是一个分子的质量，然后 Nm 给出气体的总质量， Nm/V 给出气体密度 ρ 。我们可以将压力公式重写为： $p = \frac{1}{3} \rho \overline{v^2}$ （压力仅取决于密度和分子的均方速度）。 $\overline{v^2}$ 公式的物理解释 - N 1) 更多的分子，更多的碰撞) p 1 - m 1) 更大的质量，碰撞时更大的力) p 1 - v 1) 更猛烈地撞击容器，也更频繁) p 1 - V 1) 在气体中花费更多的时间，更少与容器碰撞) p % 4.3.3 动能我们现在有两个理想气体的方程： $pV = nRT$ ，或 $pV = NkT$ 理想气体定律 $p = N/V \cdot m \overline{v^2}/3$ 压力定律比较两个方程： $pV = \frac{1}{3} N m \overline{v^2} = NkT$) $m \overline{v^2} = 3kT$ 气体中单个分子的平均动能为： $\overline{E_k} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$ 平均 K.E. 理想气体分子的热力学温度与其热力学温度成正比 - 分子速度的有用关系： v_{rms}^2 / T 回想一下我们在第 4.1.1 节中的陈述，较高的温度意味着分子的速度较高 - 我们只讨论平移 K.E. 这里分子具有这种能量，因为它们在空间中移动，总动能也可能包括旋转 K.E. 和振动 K.E. [20] [20] 经典热物理有一个重要的结果，称为能量均分定理。它指出每个独立自由度的每个分子的平均能量为 $\frac{1}{2} kT$ 。分子 41

➤ $\langle E_k \rangle = \frac{3}{2} kT$ gives the *mean*, or *average* K.E. per molecule

gas molecules exchange energies with each other upon collisions

for an individual molecule, its K.E. is not a constant

but mean K.E. is constant, which depends on temperature T only

➤ in a mixture of several gases, K.E. is shared *equally* among its components

this is because of repeated collisions between particles

though all molecules have same K.E., heavier molecules will move more slowly

Example 4.7 Air consists of oxygen (O_2 , molar mass 32 g mol^{-1}) and nitrogen (N_2 , molar mass 28 g mol^{-1}). (a) Calculate the mean translational kinetic energy of these molecules at 300 K . (b)

Estimate the typical speed for each type of the molecule.

$$\begin{aligned} \text{mean K.E. of single molecule: } \langle E_k \rangle &= \frac{3}{2} kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 \approx 6.21 \times 10^{-21} \text{ J} \\ \langle E_k \rangle &= \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} kT \Rightarrow \frac{1}{2} \frac{M}{N_A} \langle v^2 \rangle = \frac{3}{2} kT \Rightarrow \langle v^2 \rangle = \frac{3kN_A T}{M} = \frac{3RT}{M} \end{aligned}$$

$$\text{for oxygen molecule: } v_{\text{O}_2} \approx \sqrt{\frac{3 \times 8.31 \times 300}{0.032}} \approx 483 \text{ m s}^{-1}$$

$$\text{for nitrogen molecule: } v_{\text{N}_2} \approx \sqrt{\frac{3 \times 8.31 \times 300}{0.028}} \approx 517 \text{ m s}^{-1} \quad \square$$

Example 4.8 A cylinder container initially holds a gas of helium-4 at a temperature of 54°C .

(a) Find the mean square speed of these helium atoms. (b) If the temperature is raised to 540°C , find the r.m.s. speed of the atoms.

$$\text{mass of one helium-4 atom: } m = 4u = 4 \times 1.66 \times 10^{-27} \approx 6.64 \times 10^{-27} \text{ kg}$$

$$\text{at } 54^\circ\text{C: } \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} kT \Rightarrow \langle v^2 \rangle = \frac{3kT}{m} = \frac{3 \times 1.38 \times 10^{-23} \times (54 + 273)}{6.64 \times 10^{-27}} \approx 2.04 \times 10^6 \text{ m}^2 \text{ s}^{-2}$$

$$\text{note relation between } v \text{ and } T: \langle v^2 \rangle \propto T \Rightarrow \frac{\langle v'^2 \rangle}{\langle v^2 \rangle} = \frac{T'}{T} \Rightarrow v'_{\text{rms}} = \sqrt{\frac{T'}{T}} \times v_{\text{rms}}$$

$$\text{at } 540^\circ\text{C: } v'_{\text{rms}} = \sqrt{\frac{540 + 273}{54 + 273}} \times \sqrt{2.04 \times 10^6} \approx 2.25 \times 10^3 \text{ m s}^{-1} \quad \square$$

Question 4.6 A fixed mass of gas expands to twice its volume at constant temperature. (a) How does its pressure change? (b) How does mean kinetic energy change?

can move in three directions, corresponding to three translational degrees of freedom, thus its mean translational kinetic energy is $\frac{3}{2} kT$. For a polyatomic gas (each molecule consists of several atoms), apart from translational motion, it has additional rotational degrees of freedom and different vibrational modes, so its average energy can be calculated by counting the total number of degrees of freedom.

Yuhao Yang 4 IDEAL GASES $\bar{h} \epsilon_k i = \frac{3}{2} kT$ 给出了平均 K.E. 每个分子气体分子在碰撞时彼此交换能量，其 K.E. 不是常数，而是平均 K.E. 是常数，仅取决于温度 T 。在多种气体的混合物中，K.E. 其组分之间均等分配这是因为粒子之间的重复碰撞，尽管所有分子都具有相同的 K.E.，但较重的分子移动速度较慢。示例 4.7 空气由氧气 (O_2 ，摩尔质量 32 g mol^{-1}) 和氮气 (N_2 ，摩尔质量 28 g mol^{-1}) 组成。(a) 计算这些分子在 300 K 时的平均平动动能。(b) 估计每种类型分子的典型速度。意思是 K.E. 单分子： $\bar{h} \epsilon_k i = \frac{3}{2} kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 \frac{1}{4} 6.21 \times 10^{-21} \text{ J}$ $\bar{h} \epsilon_k i = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$ $\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$ $\overline{v^2} = \frac{3kT}{m}$ $\overline{v^2} = \frac{3RT}{M}$ 对于氧分子： $\overline{v^2} = \frac{3 \times 8.31 \times 300}{0.032} \frac{1}{4} 483 \text{ m s}^{-1}$ 对于氮分子： $\overline{v^2} = \frac{3 \times 8.31 \times 300}{0.028} \frac{1}{4} 517 \text{ m s}^{-1}$ 示例 4.8 一个钢瓶容器最初容纳氦 4 气体，温度为 $54 \pm \text{C}$ 。(a) 求这些氦原子的均方速度。(b) 如果温度升至 $540 \pm \text{C}$ ，求均方根值原子的速度。一个氦 4 原子的质量： $m = 4u = 4 \times 1.66 \times 10^{-27} \frac{1}{4} 6.64 \times 10^{-27} \text{ kg}$ at $54 \pm \text{C}$: $\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$ $\overline{v^2} = \frac{3kT}{m} = \frac{3 \times 1.38 \times 10^{-23} \times (54 + 273)}{6.64 \times 10^{-27}} \frac{1}{4} 2.04 \times 10^6 \text{ m}^2 \text{ s}^{-2}$ 注意 v 和 T 之间的关系： $\overline{v^2} / T$ $\overline{v^2} / T = \frac{3k}{m}$ $\overline{v^2} = \frac{3kT}{m}$ $\overline{v^2} = \frac{3RT}{M}$ $\overline{v^2} = \frac{3 \times 8.31 \times 540}{0.004} \frac{1}{4} 2.25 \times 10^8 \text{ m}^2 \text{ s}^{-2}$ 问题 4.6 固定质量的气体在恒温下膨胀至其体积的两倍。(a) 它的压力如何变化？(b) 平均动能如何变化？可以向三个方向运动，对应三个平动自由度，因此其平均平动动能为 $\frac{3}{2} kT$ 。对于多原子气体（每个分子由多个原子组成），除了平动运动外，还具有附加的旋转自由度和不同的振动模式，因此可以通过计算自由度总数来计算其平均能量。42

Question 4.7 In order for a molecule to escape from the gravitational field of the earth, it must have a speed of $1.1 \times 10^6 \text{ m s}^{-1}$ at the top of the atmosphere. (a) Estimate the temperature at which helium-4 atoms could have this speed. (b) Helium atom actually escape from top of the atmosphere at much lower temperatures, explain how this is possible.

Yuhao Yang 4 理想气体问题4.7 为了使分子逃离地球引力场，它在大气层顶部的速度必须为 $1.1 \times 10^6 \text{ m s}^{-1}$ 。(a) 估计氦 4 原子可以具有该速度的温度。(b) 氦原子实际上在低得多的温度下从大气层顶部逸出，请解释这是如何可能的。 43

CHAPTER 5

Thermodynamics

5.1 thermal physics basics

5.1.1 temperature scales

- Celsius scale (unit: °C)

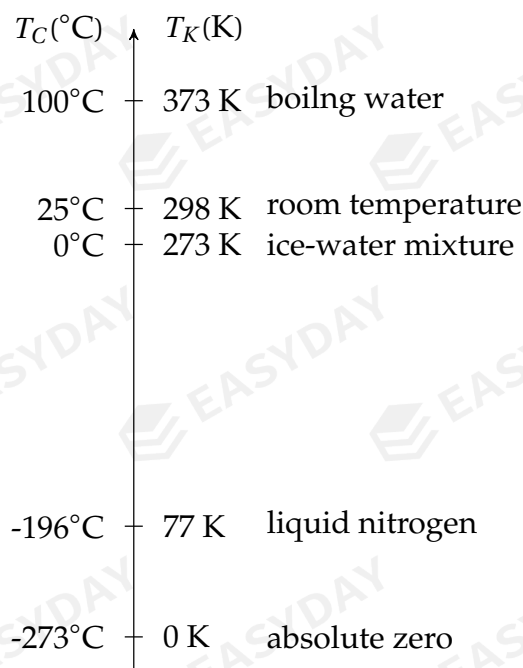
0°C defined as temperature of ice-water mixture

100°C defined as temperature of boiling water

- Kelvin scale (unit: K)

0 K (*absolute zero*) is lowest temperature possible

- conversion rule: $T_K(K) \begin{matrix} -273 \\ \rightleftharpoons \\ +273 \end{matrix} T_C(^{\circ}C)$
- change of 1°C equals change of 1 K



5.1.2 kinetic theory of matter

there are three common states of matter: solid, liquid and gas

they have very different physical properties (density, compressibility, fluidity, etc.)

but deep down, they are all composed of a large number of small molecules

in the **kinetic theory of matter**, we look at microscopic behaviour at molecular level (arrangement, motion, intermolecular forces, separation, etc.)

microscopic behaviour of molecules cause differences in *macroscopic* properties of matter

- solid: molecules close together, tightly bonded, vibrate about their positions
- liquid: molecules quite close together, vibrate but has some freedom to move about
- gas: molecules widely separated, free from neighbours, move rapidly

杨宇豪 5 热力学第 5 章热力学 5.1 热物理基础 5.1.1 温标 TC ($^{\circ}\text{C}$) TK (K) -273°C - 196°C 0°C 25°C 100°C 0 K 77 K 273 K 298 K 373 K 绝对零液氮冰水混合物室温沸水 $^{\circ}\text{C}$ 标度 (单位: $^{\circ}\text{C}$) 0°C 定义为冰水混合物的温度 100°C 定义为沸水的温度 $^{\circ}\text{C}$ 开尔文标度 (单位: K) 0 K (绝对零) 是可能的最低温度 $^{\circ}\text{C}$ 转换规则: TK (K) -273 G G G G G G G BF G G G G G G G $+273$ TC ($^{\circ}\text{C}$) $^{\circ}\text{C}$ 变化 1 $^{\circ}\text{C}$ 等于变化 1 K 5.1.2 物质运动理论物质存在三种常见状态: 固体、液体和气体, 它们具有非常不同的物理性质 (密度、可压缩性、流动性等), 但从本质上讲, 它们都是由大量小分子组成的, 在物质运动理论中, 我们在分子水平上观察微观行为 (排列、运动、分子间力、分离等)。分子的微观行为导致物质宏观性质的差异。固体: 分子紧密结合、紧密结合、围绕其位置振动。- 液体: 分子非常靠近, 振动, 但有一定的移动自由度 - 气体: 分子相距较远, 没有邻居, 移动迅速 44

5.1.3 specific latent heat

it requires heat energy to *melt* a solid or *boil* a liquid

melting and boiling usually occur at a fixed temperature

thermal energy to cause the change of state at a constant temperature is called *latent heat*

amount of latent heat needed depends on mass of substance: $Q = Lm$

we define **specific latent heat** (L) as the thermal energy required to change the state of *unit* mass of substance with no change in temperature is called

- unit of specific latent heat: $[L] = \text{J} \cdot \text{kg}^{-1}$
- specific latent heat is an *intensive* property
i.e., L does not depend on size or shape of sample, L depends on type of substance only
- for melting, L is called *specific latent heat of fusion*
for boiling, L is called *specific latent heat of vaporisation*
- latent heat is related to breaking bonds and increasing intermolecular separation
vaporisation requires larger increase in particle separation than fusion
for a given substance, $L_{\text{vapour}} > L_{\text{fuse}}$

Example 5.1 A 3.0 kW electric kettle contains 0.5 kg of water already at its boiling point.

Neglecting heat losses, determine how long it takes to boil dry. ($L_{\text{water}} = 2.26 \times 10^6 \text{ J kg}^{-1}$)

✍ heat required: $Q = mL = 0.50 \times 2.26 \times 10^6 = 1.13 \times 10^6 \text{ J}$

time needed: $t = \frac{Q}{P} = \frac{1.13 \times 10^6}{3.0 \times 10^3} \approx 380 \text{ s} \approx 6.3 \text{ min}$ □

Example 5.2 A student measures specific latent heat of fusion for ice. He uses an electric heater to melt ice but the insulation is not perfect. The experiment is carried out twice, with the heater operating at different powers. Use the data table to calculate specific latent heat of fusion.

	Power (W)	time interval (min)	mass of ice melted (g)
test 1	60	3.0	40.4
test 2	90	3.0	56.6

✍ there exists heat gain from surroundings, so effective power $P_{\text{eff}} = P_{\text{heater}} + P_{\text{sur}}$

杨宇浩 5 热力学 5.1.3 比潜热熔化固体或使液体沸腾需要热能熔化和沸腾通常发生在固定温度下恒定温度下引起物态变化的热能称为潜热所需潜热的大小取决于物质的质量： $Q = Lm$ 我们将比潜热 (L) 定义为在温度不变的情况下改变单位质量物质的状态所需的热能称为 $\hat{L}$ 比潜热单位： $[L] = \text{J} \cdot \text{kg}^{-1}$ $\hat{L}$ 比潜热是一种强度属性，即， L 不依赖于样品的大小或形状， L 仅取决于物质的类型 $\hat{L}$ 对于熔化， L 称为比熔化潜热，用于沸腾， L 称为比汽化潜热 $\hat{L}$ 潜热与断裂键有关，增加分子间分离汽化需要更大的增加对于给定物质， $L_{\text{vapour}} \approx L_{\text{fuse}}$ 的粒子分离与融合的效果相同。示例 5.1 一个 3.0 kW 的电热水壶中含有 0.5 kg 已达到沸点的水。忽略热量损失，确定煮干所需的时间。 ($L_{\text{water}} = 2.26 \times 10^6 \text{ J kg}^{-1}$) - 所需热量： $Q = mL = 0.50 \times 2.26 \times 10^6 = 1.13 \times 10^6 \text{ J}$ 所需时间： $t = Q/P = 1.13 \times 10^6 / 3.0 \times 10^3 = 380 \text{ s} \approx 6.3 \text{ min}$ 示例 5.2 学生测量冰的比熔化潜热。他用电加热器来融化冰块，但隔热效果并不完美。该实验进行两次，加热器在不同功率下运行。使用数据表计算比熔化潜热。功率 (W) 时间间隔 (min) 融化的冰块质量 (g) 测试 1 60 3.0 4 0.4 测试 2 90 3.0 56.6 - 周围环境存在热量增益，因此有效功率 $P_{\text{eff}} = P_{\text{heater}} + P_{\text{sur}}$ 45

heat energy to melt ice: $Q = mL = (P_{\text{heater}} + P_{\text{sur}})t$

$$\begin{cases} 40.4 \times L = (60 + P_{\text{sur}}) \times 3.0 \times 60 \\ 56.6 \times L = (90 + P_{\text{sur}}) \times 3.0 \times 60 \end{cases} \Rightarrow \begin{cases} L \approx 333 \text{ J g}^{-1} \\ P_{\text{sur}} \approx 14.8 \text{ W} \end{cases} \quad \square$$

Question 5.1 A student designs an experiment to determine the specific latent heat of fusion L of ice. Some ice at 0°C is heated with an electric heater. The experiment is carried out twice and the following data are obtained.

	energy supply from heater (J)	time interval (min)	mass of ice melted (g)
heater off	0	10.0	14.3
heater on	21000	5.0	70.0

(a) Suggest why two sets of readings are taken. (b) Find specific latent heat of fusion for ice.

5.1.4 specific heat capacity

heating a substance could cause an increase in its temperature

heat required is proportional to its mass m and temperature change ΔT : $Q = cm\Delta T$

we define **specific heat capacity** (c) as the thermal energy required per unit mass of substance to cause an increase of one unit in its temperature

➤ unit of specific heat capacity: $[c] = \text{J kg}^{-1} \text{K}^{-1}$ or $\text{J kg}^{-1} ^\circ\text{C}^{-1}$

➤ c is also an *intensive* property, i.e., independent of size or shape of the sample

Example 5.3 A block of 30 g ice at -20°C is added to a large cup of 270 g water at 80°C . Assume there is no energy lost, what is the final temperature of the mixture? (data: specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{K}^{-1}$, specific heat capacity of ice is $2100 \text{ J kg}^{-1} \text{K}^{-1}$, specific latent heat of ice is $3.3 \times 10^5 \text{ J kg}^{-1}$.)

✍ energy lost by hot water = energy gain by ice cube

$$\underbrace{4200 \times 0.27 \times (80 - T)}_{95^\circ\text{C water} \rightarrow T^\circ\text{C water}} = \underbrace{2100 \times 0.030 \times [0 - (-20)]}_{-20^\circ\text{C ice} \rightarrow 0^\circ\text{C ice}} + \underbrace{3.3 \times 10^5 \times 0.030}_{0^\circ\text{C ice} \rightarrow 0^\circ\text{C water}} + \underbrace{4200 \times 0.030 \times (T - 0)}_{0^\circ\text{C water} \rightarrow T^\circ\text{C water}}$$

$$90720 - 1134T = 1260 + 9900 + 126T$$

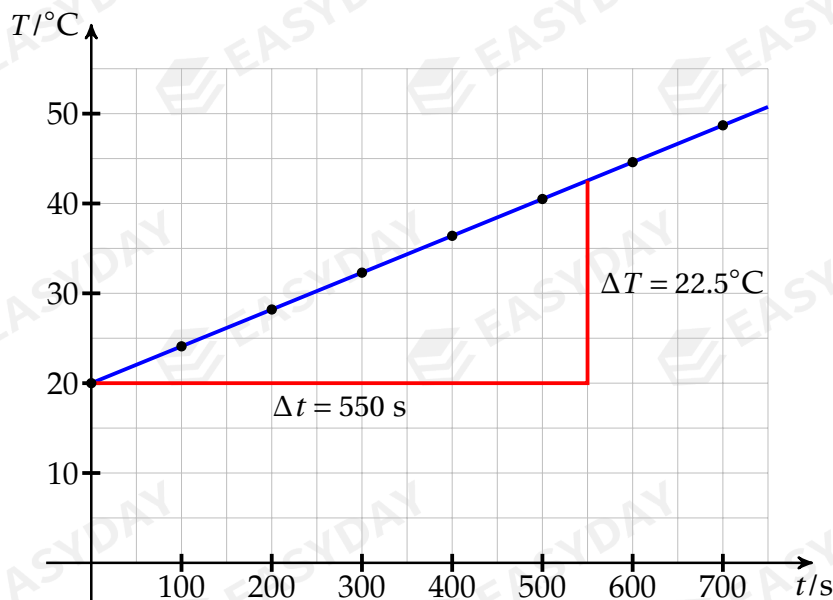
$$T = \frac{90720 - 1260 - 9900}{1134 + 126} \approx 63^\circ\text{C} \quad \square$$

Yuhao Yang 5 热力学融化冰的热能： $Q = mL = (P_{\text{heater}} + P_{\text{sur}})t$ $40.4 \times L = (60 + P_{\text{sur}}) \times 3.0 \times 60$ $56.6 \times L = (90 + P_{\text{sur}}) \times 3.0 \times 60$ $L = \frac{333}{P_{\text{sur}} - 14.8} \text{ W} \cdot \text{s} \cdot \text{g}^{-1}$ 问题 5.1 学生设计了一个实验来确定冰的比熔化潜热 L 。用电加热器加热 0°C 的一些冰。实验进行两次，得到以下数据。加热器的能量供应 (J) 时间间隔 (分钟) 融化的冰块的质量 (g) 加热器关闭 0 10.0 14.3 加热器打开 21000 5.0 70.0 (a) 说明为什么要获取两组读数。(b) 求冰的比熔化潜热。

5.1.4 比热容加热物质可能导致其温度升高所需热量与其质量 m 和温度变化成正比 ΔT : $Q = cm \Delta T$ 我们将比热容 (c) 定义为每单位质量的物质导致其温度升高一个单位所需的热能 $\hat{a}$ 比热容单位: $[c] = \text{J kg}^{-1} \text{K}^{-1}$ 或 $\text{J kg}^{-1} ^\circ\text{C}^{-1}$ $\hat{a}$ c 为还有一种强度特性，即与样品的尺寸或形状无关。实施例 5.3 将 20°C 的 30 g 冰块添加到 80°C 的一大杯 270 g 水中。假设没有能量损失，混合物的最终温度是多少？（数据：水的比热容为 $4200 \text{ J kg}^{-1} \text{K}^{-1}$ ，冰的比热容为 $2100 \text{ J kg}^{-1} \text{K}^{-1}$ ，冰的比潜热为 $3.3 \times 10^5 \text{ J kg}^{-1}$ 。）

- 热水损失的能量 = 冰块获得的能量 $4200 \times 0.27 \times (80 - T) = 95 \times 10^3 - 1134T$ $2100 \times 0.030 \times (T - (-20)) = 63T + 1260$ $95 \times 10^3 - 1134T = 63T + 1260$ $90720 - 1134T = 63T + 1260$ $90720 - 1260 = 1134T + 63T$ $89460 = 1197T$ $T = 74.7^\circ\text{C}$

Example 5.4 A 1.00 kg aluminium block is heated using an electrical heater. The current in the heater is 4.2 A and the p.d. across is 12 V. Measurements of the rising temperature are represented by the graph. Determine specific heat capacity of aluminium.



energy supplied: $Q = cm\Delta T \Rightarrow IV\Delta t = cm\Delta T \Rightarrow c = \frac{IV}{m \frac{\Delta T}{\Delta t}}$

$\frac{\Delta T}{\Delta t}$ is gradient of fitting line: $\frac{\Delta T}{\Delta t} = \frac{22.5}{550} \approx 4.09 \times 10^{-2} \text{ } ^\circ\text{C s}^{-1}$

specific heat capacity: $c = \frac{4.2 \times 12}{1.00 \times 4.09 \times 10^{-2}} \approx 1230 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ □

Question 5.2 A mixture contains 5% silver and 95% of gold by weight. Some gold is melted and the correct weight of silver is added. The initial temperature of silver is 20°C. Use the data to calculate the initial temperature of gold so that the final mixture is at melting point of gold.

	silver	gold
melting point (K)	1240	1340
specific heat capacity (solid or liquid) ($\text{J kg}^{-1} \text{ K}^{-1}$)	235	129
specific latent heat of fusion (kJ kg^{-1})	105	628

5.2 internal energy

we now consider the total energy within a thermodynamic system

molecules in a system undergo random motion, so they have kinetic energy

Yuhao Yang 5 热力学示例 5.4 使用电加热器加热 1.00 kg 的铝块。加热器中的电流为 4.2 A，p.d. 两端电压为 12V。温度升高的测量结果由图表表示。确定铝的比热容。

t / s	T / °C
100	20
200	23
300	26
400	29
500	32
600	35
700	38

Δt = 550 s ΔT = 22.5 ± 0.5 °C - 供应能量：Q = cm ΔT) I V Δt = cm ΔT) c = I V m ΔT Δt

ΔT Δt 为拟合线斜率：ΔT Δt = 22.5 550 ¼ 4.09 × 10⁻² ± 0.01 °C s⁻¹ 比热容：c = 4.2 × 12 1.00 × 4.09 × 10⁻² ¼ 1230 J kg⁻¹ ± 0.1 J kg⁻¹ °C⁻¹ ä 问题 5.2 一种混合物含有 5% 银和按重量计 95% 的黄金。熔化一些金并添加正确重量的银。银的初始温度为 20 ± 0.5 °C。使用数据计算金的初始温度，以便最终混合物处于金的熔点。银金熔点 (K) 1240 1340 比热容 (固体或液体) (J kg⁻¹ K⁻¹) 235 129 比熔化潜热 (kJ kg⁻¹) 105 628 5.2 内能

我们现在考虑热力学系统内的总能量系统中的分子进行随机运动，因此它们具有动能

there are potential energy between molecules due to intermolecular interaction

internal energy is defined as the sum of random kinetic energy of molecules and potential energy between molecules: $U = E_k + E_p$

- internal energy is a *state function* of the system

it only depends on current state of system, not on process to arrive at this state

5.2.1 kinetic energy

- internal energy counts K.E. due to random motion at molecular level

K.E. of macroscopic motion of the system as a whole is not included

- mean K.E. of molecules is directly proportional to temperature: $E_k \propto T$

K.E. of molecules depends on temperature only

higher temperature means molecules move faster, vibrate more intensively, etc.

5.2.2 potential energy

- internal energy counts P.E. due to force fields *within* the system

P.E. of the system as a whole due to *external* force fields is not included

- P.E. between molecules depends on intermolecular separation and chemical bonding

in general, greater intermolecular separation means greater P.E.^[21]: $r \nearrow \leftarrow E_p \nearrow$

breaking intermolecular bonds also causes an increase in P.E.

mean P.E. of gas > mean P.E. of liquid > mean P.E. of liquid solid

5.2.3 internal energy of ideal gas

for ideal gas, mean K.E. of one molecule: $E_k = \frac{3}{2} kT$

there is no intermolecular force, so P.E. of ideal gas is defined to be zero: $E_p = 0$

^[21] Intermolecular separation does not necessarily increase during melting processes. A typical counter example is melting of ice into water, for which intermolecular separation actually decreases (density of water > density of ice), but potential energy of the system will still increase because hydrogen bonds between H₂O molecules are broken.

杨宇豪 5 热力学由于分子间相互作用，分子间存在势能内能定义为分子随机动能与分子间势能之和： $U = E_k + E_p$ 内能是系统的状态函数，它只取决于系统当前的状态，而不取决于达到该状态的过程 5.2.1 动能 内能计入 K.E.由于分子水平上的随机运动 K.E.不包括整个系统的宏观运动 – 意思是 K.E.分子的数量与温度成正比： E_k / T K.E.分子的能量取决于温度，只有较高的温度意味着分子运动得更快，振动更剧烈等。

5.2.2 势能 内能计数 P.E.。由于系统内的力场 P.E.不包括由于外力场而导致的整个系统的损坏。一般来说，分子间的分离取决于分子间的分离和化学键合，分子间分离越大，P.E.越大[21]: r 1(E_p 1 破坏分子间键也会导致 P.E. 的增加。气体的平均 P.E. 液体的平均 P.E. 液体的平均 P.E. 液体的固体 5.2.3 理想气体的内能对于理想气体，一个分子的平均 K.E.： $E_k = \frac{3}{2} kT$ 时，没有分子间作用力，因此理想气体的 P.E. 被定义为零： $E_p = 0$ [21] 一个典型的反例是冰融化成水，分子间分离实际上会减少（水的密度 > 冰的密度），但由于 H₂O 分子之间的氢键被破坏，系统的势能仍然会增加 48。

internal energy per molecule: $U = E_k + E_p = \frac{3}{2}kT$

hence internal energy of ideal gas is purely kinetic and directly proportional to temperature

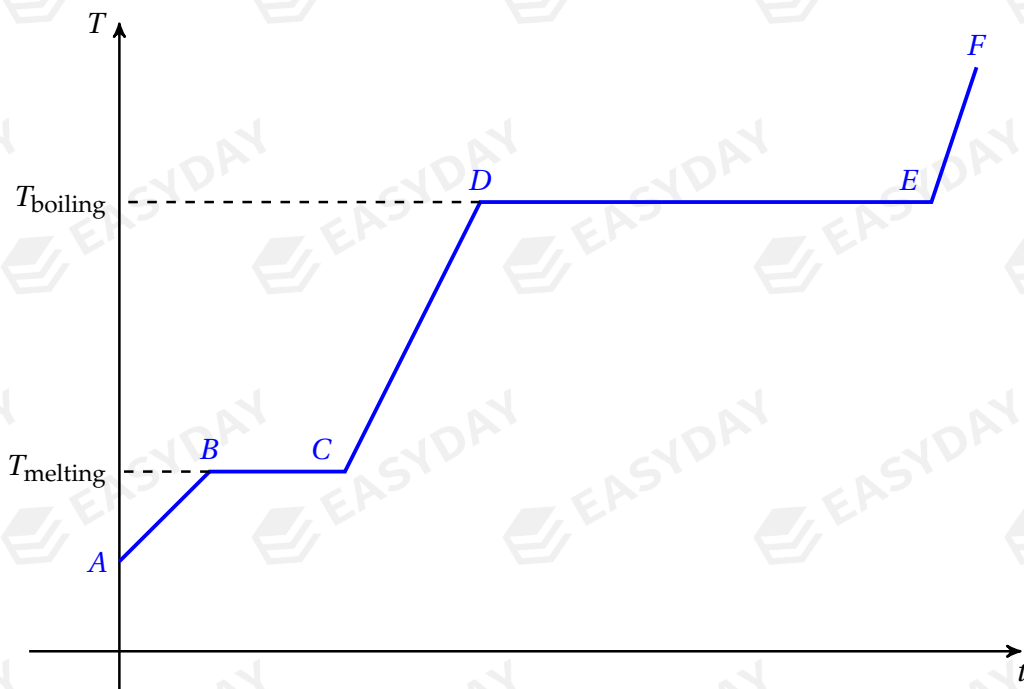
total internal energy of the gas: $U_{\text{gas}} = NU \Rightarrow U_{\text{gas}} = \frac{3}{2}NkT$

5.2.4 change of states

consider a substance being heated from its solid state

it will melt into a liquid and further vaporise into its gaseous state

we now look into the changes of internal energy during each stage



- AB (solid state): $T \uparrow \Rightarrow E_k \uparrow$, greater vibration for solid particles
but no (significant) change in mean separation^[22] $\Rightarrow$ no change in P.E.
- BC (melting): latent heat goes into breaking intermolecular bonds, $r \uparrow \Rightarrow E_p \uparrow$
but melting occurs at constant temperature^[23] $\Rightarrow$ no change in K.E.
- CD (liquid state): $T \uparrow \Rightarrow E_k \uparrow$, greater vibration and free motion
but no (significant) change in mean separation $\Rightarrow$ no change in P.E.

^[22] A typical solid material expands when it is heated, so intermolecular separation will increase slightly.

^[23] Here we talk about *pure substance*, which changes from solid into liquid at a particular temperature, called the *melting point*. But for a *mixture* of substances, melting may occur over a *range* of temperatures. It is also possible for a substance to decompose before they change states.

Yuhao Yang 5 热力学每个分子的内能： $U = E_k + E_p = \frac{3}{2} kT$ 因此，理想气体的内能是纯动能，与温度成正比气体的总内能： $U_{\text{gas}} = N U$ $U_{\text{gas}} = \frac{3}{2} N kT$ 5.2.4 状态变化考虑物质从固态被加热，它将熔化成液态并进一步汽化成气态，我们现在研究以下变化：每个阶段的内能 U A B C D E F T 沸腾 T 熔化 - AB (固态) : T 1) E_k 1, 固体颗粒振动更大，但平均分离没有 (显著) 变化 [22]) P.E 没有变化。 - BC (熔化) : 潜热会破坏分子间键， r 1) E_p 1 但在恒温下发生熔化 [23]) K.E 没有变化。 - CD (液态) : T 1) E_k 1, 更大的振动和自由运动，但平均分离没有 (显著) 变化) P.E 没有变化。 [22] 典型的固体材料受热时会膨胀，因此分子间分离会略有增加。 [23] 这里我们谈论纯物质，它在特定温度 (称为熔点) 下从固体转变为液体。但对于物质的混合物来说，熔化可能会在一定的温度范围内发生。物质也有可能改变状态之前分解。

。 49

– *DE (boiling)*: molecules break free, $r \nearrow \Rightarrow E_p \nearrow$

boiling occurs at constant temperature^[24] $\Rightarrow$ no change in K.E.

– *EF (gas state)*: $T \nearrow \Rightarrow E_k \nearrow$ particles move even faster

particles completely separated, no intermolecular force, so constant $E_p = 0$

Question 5.3 For a particular substance, why is the specific latent heat of vaporisation much greater than the specific latent heat of fusion?

evaporation

liquid changes into gas without boiling $\rightarrow$ **evaporation**

particles move randomly, i.e., they move at various speeds

some molecules move fast enough to break free

➤ *cooling effect*: evaporation causes a decrease in temperature of the liquid

most energetic molecules escaped, those remain in the liquid have less energy, $E_k \searrow \Rightarrow T \searrow$

➤ rate of evaporation increases with temperature, surface area of liquid

➤ different between boiling and evaporation

	boiling	evaporation
occurrence	throughout the liquid	at surface only
temperature	occur at boiling point	occur at any temperature
bubble formation	bubbles formed	no bubbles
rate of process	fast	slow

5.2.5 first law of thermodynamics

internal energy of a system changes upon heat transfer or doing work

first law of thermodynamics states that the increase in internal energy equals sum of heat supply to the system and work done on the system: $\Delta U = Q + W$

^[24] Again we only concern pure substances. Mixtures that boil over a range of temperatures or substance decompose before phase transition are not considered here.

Yuhao Yang 5 热力学 – DE (沸腾) : 分子断裂, r_l) E_p 1 沸腾在恒温下发生 [24]) K.E 无变化。 – E_F (气态) : T_l) E_k 1 粒子运动得更快, 完全分离的粒子, 没有分子间力, 因此恒定 $E_p = 0$ 问题 5.3 对于特定物质, 为什么比汽化潜热远大于比熔化潜热? 蒸发液体变成气体而不沸腾 – 蒸发粒子随机移动, 即它们以不同的速度移动一些分子移动得足够快, 足以挣脱 $\hat{\alpha}$ 冷却效应: 蒸发导致液体温度下降大多数能量分子逃逸, 那些留在液体中的能量较少, E_k %) T % $\hat{\alpha}$ 蒸发速率随温度的增加, 液体的表面积 $\hat{\alpha}$ 沸腾和蒸发之间的不同沸腾蒸发发生在整个液体表面, 仅温度发生在沸点发生在任何温度下气泡形成气泡形成没有气泡过程速率快慢 5.2.5 热力学第一定律系统的内能随传热或做功而变化热力学第一定律指出, 内能的增加等于向系统供应的热量和系统所做的功之和: $\Delta U = Q + W$ [24] 同样, 我们只关注纯物质。这里不考虑在一定温度范围内沸腾的混合物或物质在相变之前分解的情况。 50

➤ first law of thermodynamics is an extension of the law of conservation of energy

➤ sign conventions for Q and W

- $Q > 0$ if heat supplied to system
- $Q < 0$ if heat released by system to surroundings
- $W > 0$ if work done *on* system by external

i.e., if system is compressed and volume decreases, then $W > 0$

- $W < 0$ if system does work *against* surroundings

i.e., system expands and volume increases, then $W < 0$

➤ amount of heat energy: $Q = \begin{cases} cm\Delta T & \text{(if no change of state)} \\ Lm & \text{(during change of state)} \end{cases}$

➤ amount of work is related to pressure and change of volume

if volume changed at *constant* pressure, then $W = F\Delta s = pA\Delta s \Rightarrow W = p\Delta V$ ^[25]

if no change of volume, then no work is done

Example 5.5 A gas is heated by supplying it with 25 kJ of energy. The gas expands so that the volume increases by 0.10 m^3 . Assume the gas has a fixed pressure of 150 kPa during the process. Calculate the change in internal energy.

✍ amount of work done: $W = p\Delta V = 150 \times 0.10 = 15 \text{ kJ}$

but gas expands means work is done against surroundings, so this is negative work

change in internal energy: $\Delta U = Q + W = (+25) + (-15) = +10 \text{ kJ}$

□

Example 5.6 Use the idea of internal energy and the first law of thermodynamics, explain why boiling water requires heat supply.

✍ boiling occurs at constant temperature, so $\Delta E_k = 0$

but separation between molecules increased, so $\Delta E_p > 0$

by definition, internal energy $U = E_k + E_p$, so $\Delta U > 0$

during boiling, there is an increase in volume, so work against surroundings, $W < 0$

recall first law of thermodynamics $\Delta U = Q + W$, must have $Q > 0$

^[25] If pressure changes with volume during a thermodynamics process, then work done $W = \int p dV$. Alternatively, we can evaluate the area under a p - V graph to find the work done.

Yuhao Yang 5 热力学 – 热力学第一定律是能量守恒定律的延伸 – Q 和 W 的符号约定 $\pm Q \in \mathbb{R}$ 如果向系统提供热量 $\pm Q \in \mathbb{R}$ 如果系统向周围环境释放热量 $\pm W \in \mathbb{R}$ 如果外部对系统做功，即如果系统被压缩且体积减小，则 $W \in \mathbb{R} \pm W \in \mathbb{R}$ 如果系统确实对周围环境做功即，系统膨胀且体积增加，则 $W \in \mathbb{R}$ $\hat{=}$ 热能量： $Q = 8 > < > : \text{cm } \Delta T$ (如果没有状态变化) L_m (状态变化期间) $\hat{=}$ 功量与压力和体积变化有关，如果体积在恒定压力下变化，则 $W = F \Delta s = p A \Delta s$ $W = p \Delta V$ [25] 如果体积没有变化，则不做功示例 5.5 通过提供 25 kJ 的能量来加热气体。气体膨胀，体积增加 0.10 m^3 。假设在此过程中气体的固定压力为 150 kPa。计算内能的变化。- 所做的功量： $W = p \Delta V = 150 \times 0.10 = 15 \text{ kJ}$ 但气体膨胀意味着对周围环境做功，因此这是内能的负功变化： $\Delta U = Q + W = (+25) + (-15) = +10 \text{ kJ}$ $\hat{=}$ 例 5.6 使用内能的概念和热力学第一定律，解释原因沸水需要热量供应。- 沸腾发生在恒温下，因此 $\Delta E_k = 0$ 但分子之间的分离增加，因此 $\Delta E_p \in \mathbb{R}$ 根据定义，内能 $U = E_k + E_p$ ，因此 $\Delta U \in \mathbb{R}$ 在沸腾过程中，体积增加，因此与环境作斗争， $W \in \mathbb{R}$ 回忆热力学第一定律 $\Delta U = Q + W$ ，必须有 $Q \in \mathbb{R}$ [25] If 在热力学过程中压力随体积变化，则完成的功 $W = \int p dV$ 。或者，我们可以评估 p - V 图下的面积来找出所做的功。51

this means heat must be supplied for boiling processes $\square$

Example 5.7 When you pump up a bicycle tyre, the temperature of air inside the tyre will go up. Explain why this happens using the first law of thermodynamics.

 pumping up tyre involves compressing gas, so positive work is done: $W > 0$

for each stroke, there is little time for heat transfer, so $Q \approx 0$

according to first law of thermodynamics $\Delta U = Q + W \Rightarrow \Delta U > 0$

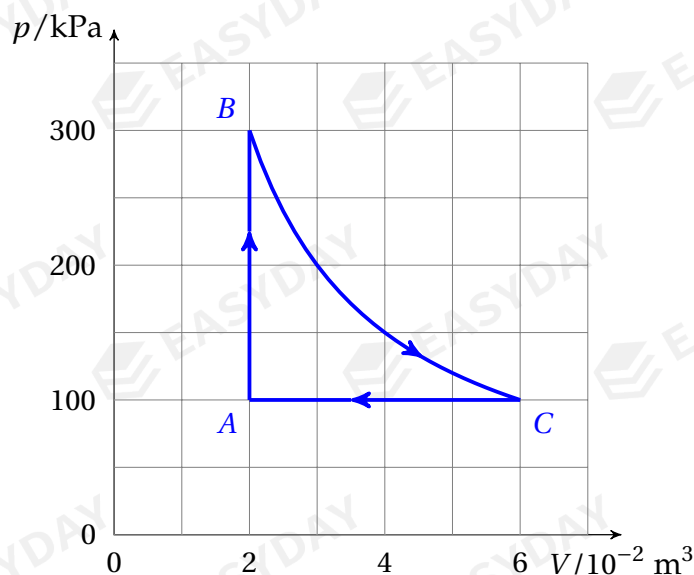
by definition, internal energy $U = E_k + E_p \Rightarrow \Delta U = \Delta E_k + \Delta E_p > 0$

but for gas, there is negligible intermolecular force, so $\Delta E_p = 0$, then must have $\Delta E_k > 0$

K.E. of molecules is proportional to temperature, higher K.E. so higher temperature $\square$

Example 5.8 An ideal gas of 0.080 mol is initially at state A and then undergoes a cycle ABCA.

The variation of its pressure p with its volume V is shown on the graph.



Temperature of state A is 300 K. The magnitude of work on gas from state B to C is 6570 J.

For each stage $A \rightarrow B$, $B \rightarrow C$ and $C \rightarrow A$ during the cycle, determine work done and heat supply to the gas, and also find the change in internal energy.

Yuhao Yang 5 热力学这意味着必须为沸腾过程提供热量。示例 5.7 当给自行车轮胎充气时，轮胎内的空气温度将会升高。使用热力学第一定律解释为什么会发生这种情况。 - 给轮胎充气涉及到压缩气体，因此会做正功：每次冲程 $W > 0$ ，几乎没有传热时间，因此根据热力学第一定律 $Q \approx 0$ $\Delta U = Q + W$ $\Delta U > 0$ 根据定义，内能 $U = E_k + E_p$ $\Delta U = \Delta E_k + \Delta E_p > 0$ 但对于气体，有是可忽略不计的分子间力，因此 $\Delta E_p = 0$ ，则必须有 $\Delta E_k > 0$ 。分子的数量与温度成正比， E_k 越高。因此温度更高。示例 5.8 0.080 mol 的理想气体最初处于状态 A，然后经历循环 ABCA。其压力 p 随体积 V 的变化如图所示。 p/kPa $V/10^{-2} \text{ m}^3$ 30 100 200 300 0 2 4 6 A B C 状态 A 的温度为 300 K。气体从状态 B 到 C 的做功大小为 6570 J。对于每个阶段 A → B，B → C 和 C → A 在循环过程中，确定所做的功和向气体提供的热量，并找出内能的变化。 52

work done depends on change in volume

$A \rightarrow B$: no change in volume, so $W_{AB} = 0$

$B \rightarrow C$: $|W_{BC}| = 6570 \text{ J}$, but expansion implies $W < 0$, so $W_{BC} = -6570 \text{ J}$

$C \rightarrow A$: $|W_{CA}| = p\Delta V_{CA} = 1 \times 10^5 \times (6 - 2) \times 10^{-2} \Rightarrow W_{CA} = +4000 \text{ J}$ (compression so $W > 0$)

change in internal energy of ideal gas depends on change in temperature

$A \rightarrow B$: same V but $p_B = 3p_A$, so $T_B = 3T_A = 900 \text{ K}$

$$\Delta U_{AB} = \frac{3}{2} Nk\Delta T_{AB} = \frac{3}{2} \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (900 - 300) \approx +5980 \text{ J}$$

$B \rightarrow C$: note that $p_B V_B = p_C V_C$, so $T_B = T_C$, no change in temperature, so $\Delta U_{BC} = 0$

$$C \rightarrow A: \Delta U_{CA} = \frac{3}{2} Nk\Delta T_{CA} = \frac{3}{2} \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (300 - 900) \approx -5980 \text{ J}$$

for cycle $ABCA$, same initial and final state, so total change in internal energy must be zero

one can check that $\Delta U_{\text{cycle}} = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$

to find supply of thermal energy, we apply first law of thermodynamics: $\Delta U = Q + W$

$$A \rightarrow B: +5980 = Q_{AB} + 0 \Rightarrow Q_{AB} = +5980 \text{ J}$$

$$B \rightarrow C: 0 = Q_{BC} + (-6570) \Rightarrow Q_{BC} = +6570 \text{ J}$$

$$C \rightarrow A: -5980 = Q_{CA} + (+4000) \Rightarrow Q_{CA} = -9980 \text{ J}$$

the table below summarises all energy changes during the cycle $ABCA$

change	W/J	Q/J	$\Delta U/\text{J}$
$A \rightarrow B$	0	+5980	+5980
$B \rightarrow C$	-6570	+6570	0
$C \rightarrow A$	+4000	-9980	-5980

□

Question 5.4 Show that when n mol of gas is heated at a fixed volume, thermal energy required to raise the temperature by 1.0 K is nR .

Question 5.5 Two identical balloons A and B hold the same amount of gas at the same initial temperature. They are given the same amount of heat. Suppose volume of A is fixed, while B is allowed to expand, compare the final temperatures of the gases in the two balloons.

杨宇豪 5 热力学 - 所做的功取决于体积 A 的变化！ B ：音量没有变化，所以 $W_{AB} = 0$ ！ C ： $\int W_{BC} = 6570 \text{ J}$ ，但展开意味着 $W < 0$ ，所以 $W_{BC} = -6570 \text{ J}$ ！ A ： $\int W_{CA} = p \Delta V = 1 \times 10^5 \times (6 - 2) \times 10^{-2} \text{ J} = +4000 \text{ J}$ （压缩，因此 $W < 0$ ）理想气体内能的变化取决于温度 A 的变化！ B ：相同的 V ，但 $p_B = 3p_A$ ，因此 $T_B = 3T_A = 900 \text{ K}$ $\Delta U_{AB} = 3/2 n k \Delta T = 3/2 \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (900 - 300) \text{ J} = +5980 \text{ J}$ ！ C ：注意 $p_B V_B = p_C V_C$ ，因此 $T_B = T_C$ ，温度没有变化，因此 $\Delta U_{BC} = 0$ ！ A ： $\Delta U_{CA} = 3/2 n k \Delta T = 3/2 \times 0.80 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23} \times (300 - 900) \text{ J} = -5980 \text{ J}$ 对于循环 ABC ，初始和最终状态相同，因此内能总变化必须为零，可以检查 $\Delta U_{\text{cycle}} = \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$ 为了找到热能的供应，我们应用热力学第一定律： $\Delta U = Q + W$ ！ B ： $+5980 = Q_{AB} + 0$ $Q_{AB} = +5980 \text{ J}$ ！ C ： $0 = Q_{BC} + (-6570)$ $Q_{BC} = +6570 \text{ J}$ ！ A ： $-5980 = Q_{CA} + (+4000)$ $Q_{CA} = -9980 \text{ J}$ 下表总结了循环 ABC 变化期间的所有能量变化 W/J Q/J $\Delta U/\text{J}$ ！ B $0 + 5980 + 5980$ B ！ $^\circ\text{C}$ $-6570 + 6570$ 0 ！ A $+4000 - 9980 - 5980$ 问题 5.4 证明当 $n \text{ mol}$ 气体以固定体积加热时，温度升高 1.0 K 所需的热能为 nR 。问题 5.5 两个相同的气球 A 和 B 在相同的初始温度下容纳相同数量的气体。它们被给予相同量的热量。假设 A 的体积固定，而 B 允许膨胀，比较两个气球中气体的最终温度。

5.3 temperature

5.3.1 temperature & thermal energy

- temperature can be considered as a *relative* measure of thermal energy
temperature can tell the *direction* of thermal energy flow
heat always (spontaneously) flows from high temperature regions to colder regions^[26]
- if two objects in contact have the same temperature, then there is no net heat transfer
the two objects are said to be in **thermal equilibrium**
- if two systems *A* and *B* are each in thermal equilibrium with a third system *C*, *A* and *B* are also in thermal equilibrium, this is called the **zeroth law of thermodynamics**^[27]

Question 5.6 A student thinks that temperature measures the amount of heat in an object. Suggest why this statement is incorrect with examples.

5.3.2 absolute zero

mean K.E. of molecules is a microscopic description of temperature *T*

minimum K.E. occurs if molecules do not move at all (completely frozen)^[28]

this corresponds to the lowest possible temperature, called **absolute zero**

- **Kelvin scale of thermodynamic temperature** is defined based on absolute zero as 0 K^[29]
- conversion rule between Celsius scale and Kelvin scale: $T_K(\text{K}) \begin{matrix} \xleftarrow{-273.15} \\ \xrightarrow{+273.15} \end{matrix} T_C(^{\circ}\text{C})$ ^[30]
- Kelvin scale is said to be an *absolute scale*

zero of Kelvin scale does not depend on property of a specific substance

^[26] This is the consequence of the *second law of thermodynamics*, which is concerned with the direction of natural processes. The law states that the total *entropy*, a quantity that counts the number of microstates of a system, of an isolated system can never decrease over time. You will learn more about entropy if you study A-Level chemistry.

^[27] This law is important for the formulation of thermal physics. The physical meaning of the law was expressed by Maxwell: "All heat is of the same kind." The zeroth law allows us to give the mathematical definition of temperature.

^[28] In the *classical* description, there is no reason not allowing a molecule to cease motion. However, due to *quantum mechanical effects*, kinetic energy of a system cannot be zero even at absolute zero.

^[29] More precisely, 0 K for absolute zero and 273.16 K for water triple point.

^[30] The numerical value of 273.15 will only be quoted in this section. For everywhere else in the notes, we will use the less precise value 273 for simplicity.

杨宇豪 5 热力学 5.3 温度 5.3.1 温度和热能 $\hat{\alpha}$ 温度可以被认为是热能的相对量度温度可以告诉热能流动的方向热量总是（自发地）从高温区域流向较冷的区域 [26] $\hat{\alpha}$ 如果两个接触的物体具有相同的温度，则不存在净热传递，则两个物体被认为处于热平衡 $\hat{\alpha}$ 如果两个系统 A 和 B 都处于热平衡状态与第三个系统 C 达到平衡，A 和 B 也处于热平衡，这称为热力学第零定律 [27] 问题 5.6 学生认为温度测量物体中的热量。用例子说明为什么这个说法不正确。 5.3.2 绝对零均值 K.E. 分子的分布是温度 T 最小值 K.E 的微观描述。如果分子根本不移动（完全冻结），则会发生这种情况 [28] 这对应于可能的最低温度，称为绝对零 $\hat{\alpha}$ 热力学温度的开尔文标度基于绝对零定义为 0 K [29] $\hat{\alpha}$ 摄氏度和开尔文标度之间的转换规则： $T_K(K) - 273.15$ $G G G G G G G G G G B F G G G G G G G G G G + 273.15 T_C(\pm C)$ [30] $\hat{\alpha}$ 开尔文标度据说是开尔文标度的绝对标度零，不依赖于特定物质的性质 [26] 这是热力学第二定律的结果，该定律与自然过程的方向有关。该定律规定，孤立系统的总熵（计算系统微观状态数量的量）永远不会随着时间的推移而减少。如果你学习 A-Level 化学，你会学到更多关于熵的知识。 [27] 该定律对于热物理的制定很重要。该定律的物理意义由麦克斯韦表述：“所有的热都是同一种类的”。第零定律使我们能够给出温度的数学定义。 [28] 在经典描述中，没有理由不允许分子停止运动。然而，由于量子力学效应，系统的动能即使绝对为零也不能为零。 [29] 更准确地说，0 K 表示绝对零，273.16 K 表示水三相点。 [30] 本节仅引用数值 273.15。对于注释中的其他地方，为了简单起见，我们将使用不太精确的值 273。 54

in contrast, zero of Celsius scale is based on properties of water

➤ it is impossible to remove any more energy from a system at 0 K (or -273.15 °C)

but there is no practicable means to bring a physical system to exactly 0 K ^[31]

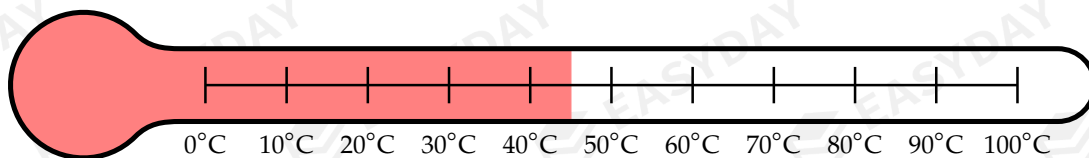
5.3.3 thermometer

a **thermometer** is a device which can be used to measure temperature

liquid-in-glass thermometer

basic principle: liquid expands in volume at higher temperature

examples include alcohol thermometer, mercury-in-glass thermometer, etc.



resistance temperature detectors (RTD)

basic principle: resistance of electronic element changes with temperature

metal wires and thermistor are both used in RTD elements

thermocouple

basic principle: difference in temperature can produce a *thermoelectric voltage* across junctions, thermocouple measures temperatures by means of this voltage

two pieces of different metal wires are joined at their ends

if there exists a temperature difference between the ends, a *thermoelectric voltage* is developed

this voltage depends on temperature difference, captured by some characteristic function

in practice, we place the *measurement* junction in an environment of unknown temperature

the other end, or the *reference* junction, is at a known temperature

^[31] This is known as the *third law of thermodynamics*, which states that it is impossible, no matter how idealized, to reduce the temperature of any closed system to absolute zero in a finite number of operations.

Yuhao Yang 5 热力学相反，零摄氏度是基于水的特性——不可能从 0 K（或 $-273.15 \pm \text{C}$ ）的系统中去除更多能量，但没有可行的方法使物理系统精确达到 0 K [31]

5.3.3 温度计

温度计是一种可用于测量温度的装置

玻璃液体温度计基本原理：

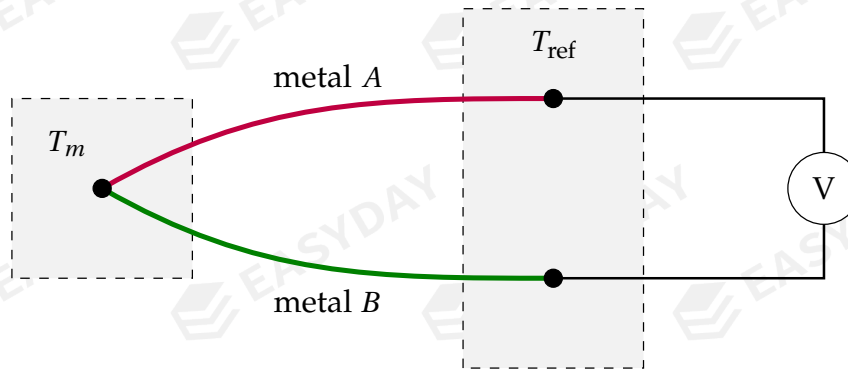
液体在较高温度下体积膨胀例如包括酒精温度计、玻璃水银温度计等

$0 \pm \text{C}$ $10 \pm \text{C}$ $20 \pm \text{C}$ $30 \pm \text{C}$ $40 \pm \text{C}$ $50 \pm \text{C}$ $60 \pm \text{C}$ $70 \pm \text{C}$ $80 \pm \text{C}$ $90 \pm \text{C}$ $100 \pm \text{C}$

电阻温度检测器（RTD）基本原理：

电子元件的电阻随温度变化

热电偶通过该电压测量温度，如果两端之间存在温差，则两根不同的金属线在其末端连接，产生热电电压，该电压取决于温差，在实践中通过某些特征函数捕获，我们将测量结置于未知温度的环境中，另一端或参考结处于已知温度[31]这被称为热力学第三定律，该定律指出，无论多么理想化，都不可能将任何封闭系统的温度降低到在有限次操作中绝对为零。 55



typical configuration of a thermocouple unit

temperature difference is deduced from the voltage reading

hence the desired temperature can be determined

➤ features of a thermometer:

- *range*: whether the thermometer can measure very low or very high temperatures
- *sensitivity*: whether a small change in temperature can be detected
- *response time*: whether changes in temperature can be immediately measured
- *linearity*: whether changes in temperature are proportional to changes in output

	liquid-in-glass	RTD wire	thermistor	thermocouple
valid range	narrow	wide	narrow	very wide
sensitivity	low	fair	high	high
response time	slow	fast	fast	very fast
linearity	good	good	limited	non-linear

Yuhao Yang 5 热力学 T_m Tref 金属 A 金属 B V 热电偶单元的典型配置从电压读数推导出温差，从而可以确定所需温度 $\hat{\alpha}$ 温度计的特点：－量程：温度计是否可以测量非常低或非常高的温度－灵敏度：是否可以检测到温度的微小变化－响应时间：是否可以立即测量温度的变化－线性：温度的变化是否与输出玻璃液体 RTD 线的变化成正比热敏电阻热电偶有效范围窄宽窄非常宽灵敏度低一般高高响应时间慢慢快快非常快线性度好好有限非线性

CHAPTER 6

Electrostatics

6.1 electric forces

6.1.1 Coulomb's law

charged objects will attract or repel one another through the electric force

Coulomb's law states that the electric force between two electrically charged particles is proportional to their charges and inversely proportional to the square of their separation:

$$F = \frac{Qq}{4\pi\epsilon_0 r^2}$$

$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ is the *permittivity of free space*

$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ is a useful constant for calculations

this law was first published by French physicist *Charles Augustin de Coulomb* in 1785

- charges Q, q in Coulomb's law are *point charges*
- for uniformly charged spheres, they can be thought as point charges
separation r is taken to be centre-to-centre distance
- symbolically, sign of F can tell direction of the electric force

for like charges (both positive or both negative), $Q_1 Q_2 > 0 \Rightarrow F > 0 \Rightarrow \text{repulsion}$

for opposite charges (one positive and one negative), $Q_1 Q_2 < 0 \Rightarrow F < 0 \Rightarrow \text{attraction}$

Example 6.1 The hydrogen atom has a radius of about 53 pm. Estimate the electric force between the proton and the orbiting electron.

$$F = \frac{Q_p Q_e}{4\pi\epsilon_0 r^2} = 8.99 \times 10^9 \times \frac{(1.60 \times 10^{-19})^2}{(53 \times 10^{-12})^2} \approx 8.2 \times 10^{-8} \text{ N} \quad \square$$

Question 6.1 Two protons are separated by a distance r . Find the ratio of the electric force to the gravitational force between them.

Yuhao Yang 6 静电第 6 章静电 6.1 电力 6.1.1 库仑定律带电物体会通过电力相互吸引或排斥库仑定律指出，两个带电粒子之间的电力与其电荷成正比，与其间距的平方成反比： $F = \frac{Qq}{4\pi\epsilon_0 r^2}$ $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ 是自由空间的介电常数 $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ 是一个有用的计算常数该定律最初由法国物理学家 Charles Augustin de Coulomb 于 1785 年发表 $\hat{r}$ 库仑定律中的电荷 Q, q 是点电荷 $\hat{r}$ 对于均匀带电球体，它们可以被视为点电荷间隔 r 取中心距 $\hat{r}$ 象征性地，符号 F 可以表明同种电荷（均为正或均为负）的电力方向， $(Q_1 Q_2 \geq 0) (F \geq 0)$ 异性电荷（一正一负）的斥力， $(Q_1 Q_2 < 0) (F < 0)$ 吸引力例 6.1 氢原子的半径约为 53 pm 。估计质子和轨道电子之间的电力。 $F = \frac{Q_p Q_e}{4\pi\epsilon_0 r^2} = 8.99 \times 10^9 \times \frac{(1.60 \times 10^{-19})^2}{(53 \times 10^{-12})^2} = 8.2 \times 10^{-8} \text{ N}$ 问题 6.1 两个质子之间的距离为 r 。求它们之间的电力与引力之比。 57

6.1.2 electric fields

to explain how charges affect each other at a distance, we introduce notion of *electric fields*

electric field is a region of space where a charged object is acted by a force

any charge Q (or several charges) can produce an electric field

any test charge q within the field will experience an electric force

Next, we will introduce the concepts of *electric field strength* and *electric potential*, and see how they are related to the force acting on a charged object and the potential energy it possesses.

You might have noticed that Coulomb's law for electrostatic forces and Newton's law of gravitation are both *inverse square laws*, it turns out that electric fields are very similar to gravitational fields in various aspects.

6.2 electric field strength

6.2.1 electric field strength

electric field strength is defined as electric force per unit positive charge: $E = \frac{F}{q}$

➤ unit of E : $[E] = \text{N C}^{-1} = \text{V m}^{-1}$ ^[32]

➤ field strength due to an isolated source of *point* charge Q

a small test charge q at distance r is acted by a force: $F = \frac{Qq}{4\pi\epsilon_0 r^2}$

field strength at this point: $E = \frac{F}{q} \Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2}$

the field is produced by Q , so field strength only depends on the source Q

➤ if the source is a charged *sphere* of radius R with uniform charge distribution

viewed from *outside* the sphere, it acts like a point charge concentrated at the centre ^[33]

^[32] You will later find in §6.4.3 the deeper reason why V m^{-1} is also a reasonable unit for field strength.

^[33] A brief explanation is given in Example 6.4.

杨宇豪 6 静电学 6.1.2 电场为了解释电荷如何在一定距离内相互影响，我们引入电场的概念。电场是一个空间区域，其中带电物体受到力的作用。任何电荷 Q （或几个电荷）都会产生电场。任何在该场内的测试电荷 q 都会受到电场力。接下来，我们将介绍电场强度和电势的概念，看看如何它们与作用在带电物体上的力及其所拥有的势能有关。你可能已经注意到，静电力的库仑定律和牛顿万有引力定律都是平方反比定律，事实证明电场在许多方面与引力场非常相似。

6.2 电场强度 6.2.1 电场强度 电场强度定义为每单位正电荷的电力： $E = F/q$ E 的单位： $[E] = \text{N C}^{-1} = \text{V m}^{-1}$ [32] $\hat{=}$ 孤立的点电荷源 Q 产生的场强在距离 r 处的小测试电荷 q 受到力的作用： $F = Qq / 4\pi\epsilon_0 r^2$ 此时的场强： $E = F/q$ $E = Q / 4\pi\epsilon_0 r^2$ 场由 Q 产生，因此场强仅取决于源 Q $\hat{=}$ 如果源是半径为 R 的带电球体，从球体外部看电荷分布均匀，则它的作用就像集中在中心的点电荷 [33] [32] 您稍后会在 § 中找到 6.4.3 V m^{-1} 也是场强合理单位的更深层次原因。 [33] 示例 6.4 中给出了简要说明。 58

therefore, $E = \frac{Q}{4\pi\epsilon_0 r^2}$ also holds for field due to charged sphere at $r > R$ ^[34]

where r is the distance from the point of interest to *centre* of the sphere

➤ field strength E is a *vector* quantity, it has a direction

to compute combined field strength due to several sources, should perform *vector sum* of contributions from each individual

➤ direction of field strength depends on the source charge Q

for positive source ($Q > 0$): field points away from the source

for negative source ($Q < 0$): field points towards the source

➤ electric force on a charge q can be found if field strength E is known

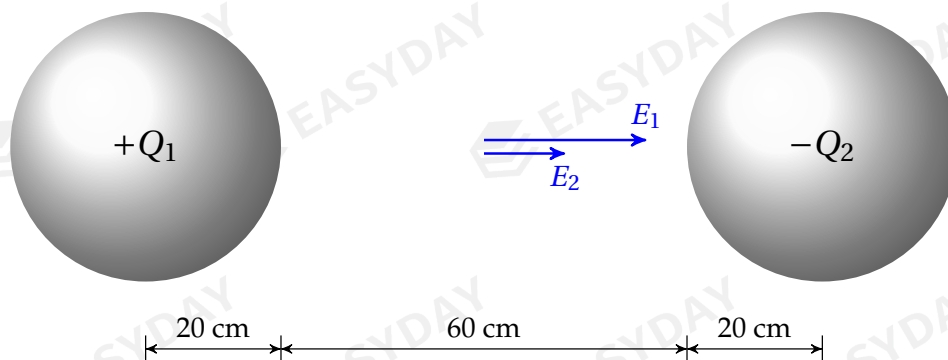
magnitude of electric force: $F = Eq$

direction of force: same direction as E if $q > 0$, but opposite to E if $q < 0$

Example 6.2 A Van de Graaff generator produces sparks when its surface electric field strength $4.0 \times 10^4 \text{ V cm}^{-1}$. If the diameter of the sphere is 40 cm, what is the charge on it?

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} \Rightarrow Q = 4\pi\epsilon_0 E r^2 = 4\pi \times 8.85 \times 10^{-12} \times 4.0 \times 10^6 \times 0.20^2 \approx 1.8 \times 10^{-5} \text{ C} \quad \square$$

Example 6.3 Two identical metal spheres of radius 20 cm carry charges $+2.0 \mu\text{C}$ and $-1.0 \mu\text{C}$ respectively. There is a 60 cm gap between them. (a) Find the electric field strength midway along the line joining their centres. (b) A dust particle carrying a charge of $-1.3 \times 10^{-8} \text{ C}$ is at this position. Find the electric force it experiences.



➤ field strengths due to the two spheres are in same direction

$$E = E_1 + E_2 = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{r_1^2} + \frac{Q_2}{r_2^2} \right) = 8.99 \times 10^9 \times \left(\frac{2.0 \times 10^{-6}}{0.25^2} + \frac{1.0 \times 10^{-6}}{0.25^2} \right) \approx 4.32 \times 10^5 \text{ N C}^{-1}$$

^[34] For electric field strength *inside* a conducting sphere, detailed discussions are given in §6.4.2.

Yuhao Yang 6 静电因此, $E = \frac{Q}{4\pi\epsilon_0 r^2}$ 也适用于 $r \in R$ [34] 处带电球体产生的场, 其中 r 是从感兴趣点到球体中心的距离。场强 E 是一个向量, 它有一个方向来计算由于多个源而产生的组合场强, 应执行每个个体贡献的矢量和。场强方向取决于源电荷 Q 对于正源 ($Q \geq 0$): 场指向远离源对于负源 ($Q < 0$): 场指向源。如果已知场强 E , 则可以找到电荷 q 上的电力大小: $F = Eq$ 力的方向: 如果 $q \geq 0$, 则与 E 方向相同, 但如果 $q < 0$, 则与 E 相反。

6.2 范德格拉夫发电机在其表面通电时产生火花场强 $4.0 \times 10^4 \text{ V cm}^{-1}$ 。如果球体的直径是40厘米, 它的电荷是多少? $E = \frac{Q}{4\pi\epsilon_0 r^2}$ $Q = 4\pi\epsilon_0 E r^2 = 4\pi \times 8.85 \times 10^{-12} \times 4.0 \times 10^4 \times 0.20^2 \approx 1.8 \times 10^{-5} \text{ C}$

示例 6.3 两个半径为 20 cm 的相同金属球分别携带 +2.0 (C 和 -1.0 (C 的电荷。它们之间有60厘米的间隙。(a) 求出它们中心连线中间的电场强度。(b) 带有 $-1.3 \times 10^{-8} \text{ C}$ 电荷的尘粒位于该位置。求它所受到的电力。

+Q1 -Q2 E_1 E_2 20 cm 60 cm 20 cm - 由于两个球体方向相同而产生的场强 $E = E_1 + E_2 = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{r_1^2} + \frac{Q_2}{r_2^2} \right) = 8.99 \times 10^9 \times \left(\frac{2.0 \times 10^{-6}}{0.25^2} + \frac{1.0 \times 10^{-6}}{0.25^2} \right) \approx 4.32 \times 10^5 \text{ N C}^{-1}$ [34]对于导电球内的电场强度, 第 6.4.2 节给出了详细讨论。 59

field strength points to the right

force on dust particle: $F = Eq = 4.32 \times 10^5 \times 1.4 \times 10^{-8} \approx 5.6 \times 10^{-3} \text{ N}$

dust particle is negatively-charged means force is opposite to field strength

so force on dust particle acts to the left

□

Question 6.2 When the charge on the Van de Graaff generator is $4.0 \times 10^{-7} \text{ C}$, the electric field strength at the sphere's surface is $2.4 \times 10^6 \text{ V m}^{-1}$. Determine the additional charge added to the sphere if the field strength at the surface becomes $3.0 \times 10^6 \text{ V m}^{-1}$.

Question 6.3 Two positively charged particles A and B are situated in a vacuum. Point P lies on the line joining the centres of the two spheres and is a distance x from A . Sketch the variation with x of electric field strength E due to the two particles.

6.2.2 electric field lines

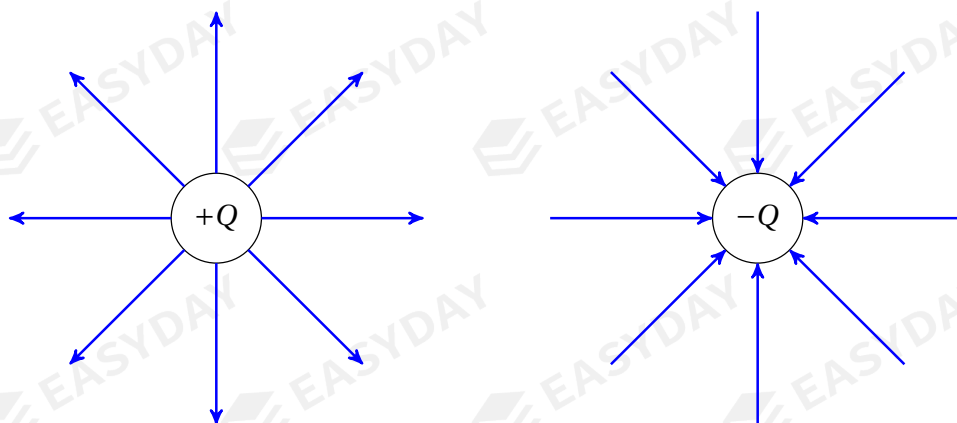
we can use **electric field lines** to visualize an electric field

➤ *arrows* of field lines show direction of the field

field lines always tend to leave positive charge, and end up at negative charges

➤ *density* or spacing of lines show strength of the field

Example 6.4 Sketch the electric field around a positively-charged sphere or a negatively charged sphere, and explain why they can be considered as point charges.



field lines of either case are *radial*, i.e., perpendicular to surface

field lines appear to start from or converge towards centre of the sphere

so charged spheres act like point charges

□

杨宇豪 6 静电场强度指向尘埃粒子上的力： $F = Eq = 4.32 \times 10^5 \times 1.4 \times 10^{-8} = 5.6 \times 10^{-3} \text{ N}$ 尘埃粒子带负电，意味着力与场强相反，因此尘埃粒子上的力作用于左侧

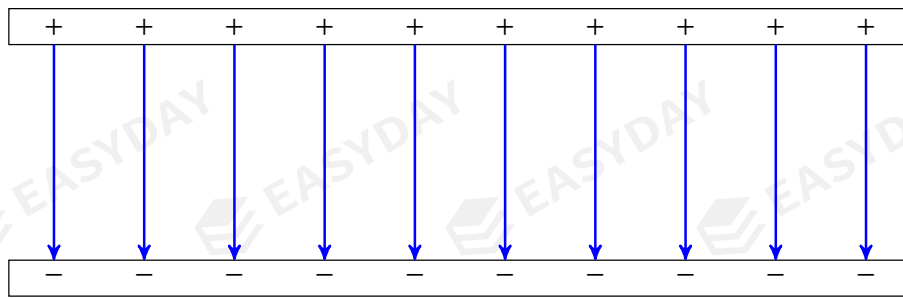
问题 6.2 当范德格拉夫发电机上的电荷为 $4.0 \times 10^{-7} \text{ C}$ ，球体表面的电场强度为 $2.4 \times 10^6 \text{ V m}^{-1}$ 。如果表面的场强变为 $3.0 \times 10^6 \text{ V m}^{-1}$ ，请确定添加到球体的附加电荷。

问题 6.3 两个带正电的粒子 A 和 B 位于真空中。点 P 位于连接两个球体中心的线上，距 A 的距离为 x 。画出两个粒子引起的电场强度 E 随 x 的变化。

6.2.2 电场线我们可以使用电场线来可视化电场 - 电场线箭头显示电场方向 电场线总是倾向于留下正电荷，最终会带负电荷 - 线的密度或间距显示电场的强度

例 6.4 画出带正电球体或带负电球体周围的电场，并解释为什么可以考虑它们作为点电荷。 $+Q -Q$ 两种情况的场线都是径向的，即垂直于表面场线似乎从球体中心开始或向球体中心汇聚，因此带电球体的行为类似于点电荷

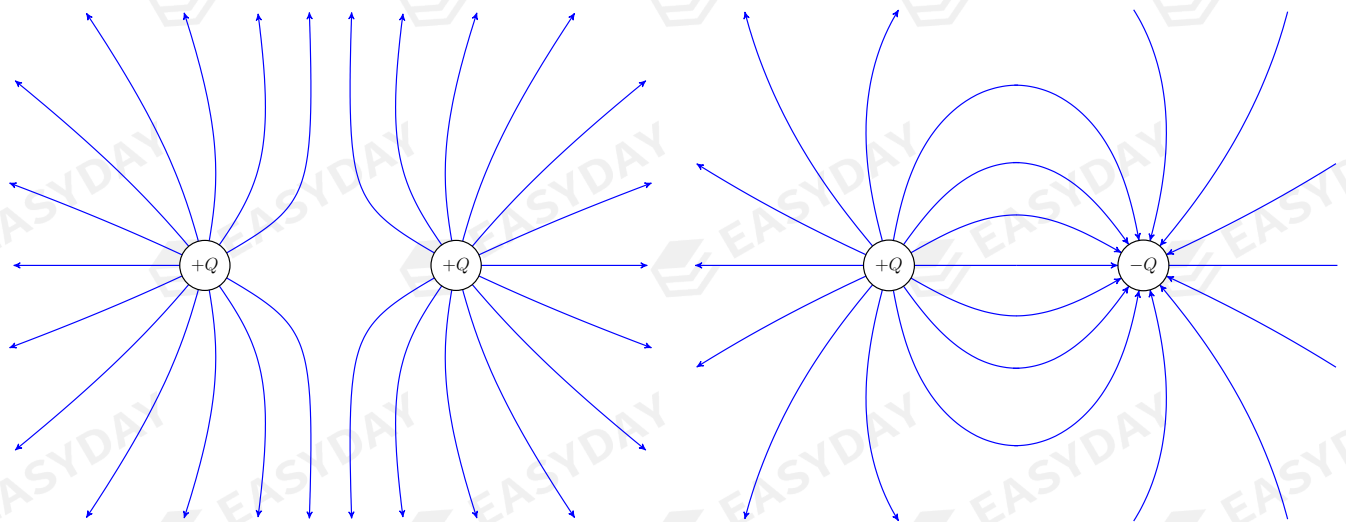
Example 6.5 Field lines between two oppositely-charged large metal plates.



field lines are *parallel* and equally spaced, so this is a *uniform* electric field

□

Example 6.6 Field pattern due to two charges of equal magnitude.



two positive charges

two opposite charges

□

6.3 potential & potential energy

6.3.1 electric potential energy

gain/loss in **electric potential energy** is defined as work done against/by electric force

(compare everything in this section with what you have learned about gravitational P.E.!)

let's start to derive the electrical P.E. between two charges Q and q separated by r

again we define $E_p = 0$ at $r = \infty$ (choice of zero potential energy, no force so no P.E.), then

electric potential energy is equal to the work done by electric force to bring a charge to a specific position from *infinity*

Yuhao Yang 6 静电例6.5 两个带相反电荷的大金属板之间的场线。 + - + - + - + - + - + - + - + - + -
场线平行且等距，因此这是一个均匀电场 ã 例 6.6 两个等量电荷产生的场模式。 + Q +Q 两个正电荷 +Q
-Q 两个相反电荷 ã 6.3 电势和势能 6.3.1 电势能增益/损失电势能定义为电力所做的功（将本节中的所有内容与您所了解的引力 P.E. 进行比较！） 让我们开始推导电势能。在由 r 分隔的两个电荷 Q 和 q 之间，我们将 $E_p = 0$ 定义为 $r = 1$ （选择零势能，没有力，因此没有 P .E.），那么电势能等于电力将电荷从无穷大带到特定位置所做的功 61

moving a test charge q from $r = \infty$ to a distance of r from Q



work done by electric force: $W = \int_{\infty}^r F dr = \int_{\infty}^r \frac{Qq}{4\pi\epsilon_0 x^2} dx = -\frac{Qq}{4\pi\epsilon_0 x} \Big|_{\infty}^r = -\frac{Qq}{4\pi\epsilon_0 r}$

since $\Delta E_p = -W$, we find: $E_p(r) - E_p(\infty) = \frac{Qq}{4\pi\epsilon_0 r}$

but $E_p(\infty) = 0$, so electric P.E. between two charges Q and q is $E_p(r) = \frac{Qq}{4\pi\epsilon_0 r}$

➤ as $r \rightarrow \infty$, $E_p \rightarrow 0$, this agrees with our definition for zero P.E. point

➤ for like charges, $Qq > 0$, so $E_p > 0$

to bring like charges closer, work must be done to overcome their *repulsion*, P.E. increases

minimum P.E. $E_p(\infty) = 0$ at infinity, so positive P.E. at finite r

➤ for opposite charges, $Qq < 0$, so $E_p < 0$

to pull opposite charges apart, work must be done to overcome their *attraction*, P.E. increases

maximum P.E. $E_p(\infty) = 0$ at infinity, so negative P.E. at finite r

➤ electric P.E. is a *scalar quantity*, sign is important

repulsion implies positive P.E., and attraction implies negative P.E.

sign of P.E. is hidden in polarities of charges

Example 6.7 To gain information about the gold nucleus ($^{197}_{79}\text{Au}$), we fire α -particles ($^4_2\alpha$) towards a thin gold foil. The size of a typical nucleus is about 10^{-14}m , what is the minimum initial speed for α -particles so that radius of gold nucleus can be determined?

🔑 as α -particle approaches the nucleus, it slows down due to the repulsive interaction

kinetic energy decreases and electric potential energy increases

if it gets close enough to the nucleus before coming to a stop, nuclear radius can be estimated

$$\text{K.E. loss} = \text{P.E. gain} \Rightarrow \underbrace{\frac{1}{2}mu^2}_0 - \underbrace{\frac{1}{2}mv^2}_0 = E_p(r) - \underbrace{E_p(\infty)}_0 \Rightarrow \frac{1}{2}mu^2 = \frac{Qq}{4\pi\epsilon_0 r}$$

$$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times u^2 = \frac{79 \times 1.60 \times 10^{-19} \times 2 \times 1.60 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times 10^{-14}} \Rightarrow u \approx 3.3 \times 10^7 \text{ m s}^{-1} \quad \square$$

Yuhao Yang 6 静电学将测试电荷 q 从 $r = 1$ 移动到距 Q 为 r 的距离，由电力完成的功： $W = \int_1^r F dr = \int_1^r \frac{Qq}{4\pi\epsilon_0 x^2} dx = -\frac{Qq}{4\pi\epsilon_0} \left[\frac{1}{x} \right]_1^r = -\frac{Qq}{4\pi\epsilon_0} \left(\frac{1}{r} - 1 \right)$ 自 $\Delta E_p = -W$ ，我们发现： $E_p(r) - E_p(1) = \frac{Qq}{4\pi\epsilon_0} \left(\frac{1}{r} - 1 \right)$ 但 $E_p(1) = 0$ ，因此电 P.E. 两个电荷 Q 和 q 之间为 $E_p(r) = \frac{Qq}{4\pi\epsilon_0} \frac{1}{r}$ 、艾普！0，这与我们对零 PE 的定义一致。点 $\hat{a}$ 对于同种电荷， $Qq \geq 0$ ，因此 $E_p \geq 0$ 要使同种电荷更接近，必须做功来克服它们的排斥力，P.E. 在无穷大处增加最小 P.E. $E_p(1) = 0$ ，因此正 P.E. 在 $\text{fin ter } \hat{a}$ 处，对于异性电荷， $Qq < 0$ ，因此 $E_p < 0$ 要将异性电荷拉开，必须做功来克服它们的吸引力 P.E. 在无穷大处增加最大 P.E. $E_p(1) = 0$ ，因此 P.E. 为负值。at finiter $\hat{a}$ 电动 P.E. 是标量，符号很重要，排斥意味着正 P.E.，吸引力意味着负 P.E.。体育标志隐藏在电荷极性中例 6.7 为了获得有关金核 (^{197}Au) 的信息，我们向薄金箔发射 α 粒子 (4He)。典型金核的尺寸约为 10^{-14}m ，那么可以确定金核半径的 α 粒子的最小初速度是多少？- 当 α 粒子接近原子核时，由于排斥相互作用，动能减小，而电势能增加，因此如果它在停止之前足够接近原子核，则其速度会减慢，可以估计核半径 K.E. 损失 = P.E. 增益 $\frac{1}{2} m u^2 - \frac{1}{2} m v^2 \Big|_{\{z\}=0} = E_p(r) - E_p(1) \Big|_{\{z\}=0} \frac{1}{2} m u^2 = \frac{Qq}{4\pi\epsilon_0} \left(\frac{1}{r} - 1 \right) \frac{1}{2} m u^2 = \frac{79 \times 1.60 \times 10^{-19} \times 2 \times 1.60 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times 10^{-14}} \frac{1}{2} m u^2 = 3.3 \times 10^7 \text{ m s}^{-1} \hat{a} 62$

Question 6.4 A metal sphere of radius 20 cm carries a charge of 5.0×10^{-7} C. A proton is sent towards the sphere at a speed of 1.8×10^6 m s⁻¹. Can the proton reach the surface of the sphere?

6.3.2 electric potential

it is also useful to define the *electric potential* for any specific point in a field

electric potential can be thought as the electric potential energy per unit charge: $V = \frac{E_p}{q}$

electric potential is the work needed to bring a unit positive charge from infinity

➤ unit: $[V] = \text{J C}^{-1} = \text{V}$

➤ electric potential due to an isolated source Q : $V = \frac{E_p}{q} = \frac{\frac{Qq}{4\pi\epsilon_0 r}}{q} \Rightarrow V = \frac{Q}{4\pi\epsilon_0 r}$

➤ potential at infinity vanishes: $V_\infty = 0$

➤ electric potential can take both signs

the sign depends on whether unit positive charge is repelled or attracted by the source

for positively-charged sources $V > 0$, while for negatively-charged sources $V < 0$

➤ electric potential is a *scalar* quantity

to find combined potential due to multiple charges, add up contributions of each charge

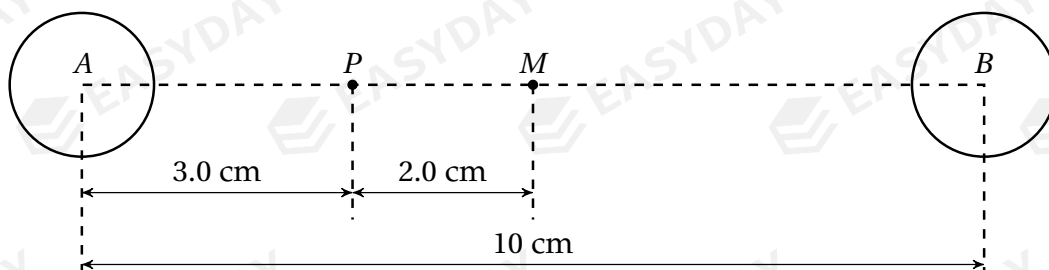
Example 6.8 An electron is accelerated from rest through a potential difference of 600 V.

Find the final speed of the electron.

✍ gain in K.E. = change in electric P.E. $\Rightarrow \frac{1}{2}mv^2 = q\Delta V$

$$v = \sqrt{\frac{2q\Delta V}{m}} = \sqrt{\frac{2.60 \times 10^{-19} \times 600}{9.11 \times 10^{-31}}} \approx 1.45 \times 10^7 \text{ m s}^{-1} \quad \square$$

Example 6.9 Two small metal spheres A and B are in a vacuum. Sphere A has charge $+20$ pC and sphere B has charge $+84$ pC. The arrangement is shown below.



Yuhao Yang 6 静电问题 6.4 半径为 20 cm 的金属球体带有 $5.0 \times 10^{-7} \text{ C}$ 的电荷。质子以 $1.8 \times 10^6 \text{ m s}^{-1}$ 的速度射向球体。质子能到达球体表面吗？

6.3.2 电势定义 场中任何特定点的电势也很有用 电势可以被认为 是每单位电荷的电势能： $V = E_p/q$ 电势是从无穷大产生单位正电荷所需的功 $\hat{a}$ 单位： $[V] = \text{J C}^{-1} = \text{V}$ $\hat{a}$ 隔离源 Q 产生的电势： $V = E_p/q = Q/q(4\pi\epsilon_0 r)$ $V = Q/4\pi\epsilon_0 r$ $\hat{a}$ 无穷远的电势消失： $V_{\infty} = 0$ $\hat{a}$ 电势可以取 两个符号，对于正电源 $V \geq 0$ ，电势取决于单位正电荷是否被源排斥或吸引，而对于负电源 $V \leq 0$ $\hat{a}$ 电势是 一个标量，用于查找由于多个电荷而产生的组合电势，将每个电荷的贡献相加 示例 6.8 电子通过 600 V 的电 势差从静止加速。求电子的最终速度。 - K.E. 增益 = 电 P.E. 的变化) $\frac{1}{2}mv^2 = q\Delta V$ $v = \sqrt{2q\Delta V/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 600 / 9.11 \times 10^{-31}} = 1.45 \times 10^7 \text{ m s}^{-1}$ $\hat{a}$ 例 6.9 两个小金属球 A 和 B 处于真空中。球体 A 的电荷为 +20 pC，球体 B 的电荷为 +84 pC。安排如下所示。 A 直径 10 厘米 3.0 厘米 2.0 厘米 63

(a) Find the electric potential at point P and point M respectively.

(b) Find the work done to move an α -particle from P to M .

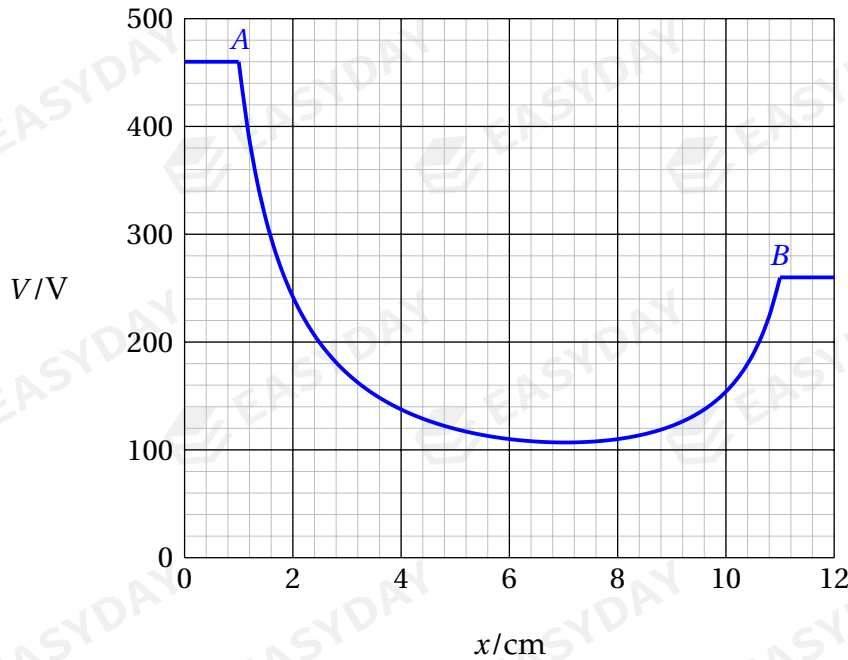
combined electric potential: $V = V_A + V_B = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_A}{r_A} + \frac{Q_B}{r_B} \right)$

at P : $V_P = 8.99 \times 10^9 \times \left(\frac{+20 \times 10^{-12}}{0.030} + \frac{+84 \times 10^{-12}}{0.070} \right) \approx 16.8 \text{ V}$

at M : $V_M = 8.99 \times 10^9 \times \left(\frac{+20 \times 10^{-12}}{0.050} + \frac{+84 \times 10^{-12}}{0.050} \right) \approx 18.7 \text{ V}$

from P to M : $W = \Delta E_p = q\Delta V = q(V_M - V_P) = 2 \times 1.60 \times 10^{-19} \times (18.7 - 16.8) \approx 6.2 \times 10^{-19} \text{ J}$ $\square$

Example 6.10 A and B are two positively-charged spheres of radius 1.0 cm. A proton P initially at rest on the surface of A moves along the line joining the centres of the two spheres. The variation with distance x from the centre of A of electric potential V at point P is given.



(a) Find the maximum speed as the proton moves from A to B .

(b) Find the speed when the proton reaches surface of B .

increase in K.E. = loss in P.E., so: $\frac{1}{2}mv^2 - 0 = q\Delta V \Rightarrow \frac{1}{2}mv^2 = q(V_A - V_P)$

maximum speed when ΔV is maximum, or $V_P = 107 \text{ V}$ becomes minimum (at $x = 7.2 \text{ cm}$)

$$\frac{1}{2} \times 1.67 \times 10^{-27} \times v_{\max}^2 = 1.60 \times 10^{-19} \times (460 - 107) \Rightarrow v_{\max} \approx 2.60 \times 10^5 \text{ m s}^{-1}$$

at surface of B , $V_P = 260 \text{ V}$ (at $x = 11.0 \text{ cm}$)

$$\frac{1}{2} \times 1.67 \times 10^{-27} \times v_B^2 = 1.60 \times 10^{-19} \times (460 - 260) \Rightarrow v_B \approx 1.96 \times 10^5 \text{ m s}^{-1} \quad \square$$

杨宇豪 6 静电 (a) 分别求 P 点和 M 点的电势。(b) 求 α 粒子从 P 移动到 M 所做的功。- 组合电势： $V = V_A + V_B = \frac{1}{4\pi\epsilon_0} (Q_A/r_A + Q_B/r_B)$ 在 P 处： $V_P = 8.99 \times 10^9 \times (+20 \times 10^{-12}/0.030 + +84 \times 10^{-12}/0.070)$ M 处： $V_M = 8.99 \times 10^9 \times (+20 \times 10^{-12}/0.050 + +84 \times 10^{-12}/0.050)$ $\frac{1}{4}$ 18.7 V 从 P 到 M： $W = \Delta E_p = q\Delta V = q(V_M - V_P) = 2 \times 1.60 \times 10^{-19} \times (18.7 - 16.8)$ $\frac{1}{4}$ 6.2×10^{-19} J $\ddot{\alpha}$ 例 6.10 A 和 B 是两个带正电的球体，半径为 1.0 cm。最初静止在 A 表面的质子 P 沿着连接两个球体中心的线移动。给出了 P 点电势 V 随距 A 中心距离 x 的变化。

x/cm	V/V
0	100
100	200
200	300
300	400
400	500
500	0
2	4
6	8
10	12

(a) 求质子从 A 移动到 B 时的最大速度。(b) 求质子到达 B 表面时的速度。= P.E. 损失，因此： $\frac{1}{2}mv^2 - 0 = q\Delta V$ $\frac{1}{2}mv^2 = q(V_A - V_P)$ ΔV 最大时的最大速度，或 $V_P = 107$ V 变为最小（在 $x = 7.2$ cm 处） $\frac{1}{2} \times 1.67 \times 10^{-27} \times v_{\max}^2 = 1.60 \times 10^{-19} \times (460 - 107)$ $v_{\max}$ $\frac{1}{4}$ 2.60×10^5 m s⁻¹ 在 B 表面， $V_P = 260$ V (在 $x = 11.0$ cm) $\frac{1}{2} \times 1.67 \times 10^{-27} \times v_B^2 = 1.60 \times 10^{-19} \times (460 - 260)$ v_B $\frac{1}{4}$ 1.96×10^5 m s⁻¹ $\ddot{\alpha}$ 64

Question 6.5 Electrical breakdown occurs when electric field strength at surface of a metal sphere exceeds $5.0 \times 10^6 \text{ N C}^{-1}$. Given that the radius of the sphere is 16 cm. What is the electric potential at the surface when electrical breakdown occurs?

Question 6.6 Two charged particles A and B are separated by 20 cm. P is a point on the line AB . Given that particle A carries charge $+7.2 \mu\text{C}$, and electric potential is zero where $AP = 5.0$ cm. Find the electric charge of B .

Question 6.7 A particle with specific charge (ratio of its electric charge to its mass) $+9.58 \times 10^7 \text{ C kg}^{-1}$ is moving towards a fixed metal sphere. The sphere has a potential of $+500 \text{ V}$. The initial speed of the particle is $3.0 \times 10^5 \text{ m s}^{-1}$ when it is a large distance from the sphere. Determine whether the particle can reach the surface of the sphere.

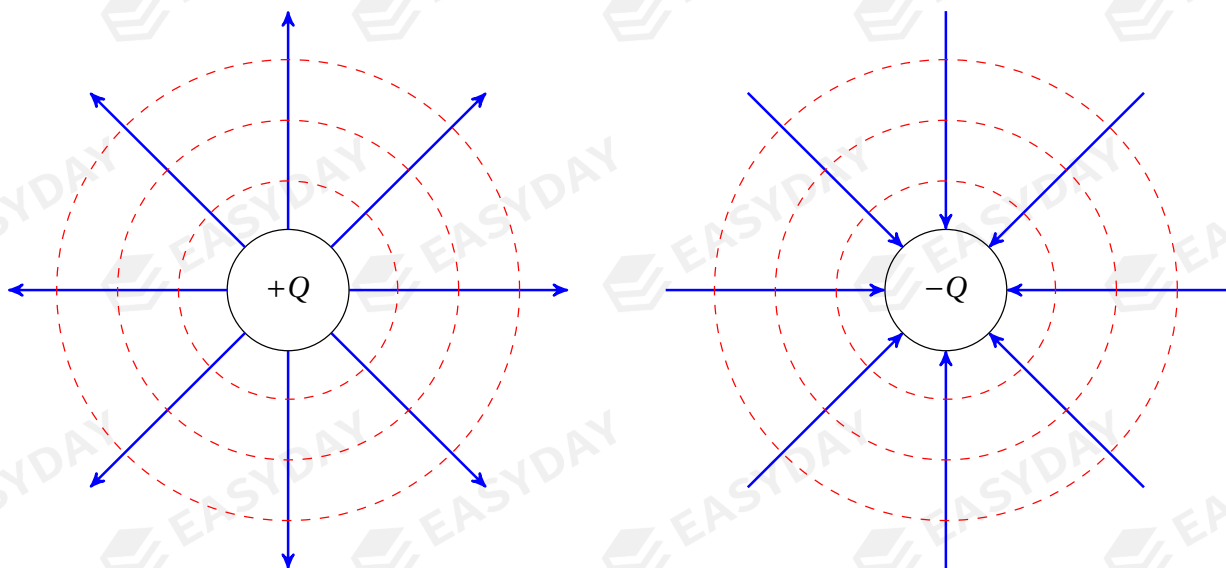
6.3.3 equipotential lines

to show potential distributions, we draw **equipotential lines** ^[35]

points on same equipotential line have constant electric potential

i.e., equipotential lines are *contour lines* of equal electric potential

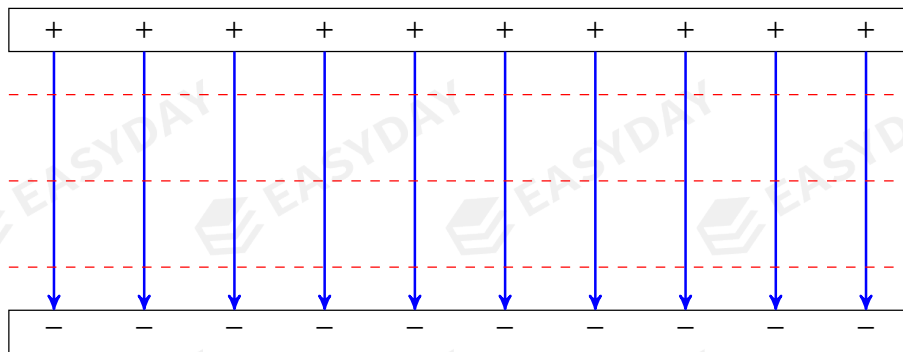
➤ for a field near a point charge, equipotential lines are a set of *concentric circles*



^[35] In three dimensions, these lines form equipotential *surfaces*.

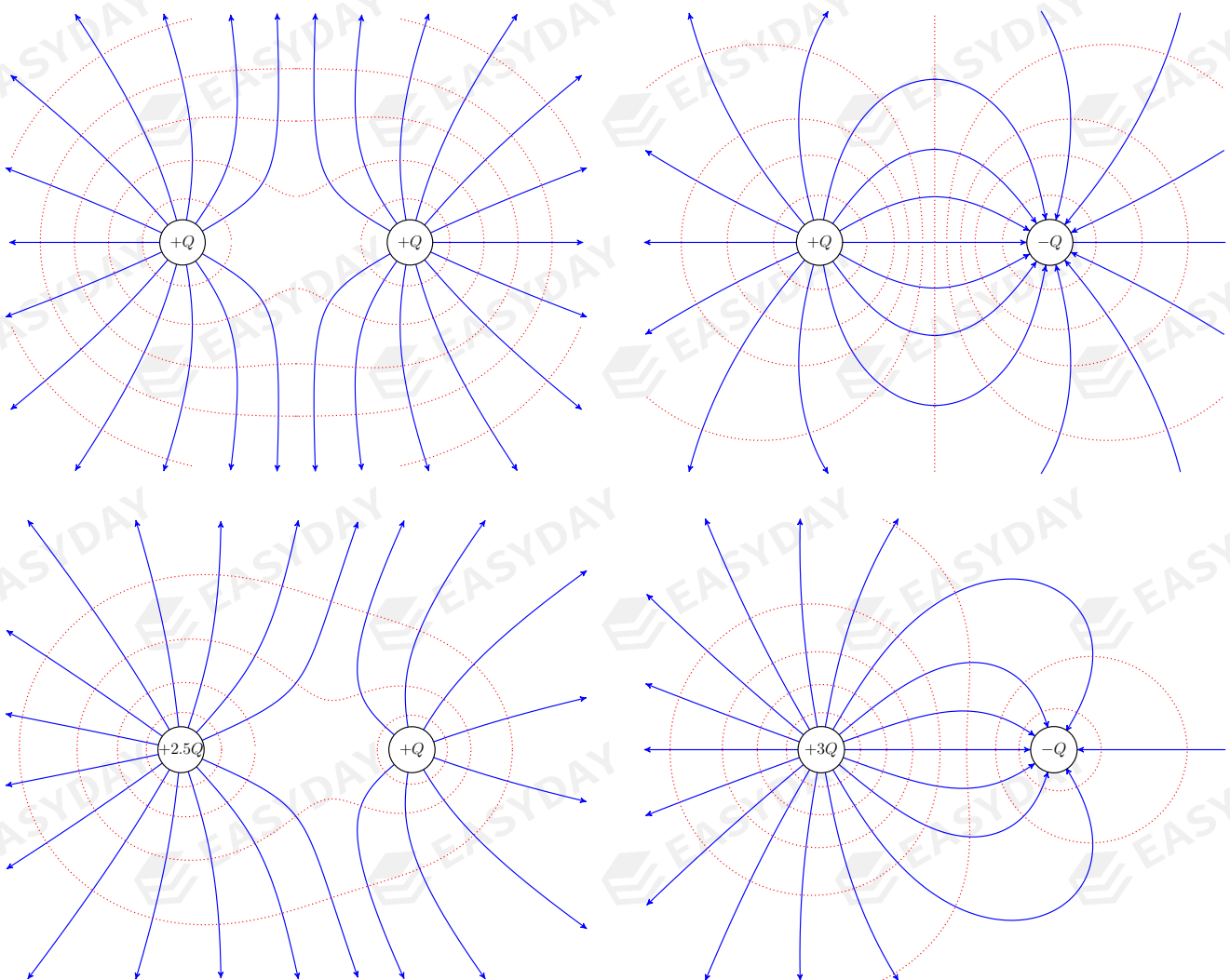
Yuhao Yang 6 静电问题6.5 当金属球表面的电场强度超过 $5.0 \times 10^6 \text{ N C}^{-1}$ 时，就会发生电击穿。假设球体的半径为 16 厘米。发生电击穿时表面电势是多少？问题 6.6 两个带电粒子 A 和 B 相距 20 厘米。P 是直线 AB 上的点。假定粒子 A 携带电荷 $+7.2 \text{ C}$ ，且 $AP = 5.0 \text{ cm}$ 处的电势为零。求 B 的电荷。问题 6.7 一个比电荷（电荷与其质量之比）为 $+9.58 \times 10^7 \text{ C kg}^{-1}$ 的粒子朝固定金属球运动。该球体的电势为 $+500 \text{ V}$ 。当粒子距离球体较远时，其初始速度为 $3.0 \times 10^5 \text{ m s}^{-1}$ 。确定粒子是否可以到达球体表面。6.3.3 等势线为了显示电势分布，我们画等势线[35]，同一条等势线上的点具有恒定的电势，即等势线是等电势的等高线。对于点电荷附近的场，等势线是一组同心圆+Q-Q [35]在三维空间中，这些线形成等势面。65

- for uniform fields, equipotential lines are a set of *parallel* straight lines

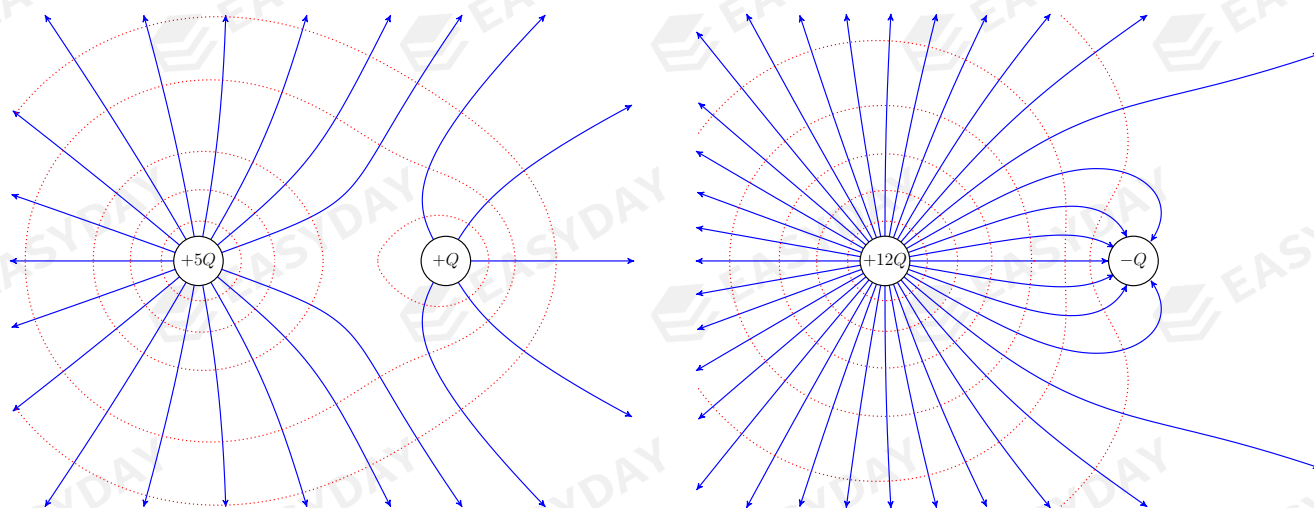


- equipotential lines are always perpendicular to the electric field lines
moving along an equipotential line requires no work done

Example 6.11 Field lines and equipotential lines due to two charges of various magnitudes



杨宇浩 6 静电 $\hat{\alpha}$ 对于均匀场，等势线是一组平行直线 $+ - + - + - + - + - + - + - + - + -$ 等势线总是垂直于沿着等势线移动的电场线，不需要做功例 6.11 两个电荷产生的场线和等势线各种幅度 $+ Q + Q + Q - Q + 2.5Q + Q + 3Q - Q$ 66



6.4 further discussions on electric fields

6.4.1 comparison with gravitational fields

both gravitational and electric force are described by inverse square law, so it follows that the mathematical language for both theories are very similar

➤ physical quantities that describe gravitational/electric fields

	vector description	scalar description
interaction between two masses/charges	force F	potential energy E_p
effect of source mass/charge	field g/E	potential φ/V

➤ comparing gravitational field with electric field ^[36]

	gravitational field	electric field	meaning
force	$F = (-)G \frac{Mm}{r^2}$	$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$	force between masses/charges
field strength	$g = \frac{F}{m} = (-)G \frac{M}{r^2}$	$E = \frac{F}{q} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$	force per unit mass/charge
potential energy	$E_p = -G \frac{Mm}{r}$	$E_p = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r}$	related to work done by force
potential	$\varphi = \frac{E_p}{m} = -G \frac{M}{r}$	$V = \frac{E_p}{q} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$	energy per unit mass/charge

^[36] Note that there is no negative mass, gravitational force always interacts attractively. This is the fundamental difference between gravitational fields and electric fields.

杨宇豪 6 静电 +5Q +Q +12Q -Q 6.4 关于电场的进一步讨论 6.4.1 与引力场的比较引力和电力都是用平方反比定律描述的，因此这两种理论的数学语言非常相似描述引力/电场的物理量矢量描述标量描述两个质量/电荷之间的相互作用力 F 势能 E_p 源质量/电荷场 g / E 电势 ϕ / V 比较引力场与电场 [36] 引力场电场含义力 $F = (-)G M m / r^2$ $F = 1 / 4\pi\epsilon_0 Q q / r^2$ 质量/电荷之间的力场强 $g = F / m = (-)G M / r^2$ $E = F / q = 1 / 4\pi\epsilon_0 Q / r^2$ 每单位质量/电荷的力势能 $E_p = -G M m / r$ $E_p = 1 / 4\pi\epsilon_0 Q q / r$ 与力势 $\phi = E_p / m = -G M / r$ $V = E_p / q = 1 / 4\pi\epsilon_0 Q / r$ 每单位质量/电荷的能量 [36] 请注意，不存在负质量，引力总是以吸引力相互作用。这是引力场和电场之间的根本区别。 67

- similarities between gravitational field and electric field
 - force and field strength both obey *inverse square laws*
 - potential energy and potential is inversely proportional to separation
 - no potential energy and no potential at infinite separation

- differences between gravitational field and electric field

mass (source of gravity) is always positive, but electric charges can be positive or negative

this fact leads to many fundamental differences between the two force fields

- electric force can be repulsive or attractive, but gravitational force is always attractive
- electric potential can take both signs, but gravitational potential is always negative

6.4.2 electric field inside conductors

consider electric field *inside* a metal conductor carrying charge Q

conductor means there are free charge carriers that can move around

but charge distribution should be stable for a charged conductor (no circulating currents)

so charge carriers must experience no force, i.e., field strength inside conductor is zero

put it the other way round, if there is an excess field, it will push free charge carriers to move around, until they are distributed so that the field inside the conductor becomes zero

moreover, there shall be no potential difference between any two points inside the conductor, otherwise charge carriers would flow, so electric potential must be constant

electric field strength is everywhere zero inside a conductor: $E = 0$

electric potential is everywhere constant inside a conductor: $V = \text{const}$

Example 6.12 Consider the electric field due to a metal sphere of radius R carrying charge Q . Plot the variation with the distance r from sphere's centre of the field strength, and the variation with r of the electric potential.

🔊 charge Q is uniformly spread out on *surface* of sphere

viewed from *outside*, the sphere appears to have all of its charge concentrated at the centre

so it can be modelled as a point charge due to its symmetric distribution of charges

杨宇浩 6 静电学 - 引力场和电场之间的相似之处 - 力和场强都遵循平方反比定律 - 势能和势能与分离成反比 - 无限分离时没有势能和势能 - 引力场和电场质量（引力源）之间的差异总是正的，但电荷可以是正的或负的，这一事实导致了两种力之间的许多基本差异场 - 电力可以是排斥力或吸引力，但万有引力总是有吸引力 - 电势可以同时具有两个符号，但万有引力总是负值

6.4.2 导体内部的电场

考虑带电荷的金属导体内部的电场 Q 导体意味着存在可以移动的自由载流子，但带电导体的电荷分布应该是稳定的（无循环电流），因此载流子必须不受力，即导体内部的场强为零反之亦然，如果有多余的场，它将推动自由电荷载流子四处移动，直到它们分布均匀，使导体内部的场变为零。此外，导体内部任何两点之间不得有电位差，否则电荷载流子会流动，因此电势必须恒定。电场强度在导体内部各处为零： $E = 0$ 导体内部电势恒定： $V = \text{const}$

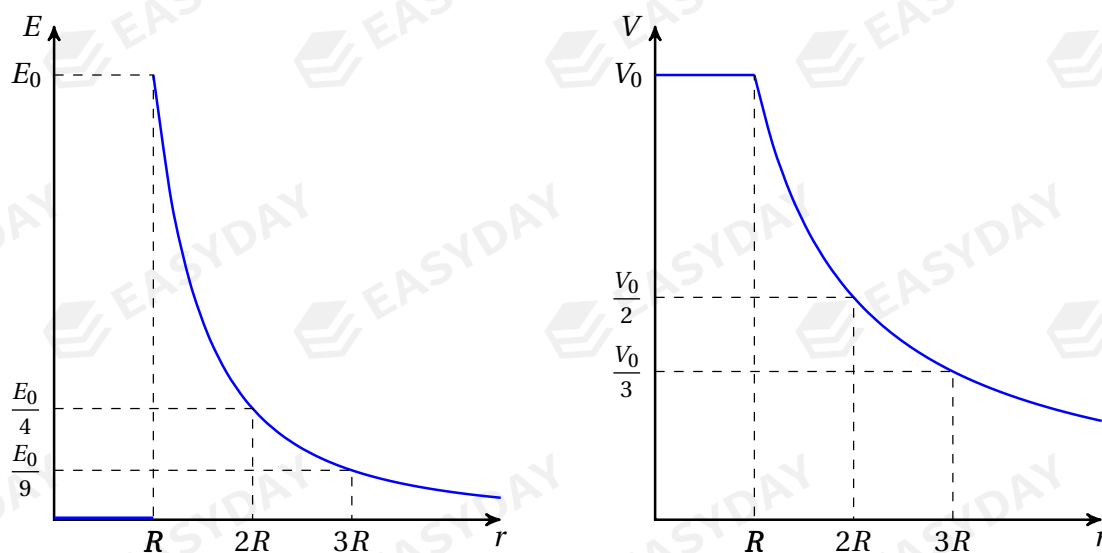
例 6.12 考虑电势半径为 R 且带有电荷 Q 的金属球产生磁场。绘制场强随距球中心距离 r 的变化，以及电势随 r 的变化。

- 从外部观察，电荷 Q 均匀分布在球体表面，球体似乎所有电荷都集中在中心，因此由于其电荷对称分布，可以将其建模为点电荷

electric field strength at distance r from sphere's centre is: $E = \frac{Q}{4\pi\epsilon_0 r^2}$ for $r > R$

electric potential at distance r from sphere's centre is: $V = \frac{Q}{4\pi\epsilon_0 r}$ for $r > R$

inside the sphere, i.e., for $r < R$, we have $E = 0$, and $V = \frac{Q}{4\pi\epsilon_0 R} = \text{const}$ □



Example 6.12: field strength and potential due to a charged metal sphere

Question 6.8 State whether the two spheres in Example 6.10 are conductors. State what feature of the potential graph supports your answer.

Question 6.9 You might have the experience that your mobile phone signal gets much weaker when you get into an elevator. Explain why this happens.

6.4.3 field strength & potential

for a small displacement Δr in an electric field, change in potential ΔV is

$$\Delta V = \frac{\Delta E_p}{q} \stackrel{\Delta E_p = -W}{=} -\frac{\Delta W}{q} = -\frac{F\Delta r}{q} \stackrel{F=Eq}{=} -E\Delta r \Rightarrow E = -\frac{\Delta V}{\Delta r}$$

as change in displacement $\Delta r \rightarrow 0$, we have $E = -\frac{dV}{dr}$

therefore we have the following theorem:

field strength is negative gradient of potential with respect to displacement

Yuhao Yang 6 静电距球体中心距离 r 处的电场强度为： $E = \frac{Q}{4\pi\epsilon_0 r^2}$ 对于 $r \geq R$ 距球体中心距离 r 处的电势为： $V = \frac{Q}{4\pi\epsilon_0 r}$ 对于球体内的 $r \leq R$ ，即，对于 $r < R$ ，我们有 $E = 0$ 和 $V = \frac{Q}{4\pi\epsilon_0 R} = \text{const}$ 。例 6.12：带电金属球产生的场强和电势问题 6.8 说明例 6.10 中的两个球是否是导体。说明潜在图的哪些特征支持您的答案。问题 6.9 您可能会有这样的经历：一进电梯，手机信号就变弱了。解释为什么会发生这种情况。6.4.3 电场中小位移 Δr 的场强和电势，电势变化 ΔV 为 $\Delta V = \Delta E \cdot q$ $\Delta E \cdot q = -W = -\Delta W = -F \cdot \Delta r = -E \cdot \Delta r$ $E = -\frac{\Delta V}{\Delta r}$ 作为位移 Δr 的变化！0，我们有 $E = -\frac{dV}{dr}$ 因此我们有以下定理：场强是电势相对于位移的负梯度

can also consider change in potential ΔV for large distance due to work done in a field

$$\Delta V = \int dV \stackrel{E=-dV/dr}{=} \int (-)E dr = - \int E dr \quad [37]$$

this gives the inverse relation: $\Delta V = - \int E dr$ [38]

➤ given a $V-r$ graph, gradient of curve gives field strength

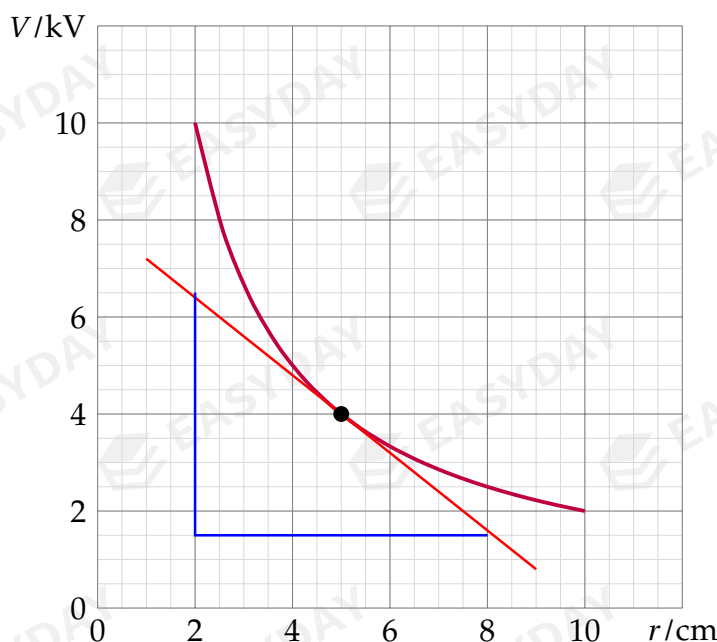
conversely, given a $E-r$ graph, area under curve gives change in potential

➤ can also write $F = -\frac{dU}{dr}$ and $\Delta U = - \int F dr$

force always acts in a direction to lower the potential energy of an object [39]

Example 6.13 The variation of electric potential near a charged object is shown on the graph.

Calculate the electric field strength at 5.0 cm from the centre of the object.



✎ draw tangent to the graph at $r = 5.0$ cm (red line), gradient of tangent gives field strength:

$$\text{gradient} = \frac{\Delta V}{\Delta r} = \frac{(1.5 - 6.5) \times 10^3}{(8.0 - 2.0) \times 10^{-2}} \approx -8.3 \times 10^4 \text{ V m}^{-1} \Rightarrow E = -\frac{\Delta V}{\Delta r} = 8.3 \times 10^4 \text{ V m}^{-1} \quad \square$$

[37] The expression is implicitly integrated from initial position to final position.

[38] These equations are correct if the charge is moving in the parallel direction to the field, i.e., the motion is along the field lines. But an object can move in all directions in the field. More rigorously, if we take the vector nature of electric field into account, we should write $\Delta V = - \int \mathbf{E} \cdot d\mathbf{r}$, and $\mathbf{E} = -\frac{\partial V}{\partial \mathbf{r}}$. (*)

[39] This result can be generalised to a very important principle of physical laws called the *least action principle*. It states that any motion of a system tends to minimise the action, a physical quantity related to the energy of the system. This fundamental law plays a crucial role in the study of theoretical physics. (*)

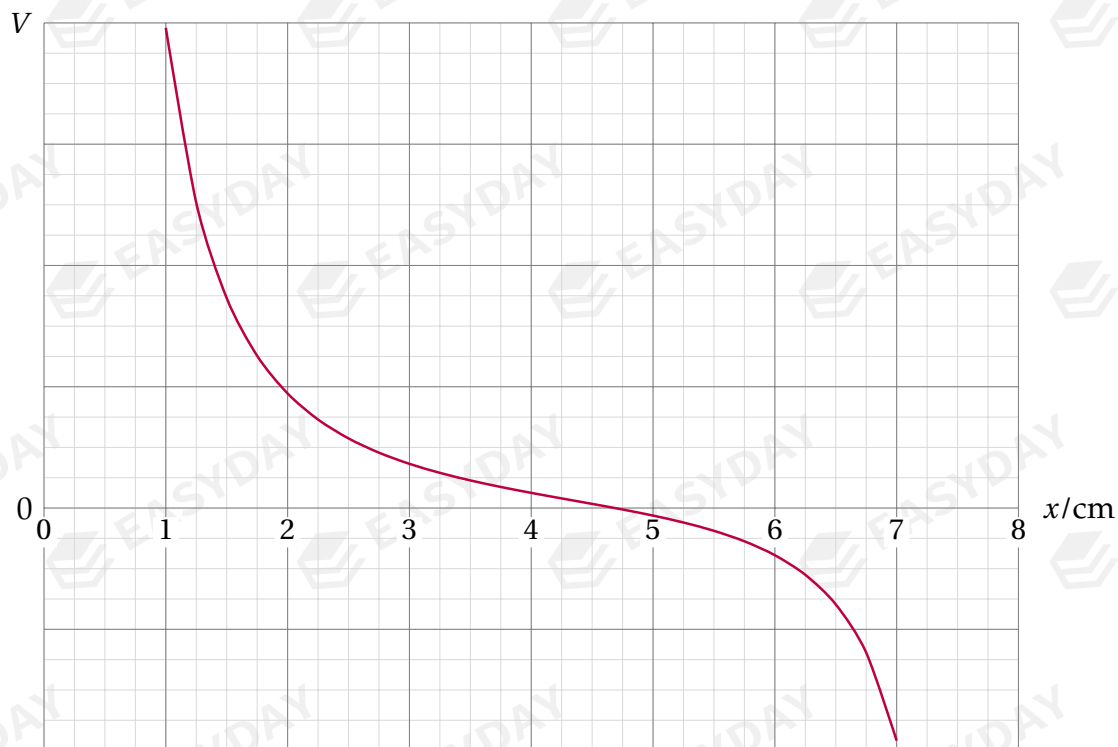
Yuhao Yang 6 静电学还可以考虑由于在场中所做的功而引起的大距离电势 ΔV 的变化 $\Delta V = \int dV = -\int E dr = -\int (-E) dr = \int E dr$ [37] 这给出了反比关系： $\Delta V = -\int E dr$ [38] 给定 $V-r$ 图，曲线梯度给出场强；反之，给定 $aE-r$ 图，曲线下面积给出电势变化 $\hat{a}$ 也可以写 $F = -dU/dr$ 和 $\Delta U = -\int F dr$ 力总是作用在降低物体势能的方向上 [39] 例 6.13 带电物体附近电势的变化如图所示。计算距物体中心 5.0 cm 处的电场强度。

r/cm	V/kV
0.0	2.2
2.4	4.4
6.6	8.8
8.8	10.0
10.0	10.0

在 $r = 5.0$ cm (红线) 处绘制图形的切线，切线梯度给出场强：梯度 $= \Delta V / \Delta r = (1.5 - 6.5) \times 10^3 / (8.0 - 2.0) \times 10^{-2} = -8.3 \times 10^4 \text{ V m}^{-1}$ $E = -\Delta V / \Delta r = 8.3 \times 10^4 \text{ V m}^{-1}$ [37] 该表达式是从初始位置到最终位置的隐式积分。[38] 如果电荷沿与场平行的方向移动，即沿场线运动，则这些方程是正确的。但物体可以在场内向各个方向移动。更严格地说，如果我们考虑电场的矢量性质，我们应该写成 $\Delta V = -\int \mathbf{E} \cdot d\mathbf{r}$ 和 $\mathbf{E} = -\nabla V$ 。(Q) [39] 这个结果可以推广到物理定律中一个非常重要的原理，称为最小作用原理。它指出系统的任何运动都会使作用最小化，作用是与系统能量相关的物理量。这一基本定律在理论物理的研究中起着至关重要的作用。(Q) 70

Question 6.10 Show that the charged object in Example 6.13 behaves like a point charge. Determine the charge it carries, and hence calculate the field strength at $r = 5.0$ cm.

Example 6.14 The variation of electrical potential along a certain line is shown. State and explain where in the field an electron will experience the greatest force.



greatest force means greatest field strength, which means maximum potential gradient

largest gradient of V - x curve at $x = 1$ cm, so greatest force at $x = 1$ cm

□

Example 6.15 electric field due to an isolated point charge

we have learned that the electric potential due to a point charge is: $V = \frac{Q}{4\pi\epsilon_0 r}$

using $E = -\frac{dV}{dr}$, we have $E = -\frac{d}{dr} \left(\frac{Q}{4\pi\epsilon_0 r} \right) = -\frac{Q}{4\pi\epsilon_0} \frac{d}{dr} \left(\frac{1}{r} \right) = \frac{Q}{4\pi\epsilon_0 r^2}$

this agrees with the expression for field strength due to an isolated charge

□

Question 6.11 For the electric field due to a charged metal sphere (see Example 6.12), convince yourself that the field strength equals negative gradient of potential at any point.

Question 6.12 We have seen the statement field strength equals negative potential gradient holds for electric fields. Does it also hold for gravitational fields?

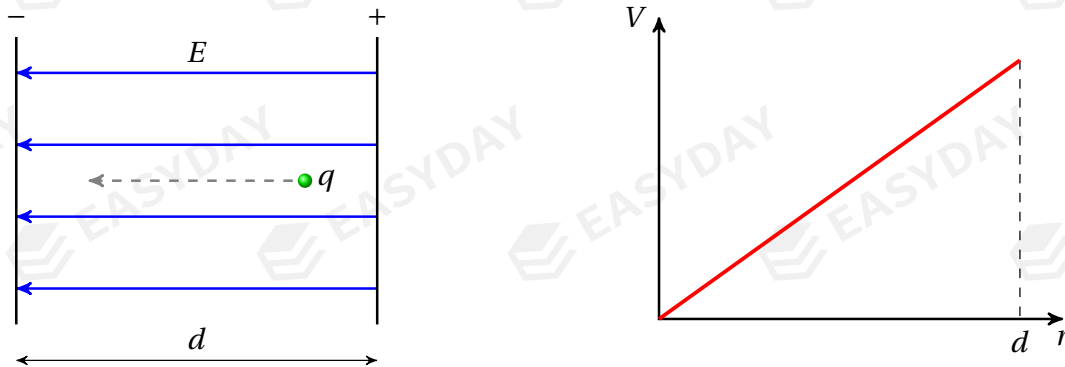
Yuhao Yang 6 静电问题 6.10 证明例 6.13 中的带电物体的行为类似于点电荷。确定它携带的电荷，从而计算 $r = 5.0 \text{ cm}$ 处的场强。例 6.14 显示了沿某条线的电势变化。陈述并解释电子在场中的哪个位置会受到最大的力。图 6.14 显示了 V 随 x 的变化。最大力意味着最大场强，这意味着最大电势梯度 $V-x$ 曲线在 $x = 1 \text{ cm}$ 处的最大梯度，因此 $x = 1 \text{ cm}$ 处的最大力。例 6.15 孤立点电荷产生的电场我们已经知道，点电荷产生的电势为： $V = \frac{Q}{4\pi\epsilon_0 r}$ 使用 $E = -\frac{dV}{dr}$ ，我们有 $E = -\frac{d}{dr} \left(\frac{Q}{4\pi\epsilon_0 r} \right) = \frac{Q}{4\pi\epsilon_0 r^2}$ 这与孤立电荷产生的场强表达式一致。问题 6.11 对于带电金属球产生的电场（见例 6.12），请确信场强等于任意点的负电位梯度。问题 6.12 我们已经看到电场强度等于负电势梯度这一说法。它也适用于引力场吗？

uniform fields revisited

given two oppositely-charged metal plates separated by a distance of d

if p.d. between the plates is V , then electric field strength between is given by $E = \frac{V}{d}$ ^[40]

we will derive this result using the theorem introduced in the last section



moving a test charge in a uniform electric field

moving a test charge q in a uniform field, work done by electric force: $W = Fd = Eqd$

change in P.E.: $\Delta E_p = -W = -Eqd$

change in potential: $\Delta V = \frac{\Delta E_p}{q} = -Ed$, or $E = -\frac{\Delta V}{d}$

plotting V - r graph, gradient of line $= -E$

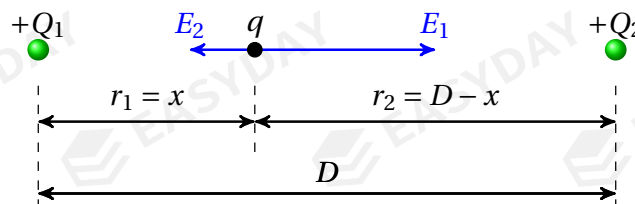
the minus sign means field strength points in the direction such that potential decreases

i.e., electric field acts from high potential to low potential

electric field due to two positive point charges

two point charges $+Q_1$, $+Q_2$ are separated by a distance of D

let's look into the electric field along the segment joining the two charges



combined potential: $V = V_1 + V_2 = \frac{Q_1}{4\pi\epsilon_0 r_1} + \frac{Q_2}{4\pi\epsilon_0 r_2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{x} + \frac{Q_2}{D-x} \right)$

^[40] You should have learned this in AS-level physics.

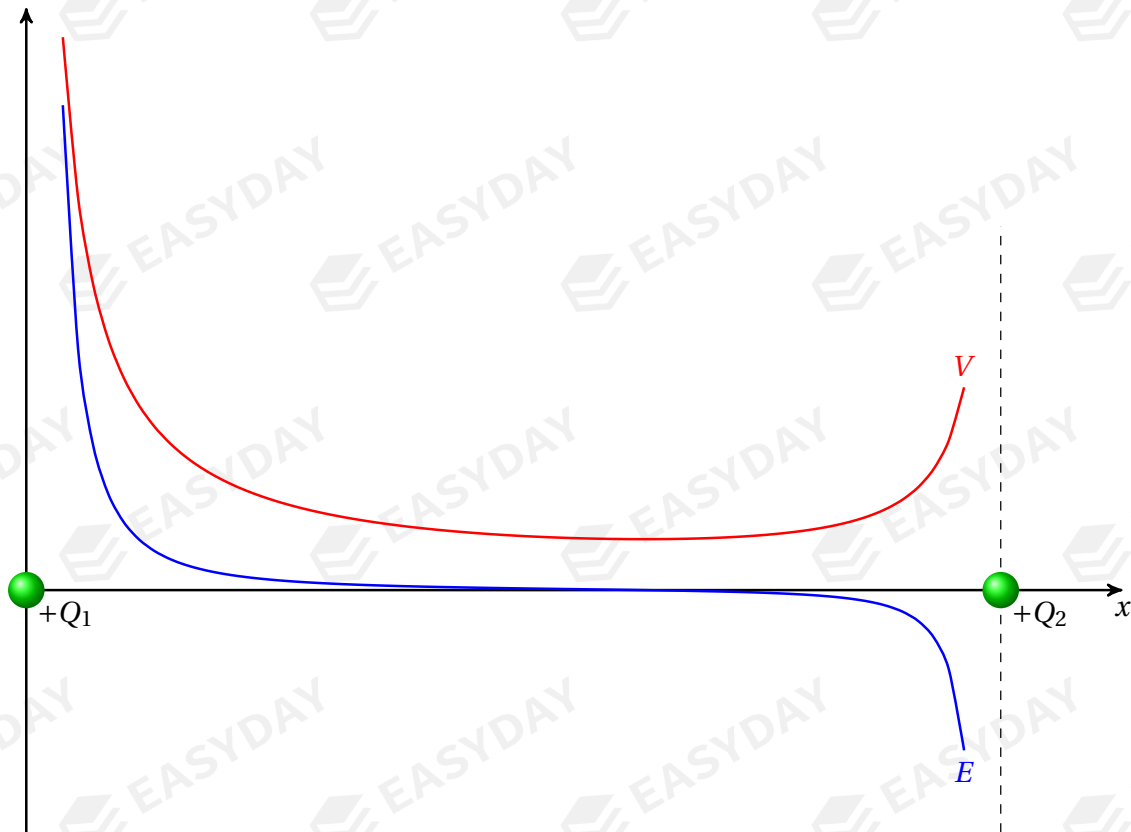
Yuhao Yang 6 重新审视静电均匀场，假设两个带相反电荷的金属板相隔距离 d （如果 p.d.）。板之间的电场强度为 V ，则板之间的电场强度由 $E = V/d$ 给出 [40] 我们将使用上一节中介绍的定理得出此结果 $E = q/d$ 在均匀电场中移动测试电荷在均匀电场中移动测试电荷 q ，由电力完成的功： $W = Fd = Eqd$ P.E. 的变化： $\Delta E_p = -Eqd$ 电势变化： $\Delta V = \Delta E_p/q = -Ed$ ，或 $E = -\Delta V/d$ 绘制 $V-r$ 图形，线的梯度 $= -E$ 负号表示场强指向电势减小的方向，即，由于两个正点电荷，电场从高电势作用到低电势电场两个点电荷 $+Q_1$ 、 $+Q_2$ 为相隔距离 D 让我们看看沿连接两个电荷的线段的电场 $+Q_1 +Q_2$ $D/r_1 = x/r_2 = D-x$ D/E_2 E_1q 组合电势： $V = V_1 + V_2 = Q_1/4\pi\epsilon_0 r_1 + Q_2/4\pi\epsilon_0 r_2 = 1/4\pi\epsilon_0 (Q_1/x + Q_2/D-x)$ [40]你应该在AS级物理中学到这一点。 72

combined field strength: $E = E_1 - E_2 = \frac{Q_1}{4\pi\epsilon_0 r_1^2} - \frac{Q_2}{4\pi\epsilon_0 r_2^2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{x^2} - \frac{Q_2}{(D-x)^2} \right)$

notice that when computing V , we carry out *scalar sum*

but for E , we carry out *vector sum*, i.e., directions of E_1 and E_2 become important

V - x graph and E - x graph for the case where $Q_1 = 3Q_2$ are sketched



Question 6.13 Convince yourself that field strength is indeed given by negative potential gradient. You may interpret it either graphically (think about gradient of tangent along the curve) or algebraically (think about the derivative of V).

Question 6.14 We have looked into the electric field between two positively-charged particles. Discuss the cases where (a) both particles are negatively charged, (b) the two particles carry opposite charges.

Yuhao Yang 6 静电组合场强： $E = E_1 - E_2 = \frac{Q_1}{4\pi\epsilon_0 r_1^2} - \frac{Q_2}{4\pi\epsilon_0 r_2^2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{x^2} - \frac{Q_2}{(D-x)^2} \right)$
) 请注意，在计算 V 时，我们执行标量和，但对于 E ，我们执行矢量和，即对于 $Q_1 = 3Q_2$ 的情况， E_1 和 E_2 的方向变得很重要。 $x \in V + Q_1 + Q_2$ 问题 6.13 说服自己，场强确实是由负电势梯度给出的。您可以用图形方式（考虑沿曲线的切线梯度）或代数方式（考虑 V 的导数）来解释它。问题 6.14 我们研究了两个带正电粒子之间的电场。讨论以下情况：(a) 两个粒子都带负电，(b) 两个粒子带有相反的电荷。 73

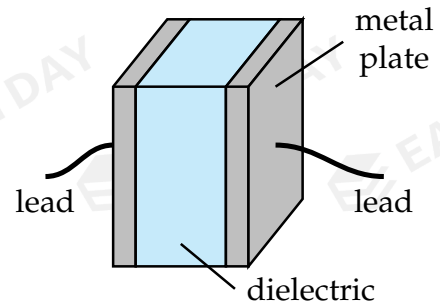
CHAPTER 7

Capacitors

7.1 capacitors: an introduction

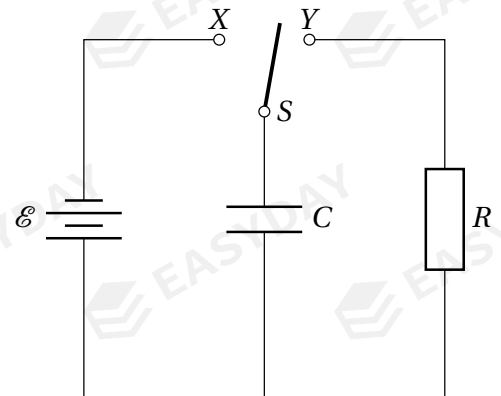
7.1.1 capacitors

capacitors are elementary electrical units widely used in electrical and electronic engineering. A typical capacitor has two conductive plates. Between the plates there is usually an insulating material called *dielectric*.



the circuit symbol for a capacitor is

if we construct an electric circuit as shown, when contact S is moved to X , the capacitor is connected to a voltage supply and becomes charged. Positive and negative charges are separated onto two plates, and they will stay where they are even if we disconnect the capacitor from the voltage supply.



if we then move S to Y , the charged capacitor discharges and drives a current through resistor R , i.e., it can act as a temporary power source^[41]. So capacitors can be used to store and release energy^[42].

^[41] Details on charging and discharging processes will be gone through in §7.4.

^[42] Other important functions of capacitors in electronic circuits include smoothing output voltage of power supplies, blocking direct current while allowing alternating current to pass, etc.

Yuhao Yang 7 电容器第 7 章电容器 7.1 电容器：简介 7.1.1 电容器引线金属板介电电容器是电气和电子工程中广泛使用的基本电气单元。典型的电容器在板之间有两个导电板，通常有一种绝缘材料，称为介质电路。电容器的符号是 E_{RC} YX S 如果我们构造一个电路，如图所示，当触点 S 移动到 X 时，电容器连接到电源并充电，正电荷和负电荷分离到两个板上，即使我们断开电容器与电源的连接，如果我们然后将 S 移动到 Y，充电的电容器放电并驱动电流通过电阻器 R，即它可以充当临时电源[41]，因此电容器可用于存储和释放能量[42] [41]有关充电和放电过程的详细信息将在第 7.4 节中介绍。[42]电容器在电子电路中的其他重要功能包括平滑电源的输出电压、阻止直流电而允许交流电通过等。74

7.1.2 mutual capacitance

to describe ability of a capacitor to store charges, we define the notion of capacitance

(mutual) capacitance of a parallel-plate capacitor is defined as the ratio of the charge stored on one plate to the potential difference across the two plates.

in a word equation: mutual capacitance $C = \frac{\text{charge on one plate } Q}{\text{p.d. } V \text{ across the plates}} \Rightarrow C = \frac{Q}{V}$

➤ unit of capacitance: **farad** ^[43] : $[C] = \text{F}$

farad a derived unit: $1 \text{ F} = 1 \text{ C} \cdot \text{V}^{-1}$

farad is a large unit, more common subunits of capacitance in use are sub-multiples of farad:

$$1 \mu\text{F} = 10^{-6} \text{ F}, \quad 1 \text{ nF} = 10^{-9} \text{ F}, \quad 1 \text{ pF} = 10^{-12} \text{ F}$$

➤ for a parallel-plate capacitor, charges on the two plates are equal but opposite

net charge on the capacitor: $Q_{\text{net}} = (+Q) + (-Q) = 0$

so we should emphasise on the notion of charge on *one* plate in the definition

➤ capacitance depends on *geometry* of the device and permittivity of the dielectric material

capacitance does not depend on electric field or potential^[44]

for example, capacitance between two metal plates is: $C = \frac{\epsilon_0 A}{d}$ ^[45]

A is area of plate, d is distance between plates, both are geometrical quantities

7.1.3 self-capacitance

there are two closely related notions of capacitance: *mutual* capacitance and *self* capacitance

the definition for capacitance given in the previous section, is actually *mutual* capacitance

on the other hand, all bodies are able to store electrical charge

^[43] The unit is named after Michael Faraday, a British physicist who developed the concept of capacitance. Faraday's other main discoveries include electromagnetic induction and electrolysis. He established the basis for the concept of the electromagnetic field in physics.

^[44] Recall the resistance of an electrical component. Resistance is defined as the ratio of p.d. to current, but the value of resistance is essentially dependent on the length, cross-sectional area and material of the component, instead of the p.d. applied or the current flowing through it.

^[45] If there is *dielectric* in between, the formula should be rewritten as $C = \frac{\epsilon A}{d}$, where ϵ is permittivity of dielectric. These formulae are not examinable by the syllabus.

Yuhao Yang 7 电容器 7.1.2 互电容描述电容器存储电荷的能力，我们定义电容（互）电容的概念平行板电容器的电容定义为一个板上存储的电荷与两个板之间的电势差之比。简而言之：互电容 $C = \frac{Q}{V}$ 跨极板) $C = \frac{Q}{V}$ 电容单位：法拉 [43] : $[C] = F$ 法拉派生单位： $1 F = 1 C \cdot V^{-1}$ 法拉是一个大单位，更常见的电容单位是法拉的约数： $1 F = 10^{-6} F$ 、 $1 nF = 10^{-9} F$ 、 $1 pF = 10^{-12} F$ 对于平行板电容器，两个板上的电荷相等，但电容器上的净电荷相反： $Q_{net} = (+Q) + (-Q) = 0$ 因此我们应该强调定义中一个板上电荷的概念 电容取决于器件的几何形状，介电材料电容的介电常数不依赖于电场或电势 [44] 例如，两个金属板之间的电容为： $C = \epsilon_0 \frac{A}{d}$ [45] A 是极板面积， d 是极板之间的距离，两者都是几何量。 7.1.3 自电容电容有两个密切相关的概念：互电容和自电容。上一节给出的电容定义实际上是互电容，所有物体都能够存储电荷 [43] 该单位以提出电容概念的英国物理学家迈克尔·法拉第 (Michael Faraday) 命名。法拉第的其他主要发现包括电磁感应和电解。他奠定了物理学中电磁场概念的基础。 [44] 回想一下电气元件的电阻。电阻定义为 $p.d.$ 的比率。电流，但电阻值本质上取决于元件的长度、横截面积和材料，而不是 $p.d.$ 应用或流过它的电流。 [45] 如果中间有电介质，则公式应重写为 $C = \epsilon \frac{A}{d}$ ，其中 ϵ 是电介质的介电常数。这些公式不受教学大纲的检验。 75

any object that can be electrically charged exhibits capacitance

we define **self-capacitance** of an object as the amount of charge that must be added to increase per unit electrical potential

in a word equation, self capacitance $C = \frac{\text{charge of object } Q}{\text{electric potential of object } V} \Rightarrow C = \frac{Q}{V}$ [46]

Example 7.1 Self-capacitance of a charged metal sphere in a vacuum

consider a metal sphere of radius R and carries an electric charge of Q

its electric potential: $V = \frac{Q}{4\pi\epsilon_0 R}$

self-capacitance of the sphere: $C_{\text{sphere}} = \frac{Q}{V} = Q \times \frac{4\pi\epsilon_0 R}{Q} \Rightarrow C_{\text{sphere}} = 4\pi\epsilon_0 R$

note that capacitance is only dependent on its geometrical property (radius R) □

Example 7.2 A conducting sphere of radius 1.0 m is situated in free space. (a) Find its capacitance. (b) In order to raise its potential to 5000 V, find the amount of charge needed.

capacitance of the sphere: $C = 4\pi\epsilon_0 R = 4\pi \times 8.85 \times 10^{-12} \times 1.0 \approx 1.11 \times 10^{-10} \text{ F}$

(here you can see farad being an impractically huge unit)

charge on sphere: $Q = CV = 1.11 \times 10^{-10} \times 5000 \approx 5.56 \times 10^{-7} \text{ C}$ □

analogy with ideal gases

an interesting analogy can be made between capacitors and ideal gases

recall an ideal gas is described by equation $pV = nRT$ [47]

compare $\begin{cases} \text{amount of charge: } Q = CV \\ \text{amount of substance: } n = \left(\frac{V}{RT}\right)p \end{cases}$

volume V of a container has a certain space *capacity*, at fixed T , pumping more gas (increase n) into system, pressure p increases

similarly, a capacitor has a certain charge capacity, adding more charge Q increases p.d. V , so the quantity C is naturally called *capacitance*

also for a container, there exists a maximum pressure which it can withstand

[46] For either mutual capacitance or self capacitance, defining equation $C = \frac{Q}{V}$ takes the same form, but you should keep in mind that Q and V represent different things in different contexts.

[47] Don't confuse voltage V with volume V !

Yuhao Yang 7 电容器任何可以带电的物体都表现出电容，我们将物体的自电容定义为为增加每单位电势而必须添加的电荷量，用一个词方程表示，自电容 $C = \text{物体的电荷 } Q / \text{物体的电势 } V$) $C = Q / V$ [46] 示例 7.1 真空中带电金属球的自电容考虑半径为 R 的金属球，并携带电荷 Q 其电势： $V = Q / 4\pi\epsilon_0 R$ 球体的自电容： $C_{\text{sphere}} = Q / V = Q \times 4\pi\epsilon_0 R / Q$) $C_{\text{sphere}} = 4\pi\epsilon_0 R$ 请注意，电容仅取决于其几何特性（半径 R ）

示例 7.2 半径为 1.0 m 的导电球体位于自由空间中。(a) 求出其电容。(b) 为了将其电势提高到 5000 V ，求所需的电荷量。

- 球体电容： $C = 4\pi\epsilon_0 R = 4\pi \times 8.85 \times 10^{-12} \times 1.0 \text{ } \frac{1}{\text{V}} = 1.11 \times 10^{-10} \text{ F}$ （在这里您可以看到法拉是一个不切实际的巨大单位）球体电荷： $Q = CV = 1.11 \times 10^{-10} \times 5000 \text{ } \frac{1}{\text{V}} = 5.56 \times 10^{-7} \text{ C}$ 与理想气体的类比电容器和理想气体之间可以进行有趣的类比回忆理想气体由方程 $pV = nRT$ [47] 描述比较 8

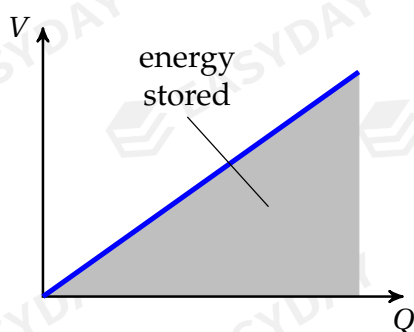
>><>>: 电荷量: $Q = CV$ 物质质量: $n = (V / RT) p$ 容器的体积 V 有一定的空间容量，在固定 T 下，将更多气体（增加 n ）泵入系统，压力 p 类似地增加，电容器有一定的电荷容量，增加更多的电荷 Q 会增加 p 。d。 V ，因此量 C 自然也称为容器的电容，存在它可以承受的最大压力 [46] 对于互电容或自电容，定义方程 $C = Q / V$ 采用相同的形式，但您应该记住 Q 和 V 在不同的上下文中代表不同的事物。[47] 不要将电压 V 与体积 V 混淆！ 76

for a capacitor, there exists a *breakdown voltage*, or *withstand voltage*, beyond which there could be sparking across the capacitor

7.2 energy stored in a capacitor

to charge a capacitor, need to push electrons off one plate and onto the other

separation of positive and negative charges requires work done $\Rightarrow$ energy is stored



since charge Q varies with p.d. V , we shall use the V - Q graph to find work done W

area under V - Q graph is equal to work done W

$$W = \frac{1}{2} QV$$

substitute $Q = CV$, energy stored in capacitor is^[48]:

$$W = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

Example 7.3 When the p.d. across a capacitor of 1.8×10^{-4} F is increased from 10 V to 20 V, how much additional energy is stored?

energy change: $\Delta W = W_f - W_i = \frac{1}{2} CV_f^2 - \frac{1}{2} CV_i^2 = \frac{1}{2} \times 1.8 \times 10^{-4} \times (20^2 - 10^2) = 2.7 \times 10^{-2}$ J $\square$

Question 7.1 A capacitor of 2500 μ F is charged to a working voltage of 18 V. (a) What is the magnitude of positive charge on its plate? (b) What is the energy stored?

Question 7.2 A capacitor initially charged to a potential difference of 16 V discharges and loses 40% of its energy. What is its new p.d.?

Question 7.3 For an isolated metal sphere of radius 30 cm situated in vacuum, what is the electric potential energy stored when charged to a potential of 120 kV?

^[48] Rigorously speaking, this electrical potential energy is stored within the *electric fields* between metal plates of capacitor.

Yuhao Yang 7 电容器对于电容器，存在击穿电压或耐受电压，超过该电压，电容器上可能会产生火花 7.2 存储在电容器中的能量为电容器充电，需要将电子从一个极板推到另一极板上正负电荷分离需要做功) 能量被存储 VQ 能量存储，因为电荷 Q 随 p.d. 变化。 V 时，我们将使用 $V-Q$ 图求出 $V-Q$ 图下的做功 W 面积等于做功 $W = \int_0^Q V dQ$ 替代 $Q = CV$ ，电容器中存储的能量为 [48]： $W = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$ 例 7.3 当 p.d. 将 $1.8 \times 10^{-4} \text{ F}$ 的电容器从 10 V 增加到 20 V ，会存储多少额外能量？ - 能量变化： $\Delta W = W_f - W_i = \frac{1}{2} CV_f^2 - \frac{1}{2} CV_i^2 = \frac{1}{2} \times 1.8 \times 10^{-4} \times (20^2 - 10^2) = 2.7 \times 10^{-2} \text{ J}$ 问题 7.1 将 2500 F 的电容器充电至 18 V 的工作电压。(a) 极板上正电荷的大小是多少？ (b) 储存的能量是多少？ 问题 7.2 最初充电至 16 V 电位差的电容器放电并损失 40% 的能量。它的新 p.d. 是什么？ 问题 7.3 对于真空中半径为 30 cm 的孤立金属球，充电至 120 kV 时储存的电势能是多少？ [48] 严格来说，这种电势能存储在电容器金属板之间的电场内。 77

7.3 capacitor networks

7.3.1 capacitors in parallel

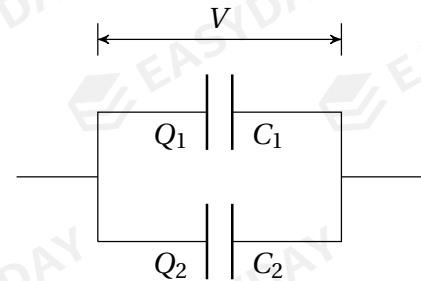
consider two capacitors connected in parallel

same p.d. V across the network: $V = V_1 = V_2$

but charge Q is shared: $Q_{\text{total}} = Q_1 + Q_2$

$$\frac{Q_{\text{total}}}{V} = \frac{Q_1}{V} + \frac{Q_2}{V}$$

$$C_{\text{total}} = C_1 + C_2$$



if three or more capacitors in parallel: $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$ $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$

➤ adding extra capacitor in parallel to a network, total capacitance will increase

explanation: when several capacitors connected in parallel, equivalent to a single capacitor with larger plates, so more charge on the plates $\Rightarrow C \uparrow$

Example 7.4 A capacitor with capacitance C_0 is charged to a p.d. V_0 . It is disconnected from the power supply, and then connected across an identical capacitor. Discuss the change in p.d., and change in energy stored in the system.

✍ initial charge $Q = C_0 V_0$, initial energy stored $W_0 = \frac{1}{2} C V_0^2$

combined capacitance: $C = C_0 + C_0 = 2C_0$

charge is conserved, so final p.d. across: $V = \frac{Q}{C} = \frac{C_0 V_0}{2C_0} = \frac{1}{2} V_0$

charge shared between capacitors, the first capacitor loses half its charge and p.d.

final energy stored in system: $W = \frac{1}{2} C V^2 = \frac{1}{2} \times 2C_0 \times \left(\frac{1}{2} V_0\right)^2 = \frac{1}{4} C_0 V_0^2 \Rightarrow W = \frac{1}{2} W_0$

half of stored energy is lost as heat when electrons flow between two capacitors □

Question 7.4 A capacitor A of capacitance C and a second capacitor B of capacitance $3C$ are connected in parallel. If a voltage is applied across the network, what is the ratio of energy stored in A to that in B ? What about the ratio of electric charge?

Yuhao Yang 7 电容器 7.3 电容器网络 7.3.1 并联电容器 C_1, C_2, V 考虑并联相同 p.d. 的两个电容器。网络上的 $V: V = V_1 = V_2$ ，但电荷 Q 是共享的： $Q_{\text{total}} = Q_1 + Q_2$ $Q_{\text{total}} V = Q_1 V + Q_2 V$ $C_{\text{total}} = C_1 + C_2$ 如果三个以上电容器并联： $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$ $Q_{\text{total}} = Q_1 + Q_2 + Q_3 + \dots$ 在网络中并联额外的电容器，总电容会增加说明：当多个电容器并联时，相当于单个电容器的极板较大，因此极板上的电荷较多) C 1 示例 7.4 电容为 C_0 的电容器被充电至 p.d. V_0 。它与电源断开，然后连接到一个相同的电容器上。讨论 p.d. 的变化以及系统中存储的能量的变化。 - 初始电荷 $Q = C_0 V_0$ ，初始存储能量 $W_0 = \frac{1}{2} C_0 V_0^2$ 组合电容： $C = C_0 + C_0 = 2C_0$ 电荷守恒，因此最终 p.d. 跨接： $V = \frac{Q}{C} = \frac{C_0 V_0}{2C_0} = \frac{1}{2} V_0$ 电容器之间共享电荷，第一个电容器失去一半电荷，p.d. 系统中存储的最终能量： $W = \frac{1}{2} C V^2 = \frac{1}{2} \times 2C_0 \times \left(\frac{1}{2} V_0\right)^2 = \frac{1}{4} C_0 V_0^2$ $W = \frac{1}{2} W_0$ 当电子在两个电容器之间流动时，存储能量的一半以热量形式损失。问题 7.4 电容 C 的电容器 A 和电容 $3C$ 的第二个电容器 B 并联。如果在网络上施加电压，A 中存储的能量与 B 中存储的能量之比是多少？那么电荷的比例呢？ 78

7.3.2 capacitors in series

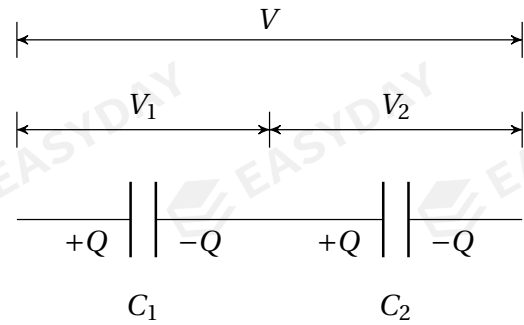
consider next two series capacitors

same charge Q on each plate: $Q = Q_1 = Q_2$ ^[49]

p.d. is shared: $V_{\text{total}} = V_1 + V_2$

$$\frac{V_{\text{total}}}{Q} = \frac{V_1}{Q} + \frac{V_2}{Q}$$

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2}$$



if three or more capacitors in series:

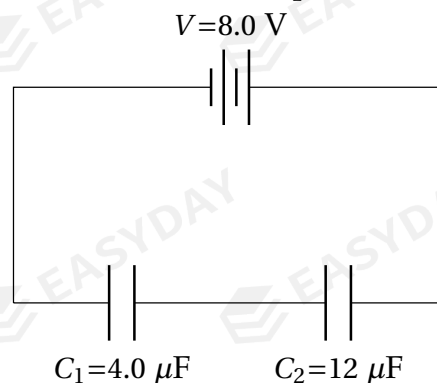
$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

$$V_{\text{total}} = V_1 + V_2 + V_3 + \dots$$

➤ adding extra capacitor in series to a network, total capacitance will decrease

explanation: when several capacitors connected in series, equivalent to a parallel-plate capacitor with greater separation, so more charge on the plates $\Rightarrow C \searrow$

Example 7.5 For the circuit shown below, find the p.d. across each of the capacitor.



✎ using result for combined capacitance, we have: $C_{\text{total}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{1}{4.0} + \frac{1}{12} \right)^{-1} = 3.0 \mu\text{F}$

charge for the network: $Q = C_{\text{total}} V_{\text{total}} = 3.0 \times 8.0 = 24 \mu\text{C}$

series network so all capacitors have same Q , so: $Q_1 = Q_2 = 24 \mu\text{C}$

p.d. across each individual capacitor: $V_1 = \frac{Q_1}{C_1} = \frac{24}{4.0} = 6.0 \text{ V}$, $V_2 = \frac{Q_2}{C_2} = \frac{24}{12} = 2.0 \text{ V}$

alternatively, we can use properties of series network: $Q_1 = Q_2$ and $V_{\text{total}} = V_1 + V_2$

we can solve simultaneous equations:
$$\begin{cases} 4.0V_1 = 12V_2 \\ V_1 + V_2 = 8.0 \end{cases} \Rightarrow \begin{cases} V_1 = 6.0 \text{ V} \\ V_2 = 2.0 \text{ V} \end{cases} \quad \square$$

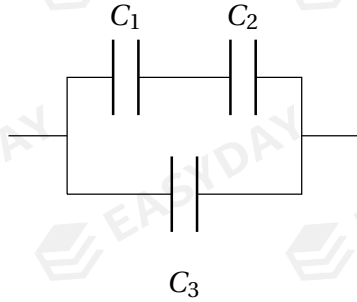
^[49] Initially, the H-shaped isolated section between capacitors is uncharged. Since no charge can enter or leave this section, its net charge must remain zero, so same charge Q on each plate of series capacitors.

Yuhao Yang 7 电容器 7.3.2 串联电容器 C_1 C_2 $-Q+Q$ $-Q+Q$ V_1 V_2 V 考虑接下来的两个串联电容器，每个板上的电荷 Q 相同： $Q = Q_1 = Q_2$ [49] p.d.共享： $V_{\text{total}} = V_1 + V_2$ $V_{\text{total}} Q = V_1 Q + V_2 Q$ $C_{\text{total}} = \frac{1}{C_1} + \frac{1}{C_2}$ 如果串联三个以上电容器： $\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$ $V_{\text{total}} = V_1 + V_2 + V_3 + \dots$ 添加额外说明：当几个电容器串联到网络中时，相当于一个平行板电容器，其间距更大，因此板上的电荷更多） $C\%$ 示例 7.5 对于下面所示的电路，求 p.d. 跨过每个电容器。 $C_1=4.0$ (F) $C_2=12$ (F) $V=8.0$ V - 使用组合电容的结果，我们得到： $C_{\text{total}} = (\frac{1}{C_1} + \frac{1}{C_2})^{-1} = (\frac{1}{4.0} + \frac{1}{12})^{-1} = 3.0$ (F) 网络电荷： $Q = C_{\text{total}} V_{\text{total}} = 3.0 \times 8.0 = 24$ (C) 串联网络，因此所有电容器具有相同的 Q ，因此： $Q_1 = Q_2 = 24$ (C) p.d. 每个电容器两端： $V_1 = \frac{Q}{C_1} = \frac{24}{4.0} = 6.0$ V， $V_2 = \frac{Q}{C_2} = \frac{24}{12} = 2.0$ V 或者，我们可以使用串联网络的属性： $Q_1 = Q_2$ 和 $V_{\text{total}} = V_1 + V_2$ 我们可以求解联立方程： $8.0 = \frac{Q}{4.0} + \frac{Q}{12}$ $V_1 + V_2 = 8.0$ $8.0 = \frac{Q}{4.0} + \frac{Q}{12}$ $8.0 = \frac{3Q + Q}{12}$ $8.0 = \frac{4Q}{12}$ $8.0 = \frac{Q}{3}$ $Q = 24$ $V_1 = \frac{24}{4.0} = 6.0$ V $V_2 = \frac{24}{12} = 2.0$ V [49] 最初，电容器之间的 H 形隔离部分不带电。由于没有电荷可以进入或离开该部分，因此其净电荷必须保持为零，因此串联电容器的每个板上的电荷 Q 相同。

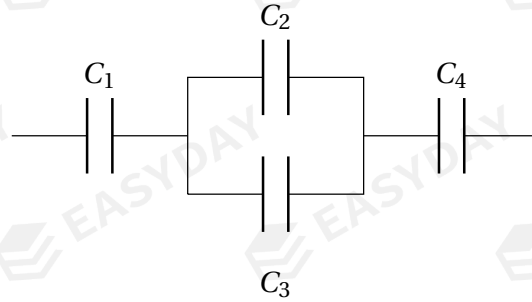
7.3.3 capacitor networks

more complicated capacitor networks can be considered as a combination of some smaller networks with capacitors in parallel or in series

Example 7.6 $C_1 = C_2 = C_3 = C_4 = 10 \mu\text{F}$, calculate the capacitance of the network (a) and (b).



(a)



(b)

$$(a) \quad C_{12} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \left(\frac{1}{10} + \frac{1}{10} \right)^{-1} = 5.0 \mu\text{F}$$

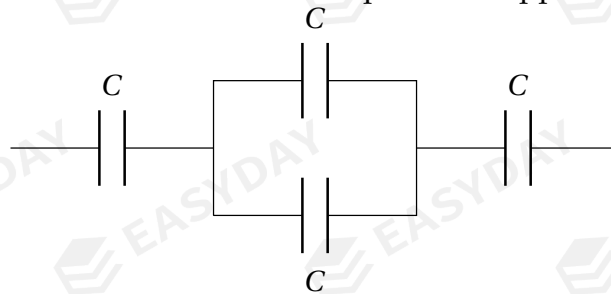
$$C_{\text{total}} = C_{12} + C_3 = 5 + 10 = 15 \mu\text{F}$$

$$(b) \quad C_{23} = C_2 + C_3 = 10 + 10 = 20 \mu\text{F}$$

$$C_{\text{total}} = \left(\frac{1}{C_1} + \frac{1}{C_{23}} + \frac{1}{C_4} \right)^{-1} = \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{10} \right)^{-1} = 4.0 \mu\text{F}$$

□

Example 7.7 Four identical capacitors are arranged as shown. Each capacitor can withstand a maximum p.d. of 12 V, what is the maximum safe p.d. to be applied between the terminals?



suppose charge on the leftmost capacitor is Q , then charge of rightmost capacitor is also Q

but for the two capacitors in parallel, charge is shared, so each has charge $\frac{Q}{2}$

p.d. across the parallel network is half of the p.d. across the two capacitors near the ends

so maximum p.d. across the terminals: $V_{\text{max}} = 12 + 6 + 12 = 30 \text{ V}$

□

Question 7.5 Using at most four capacitors of $24 \mu\text{F}$, design a network that has a combined capacitance of (a) $72 \mu\text{F}$, (b) $8 \mu\text{F}$, (c) $36 \mu\text{F}$, and (d) $18 \mu\text{F}$.

Yuhao Yang 7 电容器 7.3.3 电容器网络更复杂的电容器网络可以被认为是一些较小的网络与电容器并联或串联的组合。例 7.6 $C_1 = C_2 = C_3 = C_4 = 10 \text{ (F)}$ ，计算网络 (a) 和 (b) 的电容。

(a) $C_{12} = (1/C_1 + 1/C_2)^{-1} = (1/10 + 1/10)^{-1} = 5.0 \text{ (F)}$ 总计 $= C_{12} + C_3 = 5 + 10 = 15 \text{ (F)}$

(b) $C_{23} = C_2 + C_3 = 10 + 10 = 20 \text{ (F)}$ $C_{\text{total}} = (1/C_1 + 1/C_{23} + 1/C_4)^{-1} = (1/10 + 1/20 + 1/10)^{-1} = 4.0 \text{ (F)}$

示例 7.7 四个相同的电容器如图所示排列。每个电容器可承受最大 p.d. 12 V 的最大安全 p.d. 是多少？应用于终端之间？

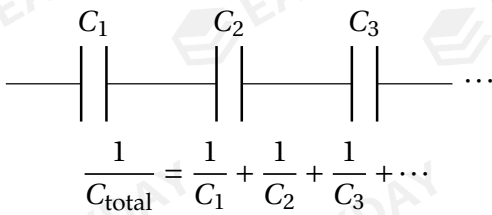
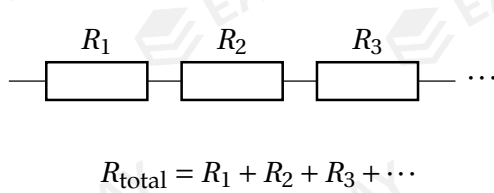
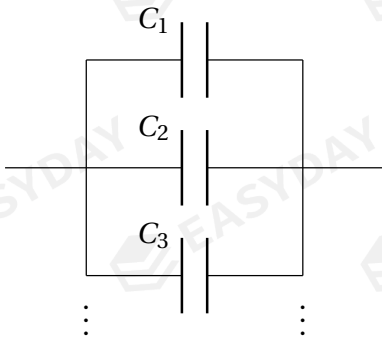
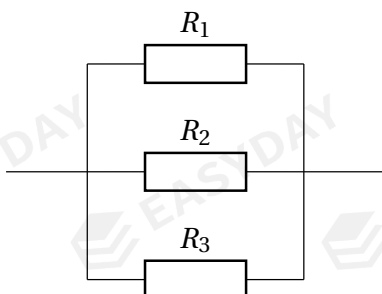
$C \ C \ C \ C$ - 假设最左边电容器的电荷是 Q ，那么最右边电容器的电荷也是 Q ，但是对于并联的两个电容器，电荷是共享的，因此每个电容器都有电荷 $Q/2$ p.d. 并行网络上的 p.d. 的一半。两端附近的两个电容器之间的最大 p.d. 端子之间： $V_{\text{max}} = 12 + 6 + 12 = 30 \text{ V}$

问题 7.5 最多使用四个 24 (F) 电容器，设计一个组合电容为 (a) 72 (F) 、(b) 8 (F) 、(c) 36 (F) 和 (d) 18 (F) 的网络。

Question 7.6 If the two networks in Example 7.6 are both connected to a supply voltage of 15 V, determine the p.d. across each individual capacitor.

7.3.4 capacitors & resistors

as a quick review, we compare capacitor networks with resistor networks

	capacitors	resistors
in series	same charge  $\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$	same current  $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$
in parallel	same p.d. across  $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$	same p.d. across  $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

Yuhao Yang 7 电容器问题 7.6 如果例 7.6 中的两个网络都连接到 15 V 的电源电压，请确定 p.d。跨每个单独的电容器。7.3.4 电容器和电阻器作为快速回顾，我们将电容器网络与电阻网络进行比较电容器电阻器串联相同充电相同电流 $C_1 C_2 C_3 \dots R_1 R_2 R_3 \dots$ $\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$ $\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$ 并联相同 p.d. 跨越同一个 p.d. $C_1 C_2 C_3 \dots R_1 R_2 R_3 \dots$ $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$ $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$

7.4 charging & discharging capacitors

in this section, we will investigate how the p.d. across a capacitor changes with time when it is being charged, and we will also look into discharging processes

7.4.1 charging phase

initial state: no charge in capacitor: $Q(0) = 0, V_C(0) = 0$

at any instant: $dQ = CdV_C$

current in circuit: $I = \frac{V_R}{R} = \frac{\mathcal{E} - V_C}{R}$

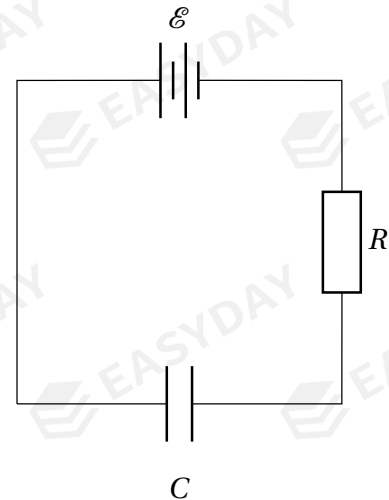
change of charge: $dQ = Idt = \frac{\mathcal{E} - V_C}{R} dt = CdV_C$

$$\frac{dt}{RC} = \frac{dV_C}{\mathcal{E} - V_C}$$

$$\int_0^t \frac{dt}{RC} = \int_0^{V_C} \frac{dV_C}{\mathcal{E} - V_C}$$

$$\frac{t}{RC} \Big|_0^t = -\ln(\mathcal{E} - V_C) \Big|_0^{V_C}$$

simplify everything, we get: $V_C = \mathcal{E} (1 - e^{-t/RC})$



- p.d. of capacitor increases at a decreasing rate when it is being charged

as electric charges are separated onto the two plates, pushing more $+Q/-Q$ onto $+ve/-ve$ plate requires more work done to overcome the repulsion $\Rightarrow$ increase in p.d. slows down

- p.d of capacitor eventually tends to the battery e.m.f.

charge will continue to flow if there exists a potential difference

when $V_C = \mathcal{E}$, no charge flow, hence charging current gradually drops to zero

7.4.2 discharging phase

capacitor initially charged with $Q(0) = Q, V_C(0) = V_0$

at any instant, $dQ = -CdV_C$ (minus sign because charge decreases during discharging)

Yuhao Yang 7 电容器 7.4 电容器的充电和放电在本节中，我们将研究 p.d. 是如何变化的。电容器在充电时随时间变化，我们还将研究放电过程 7.4.1 充电阶段 E R C 初始状态：电容器不充电： $Q(0) = 0, V_C(0) = 0$ 任意时刻： $dQ = C dV_C$ 电路中电流： $I = \frac{V_R}{R} = \frac{E - V_C}{R}$ 电荷变化： $dQ = I dt = \frac{E - V_C}{R} dt = C dV_C$
 $\frac{dV_C}{dt} = \frac{E - V_C}{RC}$ $\int_0^t \frac{dV_C}{E - V_C} = \int_0^t \frac{1}{RC} dt$ $-\ln(E - V_C) = \frac{t}{RC} - \ln(E - V_C(0))$ 简化一切，我们得到： $V_C = E - (E - V_C(0))e^{-t/RC}$ p.d. 当电容器充电时，随着电荷被分离到两个板上，电容器的电压以递减的速率增加，将更多的 $+Q/-Q$ 推到 $+ve/-ve$ 板上需要做更多的工作来克服斥力) p.d 的增加。减慢电容器的 p.d 最终趋于电池电动势。如果 $V_C = E$ 时存在电势差，则电荷将继续流动，没有电荷流动，因此充电电流逐渐降至零 7.4.2 放电阶段电容器最初以 $Q(0) = Q, V_C(0) = V_0$ 充电，在任何时刻， $dQ = -C dV_C$ (负号，因为放电过程中电荷减少) 82

also $V_C = V_R$ because C and R are in parallel

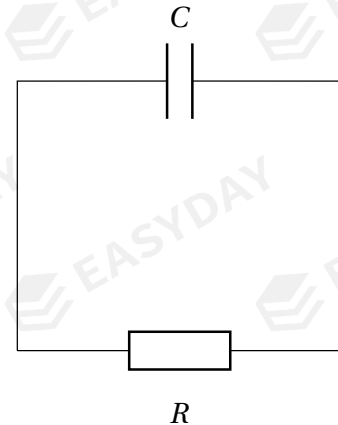
charge change: $dQ = Idt = \frac{V_C}{R} dt = -CdV_C$

$$-\frac{dt}{RC} = \frac{dV_C}{V_C}$$

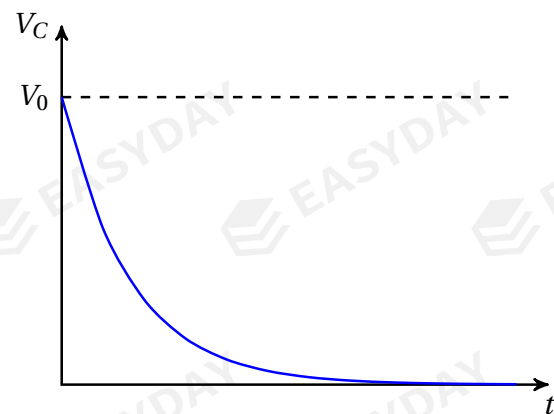
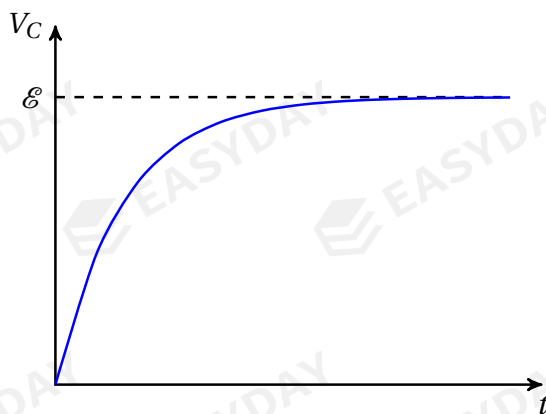
$$-\int_0^t \frac{dt}{RC} = \int_{V_0}^{V_C} \frac{dV_C}{V_C}$$

$$-\frac{t}{RC} \Big|_0^t = \ln(V_C) \Big|_{V_0}^{V_C}$$

simplify everything, we get: $V_C = V_0 e^{-t/RC}$



- p.d. of capacitor gradually drops to zero during discharging
- discharging current also gradually approaches zero



7.4.3 time constant

$\tau \equiv RC$ is called **time constant**, which determines charging and discharging rate of a capacitor

$R \nearrow \Rightarrow$ smaller charging/discharging current $\Rightarrow$ takes longer to charge/discharge

$C \nearrow \Rightarrow$ more charge to be charged/discharged $\Rightarrow$ takes longer

charging or discharging of capacitors is not instantaneous, there is always a certain delay

as a rule of thumb, after a time $t = 3 \sim 5\tau$, charging or discharging is almost complete

Yuhao Yang 7 电容器 C R 也 $V_C = V_R$ 因为 C 和 R 是并联的电荷变化： $dQ = I dt = V_C R dt = -C dV_C$
 $- dt RC = dV_C V_C - Z t 0 dt RC = Z V_C V_0 dV_C V_C - t RC \int_{t_0}^t dt = \ln(V_C / V_0)$ 简化一切，我们得到： $V_C = V_0 e^{-t / RC}$ p.d. 放电过程中电容器的容量逐渐降至零 放电电流也逐渐接近于零 $V_C E t$ 电容器的充电阶段 $V_C V_0 t$ 电容器的放电阶段 7.4.3 时间常数 $\tau = RC$ 称为时间常数，它决定电容器的充电和放电速率 R 1) 较小的充电 / 放电电流) 需要较长的充电 / 放电时间 C 1) 较多的充电 / 放电时间) 需要较长的充电或放电时间 电容器不是瞬时的，根据经验，总会有一定的延迟，经过时间 $t = 3 \gg 5\tau$ 后，充电或放电几乎完成 83

CHAPTER 8

Magnetic Fields

8.1 magnetism

8.1.1 magnets

magnetic effects are commonly seen in *magnets*

a magnet creates a magnetic field that attracts or repels other magnets

➤ polarity of magnets

a magnet has two poles, the *north pole* and the *south pole*

freely suspend a magnet, the north pole points towards earth's geographic north pole

when two magnets are brought near each other, like poles repel, and opposite poles attract

➤ we use **magnetic field lines** to graphically represent how magnetic field permeate space

by convention, fields lines emerge from north pole and go into south pole

density of field lines shows strength of magnetic field

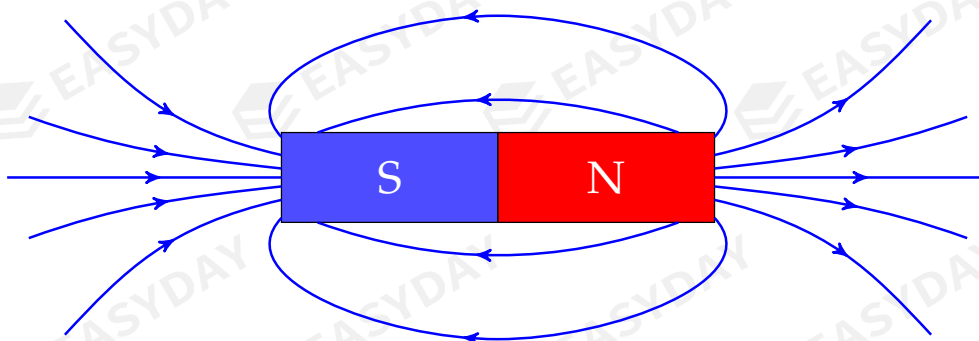
direction of field line tells how the north pole of a small compass will line up at that point

➤ strength of the field is described by a quantity called **magnetic flux density**, denoted by B

when we draw field lines, we are actually drawing the pattern of flux density B

the notion of flux density will be defined later in details in §8.2.2

Example 8.1 magnetic field around a bar magnet



Yuhao Yang 8 磁场第 8 章磁场 8.1 磁力 8.1.1 磁铁磁效应常见于磁铁中磁铁产生吸引或排斥其他磁铁的磁场 - 磁铁的极性磁铁有两个磁极，北极和南极自由悬挂磁铁，当两个磁铁彼此靠近时，北极指向地球的地理北极，就像两极相斥一样，相反磁极吸引。我们使用磁场线来图形化地表示磁场如何按照惯例渗透到空间，磁场线从北极出现并进入南极。磁场线的密度显示磁场的强度。磁场线的方向告诉小罗盘的北极将如何在该点对齐。磁场的强度由称为磁通密度的量来描述，当我们绘制磁场时用 B 表示线，我们实际上是在绘制磁通密度 B 的图案。磁通密度的概念将在稍后的第 8.2.2 节中详细定义。示例 8.1 条形磁铁周围的磁场 S N 84

Question 8.1 For two identical bar magnets placed side by side as shown, what does the magnetic field look like? Try sketching the magnetic field lines.



(a) two attracting bar magnets

(b) two repelling bar magnets

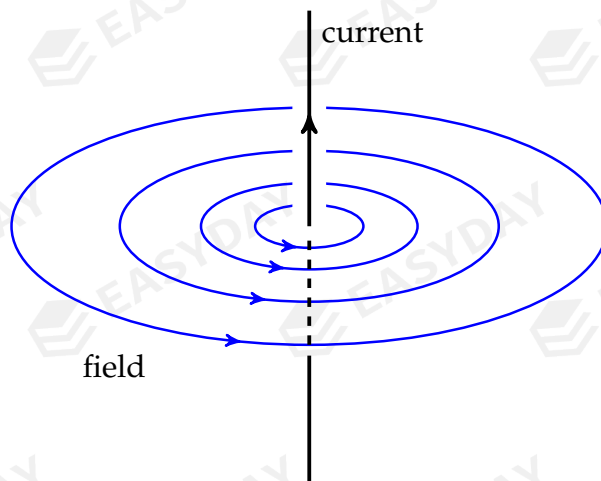
8.1.2 magnetic field due to currents

an electric current also induces a magnetic field around it

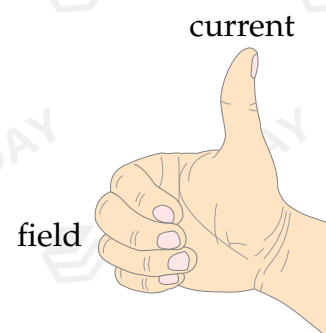
this was first discovered by Danish physicist *Hans Christian Orsted* in 1820, when he noticed the turning of a compass needle placed next to a wire carrying current.

we will look into what happens when a current flows through a straight wire or a coil

- pattern of the field can be determined using **right-hand (grip) rule**^[50]



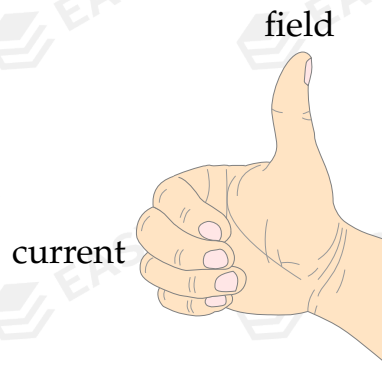
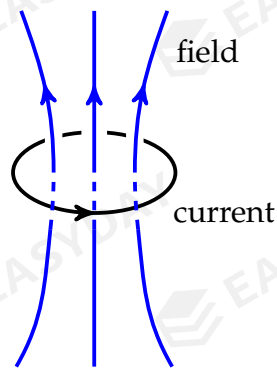
field due to a long straight current-carrying wire



- strength of the field is proportional to the current: $B \propto I$

^[50]The awesome right-hand rule illustrations below are copied from the PStricks web site: <http://tug.org/PStricks/main.cgi?file=examples>. The credits for these figures are attributed to CTAN community member *Thomas Söll*.

Yuhao Yang 8 磁场问题 8.1 如图所示，将两个相同的条形磁铁并排放置，磁场会是什么样子？尝试画出磁场线。S N S N (a) 两个吸引的条形磁铁 N S S N (b) 两个排斥的条形磁铁 8.1.2 电流产生的磁场电流也会在其周围感应出一个磁场，这是丹麦物理学家 Hans Christian Orsted 于 1820 年首次发现的，当时他注意到放置在载流电线旁边的罗盘针的转动。我们将研究当电流流过直导线或线圈时会发生什么可以使用右手（握力）定则 [50] 电流场来确定场的模式由于长直载流导线而产生电流场场的强度与电流成正比： B / I [50] 下面精彩的右手定则插图复制自 PS Tricks 网站：<http://tug.org/PSTricks/main.cgi?file=examples>。这些数字归功于 CTAN 社区成员 Thomas Söll。 85



field due to a circulating current

for both straight wires [51] and coils [52], greater current means stronger field

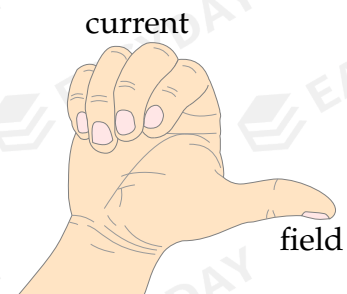
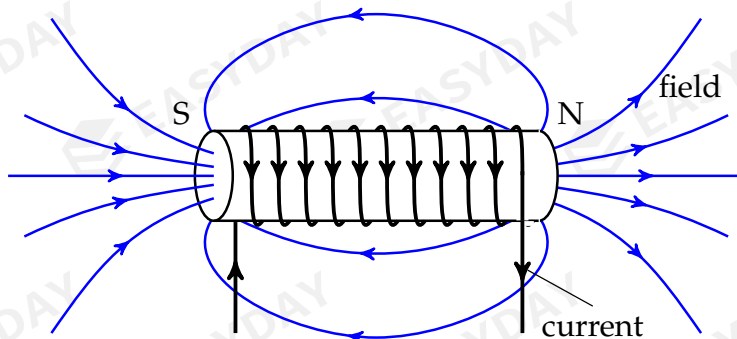
➤ strength of magnetic field can be increased with *soft iron*

this is because *ferromagnetic materials* (iron, cobalt, nickel) can attract magnetic field lines [53]

8.1.3 solenoids & electromagnets

strength and polarity of the field due to a coil can be changed easily by tuning currents, so coils are widely used to create magnetic fields where needed

a current-carrying coil is also called a **solenoid**



magnetic field around a solenoid

[51] The magnetic flux density generated by a current flowing in an infinitely long wire in free space is given by: $B = \frac{\mu_0 I}{2\pi r}$, I is the electric current, r is the perpendicular distance from the current, and $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ is a fundamental physical constant called the *vacuum permeability*. (*)

[52] The magnetic flux density at the centre of a current-carrying coil in free space is given by: $B = \frac{\mu_0 N I}{L}$, where N is the number of turns, I is the electric current flowing through the coil, L is the length of coil, and μ_0 is the vacuum permeability. (*)

[53] If the straight wire is immersed in a material with *relative permeability* μ_r , then the field becomes: $B = \frac{\mu_0 \mu_r I}{2\pi r}$. Similarly, if a material with relative permeability μ_r is present, then the magnetic flux density inside a coil becomes: $B = \frac{\mu_0 \mu_r N I}{L}$. A good magnetic material (high permeability material), such as iron, has large μ_r , and therefore can greatly intensify the magnetic field. (*)

杨宇浩 8 磁场电流场是由直导线 [51] 和线圈 [52] 的循环电流产生的，电流越大，磁场越强软铁可以增加磁场强度，这是因为铁磁材料（铁、钴、镍）可以吸引磁场线 [53] 8.1.3 螺线管和电磁铁的强度和极性线圈产生的磁场可以通过调节电流轻松改变，因此线圈被广泛用于在需要的地方产生磁场。载流线圈也称为螺线管电流场 NS 磁场螺线管周围的电流磁场 [51] 自由空间中无限长导线中的电流产生的磁通密度由下式给出： $B = \frac{\mu_0 I}{2\pi r}$ ， I 是电流， r 是与电流的垂直距离， $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ 是称为真空磁导率的基本物理常数。 (Q) [52] 自由空间中载流线圈中心的磁通密度由下式给出： $B = \frac{\mu_0 N I}{L}$ ，其中 N 是匝数， I 是流过线圈的电流， L 是线圈的长度， μ_0 是真空磁导率。 (Q) [53] 如果将直导线浸入相对磁导率 μ_r 的材料中，则磁场变为： $B = \frac{\mu_r \mu_0 I}{2\pi r}$ 。类似地，如果存在相对磁导率 μ_r 的材料，则线圈内的磁通密度变为： $B = \frac{\mu_r \mu_0 N I}{L}$ 。良好的磁性材料（高磁导率材料），例如铁，具有较大的 μ_r ，因此可以大大增强磁场。 (Q) 86

- a solenoid generates a magnetic field similar to that of a bar magnet
useful to talk about the north and south poles of a solenoid
- direction of magnetic field in a solenoid is given by the **right-hand (grip) rule**
- magnetic field produced by a solenoid can be controlled
current $I \uparrow \Rightarrow$ stronger field, also number of turns $N \uparrow \Rightarrow$ stronger field
- inserting an *iron core* inside greatly strengthens the field, this makes an *electromagnet*

Question 8.2 Describe the magnetic field due to an alternating current through a solenoid.

8.2 magnetic force on current-carrying conductor

8.2.1 magnetic force on current-carrying conductor

a current-carrying wire produces its own magnetic field, when it is surrounded by an external magnetic field, the two fields would interact $\Rightarrow$ a **magnetic force** on the conductor

- direction of magnetic force on current can be worked out with **Fleming's left-hand rule**^[54]

- magnitude of magnetic force: $F = BIL \sin \theta$

B is the magnetic flux density to be specified later

I is the current, θ is the angle between B and I

- when $B \perp I$, magnetic force $F = BIL$

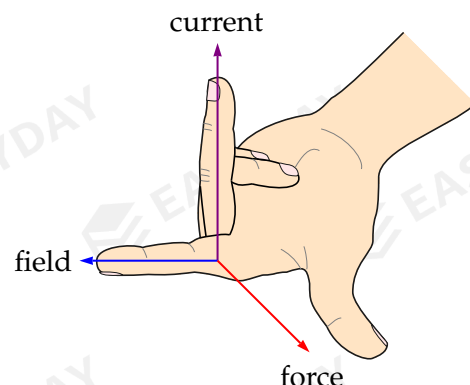
when $B \parallel I$, there is no magnetic force

when B forms angle θ with I , only the *perpendicular*

component contributes to the force, giving rise to the $\sin \theta$ factor in the formula

- magnetic force is perpendicular to both B and I

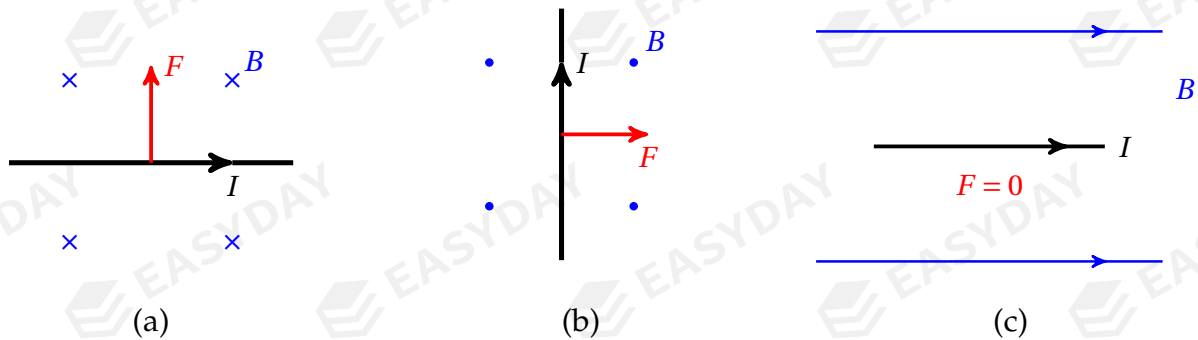
Example 8.2 Determine the direction of the magnetic force acting. Check yourself.



Fleming's left-hand rule

^[54] Figure credit again goes to Thomas Söll from the CTAN community.

Yuhao Yang 8 磁场 $\hat{a}$ 螺线管产生类似于条形磁铁的磁场，有助于讨论螺线管的北极和南极 $\hat{a}$ 螺线管中的磁场方向由右手（握力）规则给出 $\hat{a}$ 螺线管产生的磁场可以控制电流 I 1) 更强的磁场，也可以控制匝数 N 1) 更强的磁场 $\hat{a}$ 在里面插入一个铁芯会极大地增强磁场，这使得电磁铁成为问题 8.2 描述由于交流电通过螺线管而产生的磁场。8.2 载流导体上的磁力 8.2.1 载流导体上的磁力载流导线会产生自己的磁场，当它被外部磁场包围时，两个磁场会相互作用）导体上的磁力电流场力弗莱明左手定则 $\hat{a}$ 电流上的磁力方向可以用弗莱明左手定则求得[54] $\hat{a}$ 磁力大小： $F = B I L \sin \theta$ B 是稍后指定的磁通密度， I 是电流， θ 是 B 和 I 之间的角度 $\hat{a}$ 当 $B \perp I$ ，磁力 $F = B I L$ 当 $B \parallel I$ 时， B 与 I 形成角度 θ 时，没有磁力，只有垂直分量对力有贡献，从而得出公式中的 $\sin \theta$ 因子 $\hat{a}$ 磁力垂直于 B 和 I 例 8.2 确定磁力作用的方向。检查一下你自己。[54]图再次由 CTAN 社区的 Thomas Söll 提供。87



8.2.2 magnetic flux density

rewrite $F = BIL\sin\theta$ as $B = \frac{F}{IL\sin\theta}$, we can now give a formal definition for flux density B :

magnetic flux density B at a point is defined as the force acted per unit length on a conductor carrying a unit current at right angle to the magnetic field

- flux density describes the strength of a magnetic field^[55]
- B is measured in **tesla** (T)^[56]: $[B] = \text{T}$, $1 \text{ T} = 1 \text{ N} \cdot \text{A}^{-1} \cdot \text{m}^{-1}$

if a wire of 1 m normal to the magnetic field that carries a current of 1 A experiences a force of 1 N, then the magnetic flux density is 1 T

- B is a *vector* quantity, with both magnitude and direction

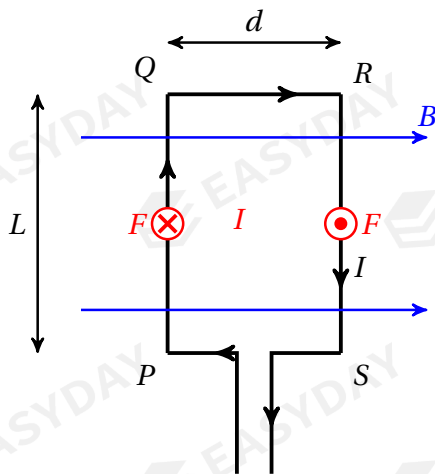
Example 8.3 A wire of 0.80 m carrying a current of 5.0 A is lying at right angles to a magnetic field, it experiences a force of 0.60 N, what is the flux density?

$$\hookrightarrow B = \frac{F}{IL} = \frac{0.60}{5.0 \times 0.80} = 0.15 \text{ T} \quad (\text{a field of over } 0.1 \text{ T is actually quite strong!}) \quad \square$$

^[55] Magnetic field strength is a different quantity, defined by $H = \frac{B}{\mu}$, with $\mu = \mu_0\mu_r$ being the *magnetic permeability* of material. The naming of field strength H and flux density B are due to historical reasons.

^[56] Tesla is a very large unit. For example, the strength of a typical refrigerator magnet is of about 10^{-3} T , even the very strong superconducting coils used in MRI are of around $1 \sim 3 \text{ T}$.

杨宇豪 8 磁场 $\times \times \times \times I F B$ (a) $I F B$ (b) $B I F = 0$ (c) 8.2.2 磁通密度将 $F = B I L \sin \theta$ 重写为 $B = F I L \sin \theta$ ，现在我们可以给出磁通密度 B 的正式定义：一点的磁通密度 B 定义为导体上单位长度所受的力承载与磁场成直角的单位电流 $\hat{\alpha}$ 磁通密度描述了磁场的强度 [55] $\hat{\alpha} B$ 以特斯拉 (T)[56] 为单位测量： $[B] = T, 1 T = 1 N - A^{-1} - m^{-1}$ 如果垂直于磁场、承载 1 A 电流的 1 m 导线受到 1 N 的力，则磁通密度为 1 T $\hat{\alpha} B$ 是一个矢量，有大小和方向例 8.3 一根 0.80 m 的导线，载有 5.0 A 的电流，与磁场成直角放置，它受到 0.60 N 的力，磁通密度是多少？ $- B = F I L = 0.60 / (5.0 \times 0.80) = 0.15 T$ （超过 0.1 T 的磁场实际上相当强！） $\ddot{\alpha}$ [55] 磁场强度是一个不同的量，由 $H = B / \mu_0$ 定义，其中 μ_0 是材料的磁导率。场强 H 和磁通密度 B 的命名是有历史原因的。[56] 特斯拉是一个非常大的单位。例如，典型冰箱磁体的强度约为 $10^{-3} T$ ，甚至 MRI 中使用的非常强的超导线圈也约为 $1 \gg 3 T$ 。88

Example 8.4 Torque on a rectangular metal frame in uniform magnetic field

a rectangular frame lies in parallel with the field

$QR, PS \parallel B$, so no force acting on these two sides

$PQ, RS \perp B$, so there is magnetic force $F = BIL$

using left-hand rule, we find F_{PQ} acts into the paper, F_{RS} acts out of the paper

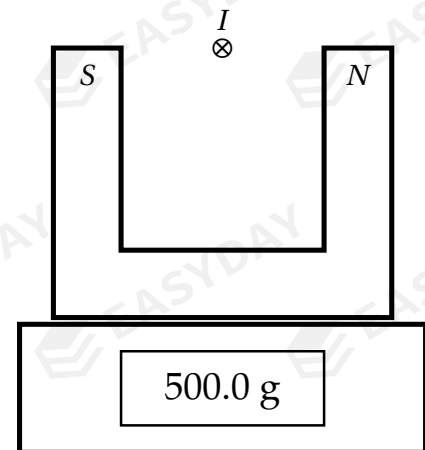
this is a pair of equal but opposite forces

they produce a torque about central axis of frame: $\tau =$

$Fd = BILd$, causing the frame to rotate^[57] □

Question 8.3 If the plane of the rectangular frame is at right angle to the magnetic field, describe the magnetic forces acting on each side and hence state what happens to the frame.

Example 8.5 A U-shaped magnet is placed on a balance with a wire suspended above it as shown. Magnetic flux density between the poles is about 0.50 T. The part of the wire that is in the field is of length 8.0 cm. The balance initially shows a reading of 500.0 g. When a current of 10 A flowing into the paper is switched on, what is the new reading on the balance?



✎ use left-hand rule, force on wire acts upwards

from *Newton's third law*, reaction force on magnet acts downwards

so there will be an increase in the balance reading

magnitude of the force: $F = BIl = 0.50 \times 10 \times 8.0 \times 10^{-2} = 0.40 \text{ N}$

change in the mass reading: $\Delta m = \frac{F}{g} = \frac{0.40}{9.81} \approx 0.0407 \text{ kg} \approx 40.7 \text{ g}$

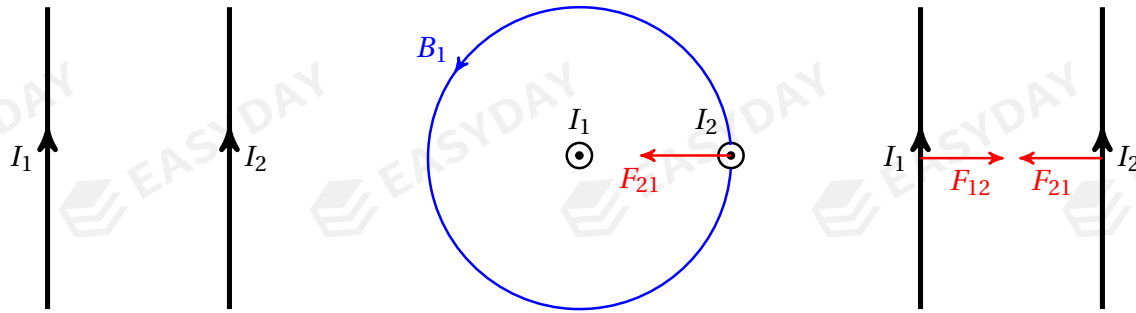
new reading on balance: $m_{\text{new}} = 500.0 + 40.7 = 540.7 \text{ g}$ □

Question 8.4 If the current in Example 8.5 is replaced by a current of 6.0 A flowing out of the plane of the paper. What is the reading on the balance?

^[57] This effect allows engineers to build *electric motors*, such as those in electric vehicles, washing machines, blenders, etc.

杨宇浩 8 磁场例8.4 均匀磁场中矩形金属框架上的扭矩 B P Q R S I FFL d I 矩形框架与磁场 QR、PS 平行，因此没有力作用在这两侧 PQ、RS，所以使用左手定则存在磁力 $F = BIL$ ，我们发现 FPQ 作用在纸内，FRS 作用在纸外，这是一对大小相等但方向相反的力，它们产生绕框架中心轴的扭矩： $\tau = Fd = BILd$ ，导致框架旋转 [57] 问题 8.3 如果矩形框架的平面与磁场成直角，请描述作用在每一侧的磁力并因此说明框架发生了什么。INS 500.0 g 示例 8.5 如图所示，将 U 形磁铁放置在天平上，并在其上方悬挂一根金属丝。磁极之间的磁通密度约为 0.50 T。位于磁场中的导线部分长度为 8.0 cm。天平最初显示的读数为 500.0 g。当通入纸的电流为 10 A 时，天平上的新读数是多少？- 使用左手定则，根据牛顿第三定律，导线上的力向上作用，磁铁上的反作用力向下作用，因此天平读数大小会增加： $F = BIL = 0.50 \times 10 \times 8.0 \times 10^{-2} = 0.40 \text{ N}$ 质量读数的变化： $\Delta m = F/g = 0.40/9.81 \approx 0.0407 \text{ kg} \approx 40.7 \text{ g}$ 天平新读数： $m_{\text{new}} = 500.0 + 40.7 = 540.7 \text{ g}$ 问题 8.4 如果将例 8.5 中的电流替换为从纸平面流出的 6.0 A 电流。天平上的读数是多少？[57]这种效应使工程师能够制造电动机，例如电动汽车、洗衣机、搅拌机等中的电动机。89

force between long straight current-carrying wires



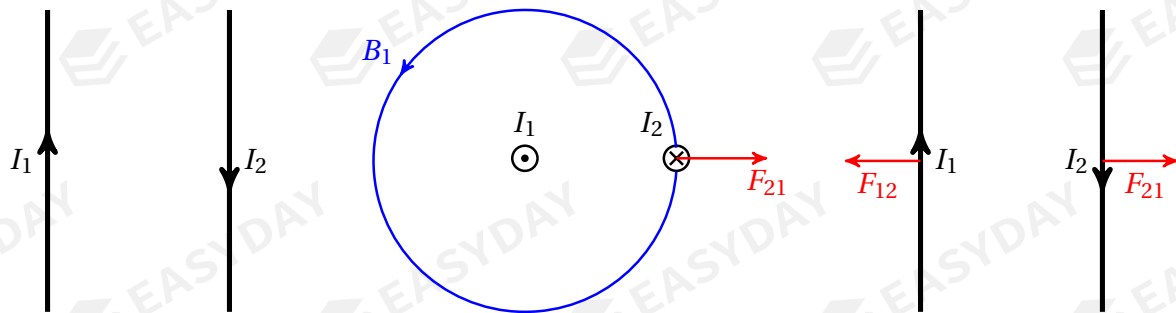
if we have two straight parallel wires carrying currents in the same direction

force acting on I_2 is due to magnetic field B_1 generated by I_1

from top view, I_1 creates a counter-clockwise B_1 , so I_2 experiences an upward field

using left-hand rule, force acting on I_2 by I_1 , denoted F_{21} ^[58], points to the left

since F_{21} and F_{12} are a pair of action-reaction, so F_{12} points to the right



similar discussions apply for I_1 and I_2 flowing in opposite directions (left as exercise)

hence we come to an conclusion: *parallel currents attract, and anti-parallel currents repel*^[59]

Question 8.5 A light coil of wire of several loops is suspended from a fixed point. When an electric current is switched on in the coil, state and explain the change in the separation between the loops.

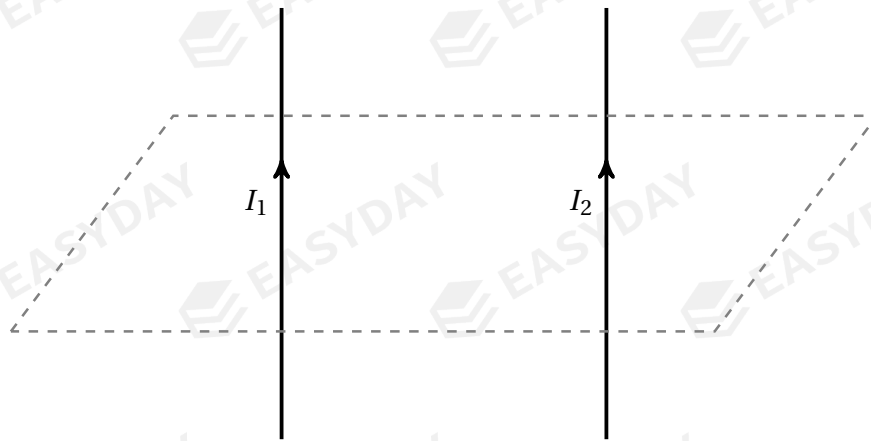
Question 8.6 Two long parallel wires carry currents of $I_1 = 3.0$ A and $I_2 = 5.0$ A in the same direction as shown. The flux density at a point at perpendicular distance r from a straight long wire carrying current I is given by $B = \frac{\mu_0 I}{2\pi r}$.

(a) Draw at least three lines to show the magnetic field due to I_1 .

^[58] F_{AB} denotes the force acted on object A by object B .

^[59] The S.I. unit for current, ampere, is defined using two long wires parallel to each other carrying the same current in the same direction. If the wires are separated by 1 m and the magnetic force experienced per metre is 2.0×10^{-7} N, then the current is of 1 A.

Yuhao Yang 8 磁场长直载流导线之间的力 $I_1 I_2 B_1 F_{21} I_1 I_2 F_{21} F_{12}$ 如果我们有两根平行的直导线，载流方向相同，则作用在 I_2 上的力是由于从俯视图中 I_1 产生的磁场 B_1 造成的， I_1 产生逆时针 B_1 ，因此使用左手定则， I_2 会经历一个向上的磁场， I_1 作用在 I_2 上的力，表示为 F_{21} [58]，指向左边，因为 F_{21} 和 F_{12} 是一对作用-反应，所以 F_{12} 指向右边 $I_1 I_2 I_1 I_2 B_1 F_{21} I_1 I_2 F_{21} F_{12}$ 类似的讨论适用于反向流动的 I_1 和 I_2 （左边作为练习）因此我们得出一个结论：平行电流吸引，反平行电流排斥 [59] 问题 8.5 A 光线圈多圈导线悬挂在一个固定点上。当线圈中接通电流时，说明并解释回路之间的间隔的变化。问题 8.6 如图所示，两条长平行导线沿相同方向承载 $I_1 = 3.0 \text{ A}$ 和 $I_2 = 5.0 \text{ A}$ 的电流。距承载电流 I 的直长导线垂直距离 r 处的磁通密度由 $B = \frac{\mu_0 I}{2\pi r}$ 给出。(a) 画至少三条线来显示 I_1 产生的磁场。 [58] F_{AB} 表示物体 B 作用在物体 A 上的力。 [59] 电流的国际单位安培是使用两条彼此平行的长导线定义的，它们在相同方向上承载相同的电流。如果导线相距 1 m ，每米受到的磁力为 $2.0 \times 10^{-7} \text{ N}$ ，则电流为 1 A 。 90



- (b) State and explain the direction of force acting on I_2 .
- (c) Given that I_1 and I_2 are separated by 4.0 cm, what is flux density due to I_1 at I_2 ?
- (d) What is the magnetic force per unit length acting on I_2 ?
- (e) What is the magnitude and the direction of the force per unit length acting on I_1 ?

Question 8.7 If the current in one of the two wires in Question 8.6 is replaced by an alternating current, then the two wires should begin to vibrate. However, the wires are not observed to move. Make some reasonable estimates and explain why the vibration is not observed.

8.3 magnetic force on charged particles

electric currents are formed by moving charges, since current-carrying wires in a magnetic field experience force, charged particle moving in magnetic field should also experience force

8.3.1 magnetic force on charged particles

starting from $F = BIl \sin \theta$, let's substitute $I = nAqv$ $\Rightarrow F = B(nAqv)l \sin \theta$

notice n is the number density of charged particles, so nAl together gives the total number

if we look at the force acting on *one* charged particle, then $nAl = 1$, so: $F = Bqv \sin \theta$

➤ magnitude of magnetic force on charged particle: $F = Bqv \sin \theta$

B is magnetic flux density, q is electric charge of the particle, v is its velocity

θ refers to the angle between B and v

Yuhao Yang 8 磁场 I1 I2 (b) 陈述并解释作用在 I2 上的力的方向。(c) 鉴于 I1 和 I2 相距 4.0 cm, I1 在 I2 处产生的磁通密度是多少? (d) I2 上单位长度的磁力是多少? (e) 作用在 I1 上的单位长度力的大小和方向是多少? 问题 8.7 如果将问题 8.6 中两根导线中的一根电流替换为交流电, 那么这两根导线应该开始振动。然而, 没有观察到电线移动。做出一些合理的估计并解释为什么没有观察到振动。

8.3 带电粒子上的磁力电流是由移动的电荷形成的, 因为磁场中载流导线会受到力, 所以在磁场中移动的带电粒子也应该受到力

8.3.1 带电粒子上的磁力从 $F = B I \sin \theta$ 开始, 让我们代入 $I = n A q v$ $F = B (n A q v) I \sin \theta$ 注意 n 是带电粒子的数密度, 因此 $n A l$ 一起得出如果我们观察作用在一个带电粒子上的力, 则为总数, 则 $n A l = 1$, 因此

: $F = B q v \sin \theta$ 带电粒子上磁力的大小: $F = B q v \sin \theta$ B 是磁通密度, q 是粒子的电荷, v 是其速度 θ 指的是 B 和 v 之间的角度 91

- direction of magnetic force can be determined using *Fleming's left-hand rule*

for $+q$, current in same direction as v

for $-q$, current in opposite direction as v

- force depends on *perpendicular component* of B

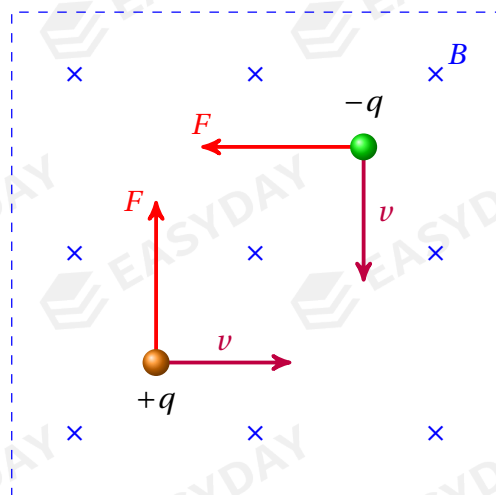
if $v \perp B$, magnetic force $F = Bqv$

if $v \parallel B$, there is no magnetic force

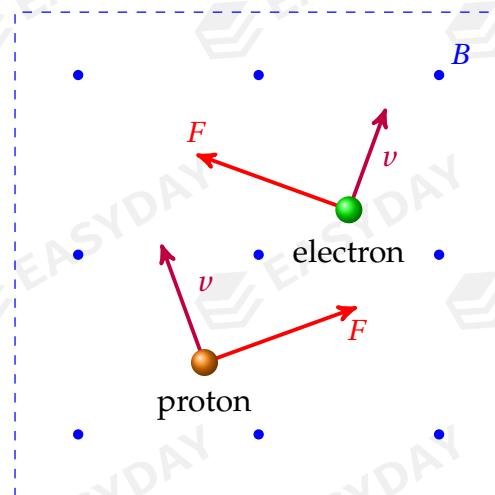
when particle moves at angle θ to B , contribution to the force only comes from the *perpendicular component*, giving rise to the $\sin\theta$ factor

- magnetic force always perpendicular to both velocity v and magnetic field B ^[60]

Example 8.6 Use the left-hand rule to find the direction of the magnetic forces acting on the following moving charges. Check yourself.



(a)



(b)

charged particles in electric & magnetic fields

in either electric field or magnetic field, charged particle may experience a force^[61]

in this section, we will compare the difference between electric field and magnetic field

^[60] Vector form of the formula: $\vec{F} = q\vec{v} \times \vec{B}$. (*)

^[61] The combination of electric and magnetic force on the charge due to electromagnetic fields is called the *Lorentz force*: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. If you have a good knowledge in maths, everything we cover in this section can be recovered from this vector equation. (*)

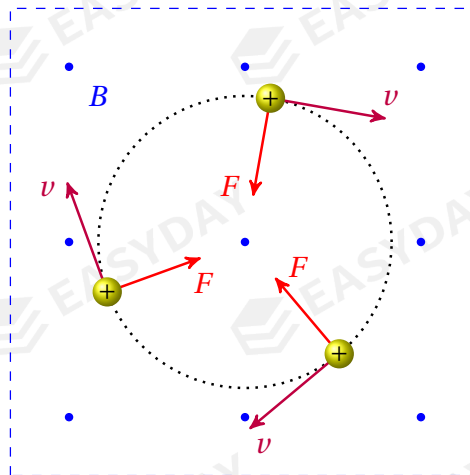
杨宇豪 8 磁场电流场力弗莱明左手定则 $\hat{\alpha}$ 磁力的方向可以使用弗莱明左手定则确定 $+q$ ，对于 $-q$ ，电流与 v 方向相同，与 v 方向相反的电流 $\hat{\alpha}$ 力取决于 B 的垂直分量，如果 $v \perp B$ ，磁力 $F = B q v$ 如果 $v \parallel B$ ，当粒子以相对于 B 的角度 θ 运动时，不存在磁力，对力的贡献仅来自于垂直分量，从而产生 $\sin \theta$ 因子 $\hat{\alpha}$ 磁力始终垂直于速度 v 和磁场 B [60] 例 8.6 使用左手定则求作用于以下移动电荷的磁力方向。检查一下你自己。

。 $\times \times \times \times \times \times \times \times v F + q v F - q B (a) v F$ 质子 $v F$ 电子 $B (b)$ 电场和磁场中的带电粒子无论在电场还是磁场中，带电粒子都可能受到力 [61] 在本节中，我们将比较电场和磁场之间的差异 [60] Vector 公式的形式： $\epsilon F = q \epsilon v \times \epsilon B$. (Q) [61] 电磁场对电荷产生的电场力和磁力的组合称为洛伦兹力： $\epsilon F = q (\epsilon E + \epsilon v \times \epsilon B)$ 。如果您有很好的数学知识，我们在本节中介绍的所有内容都可以从这个向量方程中恢复出来。(Q)9

	electric field	magnetic field
strength of the field	electric field strength E	magnetic flux density B
magnitude of force	$F_E = Eq$	$F_B = Bqv \sin \theta$
whether force depends on velocity	no dependence, acts equally on stationary or moving charges	depend on perpendicular component of velocity
direction of force	$F_E \parallel E$ same as/opposite to E for $+q/-q$	$F_B \perp B$ and $F_B \perp v$ use left-hand rule
effect of force on motion of particle	can change both magnitude and direction of velocity	only changes direction of velocity, cannot change magnitude of velocity
work done by force	work can be done, can define notions of potential and P.E.	magnetic force does no work for charged particles

8.3.2 charged particle in uniform magnetic fields

suppose a charged particle is moving at right angle to a uniform magnetic field



circular motion of a charged particle in uniform magnetic field

magnetic force F_B always at right angle to motion, so F_B keeps changing direction of velocity

uniform field means a constant force, so F_B deflects the particle at the same rate

the particle should describe a *circular* path!

杨宇豪 8 磁场电场磁场强度电场强度 E 磁通密度 B 力的大小 $F_E = E q$ $F_B = B q v \sin \theta$ 力是否与速度无关，对静止或移动电荷的作用相等，取决于力的速度方向的垂直分量 $F_E \perp E$ $F_B \perp B$ 和 $F_B \perp v$ 与 E 相同/相反，对于 $+q/-q$ 使用左手法则力对粒子运动的影响可以同时改变速度的大小和方向只能改变速度的方向，不能改变速度的大小力所做的功可以做功，可以定义势能和 P 的概念。磁力对带电粒子不起作用 8.3.2 均匀磁场中的带电粒子假设带电粒子在均匀磁场中以直角运动 $B \perp v$ $F = v F + v F + v F +$ 带电粒子在均匀磁场中的圆周运动磁力 F_B 始终与运动成直角，因此 F_B 不断改变速度方向均匀场意味着力恒定，因此 F_B 以与粒子应描述圆形路径相同的速率偏转粒子！ 9

for a charged particle moving in a uniform magnetic field, *magnetic force provides centripetal force for circular motion*: $Bqv = \frac{mv^2}{r}$, or $Bqv = m\omega^2 r$

- can solve for radius of the orbiting particles: $r = \frac{mv}{Bq}$
 - $v \uparrow \Rightarrow r \uparrow$, faster particles take larger circles
 - $B \uparrow \Rightarrow r \downarrow$, stronger magnetic field, larger centripetal force, smaller circles
 - $m \uparrow \Rightarrow r \uparrow$, larger mass, larger inertia, so larger circles
- radius of curvature relates to charge-to-mass ratio (also called *specific charge*) of the particle
rearrange the equation we have $\frac{q}{m} = \frac{v}{Br}$, which can be computed using experimental data

Example 8.7 An α -particle travelling at $2.5 \times 10^4 \text{ m s}^{-1}$ enters a region of uniform magnetic field. The field has flux density of 5.4 mT and is normal to direction of particle's velocity. What is the radius of α -particle's path?

✎ radius of circular arc: $r = \frac{mv}{Bq} = \frac{4 \times 1.66 \times 10^{-27} \times 2.5 \times 10^4}{5.4 \times 10^{-3} \times 2 \times 1.60 \times 10^{-19}} \approx 0.096 \text{ m} \approx 9.6 \text{ cm}$ □

Question 8.8 For an α -particle and a β -particle entering the same uniform magnetic field at a same speed, compare the radius of their paths.

Question 8.9 For a charged particle undergoing circular motion in a uniform magnetic field, show that its angular velocity is independent of the radius of its path.

Question 8.10 If a charged particle enters a uniform magnetic field at an angle $\theta \neq 90^\circ$, state and explain the path of this particle. (Hint: think about components of its velocity.)

8.3.3 mass spectrometer

a **mass spectrometer** is a device to measure the charge-to-mass ratio of charged particles

charged particles accelerated through an electric field: $\frac{1}{2}mv^2 = qV$

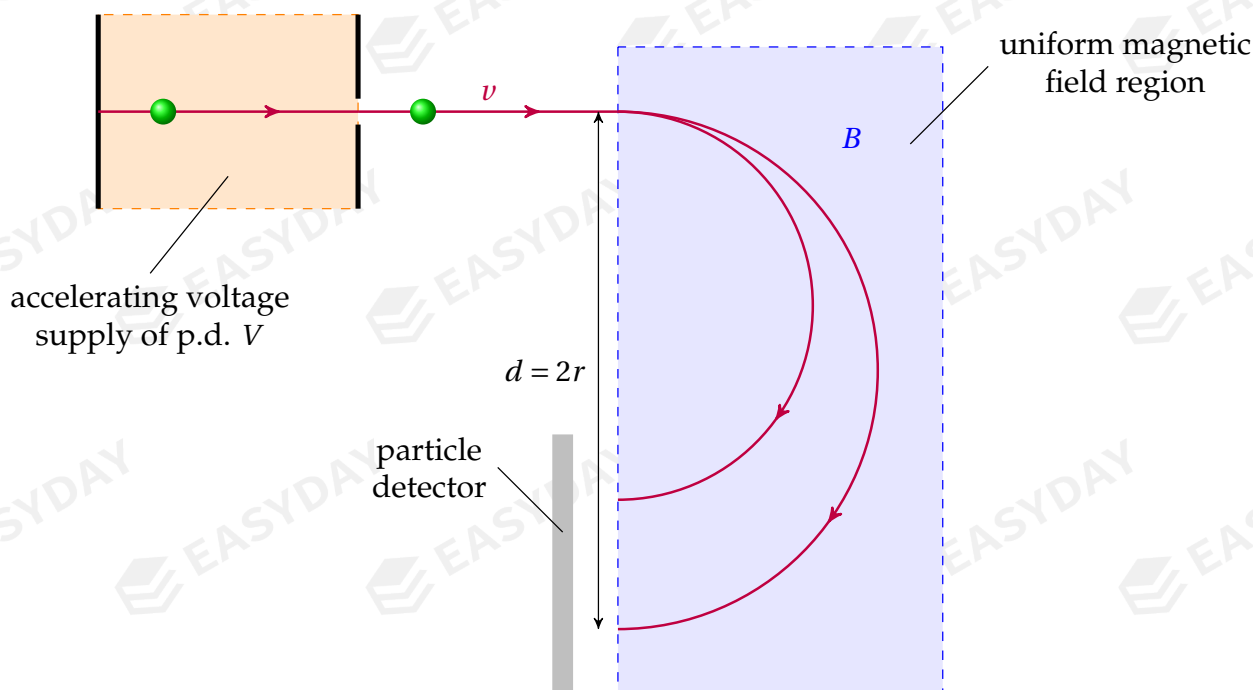
they then enter a uniform magnetic field: $Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bqr}{m}$

eliminating v : $v^2 = \frac{2qV}{m} = \frac{B^2 q^2 r^2}{m^2} \Rightarrow \frac{q}{m} = \frac{2V}{B^2 r^2}$

we can measure V, B, r in practice, so the charge-to-mass ratio $\frac{q}{m}$ is worked out

different particles have different values of $\frac{q}{m}$, unknown particles can be identified

Yuhao Yang 8 磁场对于在均匀磁场中运动的带电粒子，磁力为圆周运动提供向心力： $Bqv = mv^2/r$ ，或 $Bqv = m\omega^2 r$ 可以求解轨道粒子的半径： $r = mv/Bq$ ，较快的粒子采用较大的圆， B 较大，较大向心力，较小的圆， m 较大，较大的质量，较大的惯性，因此较大的圆曲率半径与粒子的荷质比（也称为比电荷）有关重新排列方程，我们得到 $q/m = v/Br$ ，可以使用实验数据计算例 8.7 一个 α 粒子以 $2.5 \times 10^4 \text{ m/s}$ 运动，进入均匀磁场区域。该场的通量密度为 5.4 mT ，并且垂直于粒子速度方向。 α 粒子的路径半径是多少？ - 圆弧半径： $r = mv/Bq = 4 \times 1.66 \times 10^{-27} \times 2.5 \times 10^4 / (5.4 \times 10^{-3} \times 2 \times 1.60 \times 10^{-19}) = 0.096 \text{ m} = 9.6 \text{ cm}$ 问题 8.8 对于以相同速度进入同一均匀磁场的 α 粒子和 β 粒子，比较它们的路径半径。问题 8.9 对于在均匀磁场中做圆周运动的带电粒子，证明其角速度与其路径半径无关。问题 8.10 如果带电粒子以 $\theta = 90^\circ$ 角度进入均匀磁场，请说明并解释该粒子的路径。（提示：考虑其速度的组成部分。） 8.3.3 质谱仪质谱仪是一种测量带电粒子荷质比的装置带电粒子通过电场加速： $\frac{1}{2}mv^2 = qV$ 然后它们进入均匀磁场： $Bqv = mv^2/r$ $v = Bqr/m$ 消除 v ： $v^2 = 2qV/m = B^2q^2r^2/m^2$ $q/m = 2V/B^2r^2$ 实际中我们可以测量 V 、 B 、 r ，从而计算出荷质比 q/m 不同的粒子具有不同的 q/m 值，可以识别未知粒子 94



deflection of two charged particles in a mass spectrometer

Question 8.11 In the figure above, two paths of deflected particles are shown. Give reasons why the radius of the circular path can be different.

Question 8.12 If the particles sent into the mass spectrometer are positively-charged, in which direction should the magnetic field be applied?

Question 8.13 A particle carrying a charge of $+e$ enters a uniform magnetic field of 8.8 mT at right angles with an initial speed of $1.4 \times 10^5 \text{ m s}^{-1}$. It describes a semi-circle with diameter of 66 cm . (a) Find the mass of the particle. (b) Suggest possible composition of this particle.

8.3.4 cyclotron

a **cyclotron** is a type of *particle accelerator*

the idea is to make use of a magnetic field to guide moving charges into a spiral path between accelerations by an electric field

as the particle enters and leaves the region of electric field, it gains extra energy of qV

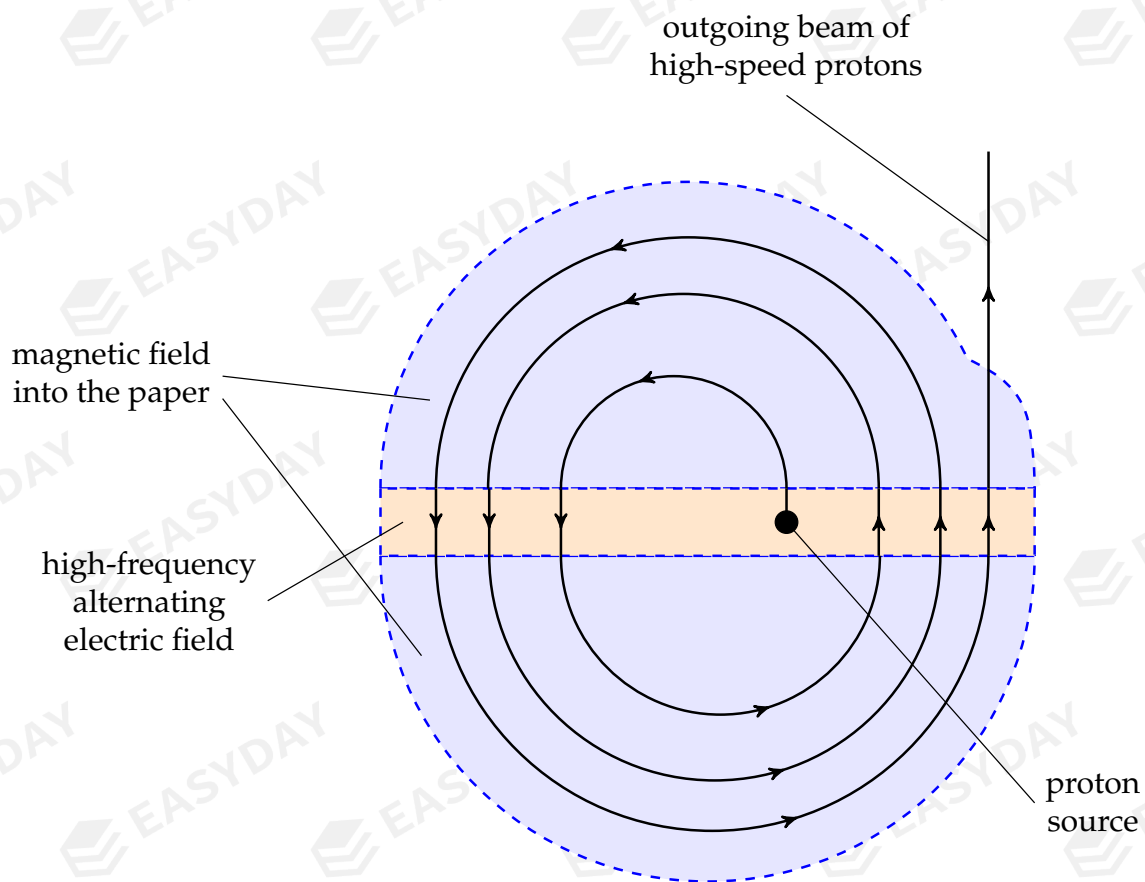
it then follows a semi-circular path under magnetic force and re-enters the electric field

polarity of the electric field is reversed so the particle continues to accelerate across the gap

energy of particle increases by qV , it then moves in a larger semi-circle in magnetic field

杨宇豪 8 磁场加速 p.d. 的电压供应 V B 均匀磁场区域 $v d = 2r$ 粒子探测器质谱仪中两个带电粒子的偏转问题 8.11 上图中，显示了偏转粒子的两条路径。给出圆形路径的半径可能不同的原因。问题 8.12 如果送入质谱仪的粒子带正电，磁场应施加在哪个方向？问题 8.13 带有 $+e$ 电荷的粒子以直角进入 8.8 mT 的均匀磁场，初始速度为 $1.4 \times 10^5 \text{ m s}^{-1}$ 。它描述了一个直径为 66 厘米 的半圆。(a) 求粒子的质量。(b) 建议该粒子的可能组成。

8.3. 4 回旋加速器回旋加速器是一种粒子加速器，其思想是利用磁场引导移动电荷进入电场加速之间的螺旋路径，当粒子进入和离开电场区域时，它获得 qV 的额外能量，然后在磁力作用下沿着半圆形路径重新进入电场，电场极性反转，因此粒子继续加速穿过间隙能量粒子增加 qV ，然后在磁场 95 中沿更大的半圆运动



protons being accelerated in a cyclotron

repeat this process, the particle leaves the exit port with very high speed

➤ cyclotron frequency

for charged particles circulating in a uniform magnetic field: $Bqv = m\omega^2 r$

time to complete one full turn is: $T = \frac{2\pi r}{v} = \frac{2\pi}{v} \times \frac{mv}{Bq} \Rightarrow T = \frac{2\pi m}{Bq}$

this shows the period is independent of radius of the circular path!

if an alternating voltage is applied at frequency $f = \frac{Bq}{2\pi m}$, particles can be accelerated continuously at just the right time when they cross the gap

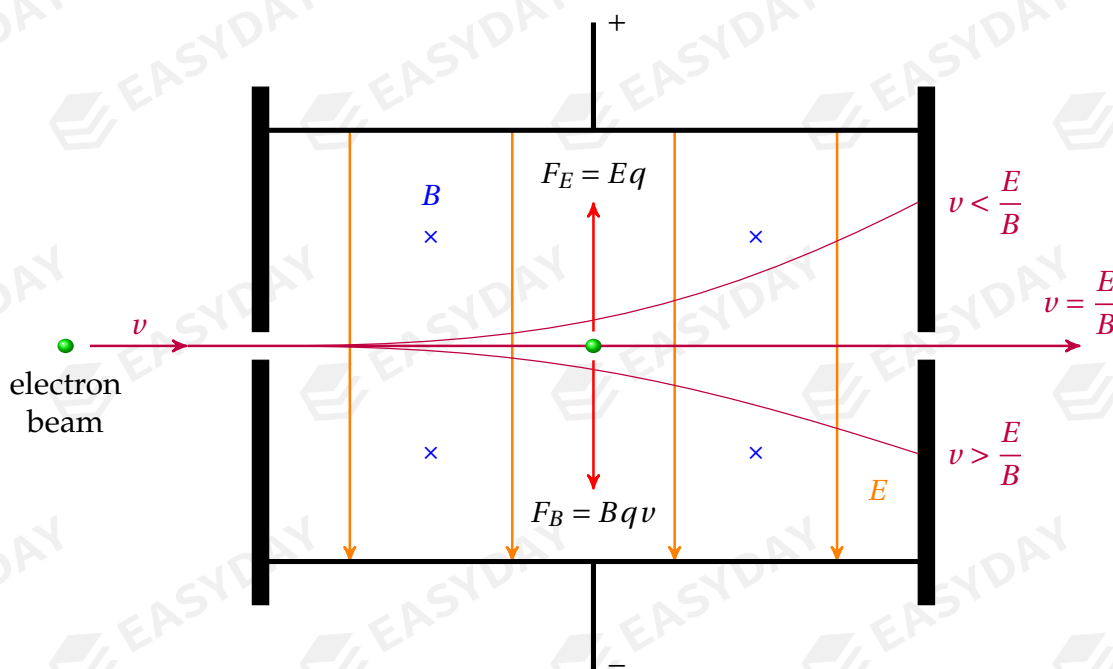
this frequency is known as the *cyclotron frequency*

Question 8.14 Protons are accelerated in a cyclotron where the magnetic field region has a uniform flux density of 1.5 T and the voltage between the gap is 20 kV. (a) Find the frequency of the voltage supply needed. (b) Find the number of circles the protons must make in order to gain an energy of 10 MeV.

杨宇浩 8 磁场磁场进入纸张高频交变电场质子源出射高速质子束质子在回旋加速器中被加速，重复此过程，粒子以非常高的速度离开出口 $\hat{a}$ 在均匀磁场中循环的带电粒子的回旋加速器频率： $B q v = m \omega 2r$ 完成一整圈的时间为： $T = 2 \pi r v = 2 \pi v \times m v B q$) $T = 2 \pi m B q$ 这表明周期与圆形路径的半径无关！如果以频率 $f = B q 2 \pi m$ 施加交流电压，粒子在穿过间隙时可以在适当的时间连续加速，该频率称为回旋加速器频率。问题 8.14 质子在回旋加速器中加速，其中磁场区域的均匀磁通密度为 1.5 T，间隙之间的电压为 20 kV。(a) 找出所需电源的频率。(b) 求出质子必须绕多少圈才能获得 10 MeV 的能量。96

8.3.5 velocity selector

velocity selector is a device to produce a beam of charged particles all with same speed v



velocity selector

let's consider a beam of electrons passing through a region where both uniform electric field and magnetic field are applied, electrons at desired speed v are *undeflected*

no net force acting on these electrons, so equilibrium between electric and magnetic force

$$F_E = F_B \Rightarrow Eq = Bqv \Rightarrow v = \frac{E}{B}$$

for particles entering the same region with higher speed, $F_B > F_E$, they deflect downwards

for particles entering the same region with lower speed, $F_B < F_E$, they deflect upwards

Question 8.15 If electrons at speed v are undeflected when they pass through the velocity selector, what about a beam of α -particles entering the same region at the same speed v ?

Question 8.16 A uniform magnetic field with flux density $6.0 \times 10^{-2} \text{ T}$ is applied out of the plane of the paper. A beam of protons are travelling into this region at a speed of $3.5 \times 10^4 \text{ m s}^{-1}$.

(a) In which direction do the protons deflect? (b) A uniform electric field is now applied in the same region so that the protons become undeviated. What is the strength and the direction of this electric field?

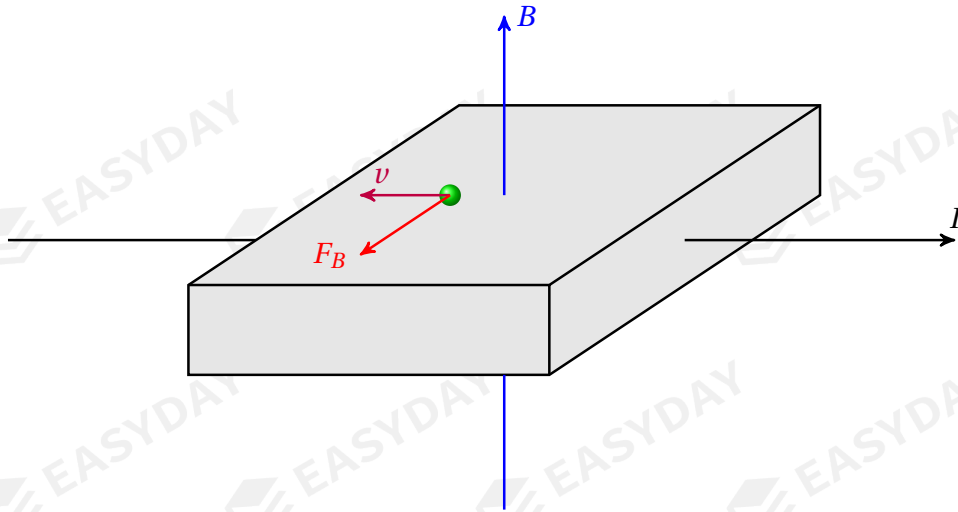
Yuhao Yang 8 磁场 8.3.5 速度选择器速度选择器是一种产生速度相同的带电粒子束的装置 $\vec{v} \times \vec{B} = E \hat{y}$ + 电子束 $\vec{v} = E/B \hat{y}$ $\vec{E} = E \hat{y}$ $\vec{B} = B \hat{z}$ $\vec{F}_E = E q \hat{y}$ $\vec{F}_B = B q \vec{v}$ 速度选择器让我们考虑一束电子穿过一个同时施加均匀电场和磁场的区域，电子处于期望的位置速度 $\vec{v}$ 未偏转没有合力作用于这些电子，因此电场力和磁力之间达到平衡 $F_E = F_B$ $E q = B q v$ $v = E/B$ 对于以较高速度进入同一区域的粒子 $F_B > F_E$ ，它们向下偏转对于以较低速度进入同一区域的粒子 $F_B < F_E$ ，它们向上偏转问题 8.15 如果速度为 v 的电子在通过速度选择器，如果一束 α 粒子以相同的速度 v 进入同一区域呢？问题 8.16 在纸平面外施加磁通密度为 $6.0 \times 10^{-2} \text{ T}$ 的均匀磁场。一束质子以 $3.5 \times 10^4 \text{ m s}^{-1}$ 的速度进入该区域。(a) 质子向哪个方向偏转？(b) 现在在同一区域施加均匀电场，使质子变得不偏离。这个电场的强度和方向是多少？ 97

8.4 Hall effect

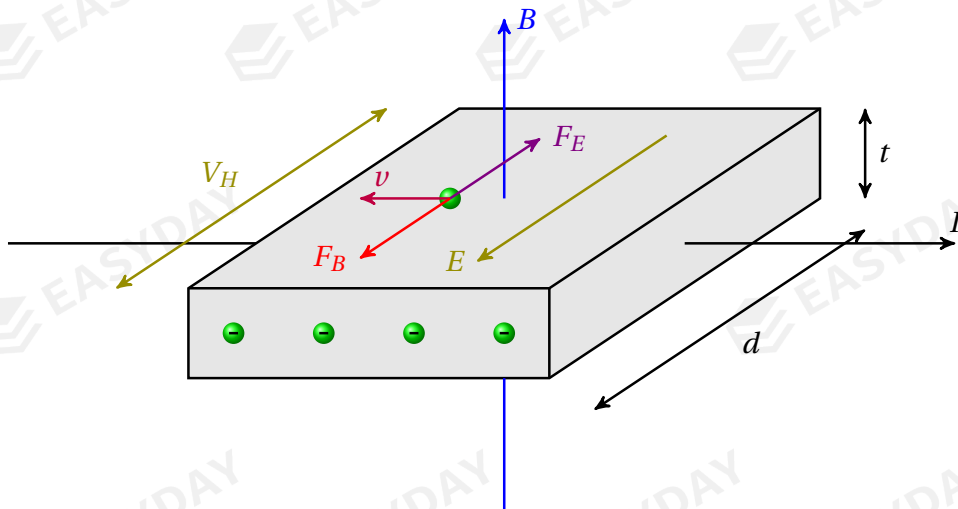
when an electric current passes through a conductor surrounded by an external magnetic field perpendicular to the current, a transverse potential difference is produced

this phenomenon is known as the **Hall effect** (discovered by American physicist *Edwin Hall* in 1879), the voltage difference is called the **Hall voltage**

suppose magnetic field goes upward, current flows to the right (see figure), then charge carriers (assumed to be electrons for now) experience a magnetic force pointing out of paper



this magnetic force causes a build-up of extra charge on front side of conductor, which produces an internal electric field E inside conductor, i.e., a potential difference V_H is formed



杨宇豪 8 磁场 8.4 霍尔效应当电流通过被垂直于电流的外部磁场包围的导体时，会产生横向电势差，这种现象称为霍尔效应（由美国物理学家埃德温·霍尔于 1879 年发现），该电压差称为霍尔电压。假设磁场向上，电流向右流动（见图），则电荷载流子（假设为电子）受到指向纸外的磁力 $\mathbf{B} \times \mathbf{v}$ 该磁力导致导体正面积聚额外电荷，从而在导体内部产生内部电场 $\mathbf{E}$ ，即形成电势差 V_H $V_H = \frac{B I v}{n e d}$ 98

as charge carriers accumulate on one side, Hall voltage opposes further migration of charges
steady V_H is established when magnetic force and electric force are balanced: $Eq = Bqv$

- strength of internal electric field is related to Hall voltage: $E = \frac{V_H}{d}$

where d is width of the conductor

- drift velocity v of particles is related to current I : $I = nAqv$

where A is cross section of conductor and n is number density of charge carriers

we now have: $\frac{V_H}{d}q = Bq\frac{I}{nAq} \Rightarrow V_H = \frac{BId}{nAq}$

note that cross section $A = dt$, where t is thickness of conductor as shown

we therefore obtain an useful expression for Hall voltage: $V_H = \frac{BId}{n(td)q} \Rightarrow V_H = \frac{BI}{ntq}$

➤ to produce noticeable V_H , we want smaller n and smaller t

- $n \downarrow \Rightarrow$ few free charge carriers $\Rightarrow$ *semi-conductors* are preferable
- $t \downarrow \Rightarrow$ small thickness $\Rightarrow$ *thin slice* of component is preferable

➤ polarity of V_H is determined by nature of charge carries

charge carriers can be *free electrons* (negatively charged) or *holes* (positively charged)

then Hall voltage induced would have opposite polarity

➤ if current I forms angle θ with flux density B , again only perpendicular component matters

expression for Hall voltage becomes: $V_H = \frac{BI}{ntq} \sin\theta$

➤ apply a fixed current in a conductor, V_H is proportional to B

so magnetic flux density can be calculated once we find V_H

one can make use of Hall effect to build a *Hall probe*, a device used to measure flux density B

when using a Hall probe, one should rotate and record the greatest voltage reading to ensure the current applied is at right angle to the external magnetic field

Question 8.17 Why is it difficult to detect Hall voltage in a thin slice of copper?

Question 8.18 A Hall probe is placed near to one end of a strong magnet. State and explain the variation in the Hall voltage as the probe is rotated for one complete revolution.

Yuhao Yang 8 磁场当载流子在一侧聚集时，霍尔电压阻止电荷进一步迁移当磁力和电力平衡时，建立稳定的 V_H ： $E_q = B q v$ - 内部电场强度与霍尔电压有关： $E = V_H d$ 其中 d 是导体宽度 - 粒子的漂移速度 v 与电流 I 相关： $I = n A q v$ 其中 A 是导体横截面， n 是载流子数密度我们现在有： $V_H d q = B q I n A q$) $V_H = B I d n A q$ 请注意，横截面 $A = d t$ ，其中 t 是导体的厚度，如图所示，因此我们获得了霍尔电压的有用表达式： $V_H = B I d n (t d) q$) $V_H = B I n t q \hat{a}$ 为了产生明显的 V_H ，我们希望更小的 n 和更小的 t - $n \#$) 很少的自由电荷载流子) 半导体是优选的 - $t \#$) 小厚度) 元件的薄片是优选的 - V_H 的极性由电荷载流子的性质决定电荷载流子可以是自由电子 (带负电) 或空穴 (带正电)，那么感应的霍尔电压将具有相反的极性 - 如果电流 I 与磁通密度 B 形成角度 θ ，同样只有垂直分量才重要霍尔电压的表达式变为： $V_H = B I n t q \sin \theta$ - 在导体中施加固定电流， V_H 与 B 成正比，因此一旦找到 V_H 就可以计算出磁通密度。可以利用霍尔效应制作一个霍尔探头，用于测量磁通密度的装置。B 使用霍尔探头时，应旋转并记录最大电压读数，以确保施加的电流与外部磁场成直角。问题 8.17 为什么在薄铜片中检测霍尔电压很困难？问题 8.18 将霍尔探头放置在强磁铁的一端附近。陈述并解释当探头旋转一整圈时霍尔电压的变化。 99

CHAPTER 9

Electromagnetic Induction

9.1 magnetic flux

magnetic flux is defined as the product of a closed area and the magnetic flux density going through it at right angles: $\phi = BA \cos \theta$

➤ unit for magnetic flux: $[\phi] = \text{Wb}$, $1 \text{ WB} = 1 \text{ T} \cdot \text{m}^2$

➤ if magnetic field is perpendicular to the area
magnetic flux simply becomes: $\phi = BA$

➤ for a coil with N turns, total magnetic flux is

$$\Phi = N\phi = NBA$$

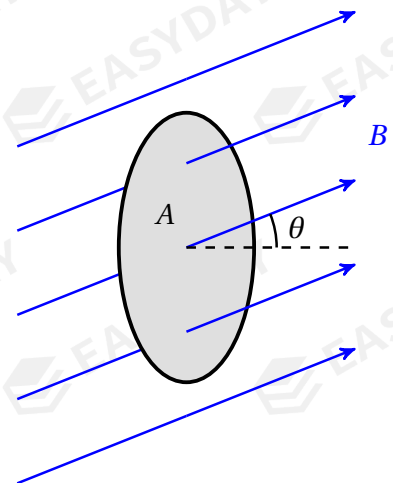
this is called **magnetic flux linkage**

➤ magnetic flux ϕ can be graphically thought as the number of field lines through area A

cutting of field lines means a change in flux

➤ change in flux can be caused by many processes, for example

- moving a magnet towards/away from a coil
- varying current in a solenoid/electromagnet
- inserting an iron core into a solenoid/electromagnet
- pushing a straight conductor through a magnetic field
- rotating a coil in a magnetic field
- ...



Question 9.1 Explain why the processes mentioned above would give rise to a change in magnetic flux.

杨宇豪 9 电磁感应第 9 章电磁感应 9.1 磁通量磁通量定义为闭合区域与以直角穿过该区域的磁通密度的乘积： $\psi = B A \cos \theta$ $A \theta B \hat{a}$ 磁通量单位： $[\psi] = \text{Wb}$, $1 \text{ WB} = 1 \text{ T} \cdot \text{m}^2$ $\hat{a}$ if 磁场垂直于区域磁通量简单地变为： $\Phi = B A \hat{a}$ 对于 N 匝线圈，总磁通量为 $\Phi = N \Phi = N B A$ 这称为磁通链 $\hat{a}$ 磁通量 Φ 可以图形化地视为穿过区域的磁力线数量磁力线的切割意味着通量的变化 $\hat{a}$ 磁通量的变化可以由许多过程引起，例如 - 将磁铁移近/移开从线圈 - 改变螺线管/电磁铁中的电流 - 将铁芯插入螺线管/电磁铁 - 将直导体推过磁场 - 在磁场中旋转线圈 - 问题 9.1 解释为什么上述过程会引起磁通量的变化。 100

9.2 laws of electromagnetic induction

electromagnetic induction is the phenomena that magnetism produces electricity

the laws of electromagnetic induction were discovered by *Michael Faraday* and *Heinrich Lenz* in the 1830s, and later described mathematically by *James Clerk Maxwell*

we will study in what conditions voltages and currents could be induced, and how to find their magnitudes and polarities

9.2.1 Faraday's law

Faraday's law states that induced e.m.f./voltage is proportional to rate of change in magnetic flux (linkage): $\mathcal{E} \propto \frac{\Delta\Phi}{\Delta t}$

➤ Faraday's law gives the *magnitude* of the induced e.m.f.

the key here is the *change* in flux: as long as flux changes, e.m.f. will be induced

➤ if $\mathcal{E}$, Φ , t are all given in SI units, this proportional relation becomes an identity: $\mathcal{E} = \frac{\Delta\Phi}{\Delta t}$

Example 9.1 A coil of 80 turns is wound tightly around a solenoid with a cross-sectional area of 35 cm^2 . The flux density at the centre of the solenoid is 75 mT. (a) What is the flux linkage in the coil? (b) The current in the solenoid is *reversed* in direction in a time of 0.40 s, what is the average e.m.f. induced?

✎ flux linkage: $\Phi = NBA = 80 \times 75 \times 10^{-2} \times 35 \times 10^{-4} = 0.21 \text{ Wb}$

average e.m.f. induced: $\mathcal{E} = \frac{\Delta\Phi}{\Delta t} = \frac{(+\Phi) - (-\Phi)}{\Delta t} = \frac{2 \times 0.21}{0.40} = 1.05 \text{ V}$ $\square$

9.2.2 Lenz's law

Lenz's law states that induced e.m.f. or current is always in the direction to produced effects that *oppose* the change in flux that produced it

➤ Lenz's law gives the *polarity* of the induced current

the key here is the nature dislikes any change in flux: effects of induced current always *oppose* the cause of its production

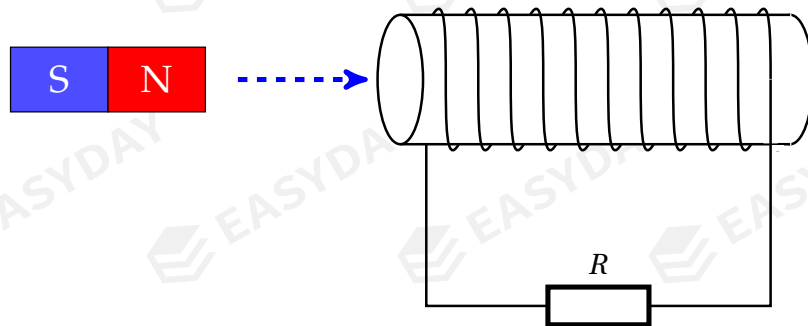
杨宇豪 9 电磁感应 9.2 电磁感应定律 电磁感应定律是磁产生电的现象。电磁感应定律是由迈克尔·法拉第和海因里希·伦茨在 1830 年代发现的，后来由詹姆斯·克拉克·麦克斯韦用数学方法描述。我们将研究在什么条件下可以感应出电压和电流，以及如何找到它们的大小和极性 9.2.1 法拉第定律 法拉第定律指出，感应电动势/电压与磁通量（联动）的变化率成正比： $\mathcal{E} / \Delta \Phi = \Delta t$ 法拉第定律给出了感应电动势的大小。这里的关键是通量的变化：只要通量变化，电动势就会变化。如果 $\mathcal{E}$ 、 Φ 、 t 均以 SI 单位给出，则该比例关系变为恒等式： $\mathcal{E} = \Delta \Phi / \Delta t$ 例 9.1 将 80 匝线圈紧紧缠绕在横截面积为 35 cm^2 的螺线管上。螺线管中心的磁通密度为 75 mT 。(a) 线圈中的磁链是什么？(b) 螺线管中的电流在 0.40 s 时间内反向，则平均电动势是多少。诱发？ - 磁链： $\Phi = N B A = 80 \times 75 \times 10^{-2} \times 35 \times 10^{-4} = 0.21 \text{ Wb}$ 平均电动势 感应电动势： $\mathcal{E} = \Delta \Phi / \Delta t = (+\Phi) - (-\Phi) / \Delta t = 2 \times 0.21 / 0.40 = 1.05 \text{ V}$ 9.2.2 楞次定律 楞次定律指出感应电动势。或者电流总是朝着产生与产生它的通量变化相反的效应的方向 楞次定律给出了感应电流的极性 这里的关键是大自然不喜欢通量的任何变化：感应电流的效应总是与其产生的原因相反 101

➤ Lenz's law is a consequence of the conservation of energy

since induced current dissipates electrical energy as heat, it must cause loss of original forms of energy possessed by the system

if the change in flux is caused by motion, then magnetic force on/by the induced current must *resist* this motion

Example 9.2 A coil of 200 turns is connected to a resistor $R = 4.0 \, \Omega$. The coil has a diameter 10 cm. Initially a bar magnet is at great distance from the coil. The magnet is then inserted into the coil and field inside coil becomes 0.40 T. The process occurs within a duration of 2.0 s.



(a) What is average induced e.m.f. in the coil? (b) What is average induced current through resistor R ? (c) In which direction does this current flow?

✍ change in flux linkage: $\Delta\Phi = NBA - 0 = 200 \times 0.40 \times \pi \times 0.050^2 \approx 0.628 \text{ Wb}$

average e.m.f. induced: $\mathcal{E} = \frac{\Delta\Phi}{\Delta t} = \frac{0.628}{2.0} \approx 0.314 \text{ V}$

average current induced: $I = \frac{\mathcal{E}}{R} = \frac{0.314}{4.0} \approx 0.0785 \text{ A}$

as magnet approaches, coil experiences an increasing flux to the right

according to Lenz's law, field due to induced current acts to the left to oppose the change

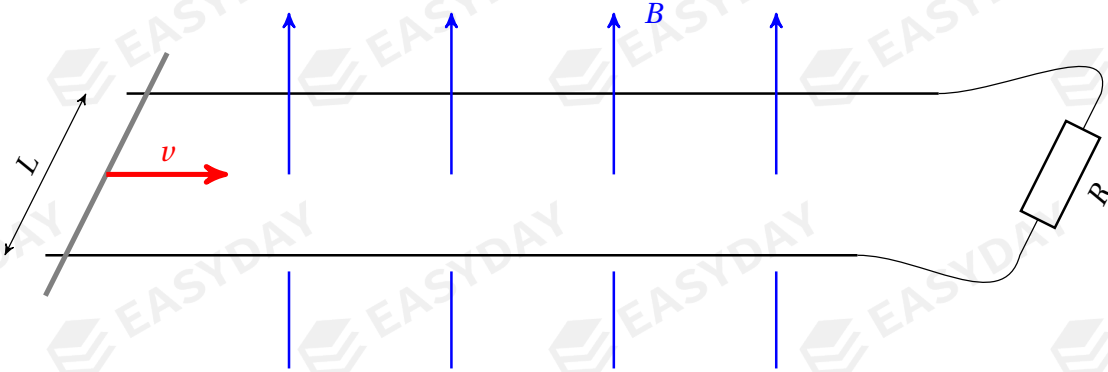
alternatively, left side of the coil should behave as a north pole to oppose motion of magnet

use right-hand rule, we find induced current flows through R to the right

Question 9.2 If the magnet is now pulled away from the coil, state and explain the direction of the induced current.

Yuhao Yang 9 电磁感应 楞次定律是能量守恒的结果，因为感应电流将电能以热的形式耗散，如果磁通量的变化是由运动引起的，则必定会导致系统所拥有的原始能量形式的损失，则感应电流上/由感应电流产生的磁力必须抵抗这种运动。示例 9.2 一个 200 匝的线圈连接到一个电阻 $R = 4.0 \, \Omega$ 。线圈的直径为 10 厘米。最初，条形磁铁距离线圈很远。然后将磁铁插入线圈中，线圈内的磁场变为 $0.40 \, \text{T}$ 。该过程发生在 2.0 秒的持续时间内。RSN (a) 什么是平均感应电动势？在线圈中？(b) 通过电阻器 R 的平均感应电流是多少？(c) 该电流朝哪个方向流动？ - 磁链变化： $\Delta \Phi = N B A - 0 = 200 \times 0.40 \times \pi \times 0.0502^2 \approx 0.628 \, \text{Wb}$ 平均电动势感应： $E = \frac{\Delta \Phi}{\Delta t} = \frac{0.628}{2.0} \approx 0.314 \, \text{V}$ 平均感应电流： $I = \frac{E}{R} = \frac{0.314}{4.0} \approx 0.0785 \, \text{A}$ 当磁铁接近时，根据楞次定律，线圈会经历向右增加的磁通量，由于感应电流作用于左侧以对抗变化，线圈的左侧应表现出当北极阻止磁体运动时，使用右手定则，我们发现感应电流通过 R 向右流动。问题 9.2 如果现在将磁体拉离线圈，请说明并解释感应电流的方向。 102

Example 9.3 Two parallel metal tracks are separated by a distance of $L = 45 \text{ cm}$ and placed in a uniform magnetic field of 0.80 T . The tracks are connected to a resistor $R = 6.0 \Omega$. A long metal rod is being pushed under an external force at $v = 3.0 \text{ m s}^{-1}$ along the tracks as shown.



- (a) What is the induced current through resistor? (b) In which direction does this current flow?
 (c) For the rod to travel at constant speed, what is magnitude of external force required?

change of flux in time interval Δt is: $\Delta\phi = \Delta(BA) = B\Delta A = BLv\Delta t$

e.m.f. induced: $\mathcal{E} = \frac{\Delta\phi}{\Delta t} \Rightarrow \boxed{\mathcal{E} = BLv}$

induced current: $I = \frac{\mathcal{E}}{R} = \frac{BLv}{R} = \frac{0.80 \times 0.45 \times 3.0}{6.0} = 0.18 \text{ A}$

according to Lenz's law, magnetic force on induced current opposes the rod's motion of cutting field lines, so magnetic force acts to the left

using Fleming's left-hand rule, we find induce current flows in anti-clockwise direction

if rod travels at constant speed, then equilibrium between external push and magnetic force

external force required: $F_{\text{ext}} = F_B = BIL = 0.80 \times 0.18 \times 0.45 \approx 0.065 \text{ N}$ $\square$

Question 9.3 In Example 9.3, can you find alternative arguments to determine the direction of the current induced as the rod cuts through the magnetic field?

Hall voltage & induced voltage in a coil

in this section, we compare readings on voltmeter connected to a Hall probe and a coil

Hall probe picks up voltage proportional to flux density: $V_H \propto B$

coil picks up induced e.m.f. proportional to *rate of change* in flux: $\mathcal{E} \propto \frac{d\Phi}{dt}$

Question 9.4 If a Hall probe is placed near a permanent magnet, what voltage do you measure?

Yuhao Yang 9 电磁感应例9.3 两条平行的金属轨道相距 $L \leq 45 \text{ cm}$ ，并放置在 0.80 T 的均匀磁场中。轨道连接到电阻 $R \leq 6.0 \Omega$ 。如图所示，一根长金属棒在 $v = 3.0 \text{ m s}^{-1}$ 的外力作用下沿着轨道被推动。

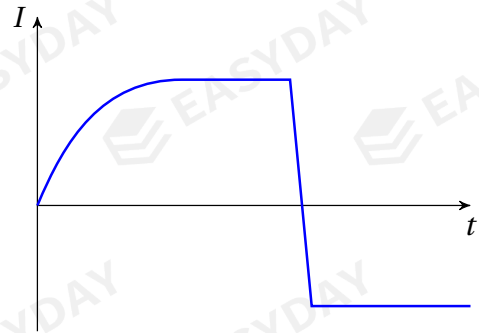
(a) 通过电阻器的感应电流是多少？ (b) 该电流朝哪个方向流动？ (c) 为使杆匀速运动，需要多大的外力？

- 时间间隔 Δt 内的通量变化为： $\Delta \Phi = \Delta (B A) = B \Delta A = B L v \Delta t$ e.m.f. 感应： $E = \Delta \Phi / \Delta t$ $E = B L v$ 感应电流： $I = E / R = B L v / R = 0.80 \times 0.45 \times 3.0 / 6.0 = 0.18 \text{ A}$ 根据楞次定律，感应电流上的磁力与杆切割磁力线的运动相反，因此根据弗莱明左手定则，磁力作用于左侧，我们求如果杆以恒定速度运动，则感应电流沿逆时针方向流动，则需要外部推力和磁力之间达到平衡： $F_{\text{ext}} = F_B = B I L = 0.80 \times 0.18 \times 0.45 = 0.065 \text{ N}$

问题 9.3 在例 9.3 中，你能找到其他参数来确定杆穿过磁场时感应电流的方向吗？霍尔电压和线圈中的感应电压在本节中，我们比较连接到霍尔探头的电压表和线圈上的读数霍尔探头拾取与磁通密度成比例的电压： V_H / B 线圈拾取感应电动势。与磁通变化率成正比： $E = d\Phi / dt$ 问题 9.4 如果将霍尔探头放置在永磁体附近，您会测量到什么电压？ 103

If a small coil is placed in the same field, state and explain whether you can measure a non-zero voltage in the coil. If not, state three different ways in which you can produce a voltage.

Question 9.5 A Hall probe is placed near one end of a solenoid that carries a varying current as shown in the graph. (a) Sketch the variation of Hall voltage with time. (b) If Hall probe is replaced by a small coil parallel to the solenoid's end, sketch the variation of the voltage induced in the coil.



9.3 applications of electromagnetic induction

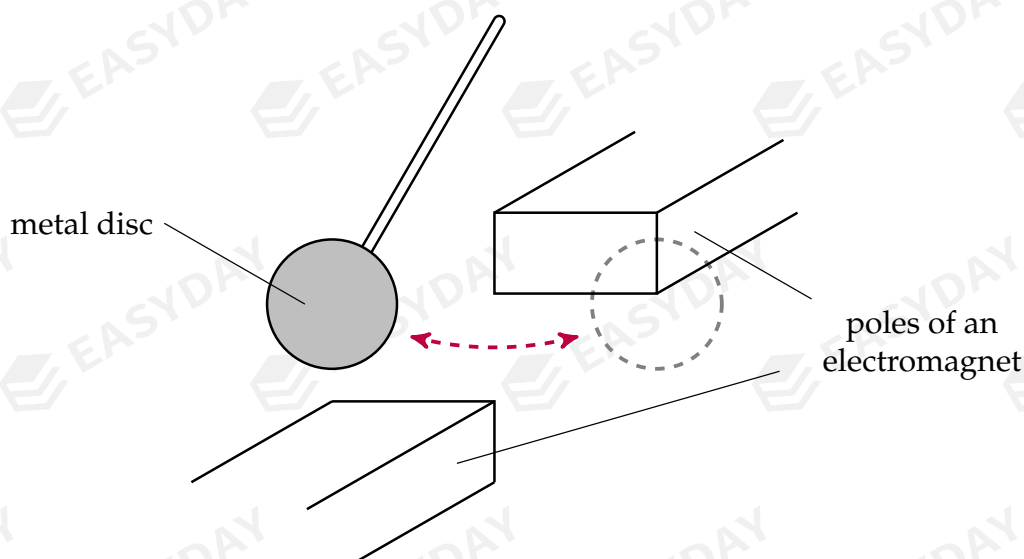
9.3.1 magnetic braking

induction brakes make use of induced current to dissipate kinetic energy of a moving object
in this section, we will look at two simple demonstrations, but the principle can be used in
braking system of high-speed trains, roller-coasters, etc.

the damped pendulum

a metal disc can swing freely between the poles of an electromagnet

when the electromagnet is switched on, the disc comes to rest very quickly



Yuhao Yang 9 电磁感应如果在同一磁场中放置一个小线圈，请说明并解释是否可以测量线圈中的非零电压。如果没有，请说明产生电压的三种不同方法。问题 9.5 将霍尔探头放置在带有变化电流的螺线管一端附近，如图所示。(a) 绘制霍尔电压随时间的变化。(b) 如果用与螺线管末端平行的小线圈代替霍尔探头，请画出线圈中感应电压的变化。9.3 电磁感应的应用 9.3.1 磁力制动感应制动器利用感应电流来耗散运动物体的动能，本节我们将看两个简单的演示，但其原理可用于高速列车、过山车等制动系统。当电磁铁接通时，金属圆盘的阻尼摆可以在电磁铁的磁极之间自由摆动，圆盘很快就静止在电磁铁的磁极上金属盘104

as disc moves in and out of field, change in flux gives rise to e.m.f induced

note that different parts of the disc experience different rates of change in flux, different e.m.f. is induced in different parts for the disc

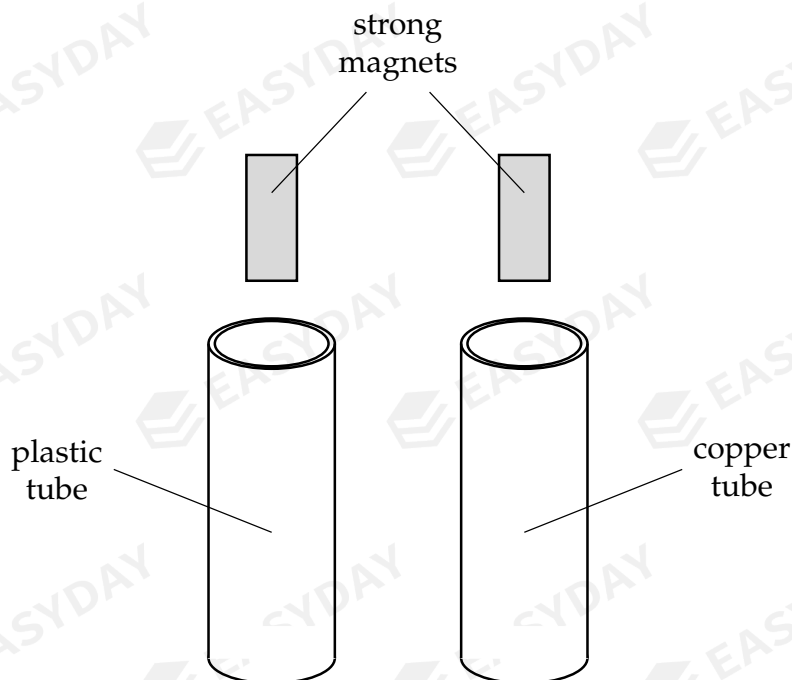
this causes circulating currents flowing in the disc, called **eddy currents**

- vibrational energy is lost as heat due to the eddy currents induced
- magnetic force on induced current causes damping

so amplitude of oscillation decreases quickly

falling magnet

suppose we drop two strong magnets down a plastic tube and a copper tube



as magnet falls, tube experiences change in flux, so e.m.f is induced

since plastic is an insulator, so no current flows in plastic tube, magnet undergoes free fall

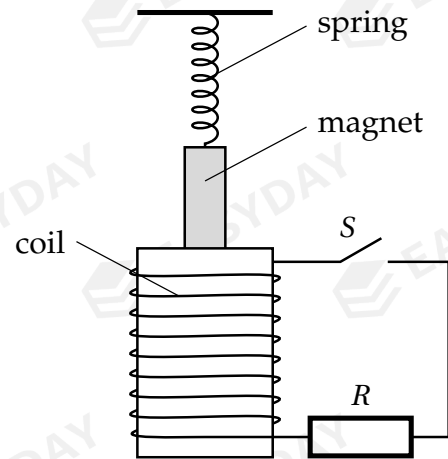
but in the conducting copper tube, eddy current is induced in the tube

- energy is lost as heat due to induced current, not all G.P.E. converted into K.E.
- induced current exerts magnetic force on magnet to oppose its motion, acceleration $a < g$

so magnet falls more slowly through the copper tube

Yuhao Yang 9 电磁感应当圆盘移入和移出磁场时，通量的变化会引起电动势感应，即圆盘的不同部分经历不同的通量变化率、不同的电动势。在圆盘的不同部分感应出这会导致在圆盘中流动的循环电流，称为涡流 - 由于涡流感应，振动能量以热量的形式损失 - 感应电流上的磁力导致阻尼，因此振荡幅度迅速减小磁铁下落假设我们将两个强磁铁放入塑料管和铜管强磁铁塑料管铜管当磁铁下落时，管子经历通量变化，因此会产生电动势，因为塑料是绝缘体，因此没有电流在塑料管中流动时，磁铁会自由落体，但在导电铜管中，管内会产生涡流 - 能量因感应电流而以热量形式损失，并非所有 G.P.E. 转换为 K.E. - 感应电流对磁体施加磁力以阻止其运动，加速度为 ζg ，因此磁体通过铜管 105 下落的速度更慢

Question 9.6 A magnet is suspended from the free end of a spring. When displaced vertically and released, the magnet can oscillate in and out of a coil (see diagram). The switch S is initially open, there is negligible change in the amplitude. However, when the switch is closed, the amplitude is seen to decrease quickly. Explain the reasons.



9.3.2 the generator

imagine a coil rotates with constant angular speed ω in a uniform magnetic field B

let's assume that the coil initially lies in parallel

to the field, i.e., $\theta = 0$ at $t = 0$

at time t , coil forms an angle $\theta = \omega t$ with the magnetic field (see diagram)

magnetic flux linkage through coil is:

$$\Phi = NBA \sin \theta = NBA \sin \omega t$$

magnitude of induced e.m.f. is:

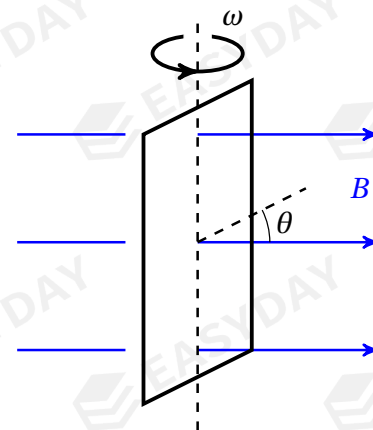
$$\mathcal{E} = \frac{d\Phi}{dt} = \omega NBA \cos \omega t$$

this is a sinusoidal voltage with maximum value: $\mathcal{E}_{\max} = \omega NBA$

for a coil rotating in a magnetic field like this, an *alternating current* is generated

this is basically how a *generator* works

in practice, it is a strong electromagnet rotating inside a large coil to generate electricity



Question 9.7 State and explain the effect on the voltage generated if the coil rotates faster.

Question 9.8 For the coil rotating at a uniform angular speed in a uniform magnetic field, is the magnetic flux in phase with the e.m.f. generated? If not, what is the phase difference?

Question 9.9 Does the generator output the maximum voltage when the rotating coil is in parallel to the magnetic field or when the coil is at right angle to the field?

Yuhao Yang 9 电磁感应 R 弹簧磁铁线圈 S 问题 9.6 磁铁悬挂在弹簧的自由端。当垂直移动并释放时，磁铁可以在线圈中振荡进出（见图）。开关 S 最初是打开的，幅度的变化可以忽略不计。然而，当开关闭合时，可以看到幅度迅速减小。解释原因。

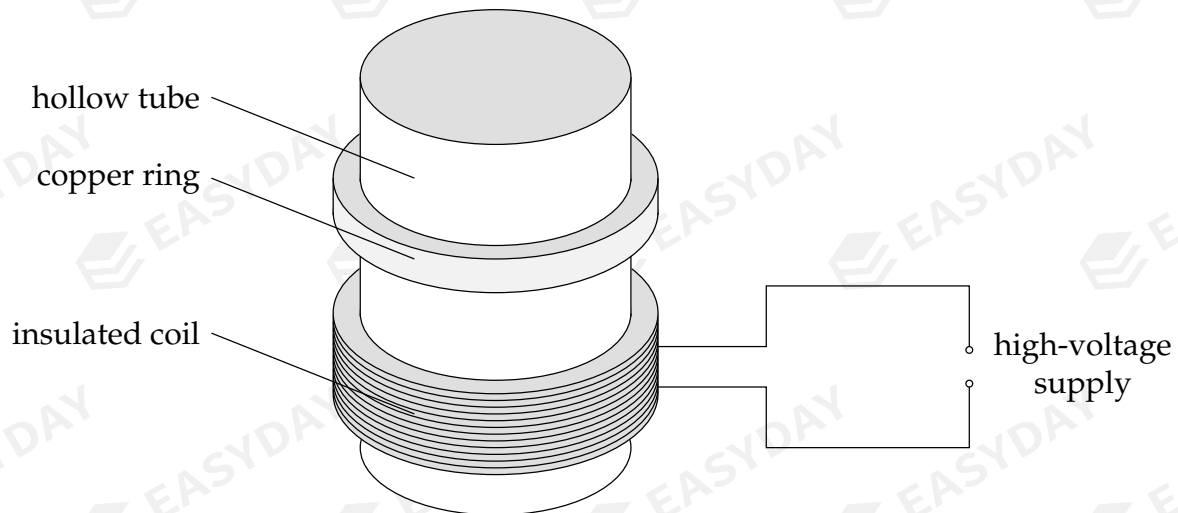
9.3.2 发电机假设线圈在均匀磁场中以恒定角速度 ω 旋转 $B \omega B \theta$ 假设线圈最初与磁场平行，即 t 时刻 $\theta = 0 = 0$ 时刻 t ，线圈与磁场形成角度 $\theta = \omega t$ （见图）穿过线圈的磁通量为： $\Phi = N B A \sin \theta = N B A \sin \omega t$ 感应电动势大小是： $E = d\Phi / dt = \omega N B A \cos \omega t$ 这是一个具有最大值的正弦电压： $E_{\max} = \omega N B A$ 对于在这样的磁场中旋转的线圈，会产生交流电这基本上就是发电机在实践中的工作原理，它是一个在大线圈内旋转以发电的强电磁体问题 9.7 说明并解释如果线圈旋转得更快，对产生的电压的影响。问题 9.8 对于在均匀磁场中以均匀角速度旋转的线圈，磁通量与电动势同相吗？生成？如果不是，相位差是多少？问题 9.9 当旋转线圈与磁场平行或与磁场成直角时，发电机输出的电压最大吗？ 106

9.3.3 electromagnetic gun

a coil of wire of many turns is wound on a hollow tube

a light copper ring that can move freely along the tube is placed on the coil

let's find out what happens to the ring when a high-voltage supply is switched on



when supply is switched on, flux density in coil greatly increases

sudden change in magnetic flux could induce a huge e.m.f. in copper ring

induced current in ring would experience a repulsive magnetic force

this repulsive force could be way larger than ring's weight, ring would jump out from tube

if the ring is replaced by a small metal sphere placed inside the tube, the same strong magnetic force arising from a sudden change in flux could fire the sphere at very high speed

Question 9.10 The coil is now connected to a stable d.c. voltage supply. If we quickly insert an iron core into the tube, what might happen to the light copper ring?

other applications

as a final remark, electromagnetic induction is widely used in many other areas as well

for those who are interested, you may research on the principles behind wireless charging, induction cooking, contactless payment technology, smart pencils for tablets or computers, etc.

杨宇浩 9 电磁感应 9.3.3 电磁枪在空心管上缠绕多匝线圈，在线圈上放置一个可以沿管自由移动的轻质铜环，我们看看接通高压电源时，环会发生什么情况空心管铜环绝缘线圈高压电源接通电源时，线圈中的磁通密度大大增加，磁通的突然变化会感应出巨大的电动势。在铜环中，环中的感应电流会受到排斥磁力，这种排斥力可能比环的重量大得多，如果用放置在管内的小金属球代替环，环就会从管中跳出，通量突然变化产生的相同强磁力可以以非常高的速度点燃球体。问题9.10线圈现在连接到稳定的直流电。电压供应。如果我们快速将铁芯插入管子中，轻铜环可能会发生什么？其他应用最后说一下，电磁感应在许多其他领域也有广泛的应用，有兴趣的人可以研究一下无线充电、感应烹饪、非接触式支付技术、平板电脑或电脑智能铅笔等背后的原理107

CHAPTER 10

Alternating Currents

a **direct current** (d.c.) flows in one direction only

an **alternating current** (a.c.) reverses its flow direction from time to time

a.c. has certain advantages than d.c., as you will see in this chapter

we will study the mathematics of a.c. and the transmission process for a.c.

10.1 sinusoidal a.c.

as we have seen in §9.3.2, currents produced from generators are naturally sinusoidal

sinusoidal a.c. is one of the most important types of a.c. waveform in electrical engineering.

for most cases in this course, we focus on a.c. that varies like a sine wave

10.1.1 sinusoidal waveform

current: $I = I_0 \sin \omega t$ / voltage: $V = V_0 \sin \omega t$

➤ I_0, V_0 are called *peak current* and *peak voltage* (amplitudes of the a.c. signal)

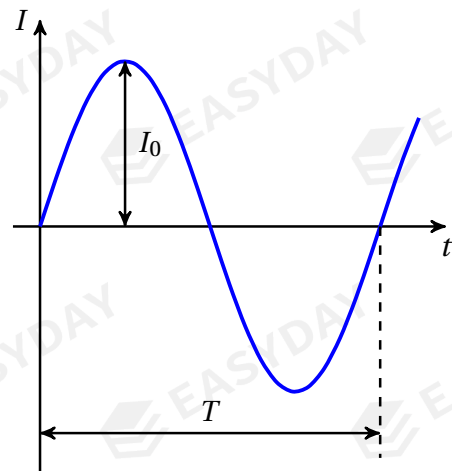
➤ ω is **angular frequency**, which describes how fast a.c. signal oscillates (same idea as for simple harmonic motion, see §3)

frequency and period of the signal are given by:

$$\omega = 2\pi f, \quad T = \frac{1}{f} = \frac{2\pi}{\omega}$$

➤ mean current: $\langle I \rangle = 0$, mean voltage: $\langle V \rangle = 0$

this is because a.c. signal fluctuates with time, positive and negative bits cancel out



Yuhao Yang 10 交流电第 10 章交流电直流电 (d.c.) 沿一个方向流动，只有交流电 (a.c.) 有时会反转其流动方向。比直流有一定的优势，正如您将在本章中看到的，我们将研究交流的数学。以及交流电的传输过程

10.1 正弦交流电正如我们在第 9.3.2 节中看到的，发电机产生的电流自然是正弦交流电。是最重要的交流电类型之一。电气工程中的波形。对于本课程的大多数情况，我们关注交流。

10.1.1 正弦波形

电流： $i = I_0 \sin \omega t$ / 电压： $V = V_0 \sin \omega t$ I_0 、 V_0 称为峰值电流和峰值电压（交流信号的幅度） ω 是角频率，它描述了交流电的速度。信号振荡（与简谐运动相同，参见第 3 节）信号的频率和周期由下式给出： $\omega = 2\pi f$, $T = 1/f = 2\pi/\omega$

平均电流： $\langle i \rangle = 0$ ，平均电压： $\langle V \rangle = 0$ 这是因为交流电。信号随时间波动，正负位相互抵消

Question 10.1 An alternating voltage has a peak value of 16 V and period of 0.10 s. Write down a mathematical equation that describes the variation of this voltage.

Question 10.2 An alternating voltage is produced from a simple generator. If the rotating speed of the coil in the generator doubles, describe quantitatively the change in the peak value and frequency of the alternating voltage.

Question 10.3 A student argues that when an alternating current is driven through a resistor, the mean current is zero, so an alternating current does not produce heating power on the resistor. State and explain whether this is correct.

10.1.2 power

electrical power dissipated in a resistor: $P = I^2 R$, or, $P = \frac{V^2}{R}$

since $I^2, V^2 \geq 0$, P can never be negative, so an a.c. can produce effective power in a resistor

note that for an a.c., P keeps changing with time, as I and V are both varying with time

in everyday life, we are more concerned about the *mean power*

mean power output for an a.c. is: $\langle P \rangle = \langle I^2 \rangle R = \frac{\langle V^2 \rangle}{R}$

so we see the necessity to introduce mean square values $\langle I^2 \rangle$ and $\langle V^2 \rangle$

let's further introduce root mean square (r.m.s.) values: $I_{\text{rms}} = \sqrt{\langle I^2 \rangle}$, and $V_{\text{rms}} = \sqrt{\langle V^2 \rangle}$

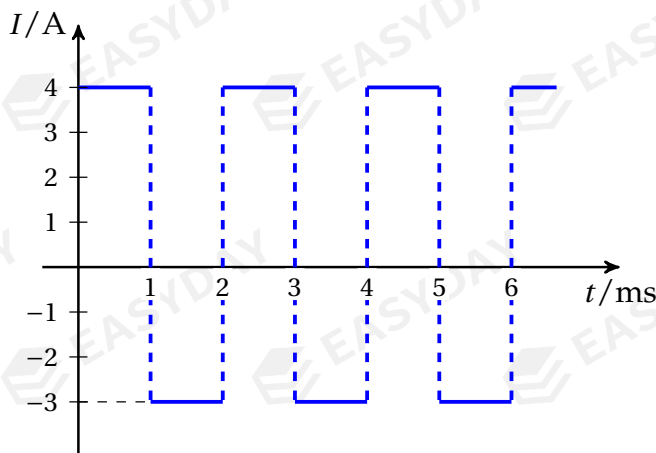
we can now write the mean power for a.c. as: $\langle P \rangle = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}$

Example 10.1 The variation with time of an alternating current in a resistor of 120Ω is shown. (a) What is the value of its r.m.s. current? (b) What is the mean power dissipated by the resistor?

$$\langle I^2 \rangle = \frac{4^2 + (-3)^2}{2} = 12.5 \text{ A}^2$$

$$I_{\text{rms}} = \sqrt{\langle I^2 \rangle} = \sqrt{12.5} \approx 3.54 \text{ A}$$

$$\langle P \rangle = I_{\text{rms}}^2 R = 12.5 \times 120 = 1500 \text{ W} \quad \square$$



Yuhao Yang 10 交流电问题 10.1 交流电压的峰值为 16 V ，周期为 0.10 s 。写下描述该电压变化的数学方程。问题 10.2 交流电压由简单的发电机产生。如果发电机中线圈的转速增加一倍，定量地描述交流电压的峰值和频率的变化。问题 10.3 一名学生认为，当交流电流过电阻器时，平均电流为零，因此交流电不会在电阻器上产生热功率。陈述并解释这是否正确。

10.1.2 功率电阻器中消耗的电功率： $P = I^2 R$ ，或 $P = V^2 / R$ 因为 I 和 V 随时间变化， P 永远不可能为负，因此交流电可以在电阻器中产生有效功率。注意，对于交流电， P 随时间变化，因为日常生活中 I 和 V 都随时间变化，我们更关心交流电的平均功率平均输出功率。是： $\langle P \rangle = \langle I^2 \rangle R = \langle V^2 \rangle / R$ 因此，我们看到引入均方值 $\langle I^2 \rangle$ 和 $\langle V^2 \rangle$ 的必要性，让我们进一步引入均方根 (r.m.s.) 值： $I_{\text{rms}} = \sqrt{\langle I^2 \rangle}$ 和 $V_{\text{rms}} = \sqrt{\langle V^2 \rangle}$ 我们现在可以写出交流的平均功率。为： $\langle P \rangle = I_{\text{rms}}^2 R = V_{\text{rms}}^2 / R$

例 10.1 显示 120Ω 电阻中交流电流随时间的变化。(a) 其均方根值是多少？当前的？(b) 电阻器消耗的平均功率是多少？

- $\langle I^2 \rangle = \frac{1}{T} \int_0^T I^2 dt = \frac{1}{6} [4^2 + (-3)^2 + \dots + (-3)^2] = 12.5\text{ A}^2$
 $I_{\text{rms}} = \sqrt{\langle I^2 \rangle} = \sqrt{12.5} = 3.54\text{ A}$
 $\langle P \rangle = I_{\text{rms}}^2 R = 12.5 \times 120 = 1500\text{ W}$

10.1.3 r.m.s. current & r.m.s. voltage

in last section, we studied the mathematical aspect of the r.m.s. value

but we still need a definition for r.m.s. current and r.m.s. voltage from a physical viewpoint

physically, r.m.s. value of an alternating current is defined as follows

r.m.s. current/voltage of an a.c. equals a steady d.c. current/voltage that delivers same average power to a resistive load

➤ for *sine waves*, r.m.s values are related to peak values by: $I_{\text{r.m.s}} = \frac{1}{\sqrt{2}} I_0$ and $V_{\text{r.m.s}} = \frac{1}{\sqrt{2}} V_0$

proof (*): total energy dissipation in one period is: $W_T = \int_0^T P dt$

so mean power can be given by: $\langle P \rangle = \frac{W_T}{T} = \frac{1}{T} \int_0^T P dt$

substitute $P = I^2 R \stackrel{I=I_0 \sin \omega t}{=} I_0^2 R \sin^2 \omega t$, we have: $\langle P \rangle = \frac{I_0^2 R}{T} \int_0^T \sin^2 \omega t dt$

the integral is carried out: $\int_0^T \sin^2 \omega t dt = \frac{1}{2} \int_0^T (1 - \cos 2\omega t) dt = \frac{1}{2} \left(t - \frac{\sin 2\omega t}{2\omega} \right) \Big|_0^T = \frac{1}{2} T$

now we have: $\langle P \rangle = I_{\text{rms}}^2 R = \frac{1}{2} I_0^2 R \Rightarrow I_{\text{rms}} = \frac{1}{\sqrt{2}} I_0$

a similar calculation for voltage would show: $V_{\text{rms}} = \frac{1}{\sqrt{2}} V_0$ $\square$

➤ it is worth pointing out that the $\frac{1}{\sqrt{2}}$ -relation for r.m.s. values only holds for sine waves

for other waveforms, e.g., square waves or triangle waves, numerical constant is different

➤ value of voltages stated for mains electricity supply usually refers to the r.m.s. value^[62]

Example 10.2 An a.c. power supply produces a sinusoidal output across a resistor of 30 Ω .

The maximum voltage is found to be 75 V. Find energy dissipated in the resistor in 2.0 minutes.

✏ r.m.s voltage: $V_{\text{rms}} = \frac{1}{\sqrt{2}} V_0 = \frac{75}{\sqrt{2}} \approx 53.0 \text{ V}$

mean power output: $\langle P \rangle = \frac{V_{\text{rms}}^2}{R} = \frac{53.0^2}{30} \approx 93.8 \text{ W}$

energy dissipation: $W = \langle P \rangle t = 93.8 \times 2.0 \times 60 \approx 1.13 \times 10^4 \text{ J}$ $\square$

^[62] Different countries have different standards. China mains electricity supplies a r.m.s. voltage of 220 V. UK uses a r.m.s. 230 V distribution system. USA has national standard of a r.m.s. 110 V voltage.

Yuhao Yang 10 交流电 10.1.3 r.m.s. 电流和均方根值在上一节中，我们研究了均方根电压的数学方面。值，但我们仍然需要有效值的定义电流和有效值从物理角度看电压，r.m.s. 交流电的值定义如下：rms 交流电的电流/电压等于稳定的直流电向电阻负载提供相同平均功率的电流/电压。对于正弦波，均方根值与峰值的关系如下： $I_{r.m.s} = I_p / \sqrt{2}$ 和 $V_{r.m.s} = V_p / \sqrt{2}$ 证明 (Q)：一个周期内的总能量耗散为： $W_T = \int_0^T P dt$ 因此平均功率可由下式给出： $\langle P \rangle = \frac{1}{T} \int_0^T P dt$ 代入 $P = I^2 R = I_0^2 \sin^2 \omega t = \frac{1}{2} I_0^2 (1 - \cos 2\omega t)$ ，可得： $\langle P \rangle = \frac{1}{2} I_0^2 R \int_0^T \sin^2 \omega t dt$ 进行积分： $\int_0^T \sin^2 \omega t dt = \frac{1}{2} \int_0^T (1 - \cos 2\omega t) dt = \frac{1}{2} [t - \frac{\sin 2\omega t}{2\omega}]_0^T = \frac{1}{2} T$ 现在我们有： $\langle P \rangle = \frac{1}{2} I_{rms}^2 R = \frac{1}{2} I_0^2 R$ $I_{rms} = I_p / \sqrt{2}$ 类似的电压计算将显示： $V_{rms} = V_p / \sqrt{2}$ 值得指出的是，rms 的 $1/\sqrt{2}$ 关系式。值仅适用于正弦波，对于其他波形，例如方波或三角波，数值常数不同。主电源规定的电压值通常指均方根值。值 [62] 示例 10.2 交流电电源通过 30Ω 电阻产生正弦输出。发现最大电压为 $75 V$ 。求 2.0 分钟内电阻器消耗的能量。- 均方根电压： $V_{rms} = V_p / \sqrt{2} = 75 / \sqrt{2} = 53.0 V$ 平均输出功率： $\langle P \rangle = V_{rms}^2 / R = 53.0^2 / 30 = 93.8 W$ 能量耗散： $W = \langle P \rangle t = 93.8 \times 2.0 \times 60 = 1.13 \times 10^4 J$ [62] 不同的国家有不同的标准。中国主电源供应均方根值电压 220V。英国使用均方根 (rms)。230V 配电系统。美国有均方根国家标准。110V 电压。110

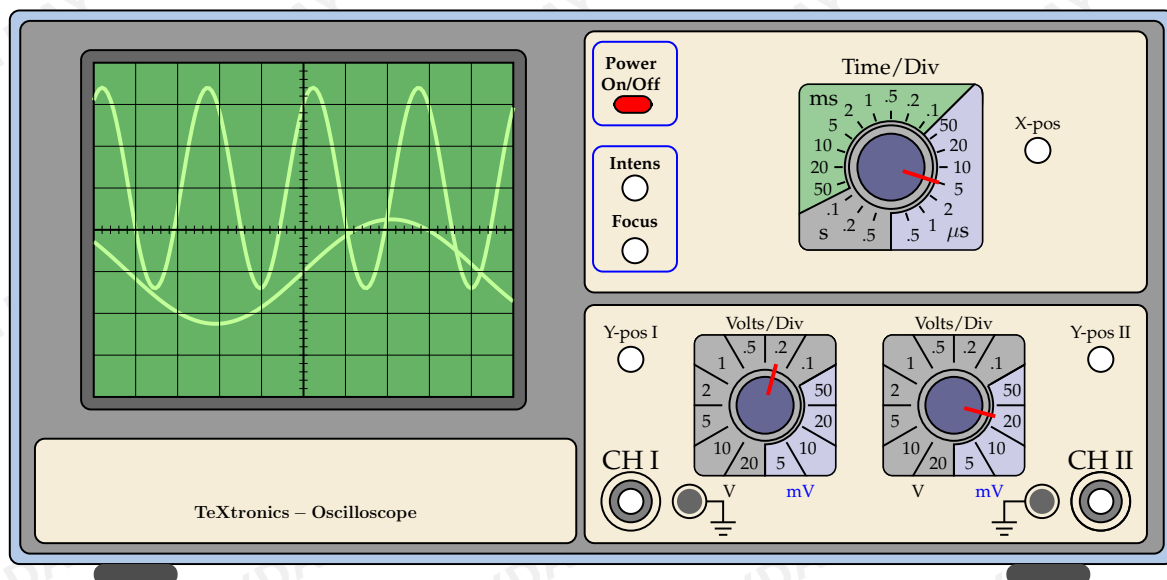
Question 10.4 An alternating voltage V is represented by the equation: $V = 310 \sin(100\pi t)$, where V and t are measured in SI units. For this voltage, find (a) the peak voltage, (b) the r.m.s voltage, (c) the frequency, (d) the mean power when it is applied across a resistor of 250Ω .

Question 10.5 What is the average power dissipated in a resistor when the alternating supply has a peak current of 5.0 A and a peak voltage of 8.0 V ?

Question 10.6 The peak value of a sinusoidal alternating current is equal to a steady direct current. When they are applied to the same load, what is the ratio of power dissipation $\frac{P_{\text{d.c.}}}{P_{\text{a.c.}}}$?

10.1.4 measurement of a.c. with an oscilloscope

to measure a varying voltage, we can use a **cathode-ray oscilloscope** (c.r.o.) (see figure^[63])



an oscilloscope is basically an electron deflection tube that uses a beam of electrons to trace the input voltage as a function of time.

electron beam is controlled by two sets of parallel plates

horizontal plates control how fast beam moves across screen, giving a horizontal time axis

a.c. voltage is applied across vertical plates, causing beam to bend upwards or downwards

when beam hits fluorescent screen, entire trace of the beam can be seen across the screen

^[63]The illustration of the oscilloscope was created by *Hugues Vermeiren*. The source code is downloaded from TeXample: <http://www.texample.net/tikz/examples/textronics-oscilloscope/>.

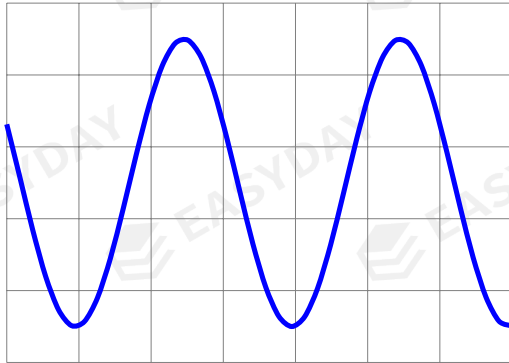
Yuhao Yang 10 交流电流问题10.4 交流电压V 由以下方程表示： $V = 310 \sin(100\pi t)$ ，其中V 和t 以SI 单位测量。对于该电压，求 (a) 峰值电压，(b) 均方根电压，(c) 频率，(d) 施加在 $250 \, \Omega$ 电阻器上时的平均功率。问题 10.5 当交流电源的峰值电流为 $5.0 \, \text{A}$ 、峰值电压为 $8.0 \, \text{V}$ 时，电阻器中消耗的平均功率是多少？问题10.6 正弦交流电的峰值等于稳定的直流电。当它们施加到相同负载时，功耗Pd.c之比是多少？帕克？10.1.4 交流电的测量如果要使用示波器测量变化的电压，我们可以使用阴极射线示波器 (c.r.o.) (见图 [63])

TeXtronic-Oscillograph CH I CH II Y-pos I Y-pos II V olts/Div V mV 50 .1.2.51 2 5 10 20 V
olts/Div V mV 50 .1.2.51 2 5 10 20 开机/关机聚焦强度 X-pos 时间/分度 s (s ms .5125102050.1.2
.5125102050.1.2.5 示波器基本上是一个电子偏转管，它使用电子束来跟踪输入电压随时间的变化。电子束由两组平行板控制。水平板控制电子束在屏幕上移动的速度，从而给出水平时间轴交流电。电压施加在垂直板上，当光束撞击荧光屏时，导致光束向上或向下弯曲，整个光束轨迹可以在屏幕上看到[63]示波器的插图由 Hugues Vermeiren 创建。源代码从TeXample下载

: <http://www.texample.net/tikz/examples/textronics-oscillscope/>。111

- to take readings from oscilloscope, remember it displays a voltage against time graph
- **voltage gain**, or **Y-gain**, tells the number of volts per vertical division
 - **time base** gives the time unit per horizontal division

Example 10.3 An oscilloscope displays an a.c. voltage signal as shown.



suppose the time base setting is 10 ms/div, and the voltage gain is 5 V/div.

4 vertical divisions from highest to lowest, so

$$\text{peak voltage: } V_0 = \frac{1}{2} \times 4 \times 5 = 10 \text{ V}$$

3 horizontal divisions between peak to peak, so

$$\text{period: } T = 3 \times 10 = 30 \text{ ms}$$

$$\text{frequency } f = \frac{1}{T} = \frac{1}{30 \times 10^{-3}} \approx 33.3 \text{ Hz} \quad \square$$

Question 10.7 In Example 10.3, if the time-base setting is 5 ms/div and the voltage gain is 2 V/div, write down an equation that represents this alternating voltage.

10.2 power supply systems

10.2.1 high-voltage transmission

electricity is sent from power stations to consumers around the country

for long-distance transmission, we need minimise energy losses

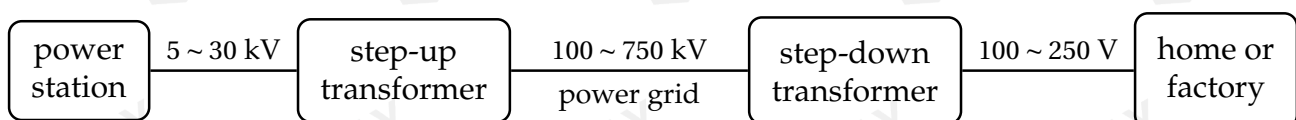
power dissipated due to resistance in cables is $I^2 R$, should transmit at low currents

since output power of a power station is fixed, low current means high voltage

so transmission at high voltages minimises energy loss in power grids

but for reasons of safety and efficiency, desirable to have low voltages at both generating end (power station) and receiving end (home or factory)

this requires converting a.c. into higher or lower voltages $\Rightarrow$ need for *transformers*



Yuhao Yang 10 交变电流 $\hat{a}$ 要从示波器中获取读数，请记住它显示电压与时间关系图 - 电压增益或 Y 增益表示每个垂直格的伏特数 - 时基给出每个水平格的时间单位示例 10.3 示波器显示交流电电压信号如图所示。假设时基设置为 10 ms/div ，电压增益为 5 V/div 。从最高到最低有 4 个垂直刻度，因此峰值电压： $V_0 = 12 \times 4 \times 5 = 10 \text{ V}$ 峰值到峰值之间有 3 个水平刻度，因此周期： $T = 3 \times 10 = 30 \text{ ms}$ 频率 $f = 1/T = 1/30 \times 10^{-3} \approx 33.3 \text{ Hz}$ 问题 10.7 在例 10.3 中，如果时基设置为 5 ms/div ，电压增益为 2 V/div ，写下表示该交流电压的方程。

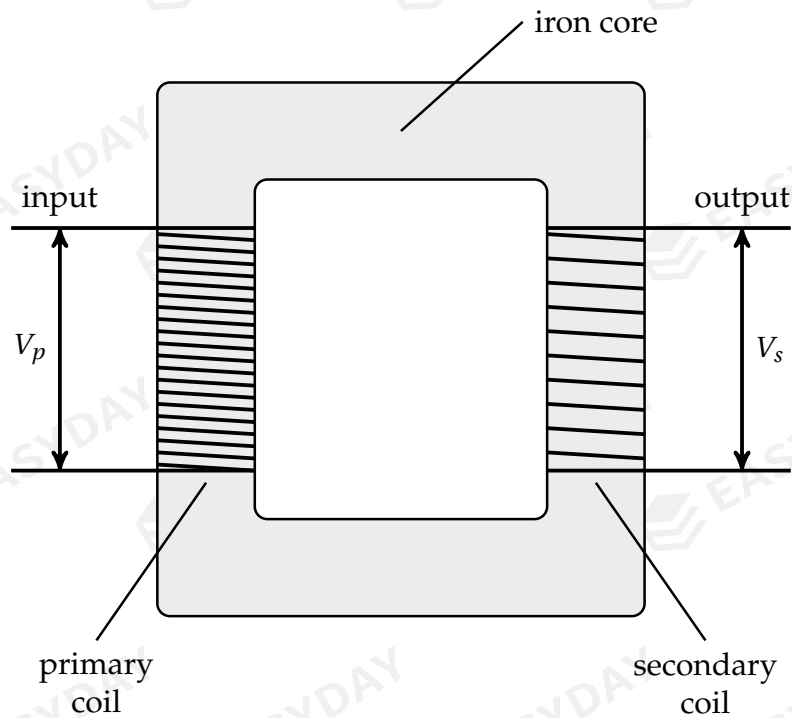
10.2 供电系统 10.2.1 高压输电是从发电站发送到全国各地的消费者进行长距离传输，我们需要最大限度地减少能量损失，由于电缆中的电阻而消耗的功率为 $I^2 R$ ，应该以低电流传输，因为电站的输出功率是固定的，小电流意味着高电压，所以高压传输可以最大限度地减少电网的能量损失，但出于安全和效率的考虑，最好在两端都采用低电压发电端（电站）和接收端（家庭或工厂）这需要转换交流电。更高或更低电压）需要变压器电站升压变压器降压变压器家庭或工厂 $5 \gg 30 \text{ kV} \gg 100 \gg 750 \text{ kV}$ 电网 $100 \gg 250 \text{ V}$ 112

10.2.2 transformers

transformer is a device that can change values of voltage for an alternating current

transformers are an important part of the national power grid

transformers are essential for transmission, distribution, and utilization of electrical energy



structure of a typical transformer

➤ principle of transformers

a.c. flowing in **primary coil** (input) produces a changing magnetic field, i.e. a changing flux

this flux is linked with **secondary coil** through iron core

an a.c. voltage with same frequency is then induced in secondary coil (output)

➤ **turns-ratio equation** for a transformer

assume no loss in magnetic flux, i.e., all field inside transformer's iron core

from Faraday's law: $V_p = N_p \frac{d\phi}{dt}$, $V_s = N_s \frac{d\phi}{dt}$, where N_p , N_s are number of turns of coils

cancel out $\frac{d\phi}{dt}$, we find: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

if $N_p < N_s$, output voltage is increased → *step-up transformers*

if $N_p > N_s$, output voltage is decreased → *step-down transformers*

杨宇浩 10 交流电 10.2.2 变压器变压器是能改变交流电压值的装置，变压器是国家电网的重要组成部分，变压器是电能的传输、分配和利用所必需的输入 V_p 输出 V_s 初级线圈次级线圈典型变压器的铁芯结构①变压器原理初级线圈（输入）中的电流产生变化的磁场，即变化的磁通，该磁通通过铁芯和交流电与次级线圈相连。然后在次级线圈（输出）中感应出相同频率的电压变压器的匝数比方程假设磁通没有损耗，即根据法拉第定律，变压器铁芯内部的所有磁场： $V_p = N_p \frac{d\Phi}{dt}$ ， $V_s = N_s \frac{d\Phi}{dt}$ ，其中 N_p 、 N_s 是线圈匝数抵消 $\frac{d\Phi}{dt}$ ，我们发现： $V_s/V_p = N_s/N_p$ 如果 $N_p < N_s$ ，输出电压增加！升压变压器如果 $N_p > N_s$ ，输出电压下降！降压变压器 113

➤ if transformer is 100% efficient, then input power equals output power

for ideal transformers: $I_p V_p = I_s V_s$

➤ in practice, there always exists losses of energy from the transformer

causes of energy loss from the transformer include

- heat produced by *eddy currents* induced in iron core

this is reduced by *laminating* the core with insulate layers

- heat generated in coils due to resistance

can use thick copper wire to minimise resistance

- leakage of magnetic flux into surroundings

transformer's core is made of a continuous loop of iron to minimise this effect

Example 10.4 An ideal transformer has 200 turns on the primary coil and 5000 turns on the secondary coil. The r.m.s. input voltage to the primary coil is 8.0 V. What is the *peak* voltage across a resistor connected to the secondary coil?

$$\frac{V_{s,0}}{V_{p,0}} = \frac{N_s}{N_p} \Rightarrow \frac{V_{s,0}}{\sqrt{2}V_{p,\text{rms}}} = \frac{N_s}{N_p} \Rightarrow \frac{V_{s,0}}{8.0 \times \sqrt{2}} = \frac{5000}{200} \Rightarrow V_{s,0} \approx 2830 \text{ V} \quad \square$$

Question 10.8 An ideal transformer has 6000 turns on its primary coil. It converts a mains supply of 220 V r.m.s. to an a.c. voltage with a peak value of 12.0 V. Find the number of turns on the secondary coil.

Question 10.9 If a *steady* d.c. voltage is applied to the input of a simple transformer, what is the output voltage produced?

Question 10.10 State and explain whether the current in the primary coil of a transformer is in *phase* with the voltage induced in the secondary coil.

10.2.3 rectifiers

some electronic equipments (e.g., your smartphone, laptop, etc.) must work with d.c.

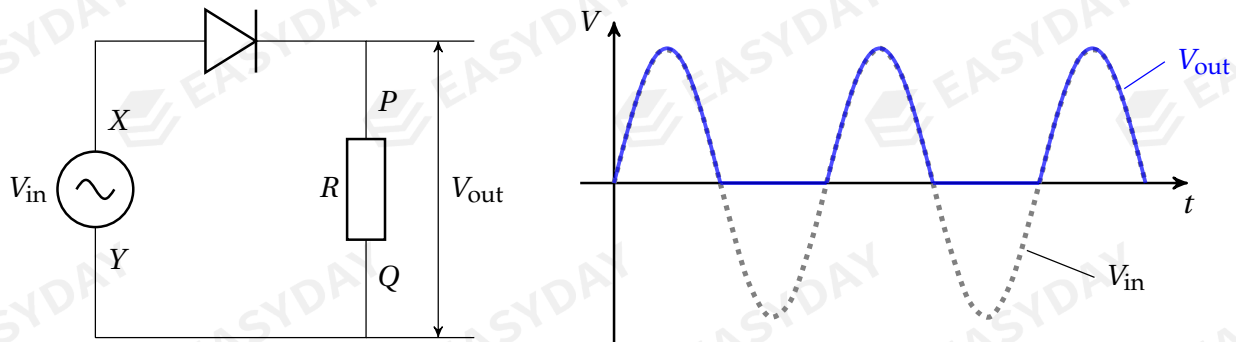
for these appliances require **rectification**, a process that converts an a.c. into a d.c.

rectification uses *diodes*, electronic components that only allow current in one direction

Yuhao Yang 10 交流 â 如果变压器效率为 100%，则理想变压器的输入功率等于输出功率： $I_p V_p = I_s V_s$ â 在实践中，总是存在变压器的能量损失变压器能量损失的原因包括 - 铁芯中感应的涡流产生的热量，可以通过在铁芯上层压绝缘层来减少 - 线圈因电阻而产生的热量可以使用粗铜线来最小化电阻 - 磁通泄漏到周围环境中变压器的铁芯由连续的铁环制成，以最大限度地减少这种影响示例 10.4 理想的变压器初级线圈有 200 匝，次级线圈有 5000 匝。均方根值初级线圈的输入电压为 8.0V。连接到次级线圈的电阻两端的峰值电压是多少？ - $V_s,0 V_{p,0} = N_s N_p$) $V_s,0 p 2 V_{p,rms} = N_s N_p$) $V_s,0 8.0 \times p 2 = 5000 200$) $V_s,0 \frac{1}{4} 2830 V$ ä 问题 10.8 理想变压器的初级线圈有 6000 匝。它将 220 V r.m.s. 的主电源转换为到交流电电压峰值为 12.0 V。求次级线圈的匝数。问题 10.9 如果一个稳定的直流电电压施加到一个简单变压器的输入端，产生的输出电压是多少？问题 10.10 说明并解释变压器初级线圈中的电流是否与次级线圈中感应的电压同相。10.2.3 整流器某些电子设备（例如您的智能手机、笔记本电脑等）必须在直流电下工作。对于这些设备，需要进行整流，这是一个将交流电转换为交流电的过程。进入华盛顿特区整流使用二极管，这是只允许一个方向电流的电子元件114

half-wave rectification

half-wave rectification uses a single diode



when input terminal X is positive, current can flow through diode, terminal P is positive

with respect to Q for load resistor R

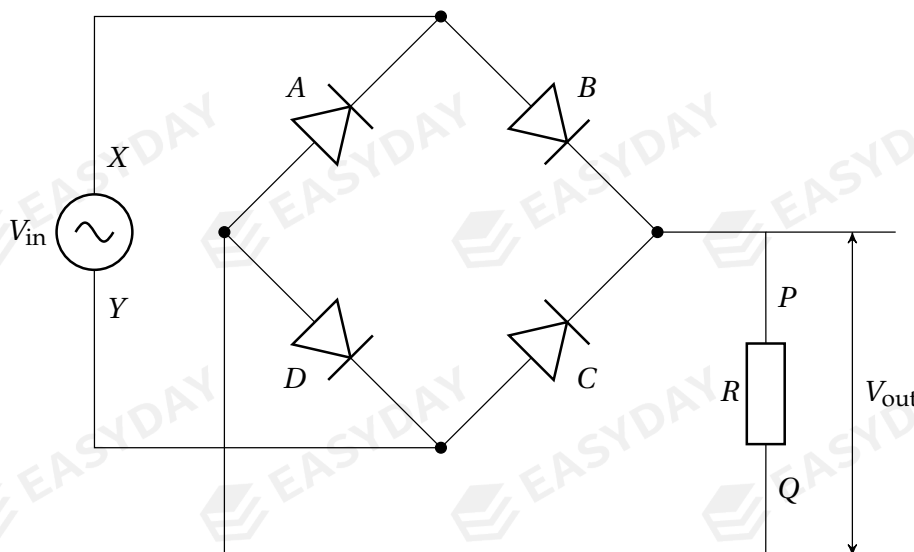
when input terminal Y is negative, flow of current is blocked, so zero output voltage on R

V_{out} across load R is in one direction only, i.e., it becomes a d.c.

but power available from half-wave rectifier is only half of supply power

full-wave rectification

full-wave rectification requires using a combination of four-diode bridge structure

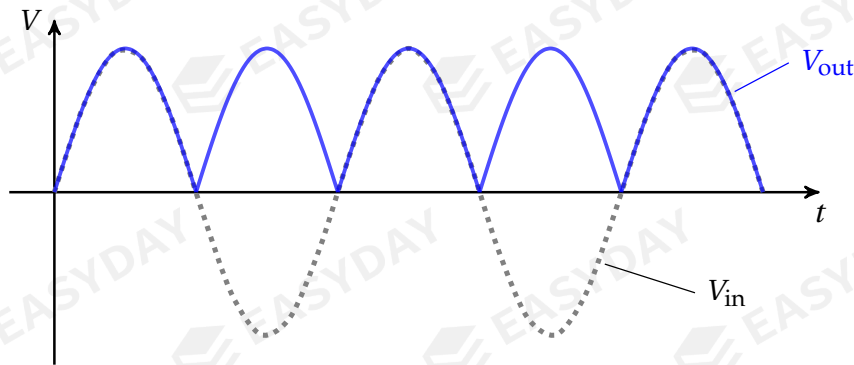


when input terminal X is positive, current flow: $X \rightarrow B \rightarrow R \rightarrow D \rightarrow Y$

when input terminal Y is positive, current flow: $Y \rightarrow C \rightarrow R \rightarrow A \rightarrow X$

Yuhao Yang 10 交流半波整流半波整流使用单个二极管 V_{in} R V_{out} X Y P Q t V V_{in} V_{out} 当输入端 X 为正时，电流可以流过二极管，负载电阻 R 的 P 端相对于 Q 为正，当输入端 Y 为负时，电流被阻止，因此 R V_{out} 上负载 R 上的零输出电压仅在一个方向，即，它变成了华盛顿特区但半波整流器提供的功率仅为电源功率的一半全波整流全波整流需要使用四二极管桥式结构的组合 V_{in} A B C D R V_{out} X Y P Q 当输入端 X 为正时，电流为： X ！乙！罗！ D ！ Y 当输入端 Y 为正时，电流为： Y ！丙！罗！一个！ X 115

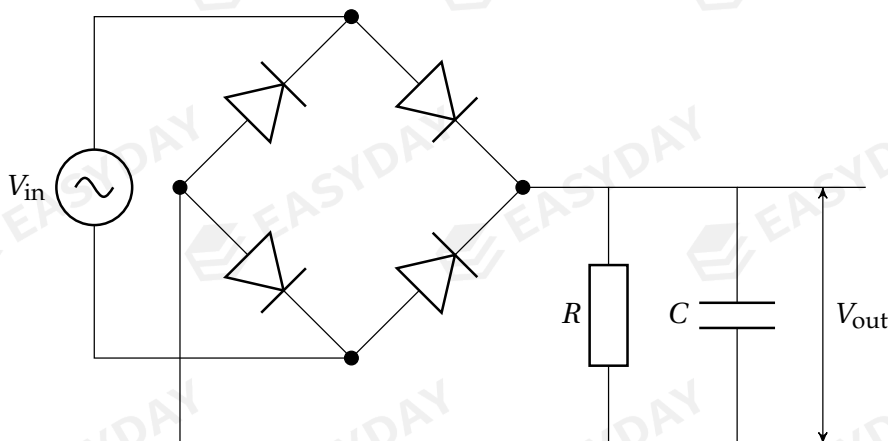
either case, resulting output current in load resistor R always flows in same direction
 terminal P is always positive with respect to Q for R no matter what polarity for V_{in}
 output voltage across R is now full-wave rectified



Question 10.11 If we want to have terminal Q to be positive with respect to P for load resistor R , how should we rebuild the bridge rectifier with the same four diodes?

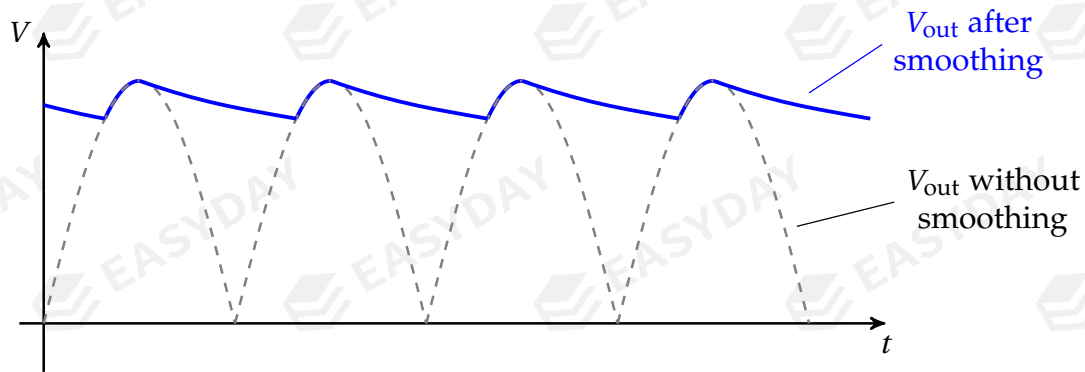
10.2.4 smoothing

note that d.c. resulting from rectification still varies with time
 to produce steady d.c. from fluctuating d.c., a process called **smoothing** is carried out
 smoothing uses *capacitors*, which are connected in *parallel* with the load resistor (see figure)



capacitor can store electrical energy as voltage output from rectifier rises
 as voltage from rectifier drops, capacitor slowly discharges and feeds energy to the load
 output voltage V_{out} across load will have less ripples over the cycles
 less fluctuation in V_{out} so we say V_{out} is now smoothed

Yuhao Yang 10 交流电无论哪种情况，负载电阻 R 中的输出电流始终以相同方向流动对于 R ，端子 P 始终相对于 Q 为正，无论 R 上的 V_{in} 输出电压是什么极性，现在都是全波整流 $t V_{in} V_{out}$ 问题 10.11 如果我们希望负载电阻 R 的端子 Q 相对于 P 为正，我们应该如何使用相同的四个二极管重建桥式整流器？ 10.2.4 平滑注意 d.c.整流产生的结果仍然随时间变化以产生稳定的直流电。对于波动的直流电，会执行一个称为平滑的过程。平滑使用电容器，这些电容器与负载电阻并联（见图） $V_{in} R C V_{out}$ 电容器可以存储电能，因为整流器输出的电压上升，而整流器电压下降，电容器缓慢放电并向负载馈送能量负载上的输出电压 V_{out} 在整个周期内的纹波将减少， V_{out} 的波动较小，因此我们说 V_{out} 现在已平滑116



➤ effect of smoothing is controlled by choice of load resistor R and smoothing capacitor C

rate of charging and discharging depend on *time constant* RC (see §7.4)

greater RC , smoothing capacitor discharges more slowly, giving less ripples

a larger capacitor for a fixed resistor usually gives better smoothing effect

but if RC is too large, charging would become very slow, capacitor might not be completely charged after each cycle, this could also result into undesirable effects

Question 10.12 If a second capacitor is added in *series* with the first smoothing capacitor, use sketches to show the changes you would expect for the output voltage.

Yuhao Yang 10 交流电流 i_V V_{out} 无平滑处理后的 V_{out} 平滑效果由负载电阻 R 和平滑电容器 C 的选择控制充电和放电速率取决于时间常数 RC （参见第 7.4 节） RC 越大，平滑电容器放电越慢，纹波越小对于固定电阻器，较大的电容器通常会提供更好的平滑效果，但如果 RC 太大，充电将变得非常慢，电容器每个周期后可能不会完全充电，这也可能导致不良影响。问题 10.12 如果添加第二个电容器与第一个平滑电容器串联，请使用草图显示您预期的输出电压变化。 11

CHAPTER 11

Quantum Physics Basics

In this chapter, we will study the laws of nature at very small scales.

We start by reviewing classical concepts like particles or waves, and then see how they break down when being applied to the microscopic world. For those observations that could not be explained with classical physics, we need *quantum physics*. The theory of quantum physics is very deep and profound, so surely way beyond the scope of our course. What we will study in this A-Level course are merely four tips of an enormous iceberg, namely

- photon theory / photoelectric effect
- matter waves / electron diffraction
- electron energy levels / atomic spectra
- band theory / electrical conductivity of solids

11.1 classical theories

in classical physics, both particle models and the wave models have been very useful

though both being successful, the two model are distinct in many aspects

to understand a phenomenon, we take either the particle picture or the wave picture

it seems that there is no way particle models would reconcile with wave models, or is it not?

11.1.1 particle models

in particle model, any system is considered to consist of particles governed by *Newtonian mechanics*, physical properties of this system are predicted by studying behaviour of particles

➤ areas of science where particle models are used to interpret and make predictions include:

- *electricity*: electric current formed by motion of charge carriers

Yuhao Yang 11 量子物理基础第11章量子物理基础在本章中，我们将研究非常小尺度的自然定律。我们首先回顾粒子或波等经典概念，然后看看它们在应用于微观世界时如何分解。对于那些无法用经典物理学解释的观察结果，我们需要量子物理学。量子物理理论非常深奥，因此肯定超出了我们课程的范围。本次 A-Level 课程我们要学习的只是冰山四角，分别是 - 光子理论/光电效应 - 物质波/电子衍射 - 电子能级/原子光谱 - 能带理论/固体电导率 11.1 经典物理学中的经典理论，粒子模型和波模型虽然都很成功，但都非常有用，这两个模型在理解一个现象时在很多方面是不同的，我们要么用粒子图，要么用波图，看来粒子模型与波是不可能调和的模型，或者不是？ 11.1.1 粒子模型在粒子模型中，任何系统都被认为是由受牛顿力学控制的粒子组成，该系统的物理性质是通过研究粒子的行为来预测的 - 使用粒子模型解释和进行预测的科学领域包括： - 电：由电荷载流子运动形成的电流 118

- *ideal gas*: pressure caused by collisions of gas molecules
- *solid*: elasticity due to interaction between solid atoms
- *radioactivity*: decay interpreted as the emission of α -/ β -particles from the nucleus
- *chemistry*: chemical reactions due to exchange of electrons between atoms and molecules
- ...

11.1.2 wave models

in the wave picture, energy is transferred via the vibration of medium or force fields

➤ phenomena that can be explained in terms of wave models include:

- *water waves*: variation in the vertical displacement of water surface
- *propagation of sound*: variation in the pressure and density of medium
- *light*: variation of electric and magnetic fields
- ...

➤ a key feature that makes waves distinct from particles is that waves can *superpose*

when two or more waves meet, they add up or cancel out to give a resultant wave

this gives rise to the characteristic properties of waves:

- *interference*: waves superpose to form a resultant wave of greater or lower amplitude
- *diffraction*: bending of waves around obstacles, or spreading out of waves through slits

11.1.3 history of light

efforts to understand the nature of light can be traced all the way back to ancient Greeks

we are not going to examine the ideas of early thinkers well over 3,000 years ago

let's skip a couple of years and jump to the ideas developed since the Scientific Revolution

➤ particle theory of light (*Issac Newton*, 1671)

key idea: light rays is comprised of a stream of massless particles called *corpuscles*

- explains straight-line propagation
- explains reflection and refraction
- explains colours of light seen in dispersion in prism (corpuscles have different colours)

杨宇豪 11 量子物理基础 - 理想气体：气体分子碰撞产生的压力 - 固体：固体原子间相互作用产生的弹性 - 放射性：衰变解释为原子核中 α -/ β -粒子的发射 - 化学：原子和分子之间电子交换引起的化学反应 - 11.1.2 波图中的波模型，能量通过介质或力场的振动传递 $\hat{O}$ 现象用波模型解释包括：- 水波：水面垂直位移的变化 - 声音的传播：介质压力和密度的变化 - 光：电场和磁场的变化 - 波与粒子的一个关键特征是，当两个或多个波相遇时，波可以叠加，它们相加或抵消，产生合成波，从而产生波的特性：- 干涉：波叠加形成更大或更低振幅的合成波 - 衍射：波绕过障碍物弯曲，或通过狭缝传播出去 11.1.3 光的历史理解光本质的努力可以追溯到古希腊人我们不打算研究 3,000 多年前的早期思想家的思想让我们跳过几年，跳到科学革命以来发展的思想 - 光的粒子理论 (Issac Newton, 1671) 关键思想：光线由流组成称为微粒的无质量粒子 - 解释直线传播 - 解释反射和折射 - 解释在棱镜色散中看到的光的颜色（微粒有不同的颜色） 119

the problems with the particle model include:

- does not agree with observations on refraction
- cannot predict the interference and diffraction of light

➤ wave theory of light (*Christian Huygens*, 1678)

key idea: light is a wave that transfers energy within a medium known as *aether*

- follows laws of reflection and refraction
- explains colours of light (light have different wavelengths)
- explains interference (Thomas Young's double slit experiment)
- explains diffraction (Poisson spot experiment)

only problem is that aether, the medium in which light lives, was not experimentally found

➤ electromagnetic theory (*James Maxwell*, 1865)

key idea: light is an *electromagnetic wave* ^[64]

- electric and magnetic fields travel through space in the form of waves at speed of light
- propagation of electromagnetic wave does not require medium, hence no need for aether

all behaviour of light known at that time could be explained with Maxwell's theory

so scientists were convinced that light travelled through space as an electromagnetic wave

11.2 photon theory

11.2.1 photoelectric effect

electromagnetic radiation incident upon a metal can cause emission of electrons from the metal surface, this is called **photoelectric effect**, ^[65] emitted electrons are called **photoelectrons**

simply put, conduction electrons in metal gain additional energy by absorbing the incoming

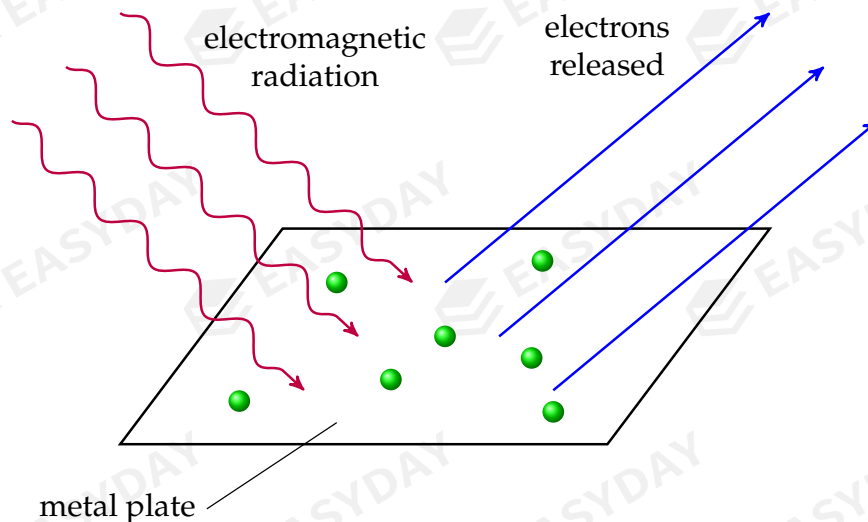
^[64] Maxwell's equations fully describe the behaviour of electric and magnetic fields. Based on the four equations that Maxwell established, he found the vibration of electric and magnetic field can propagate in space as a classical wave. The speed of electromagnetic wave is given by

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{1}{\sqrt{8.85 \times 10^{-12} \times 4\pi \times 10^{-7}}} = 3.00 \times 10^8 \text{ m s}^{-1},$$

which is the same as speed of light. This gave Maxwell the intuition that light is an electromagnetic wave.

^[65] Photoelectric effect was first observed in 1887 by *Heinrich Hertz*, who found electrodes illuminated with ultraviolet light create electric sparks. In school labs, one can shine ultraviolet radiation from a mercury lamp onto a zinc plate to cause photo-emission.

Yuhao Yang 11 量子物理学基础粒子模型的问题包括：－与折射观察结果不一致－无法预测光的干涉和衍射－光的波动理论 (Christian Huygens, 1678) 关键思想：光是一种在称为以太的介质内传递能量的波－遵循反射和折射定律－解释光的颜色（光具有不同的波长）－解释干涉（托马斯·杨的双缝）实验）－解释衍射（泊松斑实验）唯一的问题是以太，光赖以存在的介质，没有通过实验发现电磁理论（詹姆斯·麦克斯韦，1865）关键思想：光是一种电磁波 [64]－电场和磁场以波的形式以光速在空间中传播－电磁波的传播不需要介质，因此不需要以太当时已知的所有光行为都可以用麦克斯韦理论来解释，因此科学家们相信光作为电磁波在空间中传播 11.2 光子理论 11.2.1 光电效应电磁辐射入射到金属上会导致金属表面发射电子，这称为光电效应，[65]发射的电子简单地称为光电子，金属中的传导电子通过吸收传入的能量而获得额外的能量[64]麦克斯韦方程充分描述了电场和磁场的行为。根据麦克斯韦建立的四个方程，他发现电场和磁场的振动可以作为经典波在空间中传播。电磁波的速度由 $c = 1/\sqrt{\epsilon_0 \mu_0} = 1/\sqrt{8.85 \times 10^{-12} \times 4\pi \times 10^{-7}} = 3.00 \times 10^8 \text{ m s}^{-1}$ 给出，与光速相同。这让麦克斯韦直觉地认为光是一种电磁波。[65]光电效应由 Heinrich Hertz 于 1887 年首次观察到，他发现用紫外线照射的电极会产生电火花。在学校实验室中，人们可以将汞灯的紫外线照射到锌板上以引起光发射。 120



radiation, if this energy is sufficient for electrons to overcome the electrostatic attraction from the positive metal ions, they break free from the metal surface

➤ some detailed experimental observations on the photoelectric effect are:

- there exists a minimum **threshold frequency** f_0 for incident radiation
when $f < f_0$, no electrons released from metal surface
- emission of electrons is immediate when radiation is incident as long as $f > f_0$
even low-intensity light is effective
- increasing intensity has no effect on energies of electrons
- increasing intensity of incident light causes number of photoelectrons emitted to increase
- increasing radiation frequency increases electron energies

➤ wave theory of light fails to explain any of these properties, according to wave model:

- electrons could gradually build up energies by absorbing wave energy over time
so radiation at any frequency should all work
- need very intense light to have immediate effect
- greater intensity mean higher energy, electrons released should have greater K.E.
- varying frequency of radiation should have no effect on energy of electrons released

photoelectric effect sees the breakdown of the wave model, some new ideas are needed!

杨宇豪 11 量子物理学基础金属板电磁辐射电子释放辐射，如果该能量足以使电子克服金属正离子的静电引力，它们就会脱离金属表面 – 光电效应的一些详细实验观察如下： – 当 $f < f_0$ 时，入射辐射存在最小阈值频率 f_0 ，没有电子从金属表面释放 – 当辐射入射时，只要 $f \geq f_0$ 均匀，电子就会立即发射低强度的光是有效的 – 增加强度对电子的能量没有影响 – 增加入射光的强度会导致发射的光电子数量增加 – 增加辐射频率会增加电子能量 $\hat{\alpha}$ 根据波模型，光的波动理论无法解释这些属性中的任何一个： – 随着时间的推移，电子可以通过吸收波能量逐渐积累能量，因此任何频率的辐射都应该起作用 – 需要非常强的光才能立即产生效果 – 强度越大意味着能量越高，释放的电子应该具有更大的能量 K.E. – 不同频率的辐射应该对释放的电子的能量没有影响光电效应看到波模型的崩溃，需要一些新的想法！ 121

11.2.2 photon theory

photoelectric effect was explained by *Albert Einstein* in 1905 ^[66]

Einstein's revolutionary idea: when radiation delivers energy to matter, the transfer of energy is not continuous but carried in *discrete* packets, called *photons*

a **photon** is a packet (*quanta*) of electromagnetic energy

➤ energy of one photon is given by: $E = hf$, where $h = 6.63 \times 10^{-34}$ J s is the **Planck constant**

since $E \propto f$, higher/lower radiation frequency means greater/smaller photon energies

➤ wave equation $c = \lambda f$ relates frequency f of an electromagnetic wave to its wavelength λ

so energy of one photon is also given by: $E = \frac{hc}{\lambda}$

since $E \propto \frac{1}{\lambda}$, longer/shorter wavelength means lower/greater photon energies

➤ the equation $E = hf$, or $E = \frac{hc}{\lambda}$, relates a particle property with a wave property

E is the energy of one photon, treated like a single particle

but f and λ are both introduced to describe a wave, not a particle

➤ intensity of a beam of radiation is: $I = \frac{P}{A}$

total energy incident on a given area per unit time determines the radiation intensity

so intensity depends on the product of the number of photons arriving per unit time and

energy of each photon: $I \propto nhf$, or more precisely, $I = \frac{nhf}{A}$

➤ when considering energy of a photon, **electronvolt** is a useful energy unit

one electronvolt (1 eV) is work needed to make an electron travel through a p.d. of 1 V

conversion between electronvolt and joule is: $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$

Example 11.1 A laser emits red light of 650 nm at a power rating of 2.0 mW. (a) What is the energy carried by one photon? (b) How many photons are emitted per second?

✎ energy of one photon: $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{650 \times 10^{-9}} \approx 3.06 \times 10^{-19} \text{ J}$

number of photons: $N = \frac{\text{total energy output from laser}}{\text{energy of one photon}} = \frac{2.0 \times 10^{-3}}{3.06 \times 10^{-19}} \approx 6.54 \times 10^{15}$ □

^[66] Albert Einstein was awarded the Nobel Prize in physics in 1921 for 'his discovery of the law of the photoelectric effect'. This discovery led to the quantum revolution of modern physics.

杨宇豪 11 量子物理学基础 11.2.2 光子理论光电效应由阿尔伯特·爱因斯坦于 1905 年解释[66] 爱因斯坦的革命性思想：当辐射将能量传递给物质时，能量的传递不是连续的，而是以离散的包的形式进行，称为光子。光子是电磁能的包（量子）。一个光子的能量由下式给出： $E = hf$ ，其中 $h = 6.63 \times 10^{-34} \text{ J s}$ 是普朗克常数，因为 E/f ，更高/更低的辐射频率意味着更大/更小的光子能量 $\hat{\omega}$ 波动方程 $c = \lambda f$ 将电磁波的频率 f 与其波长 λ 联系起来，因此一个光子的能量也由下式给出： $E = hc/\lambda$ 因为 $E \propto 1/\lambda$ ，更长/更短的波长意味着更低/更大的光子能量 $\hat{\omega}$ 方程 $E = hf$ 或 $E = hc/\lambda$ ，将粒子属性与波属性联系起来 E 是一个光子的能量，被视为单个粒子，但引入 f 和 λ 都是为了描述波，而不是粒子 $\hat{\omega}$ 辐射束的强度为： $I = P/A$ 每单位时间入射到给定区域的总能量决定辐射强度，因此强度取决于每单位时间到达的光子数量与每个光子的能量的乘积： $I = nhf/A$ ，或更准确地说， $I = nhf/A$ $\hat{\omega}$ 当考虑光子的能量时，电子伏特是一个有用的能量单位，一电子伏特 (1 eV) 是使电子穿过 p.d. 所需的功。电子伏特和焦耳之间的 1 V 转换为： $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$ 示例 11.1 激光器以 2.0 mW 的额定功率发射 650 nm 的红光。(a) 一个光子携带的能量是多少？(b) 每秒发射多少个光子？ - 一个光子的能量： $E = hc/\lambda = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 650 \times 10^{-9} \approx 3.06 \times 10^{-19} \text{ J}$ 光子数量： $N = \text{一个光子的激光能量输出的总能量} = 2.0 \times 10^{-3} / 3.06 \times 10^{-19} \approx 6.54 \times 10^{15} \text{ s}^{-1}$ [66]阿尔伯特·爱因斯坦因“发现光电效应定律”而获得 1921 年诺贝尔物理学奖。这一发现引发了现代物理学的量子革命。

Question 11.1 Calculate the energy, in eV, of a photon of light of wavelength 440 nm.

Question 11.2 If the light source in Example 11.1 is replaced by a green laser with the same power output, how does the number of emitted photons per unit time change?

11.2.3 photoelectric effect explained

from the viewpoint of photon theory, photoelectric effect can be explained easily to release an electron from metal, it requires a minimum energy Φ , called the **work function**, for the electron to overcome attraction due to metal ions to escape from metal surface as radiation shines upon metal, photon energies are absorbed by electrons since photon energies are *discrete*, or *quantised*, the absorption is sort of all or nothing if photon energy is greater than Φ , electrons break free from metal excess energy, if any, would become kinetic energy of the free electron this is summarised in **Einstein's photoelectric equation**: $hf = \Phi + E_{k,\max}$

➤ note that we are talking about *maximum* K.E. of emitted electrons

electron emitted from the *surface* would have greatest K.E.

for electrons to be released from *below* the surface, they require more energy than work function, so less K.E. than maximum value

➤ at critical condition, incoming photon has just enough energy to release electron

so at threshold frequency f_0 , photon energy equals work function: $hf_0 = \Phi$

➤ experimental observations mentioned in §11.2.1 can now be understood

- below threshold frequency f_0 , not enough photon energy available to electron to overcome work function, so no effect for $f < f_0$
- interaction between photon and electron is *one-to-one*, so no time delay
- greater radiation intensity means more photons per unit time, so more electrons released
- greater frequency means higher photon energy, so greater K.E. for emitted electron

Example 11.2 Given that work function energy of gold is 4.9 eV. Find the longest wavelength of electromagnetic wave that could release electrons from gold.

🔗 threshold frequency: $f_0 = \frac{\Phi}{h} = \frac{4.9 \times 1.60 \times 10^{-19}}{6.63 \times 10^{-34}} \approx 1.18 \times 10^{15} \text{ Hz}$

Yuhao Yang 11 量子物理基础问题 11.1 计算波长 440 nm 的光子的能量（以 eV 为单位）。问题 11.2 如果将例 11.1 中的光源替换为具有相同功率输出的绿光激光器，则单位时间发射的光子数如何变化？ 11.2.3 从光子理论的角度解释光电效应，光电效应可以很容易地解释为从金属中释放一个电子，它需要一个最小的能量 Φ ，称为功函数，当辐射照射到金属上时，电子克服金属离子的吸引力而从金属表面逃逸，光子能量被电子吸收，因为光子能量是离散的，或者说是量子化的，如果光子能量大于 Φ ，则吸收是全有或全无，电子断裂没有金属多余的能量，如果有的话，将成为自由电子的动能，这在爱因斯坦的光电方程中进行了总结： $hf = \Phi + E_{k,max}$ - 请注意，我们正在谈论最大 K.E。发射电子的数量从表面发射的电子将具有最大的 K.E. 对于从表面以下释放的电子，它们需要比功函数更多的能量，因此 K.E 较小。在临界条件下，入射光子具有足够的能量来释放电子，因此在阈值频率 f_0 处，光子能量等于功函数： $hf_0 = \Phi$ 现在可以理解第 11.2.1 节中提到的实验观察结果 - 低于阈值频率 f_0 ，电子没有足够的光子能量来克服功函数，因此对 $f < f_0$ 没有影响 - 光子和电子之间的相互作用是一对一的，因此没有时间延迟 - 更大的辐射强度意味着更多每单位时间产生更多的光子，因此释放更多的电子 - 更高的频率意味着更高的光子能量，因此更大的 K.E. 对于发射电子示例 11.2 假设金的功函数能量为 4.9 eV。找出可以从金中释放电子的电磁波的最长波长。 - 阈值频率： $f_0 = \Phi/h = 4.9 \times 1.60 \times 10^{-19} / 6.63 \times 10^{-34} \approx 1.18 \times 10^{15} \text{ Hz}$ 123

threshold wavelength: $\lambda_0 = \frac{c}{f_0} = \frac{3.00 \times 10^8}{1.18 \times 10^{15}} \approx 2.54 \times 10^{-7} \text{ m}$ (ultraviolet light) $\square$

Question 11.3 Sodium has a work function of $3.8 \times 10^{-19} \text{ J}$. If a light of 500 nm is incident on sodium, determine whether electrons can be emitted from the surface.

Question 11.4 When electromagnetic radiation of wavelength 1200 nm is incident on a metal surface, the maximum kinetic energy of the electrons released is found to be $5.4 \times 10^{-20} \text{ J}$. What is the work function of this metal?

Question 11.5 When a beam of light of a particular frequency and intensity is shone onto a metal surface, electrons are released. If another beam of light of same intensity but higher frequency is used, what is the effect on the rate of emission of electrons from this surface?

measurement of the Planck constant and work function energy

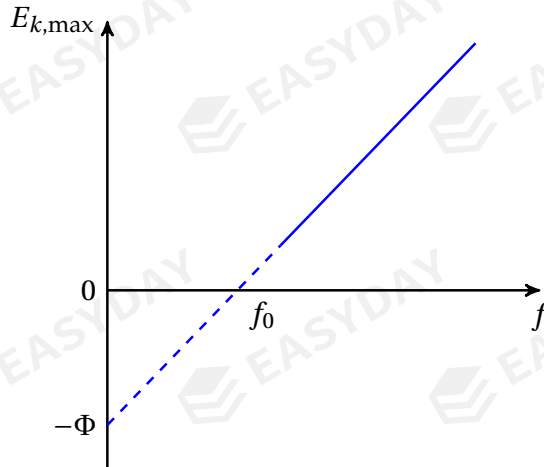
when radiation with different frequencies is incident onto a metal, we measure maximum K.E. of the electrons emitted from the surface, a set of readings ($f, E_{k,\max}$) can be found

note that the photoelectric equation can be rearranged as: $E_{k,\max} = hf - \Phi$

if we plot a graph of $E_{k,\max}$ against f , data points shall fall on a straight line

information about the Planck constant h , threshold frequency f_0 , work function Φ can all be computed with the best-fit line

- gradient = h
- y-intercept = $-\Phi$
- x-intercept = $\frac{\Phi}{h} = f_0$

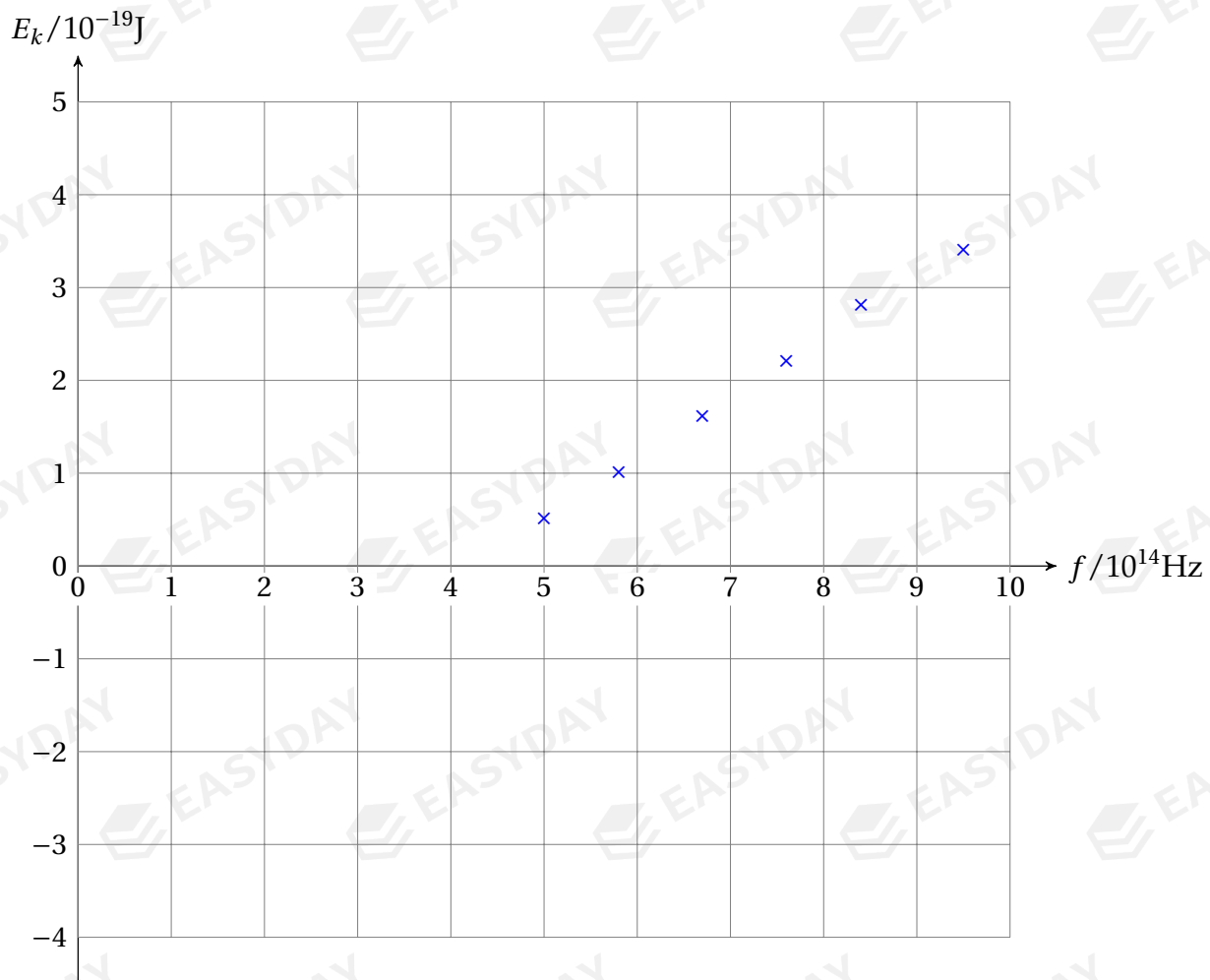


Question 11.6 If a different metal with a greater work function energy is used, describe the change for the line that shows the variation with f of $E_{k,\max}$ for this metal.

Question 11.7 Electromagnetic radiation is incident upon a metal plate. The graph shows how maximum kinetic energy E_k of emitted electrons varies with frequency f of the radiation. Use the graph to find (a) the threshold frequency, (b) a value of Planck constant.

Yuhao Yang 11 量子物理学基础 阈值波长： $\lambda_0 = c / f_0 = 3.00 \times 10^8 \text{ m/s} / 1.18 \times 10^{15} \text{ s}^{-1} = 2.54 \times 10^{-7} \text{ m}$ (紫外光) $\ddot{a}$ 问题 11.3 钠的功函数为 $3.8 \times 10^{-19} \text{ J}$ 。如果 500 nm 的光入射到钠上，确定是否电子可以从表面发射。问题 11.4 当波长为 1200 nm 的电磁辐射入射到金属表面时，释放出的电子的最大动能为 $5.4 \times 10^{-20} \text{ J}$ 。该金属的功函数是多少？问题 11.5 当特定频率和强度的光束照射到金属表面时，电子被释放。如果使用另一束强度相同但频率更高的光，对该表面的电子发射速率有何影响？当不同频率的辐射入射到金属上时，测量普朗克常数和功函数能量，我们测量最大 K.E。从表面发射的电子的数量，可以找到一组读数 ($f, E_{k,\max}$) 请注意，光电方程可以重新排列为： $E_{k,\max} = hf - \phi$ $E_{k,\max} / f - \phi / h = f_0$ 如果我们绘制 $E_{k,\max}$ 与 f 的关系图，数据点应落在一条直线上有关普朗克常数 h 、阈值频率 f_0 、功函数 ϕ 的信息都可以使用最佳拟合线 - 梯度来计算 $= h$ - y 截距 $= -\phi$ - x 截距 $= \phi / h = f_0$ 问题 11.6 如果使用具有更大功函数能量的不同金属，请描述显示该金属的 $E_{k,\max}$ 随 f 变化的线的变化。问题 11.7 电磁辐射入射到金属板上。该图显示了发射电子的最大动能 E_k 如何随辐射频率 f 变化。使用该图求出 (a) 阈值频率，(b) 普朗克常数的值。

124



11.2.4 photon momentum

in photoelectric effect, photons can knock electrons out of a metal, this suggests that photons could have *momentum*, even though they do not have mass

earliest experimental evidence of photon momentum was came from *Arthur Compton* in 1923, who studied the scattering of X-ray photons by electrons in substances^[67]

it can be shown^[68] that photon momentum is given by: $p = \frac{h}{\lambda}$

➤ photon momentum a *relativistic* momentum, as photons move at speed of light

definition for classical momentum $p = mv$ does not apply for photons

^[67] Arthur Compton was awarded the Nobel Physics Prize in 1929 for the discovery of this scattering effect, now known as *Compton scattering*.

^[68] This derives from Einstein's theory of *special relativity*, which states that energy and momentum are related by the equation: $E^2 = m_0^2 c^4 + p^2 c^2$. Photons have zero rest mass, i.e., $m_0 = 0$. This relativistic relation becomes $E = pc$ for photons. Now recall photon energy is given by $E = hf$. Rearrange the terms, we can show: $p = \frac{hf}{c} = \frac{h}{\lambda}$.

杨宇浩 11 量子物理基础 Ek /10-19J f /1014Hz0 1 2 3 4 5 6 7 8 9 10 -4 -3 -2 -1 0 1 2 3 4 5 × × × × × ×

11.2.4 光电效应中的光子动量，光子可以将电子从金属中击出，这表明光子可以光子动量的最早实验证据来自 Arthur Compton 于 1923 年，他研究了物质中电子对 X 射线光子的散射 [67]，可以证明 [68] 光子动量由下式给出： $p = h \lambda^{-1}$ 光子动量相对论动量，因为光子以光速移动经典动量的定义 $p = mv$ 不适用于光子 [67] Arthur 康普顿因发现这种散射效应（现在称为康普顿散射）而于 1929 年荣获诺贝尔物理学奖。[68] 这源自爱因斯坦的狭义相对论，该理论指出能量和动量通过以下方程相关： $E^2 = m_0^2 c^4 + p^2 c^2$ 。光子的静止质量为零，即 $m_0 = 0$ 。对于光子来说，这种相对论关系变为 $E = pc$ 。现在回想一下光子能量由 $E = hf$ 给出。重新排列这些项，我们可以显示： $p = hf/c = h/\lambda$ 。125

➤ due to exchange of momentum, electromagnetic wave can exert *radiation pressure*

forces generated by radiation pressure are negligible under everyday circumstances, but they could have noticeable effects on spacecraft in outer space and comet tails

Example 11.3 A beam of light has wavelength 600 nm, cross-sectional area 0.16 cm^2 and power 5.0 mW. The beam is normally incident onto a surface and is completely absorbed. Calculate, for a time of 1.0 s, (a) the number of photons incident onto the surface, (b) the change of total momentum of the photons, (c) the light pressure on the surface.

$$\text{energy of one photon: } E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{600 \times 10^{-9}} \approx 3.32 \times 10^{-19} \text{ J } 1.5083$$

$$\text{number of photons arriving in 1.0 s: } N = \frac{5.0 \times 10^{-3} \times 1.0}{3.32 \times 10^{-19}} \approx 1.51 \times 10^{16}$$

$$\text{momentum for one photon: } p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{600 \times 10^{-9}} \approx 1.11 \times 10^{-27} \text{ kg m s}^{-1}$$

$$\text{change of total momentum: } \Delta P = Np = 1.51 \times 10^{16} \times 1.11 \times 10^{-27} \approx 1.67 \times 10^{-11} \text{ kg m s}^{-1}$$

$$\text{average force due to these photons: } F = \frac{\Delta P}{\Delta t} = \frac{1.67 \times 10^{-11}}{1.0} \approx 1.67 \times 10^{-11} \text{ N}$$

$$\text{light pressure on surface: } p = \frac{F}{A} = \frac{1.67 \times 10^{-11}}{0.16 \times 10^{-4}} \approx 1.04 \times 10^{-6} \text{ Pa}$$

□

Question 11.8 A laser of power P is incident normally on a spot of area A , show that the pressure caused by the beam can be given by: $p = \frac{P}{cA}$.

Question 11.9 When an electron and a positron meet together, they will annihilate and produce two γ -photons: ${}_{-1}^0e + {}_{+1}^0e \rightarrow \gamma + \gamma$. Assume the electron and the positron have negligible kinetic energy before the interaction, explain why the two photons produced must move off in opposite directions with equal energies.

杨宇豪 11 量子物理基础知识电磁波由于动量交换，可以施加辐射压力辐射压力产生的力在日常情况下可以忽略不计，但对外太空的航天器和彗尾却能产生明显的影响例 11.3 一束光的波长为 600 nm，横截面积为 0.16 cm²，功率为 5.0 mW。光束通常入射到表面并被完全吸收。计算 1.0 s 时间内，(a) 入射到表面的光子数量，(b) 光子总动量的变化，(c) 表面上的光压。

一个光子的能量： $E = hc/\lambda = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 600 \times 10^{-9} = 3.32 \times 10^{-19} \text{ J}$

1.0 秒内到达的光子数量： $N = 5.0 \times 10^{-3} / 3.32 \times 10^{-19} = 1.51 \times 10^{16}$

1 个光子的动量： $p = h/\lambda = 6.63 \times 10^{-34} / 600 \times 10^{-9} = 1.11 \times 10^{-27} \text{ kg m s}^{-1}$

总动量变化： $\Delta P = Np = 1.51 \times 10^{16} \times 1.11 \times 10^{-27} = 1.67 \times 10^{-11} \text{ kg m s}^{-1}$

这些光子产生的平均力： $F = \Delta P / \Delta t = 1.67 \times 10^{-11} / 1.0 = 1.67 \times 10^{-11} \text{ N}$

表面光压： $p = F/A = 1.67 \times 10^{-11} / 0.16 \times 10^{-4} = 1.04 \times 10^{-6} \text{ Pa}$

问题 11.8 功率为 P 的激光垂直入射到面积 A 的一点上，证明光束引起的压力可由下式给出： $p = P/cA$

问题 11.9 当电子和正电子相遇时，它们会湮灭并产生两个 γ 光子： $0^{-1}e + 0^{+1}e \rightarrow \gamma + \gamma$ 。假设电子和正电子在相互作用之前具有可忽略不计的动能，解释为什么产生的两个光子必须以相等的能量向相反的方向移动。

11.3 wave-particle duality

11.3.1 matter waves

inspired by photon theory, which shows electromagnetic waves have a particulate nature, *Louis de Broglie* suggested in his 1924 PhD thesis that all matter has a wave-like nature ^[69]

wave characteristic of a particle can be represented by a wavelength

de Broglie wavelength of a matter particle is given by: $\lambda = \frac{h}{p}$, where $p = mv$ is the particle's momentum

Example 11.4 What is the wavelength of a human of 70 kg walking at around 2.0 m s^{-1} ?

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{75 \times 2} \approx 4.7 \times 10^{-36} \text{ m}$$

this wavelength is too small compared with any obstacle we encounter in everyday lives
so human bodies do not exhibit noticeable wave behaviour $\square$

Example 11.5 An electron is accelerated from rest through a potential difference of 50 V. What is the de Broglie wavelength of this electron?

K.E. of electron equals change in electric P.E., so

$$\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 1.60 \times 10^{-19} \times 50}{9.11 \times 10^{-31}}} \approx 4.19 \times 10^6 \text{ m s}^{-1}$$

$$\text{wavelength of electron: } \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 4.19 \times 10^6} \approx 1.74 \times 10^{-10} \text{ m}$$

this wavelength is comparable to scale of atomic spacing (also around 10^{-10} m)

so these electron can be *diffracted* by solid crystals $\square$

Question 11.10 An α -particle is moving with a kinetic energy of $2.4 \times 10^{-15} \text{ J}$. Find its speed, and hence find its de Broglie wavelength.

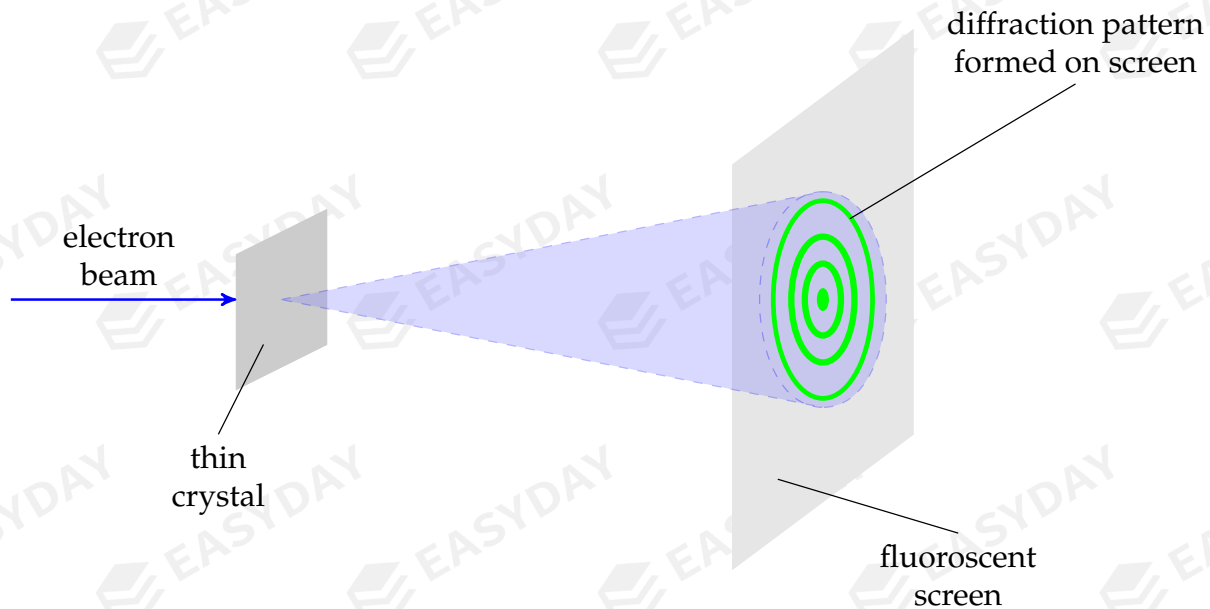
Question 11.11 If a proton and an electron are accelerated through the same voltage, (a) how would their energy compare? (b) how would their wavelength compare?

^[69] Louis de Broglie was awarded the 1929 Nobel Physics Prize 'for his discovery of the wave nature of electrons'. *Schrödinger's equation* and *Bohr's atomic model* was heavily influenced by ideas of de Broglie.

杨宇豪 11 量子物理学基础 11.3 波粒二象性 11.3.1 物质波受光子理论的启发，电磁波具有粒子性，路易斯·德布罗意在他 1924 年的博士论文中提出，所有物质都具有波状性质 [69] 粒子的波特性可以用波长德布罗意来表示物质粒子的波长由下式给出： $\lambda = h/p$ ，其中 $p = mv$ 是粒子的动量例 11.4 重 70 kg 的人以 2.0 m s^{-1} 左右的速度行走的波长是多少？ $\lambda = h/mv = 6.63 \times 10^{-34} / 75 \times 2 = 4.7 \times 10^{-36}\text{ m}$ 与我们在日常生活中遇到的任何障碍相比，这个波长太小，因此人体不会表现出明显的波行为 例 11.5 电子通过 50 V 的电势差从静止加速。该电子的德布罗意波长是多少？K.E. 电子的变化等于电 P.E. 的变化，因此 $\frac{1}{2}mv^2 = qV$ $v = \sqrt{2qV/m} = \sqrt{2 \times 1.60 \times 10^{-19} \times 50 / 9.11 \times 10^{-31}} = 4.19 \times 10^6\text{ m s}^{-1}$ 电子波长： $\lambda = h/mv = 6.63 \times 10^{-34} / 9.11 \times 10^{-31} \times 4.19 \times 10^6 = 1.74 \times 10^{-10}\text{ m}$ 这个波长与原子间距的尺度相当（也在 10^{-10} m 左右），因此这些电子可以被固体晶体衍射 问题 11.10 一个 α 粒子以 $2.4 \times 10^{-15}\text{ J}$ 的动能移动。求它的速度，从而得出它的速度求其德布罗意波长。问题 11.11 如果质子和电子通过相同的电压加速，(a) 它们的能量如何比较？(b) 它们的波长如何比较？[69] 路易斯·德布罗意“因发现电子的波动性”而荣获 1929 年诺贝尔物理学奖。薛定谔方程和玻尔原子模型深受德布罗意思想的影响。 127

11.3.2 electron diffraction

wave property of electron was confirmed by *Clinton Davisson* and *George Thomson* in 1927 they showed experimentally electrons could be diffracted by metal crystals ^[70]



electron diffraction experiment

- electron diffraction experiment shows that particles do have wave-like properties
if electrons behaved like particles, we would see a round spot of *uniform* distribution
but the actual pattern formed on *fluorescent screen* is a set of *concentric rings*
this is a typical diffraction pattern, hence proves wave properties of electrons
- each metal has a different lattice structure, so each produces a different pattern
this allows investigation of structure of matter (explore arrangements of atoms, structures of complex molecules, structure of atomic nuclei, etc.) using electron diffraction
- since wavelength of an electron is much shorter than visible light
this allows *electron microscope* to have much higher resolving power than optical microscopes

Question 11.12 If a higher p.d. is applied to accelerate the electron beam in the electron diffraction experiment, how would the pattern change?

^[70] The Nobel Prize in Physics 1937 was awarded jointly to Clinton Davisson and George Thomson 'for their experimental discovery of the diffraction of electrons by crystals'

杨宇豪 11 量子物理学基础 11.3.2 电子衍射波性质 1927 年，Clinton Davisson 和 George Thomson 证实了电子的波性质，他们通过实验表明电子可以被金属晶体衍射 [70] 薄晶体荧光屏上形成的衍射图案电子束电子衍射实验电子衍射实验表明，粒子确实具有波状性质如果电子表现得像粒子，我们会看到一个均匀的圆斑但荧光屏上形成的实际图案是一组同心环，这是典型的衍射图案，因此证明了电子的波动特性 - 每种金属具有不同的晶格结构，因此每种金属都会产生不同的图案，这允许使用电子衍射研究物质结构（探索原子的排列、复杂分子的结构、原子核的结构等） - 由于电子的波长比可见光短得多，这使得电子显微镜比光学显微镜具有更高的分辨率11.12 如果更高的 p.d.在电子衍射实验中对电子束进行加速，图案会发生怎样的变化？ [70]1937年诺贝尔物理学奖联合授予克林顿·戴维森和乔治·汤姆森，“表彰他们对晶体电子衍射的实验发现” 1

11.3.3 wave-particle duality

we have seen both particle-like and wave-like behaviour in light and electrons

	electromagnetic radiation	electron
particle-like behaviour	photoelectric effect, radiation pressure	deflection in electric/magnetic fields, decay, scattering
wave-like behaviour	interference, diffraction, Doppler effect, reflection, refraction	electron diffraction

when light/electron move through space, they behave like a wave

when light/electron interact with each other, they behave like particles

they show a two-sided nature, or a *dual* nature of being both a wave and a particle, described by either particle model or wave model under different circumstances

in fact, all matter (protons, neutrons, atoms, cells, basketballs, human body, earth, etc.) has this universal dual nature, called **wave-particle duality**

wave-particle duality addresses breakdown of classical concepts like particle or wave

to fully describe the behaviour of microscopic objects, we need *quantum mechanics* ^[71]

11.4 quantisation of electron energy levels

in a simple atomic model, electrons move around the nucleus in circular orbits

but now we understand electrons have wave properties, as an electron moves in its orbit as a wave, it can superpose with itself

only orbits in which the electron can superpose constructively with itself are preferable

so electrons are only allowed to move in certain orbits in an atom

this means can only take certain values of energy, called *energy levels*

^[71]Being a central concept of quantum theory, wave-particle duality is deeply embedded into the foundations of quantum physics. In non-relativistic quantum mechanics, all information about a particle is encoded in its *wave function*, whose evolution with time is described by the famous *Schrödinger's equation*.

杨宇浩 11 量子物理基础 11.3.3 波粒二象性我们已经看到了光和电子中的类粒子和类波行为电磁辐射电子类粒子光电效应、电/磁行为的偏转辐射压力场、衰变、散射类波行为干涉、衍射、多普勒效应、反射、折射光/电子在空间中移动时的电子衍射，当光/电子在空间中移动时，它们的行为像波相互作用，它们像粒子一样表现出双面性，或者既是波又是粒子的二元性，在不同情况下用粒子模型或波模型来描述。事实上，所有物质（质子、中子、原子、细胞、篮球、人体、地球等）都具有这种普遍的二元性，称为波粒二象性。波粒二象性解决了粒子或波等经典概念的崩溃，以充分描述微观物体的行为，我们需要量子力学[71] 11.4 简单原子模型中电子能级的量子化，电子在圆形轨道上围绕原子核运动，但现在我们知道电子具有波动特性，当电子作为波在其轨道上运动时，它只能与自身叠加，而电子只能与自身相长叠加的轨道是优选的，因此电子只能在原子中的某些轨道上运动，这意味着只能采用某些能量值，称为能级[71]，这是量子理论的一个中心概念，波粒对偶性深深植根于量子物理学的基础之中。在非相对论量子力学中，有关粒子的所有信息都编码在其波函数中，波函数随时间的演化由著名的薛定谔方程描述。 129

11.4.1 hydrogen atom (*)

theoretical explanation for electron energy levels was developed in 1913 by Danish physicist *Niels Bohr* in his theory of hydrogen atom [72]

let's look at the hydrogen atom – the simplest possible atom in nature

electrostatic attraction by proton provides cen-

tripetal force for electron to move in circles

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r} \Rightarrow v^2 = \frac{e^2}{4\pi\epsilon_0 m_e r}$$

for electron to *constructively* superpose with itself, the perimeter of its orbit must be an integer multiple of its de Broglie wavelength

$$2\pi r = n\lambda \Rightarrow 2\pi r = \frac{nh}{m_e v} \Rightarrow v = \frac{nh}{2\pi m_e r}$$

where n is positive integer 1, 2, 3, ...

compare the two equations, we can eliminate v

$$\frac{e^2}{4\pi\epsilon_0 m_e r} = \frac{n^2 h^2}{4\pi^2 m_e^2 r^2}$$

we hence find radius of allowed orbits satisfies:

$$r_n = n^2 a_0 \quad \text{where } a_0 = \frac{h^2 \epsilon_0}{\pi m_e e^2} \approx 5.29 \times 10^{-11} \text{ m}$$

so we see all allowed orbits must have a radius equal to integer multiple of a_0

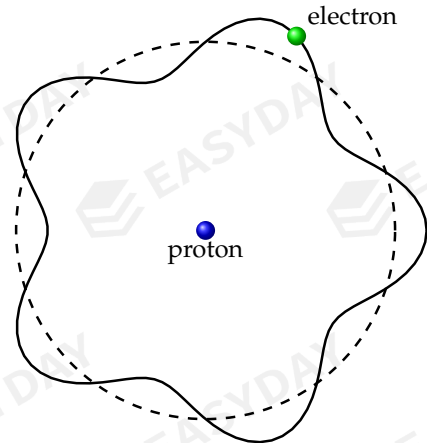
total energy possessed by the electron consist of K.E. and electric P.E.:

$$E = E_k + E_p = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = \frac{1}{2} m_e \frac{e^2}{4\pi\epsilon_0 m_e r} - \frac{e^2}{4\pi\epsilon_0 r} \Rightarrow E = -\frac{e^2}{8\pi\epsilon_0 r}$$

substitute the allowed orbital radius, we find an expression for electron energy:

$$E_n = -\frac{R_E}{n^2} \quad \text{where } R_E = \frac{m_e e^4}{8h^2 \epsilon_0^2} \approx 2.18 \times 10^{-18} \text{ J} \approx 13.6 \text{ eV}$$

this shows electron can only have specific energies in the hydrogen atom



the hydrogen atom

11.4.2 electron energy levels

it can be shown that for any atom, electrons can only have certain fixed values of energy

[72] Niels Bohr was surely one of the greatest physicists of the 20th century. He made foundational contributions to understanding atomic structure and quantum mechanics, for which he received the Nobel Physics Prize in 1922. He was also the founder of the Institute of Theoretical Physics at the University of Copenhagen, which soon became the centre of pioneering researches on quantum theory in the world.

杨宇豪 11 量子物理基础 11.4.1 氢原子 (Q) 电子能级的理论解释是丹麦物理学家 Niels Bohr 于 1913 年在他的氢原子理论中提出的[72] 让我们看看氢原子——自然界中最简单的原子质子电子氢原子受到质子的静电引力为电子提供向心力 $\frac{e^2}{4\pi\epsilon_0 r^2} = m_e v^2 / r$ $v^2 = \frac{e^2}{4\pi\epsilon_0 m_e r}$ 电子要与其自身相长叠加，其轨道周长必须是其德布罗意波长的整数倍 $2\pi r = n\lambda$ $2\pi r = nh / m_e v$ $v = nh / 2\pi m_e r$ 其中 n 为正整数 1, 2, 3, 比较两个方程，可消去 v $\frac{e^2}{4\pi\epsilon_0 m_e r} = n^2 h^2 / 4\pi^2 m_e^2 r^2$ 因此，我们发现允许轨道的半径满足： $r_n = n^2 a_0$ 其中 $a_0 = \frac{h^2 \epsilon_0 \pi m_e}{e^2} \approx 5.29 \times 10^{-11} \text{ m}$ 因此我们看到所有允许轨道的半径必须等于由 K.E. 组成的电子所拥有的总能量 a_0 的整数倍。和电 P.E.： $E = E_k + E_p = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = \frac{1}{2} m_e \frac{e^2}{4\pi\epsilon_0 m_e r} - \frac{e^2}{4\pi\epsilon_0 r}$ $E = -\frac{e^2}{8\pi\epsilon_0 r}$ 代入允许的轨道半径，我们找到电子能量的表达式： $E_n = -RE / n^2$ where $RE = \frac{m_e e^4}{8h^2 \epsilon_0^2} \approx 2.18 \times 10^{-18} \text{ J} \approx 13.6 \text{ eV}$ 这表明电子只能在氢原子中具有特定的能量 11.4.2 电子能级可以证明对于任何原子，电子只能具有某些固定的能量值 [72] Niels Bohr 无疑是 20 世纪最伟大的物理学家之一。他对理解原子结构和量子力学做出了基础性贡献，并因此获得 1922 年诺贝尔物理学奖。他也是哥本哈根大学理论物理研究所的创始人，该研究所很快成为世界量子理论先驱研究的中心。 130

we say energy of electrons in an atom is *discrete*, or **quantised**

these allowed values of energies are called **energy levels**^[73]

➤ *quantisation* means an electron can have only specific values of energy in an atom

no intermediate value between levels is allowed

➤ the lowest energy level is called the **ground state**

any level higher than the ground state is called an

excited state

in our illustration, E_1 is the ground state, while E_2 ,

E_3 , E_4 are all excited states

➤ electron energy levels are all *negative*

to pull an electron away from the nucleus, work must be done to overcome electrostatic attraction

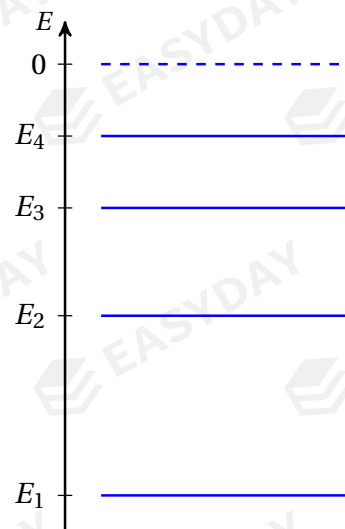
an electron at infinity would have greatest P.E., which is defined to be zero

so orbiting electrons have energies less than zero

an electron that has zero energy would become a *free* electron

➤ electrons can jump, or *transit*, between energy levels

electron transition is associated with emission or absorption of photon with the right energy



electron energy levels of some atom

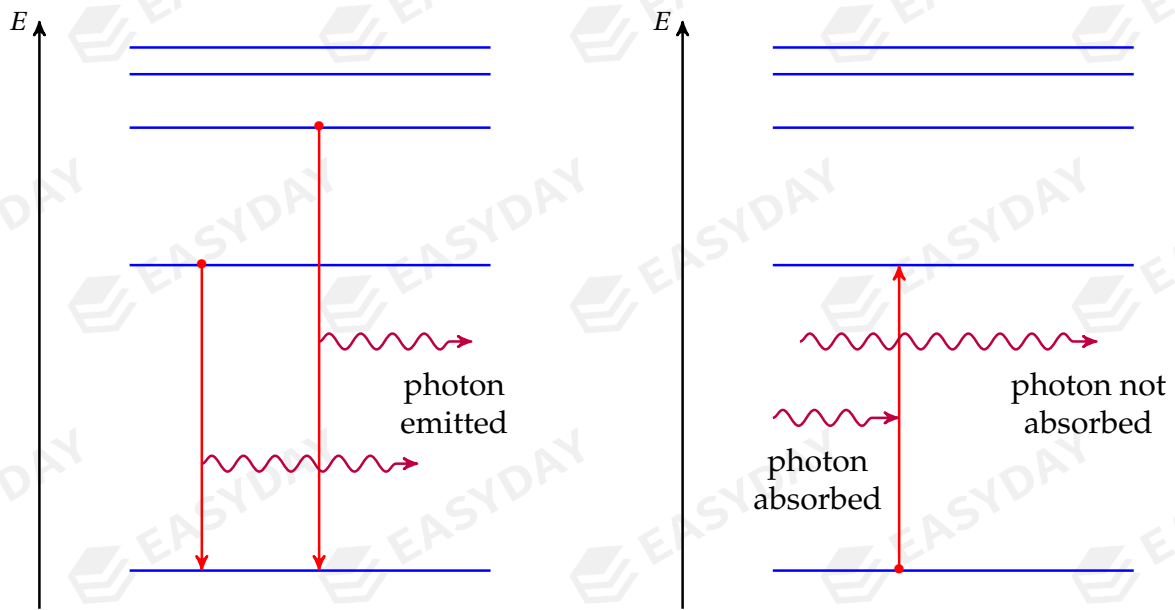
11.4.3 atomic spectrum

passing light through a *prism* or a *diffraction grating*, different wavelengths are separated, a phenomenon known as **dispersion**, this produces a **spectrum** which shows distribution of energy emitted from the source in order of wavelengths

for example, white light consists of a range of wavelengths from around 400 nm to 700 nm, so it gives a *continuous spectrum* with rainbow of colours

^[73] If you study chemistry, you shall recall this idea of discrete energy levels is related to the concepts of energy shells and sub-shells of atoms.

Yuhao Yang 11 量子物理学基础我们说原子中电子的能量是离散的，或者说是量子化的，这些允许的能量值称为能级[7 3] $E_1, E_2, E_3, E_4, \dots$ 某些原子的电子能级 $\hat{H}$ 量子化意味着原子中电子只能具有特定的能量值，不允许有能级之间的中间值 $\hat{H}$ 最低能级称为基态任何高于基态的能级为在我们的插图中称为激发态， E_1 是基态，而 E_2, E_3, E_4 都是激发态 – 电子能级均为负，要将电子拉离原子核，必须做功来克服静电引力无穷远处的电子将具有最大 $P.E.$ ，它被定义为零，因此轨道电子的能量小于零，能量为零的电子将成为自由电子 – 电子可以在能级之间跳跃或跃迁电子跃迁是相关的发射或吸收具有正确能量的光子 11.4.3 原子光谱光通过棱镜或衍射光栅，不同的波长被分离，这是一种称为色散的现象，这会产生一个光谱，显示从光源发射的能量按波长顺序的分布，例如，白光由大约 400 nm 到 700 nm 的波长范围组成，因此它给出了具有彩虹色的连续光谱 [73]如果您学习化学，您应该记得这个想法离散能级的概念与原子的能量层和子层的概念有关。 131



emission and absorption of photons as a result of electron transitions



continuous spectrum of white light

electron transition between different levels causes emission or absorption of photons

this leads to the emission spectrum and absorption spectrum of atoms

emission spectrum

hot gases of an element can release photons, giving an **emission spectrum**

this happens when an electron transits from a high energy level to a lower level

energy of the photon emitted is equal to energy difference between the two levels

since electron energy levels are discrete, only specific changes of energies are possible, so only photons with specific energies can be emitted

this means photons emitted only have specific wavelengths, or specific frequencies

so a collection of sharp and bright lines are seen in emission spectrum

- emission spectrum is a *discrete* spectrum, also called a *line* spectrum
- atoms of each element has a unique electron energy level structure

杨宇浩 11 量子物理学基础 E^2 发射的光子 E^2 光子不被吸收光子被吸收电子跃迁导致的光子的发射和吸收
白光的连续光谱电子在不同能级之间的跃迁导致光子的发射或吸收，从而产生原子的发射光谱和吸收光谱
元素的热气体可以释放光子，给出发射光谱当电子从高能级跃迁到低能级时，发射的光子的能量等于两个
能级之间的能量差，因为电子能级是离散的，只能发生特定的能量变化，因此只能发射具有特定能量的光
子，这意味着发射的光子仅具有特定的波长或特定的频率，因此在发射光谱中可以看到一系列尖锐而明亮
的线，发射光谱是离散光谱，也称为线光谱，每个元素的原子具有独特的电子能级结构¹³²



emission spectrum of the hydrogen atom

so each element produces a unique emission spectrum, leading to quite a few applications:

- explains the colour of flames when a particular chemical element is present
- allows the identification of elements in an unknown substance in chemical analysis
- explains varied colours of electric signs lighted by gas-discharge tubes

absorption spectrum

pass white light through a *cool gas*, photons with the right energies can be absorbed, giving rise to an **absorption line spectrum**

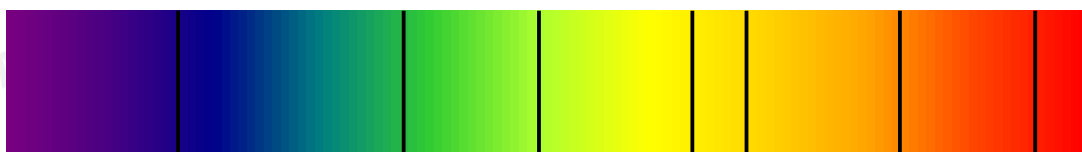
by absorbing photon, electron can transit from a low energy level to a higher level

energy of the photon absorbed must equal energy difference between the two levels

since electron energy levels are discrete, so only photons with specific energies can be absorbed, while other photons are unaffected as they pass through the gas

wavelengths of these absorbed photons will be missing in the emergent spectrum

so we would observe a set of *dark lines* appearing in a background of continuous spectrum



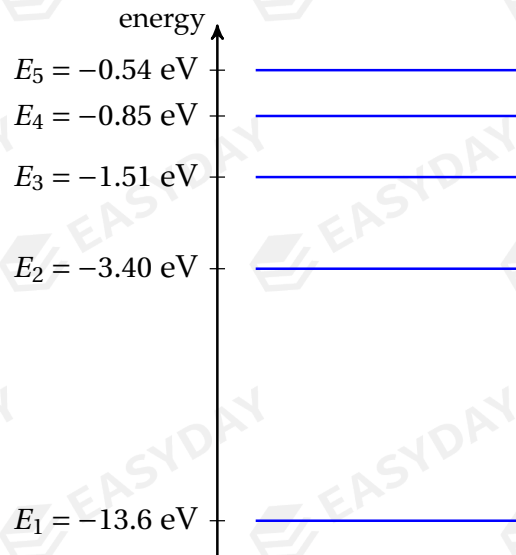
absorption spectrum for some element

- absorption spectrum is also a *discrete* spectrum, or a *line* spectrum
- as photons with the right energies are absorbed, they can be re-emitted through *de-excitation* but these photons are re-emitted in *random* directions so they would still appear dark compared with those unaffected wavelengths
- absorption spectrum is widely used in many areas
 - determine the composition of a particular substance in analytical chemistry

Yuhao Yang 11 量子物理学基础氢原子的发射光谱，因此每种元素都会产生独特的发射光谱，从而产生相当多的应用：- 解释存在特定化学元素时火焰的颜色 - 允许在化学分析中识别未知物质中的元素 - 解释气体放电管发出的电信号的不同颜色吸收光谱让白光穿过冷气体，可以吸收具有正确能量的光子，从而产生吸收线光谱通过吸收光子，电子可以从低能级跃迁到高能级。被吸收的光子的能量必须等于两个能级之间的能量差，因为电子能级是离散的，因此只有具有特定能量的光子才能被吸收，而其他光子在穿过气体波长时不受影响，这些被吸收的光子将在出射光谱中丢失，因此我们会观察到某些元素的连续光谱吸收光谱背景中出现的一组暗线 α 吸收光谱为也是离散光谱或线光谱 $\hat{\alpha}$ 当具有正确能量的光子被吸收时，它们可以通过去激发重新发射，但这些光子以随机方向重新发射，因此与那些不受影响的波长相比，它们仍然显得较暗 $\hat{\alpha}$ 吸收光谱广泛用于许多领域 - 确定分析化学中特定物质的成分 133

- determine chemical compositions of stars in astronomical spectroscopy
- explain colour of chemicals in terms of the complementary colour of photons absorbed

Example 11.6 Some electron energy levels of the hydrogen atom is shown. Find the longest and the shortest wavelength produced by electron transitions between the energy levels given.



energy levels of the hydrogen atom

energy of photon emitted equals change of electron energy level: $\frac{hc}{\lambda} = \Delta E$, or $\lambda = \frac{hc}{\Delta E}$

transition with least/greatest energy change gives rise to longest/shortest wavelength, so

$$E_5 \rightarrow E_4 \Rightarrow \lambda_{\max} = \frac{hc}{E_5 - E_4} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{(0.85 - 0.54) \times 1.60 \times 10^{-19}} \approx 4.0 \times 10^{-6} \text{ m} \quad (\text{infra-red})$$

$$E_5 \rightarrow E_1 \Rightarrow \lambda_{\min} = \frac{hc}{E_5 - E_1} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{(13.6 - 0.54) \times 1.60 \times 10^{-19}} \approx 9.5 \times 10^{-8} \text{ m} \quad (\text{ultraviolet}) \quad \square$$

Example 11.7 The emission spectrum of the hydrogen atom consists of a number of wavelengths in the visible spectrum. Given that the visible spectrum consists of light of wavelengths within the range from 380 nm to 740 nm, also use data in Example 11.6, find the energies of visible photons that can be produced by transitions between the energy levels shown.

let's first find the range of energies, in eV, for visible photons

$$E_{\max} = \frac{hc}{\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{380 \times 10^{-9}} \approx 5.23 \times 10^{-19} \text{ J} = \frac{5.23 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 3.27 \text{ eV}$$

$$E_{\min} = \frac{hc}{\lambda_{\max}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{740 \times 10^{-9}} \approx 2.69 \times 10^{-19} \text{ J} = \frac{2.69 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 1.68 \text{ eV}$$

we shall find the combinations of electron energy levels such that their difference fall within the range between 1.68 eV and 3.27 eV, by trial and error, we find three possible combinations:

Yuhao Yang 11 量子物理学基础 - 在天文光谱中确定恒星的化学成分 - 根据吸收的光子的补色解释化学物质的颜色

例 11.6 显示了氢原子的一些电子能级。求给定能级之间电子跃迁产生的最长和最短波长。能量 $E_1 = -13.6 \text{ eV}$ $E_2 = -3.40 \text{ eV}$ $E_3 = -1.51 \text{ eV}$ $E_4 = -0.85 \text{ eV}$ $E_5 = -0.54 \text{ eV}$ 氢原子的能级 - 发射的光子能量等于电子能级的变化： $hc/\lambda = \Delta E$ ，或 $\lambda = hc/\Delta E$ 跃迁，能量变化最小/最大产生最长/最短波长，所以 $E_5 \rightarrow E_4$ $\lambda_{\max} = hc/(E_5 - E_4) = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / (0.85 - 0.54) \times 1.60 \times 10^{-19} \text{ J} = 4.0 \times 10^{-6} \text{ m}$ (红外) $E_5 \rightarrow E_1$ $\lambda_{\min} = hc/(E_5 - E_1) = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / (13.6 - 0.54) \times 1.60 \times 10^{-19} \text{ J} = 9.5 \times 10^{-8} \text{ m}$ (紫外线)

例 11.7 氢原子的发射光谱由可见光谱中的多个波长组成。鉴于可见光谱由波长在 380 nm 到 740 nm 范围内的光组成，也可使用例 11.6 中的数据，找出所示能级之间跃迁产生的可见光子的能量。

- 让我们首先找到可见光子的能量范围（以 eV 为单位） $E_{\max} = hc/\lambda_{\min} = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 380 \times 10^{-9} \text{ J} = 5.23 \times 10^{-19} \text{ J} = 5.23 \times 10^{-19} / 1.60 \times 10^{-19} \text{ eV} = 3.27 \text{ eV}$ $E_{\min} = hc/\lambda_{\max} = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 740 \times 10^{-9} \text{ J} = 2.69 \times 10^{-19} \text{ J} = 2.69 \times 10^{-19} / 1.60 \times 10^{-19} \text{ eV} = 1.68 \text{ eV}$ 我们将找到电子能级的组合使它们的差异落在 1.68 eV 和 3.27 eV 之间的范围内，通过反复试验，我们发现了三种可能的组合：

$$(1) E_3 \rightarrow E_2 \Rightarrow 3.40 - 1.51 = 1.89 \text{ eV}$$

$$(2) E_4 \rightarrow E_2 \Rightarrow 3.40 - 0.85 = 2.55 \text{ eV}$$

$$(3) E_5 \rightarrow E_2 \Rightarrow 3.40 - 0.85 = 2.85 \text{ eV}$$

these emission lines are called H_α (656 nm), H_β (486 nm) and H_γ (435 nm) lines^[74]

they are three of the four the hydrogen emission lines that are visible to human eyes^[75] $\square$

Example 11.8 A white light is incident on a cloud of cool hydrogen gas. In the emergent spectrum, a dark line is observed at a wavelength of 435 nm. Determine the energy change that gives rise to this dark line.

$$\text{photon absorbed: } E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{435 \times 10^{-9}} \approx 4.57 \times 10^{-19} \text{ J} = \frac{4.57 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 2.86 \text{ eV}$$

note that $3.40 - 0.54 = 2.86 \text{ eV}$, so this dark line is due to the electron transition: $E_2 \rightarrow E_5$ $\square$

Question 11.13 If we only consider the electron energy levels given in Example 11.6, how many different wavelengths can be detected in the emission spectrum of the hydrogen atom?

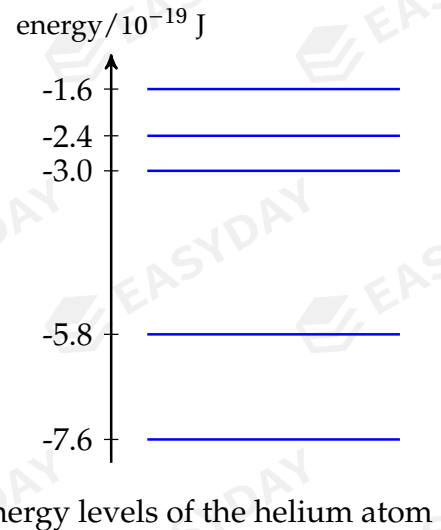
Question 11.14 Three lines are observed at wavelengths 486 nm, 656 nm, and 1880 nm in the emission spectrum of hydrogen atoms. (a) Calculate the photon energies for these wavelengths. (b) Draw a diagram with *three* labelled energy levels, and show the energy changes for the three wavelengths produced with arrows in your diagram.

^[74]The H_α line is the brightest hydrogen line in the visible range. It plays an important role in astronomy, as it can be used to study a star's surface temperature, the velocity of a distant stellar object, etc.

^[75]These lines belong to a family of the spectral lines of the hydrogen atom, known as the *Balmer lines*, named after Johann Balmer. Balmer discovered an empirical equation to predict the series in 1885, but the reason why the equation worked was eventually clarified by Neils Bohr with the atomic model which now bears his name. We now understand the Balmer lines correspond to emissions of photons by electrons jumping to the second level from higher energy states.

杨宇豪 11 量子物理基础 (1) $E_3 - E_2 = 3.40 - 1.51 = 1.89 \text{ eV}$ (2) $E_4 - E_2 = 3.40 - 0.85 = 2.55 \text{ eV}$ (3) $E_5 - E_2 = 3.40 - 0.85 = 2.85 \text{ eV}$ 这些发射线称为 $H\alpha$ (656 nm)、 $H\beta$ (486 nm) 和 $H\gamma$ (435 nm) 线 [74] 它们是人眼可见的 4 条氢发射线中的 3 条 [75] 例 11.8 白光入射到冷氢气云上。在出射光谱中，在波长 435 nm 处观察到一条暗线。确定产生这条暗线的能量变化。 - 光子吸收： $E = hc/\lambda = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 435 \times 10^{-9} = 4.57 \times 10^{-19} \text{ J} = 4.57 \times 10^{-19} / 1.60 \times 10^{-19} \text{ eV} = 2.86 \text{ eV}$ 请注意 $3.40 - 0.54 = 2.86 \text{ eV}$ ，所以这条黑线是由于电子跃迁造成的： $E_2 - E_5$ 问题 11.13 如果我们只考虑例 11.6 中给出的电子能级，在氢原子的发射光谱中可以检测到多少种不同的波长？问题 11.14 在氢原子的发射光谱中，在波长 486 nm、656 nm 和 1880 nm 处观察到三条谱线。(a) 计算这些波长的光子能量。(b) 绘制带有三个标记能级的图表，并显示图表中箭头所产生的三个波长的能量变化。 [74] $H\alpha$ 线是可见光范围内最亮的氢线。它在天文学中发挥着重要作用，因为它可以用来研究恒星的表面温度、遥远恒星物体的速度等。 [75] 这些谱线属于氢原子谱线家族，称为巴尔默线，以约翰·巴尔默的名字命名。巴尔默在 1885 年发现了一个经验方程来预测该级数，但该方程有效的原因最终由尼尔斯·玻尔用现在以他的名字命名的原子模型阐明了。我们现在了解到巴尔默线对应于电子从较高能态跃迁到第二能级所发射的光子。 135

Question 11.15 The relative cool atmosphere of the sun could give rise to dark lines in the spectrum of sunlight. One particular dark spectral line has a wavelength of 590 nm. By reference to the energy levels of the helium atom, suggest how this dark line provides evidence of the presence of helium in sun's atmosphere. You may draw an arrow to show the possible electron transition that gives rise to this dark line.



11.5 band theory (*)

11.5.1 energy bands

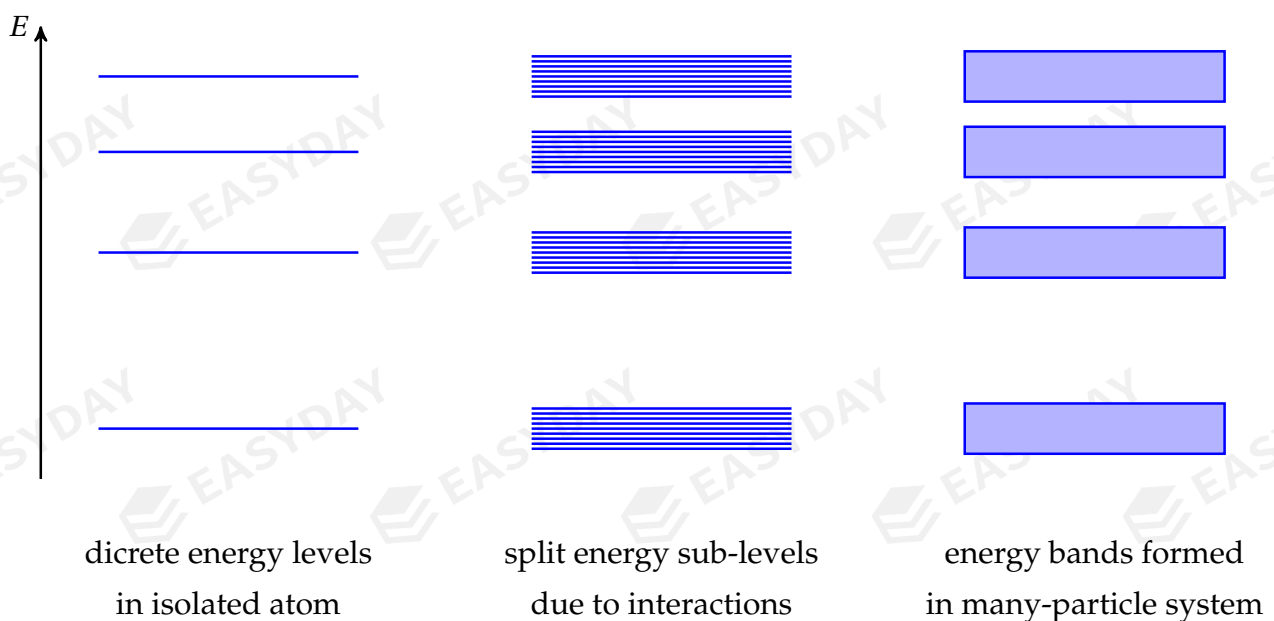
in an isolated atom, electrons have discrete energy levels

in solids, interaction between neighbouring atoms causes change of energies

original energy levels split into a band with many sub-levels

number of sub-levels in each band equals number of atoms in solid in general

due to very large number of atoms, sub-levels (seem to) form *continuous* **energy bands**



杨宇浩 11 量子物理基础 -7.6 -5.8 -3.0 -2.4 -1.6 能量/ 10^{-19} J 氢原子的能级问题 1 1.15 太阳相对凉爽的大气可能会在太阳光光谱中产生暗线。一条特定的暗光谱线的波长为 590 nm。通过参考氢原子的能级，说明这条暗线如何提供太阳大气中存在氢的证据。您可以画一个箭头来显示可能产生这条黑线的电子跃迁。

11.5 能带理论 (Q) 11.5.1 孤立原子中的能带，电子在固体中具有离散的能级，相邻原子之间的相互作用导致能量的变化原始能级分裂成具有许多子能级的能带由于原子数量非常多，每个能带中子能级的数量一般等于固体中原子的数量，子能级（似乎）形成连续的能带E孤立原子中的离散能级由于相互作用能量而分裂能量子能级在多粒子系统中形成的能带 136

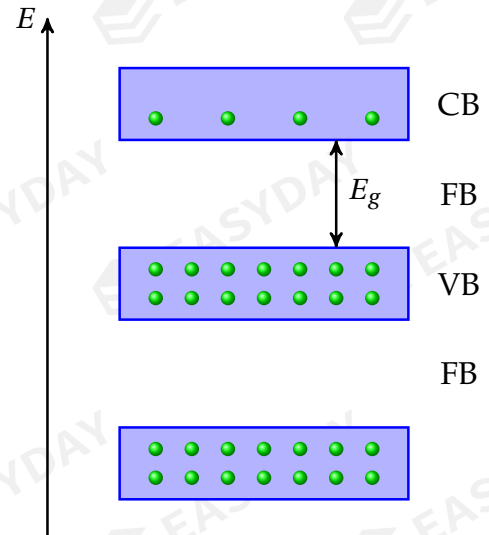
in solids, electrons fill up from lowest energy bands available (see figure for a rough idea)

forbidden bands (FB) represent energies that electrons are not allowed to take

highest fully filled band is called **valence band** (VB)

next partially filled or empty band is called **conduction band** (CB)

separation between valence band and conduction band defines **band gap** (E_g) of material



11.5.2 band theory & electrical conductivity

electrical conductivity of material can be explained by its band structure

low-energy bands below VB are usually fully filled at all times, all states are occupied, electrons cannot move freely to form electric currents, so we are not interested in low-energy bands

conductivity of material will depend on whether there are free *charge carriers* in CB and VB

metallic conductors

for typical metals, CB *overlaps* with VB, i.e., there is no band gap

there exist vacant states for conduction electrons to occupy at ease

this means conduction electrons are free to move around to form currents

as temperature rises, there is no significant change in number of free electrons

but *lattice vibration* of atoms increases, electrons become more likely to collide with vibrating atoms, so resistance of metal increases with temperature

➤ metals are *opaque* to visible light or other low-frequency radiation

width of CB for a typical metal is about 2 ~ 3 eV

low-energy photons can be absorbed by conduction electrons

visible light, infra-red, microwave therefore cannot pass through metal

➤ metals are *transparent* to high-frequency radiation such as X-rays

Yuhao Yang 11 量子物理基础 E CB VB FB FB 例如，在固体中，电子从可用的最低能带开始填充（粗略见图）禁带 (FB) 代表电子不允许占据的能量最高完全填充的能带称为价带 (VB) 下一个部分填充或空的能带称为导带 (CB) 价带之间的分离能带和导带定义了材料的带隙 (E_g) 11.5.2 能带理论和电导率材料的电导率可以通过其能带结构来解释 VB 以下的低能带通常始终被完全充满，所有态都被占据，电子不能自由移动形成电流，因此我们对低能带不感兴趣材料的电导率将取决于 CB 和 VB 金属导体中是否存在自由载流子对于典型金属，CB 重叠对于 VB，即不存在带隙，存在可供传导电子轻松占据的空态，这意味着随着温度升高，传导电子可以自由移动以形成电流，自由电子数量没有显著变化，但原子的晶格振动增加，电子更容易与振动原子碰撞，因此金属的电阻随温度增加。金属对可见光或其他低频辐射宽度不透明，典型金属的 CB 宽度约为 $2 \gg 3 \text{ eV}$ 低能光子可以被传导电子吸收，可见光、红外线、微波因此不能穿过金属 $\hat{\alpha}$ 金属对 X 射线等高频辐射是透明的 137

if X-ray photons were absorbed, electrons would take energies in forbidden band but this is not allowed, so high-frequency photons penetrate through metal

insulators

examples of insulators include glass, diamond, etc.

in normal conditions, CB of insulator is empty, energy bands up to VB are fully filled, so no movement of electron is possible, material has poor conductivity

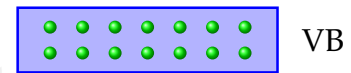
insulators also have huge band gap^{[76][77]}, so thermal excitations cannot easily make VB electrons jump into CB

insulators remain poor conductors as temperature rises

Question 11.16 Diamond has a band gap $E_g \approx 6.0$ eV. By reference to its electronic band structure, explain why diamond appears *transparent* to visible light?



CB



VB



band structure of insulator

intrinsic semiconductors

examples of an intrinsic semiconductor are silicon and germanium

band structure of an intrinsic semiconductor is similar to that of insulators

at low temperature, no free electron from the empty CB and completely occupied VB, so semi-conductors are not conductive in normal conditions

however, semiconductors have much narrower band gaps

as temperature rises, VB electrons gain thermal energy to cross band gap and enter CB

the electrons jumped into CB surely can move around to form electric currents

at the same time, VB is no longer completely filled, *holes* are formed

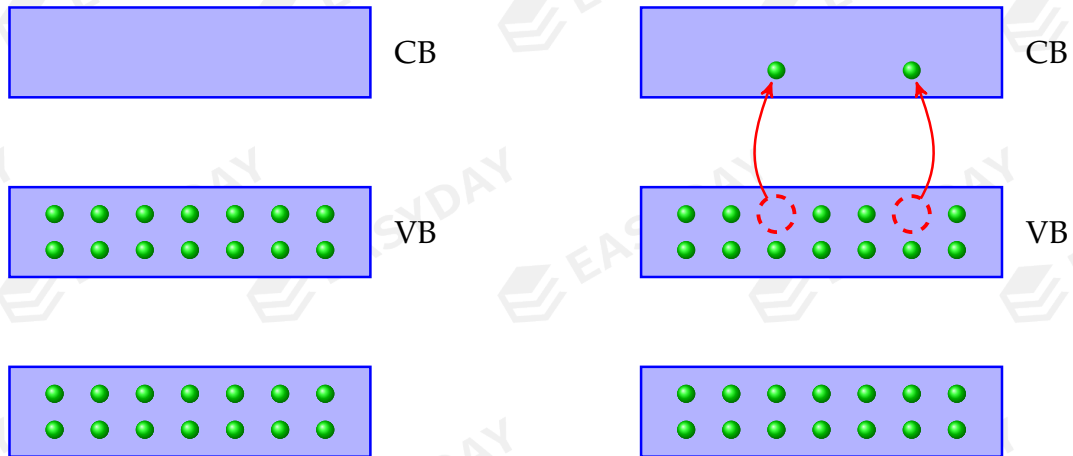
^[76]To be a little more precise, materials with a band gap $E_g \approx 5$ eV or greater are regarded as insulators.

^[77]There exist a class of insulators called *Mott insulators* which do not have large band gaps. Their poor conductivity at low temperatures is due to electron-electron interactions, which are not considered under conventional band theories.

杨宇豪 11 量子物理基础如果 X 射线光子被吸收，电子会在禁带中获取能量，但这是不允许的，所以高频光子会穿透金属绝缘体 CB 绝缘体的 VB 能带结构绝缘体的例子包括玻璃、金刚石等。在正常情况下，绝缘体的 CB 为空，直到 VB 的能带被完全充满，因此电子不可能运动，材料的性能较差电导率绝缘体也有巨大的带隙[76][77]，因此热激发不容易使 VB 电子跳入 CB 绝缘体中，随着温度升高，CB 绝缘体仍然是不良导体。问题 11.16 金刚石的带隙 $E_g \approx 6.0 \text{ eV}$ 。通过参考其电子能带结构，解释为什么钻石对可见光看起来是透明的？本征半导体的例子是硅和锗。本征半导体的能带结构在低温下与绝缘体类似，空的 CB 中没有自由电子，VB 被完全占据，所以半导体在正常情况下不导电，但是，随着温度的升高，半导体的带隙变得更窄，VB 电子获得热能穿过带隙进入 CB，电子跳入 CB 肯定可以同时移动形成电流，VB 不再完全占据[76]更精确地说，带隙 $E_g \approx 5 \text{ eV}$ 或更大的材料被视为绝缘体。[77] 存在一类称为莫特绝缘体的绝缘体，它们不具有大带隙。它们在低温下的导电性差是由于电子-电子相互作用造成的，而传统能带理论并未考虑这一点。 138

a hole is a site where an electron is missing, neighbouring electrons can move to fill the hole and leave a new hole, but this can be thought of as the hole moving about

there are now more *charge carriers* (negatively-charged electrons and positively-charged holes) available, so semiconductor has better conductivity, resistance of material would decrease



band structure of an intrinsic semiconductor at low temperature

at higher temperature VB electrons cross band gap and enter CB

➤ *lattice vibration* could also affect resistance of semiconductors

as temperature rises, vibration of atoms increases, charge carrier are more likely to get scattered, causing a decrease in conductivity

but the effect of more charge carriers is far greater than effect due to lattice vibration

so resistance of semiconductor decreases at higher temperature

Example 11.9 Silicon has a band gap $E_g \approx 1.1$ eV. State and explain whether it becomes conducting when exposed to red light of 600 nm.

✎ energy of radiation: $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{600 \times 10^{-9}} \approx 3.32 \times 10^{-19} \text{ J} \approx 2.1 \text{ eV} > E_g$

this energy is sufficient to make valence electrons in silicon to cross band gap

silicon now possesses conducting electrons and holes therefore becomes conducting □

Question 11.17 Use band theory to explain why an LDR (light-dependent resistor) has a very high resistance in the dark, but its resistance drops dramatically when being exposed to light.

杨宇豪 11 量子物理基础空穴是一个电子缺失的地方，邻近的电子可以移动来填充空穴并留下新的空穴，但这可以认为是空穴四处移动，现在有更多的载流子（带负电的电子和带正电的空穴）可用，因此半导体具有更好的导电性，材料的电阻会降低 CB VB CB VB 本征半导体的能带结构在低温下在较高温度下 VB 电子跨能带晶格振动也会影响半导体的电阻，随着温度的升高，原子振动加剧，载流子更容易分散，导致电导率下降，但载流子增多的影响远远大于晶格振动的影响，因此在较高温度下半导体的电阻降低。例 11.9 硅的带隙 $E_g = 1.1 \text{ eV}$ 。说明并解释暴露在 600 nm 红光下是否会导电。- 辐射能量： $E = h c / \lambda = 6.63 \times 10^{-34} \times 3.0 \times 10^8 / 600 \times 10^{-9} \text{ J} = 3.32 \times 10^{-19} \text{ J} = 2.1 \text{ eV}$ 例如，该能量足以使硅中的价电子穿过带隙硅现在拥有导电电子和空穴，因此变得导电 问题 11.17 使用能带理论解释了为什么 LDR（光敏电阻器）在黑暗中具有非常高的电阻，但在暴露在光线下时其电阻急剧下降。 139

CHAPTER 12

Nuclear Physics

12.1 nuclear energy

12.1.1 mass-energy equivalence

mass-energy equivalence principle states that mass and energy are equivalent to one another

this idea is represented mathematically by **Einstein's mass-energy relation**: $E = mc^2$ [78]

m is **rest mass** of the object, $c = 3.00 \times 10^8 \text{ m s}^{-1}$ is speed of light in vacuum

➤ mass-energy equivalence implies mass can be converted into pure energy

conversely, mass can be created out of energy

➤ $\Delta E = \Delta mc^2$ applies to *all* energy changes

if energy is supplied to a system, then mass of this system increases

for example, an object will have a greater mass when it is set to motion or is heated

mass is not a conserved quantity, it is *mass-energy* that is conserved in all processes

➤ if an object is stationary, its mass is called the *rest mass* m_0

mass-energy equivalence implies an object at rest still has an intrinsic *rest energy*: $E_0 = m_0 c^2$

➤ energy transformations such as chemical reactions can cause a system to lose some mass content, but this change in mass is usually negligible

[78] This might be the most famous equation in all physical sciences. When Albert Einstein wrote down the theory of *special relativity* in 1905, he found the mass m of an object with rest mass m_0 is related to the speed v at which it is moving by: $m = m_0 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$. Total energy is given by $E = mc^2 = m_0 c^2 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$.

At low speeds $v \ll c$, $E \approx m_0 c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2}\right) = m_0 c^2 + \frac{1}{2} m_0 v^2$. The second piece clearly gives the kinetic energy term, so the first piece can be thought as the rest energy stored within the mass. This is how the idea of mass-energy relation came to the great mind of Einstein.

杨宇浩 12 核物理第十二章核物理 12.1 核能 12.1.1 质能等效质能等效原理指出质量和能量彼此等效，这一概念在数学上用爱因斯坦的质能关系表示： $E = mc^2$ [78] m 是物体的静止质量， $c = 3.00 \times 10^8 \text{ m s}^{-1}$ 是物体的速度真空中的光 $\hat{a}$ 质能等价意味着质量可以反过来转化为纯能量，质量可以由能量产生 $\hat{a}$ $\Delta E = \Delta mc^2$ 适用于所有能量变化，如果向一个系统提供能量，则该系统的质量增加，例如，一个物体在运动或加热时将具有更大的质量质量不是一个转换量，它是在所有过程中守恒的质能 $\hat{a}$ 如果一个物体是静止的，它的质量是称为静止质量 m_0 质能等价意味着静止的物体仍然具有固有的静止能量： $E_0 = m_0 c^2$ $\hat{a}$ 化学反应等能量转换可能会导致系统失去一些质量含量，但这种质量变化通常可以忽略不计[78]这可能是所有物理学中最著名的方程。当阿尔伯特·爱因斯坦于 1905 年写下狭义相对论时，他发现静止质量为 m_0 的物体的质量 m 与其运动速度 v 相关： $m = m_0 \sqrt{1 - v^2/c^2}$ 。总能量由 $E = mc^2 = m_0 c^2 \sqrt{1 - v^2/c^2}$ 给出。低速时 $v \ll c$, $E \approx m_0 c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} \right) = m_0 c^2 + \frac{1}{2} m_0 v^2$. 第二部分清楚地给出了动能项，因此第一部分可以被认为是质量内存储的剩余能量。这就是质能关系的思想如何进入伟大的爱因斯坦的头脑。 140

Example 12.1 When an antiproton (antiparticle of a proton) collides with a proton, they are annihilated and two photons of equal energy are formed. (a) What is the energy of each photon? (b) What is the photon frequency?

🔑 rest energy of proton becomes photon energy:

$$E_\gamma = m_p c^2 = 1.67 \times 10^{-27} \times (3.0 \times 10^8)^2 \approx 1.50 \times 10^{-10} \text{ J}$$

photon frequency: $f = \frac{E_\gamma}{h} = \frac{1.50 \times 10^{-10}}{6.63 \times 10^{-34}} \approx 2.27 \times 10^{23} \text{ Hz}$ (high-frequency γ -photon) $\square$

Question 12.1 The combustion of one mole of solid carbon to form carbon dioxide (CO_2) at standard condition is 394 kJ. Find the change in mass for this amount of energy, and hence compare this mass change with the mass of carbon before combustion.

Question 12.2 Given that the specific heat capacity of copper is $380 \text{ J kg}^{-1} \text{ K}^{-1}$. When a copper block is heated from 300 K to 1000 K, what is the additional mass as a fraction of its rest mass?

12.1.2 binding energy

nucleons (protons and neutrons) in a nucleus bind together through the *strong nuclear force* to pull nucleons apart, work must be done to overcome the attraction so free nucleons at infinity have greater potential energy than a single nucleus since energy is equivalent to mass, free nucleons would appear more massive than when they are held together in a nucleus

this statement is supported by experimental data

useful notions in nuclear physics related to this idea can now be introduced

difference between mass of a nucleus and total mass of its constituent nucleons when separated to infinity is called the **mass defect**

energy needed to separate the nucleons in a nucleus to infinity, or equivalently, energy released when free individual nucleons combine to form a nucleus, is called the **nuclear binding energy** (E_B)

➤ for a nucleus Z with proton number Z and nucleon number A (represented by ${}_Z^AX$)

Yuhao Yang 12 核物理例12.1 当反质子（质子的反粒子）与质子碰撞时，它们湮灭并形成两个能量相等的光子。(a) 每个光子的能量是多少？(b) 光子频率是多少？- 质子的静止能量变为光子能量： $E_\gamma = m_p c^2 = 1.67 \times 10^{-27} \times (3.0 \times 10^8)^2 \frac{1}{4} = 1.50 \times 10^{-10} \text{ J}$ 光子频率： $f = E_\gamma / h = 1.50 \times 10^{-10} / 6.63 \times 10^{-34} \frac{1}{4} = 2.27 \times 10^{23} \text{ Hz}$ （高频 γ -光子）

问题12.1 在标准条件下，一摩尔固体碳燃烧形成二氧化碳（ CO_2 ）的能量为394 kJ。找出该能量的质量变化，然后将该质量变化与燃烧前的碳质量进行比较。问题12.2 假设铜的比热容为 $380 \text{ J kg}^{-1} \text{ K}^{-1}$ 。当铜块从 300 K 加热到 1000 K 时，附加质量占其静止质量的比例是多少？

12.1.2 结合能原子核中的核子（质子和中子）通过强核力结合在一起，将核子拉开，必须努力克服吸引力，因此无穷远的自由核子比单个原子核具有更大的势能，因为能量等于质量，自由核子会比它们在原子核中结合在一起时显得更大。这一说法得到了实验数据的支持。与此想法相关的核物理中有用的概念现在可以引入原子核质量和原子核质量之间的差异。其组成核子在分离到无穷大时的总质量称为将原子核中的核子分离到无穷大所需的质量缺陷能量，或者等效地，当自由单个核子结合形成原子核时释放的能量称为核结合能（EB）

对于具有质子数 Z 和核子数 A 的原子核 Z （用 ${}_Z^A X$ 表示）141

its mass defect is given by: $\Delta m = Zm_p + (A - Z)m_n - m_X$

where $m_p = 1.673 \times 10^{-27}$ kg is mass of proton, $m_n = 1.675 \times 10^{-27}$ kg is mass of neutron

➤ by definition, binding energy is equivalent to mass defect: $E_B = \Delta mc^2$

➤ nucleons in a nucleus have *negative* P.E., while free nucleons have zero P.E.

E_B is the energy required to fill this gap in order to pull nucleons apart

i.e., E_B of a nucleus equals *loss* of potential energy during its formation

➤ nuclear mass is often measured in *unified atomic mass* u, where $1 \text{ u} = 1.66 \times 10^{-27}$ kg

nuclide ${}_Z^AX$ has a nuclear mass of about Au

➤ binding energy is often measured in MeV, where $1 \text{ MeV} = 10^6 \text{ eV} = 1.60 \times 10^{-13} \text{ J}$

Example 12.2 An iron-56 nucleus (${}_{26}^{56}\text{Fe}$) has a mass of 9.288×10^{-26} kg. Calculate the nuclear binding energy per nucleon, in MeV, for ${}_{26}^{56}\text{Fe}$.

✎ mass defect: $\Delta m = 26m_p + (56 - 26)m_n - m_{\text{Fe}} = 26 \times 1.673 \times 10^{-27} + 30 \times 1.675 \times 10^{-27} - 9.288 \times 10^{-26}$

so we find $\Delta m = 8.68 \times 10^{-28}$ kg

binding energy: $E_B = \Delta mc^2 = 8.68 \times 10^{-28} \times (3.00 \times 10^8)^2 \approx 7.812 \times 10^{-11} \text{ J}$

binding energy per nucleon: $\epsilon_b = \frac{E_b}{A} = \frac{7.812 \times 10^{-11}}{56} \approx 1.395 \times 10^{-12} \text{ J}$

convert into MeV: $\epsilon_b = \frac{1.395 \times 10^{-12}}{1.60 \times 10^{-13}} \text{ MeV} \approx 8.72 \text{ MeV}$ □

Question 12.3 Show that the energy equivalent of 1.0 u is 934 MeV.

Question 12.4 Given that mass of proton is 1.007 u, mass of neutron is 1.009 u, and mass of uranium-235 nucleus is 234.992 u. Find the binding energy per nucleon of nuclide ${}_{92}^{235}\text{U}$.

Question 12.5 What is the binding energy for the hydrogen nucleus ${}_1^1\text{H}$?

Question 12.6 Why is rest mass of proton slightly larger than the unified atomic mass unit?

12.1.3 nuclear stability

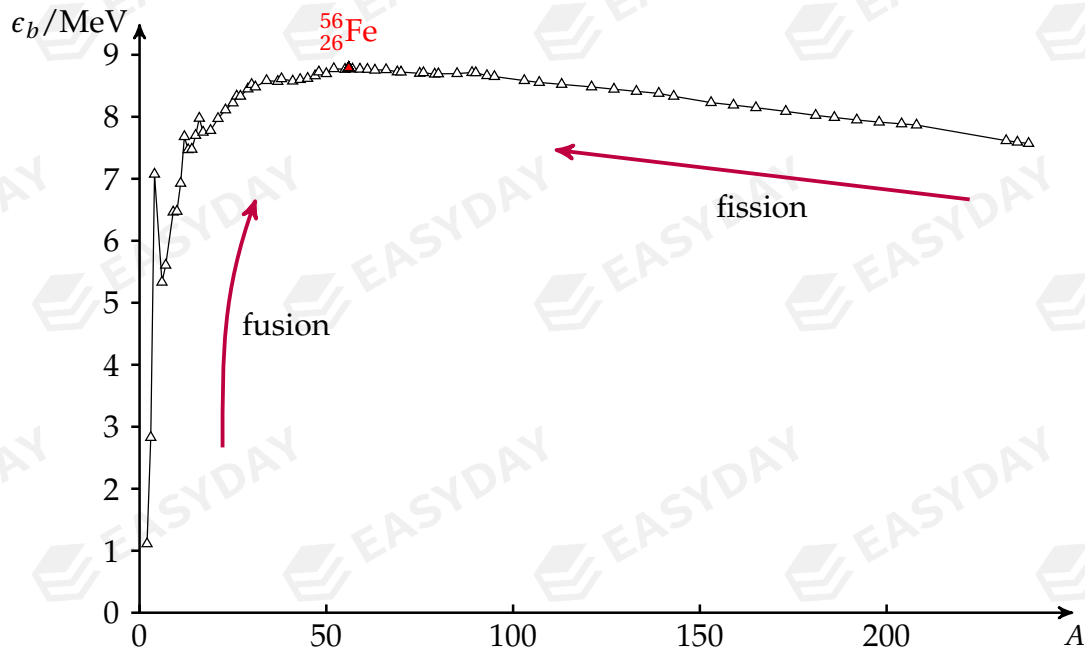
binding energy per nucleon ϵ_b is closely related to nuclear stability

binding energy per nucleon gives average energy needed to remove a nucleon from nucleus

higher ϵ_b means more difficult to pull nucleons away, so nucleus has higher stability

a graph of ϵ_b against nucleon number A can be plotted based on experimental data

Yuhao Yang 12 核物理其质量缺陷由下式给出： $\Delta m = Z m_p + (A - Z) m_n - m_X$ 其中 $m_p = 1.673 \times 10^{-27} \text{ kg}$ 是质子质量， $m_n = 1.675 \times 10^{-27} \text{ kg}$ 是中子质量。根据定义，结合能等于质量缺陷： $E_B = \Delta m c^2$ 。原子核中的核子具有负 P.E.，而游离核子具有零 P.E.。EB 是为了将核子拉开而填充该间隙所需的能量，即原子核的 EB 等于其形成过程中势能的损失。核质量通常以统一原子质量 u 来测量，其中 $1 u = 1.66 \times 10^{-27} \text{ kg}$ 。核素 ${}_Z^A X$ 的核质量约为 $A u$ 。结合能通常以 MeV 为单位测量，其中 $1 \text{ MeV} = 10^6 \text{ eV} = 1.60 \times 10^{-13} \text{ J}$ 。例 12.2 铁 56 核 (${}_{26}^{56}\text{Fe}$) 的质量为 $9.288 \times 10^{-26} \text{ kg}$ 。计算 ${}_{26}^{56}\text{Fe}$ 的每个核子的核结合能（以 MeV 为单位）。- 质量缺陷： $\Delta m = 26 m_p + (56 - 26) m_n - m_{\text{Fe}} = 26 \times 1.673 \times 10^{-27} + 30 \times 1.675 \times 10^{-27} - 9.288 \times 10^{-26}$ 所以我们找到 $\Delta m = 8.68 \times 10^{-28} \text{ kg}$ 结合能： $E_B = \Delta m c^2 = 8.68 \times 10^{-28} \times (3.00 \times 10^8)^2 = 7.812 \times 10^{-11} \text{ J}$ 每个核子结合能： $\epsilon_b = E_B / A = 7.812 \times 10^{-11} / 56 = 1.395 \times 10^{-12} \text{ J}$ 转换为 MeV： $\epsilon_b = 1.395 \times 10^{-12} / 1.60 \times 10^{-13} = 8.72 \text{ MeV}$ 。问题 12.3 证明 $1.0 u$ 的能量当量为 934 MeV 。问题 12.4 假设质子的质量为 $1.007 u$ ，中子的质量为 $1.009 u$ ，铀 ${}_{92}^{235}\text{U}$ 原子核的质量为 $234.992 u$ 。求核素 ${}_{92}^{235}\text{U}$ 每个核子的结合能。问题 12.5 氢原子核 ${}_1^1\text{H}$ 的结合能是多少？问题 12.6 为什么质子的静止质量略大于统一原子质量单位？12.1.3 每个核子的核稳定性结合能 ϵ_b 与核稳定性密切相关。每个核子的结合能给出从原子核中移除核子所需的平均能量。较高的 ϵ_b 意味着更难将核子拉走，因此原子核具有较高的稳定性。根据实验数据可以绘制 ϵ_b 与核子数 A 的关系图 142。



binding energy per nucleon ϵ_b against nucleon number A

- ${}^{56}_{26}\text{Fe}$ has greatest value of ϵ_b for all nucleus $\Rightarrow {}^{56}_{26}\text{Fe}$ is the most stable nuclide in nature
 - any physical system would evolve such that it lowers total energy
- if there could be an increase in total binding energy, nuclear reactions could occur

12.1.4 fusion

for nuclei smaller than ${}^{56}_{26}\text{Fe}$, they could combine together and release energy

fusion is the process where two light nuclei join together to form a heavier nucleus

- fusion occurs only at very *high temperatures*
- nuclei are all positively charged, for two nuclei to come close enough to fuse together, high initial K.E. is needed for them to overcome their *electrostatic repulsion*
- vast amount of energy of the sun and other *stars* are created through fusion
- extreme high temperature and high density at core of stars make fusion possible ^[79]

^[79] Even for the simplest fusion reaction, fusing hydrogen into helium, at least 1.5×10^7 K is required. This is basically how the sun shines and nurtures lives on the earth, and also many other stars in the universe power energies. For a sufficiently massive star, when it burns out of its fuel, its core collapses and becomes hot enough to start to fuse heavier elements. If a star is not able to fuse again, nuclear reaction ceases, the star collapses and becomes a *white dwarf*. For a giant star, nuclear reactions can go all the way up to ${}^{56}_{26}\text{Fe}$. At this point, no more energy can be produced through fusion. The star collapses extremely rapidly and then explodes, creating a *supernova*, during which all elements heavier than ${}^{56}_{26}\text{Fe}$

杨宇豪 12 核物理 ϵ_b/MeV A0 50 100 150 2000 1 2 3 4 5 6 7 8 9 聚变裂变 5 6 ^{26}Fe 每个核子的结合能 ϵ_b 相对于核号 A $\hat{a}$ $^{56}_{26}\text{Fe}$ 对于所有核来说 ϵ_b 值最大) $^{56}_{26}\text{Fe}$ 是最大的自然界中稳定的核素 $\hat{a}$ 任何物理系统都会演化, 如果总结合能增加, 则总能量会降低, 核反应可能发生 12.1.4 小于 $^{56}_{26}\text{Fe}$ 的核发生聚变, 它们可以结合在一起并释放能量聚变是两个轻核结合在一起形成较重核的过程 $\hat{a}$ 聚变仅在非常高的温度下发生核都带正电, 两个核足够接近以融合在一起, 高初始K.E.它们需要克服静电斥力——通过聚变产生太阳和其他恒星的大量能量, 恒星核心的极端高温和高密度使聚变成为可能[79] [79]即使是最简单的聚变反应, 将氢聚变成氦, 也至少需要 $1.5 \times 10^7 \text{ K}$ 。这基本上就是太阳如何照耀和培育地球上的生命, 也是宇宙中许多其他恒星能量的方式。对于一颗质量足够大的恒星来说, 当它的燃料燃尽时, 它的核心就会塌陷并变得足够热以开始融合较重的元素。如果恒星无法再次聚变, 核反应就会停止, 恒星就会坍缩并成为白矮星。对于巨星来说, 核反应可以一直进行到 $^{56}_{26}\text{Fe}$ 。此时, 无法通过聚变产生更多的能量。恒星极其迅速地坍塌, 然后爆炸, 产生超新星, 在此期间所有比 $^{56}_{26}\text{Fe}$ 143 重的元素

12.1.5 fission

for a nucleus larger than ${}_{26}^{56}\text{Fe}$, it can break into two or more parts and release energy

fission is the process where a massive nucleus splits into two smaller nuclei of about the same size (and is usually associated with release of several neutrons)

- note that fission requires the two product nuclei are of similar size
 - α -decay (emission of helium nucleus from an unstable nucleus) is not a fission process
- fission reactions can occur *spontaneously*, i.e. nuclei can undergo fission by themselves
 - fission can also be *induced* by bombarding the nucleus with an incident neutron
 - induced fission is used by humans to generate nuclear power or to build nuclear weapons

nuclear reactors (*)

nuclear reactors make use of energy released from induced fission reactions

thermal energy produced from fission is further converted to electrical or mechanical forms

uranium-235 (${}_{92}^{235}\text{U}$) is the most widely used fuel for fuel nuclear reactors

one of the many fission reactions of uranium-235 is: ${}_{92}^{235}\text{U} + {}_0^1\text{n} \longrightarrow {}_{56}^{141}\text{Ba} + {}_{36}^{92}\text{Kr} + 3{}_0^1\text{n}$

- each reaction releases more than one neutrons
 - they can trigger further fission reactions, making *chain reaction* possible
 - hence huge amount of energy can be released in a very short time
- rate of fission should be controlled, a runaway reaction could lead to disastrous explosion
 - control rods* (e.g., boron) are used to absorb neutrons: ${}_{5}^{10}\text{B} + {}_0^1\text{n} \longrightarrow {}_3^7\text{Li} + {}_2^4\text{He}$
 - control rods are adjusted so that one neutron per reaction goes on to produce further fission
 - in emergency, release of boron rods shuts down reactor
- reaction is expected to continue at steady rate
 - need sufficient amount of neutrons to maintain the chain reaction
 - this requires a *critical mass* for the amount of ${}_{92}^{235}\text{U}$ fuel

are formed. (*)

杨宇浩 12 核物理 12.1.5 大于 $^{56}_{26}\text{Fe}$ 的原子核裂变，它可以分裂成两个或多个部分并释放能量。裂变是一个大质量原子核分裂成两个大小相同的较小原子核的过程（通常与释放几个中子有关）。 $\hat{\alpha}$ 请注意，裂变要求两个产物原子核具有相似的尺寸。 α 衰变（发射不稳定原子核中的氦核）不是裂变过程。– 裂变反应可以自发发生，即原子核可以自行发生裂变也可以通过用入射中子轰击原子核来诱发裂变。人类利用诱发裂变来产生核能或建造核武器核反应堆。 (Q) 核反应堆利用诱发裂变反应释放的能量。裂变产生的热能进一步转换为电或机械形式。铀 235 ($^{235}_{92}\text{U}$) 是燃料核反应堆使用最广泛的燃料。铀 235 的众多裂变反应之一是： $^{235}_{92}\text{U} + ^1_0\text{n} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3^1_0\text{n}$ 。 $\hat{\alpha}$ 每个反应释放多个中子，它们可以引发进一步的裂变反应，使链式反应成为可能，因此可以在很短的时间内释放大量能量。– 裂变速率应受到控制，失控反应可能导致灾难性的爆炸。使用控制棒（例如硼）来吸收中子： $^{10}_5\text{B} + ^1_0\text{n} \rightarrow ^7_3\text{Li} + ^4_2\text{He}$ 。调整控制棒，以便每个反应产生一个中子，在紧急情况下进一步产生裂变，释放硼棒关闭反应堆。预计反应以稳定的速率继续需要足够数量的中子来维持链式反应，这需要形成 $^{235}_{92}\text{U}$ 燃料的临界质量。 (Q) 144

- only low energy neutrons (*thermal neutrons*) can be captured by ${}_{92}^{235}\text{U}$
but neutrons released through fission are very energetic
a *moderator* (e.g., water) is needed to slow down neutrons
- heat produced in fission is removed by *coolant* (e.g., water)
this heat can be used to power generators to produce electricity
- reactor is surrounded by a *shield* (e.g., a thick concrete) to prevent radiation from escaping

12.1.6 energy release from nuclear reactions

in this section, we consider only those nuclear reactions that release energy

let's write a generic nuclear reaction as: original particles $\rightarrow$ product particles + Q

two approaches to compute Q , the amount of energy release, will be introduced

method 1: use change in *mass* during the reaction

there must be a decrease in total mass to be converted into energy release

using mass-energy relation: $Q = \Delta mc^2 = (m_{\text{org}} - m_{\text{prod}})c^2$

method 2: use change of *total binding energy*

energy is released means product particles are more stable

so total binding energy of product particles is higher than that of original particles

energy released during reaction: $Q = E_{B,\text{prod}} - E_{B,\text{org}}$

Example 12.3 The sun is a huge nuclear reactor. It fuses hydrogen nuclei into helium nuclei to produce large amounts of energy. The reaction can be expressed by the formula:



(a) Evaluate the energy released in one reaction. (b) Given that the power output of the sun is roughly 4.0×10^{26} W, estimate how much hydrogen the sun burns every second. (Data for this question: $m_{\text{p}} = 1.672622 \times 10^{-27}$ kg, $m_{\text{He}} = 6.644657 \times 10^{-27}$ kg, $m_{\text{e}} = 9.11 \times 10^{-31}$ kg)

🔗 energy released in one reaction:

$$E_0 = \Delta mc^2 = (4m_{\text{H}} - m_{\text{He}} - 2m_{\text{e}})c^2 \approx 4.40 \times 10^{-29} \times (3.0 \times 10^8)^2 \approx 3.96 \times 10^{-12} \text{ J}$$

in interval Δt , sun outputs a total energy of $P\Delta t$ via a number of ΔN fusion reactions

we write $P\Delta t = \Delta NE_0$, so number of fusion reactions every second is:

Yuhao Yang 12 核物理 - ^{235}U 只能捕获低能中子（热中子），但通过裂变释放的中子能量非常大，需要慢化剂（例如水）来减慢中子速度 - 裂变产生的热量被冷却剂（例如水）带走，这些热量可用于为发电机发电 - 反应堆被屏蔽包围（例如，厚混凝土）为了防止辐射逃逸 12.1.6 本节中核反应的能量释放，我们只考虑那些释放能量的核反应让我们写一个通用的核反应为：原始粒子 $\rightarrow$ 产物粒子 + Q 将介绍两种计算 Q （能量释放量）的方法方法 1：利用反应过程中质量的变化总质量必须减少，才能使用质能关系将其转换为能量释放： $Q = \Delta mc^2 = (m_{\text{org}} - m_{\text{prod}})c^2$ 方法二：利用总结合能的变化能量释放意味着产物粒子更加稳定，因此产物粒子的总结合能高于原始粒子反应过程中释放的能量： $Q = E_{B, \text{prod}} - E_{B, \text{org}}$ 例 12.3 太阳是一个巨大的核反应堆。它将氢核聚变成氦核以产生大量的能量。该反应可用下式表示： $4\text{ }^1_1\text{H} \rightarrow \text{}^4_2\text{He} + 2\text{}^0_{+1}\text{e}$ ($^0_{+1}\text{e}$ 是正电子，电子的反粒子) (a) 计算一次反应中释放的能量。(b) 鉴于太阳的功率输出约为 $4.0 \times 10^{26} \text{ W}$ ，请估计太阳每秒燃烧多少氢气。（本题数据： $m_{\text{p}} = 1.672622 \times 10^{-27} \text{ kg}$ ， $m_{\text{He}} = 6.644657 \times 10^{-27} \text{ kg}$ ， $m_{\text{e}} = 9.11 \times 10^{-31} \text{ kg}$ ） - 一次反应释放的能量： $E_0 = \Delta mc^2 = (4m_{\text{H}} - m_{\text{He}} - 2m_{\text{e}})c^2$ $\approx 4.40 \times 10^{-29} \times (3.0 \times 10^8)^2 \approx 3.96 \times 10^{-12} \text{ J}$ 在 Δt 区间内，太阳通过若干 ΔN 聚变反应输出总能量 $P \Delta t$ 我们记为 $P \Delta t = \Delta N E_0$ ，所以每秒的聚变反应数量为：145

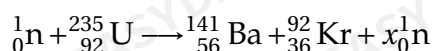
$$\frac{\Delta N}{\Delta t} = \frac{P}{E_0} = \frac{4.0 \times 10^{26}}{3.96 \times 10^{-12}} \approx 1.0 \times 10^{38} \text{ s}^{-1}$$

each reaction takes four hydrogen nuclei, so rate of hydrogen consumption:

$$\frac{\Delta M_{\text{H}}}{\Delta t} = \frac{4\Delta N}{\Delta t} \times m_{\text{H}} \approx 4 \times 1.0 \times 10^{38} \times 1.67 \times 10^{-27} \approx 6.7 \times 10^{11} \text{ kg s}^{-1}$$

the sun burns over 600 billion kilograms of hydrogen every second, just think about it! $\square$

Example 12.4 Uranium-235 nuclei ($^{235}_{92}\text{U}$) bombarded with slow neutrons can undergo nuclear reaction:



(a) Determine the number of x . (b) Using the data in

	binding energy per nucleon
$^{235}_{92}\text{U}$	7.591 MeV
$^{141}_{56}\text{Ba}$	8.326 MeV
$^{92}_{36}\text{Kr}$	8.513 MeV

the table, find the energy released for one reaction. (c) What is the change in mass, if any, before and after the reaction?

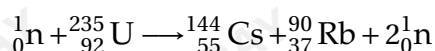
 from conservation of mass number: $1 + 235 = 141 + 92 + x \Rightarrow x = 3$

energy release during reaction equals change in total binding energy:

$$Q = E_{\text{B,Ba}} + E_{\text{B,Kr}} - E_{\text{B,U}} = 141 \times 8.326 + 92 \times 8.513 - 235 \times 7.591 = 173.277 \text{ MeV}$$

$$\text{reduction in total mass: } \Delta m = \frac{Q}{c^2} = \frac{173.277 \times 1.60 \times 10^{-13}}{(3.0 \times 10^8)^2} \approx 3.08 \times 10^{-28} \text{ kg} \quad \square$$

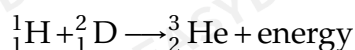
Question 12.7 Uranium-235 nuclei can undergo another fission process when being bombarded with neutrons. This is represented by:



Calculate the energy released in this reaction.

	binding energy per nucleon
$^{235}_{92}\text{U}$	7.591 MeV
$^{144}_{55}\text{Cs}$	8.212 MeV
$^{92}_{36}\text{Kr}$	8.631 MeV

Question 12.8 The *proton-proton chain reaction* is a set of nuclear reactions through which stars fuse hydrogen into helium. One intermediate reaction may be summarised as:



Calculate the energy released in this reaction.

	mass/u
^1_1H	1.00728
^2_1D	2.01356
^3_2He	3.01605

Question 12.9 Another fusion reaction in the proton-proton chain is represented by:



杨宇浩 12 核物理 $\Delta N \Delta t = P E_0 = 4.0 \times 10^{26} \times 3.96 \times 10^{-12} \times \frac{1}{4} \times 1.0 \times 10^3 \times 8 \text{ s}^{-1}$ 每个反应需要四个氢核，因此氢消耗率： $\Delta M_H \Delta t = 4 \Delta N \Delta t \times m_H \times \frac{1}{4} = 4 \times 1.0 \times 10^3 \times 8 \times 1.67 \times 10^{-27} \times \frac{1}{4} = 6.7 \times 10^{11} \text{ kg s}^{-1}$ 太阳每秒燃烧超过 6000 亿公斤氢气，想想看！

每个核子的结合能 $^{235}_{92}\text{U}$ 7.591 MeV $^{141}_{56}\text{Ba}$ 8.326 MeV $^{92}_{36}\text{Kr}$ 8.513 MeV 示例 12.4 用慢中子轰击的铀 $^{235}_{92}\text{U}$ 原子核 ($^{235}_{92}\text{U}$) 可以发生核反应： $1\text{ }^1_0\text{n} + ^{235}_{92}\text{U} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + x\text{ }^1_0\text{n}$ (a) 确定 x 的数量。(b) 使用表中的数据求出一个反应释放的能量。(c) 反应前后质量的变化（如果有）是多少？ - 根据质量数守恒定律： $1 + 235 = 141 + 92 + x$ $x = 3$ 反应过程中释放的能量等于总结合能的变化： $Q = E_{B,\text{Ba}} + E_{B,\text{Kr}} - E_{B,\text{U}} = 141 \times 8.326 + 92 \times 8.513 - 235 \times 7.591 =$ 总质量减少 173.277 MeV： $\Delta m = Q/c^2 = 173.277 \times 1.60 \times 10^{-13} / (3.0 \times 10^8)^2 = 3.08 \times 10^{-28} \text{ kg}$

每个核子结合能 $^{235}_{92}\text{U}$ 7.591 MeV $^{144}_{55}\text{Cs}$ 8.212 MeV $^{92}_{36}\text{Kr}$ 8.631 MeV 问题 12.7 铀 $^{235}_{92}\text{U}$ 原子核在受到中子轰击时可以经历另一个裂变过程。表示为： $1\text{ }^1_0\text{n} + ^{235}_{92}\text{U} \rightarrow ^{144}_{55}\text{Cs} + ^{90}_{37}\text{Rb} + 21\text{ }^1_0\text{n}$ 计算该反应中释放的能量。质量/u ^1_1H 1.007 28 ^2_1D 2.013 56 ^3_2He 3.016 05 问题 12.8 质子-质子链式反应是恒星将氢聚变成氦的一组核反应。一个中间反应可概括为： $1\text{ }^1_1\text{H} + 2\text{ }^2_1\text{D} \rightarrow 3\text{ }^3_2\text{He} + \text{energy}$ 计算该反应中释放的能量。问题 12.9 质子-质子链中的另一个聚变反应表示为： $3\text{ }^3_2\text{He} \rightarrow 4\text{ }^4_2\text{He} + 2\text{ }^1_1\text{p} + \text{energy}$

where energy released through this reaction is 12.86 MeV. Given that binding energy per nucleon of ${}^4_2\text{He}$ is 7.074 MeV, calculate the binding energy per nucleon for ${}^3_2\text{He}$.

Question 12.10 If the total mass of the product particles is greater than that of the original particles in a nuclear reaction, what can you say about the energy of the original particles?

12.2 radioactive decay

12.2.1 radioactive decay

the process of *random* and *spontaneous* emission of α -, β -, and γ -radiation from unstable nuclei is called **radioactive decay**

➤ **random** means decay events are not predictable

we cannot tell precisely when a particular nucleus is about to decay

we can only tell the *probability* of decay within a certain time

➤ **spontaneous** means rate of decay does not depend on external conditions

temperature, pressure, or has no effect on rate of radioactive decays

suppose initially a sample contains N radioactive nuclei that are about to decay

after a time interval Δt , ΔN nuclei undergo decay processes

probability of decay for nuclei in the sample during this time interval is $\frac{\Delta N}{N}$

divide by Δt , probability of decay for each nucleus per unit time can be found

since nuclear decay is spontaneous, this number is a constant, called the decay constant

decay constant is the probability for one nucleus to decay per unit time: $\lambda = \frac{\Delta N}{N\Delta t}$

➤ to describe the rate of decay, we introduce the notion of **activity**

activity is defined as the number of nuclei that undergo decay per unit time: $A = \frac{\Delta N}{\Delta t}$

this can be given in a differential form: $A = -\frac{dN}{dt}$

a minus sign is included because number of nuclei N decreases with time

Yuhao Yang 12 NUCLEAR PHYSICS 其中通过该反应释放的能量为 12.86 MeV。假设 $4\text{ }^2\text{He}$ 每个核子的结合能为 7.074 MeV，计算 $3\text{ }^2\text{He}$ 每个核子的结合能。问题 12.10 如果核反应中产物粒子的总质量大于原始粒子的质量，你能说原始粒子的能量是多少？

12.2 放射性衰变

12.2.1 放射性衰变

不稳定原子核随机自发发射 α -、 β - 和 γ -辐射的过程称为放射性衰变。随机意味着衰变事件是不可预测的，我们无法准确判断特定核即将衰变。我们只能知道在一定时间内衰变的概率 λ 。自发意味着衰变速率不依赖于外部条件（温度、压力），或者对放射性衰变速率没有影响。假设最初样品包含 N 个放射性核子在时间间隔 Δt 后即将衰变， ΔN 个核子经历衰变过程，样本中的原子核在这段时间内衰变的概率为 $\Delta N/N$ 。除以 Δt ，可求出每个原子核在单位时间内的衰变概率，因为核衰变是自发的，这个数是一个常数，称为衰变常数。衰变常数是一个原子核在单位时间内衰变的概率： $\lambda = \Delta N / (N \Delta t)$ 。来描述衰变的速率，我们引入活度概念。活度定义为每单位时间发生衰变的原子核数量： $A = \Delta N / \Delta t$ 。这可以以微分形式给出： $A = -dN/dt$ 。包括负号，因为原子核数量 N 随时间减少。

- compare with the expression for decay constant, we find: $A = \lambda N$

this equation shows that if there are more nuclei present, the sample has greater activity

- units for decay constant and activity: $[\lambda] = [A] = \text{s}^{-1}$

for activity, one decay event per unit time is defined as one *becquerel*: $1 \text{ Bq} = 1 \text{ s}^{-1}$

12.2.2 decay equations

variation of number of undecayed nuclei with time can be derived from the equation: $A = \lambda N$

recall that $A = -\frac{dN}{dt}$, so this equation is a differential equation in disguise: $-\frac{dN}{dt} = \lambda N$

this can be solved by separating variables and integrating

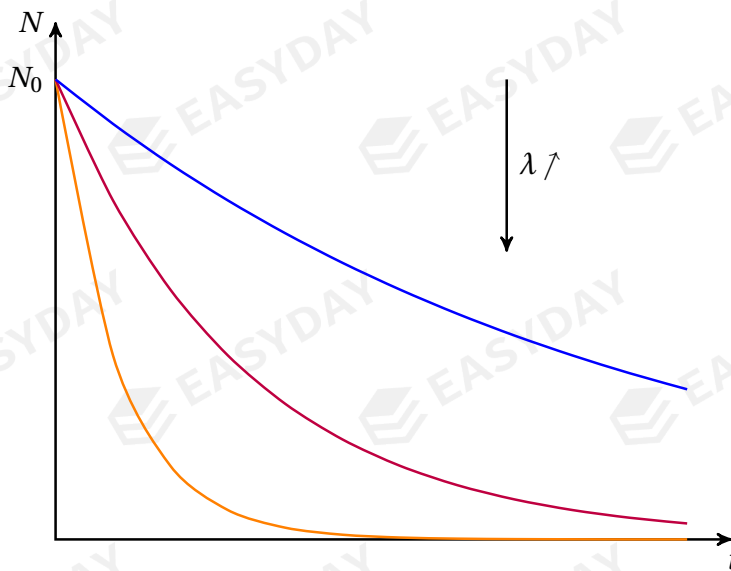
$$-\lambda dt = \frac{dN}{N} \Rightarrow -\lambda \int_0^t dt = \int_{N_0}^N \frac{dN}{N} \Rightarrow -\lambda t \Big|_0^t = \ln N \Big|_{N_0}^N \Rightarrow -\lambda t = \ln \frac{N}{N_0} \Rightarrow N = N_0 e^{-\lambda t}$$

this shows that decay events obey *exponential decay laws*

- number of undecayed nuclei at time t is given by: $N(t) = N_0 e^{-\lambda t}$

N_0 is initial number of nuclei in the sample at $t = 0$

- $\lambda \nearrow \Rightarrow$ higher probability of decay, or greater rate of decay



exponential decay law for number of nuclei: $N(t) = N_0 e^{-\lambda t}$

- since activity $A = \lambda N$, so activity varies with time as: $A(t) = A_0 e^{-\lambda t}$

Example 12.5 A sample containing 6.0×10^7 iodine-131 nuclei has an activity of 60 Bq. (a) What is the decay constant of iodine-131? (b) How many nuclei remain undecayed after 20 days?

- ✎ decay constant: $\lambda = \frac{A}{N} = \frac{60}{6.0 \times 10^7} \approx 1.0 \times 10^{-6} \text{ s}^{-1}$

杨宇豪 12 核物理 $\hat{\alpha}$ 与衰变常数的表达式进行比较，我们发现： $A = \lambda N$ 该方程表明，如果存在的原子核越多，样品的活度就越大 $\hat{\alpha}$ 衰变常数和活度的单位： $[\lambda] = [A] = \text{s}^{-1}$ 活度，每单位时间的一次衰变事件定义为 1 个贝克勒尔： $1 \text{ Bq} = 1 \text{ s}^{-1}$ 12.2.2 衰变方程未衰变核数随时间的变化可以从以下方程导出： $A = \lambda N$ 回想一下 $A = -\frac{dN}{dt}$ ，所以这个方程是一个变相的微分方程： $-\frac{dN}{dt} = \lambda N$ 这可以通过分离变量并积分来求解 $-\lambda dt = \frac{dN}{N}$ $-\lambda \int_{t_0}^t dt = \int_{N_0}^N \frac{dN}{N}$ $-\lambda t = \ln \frac{N}{N_0}$ $\ln \frac{N}{N_0} = -\lambda t$ $\frac{N}{N_0} = e^{-\lambda t}$ $N = N_0 e^{-\lambda t}$ 这表明衰变事件遵循指数衰变定律 $\hat{\alpha}$ 时间 t 时未衰变核的数量由下式给出： $N(t) = N_0 e^{-\lambda t}$ N_0 是 t 时样品中核的初始数量 $= 0 \hat{\alpha} \lambda$ 更高的衰变概率，或更大的衰变率 $N = N_0 e^{-\lambda t}$ 原子核数量的指数衰变定律： $N(t) = N_0 e^{-\lambda t}$ $\hat{\alpha}$ 由于活性 $A = \lambda N$ ，因此活性随时间变化为： $A(t) = A_0 e^{-\lambda t}$ 示例 12.5 含有 6.0×10^7 碘 131 核的样品活性为 60 Bq。(a) 碘 131 的衰变常数是多少？(b) 20 天后有多少原子核保持未衰变？ - 衰减常数： $\lambda = \frac{A}{N} = \frac{60}{6.0 \times 10^7} = 1.0 \times 10^{-6} \text{ s}^{-1}$ 148

number of remaining nuclei: $N = N_0 e^{-\lambda t} = 6.0 \times 10^7 \times e^{-1.0 \times 10^{-6} \times 20 \times 3600} \approx 1.1 \times 10^7$ $\square$

Example 12.6 $^{24}_{11}\text{Na}$, an isotope of sodium, has a decay constant of $1.28 \times 10^{-5} \text{ s}^{-1}$. Suppose a sample initially contains a mass of $9.0 \mu\text{g}$ of $^{24}_{11}\text{Na}$. Find (a) initial number of $^{24}_{11}\text{Na}$ nuclei, (b) initial activity of this sample, (c) the activity after 48 hours.

initial number of nuclei: $N_0 = \frac{9.0 \mu\text{g}}{24\text{u}} = \frac{9.0 \times 10^{-9}}{24 \times 1.66 \times 10^{-27}} \approx 2.26 \times 10^{17}$

initial activity: $A_0 = \lambda N_0 = 1.28 \times 10^{-5} \times 2.26 \times 10^{17} \approx 2.89 \times 10^{12} \text{ Bq}$

activity after 20 hours: $A = A_0 e^{-\lambda t} = 2.89 \times 10^{12} \times e^{-1.28 \times 10^{-5} \times 48 \times 3600} \approx 3.17 \times 10^{11} \text{ Bq}$ $\square$

Question 12.11 A sample of polonium-205 has an activity of $6.7 \times 10^{15} \text{ Bq}$. The decay constant of polonium-205 is known to be $1.16 \times 10^{-4} \text{ s}^{-1}$. (a) Find the number of nuclei needed to give this activity. (b) Find the mass of polonium-205 in this sample. (c) Calculate the time needed for the activity reduce to 1% of its initial value.

Question 12.12 Technetium-99m is widely used as a radioactive tracer for medical diagnostic procedures. If some of this isotope, with an activity of 900 MBq was injected into a patient. The activity is found to reduce to 56.4 MBq after 24 hours. (a) What is the decay constant of technetium-99m? (b) How many technetium-99m nuclei are still left in the patient's body?

12.2.3 half-life

it is more convenient to define a time quantity to describe how fast radioactive nuclei decay

it is useful to define half-life of a radioactive sample

half-life ($t_{1/2}$) is the mean time taken for number of radioactive nuclei in the sample, or activity of the sample, to reduce to half of its initial value

➤ half-life $t_{1/2}$ is closely related to decay constant λ

by definition, at $t = t_{1/2}$, $N = \frac{1}{2} N_0$, $A = \frac{1}{2} A_0 \Rightarrow e^{-\lambda t_{1/2}} = \frac{1}{2} \Rightarrow e^{\lambda t_{1/2}} = 2 \Rightarrow \lambda t_{1/2} = \ln 2$

➤ λ is a constant, so $t_{1/2}$ is also a constant over the lifetime of nuclear decay

half-life of a sample of isotope does not depend on initial number of nuclei or initial activity

Example 12.7 Radium-224 has a half-life of 3.63 days. (a) What is the activity from 6.0 mg of pure radium-224? (b) How many radium-224 nuclei have undergone decay after 10 days?

Yuhao Yang 12 核物理剩余原子核数： $N = N_0 e^{-\lambda t} = 6.0 \times 10^7 \times e^{-1.0 \times 10^{-6} \times 20 \times 3600} \approx 1.1 \times 10^7$ ä

示例 12.6 $^{24}_{11}\text{Na}$ ，钠的同位素，其衰变常数为 $1.28 \times 10^{-5} \text{ s}^{-1}$ 。假设样品最初含有质量为 9.0 g 的 $^{24}_{11}\text{Na}$ 。求 (a) $^{24}_{11}\text{Na}$ 核的初始数量，(b) 该样品的初始活性，(c) 48 小时后的活性。

- 初始核数： $N_0 = 9.0 \text{ g} / 24 \text{ u} = 9.0 \times 10^{-9} / 24 \times 1.66 \times 10^{-27} \approx 2.26 \times 10^{17}$ 初始活性： $A_0 = \lambda N_0 = 1.28 \times 10^{-5} \times 2.26 \times 10^{17} \approx 2.89 \times 10^{12} \text{ Bq}$ 20 小时后的活度： $A = A_0 e^{-\lambda t} = 2.89 \times 10^{12} \times e^{-1.28 \times 10^{-5} \times 48 \times 3600} \approx 3.17 \times 10^{11} \text{ Bq}$ ä 问题 12.11 钋-205 样品的活度为 $6.7 \times 10^{15} \text{ Bq}$ 。已知钋205的衰变常数为 $1.16 \times 10^{-4} \text{ s}^{-1}$ 。

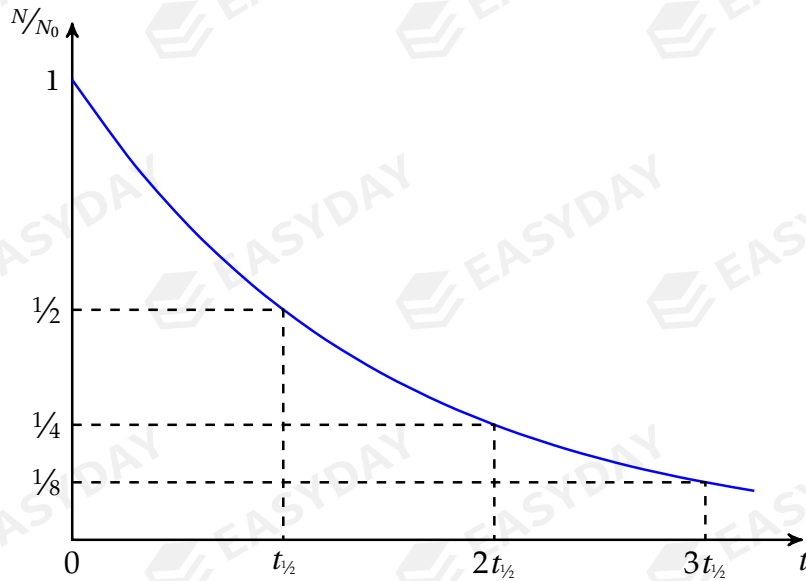
(a) 求出进行该活动所需的原子核数量。(b) 求该样品中钋205 的质量。(c) 计算活动减少到其初始值的 1% 所需的时间。

问题 12.1 2 Technetium-99m 广泛用作医疗诊断程序的放射性示踪剂。如果将一些活性为 900 MBq 的同位素注射到患者体内。24 小时后，活性降低至 56.4 MBq。

(a) Technetium-99m 的衰变常数是多少？(b) 患者体内还残留有多少个 Technetium-99m 原子核？

12.2.3 半衰期定义一个时间量来描述放射性核衰变的速度会更方便，这对于定义放射性样品的半衰期很有用。半衰期 ($t_{1/2}$) 是样品中放射性核数或样品活性减少到其初始值一半所需的平均时间。根据定义，半衰期 $t_{1/2}$ 与衰变常数 λ 密切相关，在 $t = t_{1/2}$, $N = \frac{1}{2} N_0$, $A = \frac{1}{2} A_0$) $e^{-\lambda t_{1/2}} = \frac{1}{2}$) $e^{\lambda t_{1/2}} = 2$) $\lambda t_{1/2} = \ln 2$ ä λ 是一个常数，因此 $t_{1/2}$ 也是核衰变寿命期内的常数同位素样品的半衰期不依赖于初始原子核数量或初始活度

实施例 12.7 Radium -224 的半衰期为 3.63 天。(a) 6.0 mg 纯镭 224 的活性是多少？(b) 10 天后有多少镭 224 原子核发生衰变？ 149



half-life of radioactive decay

✎ decay constant: $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3.63 \times 24 \times 3600} \approx 2.21 \times 10^{-6} \text{ s}^{-1}$

initial number of Ra-224 nuclei: $N_0 = \frac{6.0 \text{ mg}}{224 \text{ u}} = \frac{6.0 \times 10^{-6}}{224 \times 1.66 \times 10^{-27}} \approx 1.61 \times 10^{19}$

initial activity: $A_0 = \lambda N_0 = 2.21 \times 10^{-6} \times 1.61 \times 10^{19} \approx 3.57 \times 10^{13} \text{ Bq}$

number of undecayed nuclei: $N = N_0 e^{-\lambda t} = 1.61 \times 10^{19} \times e^{-2.21 \times 10^{-6} \times 10 \times 24 \times 3600} \approx 2.39 \times 10^{18}$

number of nuclei that have decayed: $\Delta N = N_0 - N = 1.61 \times 10^{19} - 2.39 \times 10^{18} \approx 1.37 \times 10^{19}$ □

Example 12.8 Living trees contain a certain percentage of $^{14}_6\text{C}$, a radioactive isotope of carbon that has a half-life of 5570 years. A sample of dead wood is found to have an activity of 0.42 Bq, while an equal mass of living wood has an activity of 1.60 Bq. Find the age of the dead wood.

✎ $A = A_0 e^{-\lambda t} = A_0 e^{-\frac{\ln 2}{t_{1/2}} t} \Rightarrow 0.42 = 1.60 e^{-\frac{\ln 2}{5570} t} \Rightarrow -\frac{\ln 2}{5570} t = \ln\left(\frac{0.42}{1.60}\right) \Rightarrow t \approx 10700 \text{ years}$ □

Question 12.13 The number of uranium-238 nuclei in a rock sample is believed to have decreased from 3.50×10^{17} to 3.27×10^{17} in 480 million years. Estimate the half-life of uranium-238.

Question 12.14 Plutonium-238 is a powerful alpha emitter with a half-life of 87.7 years. One decay of plutonium-238 releases an energy of about $9.0 \times 10^{-13} \text{ J}$. A nuclear battery containing a sealed plutonium source is implanted into patient's body to power heart pacemakers. The battery has an initial activity of $6.0 \times 10^{10} \text{ Bq}$. (a) Calculate the initial power released by the source. (b) Find the mass of plutonium required to produce this power. (c) It is required that power output to the pacemaker is at least 60% of the initial power. Calculate the time, in years,

杨宇豪 12 核物理 $N/N_0 = 1/2, 1/4, 1/8, 0, 1/2, 2/2, 1/2, 3/2, 1/2$ 放射性衰变半衰期 - 衰变常数: $\lambda = \ln 2 / t_{1/2} = \ln 2 / 3.63 \times 24 \times 3600 \text{ s} = 2.21 \times 10^{-6} \text{ s}^{-1}$ Ra-224 核初始数量: $N_0 = 6.0 \text{ mg} / 224 \text{ u} = 6.0 \times 10^{-6} / 224 \times 1.66 \times 10^{-27} \text{ kg} = 1.61 \times 10^{19}$ 初始活性: $A_0 = \lambda N_0 = 2.21 \times 10^{-6} \times 1.61 \times 10^{19} = 3.57 \times 10^{13} \text{ Bq}$ 未衰变核数: $N = N_0 e^{-\lambda t} = 1.61 \times 10^{19} \times e^{-2.21 \times 10^{-6} \times 10 \times 24 \times 3600} = 2.39 \times 10^{18}$ 已衰变核数: $\Delta N = N_0 - N = 1.61 \times 10^{19} - 2.39 \times 10^{18} = 1.37 \times 10^{19}$ 示例 12.8 活树含有一定比例的 ^{14}C , 一种碳的放射性同位素, 半衰期为 5570 年。死木样本的活性为 0.42 Bq, 而等量的活木样品的活性为 1.60 Bq。找出枯木的年龄。 - $A = A_0 e^{-\lambda t} = A_0 e^{-\ln 2 / t_{1/2} \cdot t}$ $0.42 = 1.60 e^{-\ln 2 / 5570 \cdot t}$ $-\ln 2 / 5570 \cdot t = \ln (0.42 / 1.60)$ $t = 10700 \text{ 年}$ 问题 12.13 据信, 岩石样本中的铀 ^{238}U 原子核在 4.8 亿年间从 3.50 英镑 1017 减少到 3.27 英镑 1017 。估计铀 ^{238}U 的半衰期。问题 12.14 钷 ^{238}Pu 是一种强大的 α 发射体, 半衰期为 87.7 年。钷 ^{238}Pu 的一次衰变释放出约 9.0×10^{-13} 焦耳的能量。含有密封钷源的核电池被植入患者体内, 为心脏起搏器提供动力。该电池的初始活性为 $6.0 \times 10^{10} \text{ Bq}$ 。(a) 计算源释放的初始功率。(b) 求出产生这种能量所需的钷质量。(c) 要求起搏器的输出功率至少为初始功率的 60%。计算时间, 以年为单位, 150

for which the battery provides sufficient power.

Question 12.15 Show that the variation of the number of undecayed nuclei with time t can be given by: $N(t) = N_0 \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$, where N_0 is the initial number of nuclei.

12.2.4 measurement of radioactive decay

a GeigerMüller tube, or a **GM tube**, is a device used to detect ionizing radiation

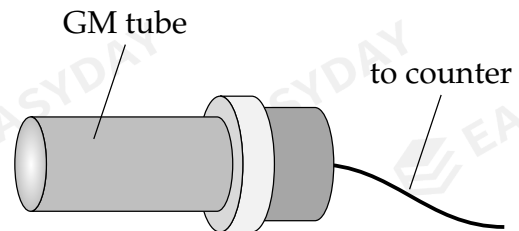
GM tube measures the number of α -particles, β -particles and γ -photons arriving per unit time

the number of decays recorded per unit time by

GM tube is called the **count rate** R

since GM tube only picks up emissions to one particular direction, but a radioactive source emits radiation in *all* directions, so count rate is a fraction of the activity of the sample

activity obeys exponential decay, so we would expect count rate to satisfy: $R = R_0 e^{-\lambda t}$



verifying the exponential decay law

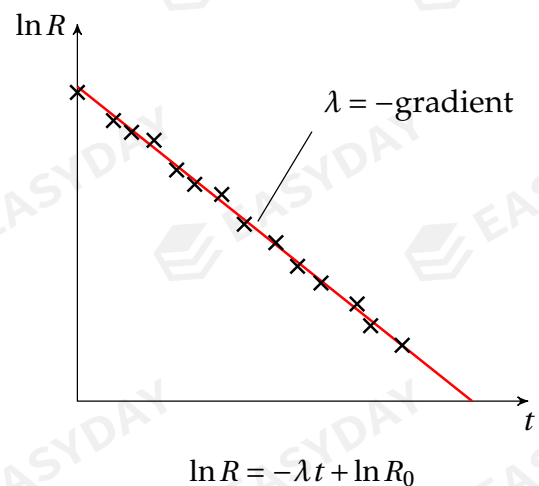
to verify the exponential decay law, we first rearrange the equation as: $\ln R = -\lambda t + \ln R_0$

if a set of measurements of R at time t are obtained, a graph of $\ln R$ against t can be plotted

if trend curve shows a straight line, the relation

$R = R_0 e^{-\lambda t}$ is then verified

decay constant λ is given by negative gradient of the best-fit line



issues with count rate

for now, we take for granted that the count rate measured is 100% accurate

杨宇豪 12 核物理电池为其提供足够的电力。问题 12.15 证明未衰变原子核数量随时间 t 的变化可由下式给出： $N(t) = N_0 (1/2)^{t/t_{1/2}}$ ，其中 N_0 是原子核的初始数量。

12.2.4 放射性衰变的测量 GM 管对抗 GeigerMüller 管，或 GM 管，是一种用于检测电离辐射的装置 GM 管测量单位时间内到达的 α 粒子、 β 粒子和 γ 光子的数量 GM 管每单位时间记录的衰变数称为计数率 R ，因为 GM 管仅拾取向一个特定方向的发射，但放射源向各个方向发射辐射，因此计数速率是样本活动服从指数衰减的分数，因此我们期望计数率满足： $R = R_0 e^{-\lambda t}$ 验证指数衰减定律 $\ln R$ $\ln R = -\lambda t + \ln R_0$ $\lambda = -\text{梯度}$ 为了验证指数衰减定律，我们首先将方程重新排列为： $\ln R = -\lambda t + \ln R_0$ 如果在时间 t 测量一组 R 的值得到，如果趋势曲线显示一条直线，则可以绘制 $\ln R$ 对 t 的图，然后验证关系 $R = R_0 e^{-\lambda t}$ 衰减常数 λ 由最佳拟合线的负梯度给出现在与计数率有关，我们理所当然地认为测量的计数率是 100% 准确的 151

but in practice, there might be a number of factors leading to measurement errors

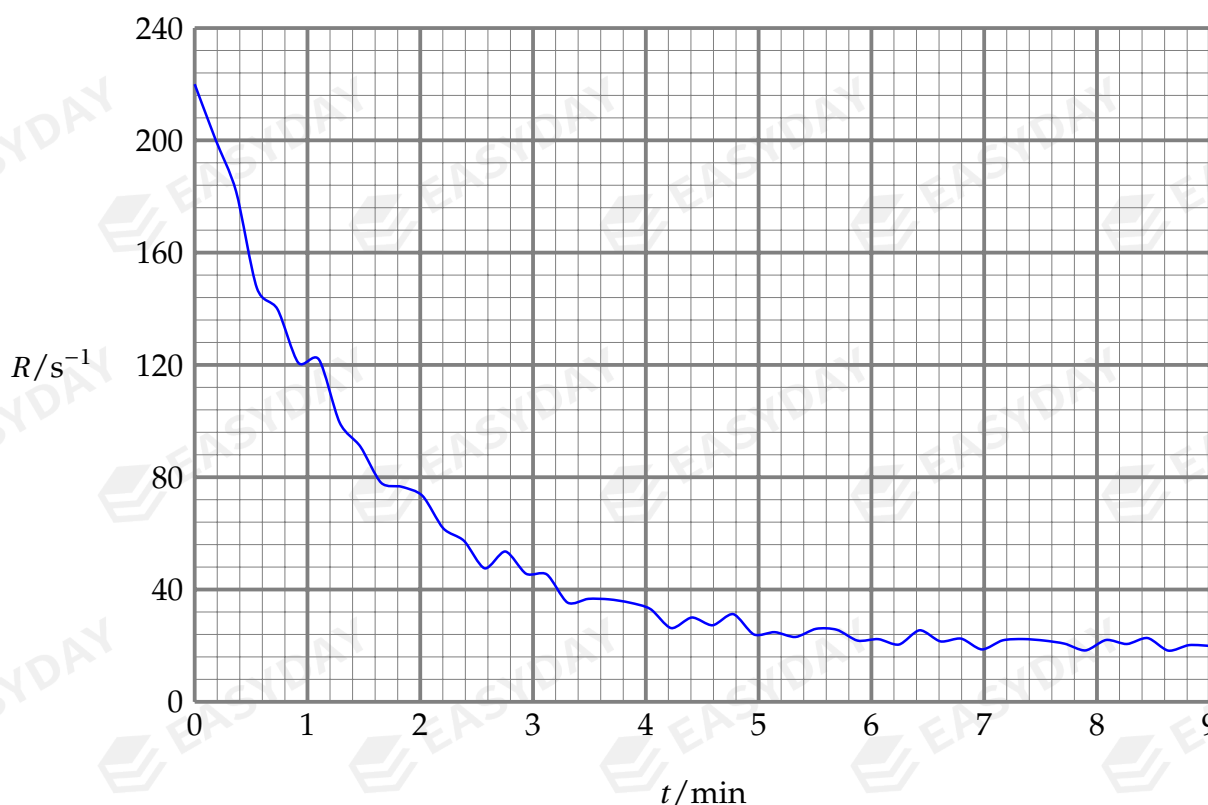
- there is always *background radiation* from earth minerals, cosmic rays, food and water, etc.

background reading must be subtracted to give a corrected count rate

- α - and β -particles emitted could be absorbed by sample itself
- product nuclei could also be radioactive, contributing to additional counts
- GM tube has *dead time*, or *resolving time*

when a count is recorded, it takes GM tube a certain time to reset for the next count

Example 12.9 A GM tube is placed close to a source of radioactive isotope. The variation with time t of the measured count rate R is shown. Determine the half-life of this isotope.



there is evidence of *background radiation* as the count rate decreases and tends to a non-zero value of 20 s^{-1} , so we need subtract this number to obtain the true count rate of sample

there are also *fluctuations* in count rate due to *random* nature of decay processes, so we need to make several calculations and take average to obtain an accurate value for half-life

let's call true count rate R_T , and call measured count rate R

as $R_T = 200 \text{ s}^{-1} \rightarrow 100 \text{ s}^{-1}$, $R = 220 \text{ s}^{-1} \rightarrow 120 \text{ s}^{-1}$, so $t_{1/2} \approx 1.0 - 0.0 \approx 1.0 \text{ min}$

Yuhao Yang 12 核物理但在实践中，可能有多种因素导致测量误差 – 总是存在来自地球矿物、宇宙射线、食物和水等的背景辐射。必须减去背景读数才能得到校正的计数率 – 发射的 α 和 β 粒子可能被样品本身吸收 – 产物核也可能具有放射性，导致额外计数 – GM 管有死区时间，或记录计数时的解析时间，需要 GM 管在一定时间后重置以进行下一次计数示例 12.9 将 GM 管放置在靠近放射性同位素源的位置。显示了测量的计数率 R 随时间 t 的变化。确定该同位素的半衰期。

R/s^{-1}	t/min
0	1
2	3
4	5
6	7
8	9
10	10
12	12
14	14
16	16
18	18
20	20
22	22
24	24
26	26
28	28
30	30
32	32
34	34
36	36
38	38
40	40
42	42
44	44
46	46
48	48
50	50
52	52
54	54
56	56
58	58
60	60
62	62
64	64
66	66
68	68
70	70
72	72
74	74
76	76
78	78
80	80
82	82
84	84
86	86
88	88
90	90
92	92
94	94
96	96
98	98
100	100
102	102
104	104
106	106
108	108
110	110
112	112
114	114
116	116
118	118
120	120
122	122
124	124
126	126
128	128
130	130
132	132
134	134
136	136
138	138
140	140
142	142
144	144
146	146
148	148
150	150
152	152

随着计数率降低并趋于非零值 $20 s^{-1}$ ，有背景辐射的证据，因此我们需要减去该数字以获得样本的真实计数率。由于衰减过程的随机性，计数率也存在波动，因此我们需要为了进行多次计算并取平均值以获得半衰期的准确值，我们将真实计数率称为 R_T ，并将测量的计数率 R 称为 $R_T = 200 s^{-1} \pm 100 s^{-1}$ ， $R = 220 s^{-1} \pm 120 s^{-1}$ ，因此 $t_{1/2} = 1.0 \pm 0.1$ 分钟 152

as $R_T = 160 \text{ s}^{-1} \rightarrow 80 \text{ s}^{-1}$, $R = 180 \text{ s}^{-1} \rightarrow 100 \text{ s}^{-1}$, so $t_{1/2} \approx 1.3 - 0.4 \approx 0.9 \text{ min}$

as $R_T = 120 \text{ s}^{-1} \rightarrow 60 \text{ s}^{-1}$, $R = 140 \text{ s}^{-1} \rightarrow 80 \text{ s}^{-1}$, so $t_{1/2} \approx 1.6 - 0.7 \approx 0.9 \text{ min}$

as $R_T = 80 \text{ s}^{-1} \rightarrow 40 \text{ s}^{-1}$, $R = 100 \text{ s}^{-1} \rightarrow 60 \text{ s}^{-1}$, so $t_{1/2} \approx 2.3 - 1.3 \approx 1.0 \text{ min}$

take average for these results, we find $t_{1/2} \approx 0.95 \text{ min}$ $\square$

杨宇豪 12 核物理 $RT = 160 \text{ s}^{-1} \pm 80 \text{ s}^{-1}$, $R = 180 \text{ s}^{-1} \pm 100 \text{ s}^{-1}$, 因此 $t_{1/2} \approx 1.3 - 0.4 \pm 0.9$ 分钟, 因为 $RT = 120 \text{ s}^{-1} \pm 60 \text{ s}^{-1}$, $R = 140 \text{ s}^{-1} \pm 80 \text{ s}^{-1}$, 因此 $t_{1/2} \approx 1.6 - 0.7 \pm 0.9$ 分钟, 因为 $RT = 80 \text{ s}^{-1} \pm 40 \text{ s}^{-1}$, $R = 100 \text{ s}^{-1} \pm 60 \text{ s}^{-1}$, 因此 $t_{1/2} \approx 2.3 - 1.3 \pm 1.0$ 分钟对这些结果取平均值, 我们发现 $t_{1/2} \approx 0.95 \text{ min} \pm 0.153$

CHAPTER 13

Medical Imaging

symptoms of disease are to be diagnosed without cutting patient open, or inserting surgical instruments into patient's body, we need **non-invasive** techniques, which include:

- X-ray/CT imaging
- ultrasonic scans
- MRI (magnetic resonance imaging)
- PET (positron emission tomography) scans]

13.1 X-ray imaging

13.1.1 nature of X-rays

X-ray is electromagnetic radiation with short wavelength λ , or high frequency f radiation with a wavelength in the range $10^{-12} \sim 10^{-8}$ m are categorised as X-rays

➤ X-rays are further categorised into *hard* and *soft* X-rays

hardness of X-ray refers to its penetrating ability

harder X-ray photons are more energetic, so more penetrating

hard X-rays have higher f and shorter λ ($10^{-13} \sim 10^{-11}$ m)

soft X-rays have slightly lower f and longer λ ($10^{-11} \sim 10^{-9}$ m)

➤ soft-X-rays are easily absorbed by patient's body, hence not contributing to imaging in medical imaging, metal sheet (e.g., aluminium) is used to filter out soft X-rays

hard X-rays can penetrate through patient's body and are more useful to form images

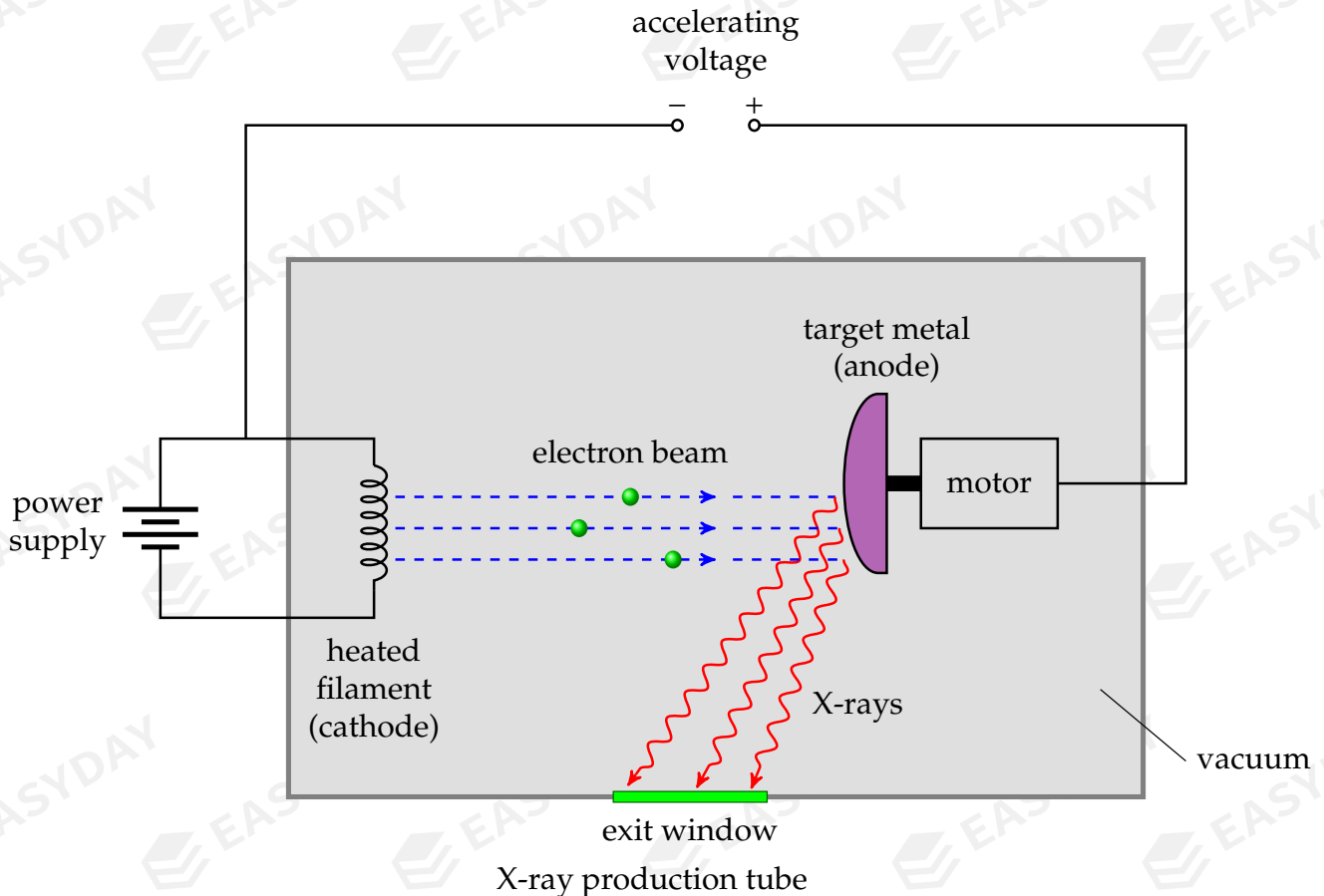
➤ there is no specific critical f/λ to distinguish X-ray from γ -ray in electromagnetic spectrum radiation emitted by *electrons* is called X-ray, radiation emitted from *nuclei* is called γ -ray

杨宇浩 13 医学影像第 13 章医学影像疾病症状的诊断需要不切开患者或将手术器械插入患者体内，我们需要非侵入性技术，包括：- X 射线/CT 成像 - 超声扫描 - MRI（磁共振成像）- PET（正电子发射断层扫描）扫描] 13.1 X 射线成像 13.1.1 X 射线的性质 X 射线是电磁辐射，短波长 λ ，或高频 f 辐射，波长在 10^{-12} » 10^{-8} m 范围内的辐射被归类为 X 射线 X X 射线又分为硬 X 射线和软 X 射线 X 射线的硬度是指其穿透能力较硬的 X 射线光子能量较高，因此更具穿透力的硬 X 射线具有较高的 f 和较短的 λ (10^{-13} » 10^{-11} m) 软 X 射线的 f 稍低和较长 λ (10^{-11} » 10^{-9} m) $\hat{a}$ 软 X 射线容易被患者身体吸收，因此对医学成像无助于成像，金属片（例如铝）用于过滤软 X 射线硬 X 射线可以穿透患者身体，更有助于成像 $\hat{a}$ 没有特定的临界 f/λ 来区分电子发射的电磁波谱辐射中的 X 射线和 γ 射线称为 X 射线，从原子核发出的辐射称为 γ 射线 154

13.1.2 production of X-rays

X-ray can be produced by smashing electrons onto a target metal

the diagram below illustrates the production of X-ray in a *X-ray tube*



under a strong electric field, electrons are pulled out from a heated cathode

these electrons are accelerated through a high p.d. towards the anode

anode is made of target metal (tungsten, cobalt, etc.)

electrons smash into anode, their K.E. are converted into the form of X-ray radiation

beams of X-rays leave the X-ray tube through an exit window

- accelerating voltage of X-ray tube V_{ac} controls energy of X-ray photons emitted
 - at higher V_{ac} , electrons have greater K.E., X-ray photons of higher energies are produced
- varying cathode current alters number of electrons flowing through filament per unit time
 - for a larger current, more electrons are pulled towards anode, producing more X-ray photons
 - so increasing current in X-ray tube increases intensity of X-ray beam
- in X-ray tube, only a tiny fraction of electron K.E. is converted into X-rays

杨宇浩 13 医学成像 13.1.2 X 射线的产生通过将电子撞击到目标金属上可以产生 X 射线下图说明了 X 射线管中 X 射线的产生 +- 加速电压电源加热灯丝（阴极） 电子束电机目标金属（阳极） 真空 X 射线出射窗强电场下的 X 射线产生管，电子从加热的阴极中被拉出这些电子被加速通过高 p.d. 朝向阳极阳极由目标金属（钨、钼等）组成，电子撞击阳极，其 K.E. 被转换成 X 射线辐射束的形式，通过出射窗离开 X 射线管 - X 射线管的加速电压 V_{ac} 控制在较高 V_{ac} 下发射的 X 射线光子的能量，电子具有更大的 K.E.，产生更高能量的 X 射线光子 - 变化的阴极电流改变单位时间内流过灯丝的电子数量，电流越大，更多的电子被拉向阳极，产生更多的 X 射线光子因此，增加 X 射线管中的电流会增加 X 射线束的强度 $\hat{\alpha}$ 在 X 射线管中，仅电子 K.E. 的一小部分。被转换成 X 射线 155

most energy is converted into thermal energy in anode metal

to avoid overheating, anode is rotated by a motor, and water can be used as coolant

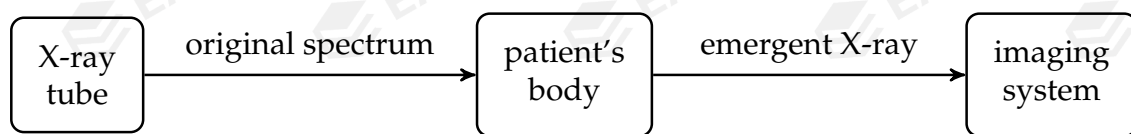
13.1.3 X-ray spectrum

to image a patient's body using X-rays, X-rays produced are sent towards the patient

X-ray emergent from patient's body can be measured and compared with original spectrum

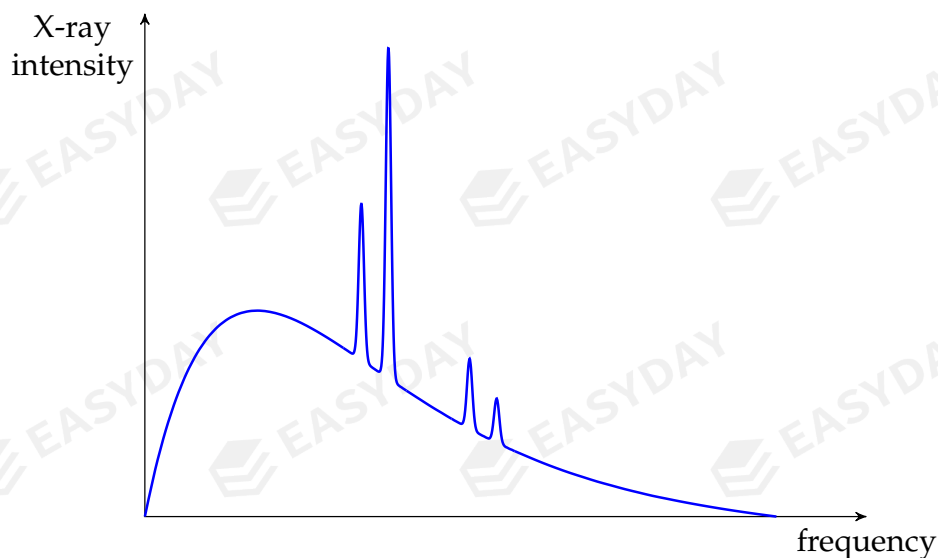
from this we can learn about absorption process of X-ray in patient's body

so information about patient's body can be obtained



so we need to study properties of *X-ray spectrum* as it is produced in X-ray tubes

the graph below shows a typical frequency spectrum of X-ray emitted from an X-ray tube



frequency spectrum of X-ray beam produced in an X-ray tube

this spectrum consists of two parts: a *continuous background*, and a few *discrete spikes*

these are due to two different mechanisms, about which we are about to discuss

braking radiation

continuous background in X-ray spectrum is due to **braking radiation**

杨宇浩 13 医学成像大部分能量在阳极金属中转化为热能，避免过热，阳极由电机旋转，水可以用作冷却剂

13.1.3 X 射线光谱利用 X 射线对患者身体进行成像，产生的 X 射线发送到患者可以测量从患者体内射出的 X 射线，并与原始光谱进行比较，由此我们可以了解 X 射线在患者体内的吸收过程，从而了解患者的信息可以获取身体 X 射线管患者身体成像系统原始频谱紧急 X 射线，因此我们需要研究 X 射线管中产生的 X 射线频谱的特性下图显示了从 X 射线管发射的 X 射线的典型频谱 X 射线管中产生的 X 射线束的 X 射线强度频率频谱该频谱由两部分组成：连续背景和一些离散尖峰，这些尖峰是由两种不同的机制引起的，我们将在其中讨论制动辐射连续背景X 射线光谱是由于制动辐射 156

high-speed electrons decelerate when they strike into the target metal

loss of K.E. of electrons are converted into electromagnetic energy, i.e. X-ray photons

no constraint on final K.E. of electrons, so there could be a *distribution* of K.E. loss

X-ray photons produced therefore are allowed to take a range of energies

so a range of frequencies of the X-ray photons, giving a broad continuous spectrum

➤ there exists a maximum frequency for emitted X-rays

X-ray photon of highest energy is produced if an electron loses all its K.E. in one collision

photon energy is proportional to frequency of radiation, so sharp cut-off at high f

Example 13.1 An X-ray tube uses accelerating voltage of 5000 V, what is the highest frequency of the X-ray produced?

✍ electrical P.E. converts into K.E. of electron, then converts into X-ray photon

$$qV_{ac} = hf_{max} \Rightarrow f_{max} = \frac{qV_{ac}}{h} = \frac{1.60 \times 10^{-19} \times 5000}{6.63 \times 10^{-34}} \approx 1.2 \times 10^{18} \text{ Hz} \quad \square$$

Question 13.1 Explain why the spectrum of X-ray would have a minimum wavelength.

characteristic radiation

discrete peaks in the X-ray spectrum arise from a process called **characteristic radiation**

incident electrons can knock out orbital electrons from inner shells in target metal

orbital electrons from higher energy levels drop to fill these vacancies

de-excitation of the electrons gives out energy as X-ray photons

electron energy levels in atoms are discrete, so only certain X-ray frequencies are emitted

➤ positions of characteristic lines are determined by target metal

each metal has its unique electron energy level, producing unique characteristic lines

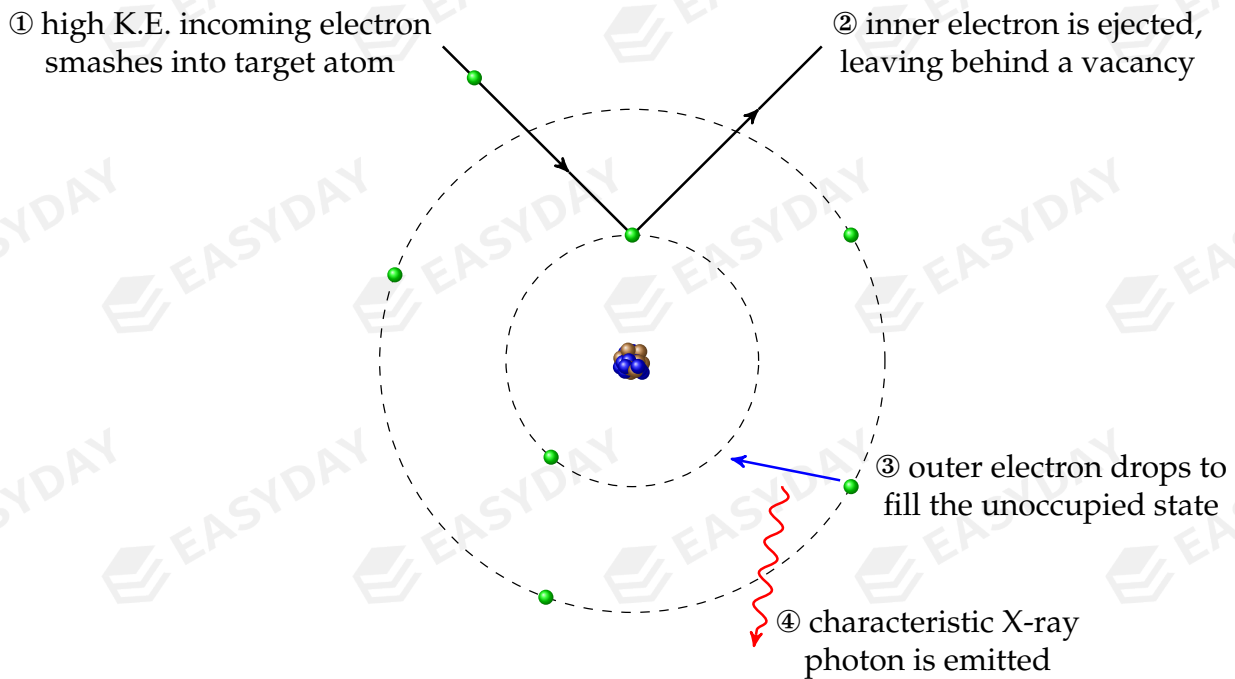
Question 13.2 If accelerating voltage in the X-ray tube is increased, suggest and explain the changes to the X-ray spectrum.

Question 13.3 Summarise the principles of production of X-rays.

13.1.4 X-ray attenuation

as an X-ray beam penetrates through a substance, its intensity decreases

Yuhao Yang 13 医学成像高速电子撞击目标时会减速 K.E. 金属损失电子转化为电磁能，即 X 射线光子对最终 K.E 没有限制。电子，因此可能存在 K.E. 的分布。因此，产生的 X 射线光子可以具有一定范围的能量，因此 X 射线光子的频率范围很大，从而提供了宽广的连续光谱 - 发射的 X 射线存在最大频率。如果电子失去所有 K.E，则会产生最高能量的 X 射线光子。在一次碰撞中，光子能量与辐射频率成正比，因此在高 f 时会出现锐截止例 13.1 X 射线管使用 50 00 V 的加速电压，产生的 X 射线的最高频率是多少？ - 电气 P.E. 转换为 K.E. 电子，然后转换为 X 射线光子 $qV_{ac} = h f_{max}$) $f_{max} = qV_{ac} / h = 1.60 \times 10^{-19} \times 5000 / 6.63 \times 10^{-34} = 1.2 \times 10^{18} \text{ Hz}$ 问题 13.1 解释为什么 X 射线光谱有一个最小波长。X 射线谱中的特征辐射离散峰源自称为特征辐射的过程入射电子可以从目标金属的内壳中击出轨道电子高能级的轨道电子下降以填充这些空位电子的去激发以 X 射线光子的形式释放能量原子中的电子能级是离散的，因此仅发射某些 X 射线频率特征线的位置由目标金属决定每个金属都有其独特的电子能级，产生独特的特征线问题 13.2 如果 X 射线管中的加速电压增加，请建议并解释 X 射线光谱的变化。问题 13.3 总结 X 射线的产生原理。13.1.4 X 射线衰减当 X 射线束穿透物质时，其强度降低 157



production of characteristic radiation

gradual decrease in X-ray intensity is called **attenuation**

change in X-ray intensity can be measured to extract information about patient's body

the law of X-ray attenuation is given by $I = I_0 e^{-\mu x}$

this shows a beam of initial intensity I_0 decreases exponentially with distance x travelled

μ is **linear attenuation constant**, or **absorption constant** of the substance

➤ the equation $I = I_0 e^{-\mu x}$ applies to *parallel* beam of X-ray only

intensity of a *divergent* beam will decrease even without absorption

➤ recall that *intensity* of radiation is defined as power transmitted per unit time: $I = \frac{P}{A}$

this is related to number of photons in the beam per unit time, given by: $\frac{\Delta N}{\Delta t} \times hf$

as each photon passes through some distance, it has certain probability to be absorbed

we may think of μ as probability for one single photon to be absorbed per unit distance

make analogy with nuclear decays, X-ray intensity should also decrease exponentially ^[80]

➤ substance with high value of μ is a good absorber of X-ray

^[80] You may have noticed the similarity between the law of X-ray attenuation ($I = I_0 e^{-\mu x}$) and nuclear decay laws ($N = N_0 e^{-\lambda t}$). The analogy is clear: intensity I of radiation is analogous to number of undecayed nuclei N , distance x is analogous to time t , and probability constant are μ and λ respectively, where λ , the decay constant, tells the probability of decay for a radioactive nucleus per unit time.

Yuhao Yang 13 医学影像 Ø 高 K.E. 入射电子撞击目标原子内部电子被弹出，留下一个空位 ⑥ 外部电子滴落以填充未占据状态发射特征 X 射线光子产生特征辐射 X 射线强度逐渐降低称为衰减 X 射线强度的变化可以测量以提取有关患者身体的信息 X 射线衰减定律由 $I = I_0 e^{-\mu x}$ 给出，这表明初始强度 I_0 的光束随行进距离 x 呈指数下降 (μ 是线性衰减常数，或物质的吸收常数 - 方程 $I = I_0 e^{-\mu x}$ 适用于 X 射线的平行光束，即使没有吸收，只有发散光束的强度也会降低 - 回想一下，辐射强度定义为每单位时间传输的功率： $I = P/A$ 这与每单位时间光束中的光子数量有关，由下式给出： $\Delta N = \Delta t \times h f$ 当每个光子穿过一定距离时，它有一定的概率被吸收，我们可以认为(μ 作为单位距离单个光子被吸收的概率，类比核衰变，X 射线强度也应呈指数下降 [80] μ 值高的物质是 X 射线的良好吸收体 [80] 您可能已经注意到 X 射线衰减定律 ($I = I_0 e^{-\mu x}$) 和核衰变定律 ($N = N_0 e^{-\lambda t}$) 之间的相似性。类比很清楚：辐射强度 I 相当于未衰变原子核的数量 N ，距离 x 相当于时间 t ，概率常数分别为 (μ 和 λ ，其中 λ 是衰变常数，表示放射性核在单位时间内衰变的概率。 158

bones are much better absorbers than soft tissues ($\mu_{\text{bone}} > \mu_{\text{tissue}}$)

with X-ray imaging techniques, bones are easily distinguished from soft tissues

➤ attenuation constant μ depends on hardness of an X-ray beam

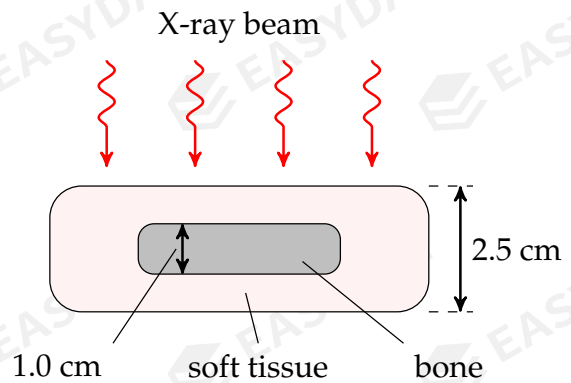
high-frequency X-ray is more penetrating, so a material has smaller μ at higher frequency

➤ can define *half-value thickness* $x_{1/2}$ (recall half-life $t_{1/2}$ for nuclear decays)

this is thickness of material at which intensity of X-ray becomes halved

it follows that: $\frac{1}{2}I_0 = I_0 e^{-\mu x_{1/2}} \Rightarrow x_{1/2} = \frac{\ln 2}{\mu}$ [81]

Example 13.2 A parallel beam of 30 keV X-ray of initial intensity I_0 is incident on a specimen of soft tissue and bone as shown. Find the emergent intensity from the specimen if (a) the beam passes through soft tissue alone, (b) the beam passes through the bone and the soft tissue.



➤ through soft tissue only:

$$I = I_0 e^{-\mu x} = I_0 e^{-0.40 \times 2.5} \approx 0.37 I_0$$

through soft tissue and bone:

$$I = I_0 e^{-\mu_m x_m} e^{-\mu_b x_b}$$

$$I_0 e^{-0.40 \times 1.5} \times e^{-2.55 \times 1.0} \approx 0.043 I_0$$

attenuation constants for 30 keV X-ray

	μ / cm^{-1}
tissue	0.40
bone	2.55

emergent intensity will be very different depending on whether it travels through bone

so there will be clear difference between bone and soft tissue in the image □

Question 13.4 Copper has a linear attenuation coefficient 0.41 mm^{-1} for X-ray radiation of energy 100 keV. If a copper filter is able to reduce the intensity of a parallel beam of X-ray by 90%, what is the thickness required?

[81] Compare with half-life of nuclear decay: $t_{1/2} = \frac{\ln 2}{\lambda}$.

Yuhao Yang 13 医学成像借助 X 射线成像技术，骨骼比软组织 (bone & tissue) 具有更好的吸收体，骨骼很容易与软组织区分开 μ 衰减常数 (取决于 X 射线束的硬度高频 X 射线更具穿透力，因此材料在较高频率下具有更小的 μ μ 可以定义半值厚度 $x_{1/2}$ (回想核衰变的半衰期 $t_{1/2}$)，这是厚度 X 射线强度减半的材料如下： $I = I_0 e^{-\mu x}$ $x_{1/2} = \ln 2 / \mu$ [81] 软组织骨骼 2.5 cm 1.0 cm 30 keV 的 X 射线束衰减常数 X 射线 (/cm⁻¹) 组织 0.40 骨骼 2.55 示例 13.2 30 keV 的平行光束如图所示，初始强度 I_0 的 X 射线入射到软组织和骨骼样本上。如果 (a) 光束仅穿过软组织，(b) 光束穿过骨骼和软组织，请求样本的出射强度。 - 仅通过软组织： $I = I_0 e^{-\mu x} = I_0 e^{-0.40 \times 2.5} = 0.37 I_0$ 通过软组织和骨骼： $I = I_0 e^{-(\mu_{\text{soft}} x_{\text{soft}} + \mu_{\text{bone}} x_{\text{bone}})} = I_0 e^{-0.40 \times 1.5 - 2.55 \times 1.0} = 0.043 I_0$ 紧急强度将非常不同，具体取决于是否穿过骨骼，因此会很清楚图像中骨骼和软组织之间的差异 μ 问题 13.4 对于能量为 100 keV 的 X 射线辐射，铜的线性衰减系数为 0.41 mm⁻¹。如果铜滤光片能够将平行 X 射线束的强度降低 90%，那么所需的厚度是多少？ [81] 与核衰变的半衰期比较： $t_{1/2} = \ln 2 / \lambda$ 。 159

13.1.5 X-ray imaging techniques

basic principle of conventional X-ray imaging: expose patient to a beam of X-ray, a fraction of the beam is absorbed by patient's body, a film or a detector captures attenuated X-rays, resulting in a black-and-white shadow image

high-quality imaging requires good sharpness, good contrast and minimal radiation dosage

image sharpness

sharpness of image means whether details on edges of structures can be seen with ease

sharpness of image is mainly determined by *width* of X-ray beam

so a narrow beam of *parallel* X-ray is desired

- parallel X-ray beam can be realized by
 - reducing size of target metal at anode of X-ray tube
 - reducing size of exit window of X-ray tube
 - using a set of lead slits to make the beam more parallel, known as *collimation*
 - using *anti-scatter screen* to reduce influence of scattering of radiation in patient's body
- good sharpness also requires patient to keep still when image is taken

image contrast

contrast means whether different structures show up with different degrees of blackening

good contrast requires a great difference in X-ray intensity

- for substances with large difference in μ , they naturally produce good contrast
- most soft tissues are poor absorbers of X-rays, not easy to distinguish them on same image
- artificial **contrast media** such as iodine (I) injections, barium (Ba) meals^[82] can be used to turn a tissue into a better absorber, therefore greatly improve contrast of image

^[82] Contrast media usually contain elements of large atomic numbers, which means each atom accommodates a large number of electrons, so it is more likely for them to interact with X-ray photons, making these atoms good absorbers of radiation.

杨宇浩 13 医学成像 13.1.5 X 射线成像技术传统 X 射线成像的基本原理：将患者暴露在 X 射线束下，一部分射线束被患者身体吸收，胶片或探测器捕获衰减的 X 射线，产生黑白阴影图像高质量的成像需要良好的清晰度、良好的对比度和最小的辐射剂量。图像的清晰度主要由 X 射线束的宽度决定，因此需要窄束的平行 X 射线——平行 X 射线束可以通过以下方式实现：减小 X 射线管阳极靶金属的尺寸；减小 X 射线管出射窗的尺寸；使用一组铅缝使光束更加平行，称为准直；使用防散射屏，以减少放射线散射对患者体内的影响；良好的清晰度还要求患者在成像时保持静止。图像对比度对比度是指不同的结构是否呈现不同程度的黑化良好的对比度需要 X 射线强度的巨大差异 $\hat{\alpha}$ 对于 (差异较大的物质，它们自然会产生良好的对比度 $\hat{\alpha}$ 大多数软组织对 X 射线的吸收性较差， 在同一图像上不易区分它们人工造影剂如碘 (I) 注射剂、钡 (Ba) 餐 [82] 可用于将组织转变为更好的吸收体，从而大大提高图像对比度[82]造影剂通常含有大原子序数的元素，这意味着每个原子容纳大量电子，因此它们更有可能与 X 射线光子相互作用，使这些原子成为良好的辐射吸收体。

160

Question 13.5 The table below shows the linear attenuation coefficient μ for X-ray beams of different energies in cortical bone and muscle. To produce an X-ray image where the bone is clearly distinguished from muscle, why is it more desirable to use low-energy X-rays?

maximum X-ray energy/keV	bone: μ/cm^{-1}	muscle: μ/cm^{-1}
30	2.55	0.40
80	0.43	0.19
200	0.25	0.14
800	0.14	0.082

dosage of radiation

X-ray radiation is ionising, hence harmful to human body

image should be developed quickly, so exposure time is minimised to reduce possible harm to reduce exposure, or **dosage** of radiation, efficient imaging systems are needed

➤ **image intensifiers** are used to reduce dosage of X-ray radiation

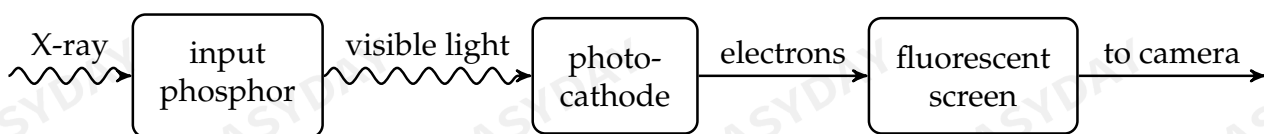


illustration of how the image is developed efficiently with an image intensifier

13.1.6 CT scans

conventional X-ray imaging only produces a 2D *shadow image*

information about *depth* of structures cannot be displayed

computerised axial tomography scan, or **CT scan**, is developed to produce 3D images

principles of CT imaging

patient's body can be thought to consist of many 2D slices

each slice can be divided into a grid of many units, called **voxels**

to obtain image of one slice, computer-programmed X-rays are exposed from all directions

detectors opposite to X-ray tube pick up transmitted signals and feed them to computer

Yuhao Yang 13 医学影像问题13.5 下表显示了不同能量的X 射线束在皮质骨和肌肉中的线性衰减系数 μ 。为了生成清晰区分骨骼和肌肉的 X 射线图像，为什么更需要使用低能量 X 射线？最大X射线能量/keV 骨骼： μ/cm^{-1} 肌肉： μ/cm^{-1}

最大X射线能量/keV	骨骼 μ/cm^{-1}	肌肉 μ/cm^{-1}
30	2.55	0.40
80	0.43	0.19
200	0.25	0.14
800	0.14	0.082

辐射剂量 X射线辐射是电离性的，因此对人体有害，应尽快显像，因此应尽量缩短曝光时间，以减少可能的危害，减少曝光或剂量辐射，需要高效的成像系统图像增强器用于减少 X 射线辐射剂量输入磷光体光阴极荧光屏 X 射线可见光电子到相机说明如何使用图像增强器有效地形成图像 13.1.6 CT 扫描传统的 X 射线成像仅产生 2D 阴影有关结构深度的图像信息无法显示计算机轴向断层扫描或 CT 扫描被开发为产生 3D CT成像的图像原理患者的身体可以被认为是由许多二维切片组成，每个切片可以分为许多单元的网格，称为体素以获得一个切片的图像，计算机编程的X射线从各个方向照射与X射线管相对的探测器拾取传输的信号并将其馈送到计算机161

computer uses emergent intensities from all directions to deduce information about each voxel, therefore a 2D slice image can be built up

this process is repeated, so images for many 2D slices are obtained

2D slice images are combined to get a comprehensive 3D image

the 3D image can be rotated and viewed from different angles

➤ advantages of CT imaging

- precise position, shape, size of structures can be shown
- tissues with similar attenuation coefficients can be distinguished

➤ disadvantages of CT imaging

- more expensive than traditional 2D radiography
- patient is exposed to higher radiation dosage

building up images (*)

we now discuss how information of each voxel is recovered from transmitted X-rays

we consider a toy model, a 2×2 grid with 4 voxels

each voxel makes its contribution to attenuation, described by numbers a , b , c and d

grid is exposed to a parallel beam of X-ray, readings from detectors are assigned to associated voxel, the voxels are reconstructed and recorded in a memory grid

this process is repeated for four different directions

all the memory grids are then added to construct a cumulative grid

knowing background reading, which is $a + b + c + d$ from all voxels, and cumulative reading, information of the original voxels can be obtained within two simple steps

step 1: subtract background reading from cumulative grid

step 2: divide the result by three

the whole process is shown in the figure, which should be self-explanatory

Yuhao Yang 13 医学影像计算机利用来自各个方向的涌现强度来推断每个体素的信息，因此可以建立 2D 切片图像，重复此过程，从而获得许多 2D 切片的图像组合 2D 切片图像以获得全面的 3D 图像 3D 图像可以旋转并从不同角度查看

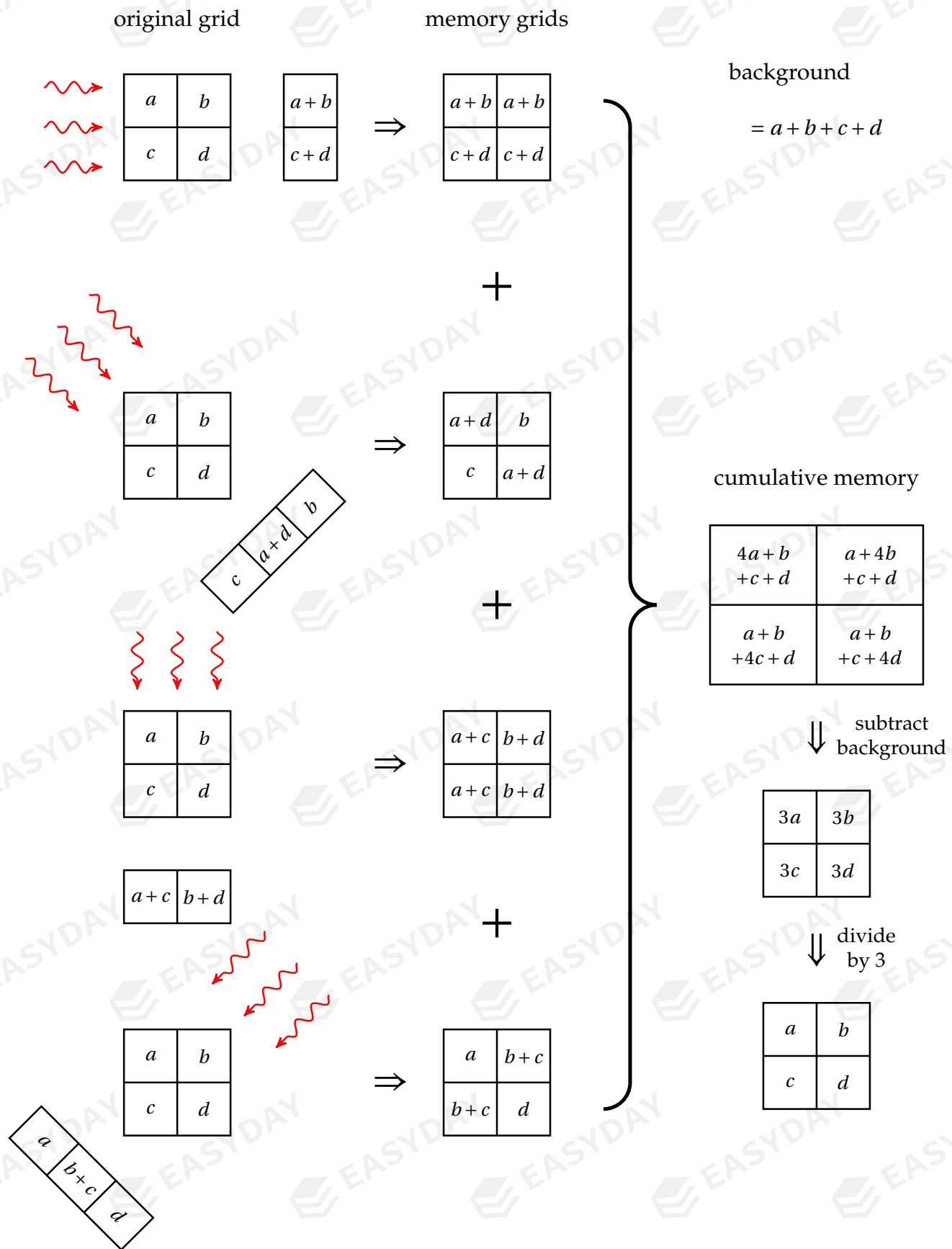
âCT 成像的优点 – 可以显示结构的精确位置、形状、大小 – 具有相似衰减的组织可以区分系数 – CT 成像的缺点 – 比传统 2D 射线照相更昂贵 – 患者暴露在更高的辐射剂量下建立图像 (Q)

我们现在讨论如何从透射 X 射线中恢复每个体素的信息我们考虑一个玩具模型，一个 2 英镑 2 的网格，有 4 个体素每个体素对衰减有贡献，用数字 a、b、c 和 d 描述网格暴露于平行的 X 射线束，读数来自探测器被分配给相关的体素，体素被重建并记录在记忆网格中，这个过程在四个不同的方向上重复，然后添加所有记忆网格以构建一个累积网格，了解背景读数，即所有体素的 $a + b + c + d$ ，并且累积读数，可以在两个简单的步骤内获得原始体素的信息

步骤1：从累积网格中减去背景读数

步骤2：将结果除以三

整个过程如图所示，这应该是不言自明的 162

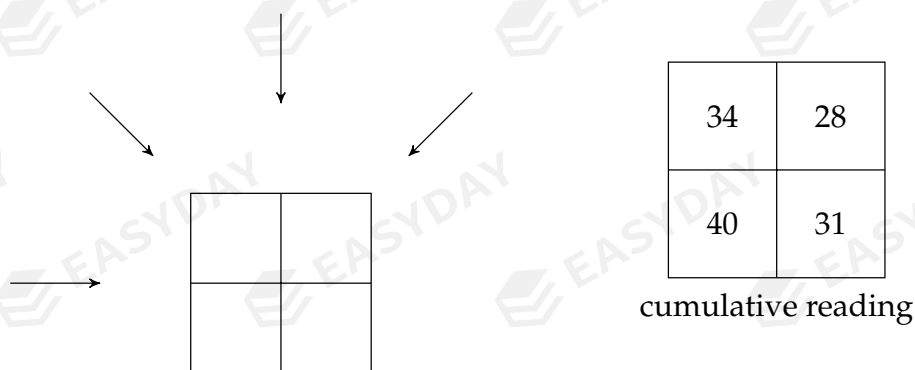


杨宇豪 13 医学影像原始网格存储网格 $cadb)cadb)cadb)cadb)+++c+da+ba+bc+da+bc+dca+dba+dcba+da+cb+da+ca+cb+db+dab+cdab+cb+cd$ 背景 $=a+b+c+d$ $4a+b+c+d$ $4b+c+d$ $a+b+4c+d$ $a+b+c+4d$ 累积内存 + 减去背景 $3a$ $3b$ $3c$ $3d$ + 除以 3 $abcd$ 恢复信息原始 2×2 体素使用从不同角度曝光获得的探测器读数 163

in practice, high-quality 2D slice image is built up from many voxels, attenuated intensities from many different directions are measured by an array of many detecting units ^[83]

huge amount of data is to be collected and processed, this requires use of power computers

Question 13.6 An X-ray image of a 2-by-2 pixel section is obtained by exposing X-rays from four different angles. Detector readings from four consecutive directions are summed to give the cumulative pattern shown. For any one direction, total of detector readings is 19. (a) Find the value of each pixel and mark them on the graph. (b) For each of the directions shown, state the readings measured by detectors.



Question 13.7 By reference to the principles of CT imaging, suggest why the radiation dosage from a CT scan is greater than a simple X-ray image.

13.2 ultrasonic scans

13.2.1 nature of ultrasound

sound wave is a *longitudinal mechanical* wave

human hearing range is around 20 Hz (low pitch) ~ 20,000 Hz (high pitch)

sound waves of frequencies higher than 20,000 Hz is **ultrasound**

➤ meaning of longitudinal waves and mechanical waves

propagation of sound waves require material medium (air, water, tissue, etc.)

longitudinal wave travels in a parallel direction to compression and rarefaction of medium

^[83] For our simple toy model, each voxel is exposed for four times and superimposed with each of the other voxels once, this is why the subtract-background-and-divide-by-three trick works.

杨宇豪 13 医学成像在实践中，高质量的2D切片图像是由许多体素建立起来的，来自许多不同方向的衰减强度是由许多检测单元的阵列测量的[83]需要收集和处理大量数据，这需要使用强大的计算机问题13.6通过从四个不同角度曝光X射线获得2×2像素截面的 X射线图像。将来自四个连续方向的检测器读数相加，得到所示的累积模式。对于任一方向，探测器读数总数为 19。(a) 找出每个像素的值并将其标记在图表上。(b) 对于所示的每个方向，说明探测器测量的读数。28 31 34 40 累计阅读数问题13.7 结合CT成像原理，说明为什么CT扫描的辐射剂量比简单的X射线图像大。13.2 超声波扫描 13.2.1 超声波的性质超声波是纵波机械波人类听觉范围约为 20 Hz（低音）» 20,000 Hz（高音）频率高于 20,000 Hz 的声波是超声波 α 纵波和机械波的含义声波的传播需要物质介质（空气、水、组织等）纵波沿与介质的压缩和稀疏平行的方向传播[83] 对于我们的简单在玩具模型中，每个体素都会曝光四次，并与其他每个体素叠加一次，这就是减去背景并除以三的技巧起作用的原因。164

- rationale for using ultrasounds in medical imaging

for a given medium, speed of sound is constant

$v = \lambda f \Rightarrow$ greater frequency means shorter wavelength, which means better *resolution*

so fine details of small structures can be detected

Question 13.8 Sound waves travel at a speed of around 1600 m s^{-1} in soft tissues. Compare the wavelength of sound waves if they have a frequency of (a) 500 Hz, (b) 10 MHz.

13.2.2 production of ultrasounds

sound can be produced by a *piezo-electric transducer*^[84]

if apply a high-frequency alternating voltage across a piezo-electric crystal, the crystal is made to vibrate, ultrasound wave is produced

- frequency of ultrasound is determined by frequency of alternating voltage: $f_{\text{sound}} = f_{\text{a.c.}}$

to generate ultrasounds, we need $f_{\text{a.c.}} > 20,000 \text{ Hz}$

- there is an optimum thickness for crystal to give out ultrasound of greatest intensity

when thickness of crystal $d = \frac{1}{2}\lambda$, amplitude of vibration is maximum^[85]

- in ultrasonic scanning, reflected sound waves are detected and processed to build images

the same piezo-electric transducer is also used to act as a sound detector

for each pulse emitted, vibration of crystal must stop quickly before reflected echo arrives

a damping mechanism is required to ensure vibration would stop in very short time

- a typical piezo-electric transducer used in ultrasonic scanning consists of several parts:

- a piece of piezo-electric crystal (quartz, PZT, PVDF, etc.)

this is the heart of the transducer, responsible for generating and detecting sound waves

- an acoustic window (made of good sound transmitter)

volume of sound generated is boosted with the acoustic window

^[84] Piezo-electricity and its use as sensing devices will be discussed in §15.1.4.

^[85] Resonance occurs when a nice *stationary wave* pattern is formed within the piezo-electric crystal. It can be shown that this occurs if the thickness of the crystal satisfies: $d = \left(n + \frac{1}{2}\right)\lambda$, where $n = 0, 1, 2, \dots$. However, additional thickness would cause greater attenuation, so the optimum thickness is $d = \frac{1}{2}\lambda$.

Yuhao Yang 13 医学成像 – 在给定介质的医学成像中使用超声波的基本原理，声速恒定 $v = \lambda f$) 频率越高意味着波长越短，这意味着分辨率越高，因此可以检测到小结构的精细细节问题 13.8 声波在软组织中以大约 1600 m s^{-1} 的速度传播。比较频率为 (a) 500 Hz、(b) 10 MHz 的声波的波长。

13.2.2 超声波的产生压电换能器可以产生声音[84] 如果在压电晶体上施加高频交流电压，使晶体振动，则产生超声波超声波的频率由交流电压的频率决定： $f_{\text{sound}} = f_{\text{a.c.}}$ 。为了产生超声波，我们需要 $f_{\text{a.c.}} \geq 20,000 \text{ Hz}$ 当晶体厚度 $d = \frac{1}{2} \lambda$ 时，晶体发出最大强度的超声波的最佳厚度，振动幅度最大 [85] 在超声波扫描中，反射声波被检测和处理以构建图像，相同的压电换能器也被用作每个发射脉冲的声音检测器，晶体的振动必须在反射回波到达之前迅速停止，需要阻尼机制以确保振动不会在很短的时间内停止 在超声波扫描中使用的典型压电换能器由几个部分组成：– 一块压电晶体（石英、PZT、PVDF 等），这是换能器的核心，负责生成和检测声波 – 声窗（由良好的声音发射器制成）产生的声音音量通过声窗增强 [84] 压电及其作为传感设备的使用将在第 15.1.4 节中讨论。[85]当压电晶体内形成良好的驻波图案时，就会发生谐振。可以证明，如果晶体的厚度满足： $d = (n + \frac{1}{2}) \lambda$ ，其中 $n = 0, 1, 2, \dots$ ，就会发生这种情况。然而，额外的厚度会导致更大的衰减，因此最佳厚度为 $d = \frac{1}{2} \lambda$ 。165

- a damping block (usually made of epoxy-resin)

this stops vibration of crystal quickly before reflected wave returns

- electrodes/connecting wires

signals can be sent to transducers from control units to generate ultrasounds

signals detected by transducer can also be sent back to processing units to build images

13.2.3 attenuation of ultrasounds

just like X-rays, intensity of ultrasound also decreases as it propagates through a substance

intensity I decreases *exponentially* with depth x : $I = I_0 e^{-\alpha x}$

I_0 is original intensity, α is called **absorption coefficient** of the medium

- ultrasonic scans mostly rely on detection of reflected waves from various boundaries
- so attenuation is not a big problem, weak echoes from deep tissues can be *amplified*

13.2.4 reflected & transmitted intensity at boundaries of structures

as ultrasound travels from one medium to another, it undergoes *reflection* and *transmission*

intensities of incident, reflected and transmitted ultrasounds satisfy the obvious relation:

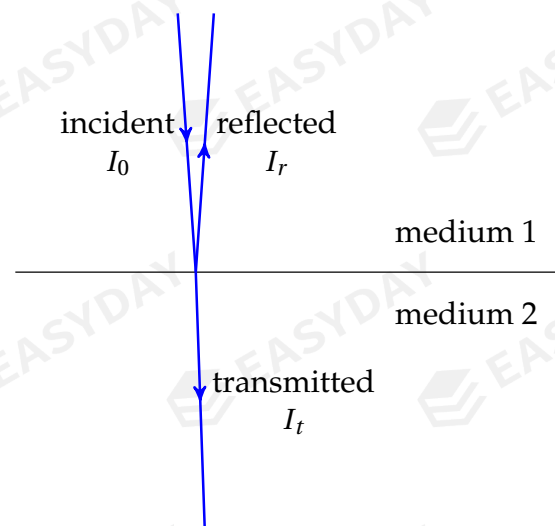
$$I_0 = I_r + I_t$$

for *normal incidence*, fraction of reflected and transmitted ultrasounds are further given by:

reflection coefficient: $\frac{I_r}{I_0} = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$

transmission coefficient: $\frac{I_t}{I_0} = 1 - \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$ [86]

where Z is the medium's *acoustic impedance*



ultrasound at boundary of two medium

[86] Proof of this relation involves solving second-order partial differential wave equations under certain boundary conditions, in which the idea of acoustic impedance is formally defined when one describes how much sound pressure is produced by vibration of acoustic medium at a given frequency.

Yuhao Yang 13 医学成像 - 阻尼块（通常由环氧树脂制成），可在反射波返回之前快速停止晶体振动 - 电极/连接线信号可以从控制单元发送到换能器以生成超声波，换能器检测到的信号也可以发送回处理单元以构建图像

13.2.3 超声波的衰减就像 X 射线一样，超声波的强度也会随着其传播通过物质而降低，强度 I 随深度 x 呈指数下降： $I = I_0 e^{-\alpha x}$ I_0 是原始强度， α 称为介质的吸收系数超声扫描主要依靠检测各个边界的反射波，因此衰减不是一个大问题，来自深层组织的微弱回波可以被放大

13.2.4 结构边界处的反射和透射强度介质 1 介质 2 入射 I_0 反射 I_r 透射 I_t 超声波从一种介质传播到两种介质边界处的超声波另一方面，它所经历的入射、反射和透射超声波的反射和透射强度满足明显的关系： $I_0 = I_r + I_t$ 对于法向入射，反射和透射超声波的分数进一步由下式给出：反射系数： $I_r / I_0 = (Z_1 - Z_2 / Z_1 + Z_2)^2$ 透射系数： $I_t / I_0 = 1 - (Z_1 - Z_2 / Z_1 + Z_2)^2$ [86] 其中 Z 是介质的声阻抗 [86] 这种关系的证明涉及在某些边界条件下求解二阶偏微分波动方程，其中当描述给定频率下声介质的振动产生多少声压时，声阻抗的概念被正式定义。166

➤ **acoustic impedance**, also called specific acoustic impedance, is defined as the product of density of medium and speed of sound wave travelling through it: $Z = \rho c$

Example 13.3 Data for air and soft tissue is given in the table below. When an ultrasound is sent from air into human body at right angles, suggest what happens.

	density/kg m ⁻³	speed of sound/m s ⁻¹
air	1.29	330
soft tissue	1060	1540

let's first find the acoustic impedance for air and soft tissue:

$$Z_{\text{air}} = \rho_{\text{air}} c_{\text{air}} = 1.29 \times 330 = 4.26 \times 10^2 \text{ kg m}^2 \text{ s}^{-1}$$

$$Z_{\text{tissue}} = \rho_{\text{tissue}} c_{\text{tissue}} = 1060 \times 1540 = 1.63 \times 10^6 \text{ kg m}^2 \text{ s}^{-1}$$

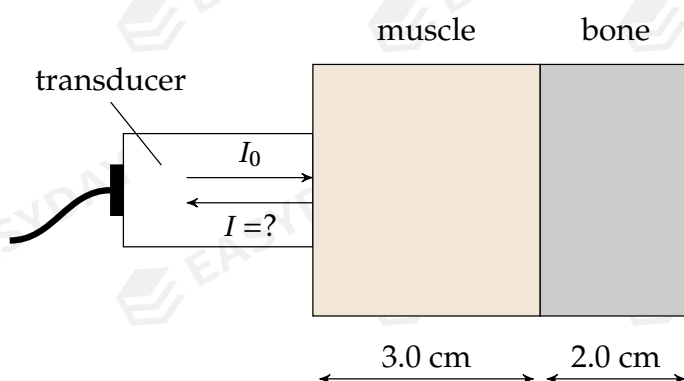
at air-tissue boundary, reflection coefficient is: $\frac{I_r}{I_i} = \left(\frac{4.26 \times 10^2 - 1.63 \times 10^6}{4.26 \times 10^2 + 1.63 \times 10^6} \right)^2 \approx 99.9\%$

almost complete reflection, so almost no ultrasound can enter human body

to overcome this problem, transducer is coupled to patient's skin using a *gel*, which is a transparent watery substance that has a similar acoustic impedance to soft tissue,

with *impedance matching*, ultrasound can get into patient's body □

Example 13.4 An ultrasonic pulse of intensity I_0 is sent into a layer of muscle, it then arrives at a muscle-bone boundary (see figure). The speed of sound c , acoustic impedance Z and absorption coefficient μ for muscle and bone are given in the table. (a) Find the intensity of the wave reflected from the muscle-bone boundary and received back at the surface of the muscle. (b) Find the arrival time of this signal after the pulse is transmitted into patient's body.



	muscle	bone
$c/\text{m s}^{-1}$	1600	4000
$Z/\text{kg m}^2 \text{ s}^{-1}$	1.7×10^6	6.4×10^6
μ/m^{-1}	23	130

Yuhao Yang 13 医学成像声阻抗，也称为比声阻抗，定义为介质密度与声波穿过介质的速度的乘积： $Z = \rho c$ 示例 13.3 空气和软组织的数据如下表所示。当超声波从空气以直角发送到人体时，请说明会发生什么。密度/ kg m^{-3} 声速/ m s^{-1} 空气 1.29 330 软组织 1060 1540 - 我们首先计算空气和软组织的声阻抗： $Z_{\text{air}} = \rho_{\text{air}} c_{\text{air}} = 1.29 \times 330 = 4.26 \times 10^2 \text{ kg m}^2 \text{ s}^{-1}$ $Z_{\text{tissue}} = \rho_{\text{tissue}} c_{\text{tissue}} = 1060 \times 1540 = 1.63 \times 10^6 \text{ kg m}^2 \text{ s}^{-1}$ 在空气组织边界处，反射系数为： $|r| = \left| \frac{4.26 \times 10^2 - 1.63 \times 10^6}{4.26 \times 10^2 + 1.63 \times 10^6} \right|^2 \approx 99.9\%$ 几乎完全反射，因此几乎为了克服超声波不能进入人体的问题，换能器使用凝胶耦合到患者的皮肤，凝胶是一种透明的水状物质，与软组织具有相似的声阻抗，通过阻抗匹配，超声波可以进入患者的体内。例 13.4 强度为 I_0 的超声波脉冲被发送到肌肉层，然后到达肌肉-骨骼边界（见图）。表中给出了肌肉和骨骼的声速 c 、声阻抗 Z 和吸收系数 α 。(a) 求从肌肉-骨骼边界反射并在肌肉表面接收回来的波的强度。(b) 求脉冲传输到患者体内后该信号的到达时间。肌肉骨 $I_0 I = ?$ 3.0 cm 2.0 cm 传感器肌肉骨

	$c/\text{m s}^{-1}$	$Z/\text{kg m}^2 \text{ s}^{-1}$	α/m^{-1}
肌肉	1600	1.7×10^6	23
骨骼	4000	6.4×10^6	130

✍ intensity received at muscle surface: $I = I_0 \times \underbrace{e^{-\mu_m x_m}}_{\text{attenuation in muscle}} \times \underbrace{\left(\frac{Z_m - Z_b}{Z_m + Z_b} \right)^2}_{\text{reflection at boundary}}$

$$I = I_0 \times e^{-23 \times 2 \times 0.030} \times \left(\frac{6.4 \times 10^6 - 1.7 \times 10^6}{6.4 \times 10^6 + 1.7 \times 10^6} \right)^2 \approx I_0 \times 0.252 \times 0.337 \approx 0.085 I_0$$

note that ultrasound travels from surface of muscle to the muscle-bone boundary, and then travels from the boundary back to the muscle surface, so it is attenuated through a total distance that equals twice the thickness of the muscle

$$\text{time taken for this return trip: } \Delta t = \frac{2x_m}{c_m} = \frac{2 \times 0.030}{1600} \approx 3.75 \times 10^{-5} \text{ s}$$

in practice, arrival time for echo waves are measured so depth of structures are found $\square$

Question 13.9 Given that a gel has an acoustic impedance of $1.65 \times 10^6 \text{ kg m}^2 \text{ s}^{-1}$. Use data in Example 13.3, calculate the reflection coefficient when ultrasound is normally incident from a gel to the human body, and hence explain why a gel is put on the patient's skin during medical diagnosis using ultrasounds.

Question 13.10 It is a good idea to use ultrasound to image the lungs? Give your reasons.

Question 13.11 In Example 13.4, if an ultrasound detector is couple to the surface of the bone at the far right, what intensity would be measured?

13.2.5 ultrasound imaging

the principles of ultrasonic imaging can be summarised as the following:

ultrasonic *pulses* produced from *piezo-electric transducer* are sent into patients body

pulses are reflected from *boundaries* of different media

reflected waves, or echoes, are received and detected by the same transducer

these signals are processed to construct the image

arrival times of the reflected waves can give information about *thickness* of structures

intensity of the reflected signals depend on acoustic impedance and attenuation coefficient of media, so can give information about *nature* of structures

➤ advantages of ultrasound imaging

– safe, no harmful radiation involved (can be used in diagnosis of fetus during pregnancy)

Yuhao Yang 13 医学成像 - 肌肉表面接收的强度： $I = I_0 \times e^{-(\alpha x_m + \frac{1}{2} \left| \frac{Z_m - Z_b}{Z_m + Z_b} \right| x_z)}$ 肌肉衰减 $\times (Z_m - Z_b / Z_m + Z_b)^2$ $| \{z\}$ 边界反射 $I = I_0 \times e^{-23 \times 2 \times 0.030 \times (6.4 \times 10^6 - 1.7 \times 10^6 / 6.4 \times 10^6 + 1.7 \times 10^6)^2} \frac{1}{4} I_0 \times 0.252 \times 0.337 \frac{1}{4} 0.085 I_0$ 请注意，超声波从肌肉表面传播到肌骨边界，然后从边界返回到肌肉表面，因此它的衰减总距离等于肌肉厚度的两倍，回程时间为： $\Delta t = 2x_m / c_m = 2 \times 0.030 \text{ m} / 1600 \text{ m/s} = 3.75 \times 10^{-5} \text{ s}$ 在实践中，测量回波的到达时间，从而找到结构的深度 $\hat{d}$ 问题 13.9 假设凝胶的声阻抗为 $1.65 \times 10^6 \text{ kg m}^{-2}\text{s}^{-1}$ 。使用例 13.3 中的数据，计算超声波通常从凝胶入射到人体时的反射系数，从而解释为什么在使用超声波进行医疗诊断时将凝胶放在患者的皮肤上。问题 13.10 使用超声波对肺部进行成像是一个好主意吗？给出你的理由。问题 13.11 在例 13.4 中，如果超声波探测器耦合到最右侧的骨头表面，将测量到什么强度？ 13.2.5 超声成像 超声成像的原理可概括如下：压电换能器产生的超声脉冲被发送到患者体内脉冲从不同介质的边界反射反射波或回波由同一换能器接收和检测这些信号被处理以构建图像反射波的到达时间可以提供有关结构厚度的信息反射信号的强度取决于介质的声阻抗和衰减系数，因此可以提供信息关于结构性质 $\hat{\alpha}$ 超声成像的优点 - 安全，不涉及有害辐射（可用于妊娠期间胎儿的诊断） 168

- provides images in real-time, almost no processing delay
- low cost, apparatuses of ultrasound scans are not expensive
- portable, apparatuses can be easily brought to a patient's bedside
- disadvantages of ultrasound imaging
 - difficult to image structures behind bones or air
 - require a highly-skilled operator to read images
- ultrasounds are also used in other areas of medical diagnosis
 - image flow of blood in vessels with Doppler ultrasonography
 - detect problems in liver, heart, kidney, abdomen, etc.
 - break up kidney stones by high-frequency resonance
 - clean teeth, medical devices, etc.

13.3 MRI (*)

13.3.1 nuclear magnetic resonance

nuclei of some atoms have an intrinsic property called **spin**

at this stage, you can think of a nucleus as if it is spinning around ^[87]

since all nuclei are positively-charged, spinning gives rise to a circulating current, magnetic fields are created around any nucleus, so spin makes nuclei behave like tiny magnets

step 1. split of energy levels

in absence of magnetic field, nuclei align randomly, any orientation shares the same energy

apply a strong magnetic field B_0 , nuclei would rearrange themselves

most nuclei line up parallel to B_0 → stable, low-energy states

^[87]This picture might be helpful to the understanding of NMR, but is not correct in the context of quantum mechanics. Spin is a *quantum* concept that has no analogy in classical physics. Spin is an *intrinsic* property of particles. For a proton, an electron, or any elementary particle, they have a fixed magnitude of a spin number, which only depends on the type of particle. This spin number has nothing to do with rotation. Spin also gives information about directions, which is even more peculiar. The spin component along any axis is *quantized*, i.e., only certain discrete values are allowed.

Yuhao Yang 13 医学成像 – 实时提供图像，几乎没有处理延迟 – 成本低，超声扫描设备并不昂贵 – 便携式，设备可以轻松带到患者床边 – 超声成像的缺点 – 难以对骨骼或空气后面的结构进行成像 – 需要高技能的操作员读取图像 – 超声还用于医疗诊断的其他领域 – 通过多普勒超声检查血管中的血流图像 – 检测问题肝脏、心脏、肾脏、腹部等 – 通过高频共振分解肾结石 – 清洁牙齿、医疗器械等 13.3 MRI (Q) 13.3.1 核磁共振某些原子的原子核在此阶段具有称为自旋的固有属性，您可以将原子核想象成正在旋转 [87]，因为所有原子核都带正电，旋转会产生循环电流，在任何原子核周围都会产生磁场，因此自旋使原子核表现得像微小的磁铁步骤1.在没有磁场的情况下分裂能级，原子核随机排列，任何方向共享相同的能量应用强磁场 B_0 ，原子核会重新排列自己大多数原子核排列平行于 B_0 ！稳定，低能态[87]这张图可能有助于理解NMR，但在量子力学的背景下是不正确的。自旋是一个量子概念，在经典物理学中没有类似的概念。自旋是粒子的固有属性。对于质子、电子或任何基本粒子，它们具有固定的自旋数大小，该大小仅取决于粒子的类型。这个旋转数与旋转无关。Spin 还提供了方向信息，这一点更加奇特。沿任何轴的自旋分量都是量化的，即只允许某些离散值。 169

a few line up anti-parallel to $B_0 \rightarrow$ unstable high-energy states

energy level of nuclei splits up when external magnetic field B_0 is switched on

note that nucleus energy levels in presence of B_0 are discrete

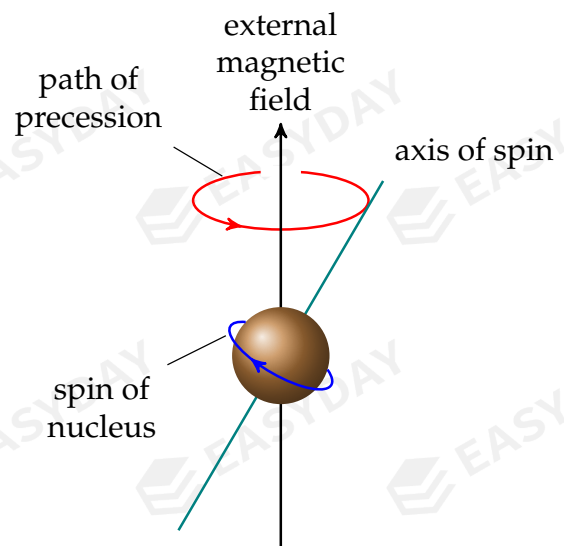
step 2. precession of nuclei

nuclei do not line up exactly along external magnetic field B_0 , axis of spin is slightly tilted

axis of spin rotates about an axis parallel to B_0 , this rotational motion is called **precession**

precession occurs at an angular frequency known as **Larmor frequency**, which is proportional to flux density B_0 of external field: $\omega_0 = \gamma B_0$

γ is the gyromagnetic ratio of the nucleus, which depends of the type of the nucleus



step 3. nuclear magnetic resonance (NMR)

radio pulses B_{RF} at same Larmor frequency are applied

low-energy nuclei absorb RF photons and flip up into higher states

this process is **nuclear magnetic resonance** [88]

Example 13.5 The gyromagnetic ratio for proton is $2.68 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$. What is the required frequency of the radiation pulse to cause magnetic resonance of hydrogen nuclei in a strong magnetic field of 1.8 T?

🔗 Larmor frequency of precession: $\omega_0 = \gamma B_0 = 2.68 \times 10^8 \times 1.8 \approx 4.82 \times 10^8 \text{ rad s}^{-1}$

resonant frequency matches Larmor frequency, so: $f = \frac{\omega_0}{2\pi} = \frac{4.82 \times 10^8}{2\pi} \approx 7.7 \times 10^7 \text{ Hz}$

this frequency is within radio frequency, so *radio pulses* should be applied □

[88] NMR was first described and measured by *Isidor Rabi* in 1938. Rabi was awarded the 1944 Nobel Physics Prize 'for his resonance method for recording the magnetic properties of atomic nuclei'

杨宇豪 13 医学成像一些排列与 B_0 反平行 - 当外部磁场 B_0 开启时，原子核的不稳定高能态能级分裂注意，存在 B_0 的原子核能级是离散的步骤 2. 原子核的进动外部磁场原子核进动自旋的自旋路径轴原子核不完全沿着外部磁场排列 B_0 ，自旋轴稍微倾斜，自旋轴围绕平行于 B_0 的轴旋转，这种旋转运动称为进动，进动发生在称为拉莫尔频率的角频率，该角频率与外场的通量密度 B_0 成正比： $\omega_0 = \gamma B_0$ γ 是原子核的旋磁比，取决于原子核的类型步骤 3。相同拉莫尔频率下的核磁共振 (NMR) 无线电脉冲 BRF 为应用的低能原子核吸收 RF 光子并翻转到更高的状态，这个过程就是核磁共振 [88] 例 13.5 质子的旋磁比为 $2.68 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$ 。在 1.8 T 的强磁场中引起氢核磁共振所需的辐射脉冲频率是多少？ - 进动的拉莫尔频率： $\omega_0 = \gamma B_0 = 2.68 \times 10^8 \times 1.8 = 4.82 \times 10^8 \text{ rad s}^{-1}$ 共振频率与拉莫尔频率匹配，因此： $f = \omega_0 / 2\pi = 4.82 \times 10^8 / 2\pi = 7.7 \times 10^7 \text{ Hz}$ 该频率在无线电频率内，因此无线电应施加脉冲 α [88] NMR 于 1938 年由伊西多·拉比 (Isidor Rabi) 首次描述和测量。拉比“因其记录原子核磁性的共振方法”而荣获 1944 年诺贝尔物理学奖

170

step 4. relaxation

once RF pulses B_{RF} are switched off, nuclei flip back to lower energy states

during the process of **relaxation**, or **de-excitation**, RF photons of certain energies are released

time taken for nuclei to relax depends on the environment of the nuclei

relaxation time can give information about nature of materials

➤ there are two processes through which nuclei could relax

– *spinlattice relaxation*, characterized by T_1 relaxation

nuclei flip so that spin orientations return to equilibrium orientation in external field B_0

energy of spinning nuclei is transferred to lattice of surrounding atoms

– *spinspin relaxation*, characterized by T_2 relaxation

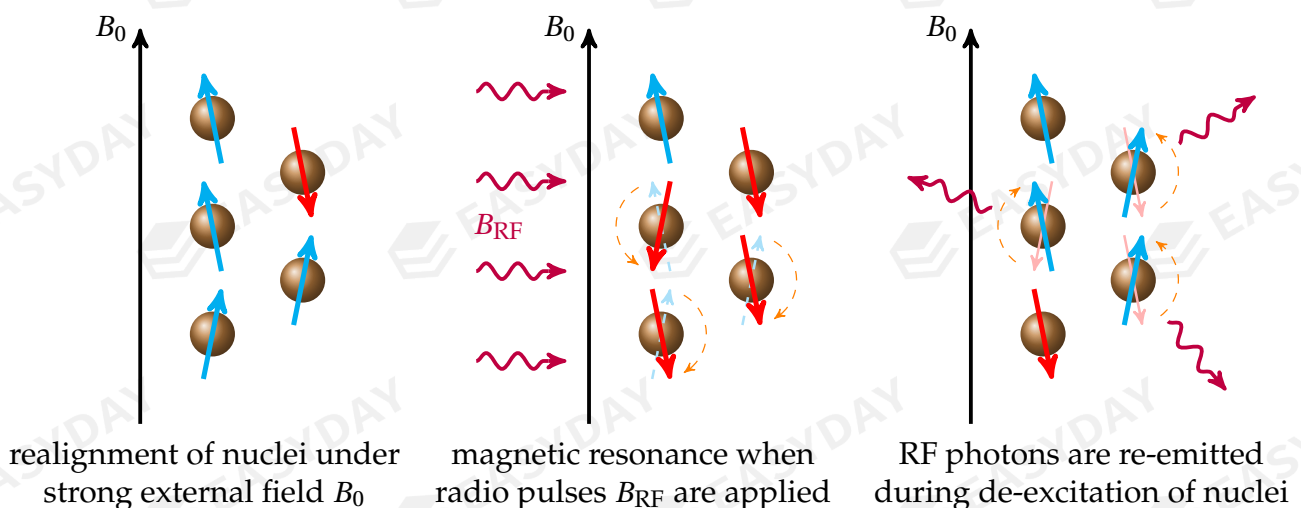
component of spin transverse to B_0 would decay exponentially to its equilibrium state

energy of spinning nuclei is transferred to other spinning nuclei

different tissues (e.g., water and fat) have distinct T_1 and T_2 relaxation times

hence various tissues can be identified based on combined information of T_1 and T_2

the whole idea of nuclear magnetic resonance and relaxation process is illustrated below



13.3.2 magnetic resonance imaging

procedures of magnetic resonance imaging (MRI) are summarised as the following:

Yuhao Yang 13 医学成像步骤 4. 弛豫一旦 RF 脉冲 BRF 关闭，原子核在弛豫或去激发过程中翻转回较低能态，释放某些能量的 RF 光子原子核弛豫所需的时间取决于原子核的环境弛豫时间可以提供有关材料性质的信息原子核可以弛豫的过程有两个 - 自旋晶格弛豫，其特征为 T1 弛豫原子核翻转，使自旋方向在外场中恢复到平衡方向 B0 自旋核的能量转移到周围原子的晶格 - 自旋弛豫，其特征是横向于 B0 的自旋的 T2 弛豫分量将指数衰减到其平衡状态自旋核的能量转移到其他自旋核不同的组织（例如水和脂肪）具有不同的 T1 和 T2 弛豫时间，因此可以根据 T1 和 T2 的组合信息来识别各种组织核磁共振和弛豫过程如下图所示 B0 强外场下原子核重新排列 B0 B0 施加无线电脉冲 BRF 时的 BRF 磁共振 B0 核去激发期间重新发射 RF 光子 13.3.2 磁共振成像磁共振成像 (MRI) 的程序总结如下： 171

large superconducting electromagnets are applied to produce *strong uniform field* B_0

nuclei then align parallel or anti-parallel to the field and *precess* at *Larmor frequency*

a set of *gradient coils* generate a *non-uniform* field that superimposes with B_0

this creates small variation in B_0 at different parts of the patient's body, so nuclei at different positions have slightly different precession frequencies

RF coils transmit *radio pulses* B_{RF} at Larmor frequency to excite nuclei

this causes *magnetic resonance* as nuclei absorb energy from B_{RF}

nuclei then *de-excite*, re-emitted signals are detected by another set of RF coils

information is sent to a powerful computer, which processes data and builds up the image

nature of the substance is determined based on *duration of relaxation*

location of nuclei is determined precisely based on the differences in *resonant frequencies*

➤ advantages of MRI

- uses radiation in radio frequency, so no known biological hazards
- good contrast between different soft tissues
- produces 3D images, 2D slice image of any cross section can be obtained easily

➤ disadvantages of MRI

- very expensive (superconducting coils must work at extremely low temperatures)
- not suitable for patients with metallic implants, prostheses, heart pacemakers, etc.
- patient must lie very still during MRI procedures
- bones do not show up clearly on an MRI image (can combine MRI scan with CT)

Question 13.12 State the main components of an MRI scanner and describe their functions.

杨宇豪 13 医学成像采用大型超导电磁体产生强均匀场 B_0 原子核，然后与场平行或反平行排列，并以拉莫尔频率进动。一组梯度线圈产生与 B_0 叠加的非均匀场，这会在患者身体的不同部位产生 B_0 的微小变化，因此不同位置的原子核具有略有不同的进动频率。RF 线圈以拉莫尔频率发射无线电脉冲 BRF 来激发原子核，这会导致磁共振，因为原子核从 BRF 原子核吸收能量，然后去激发，重新发射的信号由另一组射频线圈检测到。信息被发送到功能强大的计算机，该计算机处理数据并建立图像。根据弛豫持续时间确定物质的性质。根据共振频率的差异精确确定原子核的位置。

• MRI 的优点 – 使用射频辐射，因此没有已知的生物危害 – 不同软组织之间良好对比度 – 产生 3D 图像，可以轻松获得任何横截面的 2D 切片图像 – MRI 的缺点 – 非常昂贵（超导线圈必须在极低的温度下工作） – 不适合有金属植入物、假肢、心脏起搏器等患者 – 患者在 MRI 手术期间必须保持静止 – 骨骼在 MRI 图像上无法清晰显示（可以将 MRI 扫描与 CT 结合起来）

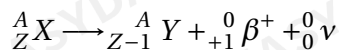
问题 13.12 说出 MRI 的主要组成部分扫描仪并描述其功能。 172

13.4 PET

13.4.1 positron

positron, also called *anti-electron*, is the *anti-particle* of electron

- mass of a positron is same as mass of an electron ($m_e = 9.11 \times 10^{-31}$ kg)
- charge of a positron is opposite to charge of an electron ($+e = 1.60 \times 10^{-19}$ C)
- positrons can be produced using radioactive β^+ -emitters



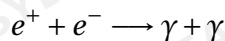
an isotope that is commonly used is fluorine-18: ${}_{9}^{18}\text{F} \longrightarrow {}_{8}^{18}\text{O} + {}_{+1}^0\beta^+ + {}_0^0\nu$

fluorine-18 is prepared by bombarding oxygen-18 with protons^[89]: ${}_{8}^{18}\text{O} + {}_1^1\text{p} \longrightarrow {}_{9}^{18}\text{F} + {}_0^1\text{n}$

Question 13.13 Another isotope of oxygen, oxygen-15 (${}_{8}^{15}\text{O}$), can undergo decay via positron emission. Write down the equation representing this decay process.

13.4.2 electron–positron annihilation

when a positron meets up with an electron, they *annihilate* and produce two γ -photons^[90]



- mass-energy is conserved during annihilation process

mass-energy of electron-positron pair transforms into electromagnetic energy of γ -photons

so energy of each γ -photon: $E_\gamma = m_e c^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 \approx 8.2 \times 10^{-14}$ J

using $E = hf$, we find frequency of each γ -photon: $f = \frac{E}{h} = \frac{8.2 \times 10^{-14}}{6.63 \times 10^{-34}} \approx 1.2 \times 10^{20}$ Hz

- total momentum is also conserved during annihilation

K.E. of electron-positron pair is usually negligible, so zero initial momentum

by conservation of momentum, total final momentum of the two γ -photons must be zero

so the two γ -photons carry equal but opposite momentum

- equal momentum means the two γ -photons have same energy

so they have same frequency and same wavelength

^[89] To send protons towards the target nuclei, we can accelerate them using a *cyclotron* (see §8.3.4).

^[90] In case you have forgotten what is meant by a photon, go back and review §11.2.2 and §11.2.4.

杨宇豪 13 医学影像 13.4 PET 13.4.1 正电子正电子，也称为反电子，是电子的反粒子 $\hat{\alpha}$ 正电子的质量与电子的质量相同 ($m_e = 9.11 \times 10^{-31} \text{ kg}$) $\hat{\alpha}$ 正电子的电荷与电子的电荷相反 ($+e = 1.60 \times 10^{-19} \text{ C}$) $\hat{\alpha}$ 可以使用放射性 β^+ 发射体 ${}^A_Z X$ 产生正电子！ ${}^A_Z X \rightarrow {}^A_{Z-1} Y + 0 + 1 \beta^+ + 0 \nu$ 常用的同位素是氟-18: ${}^{18}_9 \text{F} \rightarrow {}^{18}_8 \text{O} + 0 + 1 \beta^+ + 0 \nu$ 氟-18 是通过用质子轰击氧 18 制备的 [89]: ${}^{18}_8 \text{O} + 1 \text{ p} \rightarrow {}^{18}_9 \text{F} + 1 \text{ n}$ 问题 13.13 氧的另一种同位素氧 15 (${}^{15}_8 \text{O}$) 可以通过正电子发射进行衰变。写下代表这个衰变过程的方程。 13.4.2 正电子湮灭当正电子遇到电子时，湮灭并产生两个 γ 光子[90] $e^+ + e^- \rightarrow \gamma + \gamma$ 湮灭过程中质能守恒，正电子对的质能转化为 γ 光子的电磁能，因此每个 γ 光子的能量为： $E_\gamma = m_e c^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 8.2 \times 10^{-14} \text{ J}$ 使用 $E = hf$ ，我们发现每个 γ 光子的频率： $f = E/h = 8.2 \times 10^{-14} / 6.63 \times 10^{-34} = 1.2 \times 10^{20} \text{ Hz}$ $\hat{\alpha}$ 总动量也守恒歼灭K.E.电子-正电子对的数量通常可以忽略不计，因此动量守恒的初始动量为零，两个 γ -光子的总最终动量必须为零，因此两个 γ -光子携带相等但相反的动量 - 相等动量意味着两个 γ -光子具有相同的能量，因此它们具有相同的频率和相同的波长[89]为了将质子发送到目标核，我们可以使用回旋加速器加速它们（参见§8.3.4）。[90]如果您忘记了光子的含义，请返回并查看第 11.2.2 节和第 11.2.4 节。

- opposite momentum means the two γ -photons travel in opposite directions
so they move off from site of annihilation at 180° to each other

Question 13.14 The annihilation of a proton and an antiproton has been observed by particle physicists. Find the energy released during the process of protonantiproton annihilation.

Question 13.15 When an electron and a positron annihilate, (a) what is the wavelength of the γ -photon produced? (b) What is the momentum of the γ -photon produced?

13.4.3 positron emission tomography (PET)

for a PET scan, a small amount of tracer is injected into patient's body

tracers are molecules labelled with β^+ -emitting nuclides (e.g., fluorine-18^[91])

tracers travel around human body and accumulate at certain organs and tissues^[92]

positrons are produced through

β^+ -decay of tracers

these positrons immediately

annihilate with nearby electrons

two γ -photons are produced

and move in opposite directions

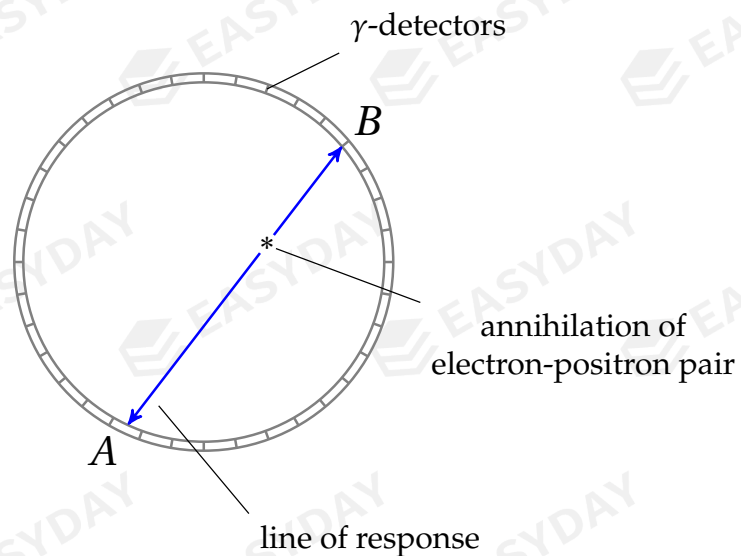
the γ -photon pair are captured

by a ring of detectors

line of response can be established

difference in arrival time can be used to find exact *location* of the annihilation event

number of events per unit time detected at one place can further tell tracer *concentration*



Example 13.6 Two γ -photons are produced from the annihilation of an electron-positron pair, and they hit the detectors A and B. Given that the two detectors are separated by 80 cm, and

^[91]The fluorine-18 nuclide can bind to a specific glucose based molecule, forming fluorodeoxyglucose (FDG, chemical formula $C_6H_{11}^{18}FO_5$), which is one of the most common tracers.

^[92]In particular, the tracers tend to accumulate in tumors. Cancer cells are more metabolically active than normal cells. They absorb the tracers at a higher rate, and therefore emit positrons at a higher rate. This effect can be seen on PET scans, allowing the doctor to detect cancer at early stages or examine the effects of cancer therapy.

杨宇浩 13 医学成像 – 相反的动量意味着两个 γ 光子以相反的方向传播，因此它们从湮灭位置以 180° 向彼此移动

问题 13.14 粒子物理学家已经观察到质子和反质子的湮灭。求质子反质子湮灭过程中释放的能量。

问题 13.15 当一个电子和一个正电子湮灭时，(a) 产生的 γ 光子的波长是多少？(b) 产生的 γ 光子的动量是多少？

13.4.3 正电子发射断层扫描 (PET) PET 扫描时，将少量示踪剂注入患者体内示踪剂是用 β^+ 发射核素（例如氟 18 [91]）标记的分子示踪剂在人体周围传播并在某些器官和组织处积聚 [92]

电子-正电子对 γ 探测器的响应湮灭线 A B 正电子通过产生示踪剂的 β^+ 衰变这些正电子立即与附近的电子湮灭产生两个 γ 光子并沿相反方向移动 γ 光子对被探测器环捕获可以建立响应线到达时间的差异可用于找到湮灭事件的精确位置在一个地方检测到的每单位时间的事件数量可以进一步判断示踪剂浓度

例 13.6 一个粒子湮灭产生两个 γ 光子电子-正电子对，它们撞击探测器 A 和 B。假设两个探测器相距 80 厘米，[91]氟 18 核素可以与特定的葡萄糖基分子结合，形成氟脱氧葡萄糖（FDG，化学式 $C_6H_{11}^{18}FO_5$ ），这是最常见的示踪剂之一。

[92]特别是，示踪剂往往会在肿瘤中积聚。癌细胞比正常细胞的代谢更加活跃。它们以更高的速率吸收示踪剂，因此以更高的速率发射正电子。这种效应可以在 PET 扫描中看到，使医生能够在早期发现癌症或检查癌症治疗的效果。

174

the arrival of the γ -photon at detector A comes 1.2 ns later than the signal picked up at detector B . On which point along the line joining A and B does the annihilation event occur?

the two γ -photons cover a total distance:

$$d = d_A + d_B = 80 \text{ cm}$$

but one photon covers a greater distance than the other:

$$d_A - d_B = c\Delta t = 3.00 \times 10^8 \times 1.2 \times 10^{-9} = 0.36 \text{ m} = 36 \text{ cm}$$

solving the two equations, we find: $d_A = 58 \text{ cm}$ and $d_B = 22 \text{ cm}$

□

Yuhao Yang 13 医学成像 γ 光子到达探测器 A 的时间比探测器 B 接收到的信号晚 1.2 ns。沿 A 和 B 连线的哪一点会发生湮灭事件？ - 两个 γ 光子覆盖的总距离为： $d = d_A + d_B = 80 \text{ cm}$ ，但一个光子比另一个光子覆盖的距离更大： $d_A - d_B = c \Delta t = 3.00 \times 10^8 \times 1.2 \times 10^{-9} = 0.36 \text{ m} = 36 \text{ cm}$ 求解这两个方程，我们发现： $d_A = 58 \text{ cm}$ 和分贝 = 22 厘米 ÷ 175

CHAPTER 14

Astrophysics

14.1 astronomical distances

14.1.1 unit of length in astronomy

distances between stars and galaxies in the universe are very large

so it is more convenient to introduce alternative units of measurement for lengths

➤ astronomical unit (AU)

an astronomical unit is the average distance between the Earth and the Sun

$$1 \text{ AU} \approx 1.50 \times 10^{11} \text{ m}$$

➤ light-year (ly)

a light-year is the distance that light can travel in one year

$$1 \text{ ly} = 3.00 \times 10^8 \text{ m s}^{-1} \times 365 \times 24 \times 3600 \text{ s} \approx 9.46 \times 10^{15} \text{ m}$$

➤ parsec (pc) (*)

one parsec is the distance at which 1 AU subtends an angle of 1 arcsec or $\frac{1^\circ}{3600}$

$$1 \text{ pc} = \frac{1.50 \times 10^{11} \text{ m}}{\frac{1^\circ}{3600} \times \frac{2\pi}{360^\circ}} \approx 3.09 \times 10^{16} \text{ m}$$

14.1.2 luminosity

luminosity (L) of an object is the total power emitted by this object

➤ unit for luminosity: $[L] = \text{W} = \text{J s}^{-1}$

➤ luminosity of a star depends on its surface temperature and its size

- temperature $T \uparrow \Rightarrow L \uparrow$

- stellar radius $R \uparrow \Rightarrow L \uparrow$

Yuhao Yang 14 天体物理学第 14 章天体物理学 14.1 天文距离 14.1.1 天文学中的长度单位宇宙中恒星和星系之间的距离非常大，因此更方便引入长度的替代测量单位天文单位 (AU) 天文单位是地球与太阳之间的平均距离 $1 \text{ AU} \approx 1.50 \times 10^{11} \text{ m}$ 光年 (ly) 一光年是光在一年内传播的距离 $1 \text{ ly} = 3.00 \times 10^8 \text{ m s}^{-1} \times 365 \times 24 \times 3600 \text{ s} \approx 9.46 \times 10^{15} \text{ m}$ 秒差距 (pc) (Q) 1 秒差距是 1 天文单位对着 1 角秒的角度的距离，或 $1 \text{ pc} = 1.50 \times 10^{11} \text{ m} \times \frac{1}{\sin(1'')} \approx 3.09 \times 10^{16} \text{ m}$ 14.1.2 光度物体的光度 (L) 是该物体发出的总功率 光度单位： $[L] = W = \text{J s}^{-1}$ 光度恒星取决于其表面温度和大小 - 温度 T) L - 恒星半径 R) $L \propto R^2 T^4$

14.1.3 radiant flux intensity

radiant flux intensity (F) is defined as the radiant power passing through a surface per unit area at right angles

- unit for radiant flux intensity: $[F] = \text{W m}^{-2}$
- F measures *observed* intensity, i.e., F gives *apparent brightness* of the object
- F depends on distance from the emitting source

if radiation through space is uniform with no absorption, then: $F = \frac{L}{4\pi d^2}$

Example 14.1 The luminosity of the sun is approximately $3.86 \times 10^{26} \text{ W}$. Given that the distance between the earth and the sun is about $1.5 \times 10^{11} \text{ m}$, what is the theoretical power of a space probe that has a solar pane of an area of 40 m^2 ?

✍ radiant flux intensity: $F = \frac{L}{4\pi d^2} = \frac{3.86 \times 10^{26}}{4\pi \times (1.5 \times 10^{11})^2} \approx 1370 \text{ W m}^{-2}$

power input: $P = FA = 1370 \times 40 \approx 5.46 \times 10^4 \text{ W}$ □

Example 14.2 The radiant flux intensity due to the sun at the position of the earth is about 2.3 times greater than that at the position of the Mars. Given that the distance between the earth and the sun is about $1.5 \times 10^{11} \text{ m}$, how far is the Mars from the sun?

✍ $F \propto \frac{1}{d^2} \Rightarrow \frac{F_E}{F_M} = \frac{d_M^2}{d_E^2} \Rightarrow \frac{d_M^2}{d_E^2} = 2.3 \Rightarrow d_M = \sqrt{2.3} \times 1.5 \times 10^{11} \approx 2.3 \times 10^{11} \text{ m}$ □

14.1.4 standard candles

astronomical objects of known luminosity are called **standard candles**

for a particular star, say X , we can look for a standard candle in the same galaxy

radiant flux intensity F of the standard candle can be measured, and its luminosity L is also known, so its distance can be found using: $d = \sqrt{\frac{L}{4\pi F}}$

distance of star X is about same as distance of standard candle, as they are in same galaxy

this is how we determine the distance of stars by the use of *standard candles*

Yuhao Yang 14 天体物理学 14.1.3 辐射通量强度 辐射通量强度 (F) 定义为以直角穿过单位面积表面的辐射功率 $\hat{\alpha}$ 辐射通量强度的单位: $[F] = \text{W m}^{-2}$ $\hat{\alpha}$ F 测量观测到的强度, 即 F 给出物体的表观亮度 $\hat{\alpha}$ 如果穿过空间的辐射均匀, 则 F 取决于距发射源的距离没有吸收, 则: $F = L / 4\pi d^2$ 例 14.1 太阳的光度大约为 $3.86 \times 10^{26} \text{ W}$ 。假设地球和太阳之间的距离约为 $1.5 \times 10^{11} \text{ m}$, 那么具有面积为 40 m^2 的太阳板的空间探测器的理论功率是多少? - 辐射通量强度: $F = L / 4\pi d^2 = 3.86 \times 10^{26} / 4\pi \times (1.5 \times 10^{11})^2 \approx 1370 \text{ W m}^{-2}$ 功率输入: $P = F A = 1370 \times 40 \approx 5.46 \times 10^4 \text{ W}$ $\hat{\alpha}$ 示例 14.2 太阳在地球的位置大约是火星位置的 2.3 倍。鉴于地球和太阳之间的距离约为 1.5 英磅 10^{11} 米, 那么火星距太阳有多远? - $F / L d^2$) $F_E / F_M = d_M^2 / d_E^2$) $d_M = d_E \sqrt{F_E / F_M} = 2.3 \times 1.5 \times 10^{11} \approx 2.3 \times 10^{11} \text{ m}$ $\hat{\alpha}$ 14.1.4 标准烛光 已知光度的天体被称为特定恒星的标准烛光, 假设 X , 我们可以在同一个星系中寻找一根标准烛光。标准烛光的辐射通量强度 F 是可以测量的, 它的光度 L 也是已知的, 因此它的距离可以通过以下公式求得: $d = \sqrt{L / 4\pi F}$ 恒星 X 的距离与标准烛光的距离大约相同, 因为它们在同一个星系中这就是我们如何使用标准烛光来确定恒星的距离 177

among the many types of standard candles you need to know two in this course:

- cepheid variables
- supernovae

cepheid variables

cepheid variables are stellar objects whose brightness varies periodically^[93]

once we know period of variation in brightness, luminosity of the star can be predicted

Example 14.3 One cepheid variable has a radiant flux intensity of $1.4 \times 10^{-16} \text{ W m}^{-2}$. It is known to have a luminosity of $1.0 \times 10^{30} \text{ W}$. Find the distance of this star in light-years.

$$\text{✎ } F = \frac{L}{4\pi d^2} \Rightarrow d = \sqrt{\frac{L}{4\pi F}} = \sqrt{\frac{1.0 \times 10^{30}}{4\pi \times 1.4 \times 10^{-16}}} \approx 2.4 \times 10^{22} \text{ m} = \frac{2.4 \times 10^{22}}{9.46 \times 10^{15}} \text{ ly} \approx 2.5 \times 10^6 \text{ ly} \quad \square$$

supernovae

luminosity of supernovae is known to be constant^[94]

➤ supernovae are extremely luminous, so suitable for measuring very large distances

Question 14.1 (a) Type Ia supernovae have luminosities at the order of 10^{36} W (approximately 5 billions of the luminosity of the sun). If the sun undergoes a supernova explosion, find the radiant flux intensity measured on earth. (b) It is estimated that the Hiroshima bomb and the Nagasaki bomb (the two nuclear bombs dropped on Japan during the World War II) released a total energy of around 100 TJ. Suppose a similar nuclear bomb completely explodes at a

^[93]The variation in the stellar brightness is related to the mass of the cepheid variable. When the star contracts, its core temperature increases, this boosts the rate of nuclear reaction so radiation pressure increases to push the stellar matter outwards against the gravity. But as the star expands, its temperature drops so its radiation pressure lowers, the gravitational attraction would tend to pull the stellar matter inwards. The period of this feedback loop between the gravitational pull and the nuclear reaction taking place in the star depends on the mass of the star, and the mass of the star determines a number of the star's features including its luminosity. Therefore, if we know the period of the brightness fluctuations, we can know about the star's mass and hence its luminosity.

^[94]To be more precise, Type Ia supernovae explosions are found to have a consistent luminosity. These events are believed to occur when a white dwarf accretes matter until its mass exceeds a critical value, known as the Chandrasekhar limit. By then its gravitational attraction can no longer be overcome by the degeneracy pressure, the star collapses and explodes. There is also Type II supernovae explosions, these occur when a star of very large mass runs out of its nuclear fuel. Radiation pressure cannot be maintained, so the star collapses rapidly towards its centre under gravity.

Yuhao Yang 14 天体物理学在本课程中，您需要了解多种类型的标准烛光中的两种：- 造父变星 - 超新星

造父变星造父变星是亮度周期性变化的恒星物体 [93] 一旦我们知道亮度变化的周期，恒星的光度就可以预测

例 14.3 一个造父变星的辐射通量强度为 $1.4 \times 10^{-16} \text{ W m}^{-2}$ 。已知它的光度为 $1.0 \times 10^{30} \text{ W}$ 。求这颗恒星的距离（以光年为单位）。 - $F = L / (4\pi d^2)$ $d = \sqrt{L / (4\pi F)} = \sqrt{1.0 \times 10^{30} / (4\pi \times 1.4 \times 10^{-16})} = 2.4 \times 10^{22} \text{ m} = 2.4 \times 10^{22} / 9.46 \times 10^{15} \text{ ly} = 2.5 \times 10^6 \text{ ly}$ ð 超新星光度已知超新星的光度是恒定的 [94] ð 超新星非常明亮，因此适合测量非常远的距离问题 14.1 (a) Ia 型超新星的光度约为 10^{36} W （大约是太阳光度的 50 亿）。如果太阳经历超新星爆炸，求在地球上测量的辐射通量强度。(b) 据估计，广岛原子弹和长崎原子弹（二战期间投在日本的两颗核弹）释放的总能量约为 100 TJ。假设一枚类似的核弹在 [93] 处完全爆炸，恒星亮度的变化与造父变星的质量有关。当恒星收缩时，其核心温度会升高，这会提高核反应速率，因此辐射压力会增加，从而将恒星物质抵抗重力向外推。但随着恒星的膨胀，其温度下降，因此其辐射压力降低，引力往往会将恒星物质向内拉。恒星中发生的引力和核反应之间的反馈回路的周期取决于恒星的质量，而恒星的质量决定了恒星的许多特征，包括其光度。因此，如果我们知道亮度波动的周期，我们就可以知道恒星的质量，从而知道它的光度。[94] 更准确地说，Ia 型超新星爆炸被发现具有一致的光度。据信，当白矮星吸积物质直至其质量超过临界值（称为查德拉斯卡极限）时，就会发生这些事件。到那时，它的引力就不再能够被简并压力所克服，恒星就会塌陷并爆炸。还有 II 型超新星爆炸，当质量非常大的恒星耗尽核燃料时就会发生这种爆炸。辐射压力无法维持，因此恒星在重力作用下迅速向其中心塌陷。 178

distance of 50 m from your home in a period of one second, what radiant flux intensity do you measure? (c) Compare the two results, which event would appear brighter?

14.2 spectrum of stars

any object at a particular temperature T emits electromagnetic radiation

radiation emitted by the object spreads into a continuous range of wavelengths λ

total intensity and typical wavelength emitted depend influentially on temperature T

for example, human body (~ 300 K) emits infra-red, cooking fire (~ 1000 K) emits visible red

stars also emit radiation with a range of wavelengths, giving rise to a characteristic spectrum

we can learn a lot about the properties of the star by looking at its spectrum

14.2.1 black body radiation

theoretically, stars can be modelled as black body emitters, whose emission spectrum at a particular temperature T can be precisely predicted

a **black body** is an idealised object that absorbs all incident radiation

➤ intensity spectrum of black body radiation has the following features:

- for given temperature, intensity tends to zero at very short or very long wavelengths
- as temperature increases, peak intensity occurs at shorter wavelength ($T \uparrow \Rightarrow \lambda_{\max} \downarrow$)
- for any λ , intensity increases with temperature ($T \uparrow \Rightarrow I \uparrow$)

➤ **Wien's law** states that wavelength of greatest intensity and surface temperature satisfies:

$$\lambda_{\max} T = b \quad \text{where Wien's displacement constant } b \approx 2.898 \times 10^{-3} \text{ m K}$$

➤ **Stefan-Boltzmann law** states that total power emitted from a black body of surface area A and surface temperature T is given by

$$L = \sigma AT^4 \quad \text{where Stefan-Boltzmann constant } \sigma \approx 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

for a star of radius R , its surface area is $A = 4\pi R^2$

then total radiant power, i.e., luminosity of a star is given by: $L = 4\pi\sigma R^2 T^4$

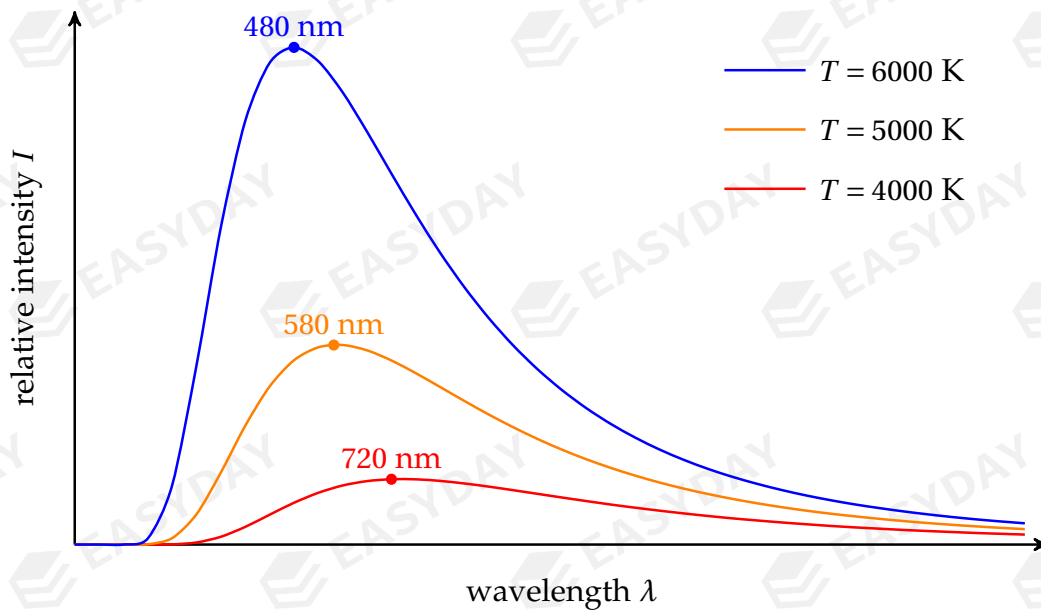
杨宇豪 14 ASTROPHYSICS 距离你家 50 m 一秒内，你测量到的辐射通量强度是多少？(c) 比较两个结果，哪个事件会显得更亮？

14.2 恒星光谱任何物体在特定温度 T 下都会发射电磁辐射物体发射的辐射会传播到连续的波长范围 λ 发射的总强度和典型波长很大程度上取决于温度 T 例如，人体 (≈ 300 K) 发射红外线，烹饪火 (≈ 1000 K) 发射可见红色恒星也发射一定波长范围的辐射，产生特征光谱，我们可以通过观察了解很多关于恒星特性的信息

14.2.1 黑体辐射的光谱理论上，恒星可以被建模为黑体发射体，其在特定温度 T 下的发射光谱可以精确预测黑体是吸收所有入射辐射的理想化物体黑体辐射的强度谱具有以下特征：

- 对于给定温度，强度在非常短或非常长的波长处趋于零 - 随着温度升高，峰值强度出现在较短的波长 (T_1) $\lambda_{\max}$ %)
- 对于任何 λ ，强度随温度增加 (T_1) I_1) $\hat{=}$ 维恩定律规定，最大强度的波长和表面温度满足： $\lambda_{\max} T = b$ 其中维恩位移常数 $b \approx 2.898 \times 10^{-3} \text{ m K}$
- $\hat{=}$ Stefan-Boltzmann 定律规定，从表面积为 A 和表面温度 T 的黑体发出的总功率由 $L = \sigma A T^4$ 给出，其中 Stefan-Boltzmann 常数 $\sigma \approx 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

对于半径为 R 的恒星，其表面积为 $A = 4\pi R^2$ 那么总辐射功率，即恒星的光度由下式给出： $L = 4\pi\sigma R^2 T^4$ 179



radiation spectrum of an ideal black body

Question 14.2 Use the values labelled on the intensity-wavelength curves on top of this page, verify that the wavelength at which the maximum radiation intensity occur satisfies Wien's law at any one of the three given temperatures.

14.2.2 surface temperature

from radiation spectrum of a star, we can identify wavelength at peak intensity

then surface temperature of the star can be determined using Wien's law

Example 14.4 The wavelength for which the maximum rate of emission from the red giant Betelgeuse is 878 nm. What is the surface temperature of the star?

$$T = \frac{b}{\lambda_{\max}} = \frac{2.898 \times 10^{-3}}{878 \times 10^{-9}} \approx 3300 \text{ K} \quad \square$$

Example 14.5 The surface temperature of the sun is about 5800 K. What is the wavelength of the peak radiation output in the solar spectrum?

$$\lambda_{\max} = \frac{b}{T} = \frac{2.898 \times 10^{-3}}{5800} \approx 5.0 \times 10^{-7} \text{ m} \approx 500 \text{ nm}$$

this wavelength lies in the range of the visible spectrum (380 ~ 740 nm)

our calculation indicates a large portion of solar radiation is visible light □

Question 14.3 It is difficult to see a human body or animals with naked eyes in a very dark environment, but night vision is possible using infra-red cameras. Explain why.

杨宇豪 14 天体物理学波长 λ 相对强度 I 720 nm 580 nm 480 nm $T = 6000\text{ K}$ $T = 5000\text{ K}$ $T = 4000\text{ K}$ 理想黑体的辐射光谱问题 14.2 使用本页顶部强度-波长曲线上标注的值，验证出现最大辐射强度的波长满足维恩定律三个给定的温度。14.2.2 根据恒星辐射光谱的表面温度，我们可以确定峰值强度处的波长，然后利用维恩定律可以确定恒星的表面温度。例 14.4 红巨星参宿四的最大发射率的波长为 878 nm。恒星的表面温度是多少？ $-T = b \lambda_{\max} = 2.898 \times 10^{-3} \frac{878 \times 10^{-9}}{1} \approx 3300\text{ K}$ 例 14.5 太阳的表面温度约为 5800 K。太阳光谱中峰值辐射输出的波长是多少？ $-\lambda_{\max} = b T = 2.898 \times 10^{-3} \frac{5800}{1} \approx 5.0 \times 10^{-7}\text{ m} \approx 500\text{ nm}$ 该波长位于可见光谱范围内 (380 » 740 nm) 我们的计算表明太阳辐射的很大一部分是可见光 问题 14.3 在非常黑暗的环境中用肉眼很难看到人体或动物，但夜视可以使用红外摄像机。解释一下为什么。 180

14.2.3 stellar radii

take a standard candle in a distant galaxy, we can calculate distance d of the galaxy

from d and radiant flux intensity F , we can find luminosity of the star: $L = 4\pi d^2 F$

from a star's spectrum, we can find surface temperature T of the star using: $\lambda_{\max} T = b$

once we find L and T , we can calculate the stellar radius R using: $L = 4\pi\sigma R^2 T^4$

Example 14.6 Aldebaran, the brightest star in the constellation Taurus, has a luminosity of 1.8×10^{29} W. The wavelength at the peak radiation intensity of its spectrum is found to be 740 nm. What is the radius of the star?

by Wien's law $\lambda_{\max} T = b$, we find surface temperature to be

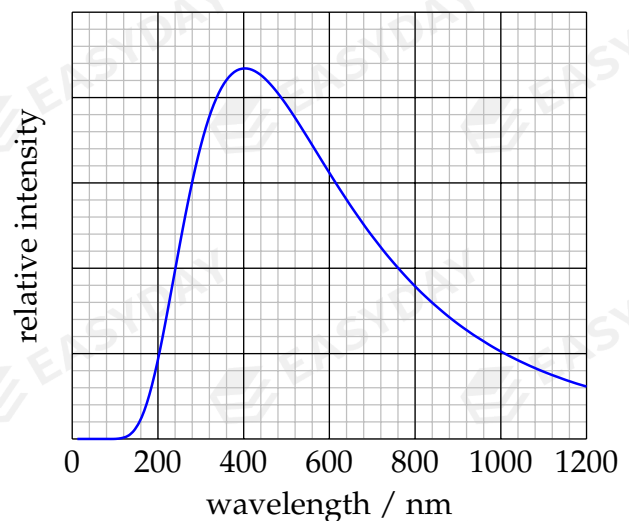
$$T = \frac{b}{\lambda_{\max}} = \frac{2.898 \times 10^{-3}}{740 \times 10^{-9}} \approx 3920 \text{ K}$$

by Stefan-Boltzmann law $L = 4\pi\sigma R^2 T^4$, so

$$1.8 \times 10^{29} = 4\pi \times 5.67 \times 10^{-8} \times 3920^4 \times R^2 \Rightarrow R \approx 3.3 \times 10^{10} \text{ m}$$

Aldebaran is about 50 times larger than the solar radius (around 7.0×10^8 m), and its appearance is red (spectrum peaked at near infra-red), this makes it a typical *red giant* star $\square$

Question 14.4 Altair, the brightest star in the constellation Aquila, is at a distance of 16.7 light years from the earth. The radiant flux intensity due to Altair measured on the earth is $1.3 \times 10^{-8} \text{ W m}^{-2}$. The intensity-wavelength curve of Altair is shown. (a) Find the luminosity of Altair. (b) Find the surface temperature of Altair. (c) Find the radius of the star.



14.3 cosmology

14.3.1 quick review: absorption spectrum

hot interior of star emits a continuous range of wavelengths

杨宇豪 14 天体物理学 14.2.3 恒星半径取一个遥远星系中的标准烛光，我们可以根据 d 和辐射通量强度 F 计算出星系的距离 d ，我们可以从恒星的光谱中找到恒星的光度： $L = 4\pi d^2 F$ ，我们可以通过找到 L 和 T 来找到恒星的表面温度 T ： $\lambda_{\max} T = b$ ，我们可以使用以下公式计算恒星半径 R ： $L = 4\pi\sigma R^2 T^4$ 示例 14.6 毕宿五是金牛座中最亮的恒星，其光度为 $1.8 \times 10^{29} \text{ W}$ 。其光谱峰值辐射强度的波长为 740 nm 。恒星的半径是多少？- 根据维恩定律 $\lambda_{\max} T = b$ ，我们发现表面温度为 $T = b / \lambda_{\max} = 2.898 \times 10^{-3} / 740 \times 10^{-9} = 3920 \text{ K}$ 根据斯蒂芬-玻尔兹曼定律 $L = 4\pi\sigma R^2 T^4$ ，因此 $1.8 \times 10^{29} = 4\pi \times 5.67 \times 10^{-8} \times 3920^4 \times R^2$ $R = \sqrt{1.8 \times 10^{29} / (4\pi \times 5.67 \times 10^{-8} \times 3920^4)} = 3.3 \times 10^{10} \text{ m}$ 毕宿五大约是太阳半径的 50 倍（约 $7.0 \times 10^8 \text{ m}$ ），其外观呈红色（光谱峰值在近红外处），这使其成为典型的红巨星 α 相对强度波长 / nm 0 200 400 600 800 1000 1200 问题 14.4 牵牛星是天鹰座中最亮的恒星，距离地球 16.7 光年。在地球上测得的 Altair 辐射通量强度为 $1.3 \times 10^{-8} \text{ W m}^{-2}$ 。显示了 Altair 的强度-波长曲线。(a) 求 Altair 的光度。(b) 求 Altair 的表面温度。(c) 求出恒星的半径。 14.3 宇宙学 14.3.1 快速回顾：恒星热内部的吸收光谱发射连续范围的波长 181

as light passes cooler outer layers of the star, photons of right energies can be absorbed
 the absorption depends on the elements present in the star's outer layers
 this gives rise to a characteristic *absorption spectrum* for the star^[95]
 the spectrum would consist of dark lines at specific wavelengths or frequencies^[96]

14.3.2 quick review: Doppler effect

but the spectral lines observed by us will be different from where they are supposed to be
 observed frequency / wavelength changes due to relative motion between star and observer
 this phenomena is known as the *Doppler effect*

- changes in spectral lines can tell whether a star / galaxy is *receding* or *approaching*
 - if wave source moves away, observed wavelength increases / frequency decreases
 - if wave source moves closer, observed wavelength decreases / frequency increases

- changes in spectral lines can also tell how fast the star is moving

you might recall observed frequency is given by: $f = f_0 \frac{c}{c \mp v}$

this formula would not be particularly useful in this part of the course

we will derive another formula for Doppler shift in the following section

- Doppler shift can only tell *radial* velocity of a star / galaxy

Doppler method cannot tell about motion in a direction that is normal to line of sight

14.3.3 redshift

data show that spectral lines of most galaxies shift in direction of increasing wavelength
 for visible spectrum, the lines shift towards red end, so this is called **redshift**

redshift of line spectrum means most galaxies are moving away

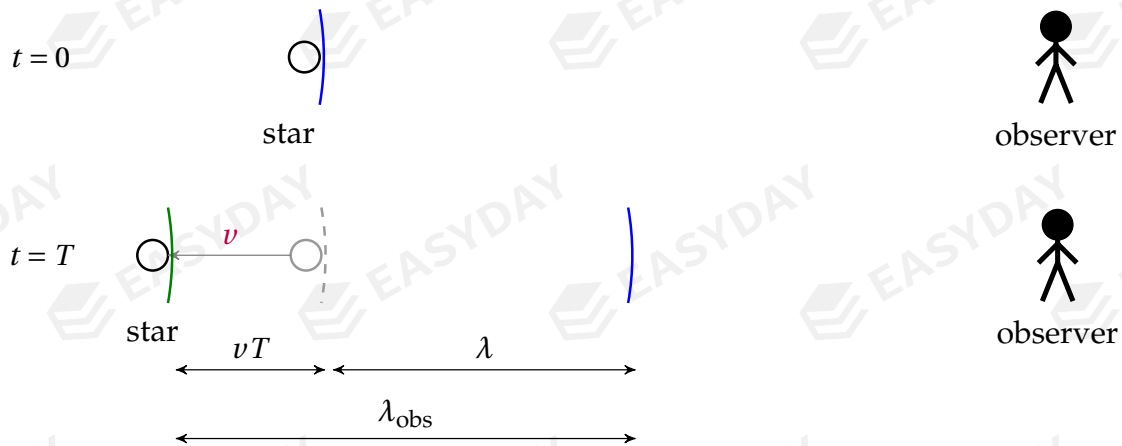
for a star / galaxy recedes at velocity v , it travels a distance of vT in one period

this causes an increase in observed wavelength: $\Delta\lambda = \lambda_{\text{obs}} - \lambda = vT$

^[95]The chemical composition of most stars is mainly hydrogen, plus a small fraction of helium, and trace amounts of heavier elements such as carbon, oxygen, etc. The presence of each element gives rise to a unique set of absorption lines.

^[96]More detailed discussions on the absorption spectrum has been given in §11.4.3.

Yuhao Yang 14 天体物理学当光穿过恒星较冷的外层时，可以吸收适当能量的光子，吸收取决于恒星外层中存在的元素，这会产生恒星的特征吸收光谱[95]该光谱将由特定波长或频率的暗线组成[96] 14.3.2快速回顾：多普勒效应，但我们观察到的谱线将与它们应该观察到的地方不同由于恒星和观察者之间的相对运动而导致频率/波长变化，这种现象被称为多普勒效应 $\hat{\alpha}$ 谱线的变化可以判断恒星/星系是否正在后退或接近 - 如果波源移开，观测到的波长增加/频率减小 - 如果波源移近，观测到的波长减小/频率增加 $\hat{\alpha}$ 谱线的变化还可以告诉恒星移动的速度，您可能记得观测到的频率由下式给出： $f = f_0 \frac{c}{c - v}$ 这个公式在这一部分中不是特别有用当然，我们将在下一节中导出多普勒频移的另一个公式 $\hat{\alpha}$ 多普勒频移只能告诉恒星/星系的径向速度多普勒方法不能告诉垂直于视线方向的运动 14.3.3 红移数据表明，大多数星系的谱线在可见光谱中向波长增加的方向移动，线向红端移动，所以这称为线谱红移意味着大多数星系对于恒星/星系正在远离以速度 v 后退，它在一个周期内行进 vT 的距离，这会导致观测到的波长增加： $\Delta\lambda = \lambda_{\text{obs}} - \lambda = vT$ [95]大多数恒星的化学成分主要是氢，加上一小部分氦，以及微量的较重元素，如碳、氧等。每种元素的存在都会产生一组独特的吸收线。 [96]第 11.4.3 节中给出了关于吸收光谱的更详细讨论。 182



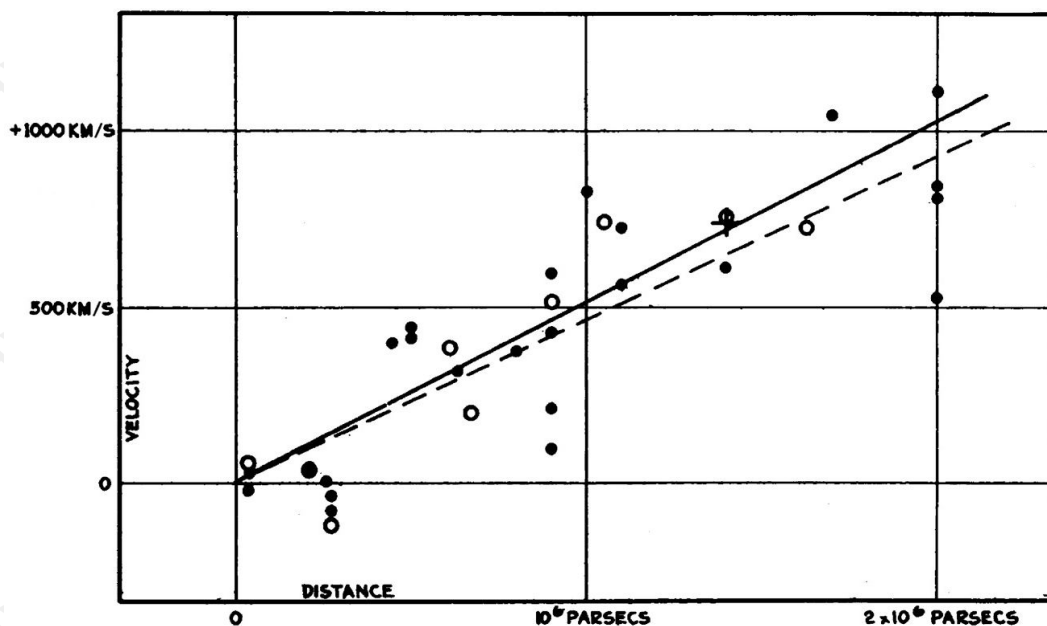
so fractional change in wavelength is: $\frac{\Delta\lambda}{\lambda} = \frac{vT}{\lambda} = \frac{vT}{cT} \Rightarrow \boxed{z \equiv \frac{\Delta\lambda}{\lambda} = \frac{v}{c}}$

the parameter z is usually called the *redshift factor*

14.3.4 recession of galaxies

Edwin Hubble discovered in 1929 that almost all galaxies are moving away

further analysis showed recession speed of a galaxy is roughly proportional to its distance



Edwin Hubble's plot of the velocity-distance relation among galaxies in his original paper

this is known as **Hubble's law**^[97]

$$\boxed{v = H_0 d}$$

Hubble constant: $H_0 \approx 2.4 \times 10^{-18} \text{ s}^{-1}$

^[97] Edwin Hubble made this conclusion based on his study of 46 galaxies. Later observations further supported the proportional relationship between the recession speed of galaxies and the distance, but the constant computed by Edwin Hubble himself was found to be greatly overestimated. Hubble's original paper can be seen here: <https://www.pnas.org/doi/full/10.1073/pnas.15.3.168>

杨宇豪 14 天体物理学恒星观测者 $t = 0$ v 恒星观测者 $\lambda v T \lambda_{\text{obs}} t = T$ 因此波长的分数变化为： $\Delta \lambda / \lambda = v T / \lambda = v T / c T = z$ $\Delta \lambda / \lambda = v / c$ 参数 z 通常称为红移因子 14.3.4 星系的退缩埃德温·哈勃在 1929 年发现，几乎所有星系正在远离进一步的分析表明，星系的退行速度大致与其距离成正比 Edwin Hubble 在其原始论文中绘制了星系之间的速度-距离关系图，这被称为哈勃定律 [97] $v = H_0 d$ 哈勃常数： $H_0 \approx 2.4 \times 10^{-18} \text{ s}^{-1}$ [97] Edwin Hubble 基于他对 46 个星系的研究得出了这一结论。后来的观测进一步支持了星系退行速度与距离之间的比例关系，但埃德温·哈勃自己计算的常数被发现被大大高估了。哈勃的原始论文可以在这里看到：<https://www.pnas.org/doi/full/10.1073/pnas.15.3.168> 183

➤ exact value of the Hubble constant is not accurately known

H_0 is believed to be between $1.5 \times 10^{-18} \text{ s}^{-1} \sim 3.5 \times 10^{-18} \text{ s}^{-1}$ (uncertainty around $\pm 40\%$!)

Example 14.7 The K-line of ionised calcium from the absorption spectrum of the galaxy NGC 4889 is measured to be 401.8 nm. The same K-line measured in a laboratory setting is known to be 393.3 nm. (a) What is the speed of the NGC 4889? (b) How long does it take for light to reach us from NGC 4889?

✏ recession speed: $v = \frac{\Delta\lambda}{\lambda} \times c = \frac{401.8 - 393.3}{393.3} \times 3.00 \times 10^8 \approx 6.48 \times 10^6 \text{ m s}^{-1}$

distance of galaxy: $d = \frac{v}{H_0} \approx \frac{6.48 \times 10^6}{2.4 \times 10^{-18}} \approx 2.7 \times 10^{24} \text{ m}$

time for light to propagate: $t = \frac{d}{c} = \frac{2.7 \times 10^{24}}{3.00 \times 10^8} \approx 9.0 \times 10^{15} \text{ s} \approx 2.9 \times 10^8 \text{ years}$

14.3.5 Big Bang theory

Hubble's law implies the universe is *expanding*

go back in time, all galaxies must be close together

this suggests the universe must begin from a highly dense state

it is now generally accepted that the universe is created from a dramatic explosion

this starting point of the universe and everything is known as the **Big Bang**^[98]

age of the universe

age of the universe can be deduced from Hubble's law

consider two galaxies started off close together during the Big Bang

assume galaxies have been moving away from one another at constant speed v for time t

distance between the two galaxies today would be $d = vt$

but this t is essentially all the time since beginning of the universe

so age of the universe: $t = \frac{d}{v} = \frac{d}{H_0 d} \Rightarrow t = \frac{1}{H_0}$

^[98]The Big Bang Theory was proposed in 1927 by Belgian astronomer and cosmologist *Georges Lemaitre*. From a theoretical view, Lemaitre deduced that the natural consequence of Einstein's theory of general relativity must be an expanding universe. In 1929, *Edwin Hubble*, who was unaware of Lemaitre's work, published his famous relation between distance and radial velocity among extra-galactic nebulae. Nevertheless, Hubble's discovery became a strong supporting evidence for the Big Bang theory.

Yuhao Yang 14 天体物理学 $\hat{a}$ 哈勃常数的准确值尚未准确得知 H_0 被认为在 $1.5 \times 10^{-18} \text{ s}^{-1} \gg 3.5 \times 10^{-18} \text{ s}^{-1}$ 之间（不确定性约为 40%！）

示例 14.7 从星系 NGC 4889 的吸收光谱中测得的肌钙 K 线为 401.8 nm。在实验室环境中测量的同一 K 线已知为 393.3 nm。

(a) NGC 4889 的速度是多少？ (b) 光从 NGC 4889 到达我们需要多长时间？ - 衰退速度： $v = \frac{\Delta \lambda}{\lambda} \times c = \frac{401.8 - 393.3}{393.3} \times 3.00 \times 10^8 \text{ m s}^{-1} = 6.48 \times 10^6 \text{ m s}^{-1}$

星系距离： $d = \frac{v}{H_0} = \frac{6.48 \times 10^6}{2.4 \times 10^{-18}} = 2.7 \times 10^{24} \text{ m}$

光时间传播： $t = \frac{d}{c} = \frac{2.7 \times 10^{24}}{3.00 \times 10^8} = 9.0 \times 10^{15} \text{ s} = 2.9 \times 10^8 \text{ 年}$

14.3.5 大爆炸理论哈勃定律意味着宇宙正在膨胀回到过去，所有星系必须靠近在一起这表明宇宙必须从高度致密的状态开始现在普遍认为宇宙是从戏剧性的状态创造出来的爆炸这个宇宙的起点和一切都被称为大爆炸[98]

宇宙年龄宇宙年龄可以从哈勃定律推导出来考虑在大爆炸期间紧密相连的两个星系假设星系在时间 t 内以恒定速度 v 相互远离今天两个星系之间的距离将是 $d = v t$ 但这个 t 本质上是自宇宙开始以来的所有时间，所以宇宙年龄： $t = \frac{d}{v} = \frac{d}{H_0 d} \Rightarrow t = \frac{1}{H_0}$ [98]

大爆炸理论由比利时天文学家和宇宙学家 Georges Lemaitre 于 1927 年提出。勒梅特从理论角度推断，爱因斯坦广义相对论的自然结果必定是宇宙膨胀。1929 年，埃德温·哈勃（Edwin Hubble）在不知道勒梅特的工作的情况下发表了他著名的河外星云距离与径向速度之间的关系。尽管如此，哈勃的发现成为大爆炸理论的有力支持证据。 184

substitute $H_0 \approx 2.4 \times 10^{-18} \text{ s}^{-1}$, one finds $t \approx 4.5 \times 10^{18} \text{ s}$, which is about 14 billion years (Gy)

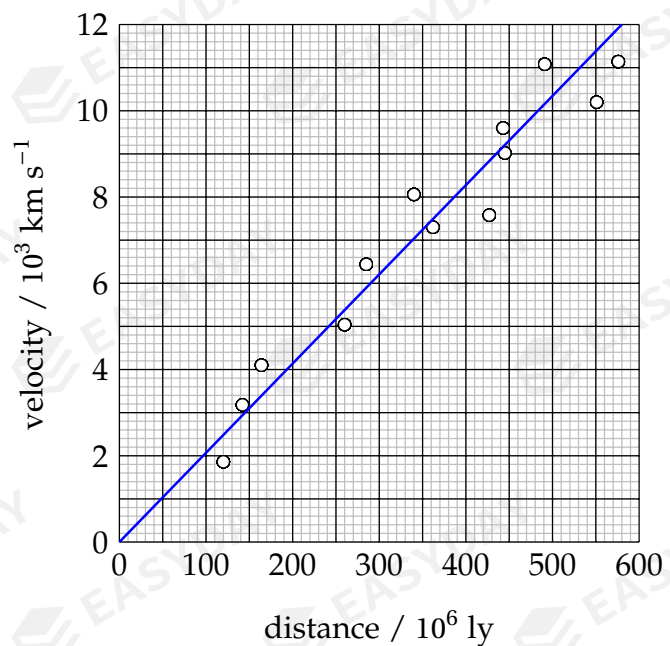
➤ there is great uncertainty in the universe's age (varying from 11 ~ 20 Gy)

two main reasons for this uncertainty are:

- our calculation is based on the assumption of constant recession speed
there are cosmological models that suggest otherwise, so different ages are predicted
- our calculation also depends on the value of H_0

but exact value of H_0 is in doubt due to large uncertainty in distance measurements

Question 14.5 The plot shows some data on the velocities and distances for a set of galaxies. A fit line has also been drawn. (a) Compute a value for the Hubble constant H_0 based on the data given. (b) Estimate the age of the universe using your calculated value for H_0 , and state any assumption that you make. (c) For a galaxy at a distance of $2 \times 10^9 \text{ ly}$, what is the fractional change in the wavelength of its spectrum?



cosmic microwave background radiation

another key evidence for the Big Bang model is the *cosmic microwave background*

after Big Bang, universe gradually cools down due to expansion

the universe today should be filled with radiation due to remnant heat from the Big Bang

theoretical calculation predicts that the universe should now have a temperature of 2.7 K

this suggests a background radiation peaked at around 1 mm (microwave band)

this radiation is therefore called **cosmic microwave background** (CMB)^[99]

^[99] Ralph Alpher, George Gamow and Robert Herman suggested the idea of CMB in theory in the 1940s. In 1965, Robert Dicke took up the problem and led a team of physicists at Princeton University to hunt for the CMB signal. In the same year, Arno Penzias and Robert Wilson at the Bell Laboratories built a large radio antenna for a communication satellite in New Jersey, while they failed in all attempts to get rid of the 'background noise' that came from all directions.

Yuhao Yang 14 ASTROPHYSICS 替代 $H_0 \approx 2.4 \times 10^{-18} \text{ s}^{-1}$ ，发现 $t \approx 4.5 \times 10^{18} \text{ s}$ ，大约是 140 亿年 (Gy) 宇宙年龄存在很大的不确定性 (从 11 到 20 Gy 变化) 这种不确定性的两个主要原因是：- 我们的计算是基于恒定衰退速度的假设，有宇宙学模型提出相反的建议，因此预测了不同的年龄 - 我们的计算也取决于 H_0 的值，但由于距离测量的很大不确定性， H_0 的确切值存在疑问

速度 / 10^3 km s^{-1} 距离 / 10^6 ly 0 2 4 6 8 10 12 0 100 200 300 400 500 600 问题 14.5 该图显示了有关一组星系的速度和距离。还绘制了一条拟合线。(a) 根据给定的数据计算哈勃常数 H_0 的值。(b) 使用计算出的 H_0 值估计宇宙的年龄，并说明您所做的任何假设。(c) 对于距离为 $2 \times 10^9 \text{ ly}$ 的星系，其光谱波长的分数变化是多少？

宇宙微波背景辐射大爆炸模型的另一个关键证据是大爆炸后的宇宙微波背景，宇宙由于膨胀而逐渐冷却，由于大爆炸的余热，今天的宇宙应该充满辐射。理论计算预测宇宙现在应该有 2.7 K 的温度，这表明背景辐射峰值在 1 毫米左右 (微波波段)，因此这种辐射被称为宇宙微波背景 (CMB) [99] [99] Ralph Alpher、George Gamow 和 Robert Herman 在 20 世纪 40 年代在理论上提出了 CMB 的想法。1965 年，罗伯特·迪克 (Robert Dicke) 着手解决这个问题，并带领普林斯顿大学的物理学家团队寻找宇宙微波背景信号。同年，贝尔实验室的阿诺·彭齐亚斯和罗伯特·威尔逊为新泽西州的一颗通信卫星建造了一个大型无线电天线，尽管他们尝试消除来自四面八方的“背景噪音”，但都失败了。 185

CMB radiation have been observed by space telescopes from all directions of cosmos^[100]

CMB corresponds to a temperature of around 3 K, which is pretty close to the prediction

Question 14.6 Use Wien's law to show that the peak intensity of the electromagnetic radiation given off by a black body at a temperature of 3 K is around 1 mm.

14.3.6 fate of the universe (*)

whether the universe will expand forever depends on strength of gravity

fate of universe narrows down to how density ρ of universe compares to critical density ρ_c

- if $\rho > \rho_c$, force of gravity dominates as there is sufficient matter

expansion will slow down, then contract back inwards

this leads to a *closed* universe, eventually ending in a *Big Crunch*

- if $\rho < \rho_c$, there is not enough gravity to stop expansion

expansion continues forever, leading to an *open* universe

but the big problem is, it is difficult to determine average density of our universe

➤ issue of *dark matter*

mass of galaxies suggested by luminosity calculation cannot provide the centripetal force needed to keep galaxies rotating

galaxies must contain mass that does not give off light, known as **dark matter**

➤ issue of *dark energy*

observation shows expansion of universe is not slowing down

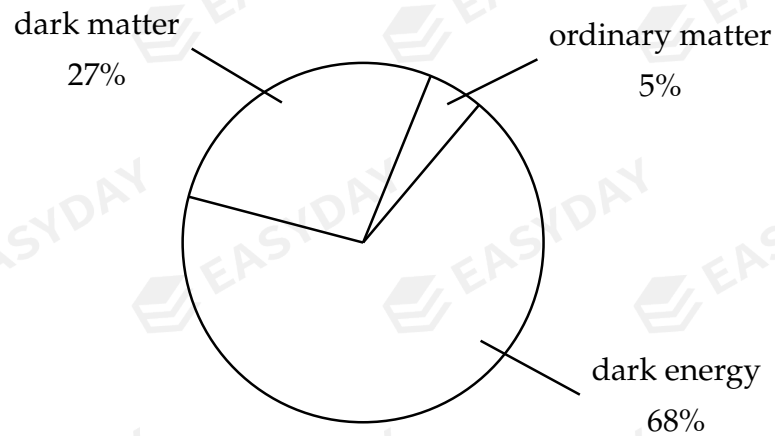
it is suggested that cosmic acceleration is caused by **dark energy**

When Princeton's team heard about the Bell Lab's result, they realized at once that the CMB had already been found. As a result, two important articles were published: one by Penzias and Wilson about the 'noise', the other by Dicke's team explaining its true nature. Penzias and Wilson did not interpret what they had found in their article, nor did they have any intention to look for CMB in the first place. Nevertheless, they received the Nobel Prize in Physics in 1978, for their accidental discovery of CMB. What about Dicke and his team? Quoting *Bill Bryson's A Short History of Nearly Everything*, the Princeton researchers got only sympathy.

^[100] Three major CMB missions include: (1) the Cosmic Background Explorer (COBE), launched by NASA in 1989, (2) the Wilkinson Microwave Anisotropy Probe (WMAP), launched by NASA in 2001 to study fluctuations in CMB in greater detail, (3) the Planck satellite, launched by ESA in 2009 to study CMB fluctuations with even greater sensitivity. These studies would allow scientists to trace back to what happened immediately after the Big Bang, and the understanding of the early universe is crucial for the

杨宇豪 14 天体物理学宇宙各个方向的太空望远镜都观测到了 CMB 辐射 [100] CMB 对应的温度约为 3 K，与问题 14.6 的预测非常接近。利用维恩定律可知，黑体在 3 K 温度下发出的电磁辐射峰值强度约为 1 mm。

14.3.6 宇宙的命运 (Q) 宇宙是否会永远膨胀取决于引力的强度宇宙的的命运缩小到宇宙的密度 ρ 与临界密度 ρ_c 相比如何 – 如果 $\rho \leq \rho_c$ ，则重力占主导地位，因为有足够的物质膨胀将减慢，然后向内收缩，这导致宇宙封闭，最终以大挤压结束 – 如果 $\rho > \rho_c$ ，则没有足够的引力来停止膨胀膨胀永远持续下去，导致宇宙开放，但最大的问题是，很难确定我们宇宙的平均密度——光度计算得出的暗物质质量问题无法提供保持星系旋转所需的向心力，星系必须包含不发光的质量，称为暗物质——暗能量观测问题表明宇宙膨胀并没有减慢，这表明宇宙加速是由暗能量引起的当普林斯顿的团队听到听到贝尔实验室的结果，他们立刻意识到宇宙微波背景已经被发现了。结果，发表了两篇重要文章：一篇是彭齐亚斯和威尔逊关于“噪音”的文章，另一篇是迪克团队解释其真实本质的文章。彭齐亚斯和威尔逊没有解释他们在文章中发现的内容，也没有任何意图首先寻找 CMB。尽管如此，他们还是因意外发现了宇宙微波背景而获得了 1978 年诺贝尔物理学奖。迪克和他的团队呢？普林斯顿大学的研究人员引用比尔·布莱森 (Bill Bryson) 的《几乎一切的简史》，只得到了同情。[100] 三个主要的 CMB 任务包括：(1) 宇宙背景探测器 (COBE)，由 NASA 于 1989 年发射；(2) 威尔金森微波各向异性探测器 (WMAP)，由 NASA 于 2001 年发射，用于更详细地研究 CMB 波动；(3) 普朗克卫星，由 ESA 于 2009 年发射，用于以更高的灵敏度研究 CMB 波动。这些研究将使科学家能够追溯到大爆炸后立即发生的事情，而了解早期宇宙对于 186



- it turns out ordinary matter only contributes to a very small part of the universe
great majority of the total mass-energy content of cosmos is dark matter and dark energy
but nature of dark matter and dark energy are not yet understood

understanding of the large-scale cosmic structures that we see today.

杨宇豪 14 天体物理学普通物质 5% 暗能量 68% 暗物质 27% – 事实证明，普通物质只占宇宙的一小部分，宇宙总质能含量的大部分是暗物质和暗能量，但暗物质和暗能量的本质尚未被了解，对我们今天看到的大尺度宇宙结构的理解。 187

CHAPTER 15

Electronics (*)

electronic sensors basically consist of three parts

change in some physical property (temperature, pressure, etc.) is picked up by a **sensor**

electrical signal is then fed to **processing units**

processor produces an output voltage to drive **output devices** to indicate this change

15.1 sensors

15.1.1 thermistor

thermistor is an electrical component with a resistance that varies with temperature

thermistors usually have a *negative temperature coefficient*, or NTC in short, which means their resistance decreases as temperature rises

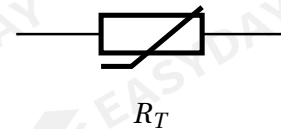
in A-Levels, we consider NTC thermistors only

➤ resistance-temperature relation is *non-linear*

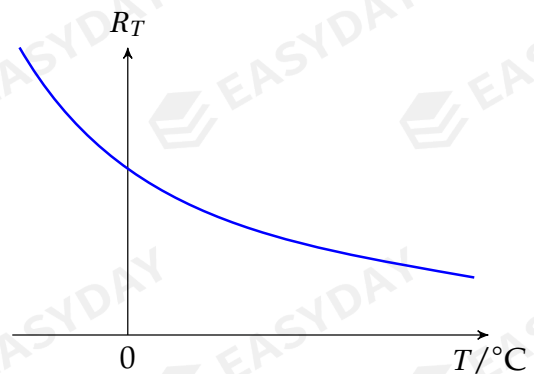
if a graph is sketched, the curve looks like an exponential decay

➤ behaviour of NTC thermistors can be explained in terms of *band theory* (see §11.5.2)

at low temperature, electrons are bound to atoms and cannot move freely; at high temperature, electrons gain energy and break free from atoms, so more free electrons available to conduct electricity



electric symbol for thermistor



考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

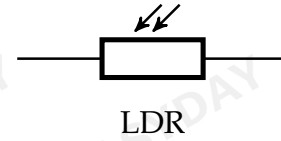
Yuhao Yang 15 ELECTRONICS (Q) CHAPTER 15 Electronics (Q) 电子传感器基本上由三部分组成一些物理属性的变化 (温度、压力等) 由传感器拾取电信号然后馈送到处理单元处理器产生输出电压来驱动输出设备来指示这种变化 15.1 传感器 15.1.1 热敏电阻 RT 热敏电阻的电气符号 $T/\pm C$ RT 热敏电阻是一种具有重新定义的电气元件。电阻随温度变化的热敏电阻通常具有负温度系数, 简称 NTC, 这意味着它们的电阻随着 A 级温度升高而降低, 我们仅考虑 NTC 热敏电阻 - 如果绘制图表, 电阻-温度关系是非线性的, 曲线看起来像指数衰减 - NTC 热敏电阻的行为可以用能带理论 (参见第 11.5.2 节) 在低温下解释, 电子与原子结合并且不能自由行动; 在高温下, 电子获得能量并脱离原子, 因此有更多的自由电子可用于导电 188

15.1.2 LDR

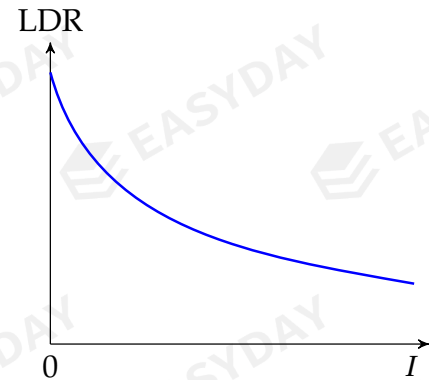
light-dependent resistor, or **LDR**, is an electrical component with a resistance that can change with intensity of light

resistance of LDR decreases as light level increases

- resistance of LDR varies *non-linearly* with intensity
- behaviour of LDR is also explained with *band theory*
bound electrons acquire energy from radiation and become conducting, greater light intensity means more free electrons available
- in dark, typical resistance of LDR $\approx 10^5 \sim 10^7 \Omega$
in bright, typical resistance of LDR $\approx 10^2 \sim 10^3 \Omega$



electric symbol for LDR



15.1.3 strain gauge

to measure width of a narrow crack, a device called **strain gauge** is widely used in industry

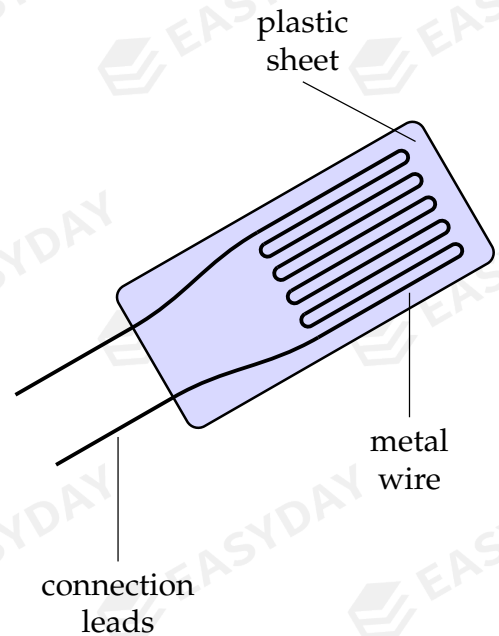
a metal-wire strain gauge consists of a folded long wire sealed on a thin plastic box

as crack widens, strain gauge gets stretched, length of wire increases, causing a change in its resistance

resistance of a metal wire is given by: $R = \frac{\rho l}{A}$

increase in resistance is proportional to increase in length of wire (for small increments)

- if change in cross-sectional area of wire is negligible
increase in resistance: $\Delta R = \frac{\rho \Delta l}{A} \Rightarrow \frac{\Delta R}{R} = \frac{\Delta l}{l}$
- if cross-sectional area of wire changes with its length
total volume of wire $V = Al$ should remain constant



structure of a strain gauge

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇浩 15 电子 (Q) 15.1.2 LDR LDR LDR 的电气符号 10 LDR 光敏电阻器，或称 LDR，是一种阻值随光强而变化的电子元件，LDR 的阻值随着光强的增加而减小，LDR 的阻值随光强呈非线性变化，LDR 的行为也可以用能带理论来解释。束缚电子从辐射中获取能量并变得导电，光强越大，可用的自由电子越多。在黑暗中，典型电阻为 $LDR \sim 10^5 \sim 10^7 \Omega$ 在明亮中，典型电阻为 $LDR \sim 10^2 \sim 10^3 \Omega$ 15.1.3 应变片连接引线应变片的金属线塑料片结构用于测量狭窄裂缝的宽度，一种称为应变片的装置在工业中广泛使用。金属线应变片由折叠的长线组成，密封在薄塑料盒上，随着裂缝变宽，应变片被拉伸，线的长度增加，导致金属线电阻的变化由下式给出： $R = \rho l / A$ 电阻的增加与线长度的增加成正比（对于小增量）如果线的横截面积的变化可以忽略不计电阻的增加： $\Delta R = \rho \Delta l / A$ $\Delta R / R = \Delta l / l$ 如果线的横截面积随其长度变化 $V = Al$ 线的总体积应保持恒定 189

it can be shown that ^[101] the percentage increase in resistance is:

$$\frac{\Delta R}{R} = \frac{2\Delta l}{l}$$

notice the differences, change in length causes increase in resistance, reduction in cross section makes the same contribution, hence an extra factor of two.

15.1.4 piezo-electric transducer

piezo-electric transducer transforms mechanical energy into electrical energy, or vice versa when a piezo-electric material ^[102] is compressed, a p.d. is produced across it conversely, p.d applied across piezo-electric material can cause change in shape since its discovery, piezo-electricity was exploited in many useful applications

➤ piezo-electric transducer can be used to generate sound waves

apply an a.c. voltage, material is forced to vibrate at same frequency as a.c.

this causes compression and rarefaction of air, producing sound waves

so piezo-electric transducer is the key component in *loudspeakers* and *buzzers*

➤ piezo-electric transducer can also be used to detect sound waves

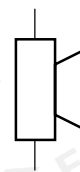
exposed to sounds, variation in air pressure produces an a.c. voltage across the device

this signal can be measured and processed to represent original sound wave

so piezo-electric transducer is also found in *microphones*



symbol for a microphone



symbol for a loudspeaker

➤ mechanism of piezo-electricity is explained on molecular level

piezo-electric materials consist of *polarised* molecules, also called *dipoles*

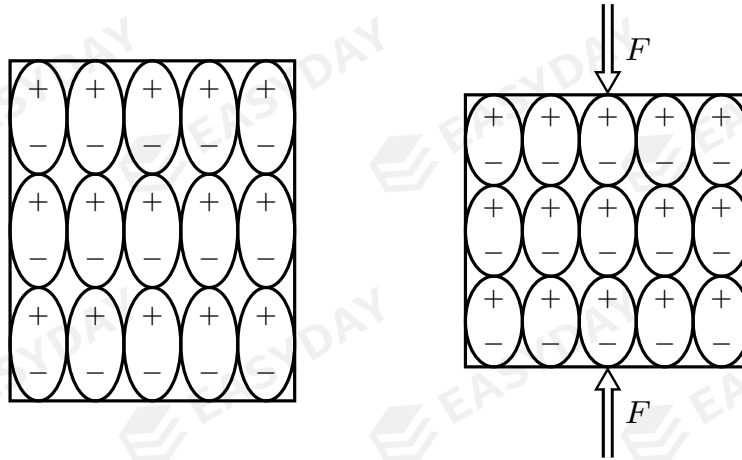
^[101] To keep $V = Al$ constant, for small change in l , one has $\frac{\Delta A}{A} \approx \frac{\Delta l}{l}$. Resistance of the stretched wire is: $R + \Delta R = \frac{\rho(l + \Delta l)}{A - \Delta A} = \frac{\rho l}{A} \left(1 + \frac{\Delta l}{l}\right) \left(1 - \frac{\Delta A}{A}\right)^{-1} \approx R \left(1 + \frac{\Delta l}{l}\right) \left(1 + \frac{\Delta l}{l}\right)^{-1}$. For $x \ll 1$, we can use binomial expansion $(1 + x)^n = 1 + nx + \dots$. Drop higher order terms, we find to the first-order approximation that: $R + \Delta R \approx R \left(1 + \frac{\Delta l}{l}\right) \left(1 + \frac{\Delta l}{l}\right) \approx R \left(1 + \frac{2\Delta l}{l}\right)$, so $\Delta R = 2R \frac{\Delta l}{l}$, it then follows that $\frac{\Delta R}{R} = \frac{2\Delta l}{l}$.

^[102] Materials that exhibit piezo-electricity include quartz, certain ceramics, various proteins, etc.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 可以证明 [101] 电阻增加的百分比为 : $\Delta R/R = 2\Delta l/l$ 注意差异, 长度的变化会导致电阻的增加, 横截面的减小会产生相同的贡献, 因此需要额外的因子 2。15.1.4 压电换能器当压电材料[102]被压缩时, 压电换能器将机械能转换为电能, 反之亦然。相反, 施加在压电材料上的 p.d 可以引起形状的变化, 自从它被发现以来, 压电被用于许多有用的应用中——压电换能器可用于产生声波, 应用交流电。电压, 材料被迫以与交流电相同的频率振动。这会导致空气压缩和稀疏, 产生声波, 因此压电换能器是扬声器和蜂鸣器的关键部件。压电换能器还可用于检测暴露在声音中的声波, 气压的变化会产生交流电。设备上的电压可以测量和处理该信号以表示原始声波, 因此压电换能器也可以在麦克风符号中找到, 用于扬声器的麦克风符号。压电机在分子水平上进行解释压电材料由极化分子组成, 也称为偶极子[101]为了保持 $V = Al$ 恒定, 对于 l 的微小变化, 有 $\Delta A/A = -1/4 \Delta l/l$ 。拉伸导线的电阻为 : $R + \Delta R = \rho(l + \Delta l)/(A - \Delta A) = \rho l A (1 + \Delta l/l) (1 - \Delta A/A)^{-1} = R (1 + \Delta l/l) (1 - \Delta l/l)^{-1}$ 。对于 $x \ll 1$, 我们可以使用二项展开式 $(1 + x)^n = 1 + nx + \dots$ 去掉高阶项, 我们发现一阶近似为 : $R + \Delta R = R (1 + \Delta l/l) (1 + \Delta l/l) = R (1 + 2\Delta l/l)$, 因此 $\Delta R = 2R \Delta l/l$, 则 $\Delta R/R = 2\Delta l/l$ 。[102]表现出压电性的材料包括石英、某些陶瓷、各种蛋白质等。190

when being compressed, dipoles rearrange themselves, causing variation of charge density, so small voltage is generated (see illustration)



15.2 processors

15.2.1 potential divider

sensors such as thermistor and LDR only produce changes in resistance, but output devices are driven by voltages → **potential divider** circuit can be used

a potential divider consists of two resistors in series connected to a fixed voltage supply

voltage is shared between the resistors: $V_0 = V_1 + V_2$

p.d. across each resistor: $V_1 = IR_1$, $V_2 = IR_2$

but same current I flows through R_1 and R_2 , so $\frac{V_1}{V_2} = \frac{R_1}{R_2}$

this shows voltage is shared in proportion to resistance

for each resistor: $V_1 = \frac{R_1}{R_1 + R_2} V_0$, $V_2 = \frac{R_2}{R_1 + R_2} V_0$

➤ if R_1 increases, while R_2 is fixed, then $V_1 \uparrow$, $V_2 \downarrow$

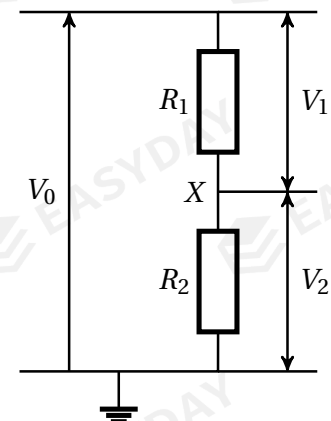
if R_1 decreases, while R_2 is fixed, then $V_1 \downarrow$, $V_2 \uparrow$

➤ sometimes we need specify *electric potential* at a given point

potential at any point is measured with respect to a reference point

this point is called **earth** or **ground**, with a potential defined to be zero

for example, potential at X is essentially equal to voltage V_2

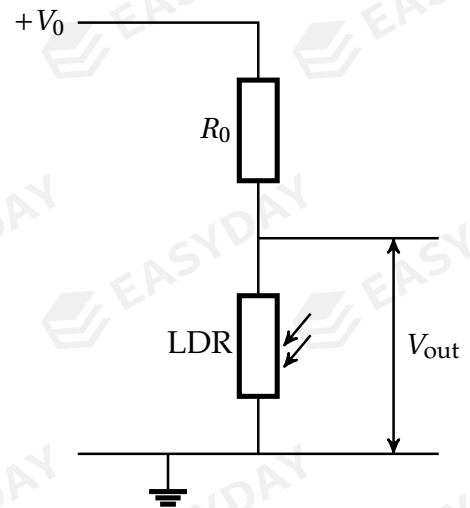


考纲提示：Chapter 15 Electronics：旧版拓展内容；2022 起不再属于 9702 A Level，当前 Paper 4 不考。

[illegible]

Example 15.1 For the sensing circuit shown, suggest how the output voltage changes as light level increases.

intensity of light increases, resistance of LDR decreases
 current in circuit increases, p.d. across R_0 increases
 so p.d. across LDR becomes smaller, V_{out} would decrease
 from $V_{\text{out}} = \frac{\text{LDR}}{\text{LDR} + R_0} V_0$, one can also see V_{out} decreases
 when resistance of LDR becomes lower
 so V_{out} decreases as light intensity increases $\square$



Question 15.1 In Example 15.1, if the fixed resistor has $R_0 = 8.0 \text{ k}\Omega$, and the LDR is in an environment such that it has a resistance of $12 \text{ k}\Omega$. The voltage supply $V_0 = +15 \text{ V}$. Find V_{out} .

Question 15.2 Design a sensing circuit that produces a greater output voltage as the temperature of the sensor rises. Explain the operation of your circuit.

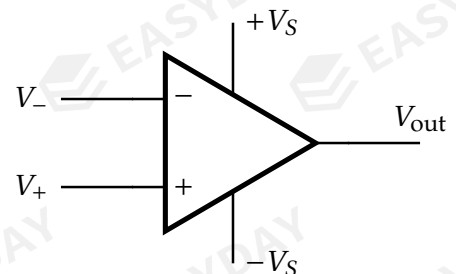
15.2.2 op-amp: basics

to amplify small input into large output signals, we need **operational amplifiers (op-amp)**^[103]

op-amp is building block for many electronic devices

➤ an op-amp has several terminals:

- V_+ : non-inverting input
- V_- : inverting input
- $\pm V_S$: positive/negative power supply voltages
- V_{out} : output voltage



circuit symbol for op-amp

➤ output voltage of op-amp is given by: $V_{\text{out}} = G_0(V_+ - V_-)$

G_0 is value of amplification of an open-loop^[104] amplifier, called **open-loop gain**

➤ value of G_0 is not well controlled in manufacturing processes

G_0 may also depend on temperature of op-amp, frequency of input signals, etc.

^[103] We are not concerned with what is inside an operational amplifier. An op-amp contains several *transistors*, and you are not required to know how they work in the A-Level syllabus. You only need to know what output voltage an op-amp would produce to drive output devices. If you want to learn more details on the functions of op-amps, you need to study a higher level course on *Electronics*.

^[104] It is called an open loop in the sense that there is no loop linking V_{out} to the input of the op-amp.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 电子 (Q) +V0 R0 LDR Vout 示例 15.1 对于所示的传感电路, 建议输出电压如何随着光照强度的增加而变化。 - 光强度增加, LDR 电阻减小, 电路中的电流增加, p.d. R0 增加, 因此 p.d. LDR 两端的电压变小, Vout 会从 $V_{out} = \frac{R_0}{R_0 + R_{LDR}} V_0$ 减小, 我们还可以看到, 当 LDR 的电阻变低时, Vout 也会减小, 因此 Vout 随着光强度的增加而减小。 ã 问题 15.1 在例 15.1 中, 如果固定电阻器的 $R_0 = 8.0 \text{ k}\Omega$, 并且 LDR 处于电阻为 $12 \text{ k}\Omega$ 的环境中。电源电压 $V_0 = +15 \text{ V}$ 。求出 V_{out} 。问题 15.2 设计一个传感电路, 当传感器温度升高时, 该电路会产生更大的输出电压。解释一下你的电路的工作原理。 1 5.2.2 运算放大器: 将小输入放大为大输出信号的基础知识, 我们需要运算放大器 (op-amp) [103] - +V+ V - Vout -VS +VS 运算放大器的电路符号运算放大器是许多电子设备的构建模块 ã 运算放大器有多个端子: - V + : 同相输入 - V- : 反相输入 - \$VS : 正/负电源电压 - Vout : 输出电压运算放大器的输出电压由下式给出: $V_{out} = G_0(V_+ - V_-)$ G_0 是开环 [104] 放大器的放大值, 称为开环增益 ã G_0 的值在制造过程中不能很好控制 G_0 还可能取决于运算放大器的温度、输入信号的频率等。 [103]我们不关心运算放大器内部的内容。运算放大器包含多个晶体管, A-Level 课程大纲中并不要求您了解它们的工作原理。您只需要知道运算放大器会产生什么输出电压来驱动输出设备。如果您想了解有关运算放大器功能的更多详细信息, 您需要学习更高级别的电子学课程。 [104]它被称为开环, 因为没有环路将 Vout 连接到运算放大器的输入。 192

➤ output voltage of op-amp cannot exceed supply voltages

when V_{out} calculated is greater than V_S , the op-amp becomes **saturated**, true V_{out} of op-amp is close to $+V_S$ or $-V_S$ depending on polarity of V_{out}

➤ all voltages in amplifier circuits are measured with respect to a reference level set at *earth*

properties of ideal op-amps

an ideal op-amp is considered to have the following properties

- *infinite open-loop gain* ($G_0 \rightarrow \infty$)

want large V_{out} for small V_{in} , value of amplification should be as large as possible^[105]

- *infinite input resistance/impedance*^[106]

want V_{in} to op-amp to be as large as possible

so large resistance between input terminals of op-amp^[107]

a consequence is that (almost) no current flows into or leaves the input terminals

- *zero output resistance/impedance*

output impedance of op-amp acts like internal resistance of batteries

want largest possible V_{out} to load, so small resistance for output terminal^[108]

- *infinite bandwidth*

bandwidth refers to range of frequencies of input voltages that are amplified by same amount

want same gain for all input frequencies, so infinite bandwidth is desired

- *infinite slew rate*: V_{out} changes immediately with V_{in} , response is instantaneous

- *zero noise*: V_{out} does not change with disturbance due to external conditions

^[105] Open-loop gain of an actual op-amp: $G_0 \approx 10^5 \sim 10^7$.

^[106] Impedance is an extension of the concept of resistance to a.c. circuits. You might just think of impedance of an electrical component as its ability to oppose currents.

^[107] Typical input impedance of an actual op-amp: $10^6 \Omega$ or higher.

^[108] Typical output impedance of an actual op-amp: $10^1 \sim 10^2 \Omega$.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) â 运算放大器的输出电压不能超过电源电压，当计算的 V_{out} 大于 V_S 时，运算放大器会饱和，运算放大器的真实 V_{out} 接近 $+V_S$ 或 $-V_S$ ，具体取决于 V_{out} 的极性 â 放大器电路中的所有电压均相对于理想运算放大器的接地特性设置的参考电平进行测量。理想的运算放大器被认为具有以下特性 ± 无穷大开环增益 ($G \rightarrow \infty$) 需要大 V_{out} ，小 V_{in} ，放大值应尽可能大 [105] ± 无限输入电阻/阻抗 [106] 希望运算放大器的 V_{in} 尽可能大，因此运算放大器输入端子之间的电阻较大 [107] 结果是（几乎）没有电流流入或离开输入端子 ± 零输出电阻/阻抗运算放大器的输出阻抗就像内阻的电池需要尽可能大的 V_{out} 负载，因此输出端的电阻要小 [108] ± 无限带宽带宽是指放大相同量的输入电压的频率范围，所有输入频率都需要相同的增益，因此需要无限带宽 ± 无限压摆率： V_{out} 随 V_{in} 立即变化，响应是瞬时的 ± 零噪声： V_{out} 不随外部条件的干扰而变化 [105] 实际运算放大器的开环增益： $G \approx 10^5 \gg 10^7$ 。 [106] 阻抗是交流电阻概念的延伸。电路。您可能只是将电气元件的阻抗视为其抵抗电流的能力。 [107] 实际运算放大器的典型输入阻抗： $10^6 \Omega$ 或更高。 [108] 实际运算放大器的典型输出阻抗： $10^1 \gg 10^2 \Omega$ 。 193

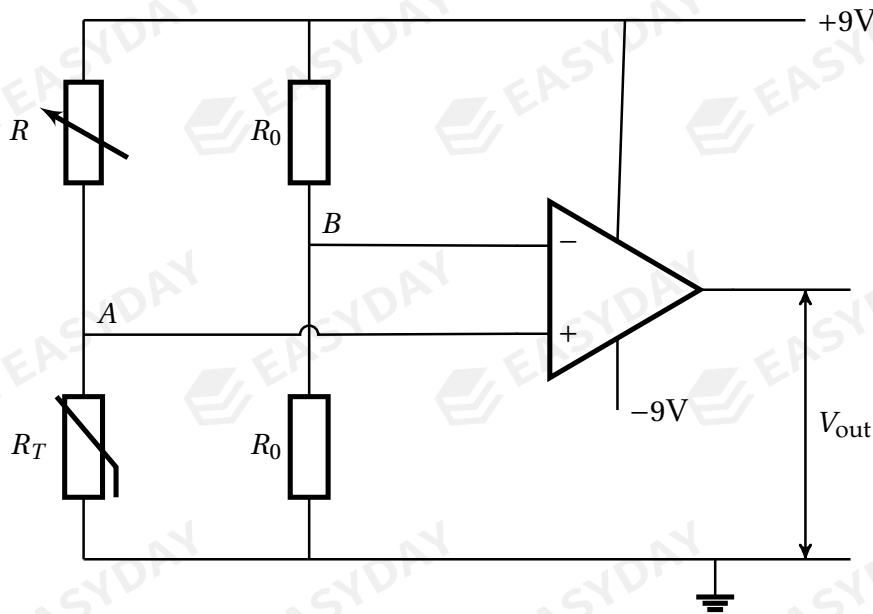
15.2.3 voltage comparator

for an open-loop op-amp, since gain G_0 is very large, so as long as there is a small difference in V_+ and V_- , $V_{\text{out}} = G_0(V_+ - V_-) \rightarrow \pm\infty$, this causes V_{out} to saturate

so $V_{\text{out}} = \pm V_S$ depending on which one of V_+ and V_- is larger, this makes a comparator

a **voltage comparator** compares two input voltages V_+ and V_- , and produces a V_{out} to indicate which one is greater: if $V_+ > V_-$, then $V_{\text{out}} = +V_S$; if $V_+ < V_-$, then $V_{\text{out}} = -V_S$

Example 15.2 A circuit that incorporates an ideal op-amp is shown. Given that $R_0 = 10 \text{ k}\Omega$, variable resistor is set to $R = 20 \text{ k}\Omega$, and the thermistor has a resistance of $R_T = 40 \text{ k}\Omega$ at 20°C , and $R_T = 10 \text{ k}\Omega$ at 50°C . Describe the change in V_{out} as temperature rises from 20°C to 50°C .



🔗 this is a comparator whose input voltages are determined from two potential divider circuits

resistor R_0 sets a constant potential for reference: $V_- = V_B = \frac{10}{10+10} \times (+9) = +4.5 \text{ V}$

p.d. across R_T determines V_A , which is fed into V_+ to be compared with V_-

at 20°C : $V_+ = V_A = \frac{40}{40+20} \times (+9) = +6.0 \text{ V}$, so $V_+ > V_- \Rightarrow V_{\text{out}} = +9 \text{ V}$

at 50°C , $V_+ = V_A = \frac{10}{10+20} \times (+9) = +3.0 \text{ V}$, so $V_+ < V_- \Rightarrow V_{\text{out}} = -9 \text{ V}$

as temperature goes from 20°C to 50°C , V_{out} initially maintains at $+9 \text{ V}$

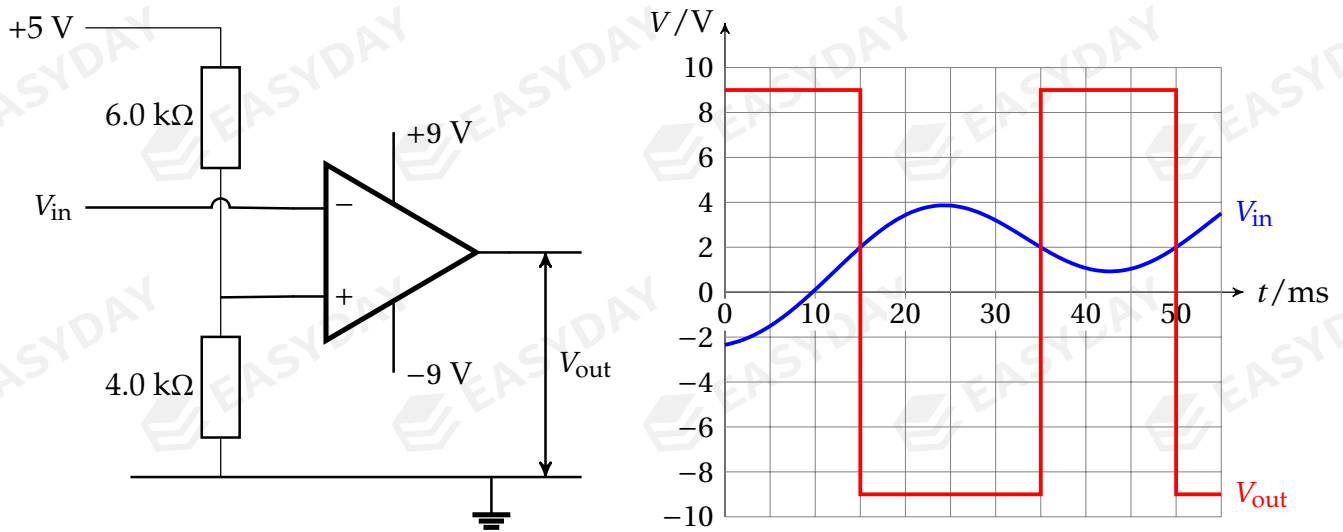
it switches over to -9 V at some critical temperature then stays at -9 V until 50°C

□

考纲提示 · Chapter 15 Electronics: 旧版拓展内容; 2022 起不再属于 9702 A Level, 当前 Paper 4 不考。

余浩阳 15 电子(四) 15.2.3 电压比较器对于开环运放, 由于增益 G_0 很大, 所以只要 V_+ 和 V_- 有微小的差别, $V_{out} = G_0(V_+ - V_-)$! §1, 这会导致 V_{out} 饱和, 因此 $V_{out} = \pm V_S$ 取决于 V_+ 和 V_- 中哪一个较大, 这使得比较器成为电压比较器, 比较两个输入电压 V_+ 和 V_- , 并产生一个 V_{out} 来指示哪一个更大: 如果 $V_+ > V_-$, 则 $V_{out} = +V_S$; 如果 $V_+ < V_-$, 则 $V_{out} = -V_S$; 如果 $V_+ = V_-$, 则 $V_{out} = 0$; if $V_+ < V_-$, 则 $V_{out} = -V_S$ 例 15.2 显示了包含理想运算放大器的电路。假定 $R_0 \rightarrow 10\text{ k}\Omega$, 可变电阻器设置为 $R \rightarrow 20\text{ k}\Omega$, 热敏电阻在 20°C 时的电阻为 $R_T \rightarrow 40\text{ k}\Omega$, 在 50°C 时为 $R_T \rightarrow 10\text{ k}\Omega$ 。描述当温度从 20°C 升至 50°C 时 V_{out} 的变化。 +9V R_T R R_0 R_0 - +A B -9V V_{out} - 这是一个比较器, 其输入电压由两个分压器电路确定, 电阻器 R_0 设置恒定电势作为参考: $V_- = V_B = 10/10 + 10 \times (+9) = +4.5\text{ V p.d.}$ R_T 确定 V_A , 将其输入 V_+ 与 20°C 时的 V_- 进行比较: $V_+ = V_A = 40/40 + 20 \times (+9) = +6.0\text{ V}$, 因此 $V_+ > V_-$) $V_{out} = +9\text{ V}$ at 50°C , $V_+ = V_A = 10/10 + 20 \times (+9) = +3.0\text{ V}$, 因此 $V_+ < V_-$) $V_{out} = -9\text{ V}$ 当温度从 20°C 升至 50°C 时, V_{out} 最初保持在 $+9\text{ V}$, 在某个临界温度下切换到 -9 V , 然后保持在 -9 V , 直到 50°C 194

Example 15.3 The diagram shows a circuit incorporating an ideal op-amp. The variation with time for input voltage V_{in} is shown. Sketch the variation of the output voltage.



potential at non-inverting input: $V_+ = \frac{4.0}{4.0 + 6.0} \times (+5) = +2 \text{ V}$

potential at inverting input is equal to input voltage: $V_- = V_{in}$

for V_{in} greater than +2 V, $V_- > V_+$, then $V_{out} = -9 \text{ V}$

for V_{in} less than +2 V, $V_- < V_+$, then $V_{out} = +9 \text{ V}$

variation of V_{out} can then be plotted, which shows the form of a square wave $\square$

15.2.4 negative feedback

in practical uses, we may desire V_{out} to reflect small changes in V_{in} , instead of stuck at $\pm V_S$

there are a few problems with open-loop amplifiers that we need to deal with:

- gain of amplifier is too high

V_{out} almost always saturates, so V_{out} cannot change continuously with V_{in}

- open-loop gain G_0 of practical op-amp is affected by temperature

different V_{out} for the same V_{in} at different temperatures is not convenient

- open-loop gain G_0 is a function of input frequency

different input frequencies are not always amplified by same amount

this would cause distortion of original signal, bandwidth would be limited

one way to improve properties of amplifier circuits is to introduce negative feedback

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 示例 15.3 该图显示了包含理想运算放大器的电路。显示了输入电压 V_{in} 随时间的变化。画出输出电压的变化。

$- + V_{in} -9\text{ V } +9\text{ V } V_{out} 4.0\text{ k}\Omega \ 6.0\text{ k}\Omega \ +5\text{ V } V/V \ t/\text{ms}$
 $0 \ 10 \ 20 \ 30 \ 40 \ 50 \ -10 \ -8 \ -6 \ -4 \ -2 \ 0 \ 2 \ 4 \ 6 \ 8 \ 10$

$V_{in} \ V_{out}$ - 同相输入端电位 : $V_{+} = 4.0 \ 4.0 + 6.0 \times (+5)$ 反相输入端的 $= +2\text{ V}$ 电势等于输入电压 : $V_{-} = V_{in}$ (V_{in} 大于 $+2\text{ V}$ 时, $V_{-} = V_{+}$) , 然后 $V_{out} = -9\text{ V}$ (V_{in} 小于 $+2\text{ V}$ 时, $V_{-} = V_{+}$) , 然后 $V_{out} = +9\text{ V}$ V_{out} 的变化可以绘制出来, 这显示了方波的形式

15.2 .4 实际使用中的负反馈, 我们可能希望 V_{out} 反映 V_{in} 的微小变化, 而不是停留在 $\pm V_S$ 上, 开环放大器有一些问题需要我们处理 :

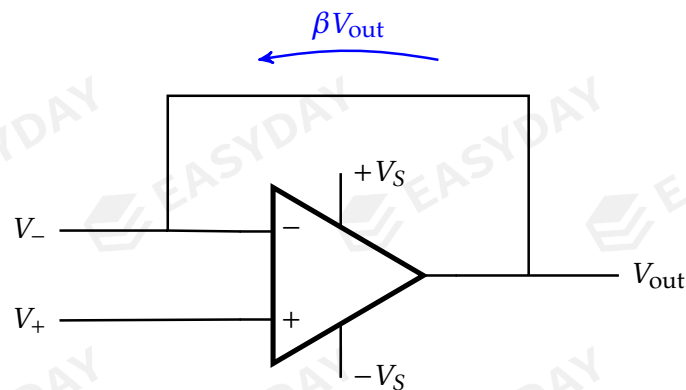
- 放大器增益太高 V_{out} 几乎总是饱和, 因此 V_{out} 不能随 V_{in} 连续变化
- 实际运算放大器的开环增益 G_0 受温度影响不同温度下相同 V_{in} 的不同 V_{out} 并不方便
- 开环增益 G_0 是输入频率的函数不同的输入频率并不总是被放大同样的量, 这会导致原始信号失真, 带宽将受到限制改善放大器电路性能的一种方法是引入负反馈

195

feedback means sending a portion of output voltage back to the input terminals

a feedback network is also called a *closed loop*

if a fraction of output V_{out} is sent back to inverting input V_- such that it opposes changes of input, this is called **negative feedback**



negative feedback mechanism

let's look into the effect of introducing negative feedback into the amplifier circuit^[109]

assume a fraction of output βV_{out} ($0 < \beta \leq 1$) is fed back to inverting input of op-amp, then combined voltage to this terminal would be the sum of original V_- and the feedback βV_{out} , the equation $V_{\text{out}} = G_0(V_+ - V_-)$ for open-loop circuit shall be rewritten, hence we have

$$V_{\text{out}} = G_0 \left[V_+ - (V_- + \beta V_{\text{out}}) \right] \Rightarrow (1 + G_0 \beta) V_{\text{out}} = G_0 (V_+ - V_-) \Rightarrow V_{\text{out}} = \frac{G_0}{1 + G_0 \beta} (V_+ - V_-)$$

overall gain of negative feedback amplifier is: $G = \frac{G_0}{1 + G_0 \beta}$

this gain is less than open-loop gain G_0 , so gain is reduced

further notice that $G_0 \beta \gg 1$, we find $G \approx \frac{G_0}{\beta G_0} = \frac{1}{\beta}$

this shows for negative-feedback amplifier, as long as open-loop gain G_0 is sufficiently large, overall gain G depends on β only, i.e., only depends on how much output is sent back to V_-

β can be controlled by a set of fixed resistors, which are not sensitive to change of frequency or temperature, so G remains constant for a wider range of frequencies and temperatures

^[109] Discussions in this section are not required by the syllabus. If you do not have the appetite, you may just skip to the conclusions.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 反馈意味着将一部分输出电压发送回输入端子。反馈网络也称为闭环，如果输出 V_{out} 的一部分被发送回反相输入 V_- ，从而反对输入的变化，这称为负反馈 $- +V_+ V_- V_{out} -V_S +V_S \beta V_{out}$ 负反馈机制让我们看看在放大器电路中引入负反馈的效果[109]假设输出 βV_{out} ($0 < \beta < 1$) 的一小部分反馈到运放的反相输入端，则该端的组合电压将是原始 V_- 与反馈 βV_{out} 之和，开环电路的方程 $V_{out} = G_0(V_+ - V_-)$ 将被重写，因此我们有 $V_{out} = G_0 h (V_+ - (V_- + \beta V_{out}))$ $(1 + G_0 \beta) V_{out} = G_0(V_+ - V_-)$ $V_{out} = G_0 / (1 + G_0 \beta) (V_+ - V_-)$ 负反馈放大器的总增益为： $G = G_0 / (1 + G_0 \beta)$ 该增益小于开环增益 G_0 ，因此增益减小，进一步注意 $G_0 \beta \gg 1$ ，我们发现 $G \approx 1 / \beta$ 对于负反馈放大器，只要开环增益 G_0 足够大，总增益 G 仅取决于 β ，即仅取决于发送回 V_- 的输出量。 β 可以由一组固定电阻控制，这些电阻对频率或温度的变化不敏感，因此 G 在更广泛的频率和温度范围内保持恒定 [109] 本节中的讨论不是教学大纲所要求的。如果你没有胃口，你可以直接跳到结论。 196

we conclude that introducing negative feedback has the following benefits:

- reduced gain
- more stable gain
- increased bandwidth
- less distortion

the golden rule

with negative feedback, we expect that V_{out} is able to change correspondingly with V_{in}

for any op-amp, $V_{\text{out}} = G_0(V_+ - V_-)$, but op-amp has very large open-loop gain G_0

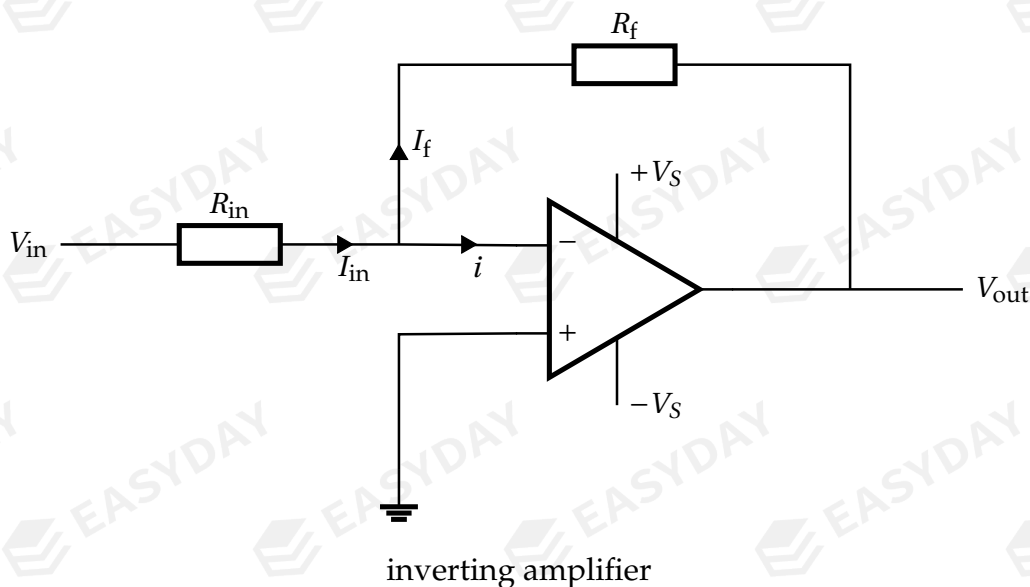
if V_{out} does not saturate, one must have $V_+ - V_- \approx 0$

this can be summarised as the *golden rule*: output of op-amp attempts to do whatever is necessary to make the difference between the two input voltages zero

note that op-amp can do this only if there is negative feedback

15.2.5 inverting amplifier

one type of widely-used amplifier circuit is called **inverting amplifier**



V_{out} is related to V_{in} by $V_{\text{out}} = \left(-\frac{R_f}{R_{\text{in}}}\right) V_{\text{in}}$, where gain of the amplifier is $G = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_{\text{in}}}$

to show this relation, we need two approximations

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 我们得出结论, 引入负反馈有以下好处: $\pm$ 增益降低 $\pm$ 增益更稳定 $\pm$ 带宽增加 $\pm$ 失真度降低负反馈的黄金法则, 我们期望对于任何运放, V_{out} 都能随 V_{in} 相应变化, $V_{out} = G_0(V_+ - V_-)$, 但运放有很大的开环增益 G_0 , 如果 V_{out} 不饱和, 则必须有 $V_+ - V_- \rightarrow 0$ 这可以概括为黄金法则: 运算放大器的输出尝试采取一切必要措施使两个输入电压之间的差异为零。请注意, 运算放大器只有在存在负反馈时才能做到这一点。

15.2.5 反相放大器一种广泛使用的放大器电路称为反相放大器 $-$ R_{in} V_{in} V_{out} $-V_S + V_S R_f / R_{in}$ 如果反相放大器 V_{out} 相关通过 $V_{out} = (-R_f / R_{in}) V_{in}$, 其中放大器的增益为 $G = V_{out} / V_{in} = -R_f / R_{in}$ 为了显示这种关系, 我们需要两个近似值 197

➤ virtual earth approximation: if V_{out} does not saturate, must have $V_+ \approx V_-$, but V_+ is earthed, i.e., $V_+ = 0$, so $V_- \approx 0$, referred to as a **virtual earth**

➤ small current approximation: because of the very large input impedance of op-amp, current flowing into inverting input is negligible, i.e., $i \approx 0$

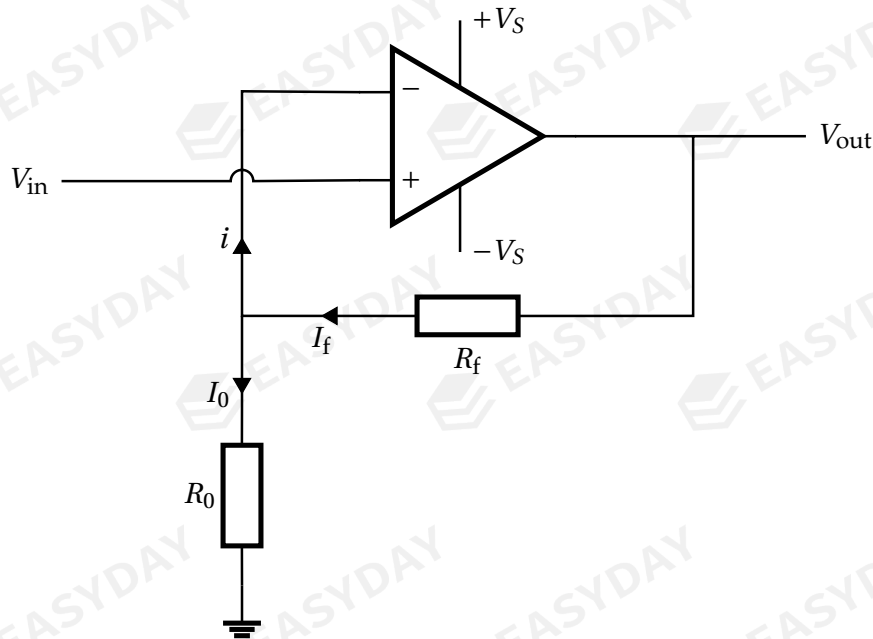
therefore currents through R_{in} and R_f are approximately the same: $I_{in} = I_f$

from this equation, we can relate V_{out} to V_{in} :

$$\frac{V_{in} - V_-}{R_{in}} = \frac{V_- - V_{out}}{R_f} \Rightarrow \frac{V_{in} - 0}{R_{in}} = \frac{0 - V_{out}}{R_f} \Rightarrow V_{out} = \left(-\frac{R_f}{R_{in}}\right) V_{in}$$

15.2.6 non-inverting amplifier

another type of common amplifier circuit is called **non-inverting amplifier**



non-inverting amplifier

for this circuit, V_{out} is given by: $V_{out} = \left(1 + \frac{R_f}{R_0}\right) V_{in}$, where overall gain $G = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_0}$

to show this, recall current into input of op-amp $i \approx 0$ due to its large input impedance, so

$$I_0 = I_f \Rightarrow \frac{V_- - 0}{R_0} = \frac{V_{out} - V_-}{R_f}$$

again, if V_{out} is not saturated, $V_- \approx V_+ = V_{in}$, then we find

$$\frac{V_{in}}{R_0} = \frac{V_{out} - V_{in}}{R_f} \Rightarrow R_0 V_{out} - R_0 V_{in} = R_f V_{in} \Rightarrow V_{out} = \left(1 + \frac{R_f}{R_0}\right) V_{in}$$

考纲提示 · Chapter 15 Electronics: 旧版拓展内容; 2022 起不再属于 9702 A Level, 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) $\hat{a}$ 虚拟地近似: 如果 V_{out} 不饱和, 则必须有 $V_+ \approx V_-$, 但 V_+ 接地, 即 $V_+ = 0$, 所以 $V_- \approx 0$, 称为虚拟地 $\hat{a}$ 小电流近似: 由于运放的输入阻抗非常大, 流入反相输入的电流可以忽略不计, 即, $i \approx 0$, 因此流过 R_{in} 和 R_f 的电流大致相同: $i_{in} = i_f$ 根据此等式, 我们可以将 V_{out} 与 V_{in} 联系起来: $V_{in} - V_- / R_{in} = V_- - V_{out} / R_f$ $V_{in} - 0 / R_{in} = 0 - V_{out} / R_f$ $V_{out} = (-R_f / R_{in}) V_{in}$ 15.2.6 非反相放大器另一种常见类型放大器电路称为同相放大器 $- +V_{in} R_f V_{out} - V_S + V_S R_0$ 如果 I_0 是该电路的同相放大器, 则 V_{out} 由下式给出: $V_{out} = (1 + R_f / R_0) V_{in}$, 其中总增益 $G = V_{out} / V_{in} = 1 + R_f / R_0$ 为了说明这一点, 回想一下进入运算放大器 $i \approx 0$ 输入的电流, 因为其电流很大输入阻抗, 所以 $I_0 = i_f$ $V_- - 0 / R_0 = V_{out} - V_+ / R_f$ 再次, 如果 V_{out} 未饱和, $V_- \approx V_+ = V_{in}$, 则我们找到 $V_{in} / R_0 = V_{out} - V_{in} / R_f$ $R_0 V_{out} - R_0 V_{in} = R_f V_{in}$ $V_{out} = (1 + R_f / R_0) V_{in}$ 198

remarks on inverting & non-inverting amplifiers

inverting and non-inverting amplifiers are starting point for more complex circuits

before going through detailed examples, let's briefly comment on the results we have found

$$\text{for inverting amplifier: } G = \frac{-R_f}{R_{in}} \quad \text{for non-inverting amplifier: } G = 1 + \frac{R_f}{R_0}$$

- circuit's overall gain for both amplifiers only depend on choice of resistors

so gain can now be easily tuned, circuit design becomes more convenient

- no parameter of op-amp shows up in final expressions for overall gain

gain is determined by feedback network, rather than by op-amp's characteristics

since value of resistors does not change significantly with temperature or frequency

this again means gain becomes now more stable with negative feedback

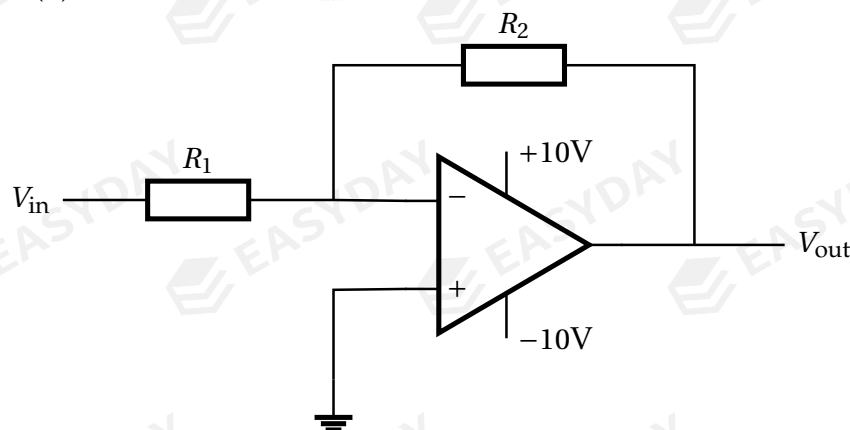
- for inverting amplifiers, $G < 0$, V_{in} and V_{out} always out of phase (opposite signs)

for non-inverting amplifiers, $G > 0$, V_{in} and V_{out} always in phase (same sign)

- the proportional relation between V_{in} and V_{out} only holds if op-amp is not saturated

if we find V_{out} greater than supply voltage from a calculation, true V_{out} equals $\pm V_S$

Example 15.4 In the inverting amplifier circuit below, $R_1 = 10 \text{ k}\Omega$, $R_2 = 50 \text{ k}\Omega$. (a) Find V_{out} when $V_{in} = +1.0 \text{ V}$. (b) Find V_{out} when $V_{in} = -4.0 \text{ V}$.



➤ overall gain of the amplifier: $G = -\frac{R_2}{R_1} = -\frac{50\text{k}}{10\text{k}} = -5.0$

for $V_{in} = +1.0 \text{ V}$: $V_{out} = (-5.0) \times (+1.0) = -5.0 \text{ V}$

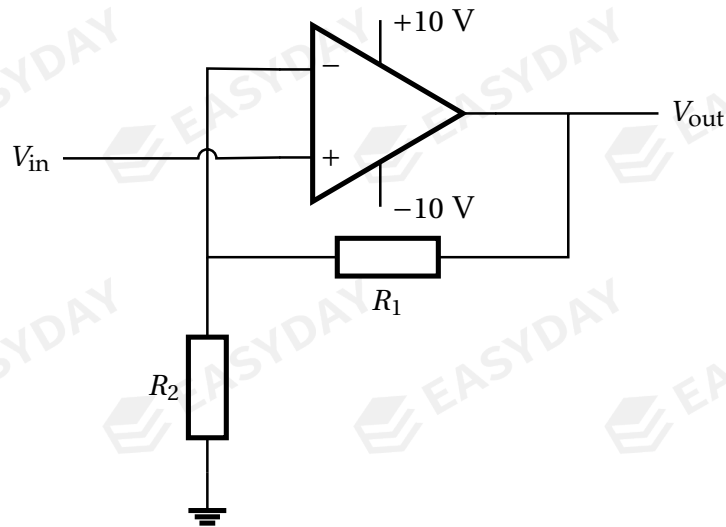
for $V_{in} = -4.0 \text{ V}$: $V_{out} = (-5.0) \times (-4.0) = +20 \text{ V}$, output becomes saturated, so $V_{out} = +10 \text{ V}$ □

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 关于反相和非反相放大器的评论反相和非反相放大器是更复杂电路的起点，在进行详细示例之前，让我们简要评论一下我们为反相放大器找到的结果： $G = -R_f / R_{in}$ 对于非反相放大器： $G = 1 + R_f / R_0$ 电路的整体两个放大器的增益仅取决于电阻器的选择，因此现在可以轻松调整增益，电路设计变得更加方便——最终表达式中没有显示运算放大器的参数，总体增益由反馈网络决定，而不是由运算放大器的特性决定，因为电阻值不会随温度或频率显著变化，这再次意味着负反馈增益现在变得更加稳定——对于反相放大器， $G < 0$ 、 V_{in} 和 V_{out} 总是异相（符号相反）对于非反相放大器， $G \geq 0$ ， V_{in} 和 V_{out} 始终同相（符号相同）只有当我们通过计算发现 V_{out} 大于电源电压，且运算放大器未饱和时， V_{in} 和 V_{out} 之间的比例关系才成立，真实的 V_{out} 等于 V_{sat} 示例 15.4 在下面的反相放大器电路中， $R_1 = 10 \text{ k}\Omega$ ， $R_2 = 50 \text{ k}\Omega$ 。(a) 当 $V_{in} = +1.0 \text{ V}$ 时求出 V_{out} 。(b) 当 $V_{in} \geq 4.0 \text{ V}$ 时查找 V_{out} 。

放大器的总增益： $G = -R_2 / R_1 = -50\text{k} / 10\text{k} = -5.0$ ($V_{in} = +1.0 \text{ V}$) : $V_{out} = (-5.0) \times (+1.0) = -5.0 \text{ V}$ ($V_{in} = -4.0 \text{ V}$) : $V_{out} = (-5.0) \times (-4.0) = +20 \text{ V}$ ，输出饱和，因此 $V_{out} = +10 \text{ V}$ 199

Example 15.5 In the non-inverting amplifier circuit below, $R_1 = 48 \text{ k}\Omega$, $R_2 = 6.0 \text{ k}\Omega$. (a) Find V_{out} when $V_{\text{in}} = +0.5 \text{ V}$. (b) Find V_{out} when $V_{\text{in}} = -3.0 \text{ V}$.

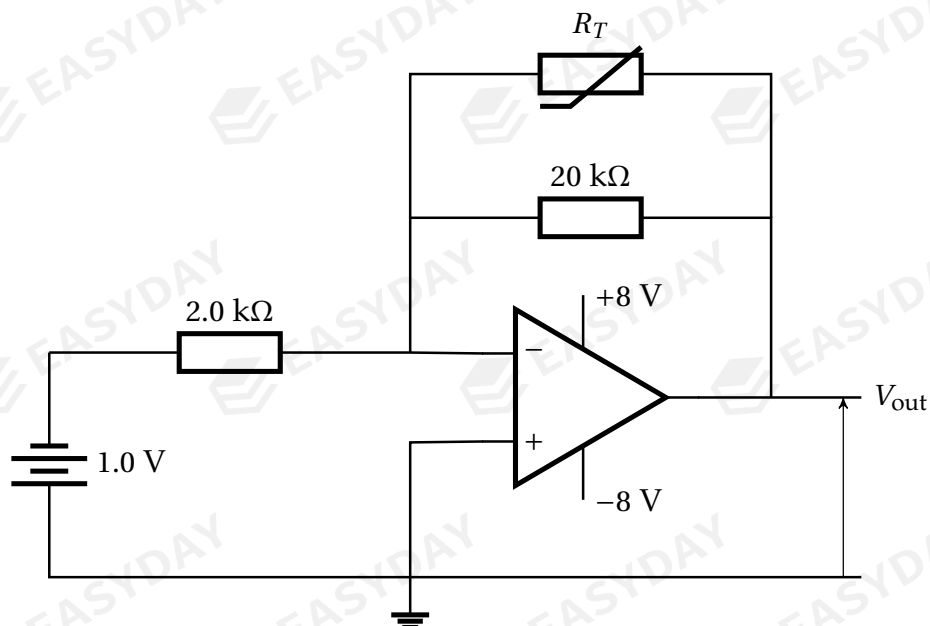


overall gain of the amplifier: $G = 1 + \frac{R_1}{R_2} = 1 + \frac{48\text{k}}{6.0\text{k}} = 9.0$

for $V_{\text{in}} = +0.5 \text{ V}$: $V_{\text{out}} = 9.0 \times (+0.5) = +4.5 \text{ V}$

for $V_{\text{in}} = -3.0 \text{ V}$: $V_{\text{out}} = 9.0 \times (-3.0) = -27 \text{ V}$, output becomes saturated, so $V_{\text{out}} = -10 \text{ V}$ $\square$

Example 15.6 An amplifier circuit incorporating an ideal op-amp is shown.



Initially, the thermistor has a resistance of $20 \text{ k}\Omega$. (a) Find the gain of this circuit. (b) Find the output voltage. (c) State and explain how V_{out} changes as temperature rises. (d) The thermistor is then removed, find the new output voltage.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 示例 15.5 在下面的同相放大器电路中, $R_1 = 48 \text{ k}\Omega$, $R_2 = 6.0 \text{ k}\Omega$ 。

(a) 当 $V_{in} = +0.5 \text{ V}$ 时求出 V_{out} 。

(b) 当 $V_{in} \geq 3.0 \text{ V}$ 时查找 V_{out} 。

– + V_{in} R_1 V_{out} –10 V +10 V R_2 - 放大器的总增益 : $G = 1 + \frac{R_1}{R_2} = 1 + \frac{48\text{k}}{6.0\text{k}} = 9.0$ ($V_{in} = +0.5 \text{ V}$) : $V_{out} = 9.0 \times (+0.5) = +4.5 \text{ V}$ ($V_{in} = -3.0 \text{ V}$) : $V_{out} = 9.0 \times (-3.0) = -27 \text{ V}$, 输出饱和, 因此 $V_{out} = -10 \text{ V}$ 。

例 15.6 显示了包含理想运算放大器的放大器电路。

– + $2.0 \text{ k}\Omega$ V_{out} – 8 V +8 V $20 \text{ k}\Omega$ 1.0 V R_T 最初, 热敏电阻的电阻为 $20 \text{ k}\Omega$ 。

(a) 求该电路的增益。

(b) 求出输出电压。

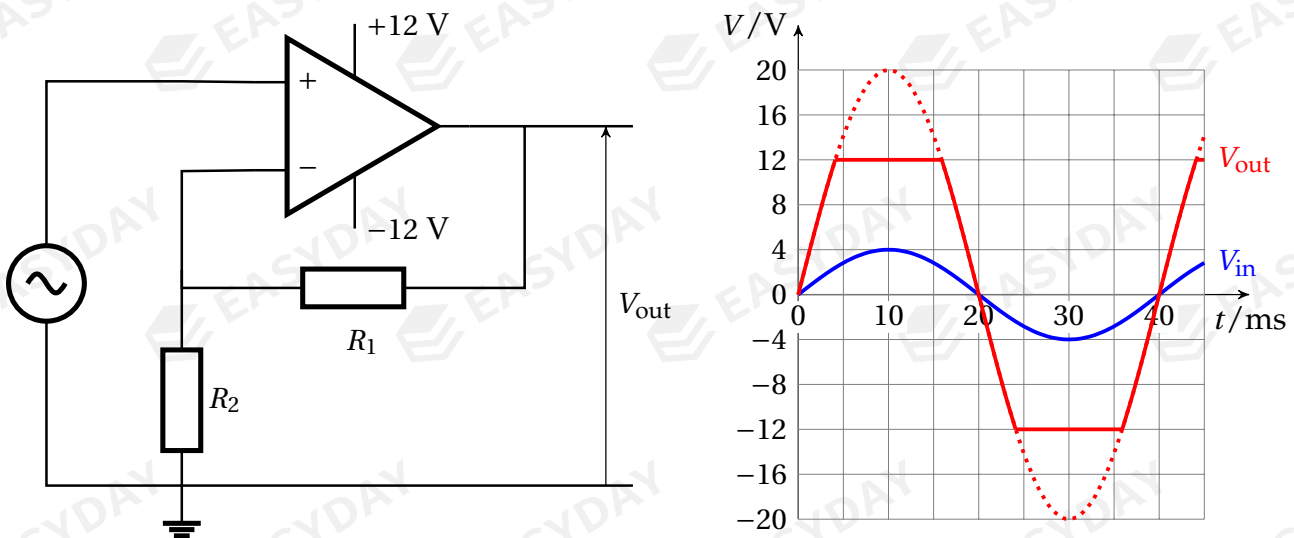
(c) 陈述并解释 V_{out} 如何随着温度升高而变化。

(d) 然后移除热敏电阻, 找到新的输出电压。

200

- (a) this is an inverting amplifier (there is negative feedback loop and V_{out} is fed into V_-)
 resistance of feedback loop: $R_f = \left(\frac{1}{20\text{k}} + \frac{1}{20\text{k}} \right)^{-1} = 10\text{k}\Omega$
 gain of amplifier: $G = -\frac{R_f}{R_{\text{in}}} = -\frac{10\text{k}}{2.0\text{k}} \Rightarrow G = -5.0$
- (b) V_{in} is at negative terminal of the battery, which is 1.0 V lower than the positive terminal but positive terminal is earthed, so $V_{\text{in}} = -1.0\text{ V}$
 output voltage: $V_{\text{out}} = GV_{\text{in}} = (-5.0) \times (-1.0) \Rightarrow V_{\text{out}} = +5.0\text{ V}$
- (c) as temperature increases, resistance of thermistor decreases
 total resistance of feedback loop becomes smaller
 gain of the circuit then becomes less negative, so V_{out} will decrease
- (d) gain of amplifier after removal of thermistor: $G' = -\frac{R'_f}{R_{\text{in}}} = -\frac{20\text{k}}{2.0\text{k}} \Rightarrow G' = -10$
 new output voltage can be computed: $V'_{\text{out}} = G'V_{\text{in}} = (-10) \times (-1.0) = +10\text{ V}$
 but voltage supply to op-amp is $\pm 8\text{ V}$, op-amp saturates, so $V'_{\text{out}} = +8\text{ V}$ $\square$

Example 15.7 A circuit incorporating an ideal op-amp is shown below. The two fixed resistors have $R_1 = 36\text{ k}\Omega$ and $R_2 = 9.0\text{ k}\Omega$. Variation of V_{in} from a voltage source is shown with the blue curve. On the same graph, plot how V_{out} varies with time.



- (a) this is a non-inverting amplifier, whose voltage gain is: $G = 1 + \frac{R_1}{R_2} = 1 + \frac{36\text{k}}{9.0\text{k}} = +5.0$
 so $V_{\text{out}} = +5.0 \times V_{\text{in}}$, with which we attempt to plot V_{out} as the red dotted curve
 but V_{out} will saturate at voltage of power supply, which is $\pm 12\text{ V}$
 taking this into account, we can plot the true V_{out} as the red solid curve $\square$

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) - (a) 这是一个反相放大器 (有负反馈环路, V_{out} 馈入 V_-) 反馈环路电阻: $R_f = (1\ 20k + 1\ 20k) = 10k\ \Omega$ 放大器增益: $G = -R_f/R_{in} = -10k/2.0k$ $G = -5.0$ (b) V_{in} 位于电池负极, 比正极低 $1.0\ V$, 但正极接地, 因此 $V_{in} = -1.0\ V$ 输出电压: $V_{out} = GV_{in} = (-5.0) \times (-1.0)$ $V_{out} = +5.0\ V$ (c) 随着温度升高, 热敏电阻的电阻减小, 反馈环路的总电阻变小, 电路的增益变小 (d) 去掉热敏电阻后放大器的增益: $G_0 = -R_{0f}/R_{in} = -20k/2.0k$ $G_0 = -10$ 可以计算新的输出电压: $V_{0\ out} = G_0 V_{in} = (-10) \times (-1.0) = +10\ V$ 但运算放大器的供电电压为 $\pm 8\ V$, 运算放大器饱和, 因此 $V_{0\ out} = +8\ V$

ä 示例 15.7 包含理想运算放大器的电路如下所示。两个固定电阻器的 $R_1 \times 36\ k\ \Omega$ 和 $R_2 \times 9.0\ k\ \Omega$ 。蓝色曲线显示了电压源的 V_{in} 变化。在同一张图表上, 绘制 V_{out} 如何随时间变化的图。

$-12\ V$ $+12\ V$ $+ - V_{out}$

R_2 R_1 V/V t/ms $0\ 10\ 20\ 30\ 40\ -20\ -1\ 6\ -12\ -8\ -4\ 0\ 4\ 8\ 12\ 16\ 20$ V_{in} V_{out} - 这是一个同相放大器, 其电压增益为: $G = 1 + R_1/R_2 = 1 + 36k/9.0k = +5.0$ 因此 $V_{out} = +5.0 \times V_{in}$, 我们尝试将 V_{out} 绘制为红色虚线曲线, 但 V_{out} 将在电源电压 ($\pm 12\ V$) 处饱和, 考虑到这一点, 我们可以将真实 V_{out} 绘制为红色实线曲线

ä 201

Question 15.3 An amplifier circuit is used to monitor light intensity. The magnitude of its gain is required to increase as light intensity increases. Suggest how this can be done.

Question 15.4 Show that the output voltage from an inverting amplifier is $V_{\text{out}} = \left(-\frac{R_f}{R_{\text{in}}}\right) V_{\text{in}}$.

Question 15.5 Show that the output voltage from a non-inverting amplifier is $V_{\text{out}} = \left(1 + \frac{R_f}{R_0}\right) V_{\text{in}}$.

15.3 output devices

15.3.1 LED

a **diode** is a device that only allows current to flow in one direction

LED, or light-emitting diode, is able to give off light when current passes through it

LEDs are efficient light emitters and they come in many different colours

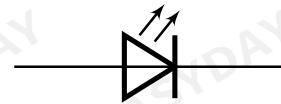
LEDs can therefore be used as indicators to show the *polarity* of V_{out} from processing units

a partial circuit incorporating two LEDs used as output devices is shown

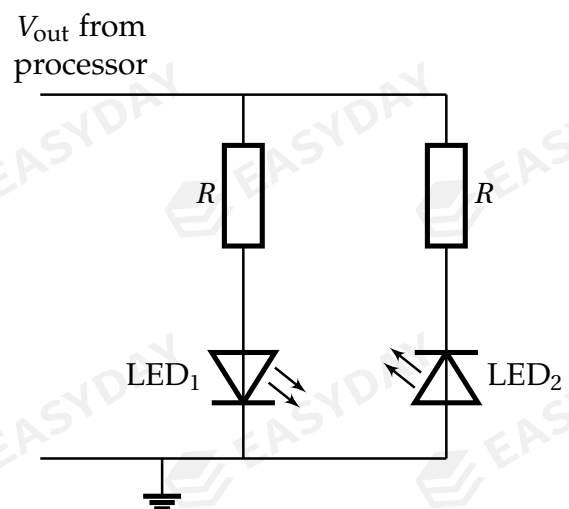
when $V_{\text{out}} > 0$, LED 1 emits light, LED 2 is off

when $V_{\text{out}} < 0$, LED 2 emits light, LED 1 is off

protector resistors R are included to ensure LED units not to be broken by large V_{out}



electric symbol for an LED



15.3.2 relay

relay is an electromagnetic switch which enables the use of a current in one circuit (control circuit) to switch on another current in a second circuit (controlled circuit)

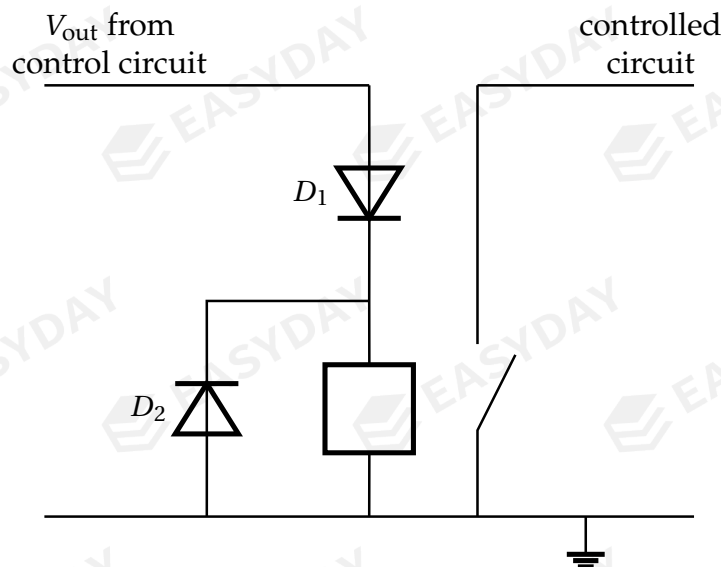
a relay consists of an electromagnet, a moveable arm and switch contacts

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 电子 (Q) 问题 15.3 使用放大器电路来监测光强度。其增益的大小需要随着光强度的增加而增加。建议如何做到这一点。问题 15.4 证明反相放大器的输出电压为 $V_{out} = (-R_f R_{in}) V_{in}$ 。问题 15.5 证明同相放大器的输出电压为 $V_{out} = (1 + R_f R_0) V_{in}$ 。15.3 输出设备 15.3.1 LED 二极管是一种只允许电流在一个方向流动的设备 LED 的电气符号 LED2 RR LED1 来自处理器的 V_{out} LED 或发光二极管，当电流通过时能够发光 LED 是高效的发光体，并且有多种不同的颜色 LED 因此可以用作指示器来显示来自处理单元的 V_{out} 的极性当 V_{out} 时，显示包含两个用作输出设备的 LED 的部分电路 $V_{out} \geq 0$ ，LED 1 发光，LED 2 在 $V_{out} < 0$ 时关闭，LED 2 发光，LED 1 关闭包括保护电阻 R ，以确保 LED 单元不会被大 V_{out} 损坏 15.3.2 继电器继电器是一种电磁开关，可以利用一个电路（控制电路）中的电流来接通第二个电路（受控电路）中的另一种电流。继电器由电磁体、可移动臂和开关触点组成 202

current flowing in the coils of the electromagnet induces a magnetic field to attract the moveable arm, which then connects the contacts and completes the second circuit

- relay is used for many purposes, some of which are listed below
 - switching large voltage/current by means of a small voltage/current
 - high voltage isolation
 - remote switching



output from processing units connected to a relay

- in practice, a relay is usually connected with two *diodes* for the following reasons:
 - diode D_1 ensures relay is triggered only for positive V_{out}
without D_1 , relay would operate for currents flowing in either direction
 - diode D_2 protects sensitive processing units from large currents induced from relay coil
when control circuit is turned off, large currents could be induced from relay coil
this current might damage an op-amp or other sensitive components
 D_2 opens a path for induced current to flow (a short circuit across coil)

15.3.3 calibrated meters

we can use sensing devices to monitor a specific physical quantity, say x

x could be temperature, light level, sound intensity, etc.

change in x would cause change in output voltage from the processing units

考纲提示 · Chapter 15 Electronics：旧版拓展内容；2022 起不再属于 9702 A Level，当前 Paper 4 不考。

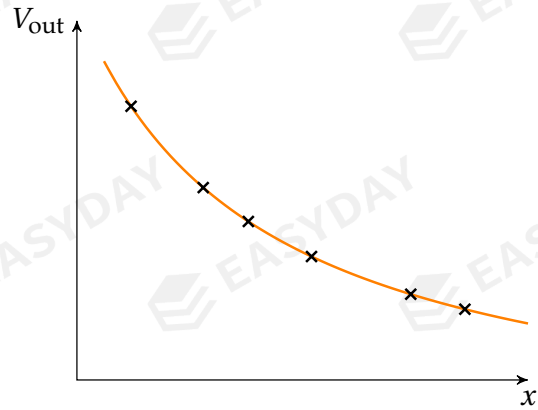
Yuhao Yang 15 ELECTRONICS (Q) 电磁铁线圈中流动的电流会产生磁场，吸引可动臂，然后连接触点并完成第二个电路 — 继电器有多种用途，其中一些如下所列 — 通过小电压/电流来切换大电压/电流 — 高压隔离 — 远程切换 D1 D2 来自控制电路的 V_{out} 控制连接到继电器的处理单元的电路输出 — 在实践中，继电器通常与两个二极管连接，原因如下： — 二极管 D1 确保继电器仅在没有 D1 的情况下为正 V_{out} 触发，继电器将在任一方向流动的电流下工作 — 二极管 D2 保护敏感处理单元免受控制电路关闭时继电器线圈感应出的大电流的影响，继电器线圈可能会感应出大电流，该电流可能会损坏运算放大器或其他敏感元件 D2 打开感应电流流动的路径（线圈短路） 15.3.3 校准我们可以使用传感设备来监测特定的物理量，例如 x x 可以是温度、光照水平、声音强度等。 x 的变化将导致处理单元 203 的输出电压发生变化

for convenience, we would hope value of x can be directly read off from a voltmeter, so we need associate a value of V_{out} with a value for x

V_{out} from processors is usually *non-linear* in x , so one must **calibrate** scale of meter in terms of x ,

calibration can be done according to a curve of measured data, called *calibration curve*

scale will be non-uniform, but this is doable



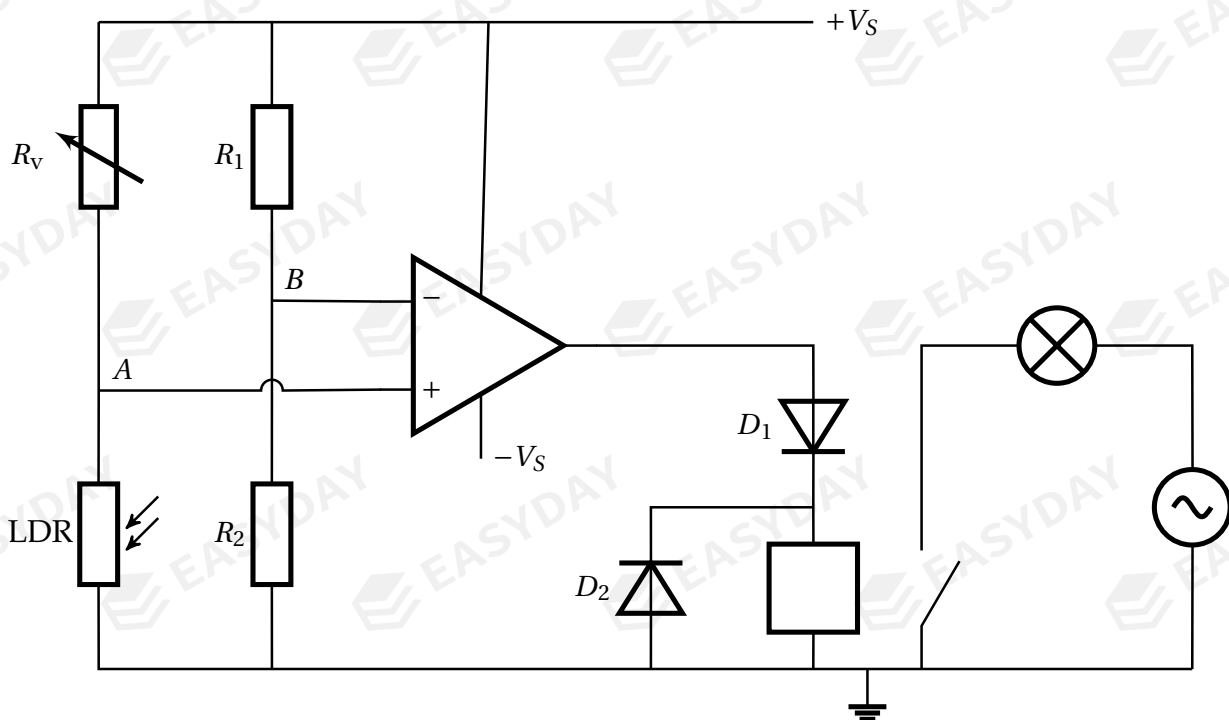
example of a calibration curve

15.4 practical electronic circuits

the many things we learned so far can be put together to build functional electronic circuits
in last part of this section, we will see a few examples of practical circuits

15.4.1 automatic illumination system

the circuit below enables automatic switching of lamp when it gets dark



this circuit consists of a comparator, whose inputs are provided from two potential dividers

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) V_{out} x 校准曲线的例子为了方便起见,我们希望 x 的值可以直接从电压表中读出,所以我们需要将 V_{out} 的值与 x 的值关联起来。处理器的 V_{out} 通常在 x 上是非线性的,所以必须用 x 来校准仪表的刻度,可以根据测量数据的曲线进行校准,称为校准曲线刻度将不均匀,但这是可行的 15.4 实用电子电路到目前为止我们学到的很多东西可以放在一起构建功能电子电路,在本节的最后部分,我们将看到一些实用电路的示例 15.4.1 自动照明系统下面的电路可以在天黑时自动切换灯 +VS R_v LDR R_1 R_2 - +A B D1 D2 -VS 该电路由一个比较器组成,其输入由两个分压器 204 提供

output from the comparator could trigger a relay that controls an illumination circuit

fixed resistors R_1 and R_2 provide a constant potential to V_- for reference

magnitude of V_+ depends on the voltage share on LDR, which depends on light conditions

when in bright, LDR has low resistance, V_+ is low, so $V_+ < V_-$, V_{out} from op-amp is negative
current through relay is blocked by diode D_1 , lamp circuit is off

as light level falls, resistance of LDR increases, V_+ then rises

at a certain light level, V_+ becomes greater than V_- , V_{out} from op-amp becomes positive

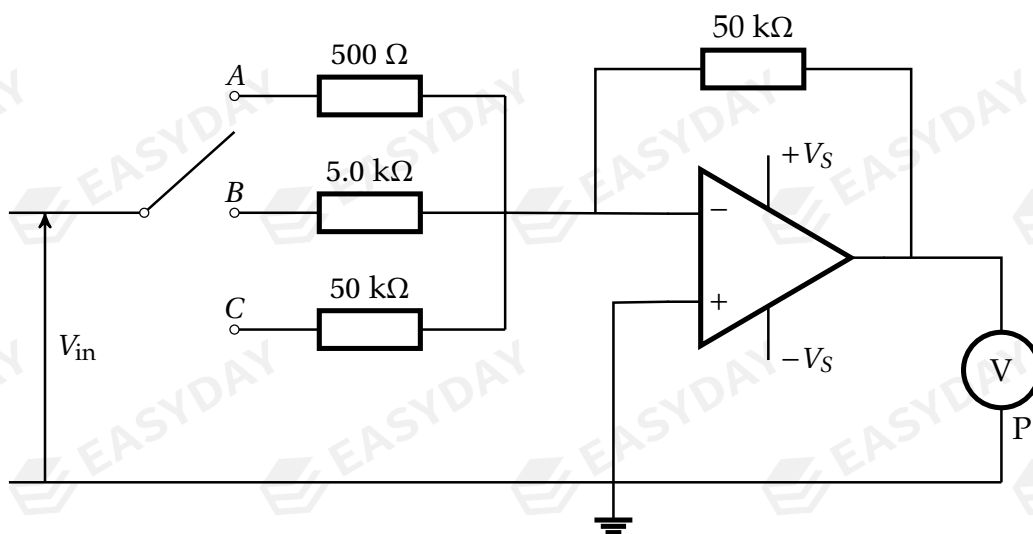
current in relay triggers the switch to close, the lamp is switched on

variable resistor R_v can be set to determine the critical light level of switch-over

15.4.2 multi-range voltmeter

the inverting amplifier can be adapted to make a multi-range voltmeter

positive connection of voltmeter is at P so that positive reading is obtained for $+V_{in}$



in this example, V_{in} can be sent into the amplifier through three different channels

you can check that each channel has a different value of gain: $G_A = -100$, $G_B = -10$, $G_C = -1.0$

if the voltmeter connected to the output of op-amp has a range of $0 \sim 10$ V, then for V_{in} less than 0.1 V, or in the range $0.1 \sim 1.0$ V, or in the range $1.0 \sim 10$ V, they can be sent through channel A, B or C respectively to get a noticeable deflection on the voltmeter

考纲提示 · Chapter 15 Electronics：旧版拓展内容；2022 起不再属于 9702 A Level，当前 Paper 4 不考。

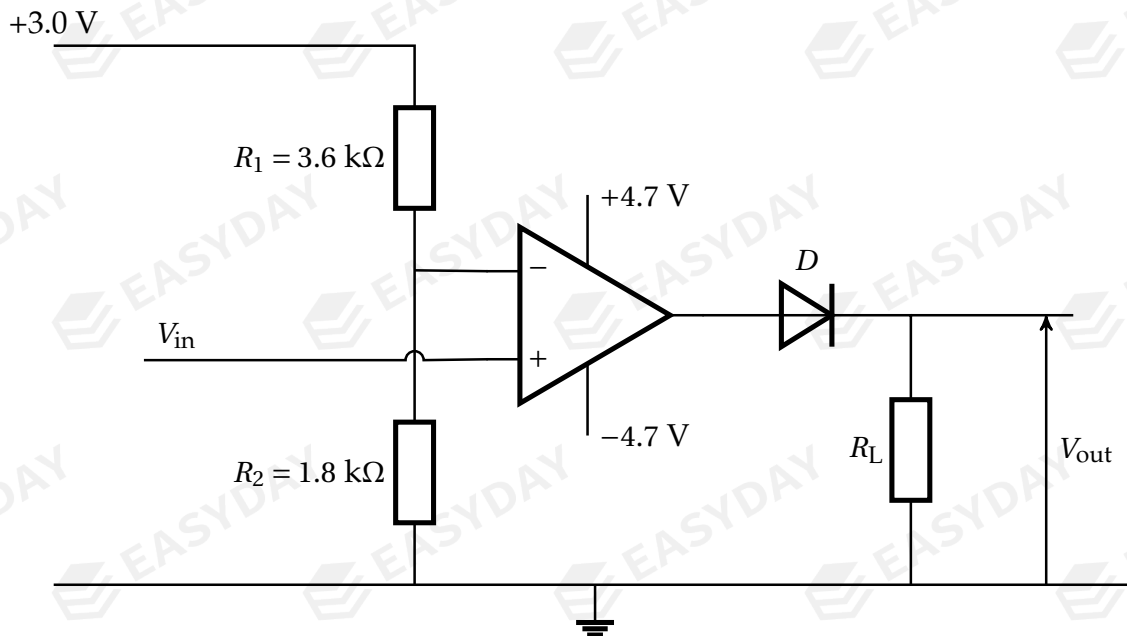
Yuhao Yang 15 ELECTRONICS (Q) 比较器的输出可触发控制照明电路的继电器固定电阻器 R_1 和 R_2 为 V_- 提供恒定电势，作为参考 V_+ 的大小取决于 LDR 上的分压，这取决于光照条件，当明亮时，LDR 电阻较低， V_+ 较低，因此 $V_+ < V_-$ ，来自运放的 V_{out} 为负电流，通过继电器的电流被二极管 D_1 阻挡，灯电路在光照强度时关闭下降，LDR 电阻增大， V_+ 然后在一定光照强度下上升， V_+ 变得大于 V_- ，运算放大器的 V_{out} 变为继电器中的正电流，触发开关关闭，灯打开可变电阻 R_v 可设置来确定切换的临界光照强度

15.4.2 多量程电压表反相放大器可用于制作多量程电压表电压表的正极连接在 P 处，以便获得正读数 $+V_{in}$
 $- + V_- V_S + V_S 500 \Omega 5.0 k\Omega 50 k\Omega A B C 50 k\Omega V_{in} P$ 在本例中， V_{in} 可以通过三个不同的通道发送到放大器，您可以检查每个通道是否具有不同的增益值： $G_A = -100$ 、 $G_B = -10$ 、 $G_C = -1.0$ 如果连接到运算放大器输出的电压表有一个范围 $0 \gg 10 V$ ，则对于 V_{in} 小于 $0.1 V$ ，或在 $0.1 \gg 1.0 V$ 范围内，或在 $1.0 \gg 10 V$ 范围内，它们可以分别通过通道 A、B 或 C 发送，以在电压表 205 上获得明显的偏转

15.4.3 digital signal regenerator

the circuit given below can be used to regenerate a digital signal ^[110]

suppose D is a silicon diode which develops a p.d. drop of 0.7 V when it is forward biased



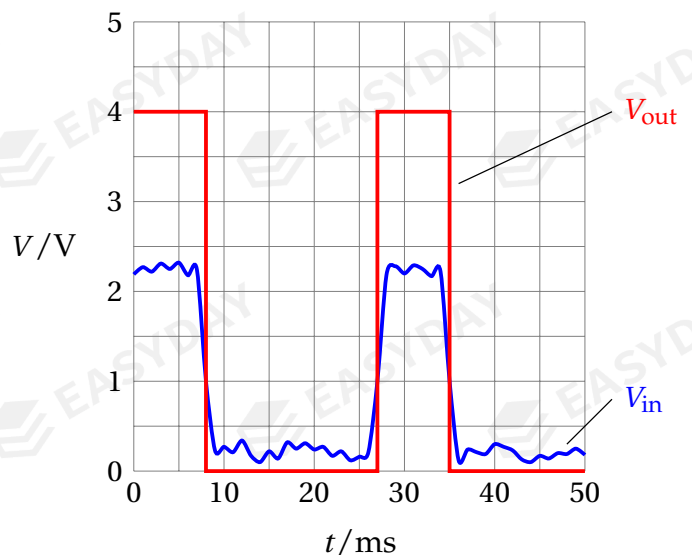
$V_- = \frac{1.8}{1.8 + 3.6} \times (+3.0) = +1.0$ V gives a constant potential to be compared with V_+

when $V_{in} > 1.0$ V, i.e., $V_+ > V_-$, output from op-amp is +4.7 V, diode D conducts and gets its voltage share of 0.7 V, the other 4.0 V is dropped across the load R_L , so $V_{out} = +4.0$ V

when $V_{in} < 1.0$ V, i.e., $V_+ < V_-$, output from op-amp is -4.7 V, diode D does not conduct, no current flows through R_L , then $V_{out} = 0$

if a *digital* signal is transmitted over a long distance, it may suffer from loss in power and noise, we would want to amplify the signal and remove the noise from it

this circuit can take in this signal and produce an output that removes the fluctuations due to noise and hence regenerates the original digital signal with only highs and lows



^[110] The use of digital signals in communication systems will be discussed in detail in §16.2.

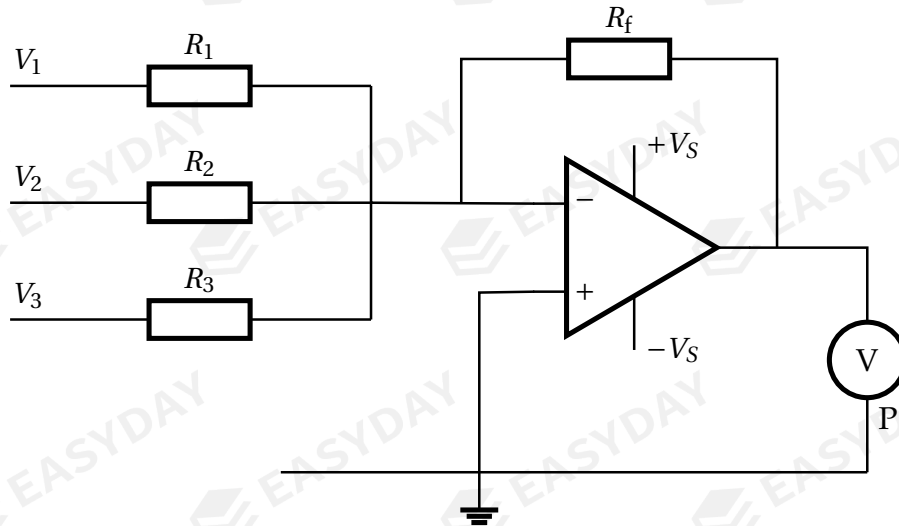
考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) 15.4.3 数字信号再生器下面给出的电路可用于再生数字信号 [110] 假设 D 是一个硅二极管, 它产生一个 p.d. 信号。正向偏置时电压降 0.7 V $+3.0\text{ V} - +V_{in}$ D RL $-4.7\text{ V} +4.7\text{ V}$ $R_1 = 3.6\text{ k}\Omega$ $R_2 = 1.8\text{ k}\Omega$ V_{out} 0 10 20 30 40 500 1 2 3 4 5 t/ms V /V V_{in} V_{out} $V_- = 1.8$ $1.8 +$ 当 $V_{in} \leq 1.0\text{ V}$ 时, $3.6 \times (+3.0) = +1.0\text{ V}$ 提供与 V_+ 比较的恒定电势, 即 $V_+ \leq V_-$, 运算放大器的输出为 $+4.7\text{ V}$, 二极管 D 导通并获得其 0.7 V 的电压份额, 另外 4.0 V 压降在负载 RL 上, 因此当 $V_{in} \leq 1.0\text{ V}$ 时, $V_{out} = +4.0\text{ V}$ 1.0 V , 即 $V_+ \leq V_-$, 运算放大器的输出为 -4.7 V , 二极管 D 不导通, 没有电流流过 RL, 则 $V_{out} = 0$ 如果数字信号长距离传输, 可能会遭受功率损失和噪声, 我们希望放大信号并消除其中的噪声, 该电路可以接收该信号并产生一个输出, 消除噪声引起的波动, 从而消除噪声引起的波动。重新生成仅具有高电平和低电平的原始数字信号 [110] 通信系统中数字信号的使用将在第 16.2 节中详细讨论。 206

15.4.4 summing amplifier

circuits incorporating op-amp can carry out many mathematical operations

the circuit we are going to introduce here, called the *summing amplifier*, is able to produce a voltage output as a *weighted sum* of several input signals ^[111]



this is basically a variation of the inverting amplifier

it can be shown that the output voltage from op-amp is given by: $V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$

positive connection of voltmeter is at *P* so positive reading is obtained for $V_1, V_2, V_3 > 0$

so reading on voltmeter is: $V = R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$, a weighted sum of inputs V_1, V_2 , and V_3

➤ in the case where $R_1 = R_2 = R_3 = R_f$, the reading simply becomes: $V = V_1 + V_2 + V_3$

this circuit produces the algebraic sum of the inputs, so it becomes a *voltage adder*

➤ in the case where $R_f = 4R_1 = 2R_2 = R_3$, the reading becomes: $V = 4V_1 + 2V_2 + V_3$

if V_1, V_2, V_3 are only allowed to be 0 or 1, they can represent a three-bit binary number

this circuit can convert this binary number into its decimal value

so what we have here is a *binary-to-decimal converter*

^[111] In a university-level electronic engineering course, you will see that op-amp can be used to build difference amplifiers, differentiator and integrator amplifiers (by including a capacitor), logarithmic amplifiers (by including a diode) and many other useful applications. Those who have interest in engineering science are encouraged to research on these topics.

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 电子 (Q) 15.4.4 包含运算放大器的求和放大器电路可以执行许多数学运算，我们在此介绍的电路称为求和放大器，能够产生电压输出作为多个输入信号的加权和 [111] $- + V - V_S + V_S V_1 R_1 V_2 R_2 V_3 R_3 R_f P$ 这基本上是反相的变体可以看出运算放大器的输出电压由下式给出： $V_{out} = -R_f (V_1 R_1 + V_2 R_2 + V_3 R_3)$ 电压表的正极连接在 P 处，因此 $V_1, V_2, V_3 \neq 0$ 获得正读数，因此电压表上的读数为： $V = R_f (V_1 R_1 + V_2 R_2 + V_3 R_3)$ ，输入的加权和 V_1 、 V_2 和 V_3 在 $R_1 = R_2 = R_3 = R_f$ 的情况下，读数变为： $V = V_1 + V_2 + V_3$ 该电路产生输入的代数和，因此它成为电压加法器 在 $R_f = 4R_1 = 2R_2 = R_3$ 的情况下，读数变为： $V = 4V_1 + 2V_2 + V_3$ 如果 V_1 、 V_2 、 V_3 只允许为 0 或 1，它们可以表示一个三位二进制数，该电路可以将这个二进制数转换为十进制值，所以我们这里有一个二进制到十进制转换器[111]在大学水平的电子工程课程中，您将看到运算放大器可用于构建差分放大器、微分器和积分放大器（通过包括电容器）、对数放大器（通过包括二极管）和许多其他有用的应用。我们鼓励那些对工程科学感兴趣的人研究这些主题。

。 207

Question 15.6 For the summing amplifier introduced, if $R_f = R_1 = 10 \text{ k}\Omega$, $R_2 = 1.0 \text{ k}\Omega$, and $R_3 = 100 \text{ }\Omega$, find an expression for the voltage reading displayed on the voltmeter, and hence suggest the function of this circuit.

Question 15.7 Design a non-inverting summing amplifier where the inputs are sent into the non-inverting input of the op-amp. (*)

Question 15.8 Design a circuit for a house such that a fire alarm can be automatically triggered when the temperature in the house becomes too high. (*)

考纲提示 · Chapter 15 Electronics : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 15 ELECTRONICS (Q) Question 15.6 对于所介绍的求和放大器, 如果 $R_f = R_1 = 10\text{ k}\Omega$, $R_1 = 1.0\text{ k}\Omega$, $R_3 = 100\text{ }\Omega$, 找到电压表上显示的电压读数的表达式, 从而提出该电路的功能。问题 15.7 设计一个同相求和放大器, 其中输入被发送到运算放大器的同相输入。(Q) 问题15.8 为房屋设计一个电路, 当房屋内温度过高时, 可以自动触发火灾警报。(Q)208

CHAPTER 16

Telecommunication (*)

telecommunication concerns transmitting information by electromagnetic means

engineers focus on solving the problem of transmitting large volumes of information over long distances with minimal loss of signal strength and distortion due to noise

in this chapter, we will study three of the key components in a communication system ^[112]

- *modulator and demodulator*

variation of an information signal is represented by a carrier wave through modulation

original information can be extracted through the reverse process called demodulation

- *analogue-to-digital and digital-to-analogue converters*

analogue signals are digitized to archive more reliable and efficient transmission

digital signals are then converted back to analogue form reproduce original signal

- *communication channel*

communication channel is a medium through which signals are transmitted

each type of communication channel has its relative merits and disadvantages

16.1 modulation

audio signals can be converted into a radio signal for transmission

rather than being transmitting directly as a radio wave, information signal is usually en-

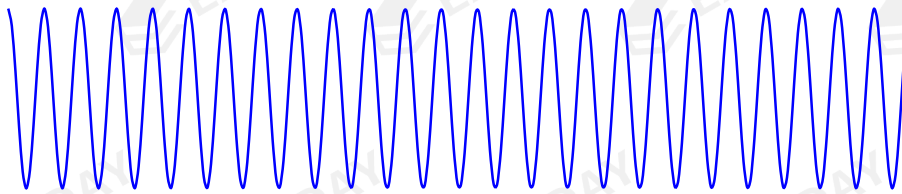
^[112] There are other important processes in telecommunication. For example, digital signals are usually passed through an *encoder* so that redundant information is removed and extra data is introduced to check for errors. Modulated signals are usually *multiplexed*, so multiple users can share the same communication channel. Common multiplexing schemes include FDMA, TDMA, CDMA, etc.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

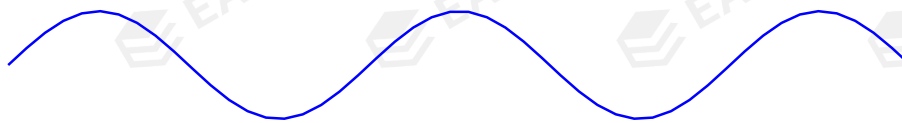
Yuhao Yang 16 电信 (Q) 第 16 章电信 (Q) 电信涉及通过电磁方式传输信息工程师重点解决长距离传输大量信息的问题，并在本章中将信号强度损失和噪声造成的失真最小化，我们将研究通信系统中的三个关键组件 [112] ± 调制器和解调器信息信号的变化通过调制由载波表示原始信息可以通过逆过程提取解调 ± 模数和数模转换器模拟信号被数字化以存档更可靠和高效的传输数字信号然后转换回模拟形式再现原始信号 ± 通信通道通信通道是传输信号的媒介每种类型的通信通道都有其相对的优点和缺点 16.1 调制音频信号可以转换为无线电信号进行传输，而不是直接作为无线电波传输，信息信号通常是 en - [112] 还有其他重要的内容电信中的流程。例如，数字信号通常通过编码器传递，以便去除冗余信息并引入额外的数据来检查错误。调制信号通常是多路复用的，因此多个用户可以共享相同的通信信道。常见的复用方案包括 FDMA、TDMA、CDMA 等 209

coded into a carrier wave for better transmission

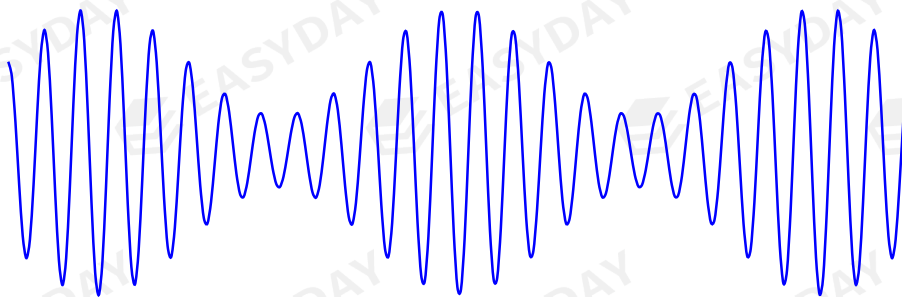
modulation means varying the amplitude or the frequency of a carrier wave in synchrony with the displacement of an information signal



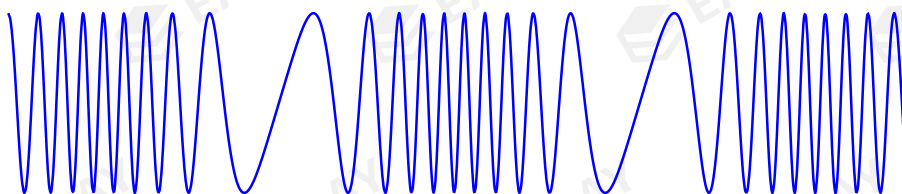
carrier wave
(before modulation)



information signal



amplitude modulated
carrier wave
(AM wave)



frequency modulated
carrier wave
(FM wave)

modulation of a carrier wave in accordance with an information signal

➤ reasons for modulation

- carrier wave has shorter wavelength, so shorter aerial is possible
- each signal can be transmitted with a different carrier frequency
so multiple signals can be transmitted at the same time without interfering each other
- modulated carrier wave is less affected by noise
- transmission over long distance is possible

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 编码到载波中以实现更好的传输调制是指与信息信号载波的位移同步改变载波的幅度或频率 (调制前) 信息信号调幅载波 (AM 波) 调频载波 (FM 波) 根据信息信号对载波进行调制
调制原因 - 载波波长较短, 因此可以使用更短的天线 - 每个信号可以使用不同的载波频率进行传输因此多个信号可以同时传输而不会互相干扰 - 调制载波受噪声影响较小 - 可以进行长距离传输 210

16.1.1 amplitude modulation

for amplitude modulated carrier wave, frequency is unchanged

amplitude varies with time according to displacement of information signal

waveform of information signal determines the *envelope* of the modulated wave

spectrum of AM wave

amplitude modulated carrier wave is found to contain more than one frequencies

mathematically, each AM wave can be considered as the sum of several simple sine waves

proof (*): suppose the original carrier wave takes the form: $x_c(t) = A_c \sin \omega_c t$

it is to be amplitude modulated by an information signal: $x_m(t)$

modulated wave has a new amplitude that varies with time: $A'_c(t) = A_c + x_m(t)$

for an audio signal of single frequency, say $x_m(t) = A_m \cos \omega_m t$, the AM wave takes the form:

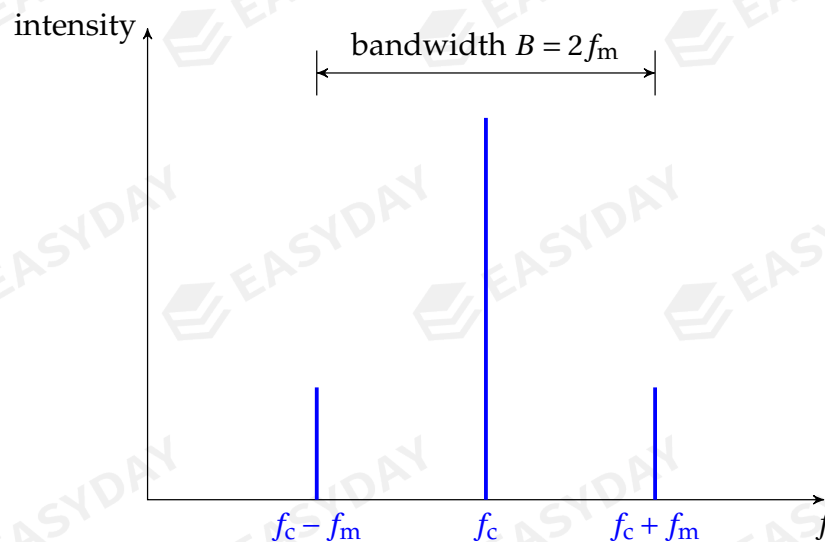
$$x'_c(t) = (A_c + A_m \cos \omega_m t) \sin \omega_c t = A_c \sin \omega_c t + A_m \sin \omega_c t \cos \omega_m t$$

using trigonometric identity: $\sin \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$, we find:

$$x'_c(t) = A_c \sin \omega_c t + \frac{1}{2} A_m \sin(\omega_c + \omega_m) t + \frac{1}{2} A_m \sin(\omega_c - \omega_m) t$$

AM wave is a combination of three components of frequencies f_c , $f_c + f_m$ and $f_c - f_m$ □

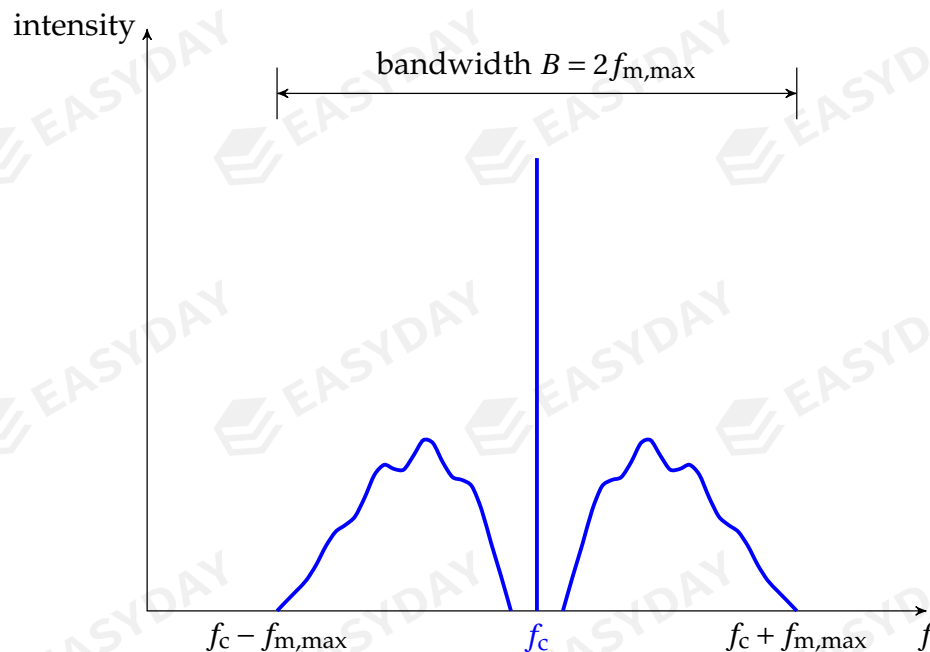
- for an information signal of *single* frequency, we can draw the spectrum for the AM wave
- apart from original carrier frequency, there are two more frequencies $f_c \pm f_m$, called *sidebands*
- positions of sideband is determined by frequency of the information signal f_m



考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇豪 16 电信 (Q) 16.1.1 调幅对于调幅载波, 频率不变, 幅度随信息信号波形的位移而变化信息信号的波形决定了 AM 波的调制波频谱的包络调幅载波在数学上可以被认为包含多个频率, 每个 AM 波可以被认为是几个简单的正弦波之和证明(Q): 假设原始载波的形式为: $x_c(t) = A_c \sin \omega_c t$ 它将由信息信号进行幅度调制: $x_m(t)$ 调制波具有随时间变化的新幅度: $A_0 c(t) = A_c + x_m(t)$ 对于单频音频信号, 例如 $x_m(t) = A_m \cos \omega_m t$, AM 波采用以下形式: $x_0 c(t) = (A_c + A_m \cos \omega_m t) \sin \omega_c t = A_c \sin \omega_c t + A_m \sin \omega_c t \cos \omega_m t$ 使用三角恒等式: $\sin \alpha \sin \beta = \frac{1}{2} \times [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$ Q, 我们发现: $x_0 c(t) = A_c \sin \omega_c t + \frac{1}{2} A_m \sin(\omega_c + \omega_m)t + \frac{1}{2} A_m \sin(\omega_c - \omega_m)t$ AM 波是频率 f_c 、 $f_c + f_m$ 和 $f_c - f_m$ 三个分量的组合, 对于单一频率的信息信号, 我们可以绘制出 AM 波的频谱, 除了原始载频外, 还有两个频率 $f_c \pm f_m$, 称为边带边带的位置由信息信号 f_m 强度的频率决定 $f_c - f_m$ $f_c + f_m$ 带宽 $B = 2 f_m$ 211

- information signal could contain a *range* of frequencies (e.g., music signal)
the two sidebands would spread into two symmetrical humps



- we define **bandwidth** as the range of frequencies occupied by the modulated signal
 - for a single-frequency information signal, bandwidth of AM wave is:

$$B = (f_c + f_m) - (f_c - f_m) \Rightarrow B = 2f_m$$

- for a multi-frequency information signal, bandwidth of AM wave is:

$$B = (f_c + f_{m,max}) - (f_c - f_{m,max}) \Rightarrow B = 2f_{m,max}$$

for sound signals, $f_m \approx 10^3$ Hz, so bandwidth is around a few kHz

compared with frequency of radio waves, bandwidth of AM is quite small

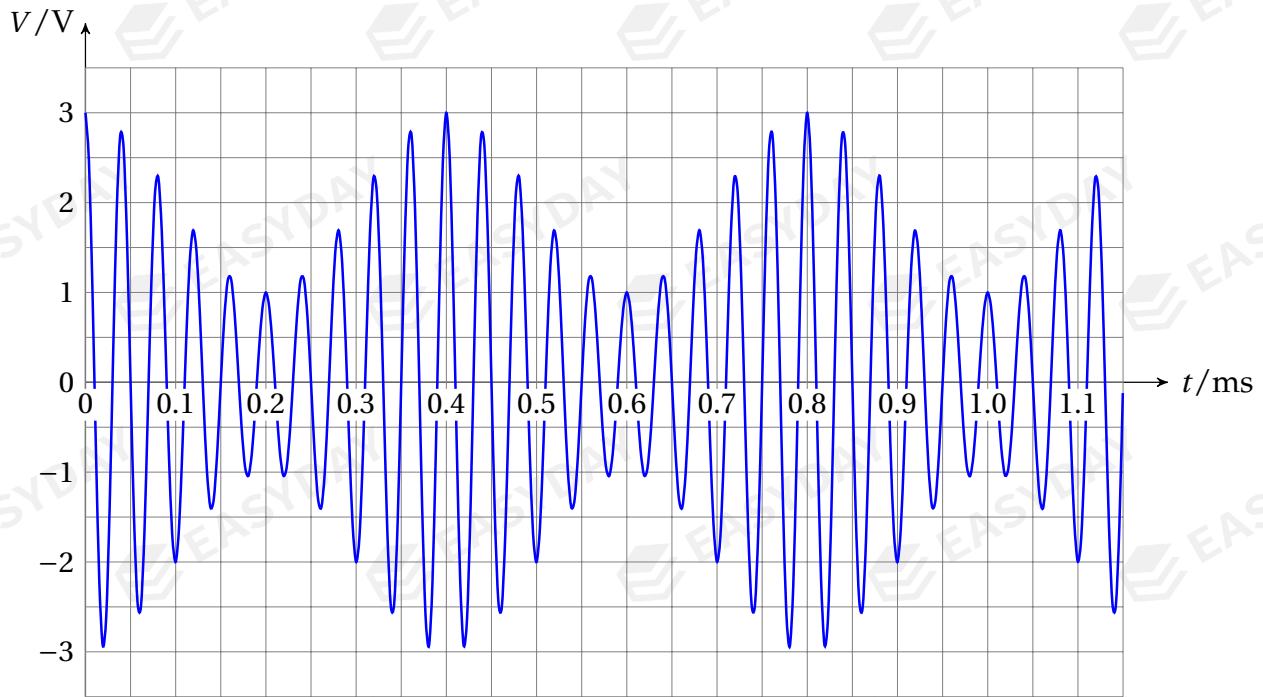
- size of bandwidth determines number of channels available in a given frequency range
smaller bandwidth for AM means more channels are possible

Example 16.1 A radio station is broadcasting several audio signals with same bandwidth.

Wavelengths of carrier waves available for this radio station ranges from 3.0×10^3 m to 1.5×10^4 m. The variation of voltage of a modulated signal is shown. (a) State the frequency of unmodulated carrier wave and the frequency of the signal wave. (b) Find the bandwidth of this signal. (c) Find the maximum number of radio channels that could be broadcast at the same time for this station.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) â 信息信号可以包含一定范围的频率 (例如音乐信号) , 两个边带将扩展成两个对称的驼峰强度 $f_c \pm f_m$, $f_c + f_{m,\max}$ 和 $f_c - f_{m,\max}$, 最大带宽 $B = 2 f_{m,\max}$ â 我们将带宽定义为调制信号占用的频率范围 - 对于单频信息信号, AM 波的带宽为: $B = (f_c + f_m) - (f_c - f_m) = 2 f_m$ - 对于多频信息信号, AM 波的带宽为: $B = (f_c + f_{m,\max}) - (f_c - f_{m,\max}) = 2 f_{m,\max}$ 对于声音信号, $f_m \leq 10^4 \text{ Hz}$, 因此与无线电波频率相比, 带宽在几 kHz 左右, AM 的带宽相当小, 带宽的大小决定了给定频率范围内可用的通道数量。AM 的带宽越小, 意味着可以使用的通道越多。示例 16.1 一个广播电台正在广播多个具有相同带宽的音频信号。该广播电台可用的载波波长范围为 $3.0 \times 10^3 \text{ m}$ 至 $1.5 \times 10^4 \text{ m}$ 。显示了调制信号的电压变化。(a) 说出未调制载波的频率和信号波的频率。(b) 找出该信号的带宽。(c) 找出该电台可以同时广播的无线电频道的最大数量。212



frequency of carrier wave: $f_c = \frac{1}{T_c} = \frac{1}{0.04 \text{ ms}} = 25 \text{ kHz}$

frequency of audio signal: $f_m = \frac{1}{T_m} = \frac{1}{0.40 \text{ ms}} = 2.5 \text{ kHz}$

bandwidth of the signal: $B = 2f_m = 2 \times 2.5 = 5.0 \text{ kHz}$

range of available carrier frequencies is found using $f = \frac{c}{\lambda}$:

maximum carrier frequency: $f_{\max} = \frac{c}{\lambda_{\min}} = \frac{3.0 \times 10^8}{3.0 \times 10^3} = 1.0 \times 10^5 \text{ Hz} = 100 \text{ kHz}$

minimum carrier frequency: $f_{\min} = \frac{c}{\lambda_{\max}} = \frac{3.0 \times 10^8}{1.5 \times 10^4} = 2.0 \times 10^4 \text{ Hz} = 20 \text{ kHz}$

number of radio channels: $n = \frac{f_{\max} - f_{\min}}{B} = \frac{100 - 20}{5.0} = 16$ □

Question 16.1 An amplitude modulated radio wave carries an audio signal of a single frequency. In the power spectrum, it is found to contain three frequencies: 45 kHz, 50 kHz, and 55 kHz. (a) What is the frequency of the carrier wave? (b) What is the frequency of the audio signal? (c) What is the bandwidth for this AM wave?

Question 16.2 A sinusoidal carrier wave has a frequency of 200 kHz. It is amplitude modulated for the broadcast of music that has frequencies ranging between 30 Hz and 4000 Hz. Sketch the power spectrum for the transmitted signal, hence or otherwise, find the bandwidth for this signal. You should label the frequency values on your graph.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇豪 16 电信 (Q) V / V t/ms 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 -3 -2 -1 0 1 2 3 - 载波频率 : $f_c = 1/T_c = 1/0.04 \text{ ms} = 25 \text{ kHz}$ 频率 : $f_m = 1/T_m = 1/0.40 \text{ ms} = 2.5 \text{ kHz}$ 带宽 : $B = 2f_m = 2 \times 2.5 = 5.0 \text{ kHz}$ 可用载波频率范围可使用 $f = c/\lambda$ 找到 : 最大载波频率 : $f_{\max} = c/\lambda_{\min} = 3.0 \times 10^8 / 3.0 \times 10^3 = 1.0 \times 10^5 \text{ Hz} = 100 \text{ kHz}$ 最小载波频率 : $f_{\min} = c/\lambda_{\max} = 3.0 \times 10^8 / 1.5 \times 10^4 = 2.0 \times 10^4 \text{ Hz} = 20 \text{ kHz}$ 无线电频道数量 : $n = (f_{\max} - f_{\min})/B = (100 - 20)/5.0 = 16$ 问题 16.1 调幅无线电波携带单一频率的音频信号。在功率谱中, 发现包含三个频率 : 45 kHz、50 kHz 和 55 kHz。(a) 载波的频率是多少? (b) 音频信号的频率是多少? (c) 该 AM 波的带宽是多少? 问题 16.2 正弦载波的频率为 200 kHz。它经过调幅, 用于播放频率范围在 30 Hz 到 4000 Hz 之间的音乐。绘制传输信号的功率谱, 从而或以其他方式找到该信号的带宽。您应该在图表上标记频率值。 213

16.1.2 frequency modulation

for frequency modulated carrier wave, amplitude is unaltered

frequency varies with time according to displacement of information signal

➤ shift in frequency of carrier wave is determined by a parameter called *frequency deviation*

frequency deviation is usually given in kHz V^{-1}

this means if displacement of information wave changes by one volt, frequency deviation gives the amount of change in frequency for the carrier wave

➤ FM wave give very complicated spectrum [113] [114]

as a consequence, each FM signal occupies very large bandwidth

Example 16.2 A carrier wave of amplitude 5.0 V and frequency 80 kHz is modulated in frequency by a sinusoidal information wave of amplitude 3.0 V and frequency 4000 Hz. The frequency deviation of the carrier wave is 2.0 kHz V^{-1} . Describe the variation of this frequency modulated carrier wave.

✎ amplitude of modulated wave remains unchanged, so it is still 5.0 V

as displacement of information wave varies between +3.0 V to -3.0 V, maximum frequency shift for the carrier wave is: $\pm 3.0 \times 2.0 = \pm 6.0 \text{ kHz}$

maximum frequency of FM wave: $f_{\max} = 80 + 6.0 = 86 \text{ kHz}$

minimum frequency of FM wave: $f_{\min} = 80 - 6.0 = 74 \text{ kHz}$

so frequency of modulated wave changes from a maximum value of 86 kHz to a minimum value of 74 kHz and then back to the maximum value repeatedly at a rate of 4000 s^{-1} □

Question 16.3 A carrier wave has a frequency of 900 kHz. It is frequency modulated by an audio signal of frequency 6.0 kHz and amplitude 2.5 V. The frequency deviation for modulation

[113] FM spectrum is so complicated that you are only supposed to know it is complicated. You will not be asked to sketch a power spectrum for FM wave in A-Levels.

[114] If a carrier wave $x_c = A_c \sin \omega_c t$ is to be frequency modulated by an information signal x_m , the FM wave takes the form: $x_c(t) = A_c \sin \left[\omega_c t + f_\Delta \int_0^t x_m t(t) dt \right]$, where f_Δ is the frequency deviation factor. In particular, for an information signal of single frequency $x_m = A_m \cos \omega_m t$, it can be shown that the FM wave becomes $x_c(t) = A_c \sin \left[\omega_c t + \frac{A_m f_\Delta}{\omega_m} \sin \omega_m t \right]$. Judging from the mathematical descriptions, FM wave is obviously much more complicated than the AM wave.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 16.1.2 频率调制对于调频载波, 幅度是根据信息信号的位移而随时间变化的不变频率 $\hat{\omega}$ 载波频率的偏移由称为频偏的参数决定频偏通常以 kHz V 为单位给出 -1 这意味着如果信息波的位移改变一伏, 频偏给出了载波频率的变化量 $\hat{\omega}$ FM 波给出非常复杂的频谱 [113] [114] 因此, 每个 FM 信号占用非常大的带宽示例 16.2 幅度为 5.0 V、频率为 80 kHz 的载波由幅度为 3.0 V、频率为 4000 Hz 的正弦信息波进行频率调制。载波的频率偏差为 2.0 kHz V⁻¹。描述该调频载波的变化。 - 调制波幅度不变, 信息波位移在 +3.0 V 至 -3.0 V 之间变化, 仍为 5.0 V, 载波最大频移为: $3.0 \times 2.0 = 6.0$ kHz FM 波最大频率: $f_{\max} = 80 + 6.0 = 86$ kHz FM 波最小频率: $f_{\min} = 80 - 6.0 = 74$ kHz, 因此调制波的频率从最大值 86 kHz 变化到最小值 74 kHz, 然后以 4000 s 的速率反复回到最大值 -1 问题 16.3 载波的频率为 900 kHz。它由频率为 6.0 kHz、幅度为 2.5 V 的音频信号进行频率调制。调制[113]FM频谱的频率偏差非常复杂, 以至于您只需要知道它是复杂的。在 A-Level 课程中, 您不会被要求绘制 FM 波的功率谱。 [114] 如果载波 $x_c = A_c \sin \omega_c t$ 由信息信号 x_m 进行频率调制, 则 FM 波的形式为: $x_c(t) = A_c \sin \left(\omega_c t + f \Delta \int x_m(t) dt \right)$, 其中 $f \Delta$ 是频率偏差因子。特别地, 对于单频 $x_m = A_m \cos \omega_m t$ 的信息信号, 可以证明 FM 波变为 $x_c(t) = A_c \sin \left(\omega_c t + \frac{A_m f \Delta}{\omega_m} \sin \omega_m t \right)$ 。从数学描述来看, FM 波显然比 AM 波复杂得多。 214

is 30 kHz V⁻¹. (a) Find the maximum and minimum frequency of the modulated wave. (b) Describe how the frequency of the modulated wave changes.

16.1.3 comparison between AM and FM

➤ relative advantages of AM transmission

- electronics for AM transmission is less complicated, so cheaper radio sets
- narrower bandwidth for AM channel, so more channels are available in a given range
- AM wave has lower frequencies, hence longer wavelengths
this means AM waves can better diffract around barriers (e.g., buildings, mountains)
each AM transmitter covers a greater area, so fewer AM transmitters required than FM

➤ relative advantages of FM transmission

- FM transmission is less affected by noise
noise would superimpose with transmitted signal, so amplitude is altered by noise
but for FM, variation of information signal is represented by variation in frequency
- greater bandwidth for FM, so better reproduction quality of sound
- FM transmission is more energy efficient
for AM waves, information is carried in the sidebands
but the sidebands only take a small fraction of total power

Question 16.4 Suggest two reasons why FM broadcasting is more costly than AM.

16.2 digital transmission

16.2.1 analogue & digital signals

an **analogue signal** a continuous signal that can take any value

a **digital signal** is a discrete signal that can only take 0's and 1's (or highs and lows), no intermediate value is allowed

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 为 30 kHz V -1 。 (a) 找出调制波的最大和最小频率。 (b) 描述调制波的频率如何变化。 16.1.3 AM 和 FM 之间的比较 AM 传输的相对优势 - AM 传输的电子设备不太复杂, 因此收音机更便宜 - AM 频道的带宽更窄, 因此在给定范围内可以使用更多的频道 - AM 波的频率较低, 因此波长更长, 这意味着 AM 波可以更好地绕障碍物 (例如建筑物、山脉) 进行衍射每个 AM 发射机覆盖的面积更大, 因此比 FM 需要的 AM 发射机更少 - FM 传输的相对优势 - FM 传输受噪声影响较小会与传输的信号叠加, 因此幅度会因噪声而改变, 但对于 FM, 信息信号的变化由频率的变化来表示 - FM 的带宽更大, 因此声音的再现质量更好 - FM 传输对于 AM 波来说更节能, 信息在边带中携带, 但边带只占总功率的一小部分问题 16.4 提出 FM 广播比 AM 更昂贵的两个原因。 16.2 数字传输 16.2.1 模拟和数字信号模拟信号可以取任意值的连续信号数字信号是只能取 0 和 1 (或高位和低位) 的离散信号, 不允许有中间值 215

16.2.2 regeneration

there are two major problems during the transmission of a signal: attenuation and noise

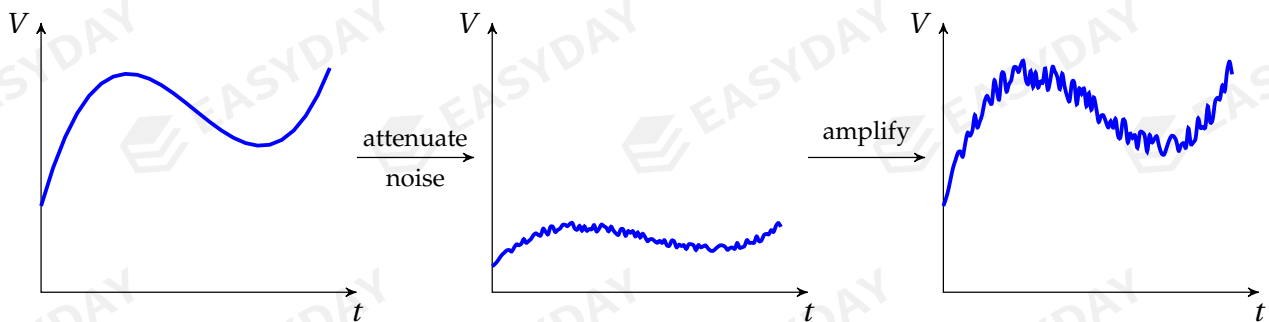
attenuation means gradual loss of signal power/intensity during transmission

noise refers to the unwanted random interference that distorts the signal

at the receiver end, we would want to remove noise and amplify the attenuated signal, so original signal is recovered, this process is called **regeneration**

➤ for an analogue signal, noise can not be distinguished

so noise will be amplified together with the weakened signal

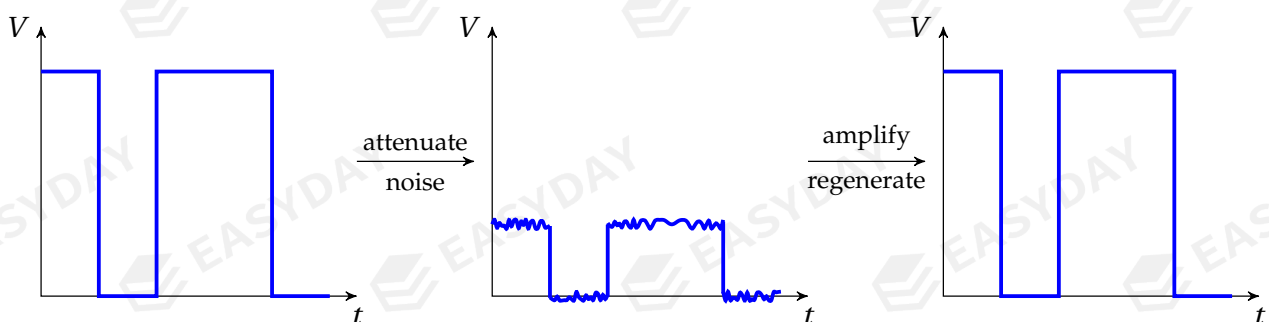


noise in an analogue system is amplified along with the weakened signal

➤ for a digital signal, it is either 0 or 1, any fluctuation must be due to noise

so noise can easily be removed from a digital signal, signal is regenerated

circuit for a simple regenerator amplifier is introduced in §15.4.3



noise in a digital system can be removed through regeneration

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇浩 16 电信 (Q) 16.2.2 再生信号在传输过程中存在两个主要问题：衰减和噪声衰减是指传输过程中信号功率/强度逐渐损失噪声是指无用的随机干扰，使接收端的信号失真，我们需要去除噪声并放大衰减的信号，从而恢复原始信号，这个过程称为再生 $\hat{\alpha}$ 对于模拟信号，噪声无法区分，因此噪声将与减弱的信号一起被放大 $V_t V_t V_t$ 衰减噪声放大模拟系统中的噪声与减弱的信号一起被放大 $\hat{\alpha}$ 对于数字信号，它是 0 或 1，任何波动都必须由噪声引起，因此噪声可以很容易地从数字信号中去除，第 15.4.3 节中介绍了简单再生放大器的信号再生电路。 V_t 衰减噪声放大再生数字系统中的噪声可以通过再生消除

advantage of digital transmission

transmission of data in digital form brings many advantages:

- noise can be removed through regeneration
- only two states are involved, so more reliable and easier electronics
- digital signals can be encrypted, so better security
- extra data can be added, so errors can be checked
- bandwidth is small, so rate of data transmission is higher
- data in digital forms are stored and processed more easily

16.2.3 binary numbers

most signals that make up speech, music, radio, and television are analogue signals

for more reliable and more efficient transmission of data, it is necessary to convert analogue signals into digital forms

this requires the use of 0's and 1's to represent different voltage levels

we next study the binary number system in which a number is expressed in 0's and 1's

- each digit in a binary number is called a *bit*
- for a binary number, each bit represents an increasing power of 2

the bit on left-hand/right-hand side of a binary number has highest/lowest value, called the *most/least significant bit*, or MSB/LSB in short

Example 16.3 Convert the binary numbers into decimal numbers: (a) 11001 (b) 10110.

$$\text{11011}_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 2^4 + 2^3 + 2^0 = 16 + 8 + 1 = 25$$

$$\text{110110}_2 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2^5 + 2^4 + 2^2 + 2^1 = 32 + 16 + 4 + 2 = 54 \quad \square$$

Example 16.4 Convert the decimal numbers into binary numbers: (a) 13 (b) 28.

$$\text{13} = 8 + 4 + 1 = 2^3 + 2^2 + 2^0 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = \text{1101}_2$$

$$\text{28} = 16 + 8 + 4 = 2^4 + 2^3 + 2^2 = 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = \text{11100}_2 \quad \square$$

- a n -bit binary system can represent 2^n different numbers

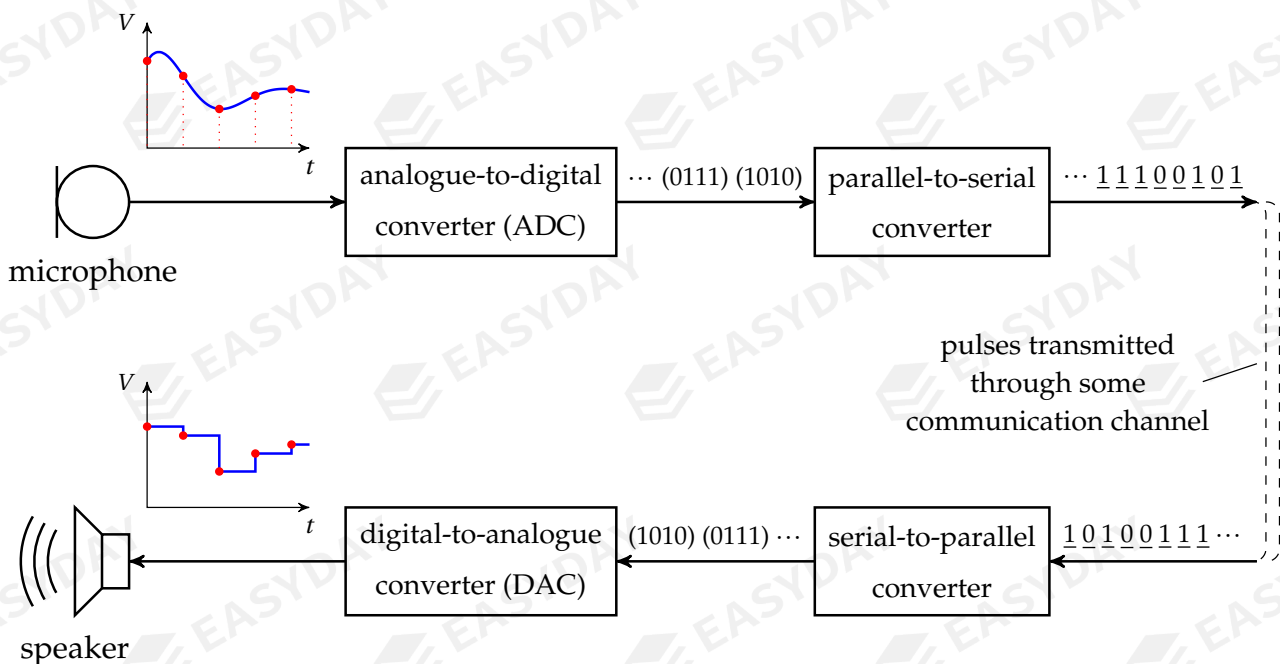
increasing the number of bits improves the precision of representation of voltage levels

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 数字传输的优点以数字形式传输数据带来许多优点：± 可以通过再生消除噪声 ± 仅涉及两种状态，因此电子学更加可靠和容易 ± 数字信号可以加密，因此安全性更好 ± 可以添加额外数据，因此可以检查错误 ± 带宽较小，因此数据传输速率较高 ± 数字形式的数据更容易存储和处理 16.2.3 二进制数大多数构成语音、音乐、无线电和无线电信号的信号电视是模拟信号，为了更可靠、更有效地传输数据，有必要将模拟信号转换为数字形式，这需要使用 0 和 1 来表示不同的电压电平。我们接下来研究二进制数系统，其中数字用 0 和 1 表示——二进制数中的每一位数字称为一个位——对于二进制数，每一位代表 2 的递增幂，二进制数的左/右手边的位有最高/最低值，称为最高/最低有效位，简称 MSB/LSB 例 16.3 将二进制数转换为十进制数：(a) 11001 (b) 10110. - 110112 = $1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 24 + 23 + 20 = 16 + 8 + 1 = 25$ 1101102 = $1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 25 + 24 + 22 + 21 = 32 + 16 + 4 + 2 = 54$ 例 16.4 将十进制数转换为二进制数：(a) 13 (b) 28. - 13 = $8 + 4 + 1 = 2^3 + 2^2 + 2^0 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1101_2$ 28 = $16 + 8 + 4 = 2^4 + 2^3 + 2^2 = 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 = 11100_2$ 例 16.5 n 位二进制系统可以表示 2^n 个不同的数字，增加位数可以提高电压电平表示的精度 217

16.2.4 analogue-to-digital conversion

digital transmission of an audio signal is represented by the diagram below:



transmission and reproduction of an audio signal in digital forms

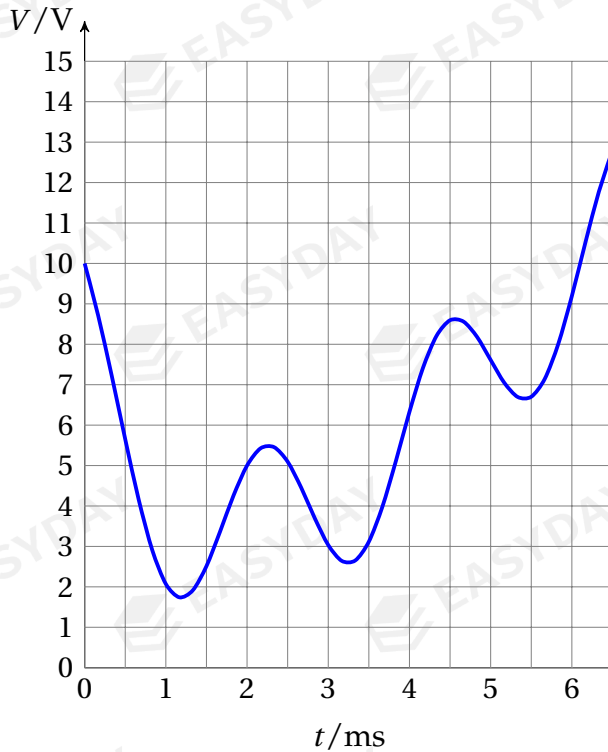
- audio signal collected by a *microphone* is an analogue signal
it must be converted into a digital signal for more reliable transmission
- **analogue-to-digital converter** (ADC) takes samples of the signal at regular time intervals
each sample value is then converted into a binary number according to voltage levels
- **parallel-to-serial converter** takes all bits of a binary number produced by ADC
each bit is then transmitted one after another
- at receiver end, the process is reversed to reproduce original audio signal
- quality of reproduction can be improved by two means:
 - increasing sampling frequency, so step width is reduced
high-frequency components in original signal can be detected
 - increasing number of bits for the binary system, so step height is reduced
smaller variations in voltages can be represented more precisely

Example 16.5 An analogue signal is passed through a analogue-to-digital converter for digital

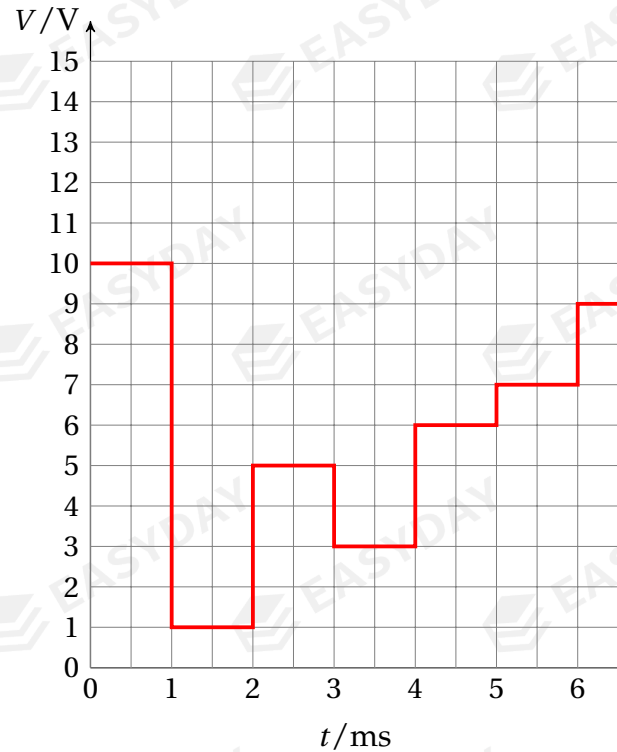
考纲提示 · Chapter 16 Telecommunication：旧版拓展内容；2022 起不再属于 9702 A Level，当前 Paper 4 不考。

杨宇豪 16 电信 (Q) 16.2.4 音频信号的模数转换数字传输如下图所示：V t ϕ - (0111) (10 10) - - - 1 1 1 0 0 1 0 1 麦克风模数转换器 (ADC) 并串转换器通过某些通信通道传输的脉冲 1 0 1 0 0 1 1 1 - - - (1010) (0111) - - - V t 数模转换器 (DAC) 串并转换器扬声器以数字形式传输和再现音频信号 $\hat{\alpha}$ 麦克风收集的音频信号是模拟信号，必须将其转换为数字信号才能更可靠地传输 $\hat{\alpha}$ 模数转换器 (ADC) 定期采样信号然后根据电压电平将每个样本值转换为二进制数 ϵ 并串转换器获取 ADC 产生的二进制数的所有位，然后逐位传输 ϵ 在接收端，过程相反以再现原始音频信号 ϵ 可以通过两种方式提高再现质量：- 增加采样频率，从而减小步长，可以检测到原始信号中的高频分量 - 增加二进制系统的位数，因此减小步长可以更精确地表示较小的电压变化示例 16.5 模拟信号通过模数转换器转换为数字 218

transmission. Samples are taken at a frequency of 1.0 kHz. Each sample is then converted into a 4-bit binary number. (a) State the values of all samples taken for this signal. (c) Draw the pattern of the reproduced signal.



original analogue signal



reproduced signal

🔗 sampling frequency is 1.0 kHz, so samples are taken every 1.0 ms

four-bit binary system is implemented, so $2^4 = 16$ voltage steps are represented

these voltage levels are 0, 1, 2, ..., 15 V

for a sample value between two voltage levels, it is rounded down

all samples taken for this signal can be summarised in the following table:

t/ms	0	1	2	3	4	5	6
sample value/V	10.0	2.0	5.0	3.0	6.3	7.6	9.2
voltage level/V	10	2	5	3	6	7	9
digital number	1010	0010	0101	0011	0110	0111	1001

these digital numbers are used to reproduce the analogue signal as shown

□

16.3 communication channels

signals can be transmitted through various communication channels

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇豪 16 TELECOMMUNICATION (Q) 传输。采样频率为 1.0 kHz。然后将每个样本转换为 4 位二进制数。
 。(a) 说明该信号的所有样本的值。(c) 绘制再现信号的模式。V/V t/ms 原始模拟信号 0 1 2 3 4 5 6 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 V/V t/ms 再现信号 0 1 2 3 4 5 6 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 - 采样频率为 1.0 kHz, 因此每 1.0 ms 采样一次, 采用四位二进制系统, 因此表示 $2^4 = 16$ 个电压步长, 对于两个电压电平之间的采样值, 这些电压电平为 0, 1, 2, 15 V, 向下舍入, 该信号的所有采样可总结在下表中:

t/ms	0	1	2	3	4	5	6	采样值/V	10.0	2.0	5.0	3.0	6.3	7.6	9.2	电压电平/V	10	2	5	3	6	7	9	数字
0	1	2	3	4	5	6	0	1	2	3	4	5	6	0	1	2	3	4	5	6	0	1	2	3

0011 0110 0111 1001 这些数字用于再现模拟信号, 如图所示 ä 16.3 通信通道信号可以通过各种通信方式传输

we next look at relative advantages and disadvantages of each channel

16.3.1 wire pairs

a **wire pair** consists of two insulated wires twisted around each other^[115]

- advantages of wire pairs: low cost, easy to install
- disadvantages of wire pairs
 - affected by noise (unwanted signals are easily picked up)
 - exist cross-linking (signal in one circuit easily induces a copy in a nearby circuit)
 - low security (easy to tap into a wire pair)
 - high attenuation (wire pair radiates electromagnetic waves when current in wire varies)
- applications: telephone to socket, amplifier to loudspeaker, etc.

16.3.2 coaxial cables

coaxial cable is a development of the wire pair

coaxial cable consists of a central core shielded by an outer conductor which is *earthed*

central core serves as the transmitting wire

outer conductor acts as signal return path and prevents interference getting into the core

- advantages of coaxial cables:
 - less noise and cross-linking (central core is shielded by outer conductor)
 - less attenuation (radiation of electromagnetic waves is prevented)
 - greater bandwidth, so higher rate of data transmission
 - higher security (more difficult to tap in than a wire pair)
- applications: television to aerial, computer to projector, landline to local exchange, etc.

^[115]The idea behind the wire pair is that, by putting the two wires very close together, they are exposed to noise equally on average. So noise can be cancelled out at receiver by taking the difference signal. Compared with a single wire, twisting reduces interference, but still, wire pair is more affected by noise compared with other channels of communication.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇豪 16 电信 (Q) 接下来我们看看各个通道的相对优缺点 16.3.1 线对线对由两根绝缘线绞合而成 [115] ã 线对的优点 : 成本低 , 易于安装 ã 线对的缺点 - 受噪声影响 (不需要的信号很容易被拾取) - 存在交叉链接 (一个电路中的信号很容易在附近电路中引起复制) - 安全性低 (易于接入线对) - 高衰减 (当线中电流变化时 , 线对辐射电磁波) ã 应用 : 电话到插座、放大器到扬声器等。 16.3.2 同轴电缆同轴电缆是线对的发展同轴电缆由中心线芯组成 , 外导体屏蔽 , 中心线接地中心线作为传输线外导体充当信号返回路径 , 防止干扰进入线芯 ã 同轴电缆的优点 : - 噪音较小和交联 (中心芯被外导体屏蔽) - 衰减更小 (防止电磁波辐射) - 带宽更大 , 因此数据传输速率更高 - 安全性更高 (比线对更难窃听) ã 应用 : 电视到天线、计算机到投影仪、固定电话到本地交换机等。 [115]线对背后的想法是 , 通过将两条线非常靠近地放在一起 , 它们平均暴露在噪声下。因此 , 可以通过获取差分信号在接收器处消除噪声。与单线相比 , 绞合可以减少干扰 , 但与其他通信通道相比 , 线对更容易受到噪声的影响。 220

16.3.3 optic fibres

optic fibre includes a core surrounded by a transparent cladding material

the core has a higher refraction index than the cladding, so this allows light to be kept in the core by *total internal reflection*

optic fibre can hence be used as a waveguide to transmit light signals

➤ advantages of optic fibres

- very low attenuation (light is totally internally reflected, so it stays in core)
- very little noise (light is immune to electromagnetic radiation)
- very large bandwidth (light has much higher frequencies than electrical signals)
- ideal for digital transmission (light pulses can be switched on and off rapidly)
- low weight and low cost (made of glass or plastic fibres)

➤ applications: internet broadband, transcontinental communication, etc.

16.3.4 radio waves

radio waves are part of electromagnetic spectrum with wavelengths $\lambda \approx 10 \text{ cm} \sim 10 \text{ km}$

radio waves can be used to transmit signals through different paths

➤ surface wave ($\lambda \approx 0.1 \sim 10 \text{ km}$, $f \approx 30 \text{ kHz} \sim 3 \text{ MHz}$)

surface waves travel close to surface of earth

long wavelength means they can diffract around obstacles

➤ sky wave ($\lambda \approx 10 \sim 100 \text{ m}$, $f \approx 3 \sim 30 \text{ MHz}$)

sky waves can be reflected by the *ionosphere* (a layer of charged particles in atmosphere)

sky waves travel over large distances via multiple reflections by ionosphere and ground

sky wave depends on atmosphere conditions so reception is not very reliable

➤ space wave ($\lambda \approx 10 \text{ cm} \sim 10 \text{ m}$, $f \approx 30 \text{ MHz} \sim 3 \text{ GHz}$)

short-wave radio waves can penetrate atmosphere and reach satellites

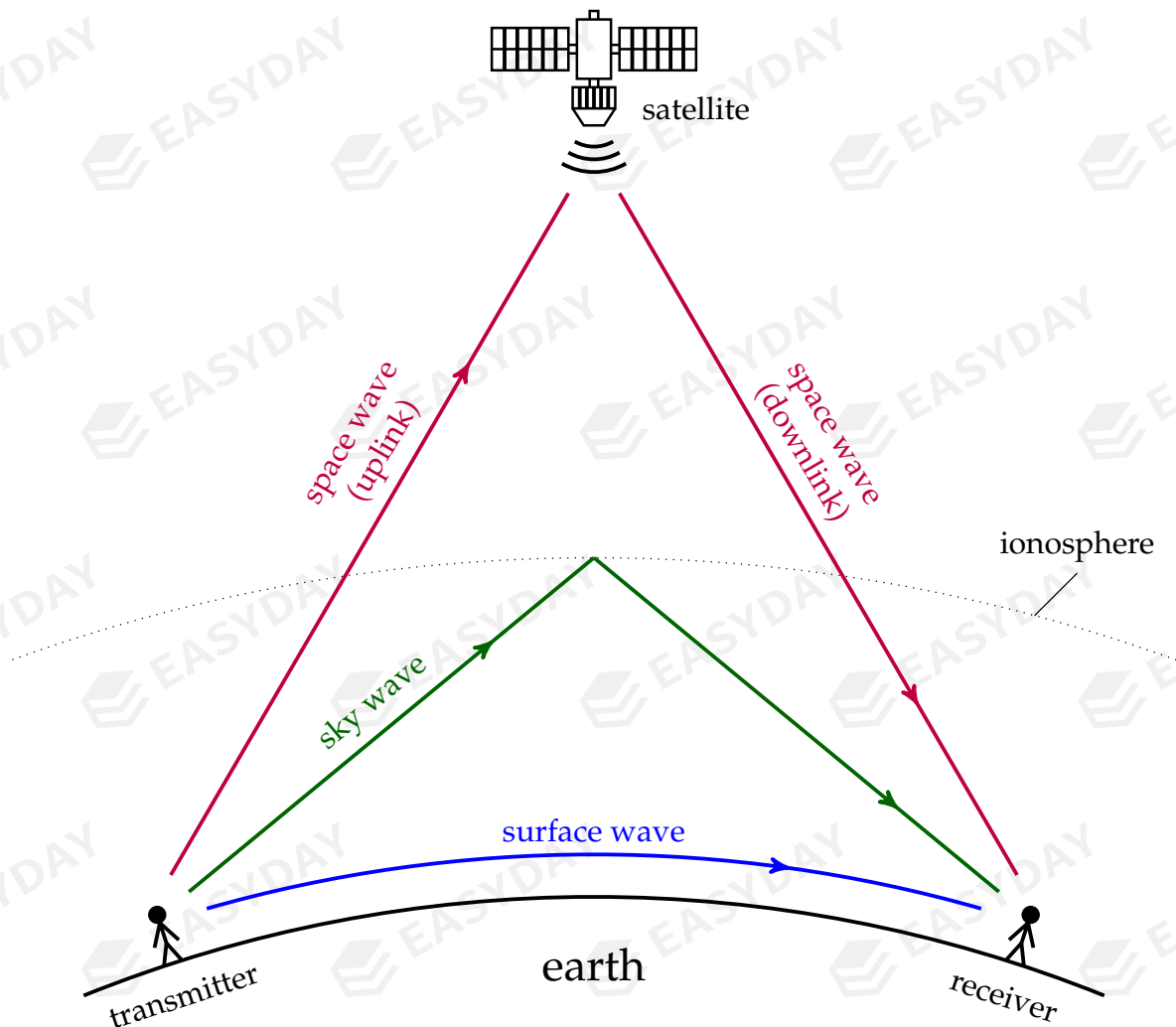
satellite can regenerate the signal and re-transmit them back to receiver

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 16.3.3 光纤光纤包括一个被透明包层材料包围的纤芯，该纤芯的折射率比包层高，因此这使得光可以通过全内反射保留在纤芯中，因此光纤可以用作传输光信号的波导。光纤的优点 - 衰减非常低（光完全在内部传播） - 非常小的噪声（光不受电磁辐射的影响） - 非常大的带宽（光的频率比电信号高得多） - 数字传输的理想选择（光脉冲可以快速打开和关闭） - 重量轻、成本低（由玻璃或塑料纤维制成） - 应用：互联网宽带、跨大陆通信等。

16.3.4 无线电波无线电波是电磁频谱的一部分，波长为 λ $10\text{ cm} \gg 10\text{ km}$ 无线电波可用于通过不同路径传输信号。表面波 ($\lambda \frac{1}{4} 0.1 \gg 10\text{ km}$, $f \frac{1}{4} 30\text{ kHz} \gg 3\text{ MHz}$) 表面波靠近地球表面传播长波长意味着它们可以绕着障碍物衍射。天波 ($\lambda \frac{1}{4} 10 \gg 100\text{ m}$, $f \frac{1}{4} 3 \gg 30\text{ MHz}$) 天波可以被电离层（大气中的一层带电粒子）反射天波在电离层上传播通过电离层和地面天波的多次反射实现长距离取决于大气条件，因此接收不是很可靠。空间波 ($\lambda \frac{1}{4} 10\text{ cm} \gg 10\text{ m}$, $f \frac{1}{4} 30\text{ MHz} \gg 3\text{ GHz}$) 短波无线电波可以穿透大气层并到达卫星卫星可以重新生成信号并将其重新传输回接收器 221

space waves are also used on surface of earth, but their short wavelength means they cannot diffract sufficiently, so transmission to receiver has to be *line-of-sight*



use of radio waves in telecommunication

communication satellites

- communication by satellite has several advantages compared with sky waves
 - satellite amplifies the signal for its return to surface, reception signal is stronger
 - space wave uses higher frequencies, this allows for greater bandwidth so more information can be transmitted per unit time
 - sky wave lacks stability, but communication by satellite is consistent
- uplink and downlink for satellite communication must use different frequencies

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

杨宇豪 16 电信 (Q) 空间波也在地球表面使用,但其波长短意味着它们不能充分衍射,因此必须以视距方式向接收器传输表面波天波空间波(上行链路)空间波(下行链路)卫星电离层发射机接收机地球在电信通信卫星中使用无线电波 - 卫星通信与天波相比有几个优点 - 卫星放大信号返回地面,接收信号更强 - 空间波使用更高的频率,这允许更大的带宽,因此单位时间可以传输更多的信息 - 天波缺乏稳定性,但卫星通信是一致的 - 卫星通信的上行和下行必须使用不同的频率 222

since uplink is greatly attenuated, downlink must be greatly amplified

to avoid uplink being swamped by downlink signals, different carrier frequencies are used

➤ one type of useful satellites are the *geostationary satellites*

geostationary satellite always stays in same position with respect to a ground observer

it has a period of 24 hours, moves from west to east in an equatorial orbit

- advantages

relative at rest, so no need to track, satellite dish can be fixed

- disadvantages

orbit is very high (radius of orbit is over 6 times of radius of earth), so long time delay

orbit is above equator, so cannot cover areas of high latitudes

➤ another type of useful satellites are the *polar satellites*

polar satellites pass over the two poles of the earth at around 1000 km above ground

- advantages

lower orbit, so shorter time delay

give better details for surface observation (e.g., street map, weather forecast, spy, etc.)

- disadvantages

not synchronous with ground observers, so satellite needs to be tracked

channel must be swapped between satellites for continuous communication

16.3.5 microwaves

microwaves have higher frequencies ($f \approx 3 \sim 300$ GHz) than radio waves ^[116]

high frequency means higher bandwidth, greater rate of data transmission is possible

but their wavelengths are shorter ($\lambda \approx 0.1 \sim 10$ cm), so not very able to diffract around barriers, transmission must be line-of-sight

➤ application: Wi-Fi, bluetooth connections, mobile phone networks, satellite links, etc.

^[116] There is no clear boundary of frequency (or wavelength) between radio waves and microwaves. Usually we take a few GHz (or a few cm) as the borderline.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 由于上行链路大大衰减, 下行链路必须大大放大, 以避免上行链路被下行链路信号淹没, 使用不同的载波频率 - 一种有用的卫星是对地静止卫星对地静止卫星始终相对于地面观测者保持在同一位置它有 24 小时的周期, 在赤道轨道上从西向东移动 - 相对静止时的优势, 因此无需轨道, 卫星天线可以固定 - 缺点轨道非常高 (轨道半径超过地球半径的 6 倍), 因此长时间延迟轨道在赤道以上, 因此无法覆盖高纬度地区 - 另一种有用的卫星是极地卫星, 极地卫星在距地面约 1 000 公里处越过地球的两极 - 优点是轨道较低, 因此较短的时间延迟为地面观测提供更好的细节 (例如街道地图、天气预报、间谍、等) - 缺点是与地面观测者不同步, 因此需要跟踪卫星, 必须在卫星之间交换信道以进行连续通信 16.3.5 微波微波的频率 ($f \sim 300 \text{ GHz}$) 比无线电波 [116] 高频率意味着更高的带宽, 可以实现更高的数据传输速率, 但它们的波长较短 ($\lambda \sim 0.1 \sim 10 \text{ cm}$), 因此不太能够绕过障碍物、传输必须是视距 $\hat{\alpha}$ 应用: Wi-Fi、蓝牙连接、移动电话网络、卫星链路等。[116]无线电波和微波之间没有明确的频率 (或波长) 界限。通常我们以几GHz (或几厘米) 作为界限。 223

causes of attenuation in communication channels

each communication channel suffers some degree of power loss

- wire pair/coaxial cable: heat loss due to resistance and radiation of energy
- optic fibre: scattering due to impurities and defects in optic fibre^[117]
- radio waves/microwaves: absorption by atmosphere/air molecules

16.4 the decibel scale

in telecommunication, engineers often want to evaluate the following:

- how much signal power is lost during transmission
- how the received power compares with noise level
- by what extent a signal is intensified if it is sent into an amplifier

these all involve computing ratios of powers of some kind, which are usually very large

engineers introduce a logarithmic scale to express power ratios, known as the **decibel scale**

with the decibel scale, power ratio $\frac{P_1}{P_2}$ is now expressed as $10\log \frac{P_1}{P_2}$ ^[118]

in electrical engineering, signal attenuation (degree of power decrease), signal-to-noise ratio and gain of amplifier (value of amplification) are usually given in decibels:

- $\text{attenuation} = 10\log \frac{P_{\text{in}}}{P_{\text{out}}}$
- $\text{signal-to-noise ratio} = 10\log \frac{P_{\text{out}}}{P_{\text{noise}}}$
- $\text{gain of amplifier} = 10\log \frac{P_{\text{out}}}{P_{\text{in}}}$

➤ unit for the decibel scale: dB

ratio in dB is actually unit free

dB is simply a reminder that we are using a logarithmic scale to express a ratio

➤ rationales for using the decibel scale

^[117] In practice, infra-red is usually used in optic fibres rather than visible light. Infra-red has longer wavelength, so they have better diffraction ability. Less scattering effect means attenuation is lower.

^[118] This is the logarithm to base 10, which can also be written as $\log_{10}$ or $\lg$.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) 通信通道衰减的原因每个通信通道都会遭受一定程度的功率损耗 – 线对/同轴电缆 : 由于电阻和能量辐射而产生的热损失 – 光纤 : 由于光纤中的杂质和缺陷导致的散射 [117] – 无线电波/微波 : 大气/空气分子的吸收 16.4 电信中的分贝标度, 工程师通常希望评估以下内容 : – 信号量有多少传输过程中功率损失 – 接收到的功率如何与噪声电平进行比较 – 如果信号被发送到放大器, 信号会被增强到什么程度这些都涉及计算某种功率比, 通常非常大的工程师引入对数标度来表示功率比, 称为分贝标度, 功率比 P_1/P_2 现在在电气工程中表示为 $10 \log P_1/P_2$ [118], 信号衰减 (功率的程度) 减小, 放大器的信噪比和增益 (放大值) 通常以分贝为单位 : – 衰减 = $10 \log P_{in}/P_{out}$ – 信噪比 = $10 \log P_{out}/P_{noise}$ – 放大器增益 = $10 \log P_{out}/P_{in}$ 分贝标度单位 : dB 比率实际上是无单位的 dB 只是提醒我们使用对数标度表达比率 $\hat{a}$ 使用分贝标度的基本原理 [117] 实际上, 光纤中通常使用红外线而不是可见光。红外线的波长较长, 因此具有较好的衍射能力。散射效应越小意味着衰减越低。 [118] 这是以 10 为底的对数, 也可以写为 $\log_{10}$ 或 $\lg$ 。 224

- ratios given in dB give more manageable numbers
- ratios given in dB can be added or subtracted [119]

Example 16.6 A signal of power 0.040 mW is sent into an amplifier. The gain of the amplifier is 36 db. What is the power for the output signal?

$$\Rightarrow 10 \log \frac{P_{\text{out}}}{P_{\text{in}}} = 36 \Rightarrow 10 \log \frac{P_{\text{out}}}{0.040} = 36 \Rightarrow P_{\text{out}} = 10^{3.6} \times 0.040 \approx 159 \text{ mW} \quad \square$$

Example 16.7 An optic fibre is known to have an attenuation of 0.12 dB km^{-1} . The transmitted signal sent into the optic fibre has a power of 50 mW. The noise level at receiver end is $8.0 \mu\text{W}$. If the signal-to-noise ratio at receiver is at least 25 dB, what is the maximum length of the fibre?

let's first find minimum reception power from the signal-to-noise ratio:

$$10 \log \frac{P_{\text{out}}}{P_{\text{noise}}} = 25 \Rightarrow \log \frac{P_{\text{out}}}{8.0 \times 10^{-6}} = 2.5 \Rightarrow P_{\text{out}} = 10^{2.5} \times 8.0 \times 10^{-6} \approx 2.53 \times 10^{-3} \text{ W}$$

$$\text{maximum attenuation in channel} = 10 \log \frac{P_{\text{in}}}{P_{\text{out}}} = 10 \log \frac{50 \times 10^{-3}}{2.53 \times 10^{-3}} \approx 130 \text{ dB}$$

$$\text{maximum length of optic fibre: } L = \frac{130}{0.12} \approx 108 \text{ km} \quad \square$$

Example 16.8 An electrical signal of 60 mW is transmitted along a cable of length 94 km. The cable has an attenuation of 3.2 dB km^{-1} . Repeater amplifiers of gain of 27 dB are placed every 8.0 km along the cable. What is the power of the received signal?

total attenuation in cable: $94 \times 3.2 = 300.8 \text{ dB}$

$$\text{number of amplifiers: } \frac{94}{8.0} = 11.75 \Rightarrow 11 \text{ amplifiers are placed}$$

$$\text{total gain due to amplifiers: } 11 \times 27 = 297 \text{ dB}$$

$$\text{so net attenuation during transmission: } 300.8 - 297 = 3.8 \text{ dB}$$

$$10 \log \frac{P_{\text{in}}}{P_{\text{out}}} = 3.8 \Rightarrow \log \frac{60}{P_{\text{out}}} = 0.38 \Rightarrow P_{\text{out}} = \frac{60}{10^{0.38}} \approx 25 \text{ mW} \quad \square$$

Question 16.5 A signal is transmitted through a cable of total length of 5.0 km. The signal attenuation per unit length is this cable is 12 dB km^{-1} . If the reception power is $2.9 \times 10^{-7} \text{ W}$, find the power of the input signal.

Question 16.6 An input signal of power 28 mW is passed along two cables followed by an

[119] This follows from the property of the logarithmic function: $\log x + \log y = \log(xy)$.

Note that: $10 \log \frac{P_1}{P_2} + 10 \log \frac{P_2}{P_3} = 10 \log \left(\frac{P_1}{P_2} \times \frac{P_2}{P_3} \right) = 10 \log \frac{P_1}{P_3}$. This means if we know the power ratio of P_1 to P_2 in dB and the power ratio of P_2 to P_3 in dB, we can immediately add the two numbers together to get the the power ratio of P_1 to P_3 in dB.

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 电信 (Q) – 以 dB 给出的比率给出了更易于管理的数字 – 以 dB 给出的比率可以相加或相减 [119] 示例 16.6 功率为 0.040 mW 的信号被发送到放大器。放大器的增益为 36 dB。输出信号的功率是多少？ $-10 \log P_{\text{out}} P_{\text{out}} = 36 \Rightarrow 10 \log P_{\text{out}} 0.040 = 3.6 \Rightarrow P_{\text{out}} = 10^{3.6} \times 0.040 \approx 159 \text{ mW}$ 示例 16.7 已知光纤的衰减为 0.12 dB km^{-1} 。发送到光纤中的传输信号的功率为 50 mW。接收器端的噪声水平为 8.0 (W。如果接收器处的信噪比至少为 25 dB，则光纤的最大长度是多少？– 让我们首先从信噪比中找到最小接收功率： $10 \log P_{\text{out}} P_{\text{noise}} = 25 \Rightarrow 10 \log P_{\text{out}} 8.0 \times 10^{-6} = 2.5 \Rightarrow P_{\text{out}} = 10^{2.5} \times 8.0 \times 10^{-6} \approx 2.53 \times 10^{-3} \text{ W}$ 通道中最大衰减 = $10 \log P_{\text{in}} P_{\text{out}} = 10 \log 50 \times 10^{-3} 2.53 \times 10^{-3} \approx 130 \text{ dB}$ 光纤最大长度： $L = 130 / 0.12 \approx 108 \text{ km}$ 示例 16.8 沿长度为 94 km 的电缆传输 60 mW 的电信号。该电缆的衰减为 3.2 dB km^{-1} 。沿电缆每 8.0 公里放置一个增益为 27 dB 的中继放大器。接收信号的功率是多少？– 电缆总衰减： $94 \times 3.2 = 300.8 \text{ dB}$ 放大器数量： $94 / 8.0 = 11.75 \Rightarrow$ 放置 11 个放大器放大器产生的总增益： $11 \times 27 = 297 \text{ dB}$ 因此传输期间的净衰减： $300.8 - 297 = 3.8 \text{ dB}$ $10 \log P_{\text{in}} P_{\text{out}} = 3.8 \Rightarrow 10 \log 60 P_{\text{out}} = 0.38 \Rightarrow P_{\text{out}} = 60 / 100.38 \approx 25 \text{ mW}$ 问题 16.5 信号通过总长度为 5.0 km 的电缆传输。该电缆每单位长度的信号衰减为 12 dB km^{-1} 。如果接收功率为 $2.9 \times 10^{-7} \text{ W}$ ，求输入信号的功率。问题 16.6 功率为 28 mW 的输入信号沿着两根电缆传递，然后是 [119] 这由对数函数的性质得出： $\log x + \log y = \log(xy)$ 。请注意： $10 \log P_1 P_2 + 10 \log P_2 P_3 = 10 \log (P_1 P_2 \times P_2 P_3) = 10 \log P_1 P_3$ 。这意味着，如果我们知道 P_1 与 P_2 的功率比（以 dB 为单位）和 P_2 与 P_3 的功率比（以 dB 为单位），我们可以立即将这两个数字相加，得到 P_1 与 P_3 的功率比（以 dB 为单位）。225

amplifier at the receiver end for output. If the attenuation in the cables are 29 dB and 38 dB respectively, and the amplifier has a gain of 43 dB. What is the output power from the amplifier?

Question 16.7 Input signal of power 12 mW is sent along an optic fibre. The noise level at the receiver end of the optic fibre is $0.75 \mu\text{W}$. It is required that the signal-to-noise ratio in the receiver must not fall below 35 dB. For this transmission, the maximum uninterrupted length of the optic fibre is 48 km. (a) What is the minimum acceptable power of the signal at the receiver? (b) What is the attenuation per unit length of the fibre?

考纲提示 · Chapter 16 Telecommunication : 旧版拓展内容 ; 2022 起不再属于 9702 A Level , 当前 Paper 4 不考。

Yuhao Yang 16 TELECOMMUNICATION (Q) 放大器在接收端用于输出。如果电缆中的衰减分别为 29 dB 和 38 dB, 放大器的增益为 43 dB。放大器的输出功率是多少? 问题 16.7 功率为 12 mW 的输入信号通过光纤发送。光纤接收器端的噪声水平为 0.75 W。要求接收器的信噪比不得低于 35 dB。对于这种传输, 光纤的最大不间断长度为 48 公里。(a) 接收器可接受的最小信号功率是多少? (b) 光纤单位长度的衰减是多少? 226

Appendix A. Glossary

1. circular motion

- *radian*: angle subtended at centre of circle by arc equal in length to the radius
- *angular displacement*: angle through which an object moving along a circular path
- *angular velocity*: rate of change in angular displacement
- *centripetal acceleration*: acceleration of an object moving in a circle that always points towards the centre of the circle
- *centripetal force*: resultant force acting on an object moving in a circle, this force always points towards the centre of the circle

2. gravitational fields

- *Newton's law of gravitation*: gravitational force between two point objects is directly proportional to the product of their masses, and inversely proportional to the square of their separation ($F \propto \frac{m_1 m_2}{r^2}$)
- *gravitational field*: a region of space in which a mass experiences a force
- *direction of gravitational field (at a point)/direction of gravitational field line*: direction of force acting on a test mass (at that point)
- *gravitational field strength*: gravitational force per unit mass ($g = \frac{F}{m}$)
- *gravitational potential (at a point)*: work done by to bring unit mass from infinity (to this point)

3. oscillations

- *oscillation*: an object moves back and forth about its equilibrium position
- *simple harmonic motion*: acceleration is proportional to displacement from the equilibrium position, and is opposite to the displacement ($a = -\omega^2 x$)
- *free oscillation*: an oscillation at its own natural frequency with no external driving force/no energy gain or loss/constant amplitude
- *forced oscillation*: an oscillation under an external driving force

杨宇豪词汇表附录 A. 术语表 1. 圆周运动 • 弧度：长度等于半径的圆弧在圆心处所成的角度 • 角位移：物体沿圆周路径运动的角度 • 角速度：角位移的变化率 • 向心加速度：物体做圆周运动且始终指向圆心的加速度 • 向心力：作用在做圆周运动的物体上的合力，此力总是指向圆心 2. 引力场 • 牛顿万有引力定律：两点物体之间的引力与它们质量的乘积成正比，与它们的距离的平方成反比 ($F = G \frac{m_1 m_2}{r^2}$) • 引力场：质量在其中受到力的空间区域 • 引力场方向（在一点）/引力场方向引力场线：作用在测试质量上的力方向（在该点） • 引力场强度：每单位质量的引力 ($g = F/m$) • 引力势（在某一点）：单位质量从无穷大（到该点）所做的功 3. 振荡 • 振荡：物体围绕其平衡位置来回移动 • 简谐振动：加速度与距平衡位置的位移成正比，为与位移相反 ($a = -\omega^2 x$) • 自由振荡：在没有外部驱动力/没有能量增益或损失/恒定振幅的情况下以其自身固有频率进行的振荡 • 受迫振荡：在外部驱动力下的振荡 227

- *kinematic properties of simple harmonics*: maximum speed but zero acceleration at equilibrium position, maximum acceleration but zero speed at maximum displacement
- *damping*: resistive forces causing gradual decrease in amplitude/loss of total energy
- *effects of damping*: amplitude/total energy of oscillation decreases exponentially, frequency/period is almost unchanged (or slightly reduced)
- *resonance*: when frequency of driving force is close to natural frequency of an oscillator, amplitude of oscillation becomes maximum

4. thermal physics

- *internal energy*: sum of the random distribution of kinetic energy and potential energy of molecules ($U = E_k + E_p$)
- *thermal energy*: energy transferred from one object to another because of a difference in their temperatures
- *thermal equilibrium*: when two objects in contact have the same temperature, there is no net flow of thermal energy
- *specific heat capacity*: thermal energy required per unit mass to cause a unit change in temperature ($c = \frac{Q}{m\Delta T}$)
- *specific latent heat*: thermal energy required per unit mass to change the state of matter with no change in temperature ($L = \frac{Q}{m}$)
- *first law of thermodynamics*: change in internal energy of a thermodynamic systems equals the heat supplied and external work done on the system ($\Delta U = Q + W$)

5. ideal gases

- *ideal gas*: a gas that obeys the equation $pV = nRT$ or $pV = NkT$ at any pressure p , volume V and thermodynamic temperature T
- *assumptions of ideal gas model*: gas molecules are widely separated, so no intermolecular forces; collisions between molecules are perfectly elastic; volume of molecules is negligible compared to volume occupied by the gas; molecules are in continuous random motion; molecules travel in straight lines between collisions, etc.

杨宇豪词汇表 • 简谐波的运动特性：平衡位置处最大速度但加速度为零，最大位移处加速度最大但速度为零 • 阻尼：导致振幅逐渐减小/总能量损失的阻力 • 阻尼效应：振幅/总振动能量呈指数下降，频率/周期几乎不变（或略有减小） • 共振：当驱动力的频率接近振荡器的固有频率时，振动振幅最大 4. 热物理 • 内能：总和分子动能和势能的随机分布 ($U = E_k + E_p$) • 热能：由于温度差异而从一个物体传递到另一个物体的能量 • 热平衡：当两个物体接触时具有相同的温度时，没有净热能流动 • 比热容：每单位质量引起单位温度变化所需的热能 ($c = Q / m \Delta T$) • 比潜热：每单位所需的热能温度不变时改变物质状态的质量 ($L = Q / m$) • 热力学第一定律：热力学系统内能的变化等于系统提供的热量和外部所做的功 ($\Delta U = Q + W$) 5. 理想气体 • 理想气体：在任何压力 p 、体积 V 和热力学温度 T 下都遵循方程 $pV = nRT$ 或 $pV = NkT$ 的气体 • 理想气体模型的假设：气体分子间隔很远，因此没有分子间作用力；分子间的碰撞是完全弹性的；与气体所占的体积相比，分子的体积可以忽略不计；分子作连续的随机运动；分子在碰撞等之间沿直线运动。228

- *cause of gas pressure*: gas molecules move randomly, as they hit the wall of the container and rebound, the change in momentum gives rise to a force, contribution of many molecules produces a pressure on the container
- *relating kinetic energy with temperature*: mean kinetic energy of particles of an ideal gas is directly proportional to its thermodynamic temperature ($\langle E_k \rangle = \frac{3}{2} kT$)

6. electric fields

- *Coulomb's law*: electric force between two point objects is proportional to the product of their electric charges, and the inverse square of their separation ($F \propto \frac{q_1 q_2}{r^2}$)
- *electric field*: an area of space in which a charged object experiences a force
- *direction of electric field (at a point)/direction of electric field line*: direction of force acting on a positive charge (at that point)
- *electric field strength*: electric force per unit positive charge ($E = \frac{F}{q}$)
- *electric potential (at a point)*: work done to bring a unit positive charge from infinity (to that point)
- *similarities between gravitational and electric fields*: both forces obey inverse square law; field strength inversely proportional to distance squared; potential inversely proportional to distance; potential is zero at infinity, etc.
- *differences between gravitational and electric fields*: gravity is always attractive, but electric field can be either attractive or repulsive; gravitational potential is always negative, while electric potential can take both signs, etc.
- *relating field strength and potential*: field strength equals (negative) gradient of potential ($E = -\frac{dV}{dr}$ for electric fields, $g = -\frac{d\phi}{dr}$ for gravitational fields)
- *electric field inside charged conductor/metal*: charge distribution is stable, no electric force on charge carriers/free electrons, so field strength inside conductor must be zero; moving charge inside conductor requires no work done, so potential is constant

7. capacitance

- *capacitance of a parallel-plate capacitor*: ratio of charge on one plate of a capacitor to its p.d.

杨宇浩词汇表 • 气体压力的成因：气体分子随机运动，撞击容器壁并反弹，动量的变化产生力，许多分子的贡献在容器上产生压力 • 动能与温度的关系：理想气体粒子的平均动能与其热力学温度成正比 ($\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$) 6. 电场 • 库仑定律：两点物体之间的电力为正比于它们的电荷乘积及其间距的平方反比 ($F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$) • 电场：带电物体在其中受到力的空间区域 • 电场方向（在某一点）/电场线方向：作用在正电荷上的力方向（在该点） • 电场强度：每单位正电荷的电力 ($E = F/q$) • 电势（在某一点）：将单位正电荷从无穷大（到该点）所做的功 • 引力场和电场之间的相似之处：两种力都遵循平方反比定律；场强与距离的平方成反比；电势与距离成反比；无穷远时电势为零，等等。 • 引力场和电场之间的差异：引力总是有吸引力的，但电场可以是吸引的，也可以是排斥的。引力势总是负的，而电势可以取两个符号等。 • 场强和电势的关系：场强等于（负）电势梯度（电场为 $E = -dV/dr$ ，引力场为 $g = -d\phi/dr$ ） • 带电导体/金属内部的电场：电荷分布稳定，电荷载流子/自由电子上没有电力，因此场导体内部的强度必须为零；在导体内移动电荷不需要做功，因此电势恒定 7. 电容 • 平行板电容器的电容：电容器一个板上的电荷与其 p.d. 的比率。

229

between the plates ($C = \frac{Q}{V}$)

- *function of capacitors*: store and release energy, separate charges, smooth voltages, block d.c. signals, etc.
- *how a capacitor stores energy*: to charge a capacitor, positive and negative charges are pushed onto each plate, work must be done to separate the charges, this converts to electrical energy stored in capacitor
- *explaining the decrease in p.d. for a discharging capacitor*: as p.d. across capacitor decreases, discharging current it drives through resistor decreases, rate of discharging decreases, so rate of decrease in p.d. decreases
- *time constant and rate of charging/discharging*: greater time constant $\tau = RC$ means slower rate of charging/discharging: greater resistance R then charging/discharging current becomes smaller; greater capacitance C then more charges must be transferred

8. magnetic field

- *magnetic field*: a region of space where a magnet/a moving charge/a current-carrying conductor experiences a force
- *magnetic flux density*: magnetic force per unit length acting on a wire carrying unit current at right angle to the magnetic field ($B = \frac{F}{Il}$)
- *motion of charged particles in uniform magnetic fields*: magnetic force acts at right angle to particle's velocity, so direction of motion keeps changing; uniform field so this force is constant; hence magnetic force provides centripetal force for circular/spiral motion
- *velocity selector*: charged particles pass through a region where a uniform electric field and a uniform magnetic field are applied at right angles, if electric and magnetic forces on the particles are equal but opposite ($Eq = Bqv$), these particles moving at $v = \frac{E}{B}$ are not deflected
- *cause of Hall effect*: charge carriers are deflected by magnetic force, causing an accumulation of electric charges on one side of the material, hence a voltage is established
- *how a stable Hall voltage is established*: redistribution of charges causes an internal electric field, electric force on charge carriers opposes deflection of charge carriers due to magnetic

Yuhao Yang 术语表极板之间 ($C = Q/V$) • 电容器的功能：储存和释放能量、单独充电、平滑电压、阻断直流。信号等 • 电容器如何存储能量：要给电容器充电，正电荷和负电荷被推到每个极板上，必须做功来分离电荷，这会转换为存储在电容器中的电能 • 解释 p.d 的减小。对于放电电容器：如 p.d. 电容器两端的电容减小，它驱动通过电阻的放电电流减小，放电速率减小，因此 p.d 的减小速率。减小 • 时间常数和充电/放电速率：时间常数 $\tau = RC$ 越大，充电/放电速率越慢：电阻 R 越大，则充电/放电电流变小；电容 C 越大，则必须转移的电荷越多 8. 磁场 • 磁场：磁体/移动电荷/载流导体受到力的空间区域 • 磁通密度：每单位长度的磁力作用在与磁场成直角的载有单位电流的导线上 ($B = F/I$) • 带电粒子在均匀磁场中的运动：磁力与磁场成直角作用物体的速度，所以运动方向不断变化；均匀场，因此该力是恒定的；因此，磁力为圆周/螺旋运动提供向心力 • 速度选择器：带电粒子穿过一个以直角施加均匀电场和均匀磁场的区域，如果粒子上的电场力和磁力大小相等但方向相反 ($E q = B q v$)，这些以 $v = E/B$ 运动的粒子不会偏转 • 霍尔效应的原因：载流子被磁力偏转，导致电荷在材料的一侧积累，因此建立电压 • 如何建立稳定的霍尔电压：电荷的重新分布导致内部电场，电荷载流子上的电力抵抗由于磁场导致的电荷载流子的偏转 230

force, the two forces tend to reach equilibrium, Hall voltage then stabilises

- *how to increase Hall voltage*: increase magnetic field/current (larger magnetic force, so greater p.d. needed to obtain equilibrium); use semi-conductor (smaller charge number density n); use a thin slice (smaller thickness t) ($V_H = \frac{BI}{ntq}$)
- *specific charge*: ratio of electric charge of a particle to its mass

9. electromagnetic induction

- *magnetic flux*: magnetic flux density $\times$ closed area at right angle to the field ($\phi = BA$)
- *magnetic flux linkage*: number of turns $\times$ magnetic flux density $\times$ closed area at right angles to the field ($\Phi = NBA$)
- *Faraday's law*: induced e.m.f. is proportional to the rate of change in magnetic flux (linkage) ($\mathcal{E} = \frac{\Delta\Phi}{\Delta t}$)
- *Lenz's law*: induced e.m.f./current is always in a direction to produce opposing effects to the original change in magnetic flux
- *eddy current*: when a metal disc/a bulk conductor experiences changes in magnetic flux/cutting of field lines, an e.m.f. is induced; if non-uniform cutting over whole disc/conductor, different e.m.f. is induced in different parts, leading to eddy currents
- *effects of induced currents*: magnetic force on induced currents opposes motion; induced currents dissipate energy as heat loss

10. alternating current

- *alternating current*: a current whose direction changes with time
- *r.m.s. voltage/current*: r.m.s. voltage/current of an alternating current equals the value of a steady voltage/current that produces the same mean power
- *principle of transformers*: alternating current in primary coils generates a changing magnetic flux, this flux is linked with secondary coils through transformer's iron core, according to Faraday's law, an output e.m.f. is induced in secondary coils
- *why transmitting electricity at high voltages*: high voltage means smaller current in cables (for same power output from power station), so less power loss during transmission

杨宇豪词汇表力，两个力趋于平衡，霍尔电压趋于稳定 • 如何增加霍尔电压：增加磁场/电流（磁力越大，需要越大的p.d.才能达到平衡）；使用半导体（较小的电荷数密度n）；使用薄片（较小厚度 t） ($V_H = B I / n t q$) • 比电荷：粒子的电荷与其质量之比 9. 电磁感应 • 磁通量：磁通密度 $\times$ 与磁场成直角的闭合面积 ($\phi = B A$) • 磁通链：匝数 $\times$ 磁通密度 $\times$ 与磁场成直角的闭合面积 ($\Phi = N B A$) • 法拉第定律：感应电动势与磁通量的变化率（联动）成正比 ($\mathcal{E} \propto \Delta \Phi / \Delta t$) • 楞次定律：感应电动势/电流总是在一个方向上产生与原始磁通量变化相反的效应 • 涡流：当金属盘/块体导体经历磁通量/磁力线切割的变化时，电动势会产生与磁通量变化率相反的方向。被诱导；如果整个圆盘/导体上的切割不均匀，则不同的电动势在不同部位产生感应，导致涡流。 • 感应电流的影响：感应电流上的磁力阻碍运动；感应电流以热损失的形式消耗能量 10. 交流电 • 交流电：方向随时间变化的电流 • 均方根值。电压/电流：均方根交流电的电压/电流等于产生相同平均功率的稳定电压/电流值 • 变压器原理：初级线圈中的交流电产生变化的磁通量，该磁通量通过变压器的铁芯与次级线圈相连，根据法拉第定律，输出电动势为是在次级线圈中感应的 • 为什么要在高压下传输电力：高压意味着电缆中的电流更小（对于电站输出的相同功率），因此传输过程中的功率损耗更少 231

- *advantages of alternating currents in electricity supply*: a.c. voltages can be easily changed by transformers; a.c. voltages can be easily created from generators
- *rectification*: converting an alternating current into a direct current (using diodes)
- *smoothing*: reduce ripples/fluctuations of output voltage (using capacitor)
- *how to increase the amount of smoothing*: use larger capacitor for same load

11. quantum physics

- *photon*: a packet/quanta of electromagnetic energy
- *Einstein relation*: photon energy is proportional to frequency of radiation ($E = hf$)
- *photoelectric effect*: when electromagnetic radiation shines upon a metal, electrons absorb photon energy and break free from metal surface
- *threshold frequency*: minimum frequency of incoming radiation/photons to release electrons from metal surface
- *work function (energy)*: minimum energy (of a photon) required for an electron to be emitted from surface of metal
- *intensity of light*: power of radiation per unit cross sectional area, which is proportional to number of incoming photons per unit time and frequency of radiation ($I \propto N \times hf$)
- *evidence provided by photoelectric effect for photon theory of light*: there exists a threshold frequency below which no electron is emitted; (maximum) kinetic energy of emitted electrons depends on frequency; electron emission rate depends on intensity of incident radiation; emission of electrons is immediate as soon as radiation is incident, etc.
- *why emitted photoelectrons have a maximum kinetic energy*: electrons with maximum K.E. are released from the surface, for electrons below the surface, additional energy is required to bring them to surface
- *why emission of electrons is immediate*: interaction between photon and electron is one-to-one, so no time delay
- *energy level (of electrons)*: electrons in an atom can only take certain values of energies, no intermediate state allowed

Yuhao Yang 词汇表 • 交流电在供电方面的优势：a.c.电压可以通过变压器轻松改变；交流电发电机可以轻松产生电压 • 整流：将交流电转换为直流电（使用二极管） • 平滑：减少输出电压的纹波/波动（使用电容器） • 如何增加平滑量：对于相同的负载使用更大的电容器 11. 量子物理学 • 光子：电磁能的包/量子 • 爱因斯坦关系：光子能量与辐射频率成正比 ($E = hf$) • 光电效应：当电磁辐射照射到物体上时金属，电子吸收光子能量并脱离金属表面 • 阈值频率：从金属表面释放电子的入射辐射/光子的最小频率 • 功函数（能量）：从金属表面发射电子所需的最小能量（光子） • 光强度：每单位横截面积的辐射功率，与每单位时间的入射光子数和辐射频率 ($I / N \times hf$) 成正比 • 光电效应为光子理论提供的证据光：存在一个阈值频率，低于该频率则不会发射电子；发射电子的（最大）动能取决于频率；电子发射率取决于入射辐射的强度；一旦辐射入射，电子就会立即发射，等等。 • 为什么发射的光电子具有最大动能：具有最大 K.E 的电子。从表面释放，对于表面以下的电子，需要额外的能量才能将它们带到表面 • 为什么电子发射是立即的：光子和电子之间的相互作用是一对一的，因此没有时间延迟 • （电子的）能级：原子中的电子只能采取一定的能量值，不允许中间态 232

- *relate line spectrum to discrete energy levels*: each line in the spectrum represents photon of a specific energy, photons are emitted/absorbed as a result of energy changes for electrons, specific energy changes so discrete/quantised energy levels for electrons
- *emission spectrum*: electrons of an excited/hot gas transit/jump from high energy levels to low levels, photons with energies equal to change in electron energy levels are released, so only photons with specific wavelengths/frequencies are emitted, giving a set of discrete bright lines
- *absorption spectrum*: passing a white light through a cool gas, electrons absorb photons of correct energies to transit/jump from low energy levels to high levels, only photons with specific wavelengths/frequencies are absorbed, these photons are re-emitted in all directions via de-excitation, giving a set of discrete dark lines in a continuous background
- *de Broglie wavelength*: wavelength of a particle moving with a certain momentum/speed ($\lambda = \frac{h}{p}$)
- *electron diffraction*: passing an electron beam through a thin metal crystal, a set of concentric rings are observed on the fluorescent screen, such diffraction patterns show the wave-like behaviour of electrons

12. nuclear physics

- *mass-energy equivalence*: mass is equivalent to energy ($E = mc^2$)
- *binding energy*: minimum energy required to separate the nucleons (protons and neutrons) in a nucleus to infinity
- *mass defect*: difference between mass of a nucleus and total mass of its constituent nucleons at infinity
- *binding energy and stability of nucleus*: larger binding energy per nucleon means higher stability for nucleus (more difficult to break up this nucleus)
- *binding energy and nuclear reaction*: a nuclear reaction is energetically possible if there is an increase in total binding energy after the reaction
- *nuclear fission*: a large nucleus splits into two smaller nuclei of similar sizes and several

杨宇豪词汇表 • 将线谱与离散能级联系起来：谱中的每条线代表特定能量的光子，光子由于电子能量变化而发射/吸收，特定能量变化，因此电子的离散/量子化能级 • 发射谱：激发/热气体的电子从高能级跃迁/跳跃到低能级，能量等于电子能级变化的光子被释放，因此只有发射特定波长/频率，给出一组离散的亮线 • 吸收光谱：让白光穿过冷气体，电子吸收正确能量的光子，以从低能级传输/跳跃到高能级，只有特定波长/频率的光子被吸收，这些光子通过去激发向各个方向重新发射，在连续背景中给出一组离散的暗线 • 德布罗意波长：以一定速率运动的粒子的波长动量/速度 ($\lambda = h p$) • 电子衍射：电子束穿过薄金属晶体，在荧光屏上观察到一组同心环，这种衍射图案显示电子的波状行为 12. 核物理 • 质能等价：质量等于能量 ($E = mc^2$) • 结合能：将原子核中的核子（质子和中子）分离到无穷大所需的最小能量 • 质量缺陷：无穷远处原子核的质量与其组成核子的总质量之间的差值 • 原子核的结合能和稳定性：每个核子的结合能越大，意味着原子核的稳定性越高（更难分裂该原子核） • 结合能和核反应：如果反应后总结结合能增加，则核反应在能量上是可能的 • 核裂变：一个大原子核分裂成两个大小相似的较小原子核和几个 233

neutrons

- *nuclear fusion*: two light nuclei combine into a heavier/larger nucleus
- *why fusion requires very high temperature*: for positively-charged nuclei to get close and fuse together, high kinetic energy is needed to overcome their electric repulsion, high K.E. so high temperature
- *radioactive decay*: spontaneous and random emission of ionising radiation (α -, β - or γ -radiation) from unstable nuclei
- *spontaneity*: rate of decay is not affected by external conditions (such as temperature, pressure, etc.)
- *randomness*: one cannot exactly predict when a particular nucleus will decay
- *activity*: rate at which nuclei decay/number of decay events per unit time
- *decay constant*: probability of a nucleus to decay per unit time
- *half-life*: mean time taken for the number/activity of nuclei to decrease to half of the initial value
- *count rate*: number of radioactive particles registered by a detector per unit time
- *how count rate differs from activity of a sample*: count rate only measures a fraction of activity; there exists background radiation; radiation emitted may be absorbed by air/sample before reaching the detector; product nuclei may be also radioactive; there is dead time for the detector, etc.

13. medical imaging

- *hardness of X-ray beam*: penetrating ability of beam, X-ray photons of higher energy/frequency have greater hardness
- *production of X-ray*: electrons accelerated through a high voltage strike into a target metal, when electrons decelerate, part of kinetic energy loss converts into X-ray photons
- *how to increase the hardness of X-ray*: increasing the accelerating voltage: this leads to more energetic X-ray photons, so X-ray beam becomes more penetrating
- *X-ray spectrum - braking radiation*: incident electrons bombarding target metal have a dis-

杨宇浩词汇表中子 • 核聚变：两个轻的原子核结合成一个较重/较大的原子核 • 为什么聚变需要非常高的温度：带正电的原子核要靠近并聚变在一起，需要高动能来克服它们的电斥力、高 K.E 和高 K.E。高温 • 放射性衰变：不稳定原子核自发且随机地发射电离辐射 (α -、 β - 或 γ - 辐射) • 自发性：衰变速率不受外部条件 (如温度、压力等) 影响 • 随机性：无法准确预测特定原子核何时衰变 • 活度：原子核衰变速率/单位时间衰变事件数量 • 衰变常数：单位时间原子核衰变的概率 • 半衰期：平均时间原子核的数量/活度减少到初始值的一半时所采取的方法 • 计数率：每单位时间探测器记录的放射性粒子数量 • 计数率与样品活度有何不同：计数率仅测量活度的一小部分；存在背景辐射；发出的辐射在到达检测器之前可能会被空气/样品吸收；产物核也可能具有放射性； 13. 医学成像 • X 射线束的硬度：光束的穿透能力，能量/频率越高的 X 射线光子硬度越大 • X 射线的产生：电子通过高压撞击目标金属加速，当电子减速时，部分动能损失转化为 X 射线光子 • 如何提高 X 射线的硬度：增加加速电压：这会产生更多能量的 X 射线光子，因此 X 射线光束变得更具穿透力 • X 射线光谱 - 制动辐射：入射电子轰击目标金属会产生放电 - 234

tribution of deceleration/K.E. loss, producing X-ray photons with a distribution of energies/frequencies, corresponding to a continuous background spectrum

- *X-ray spectrum - characteristic radiation*: incident electrons excite electrons from inner shells of the target metal atoms, de-excitation of electrons from higher levels emits X-rays of particular frequencies, electron energy levels are discrete so this gives rise to a discrete characteristic spectrum
- *why filtering out low-frequency X-rays*: low-frequency X-ray photons are not very penetrating and are mostly absorbed by patient's body, they do not contribute to imaging but might cause tissue damages, so should filter out using aluminium plates
- *sharpness*: ease with which edges of structures can be seen on an image
- *contrast*: difference in blackening/colour/intensity for different structures on an image
- *how to improve sharpness of X-ray images*: reduce the size of anode of the X-ray tube; collimate beam by a set of lead slits; use anti-scatter screen; let patient stay still, etc.
- *how to improve contrast of X-ray images*: use contrast media (iodine-based media, barium meal, etc.) to turn the tissue into a better absorber of X-rays
- *attenuation of X-ray/ultrasound*: decrease in intensity of X-ray/ultrasound as it passes through a substance
- *absorption/attenuation coefficient*: as a parallel beam of radiation/wave passes through matter, its intensity I decreases as $I = I_0 e^{-\mu x}$, where I_0 is the initial incident intensity, x is the distance the beam has travelled, μ represents the absorption/attenuation coefficient
- *production of ultrasounds*: apply a high-frequency alternating voltage across a piezo-electric crystal of optimum thickness ($d = \frac{\lambda}{2}$), it vibrates and produces ultrasound waves with greatest intensity/amplitude
- *detection of ultrasounds*: ultrasound exerts a varying pressure onto a piezo-electric crystal and generates a varying voltage across the crystal, this voltage can be measured
- *(specific) acoustic impedance*: density of a medium $\times$ speed of sound wave travelling in that medium ($Z = \rho c$)
- *impedance matching/why using gel in ultrasonic scans*: reflection coefficient for air-tissue bound-

杨宇豪术语表减速贡献/K.E.损耗，产生具有能量/频率分布的 X 射线光子，对应于连续的背景谱 • X 射线谱 - 特征辐射：入射电子从目标金属原子的内壳激发电子，较高能级的电子去激发会发射特定频率的 X 射线，电子能级是离散的，因此会产生离散的特征谱 • 为什么要滤除低频 X 射线：低频 X 射线光子的穿透力不强，大部分被患者身体吸收，对成像没有帮助，但可能会造成组织损伤，因此应使用铝板过滤掉。通过一组铅狭缝准直光束；使用防散射屏；让患者保持静止等 • 如何提高 X 射线图像的对比度：使用造影剂（碘基介质、钡餐等），使组织成为更好的 X 射线吸收体 • X 射线/超声波的衰减：X 射线/超声波穿过物质时强度降低 • 吸收/衰减系数：当平行的辐射束/波穿过物质时，其强度 I 随着 I_0 的变化而减小 $I = I_0 e^{-\mu x}$ ，其中 I_0 是初始入射强度， x 是光束行进的距离， μ 代表吸收/衰减系数 • 超声波的产生：在最佳厚度 ($d = \lambda/2$) 的压电晶体上施加高频交流电压，压电晶体振动并产生具有最大强度/振幅的超声波 • 超声波的检测：超声波对压电晶体施加变化的压力并产生变化的电压通过晶体，可以测量该电压 • (比) 声阻抗：介质的密度 $\times$ 声波在该介质中传播的速度 ($Z = \rho c$) • 阻抗匹配/为什么在超声波扫描中使用凝胶：空气组织边界的反射系数 - 235

ary is nearly 100%, waves are almost completely reflected and cannot enter patient's body, but at gel-tissue boundary, we have almost complete transmission, so transducer is coupled to skin using gel

- *positron*: anti-particle of electron (same mass as electron, but carrying a charge of $+e$)
- *annihilation*: when a particle meets its anti-particle, their mass-energy is converted into electromagnetic energy/photons
- *why γ -photons produced in annihilation events have same energy but move in opposite directions*: conservation of total momentum requires sum of the momentum for the γ -photon pair must be zero, so two γ -photons carry equal but opposite momenta, equal momentum means equal energy, and opposite momentum means the two γ -photons travel at 180° to each other
- *tracer/radiotracer*: β^+ -emitting radioactive nuclides attached/tagged to molecules that are taken up by human body
- *line of response*: straight line drawn between the two detectors detecting the γ -photon pair produced by electron-positron annihilation
- *principles of CT imaging, ultrasound imaging, PET imaging*: see appendix

14. astrophysics

- *light-year*: distance travelled by light in vacuum in one year
- *luminosity*: total radiation power output from an object/star
- *radiant flux intensity*: power transmitted/received at right angles per unit area
- *standard candle*: star with a known luminosity, can be used to determine the distance of galaxies
- *examples of standard candles*: cepheid variable, type IA supernova
- *black body*: an idealised object that absorbs all electromagnetic radiation
- *black body radiation*: a black body emits electromagnetic radiation with a range of wavelengths (depending on its temperature)
- *Wien's law*: wavelength at the peak intensity of a black body spectrum is inversely proportional to the surface temperature ($\lambda_{\text{peak}} T = \text{constant}$)

Yuhao Yang 术语表接近100%，波几乎完全反射，无法进入患者体内，但在凝胶组织边界，我们几乎完全传输，因此换能器使用凝胶耦合到皮肤 • 正电子：电子的反粒子（与电子质量相同，但携带 +e 的电荷） • 湮灭：当粒子遇到其反粒子时，其质能转换为电磁能/光子 • 为什么是 γ 光子湮没事件中产生的能量相同，但运动方向相反：总动量守恒要求 γ 光子对的动量总和必须为零，因此两个 γ 光子具有相等但方向相反的动量，动量相等意味着能量相等，动量相反意味着两个 γ 光子以 180° 相互传播 • 示踪剂/放射性示踪剂：附着/标记到人体吸收的分子上的 β^+ 发射放射性核素 • 响应线：两个探测器之间绘制的直线，检测正负电子湮灭产生的 γ 光子对 • CT 成像、超声成像、PET 成像原理：参见附录 14 天体物理学 • 光年：光在真空中一年内传播的距离 • 光度：物体/恒星的总辐射功率输出 • 辐射通量强度：每单位面积以直角发射/接收的功率 • 标准烛光：已知光度的恒星，可用于确定星系距离 • 标准烛光示例：造父变星，IA 型超新星 • 黑体：吸收所有电磁辐射的理想化物体 • 黑体辐射：黑体发射一定波长的电磁辐射（取决于其温度） • 维恩定律：黑体光谱峰值强度处的波长与表面温度成反比（ $\lambda_{\text{peak}}T = \text{常数}$ ） 236

- *redshift*: line spectra of stars/galaxies appear to have longer wavelengths due to their recession motion
- *how redshift is determined*: spectrum of a star contains characteristic absorption lines that depend on its chemical composition, by comparing the spectral lines of the star with the absorption spectrum of the same element as measured in a lab, the amount of redshift can be determined
- *Hubble's law*: recession speed of galaxies is proportional to the distance ($v = H_0 d$)
- *Big Bang*: an event where the universe was created from an extremely hot and dense state
- *evidence for the Big Bang Theory*: Hubble's law / expansion of the universe, cosmic microwave background radiation
- *how Hubble's law leads to the Big Bang model*: Hubble's law suggests more distant galaxies are moving away at higher speeds, this implies an expanding universe, so in the past the galaxies used to be very close together, the universe should have started from a hot and dense state
- *cosmic microwave background (CMB) radiation*: leftover/remnant radiation from the Big Bang that fills the entire universe: as universe expands, average temperature drops, and this should give a spectrum with peak intensity at the microwave band

杨宇豪词汇表 • 红移：恒星/星系的线光谱由于其退行运动而显得具有更长的波长 • 如何确定红移：恒星的光谱包含取决于其化学成分的特征吸收线，通过将恒星的光谱线与实验室测量的相同元素的吸收光谱进行比较，可以确定红移量 • 哈勃定律：星系的退行速度与距离成正比 ($v = H_0 d$) • 大爆炸：宇宙从极热和致密状态产生的事件 • 大爆炸理论的证据：哈勃定律/宇宙膨胀、宇宙微波背景辐射 • 哈勃定律如何得出大爆炸模型：哈勃定律表明更遥远的星系正在以更高的速度远离，这意味着宇宙正在膨胀，所以在过去，星系曾经非常接近，宇宙应该从炎热和致密的状态开始 • 宇宙微波背景(CMB) 辐射：充满整个宇宙的大爆炸的剩余辐射：随着宇宙膨胀，平均温度下降，这应该给出在微波波段 237 处具有峰值强度的光谱

Medical Physics Principles

production of X-ray beams

high-speed electrons accelerated in an electric field smash into an anode target metal

electrons decelerate and lose kinetic energy to give out X-ray photons

X-ray spectrum shows a few discrete spikes superimposed on a continuous background

there is a range of deceleration for electrons, so X-ray photons produced have a distribution of energies, so a continuous background spectrum (braking radiation)

orbiting electrons in metal atom can be knocked out, de-excitation of electrons from higher levels gives X-ray photons of specific energies, so a discrete line spectrum (characteristic radiation)

principle of CT (computerised tomography) scans

patients body is considered to consist of many 2D slices, each slice contains many voxels

X-rays are incident from different angles for one slice, transmitted intensities are used to give information about each voxel with aid of a computer, 2D slice image is obtained

repeat the process, many 2D slice images obtained can be combined to form a 3D image

3D image can be rotated and viewed from different angles

production of ultrasound

apply alternating voltages across piezo-electric crystal/quartz

crystal is made to vibrate and give out sound waves

if frequency of a.c. voltage is higher than 20,000 Hz, waves produced are ultrasounds

if crystal is cut into an optimum thickness (half wavelength of ultrasound), it can resonate to give ultrasound with maximum intensity

detection of ultrasound

ultrasound exerts a varying pressure onto a piezo-electric crystal/quartz

杨宇豪术语表医学物理原理产生X射线束高速电子在电场中加速撞击阳极目标金属电子减速并失去动能以发出X射线光子X射线光谱显示在连续背景上叠加了一些离散尖峰电子有一定的减速范围，因此产生的X射线光子具有能量分布，因此可以得到围绕金属原子中电子运行的连续背景光谱（制动辐射）较高能级的电子去激发产生特定能量的X射线光子，因此CT（计算机断层扫描）扫描患者身体的离散线谱（特征辐射）原理被认为由许多2D切片组成，每个切片包含许多体素X射线从一个切片的不同角度入射，透射强度用于借助计算机提供有关每个体素的信息，2D切片图像是重复该过程，获得的许多2D切片图像可以组合形成3D图像3D图像可以旋转并从不同角度观看超声波的产生在压电晶体/石英晶体上施加交流电压，如果交流频率为100° C，则使压电晶体/石英晶体振动并发出声波。电压高于20,000 Hz，产生的波是超声波如果晶体被切割成最佳厚度（超声波的半波长），它可以共振产生具有最大强度的超声波超声波检测超声波对压电晶体/石英施加不同的压力 238

a varying voltage is generated across the crystal, which is detected and measured
amplitude of this voltage gives information about intensity of the ultrasound
frequency of this voltage gives information about frequency of the ultrasound

principle of ultrasonic scans

ultrasonic pulses produced from piezo-electric transducer are sent into patients body
reflected waves/echoes from boundaries of different media are received and detected by the
same transducer, signals are processed and displayed
arrival times of reflected waves can give information about depth/thickness of structures
intensity of the reflected signals depend on acoustic impedance and attenuation coefficient
of media, so can give information about nature of tissues

PET imaging

tracer with β^+ -emitter is injected into patient's body
tracer accumulates in certain organ/tissue, and emits positrons
positrons annihilate with nearby electrons to produces γ -photon pairs
 γ -photons travel in opposite directions to each other
 γ -photons are captured by a ring of detectors, line of response is established
difference in arrival time can be used to determine exact location of annihilation event
number of events per unit time gives information about concentration of the tracer

Yuhao Yang 术语表晶体上会产生变化的电压，检测并测量该电压的幅度，可提供有关超声波强度的信息该电压的频率可提供有关超声波频率的信息超声波扫描原理压电换能器产生的超声波脉冲被发送到患者体内来自不同介质边界的反射波/回声由同一换能器接收和检测，信号经过处理并显示反射波的到达时间可提供有关结构深度/厚度的信息反射强度信号取决于声阻抗和介质的衰减系数，因此可以提供有关组织性质的信息将带有 β^+ 发射器的 PET 成像示踪剂注入患者体内示踪剂在某些器官/组织中积聚，并发射正电子正电子与附近的电子湮灭，产生 γ 光子对 γ 光子以彼此相反的方向传播 γ 光子被探测器环捕获，建立响应线到达时间的差异可表示为用于确定湮灭事件的准确位置每单位时间的事件数量给出了有关示踪剂浓度的信息

239

Appendix B. List of Formulae

$\omega = 2\pi f = \frac{2\pi}{T}$	angular speed/frequency and period
$v = \omega r$	relating linear speed with angular speed for circular motion
$F_c = \frac{mv^2}{r}$ or $m\omega^2 r$	centripetal force required for an object undergoing circular motion
$F = \frac{GMm}{r^2}$	gravitational force between two masses at distance r , called Newton's law of gravitation
$g = \frac{F}{m}$	defining equation for gravitational field strength
$g = \frac{GM}{r^2}$	gravitational field strength due to point mass M at distance r
$E_p = -\frac{GMm}{r}$	gravitational potential energy between two point masses separated by r
$\varphi = \frac{E_p}{m}$	defining equation for gravitational potential
$\varphi = -\frac{GM}{r}$	gravitational potential at position r due to mass M
$\frac{GMm}{r^2} = m\omega^2 r$	gravity provides centripetal force for planets/satellites
$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0$	to find minimum speed to escape from gravitational field to infinity
$a = -\omega^2 x$	defining equation of simple harmonic motion (S.H.M.)
$x = x_0 \sin(\omega t + \phi)$	variation of displacement of S.H.M. with time
$v_{\max} = \omega x_0$	maximum speed of S.H.M. oscillator
$v = \omega \sqrt{x_0^2 - x^2}$	speed-displacement relation for S.H.M. oscillator
$\omega = \sqrt{\frac{k}{m}}$	angular frequency of a mass-spring oscillator
$E = \frac{1}{2}m\omega^2 x_0^2$	total energy of S.H.M. oscillator
$U = E_k + E_p$	internal energy as sum of kinetic and potential energy
$\Delta U = Q + W$	first law of thermodynamics: change in internal energy equals heat and work done to system
$Q = cm\Delta T$	heat to raise temperature of substance without phase transition
$Q = Lm$	latent heat to change state of substance
$W = p\Delta V$	work done by/on gas at constant pressure
$pV = nRT$	ideal gas equation in terms of amount of substance n
$pV = NkT$	ideal gas equation in terms of total number of molecules N

Yuhao Yang 公式列表附录 B. 公式列表 $\omega = 2\pi f = 2\pi/T$ 角速度/频率和周期 $v = \omega r$ 将线速度与圆周运动的角速度联系起来 $F_c = mv^2/r$ 或 $m\omega^2 r$ 进行圆周运动的物体所需的向心力 $F = G M m/r^2$ 距离为 r 的两个质量之间的引力, 称为牛顿定律引力 $g = F/m$ 定义引力场强度方程 $g = G M/r^2$ 由距离 r 处的点质量 M 引起的引力场强度 $E_p = -G M m/r$ 相距 r 的两个点质量之间的引力势能 $\phi = E_p/m$ 引力势定义方程 $\phi = -G M/r$ 由于质量 M 引起的位置 r 处的引力势能 $G M m/r^2 = m\omega^2 r$ 引力为行星/卫星提供向心力 $1/2 mv^2 - G M m/r = 0$ 找到从引力场逃逸到无穷大的最小速度 $a = -\omega^2 x$ 定义简谐振动方程 (S.H.M.) $x = x_0 \sin(\omega t + \phi)$ S.H.M. 位移变化随着时间 $v_{\max} = \omega x_0$ S.H.M. 的最大速度 S.H.M. 振荡器 $v = \omega \sqrt{x_0^2 - x^2}$ 速度-位移关系 振荡器 $\omega = \sqrt{k/m}$ 质量弹簧振荡器的角频率 $E = 1/2 m\omega^2 x_0^2$ S.H.M. 的总能量 振荡器 $U = E_k + E_p$ 内能为动能和势能之和 $\Delta U = Q + W$ 热力学第一定律: 内能的变化等于热量和对系统所做的功 $Q = cm \Delta T$ 使物质温度升高而无相变的热量 $Q = Lm$ 改变物质状态的潜热 $W = p \Delta V$ 恒压下气体所做的功 $pV = nRT$ 以物质的量 n pV 表示的理想气体方程 $= N kT$ 以分子总数 N 表示的理想气体方程 N 240

$N = nN_A = \frac{m}{M}N_A$	number of molecules N , amount of substance n , mass of one molecule m and molar mass M
$p = \frac{Nm\langle v^2 \rangle}{3V}$	pressure of ideal gas determined by r.m.s. speed of particles
$p = \frac{1}{3}\rho\langle v^2 \rangle$	ideal gas pressure in terms of r.m.s. speed of particles and gas density
$E_k = \frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}kT$	mean kinetic energy of particles at a particular temperature
$F = \frac{Qq}{4\pi\epsilon_0 r^2}$	electric force between two point charges, called Coulomb's law
$E = \frac{F}{q}$	defining equation for electric field strength
$E = \frac{V}{d}$	electric field strength between two oppositely-charged metal plates
$E = \frac{Q}{4\pi\epsilon_0 r^2}$	electric field strength due to point charge Q at distance r
$E_p = \frac{Qq}{4\pi\epsilon_0 r}$	electric potential energy between two point charges separated by r
$V = \frac{E_p}{q}$	defining equation for electric potential
$V = \frac{Q}{4\pi\epsilon_0 r}$	electric potential at position r due to point charge Q
$E = -\frac{dV}{dr}, g = -\frac{d\phi}{dr}$	electric/gravitational field strength equals negative potential gradient
$C = \frac{Q}{V}$	defining capacitance as ratio of charge to potential
$C = C_1 + C_2 + \dots$	capacitance of a network of capacitors in parallel
$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	capacitance of a network of capacitors in series
$C = 4\pi\epsilon_0 R$	capacitance of a conducting sphere of radius R
$W = \frac{QV}{2}$ or $\frac{CV^2}{2}$ or $\frac{Q^2}{2C}$	energy stored in a capacitor
$I = I_0 e^{-\frac{t}{RC}}$	variation of current when charging/discharging a capacitor
$V = V_0 \left(1 - e^{-\frac{t}{RC}}\right)$	increase in voltage when a capacitor is being charged
$V = V_0 e^{-\frac{t}{RC}}$	drop in voltage when a capacitor is being discharged
$F = BIl(\sin\theta)$	magnetic force on current in straight wire (at angle θ) to magnetic field
$F = Bqv(\sin\theta)$	magnetic force on moving charge (at angle θ) in magnetic field
$Bqv = \frac{mv^2}{r}$	magnetic force provides centripetal force for particle moving in uniform magnetic field
$qV = \frac{1}{2}mv^2$	kinetic energy gain of charged particle through accelerating voltage
$qE = Bqv$	electric force on charged particles in equilibrium with magnetic force in velocity selector
$\phi = BA(\cos\theta)$	magnetic flux through cross-sectional area (at angle θ to normal of this area)

Yuhao Yang 公式列表 $N = nN_A = m/M$ N 分子数 n 、物质的量 n 、一个分子的质量 m 和摩尔质量 M $p = N m \overline{v^2} / 3$ 由均方根确定的理想气体的 3V 压力。粒子速度 $p = 1/3 \rho \overline{v^2}$ 以均方根表示的理想气体压力
 子速度和气体密度 $E_k = 1/2 m \overline{v^2} = 3/2 kT$ 特定温度下粒子的平均动能 $F = Qq / 4\pi\epsilon_0 r^2$ 两个点电荷之间的
 电力，称为库仑定律 $E = F/q$ 电场强度的定义方程 $E = V/d$ 两个带相反电荷的金属板之间的电场强度 $E = Q / 4\pi\epsilon_0 r^2$
 距离 r $E_p = Qq / r^2$ 处的点电荷 Q 产生的电场强度 $4\pi\epsilon_0 r$ 由 r 分隔的两个点电荷之间的电势能 $V = E_p / q$
 定义电势方程 $V = Q / 4\pi\epsilon_0 r$ 位置 r 处由于点电荷 Q $E = -dV/dr$ 产生的电势 $Q E = -dV/dr$, $g = -d\phi/dr$
 电场/重力场强度等于负电势梯度 $C = Q/V$ 将电容定义为电荷与电势之比 $C = C_1 + C_2 + \dots$ 并联电容器网络的电容
 $1/C = 1/C_1 + 1/C_2 + \dots$ 串联电容器网络的电容 $C = 4\pi\epsilon_0 R$ 半径为 R $W = QV/2$ 或 $CV^2/2$ 或 $Q^2/2C$ 能量的导电球体的电容
 存储在电容器中的电流 $I = I_0 e^{-t/RC}$ 电容器充电/放电时的电流变化 $V = V_0 (1 - e^{-t/RC})$ 电容器充电时的电压增加
 $V = V_0 e^{-t/RC}$ 电容器放电时的电压下降 $F = B I l (\sin \theta)$ 与磁场成直角 (θ 角) 的电流上的磁力 $F = B q v (\sin \theta)$
 磁场中移动电荷 (角度 θ) 上的磁力 $B q v = m v^2 / r$ 磁力为在均匀磁场中运动的粒子提供向心力 $qV = 1/2 m v^2$
 带电粒子通过加速电压获得动能 $qE = B q v$ 与速度选择器中的磁力平衡的带电粒子上的电力 $\phi = B A (\cos \theta)$ 通过横截面积的磁通量 (与该区域法线成角度 θ) 241

$\Phi = NBA$	magnetic flux linkage through coil with N turns
$\mathcal{E} = \frac{d\Phi}{dt}$	Faraday's law, induced e.m.f. equals changing rate of flux (linkage)
$\mathcal{E} = Blv$	induced e.m.f. as wire cuts across magnetic field lines at certain speed
$I = nAvq$	current produced by charge carriers with number density n at mean drift velocity v
$V_H = Bvd$	Hall voltage in a material with depth d when charge carriers move with drift velocity v
$V_H = \frac{BI}{ntq}$	Hall voltage across bulk conductor with thickness t when a current flows through it
$V = V_0 \sin \omega t$	an expression (not generic) for sinusoidal alternating voltage
$V_{\text{r.m.s}} = \frac{V_0}{\sqrt{2}}$	r.m.s. voltage of sinusoidal alternating current
$P = \frac{V_{\text{r.m.s}}^2}{R}$ or $I_{\text{r.m.s}}^2 R$	mean power output of alternating current to load resistor
$\frac{V_s}{V_p} = \frac{N_s}{N_p}$	turns-ratio equation for transformers
$I_p V_p = I_s V_s$	current-voltage relation for ideal transformer (no power loss)
$E = hf$ or $\frac{hc}{\lambda}$	energy of single photon, called Einstein's relation
$p = \frac{h}{\lambda}$	momentum of a photon
$hf = \Phi + E_{k,\text{max}}$	photoelectric equation: photon energy = work function + K.E. of emitted electron
$E_{\text{high}} - E_{\text{low}} = hf$	energy level change of electrons and emission/absorption of photons
$\lambda = \frac{h}{p} = \frac{h}{mv}$	de Broglie wavelength of particle associated with momentum
$E = mc^2$	Einstein's mass-energy equivalence
$E_B = \Delta mc^2$	nuclear binding energy and mass defect
$A = \frac{\Delta N}{\Delta t}$	activity defined as rate of radioactive decay
$A = \lambda N$	activity related to decay constant
$N = N_0 e^{-\lambda t}$	number of undecayed nuclei decreases exponentially with time
$A = A_0 e^{-\lambda t}$	activity of radioactive sample decreases exponentially with time
$t_{1/2} = \frac{\ln 2}{\lambda}$	half-life and decay constant
$I = I_0 e^{-\mu x}$	attenuation of X-rays/ultrasound through medium of distance x
$x_{1/2} = \frac{\ln 2}{\mu}$	half-value thickness and attenuation coefficient
$Z = \rho c$	acoustic impedance for a medium

Yuhao Yang 公式列表 $\Phi = N B$ 通过 N 匝线圈的磁通链 $E = d\Phi / dt$ 法拉第定律，感应电动势。等于磁通变化率（联动） $E = B l v$ 感应电动势。当导线以一定速度穿过磁场线时 $I = n A v q$ 数量密度为 n 的电荷载流子在平均漂移速度下产生的电流 $v_H = B v_d$ 当载流子以漂移速度移动时，深度为 d 的材料中的霍尔电压 $v_H = B I n t q$ 当电流流过厚度为 t 的体导体时的霍尔电压 $V = V_0 \sin \omega t$ 正弦交流电压 $V_{r.m.s}$ 的表达式（非通用） $= V_0 / \sqrt{2}$ 正弦交流电压 $P = V_{r.m.s} I_{r.m.s}$ 或 $I^2 R$ 负载电阻的交流电平均功率输出 $V_s V_p = N_s N_p$ 变压器匝数比方程 $I_p V_p = I_s V_s$ 理想变压器的电流电压关系（无功率损耗） $E = h f$ 或 $h c / \lambda$ 单光子能量，称为爱因斯坦关系式 $p = h / \lambda$ 光子动量 $h f = \Phi + E_{k,max}$ 光电方程：光子能量 = 功函数 + $K.E.$ 发射电子的 $E_{high} - E_{low} = h f$ 电子的能级变化和光子的发射/吸收 $\lambda = h / p = h / m v$ 与动量相关的粒子的德布罗意波长 $E = m c^2$ 爱因斯坦的质能等价 $E_B = \Delta m c^2$ 核结合能和质量缺陷 $A = \Delta N / \Delta t$ 活度定义为放射性衰变率 $A = \lambda N$ 与衰变常数相关的活度 $N = N_0 e^{-\lambda t}$ 未衰变核数随时间呈指数下降 $A = A_0 e^{-\lambda t}$ 放射性样品的活度随时间呈指数下降 $t_{1/2} = \ln 2 / \lambda$ 半衰期和衰变常数 $I = I_0 e^{-(\mu x)}$ X 射线/超声波通过距离 x $t_{1/2} = \ln 2 / \mu$ (半值厚度和衰减系数 $Z = \rho c$ 介质声阻抗 242

$$\frac{I_r}{I_0} = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$$

ratio of reflected intensity to incident intensity at boundary of media

$$F = \frac{P}{A}$$

definition of radiant flux intensity

$$F = \frac{L}{4\pi d^2}$$

radiant flux intensity related to luminosity L and distance d

$$\lambda_{\text{peak}} T = b$$

Wien's law, relating a star's spectrum to its surface temperature

$$L = \sigma T^4 \times 4\pi R^2$$

Stefan-Boltzmann law, relating a star's luminosity to its surface temperature and its size

$$z = \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

redshift of line spectra due to recession of stars/galaxies

$$v = H_0 d$$

Hubble's law, recession speed of galaxy is approximately proportional to the distance

$$t = \frac{1}{H_0}$$

estimation of age of universe

Yuhao Yang 公式列表
 $I_r/I_0 = (Z_1 - Z_2 / Z_1 + Z_2)^2$ 介质边界处反射强度与入射强度之比
 $F = P$ 辐射通量强度的定义
 $F = L / 4\pi d^2$ 与光度 L 和距离 d 相关的辐射通量强度
 $d\lambda_{\text{peak}}T = b$ 维恩定律，将恒星的光谱与其表面联系起来
温度 $L = \sigma T^4 \times 4\pi R^2$ Stefan-Boltzmann 定律，将恒星的光度与其表面温度及其大小联系起来
 $z = \Delta\lambda / \lambda = v / c$ 由于恒星/星系退行而引起的线光谱红移
 $v = H_0 d$ 哈勃定律，星系的退行速度大约与距离 $t = 1 / H_0$ 宇宙年龄估计 243

Appendix C. Physical Constants

acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$
speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ $\left(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1} \right)$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
unified atomic mass unit	$1\text{u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of proton	$m_p = 1.673 \times 10^{-27} \text{ kg}$
rest mass of neutron	$m_n = 1.675 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Wien displacement constant	$b = 2.90 \times 10^{-3} \text{ m K}$
Stefan-Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

杨宇浩物理常数附录 C. 物理常数自由落体加速度 $g = 9.81 \text{ m s}^{-2}$ 真空中光速 $c = 3.00 \times 10^8 \text{ m s}^{-1}$ 万有引力常数 $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ 自由空间介电常数 $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ ($1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ m F}^{-1}$) 基本电荷 $e = 1.60 \times 10^{-19} \text{ C}$ 统一原子质量单位 $1u = 1.66 \times 10^{-27} \text{ kg}$ 质子静止质量 $m_p = 1.673 \times 10^{-27} \text{ kg}$ 中子静止质量 $m_n = 1.675 \times 10^{-27} \text{ kg}$ 电子静止质量 $m_e = 9.11 \times 10^{-31} \text{ kg}$ 阿伏加德罗常数 $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ 摩尔气体常数 $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$ 玻尔兹曼常数 $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ 普朗克常数 $h = 6.63 \times 10^{-34} \text{ J s}$ 维恩位移常数 $b = 2.90 \times 10^{-3} \text{ m K}$ Stefan-Boltzmann 常数 $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$